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(54) **HETERODIMERIC ANTIBODIES THAT BIND CD3 AND CLDN6**

(71) Applicant: **Xencor, Inc.**, Pasadena, CA (US)

(72) Inventors: **Matthew S. Faber**, Glendora, CA (US); **Sung-Hyung Lee**, San Gabriel, CA (US); **Yoon Kyung Kim**, Pomona, CA (US); **Jing Qi**, Arcadia, CA (US); **Kendra N. Avery**, Hawthorne, CA (US); **Seung Y. Chu**, Upland, CA (US); **Alex Nisthal**, Monrovia, CA (US); **Matthew J. Bennett**, Monrovia, CA (US); **John R. Desjarlais**, Pasadena, CA (US); **Chad Borchert**, Nowthen, MN (US)

(73) Assignee: **Xencor, Inc.**, Pasadena, CA (US)

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Primary Examiner — Nelson B Moseley, II

(74) *Attorney, Agent, or Firm* — Louis-Vu T. Nguyen; Christopher J. Betti; Morgan, Lewis & Bockius, LLP

(57) **ABSTRACT**

Provided herein are novel CLDN6 binding domains, and anti-CLDN6xanti-CD3 antibodies that include such CLDN6 binding domains. Also provided herein are methods of using such antibodies for the treatment of CLDN6-associated cancers.

4 Claims, 137 Drawing Sheets

Specification includes a Sequence Listing.

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Figure 1A

Monomer 1	Monomer 2
F405A	T394F
S364D	Y349K
S364E	L368K
S364E	Y349K
S364F	K370G
S364H	Y349K
S364H	Y349T
S364Y	K370G
T411K	K370E
V397S/F405A	T394F
K370R/T411K	K370E/T411E
L351E/S364D	Y349K/L351K
L351E/S364E	Y349K/L351K
L351E/T366D	L351K/T366K
P395T/V397S/F405A	T394F
S364D/K370G	S364Y/K370R
S364D/T394F	Y349K/F405A
S364E/F405A	Y349K/T394F
S364E/F405S	Y349K/T394Y
S364E/T411E	Y349K/D401K
S364H/D401K	Y349T/T411E
S364H/F405A	Y349T/T394F
S364H/T394F	Y349T/F405A
Y349C/S364E	Y349K/S354C
L351E/S364D/F405A	Y349K/L351K/T394F
L351K/S364H/D401K	Y349T/L351E/T411E
S364E/T411E/F405A	Y349K/T394F/D401K
S364H/D401K/F405A	Y349T/T394F/T411E
S364H/F405A/T411E	Y349T/T394F/D401K

Figure 1B

Monomer 1	Monomer 2
K370E/T411D	T411K
L368E/K409E	L368K
Y349T/T394F/S354C	S364H/F405A/Y349C
T411E	D401K
T411E	D401R/T411R
Q347E/K360E	Q347R
L368E	S364K
L368E/K370S	S364K
L368E/K370T	S364K
L368E/D401R	S364K
L368E/D401N	S364K
L368E	E357S/S364K
L368E	S364K/K409E
L368E	S364K/K409V
L368D	S364K
L368D/K370S	S364K
L368D/K370S	S364K/E357L
L368D/K370S	S364K/E357Q
T411E/K360E/Q362E	D401K
K370S	S364K
L368E/K370S	S364K/E357Q
K370S	S364K/E357Q
T411E/K360D	D401K
T411E/K360E	D401K
T411E/Q362E	D401K
T411E/N390D	D401K
T411E	D401K/Q347K
T411E	D401K/Q347R
T411E/K360D/Q362E	D401K

Figure 1C

Monomer 1	Monomer 2
T411E/K360E/N390D	D401K
T411E/Q362E/N390D	D401K
T411E/Q347R	D401K/K360D
T411E/Q347R	D401K/K360E
T411E/K360	D401K/Q347K
T411E/K360D	D401K/Q347R
T411E/K360E	D401K/Q347K
T411E/K360E	D401K/Q347R
T411E/S364K	D401K/K370S
T411E/K370S	D401K/S364K
Q347E	E357Q
Q347E	E357Q/Q362K
K360D/Q362E	Q347R
K360D/Q362E	D401K
K360D/Q362E	Q347R/D401K
K360E/Q362E	Q347R
K360E/Q362E	D401K
K360E/Q362E	Q347R/D401K
Q362E/N390D	D401K
Q347E/K360D	D401N
K360D	Q347R/N390K
K360D	N390K/D401N
K360E	Y349H
K370S/Q347E	S364K
K370S/E357L	S364K
K370S/E357Q	S364K
K370S/Q347E/E357L	S364K
K370S/Q347E/E357Q	S364K

Figure 1D

Monomer 1	Monomer 2
L368D/K370S/Q347E	S364K
L368D/K370S/E357L	S364K
L368D/K370S/E357Q	S364K
L368D/K370S/Q347E/E357L	S364K
L368D/K370S/Q347E/E357Q	S364K
L368E/K370S/Q347E	S364K
L368E/K370S/E357L	S364K
L368E/K370S/E357Q	S364K
L368E/K370S/Q347E/E357L	S364K
L368E/K370S/Q347E/E357Q	S364K
L368D/K370T/Q347E	S364K
L368D/K370T/E357L	S364K
L368D/K370T/E357Q	S364K
L368D/K370T/Q347E/E357L	S364K
L368D/K370T/Q347E/E357Q	S364K
L368E/K370T/Q347E	S364K
L368E/K370T/E357L	S364K
L368E/K370T/E357Q	S364K
L368E/K370T/Q347E/E357L	S364K
L368E/K370T/Q347E/E357Q	S364K
T411E/Q362E	D401K/T411K
T411E/N390D	D401K/T411K
T411E/Q362E	D401R/T411R
T411E/N390D	D401R/T411R
Y407T	T366Y
F405A	T394W
T366Y/F405A	T394W/Y407T
Y407A	T366W
T366S/L368A/Y407V	T366W
T366S/L368A/Y407V/Y349C	T366W/S354C
T366S/L368A/Y407V/S354C	T366W/Y349C

Figure 1E

Monomer 1	Monomer 2
K392D/K409D	E356K/D399K
K370D/K392D/K409D	E356K/E357K/D399K
I199T/N203D/K247Q/R355Q/N384S/K392N/V397M/Q419E/K447_	Q196K/I199T/P217R/P228R/N276K
I199T/N203D/K247Q/R355Q/N384S/K392N/V397M/Q419E/K447_	Q196K/I199T/N276K
N384S/K392N/V397M/Q419E	N276K
D221E/P228E/L368E	D221R/P228R/K409R
C220E/P228E/L368E	C220R/E224R/P228R/K409R
F405L	K409R
T366I/K392M/T394W	F405A/Y407V
T366V/K409F	L351Y/Y407A
T366A/K392E/K409F/T411E	D399R/S400R/Y407A
L351K	L351E
I199T/N203D/K247Q/R355Q/Q419E/K447_	Q196K/I199T/P217R/P228R/N276K
I199T/N203D/K247Q/R355Q/Q419E/K447_	Q196K/I199T/N276K
I199T/N203D/K274Q/R355Q/N384S/K392N/V397M/Q419E/K447_	
N208D/Q295E/N384D/Q418E/N421D	
N208D/Q295E/Q418E/N421D	
Q196K/I199T/P217R/P228R/N276K	
Q196K/I199T/N276K	
E269Q/E272Q/E283Q/E357Q	
E269Q/E272Q/E283Q	
E269Q/E272Q	
E269Q/E283Q	
E272Q/E283Q	
E269Q	

Figure 2

<u>Variant constant region</u>	<u>Substitutions</u>
pI-ISO(-)	I199T/N203D/K274Q/R355Q/N384S/K392N/N397M/Q419E/K447_
pI_ISO(-)-Fc only	K274Q/R355Q/N384S/K392N/V397M/Q419E/K447_
pI_(-)_isosteric_A	N208D/Q295E/N384D/Q418E/N421D
pI_(-)_isosteric_A-Fc only	Q295E/N384D/Q418E/N421D
pI_(-)_isosteric_B	N208D/Q295E/Q418E/N421D
pI_(-)_isosteric_B-Fc only	Q295E/Q418E/N421D
pI_ISO(+RR)	Q196K/I199T/P217R/P228R/N276K
pI_ISO(+RR)-Fc only	P217R/P228R/N276K
pI_ISO(+)	Q196K/I199T/N276K
pI_ISO(+)-Fc only	N276K
pI_(+)_isosteric_A	E269Q/E272Q/E283Q/E357Q
pI_(+)_isosteric_B	E269Q/E272Q/E283Q
pI_(+)_isosteric_E269Q/E272Q	E269Q/E272Q
pI_(+)_isosteric_E269Q/E283Q	E269Q/E283Q
pI_(+)_isosteric_E272Q/E283Q	E272Q/E283Q
pI_(+)_isosteric_E269Q	E269Q

Figure 3

Ablation Variants

G236R
S239G
S239K
S239Q
S239R
V266D
S267K
S267R
H268K
E269R
299R
299K
K322A
A327G
A327L
A327N
A327Q
L328E
L328R
P329A
P329H
P329K
A330L
A330S/P331S
I332K
I332R
V266D/A327Q
V266D/P329K
S267R/A327Q
S267R/P329K
G236R/L328R
E233P/L234V/L235A/G236_/S239K
E233P/L234V/L235A/G236_/S267K
E233P/L234V/L235A/G236_/S239K/A327G
E233P/L234V/L235A/G236_/S267K/A327G
E233P/L234V/L235A/G236_
S239K/S267K
267K/P329K

Figure 4

Heavy Chain 1 (-) e.g. Fab-Fc	Heavy Chain 2 (+) e.g. scFv-Fc or Fab-scFv-Fc
	C220S
Heterodimer variants L368D/K370S	Heterodimer variants S364K/E357Q
Isosteric pl substitutions N208D/Q295E/N384D/Q418E/N421D	
FcKO E233P/L234V/L235A/G236_/S267K	FcKO E233P/L234V/L235A/G236_/S267K
±M428L/N434S	±M428L/N434S

Figure 5

Positive Charged scFv Linkers

Name	Sequence	Length	Charge	SEQ ID NO:
Gly-Ser 15	GGGGSGGGSGGGGS	15	0	6
Whitlow linker	GSTSGSGKPGSGEGSTKG	18	+1	7
6paxA_1 (+A)	IRPRAIGGSKPRVA	14	+4	8
+B	GKGGSGKGGSGKGGGS	15	+3	9
+C	GGKSGGGKSGGGKGS	15	+3	10
+D	GGGKSGGGKSGGGKS	15	+3	11
+E	GKGKSGKGKSGKGKS	15	+6	12
+F	GGGKSGGGKSGKGGGS	15	+3	13
+G	GKPGSGKPGSGKPGS	15	+3	14
+H	GKPGSGKPGSGKPGSGKPGS	20	+4	1
+I	GKGKSGKGKSGKGKSGKGKS	20	+8	16

Negative Charged scFv Linkers

Name	Sequence	Length	Charge	SEQ ID NO:
Gly-Ser 20	GGGGSGGGSGGGSGGGGS	20	0	17
3hsc_2 (-A)	STAGDTHLGGEDFD	14	-4	18
-B	GEGGSGEGGSGEGGS	15	-3	19
-C	GGEGSGEGGSGEGGS	15	-3	20
-D	GGGESGGGESGGGES	15	-3	21
-E	GEGESGEGESGEGES	15	-6	22
-F	GGGESGGEGSGEGGS	15	-3	23
-G	GEGESGEGESGEGESGEGES	20	-8	24

Additional scFv Linkers

GGGGSGGGSGGGGS	SEQ ID NO:25
GGGGSGGGSGGGSGGGGS	SEQ ID NO:26
GSTSGSGKPGSGEGSTKG	SEQ ID NO:27
PRGASKSGSASQTGSAPGS	SEQ ID NO:28
GTAAAGAGAAGGAAAGAAG	SEQ ID NO:29
GTSGSSGSGSGSGSGGGG	SEQ ID NO:30
GKPGSGKPGSGKPGSGKPGS	SEQ ID NO:15

Figure 6

Useful domain linkers

<u>Name</u>	<u>Sequence</u>	<u>SEQ ID NO:</u>
(GGGGS) ₁ or GGGGS	GGGGS	5
(GGGGS) ₂	GGGGSGGGGS	31
(GGGGS) ₃	GGGGSGGGGSGGGGS	32
(GGGGS) ₄	GGGGSGGGGSGGGGSGGGGS	33
(GGGGS) ₅	GGGGSGGGGSGGGGSGGGGSGGGGS	34
(GGGGS) ₆	GGGGSGGGGSGGGGSGGGGSGGGGSGGGGS	35
(GGGGS) ₇	GGGGSGGGGSGGGGSGGGGSGGGGSGGGGSGGGGS	36
(GGGGA) ₁ or GGGGA	GGGGA	37
(GGGGA) ₂	GGGGAAGGGGA	38
(GGGGA) ₃	GGGGAAGGGGAGGGGA	39
(GGGGA) ₄	GGGGAAGGGGAGGGGAGGGGA	40
(GGGGA) ₅	GGGGAAGGGGAGGGGAGGGGAGGGGA	41
(GGGGA) ₆	GGGGAAGGGGAGGGGAGGGGAGGGGAGGGGA	42
(GGGGA) ₇	GGGGAAGGGGAGGGGAGGGGAGGGGAGGGGAGGGGA	43
30AA-linker	DPALVHQRPAAPPGGGSGGGSGGGSGGG	44
(GKPGS) ₁ or GKPGS	GKPGS	45
(GKPGS) ₅	GKPGSGKPGSGKPGSGKPGSGKPGS	46
(GKPGS) ₆	GKPGSGKPGSGKPGSGKPGSGKPGSGKPGS	47
(GGGES) ₁ or GGGES	GGGES	48
"half hinge"	KTHTCPPCP	49
"full hinge C220S variant"	EPKSSDKTHTCPPCP	50
"flex half hinge"	GGGSGGGGSKTHTCPPCP	51
"charged half hinge1"	GKPGSGKPGSKTHTCPPCP	52
"charged half hinge2"	GKPGSKTHTCPPCP	53

Figure 7A

1 + 1 Fab-scFv-Fc Backbone 1**>Fab-Fc Side (SEQ ID NO: 54)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDVFS
CSVMHEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 55)

EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFS
CSVMHEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 2**>Fab-Fc Side (SEQ ID NO: 56)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDVFS
CSVMHEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 57)

EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFS
CSVMHEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 3**>Fab-Fc Side (SEQ ID NO: 58)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCEVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDVFS
CSVMHEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 59)

EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFS
CSVMHEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 4**>Fab-Fc Side (SEQ ID NO: 60)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNEVSLTCLVKGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSFFLYSKLEVDKSRWEQGDVFS
CSVMHEALHNHYTQKSLSLSPGK

Figure 7B

>scFv-Fc Side (SEQ ID NO: 61)

EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTPPVLDSDGSGFFLYSKLTVDKSRWQQGNVVFSCSVMEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 5**>Fab-Fc Side (SEQ ID NO: 62)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSRDELTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSGFFLYSKLTVDKSRWEQGDVVFSCS
VMHEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 63)

EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDQLTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTPPVLDSDGSGFFLYSKLTVDKSRWQQGNVVFSCSVMEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 6**>Fab-Fc Side (SEQ ID NO: 64)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYASTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSGFFLYSKLTVDKSRWEQGDVVFSCS
VMHEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 65)

EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYASTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREQMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTPPVLDSDGSGFFLYSKLTVDKSRWQQGNVVFSCSVMEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 7**>Fab-Fc Side (SEQ ID NO: 66)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYSSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSGFFLYSKLTVDKSRWEQGDVVFSCS
VMHEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 67)

EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYSSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREQMTKNQVSLTCLVK
GFYPSDIAVEWESNGQPENNYKTPPVLDSDGSGFFLYSKLTVDKSRWQQGNVVFSCSVMEALHNHYTQKSLSLSPGK

Figure 7C

1 + 1 Fab-scFv-Fc Backbone 8**>Fab-Fc Side (SEQ ID NO: 68)**

ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVFSSSLG
TKYTCNVDHKPSDTKVDKRVESKYGPPCPPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVDSQEDPEVQ
FNWYVDGVEVHNAKTKPREEEFNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTIKAKGQPREPQVYTL
PPSQEEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSFFLYSRLTVDKSRWEQGVDFSCSV
HEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 69)

ESKYGPPCPPAPEFLGGPSVFLFPPKPKDTLMISRTPEVTCVVDVDSQEDPEVQFNWYVDGVEVHNAKTKPREEQ
FNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKGLPSSIEKTIKAKGQPREPQVYTLPPSQEQMTKNQVKLTCLVKGF
YPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSRLTVDKSRWQEGNVFSCSVMEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 9**>Fab-Fc Side (SEQ ID NO: 70)**

ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVFSSNFG
TQTYTCNVDHKPSDTKVDKTVKCCVECPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVQF
NWYVDGVEVHNAKTKPREEEFNSTFRVSVLTVVHQDWLNGKEYKCKVSNKGLPAPIEKTISKTKGQPREPQVYTL
PSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPMLDSDGSFFLYSKLTVDKSRWEQGVDFSCSV
HEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 71)

ERKCCVECPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVQFNWYVDGVEVHNAKTKPREEQF
NSTFRVSVLTVVHQDWLNGKEYKCKVSNKGLPAPIEKTISKTKGQPREPQVYTLPPSREQMTKNQVKLTCLVKGF
YPSDIAVEWESNGQPENNYKTPPMLDSDGSFFLYSKLTVDKSRWQQGNVSCSVMEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 10**>Fab-Fc Side (SEQ ID NO: 72)**

ASTKGPSVFPLAPCSRSTSESTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVFSSNFG
TQTYTCNVDHKPSDTKVDKTVKCCVECPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVQF
NWYVDGVEVHNAKTKPREEEFNSTFRVSVLTVVHQDWLNGKEYKCKVSNKGLPAPIEKTISKTKGQPREPQVYTL
PSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPMLDSDGSFFLYSKLTVDKSRWEQGVDFSCSV
HEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 73)

ERKCCVECPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVQFNWYVDGVEVHNAKTKPREEQF
NSTFRVSVLTVVHQDWLNGKEYKCKVSNKGLPAPIEKTISKTKGQPREPQVYTLPPSREQMTKNQVKLTCLVKGF
YPSDIAVEWESNGQPENNYKTPPMLDSDGSFFLYSKLTVDKSRWQQGNVSCSVMEALHNHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 11**>Fab-Fc Side (SEQ ID NO: 74)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVFSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQV
YTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGVDFSC
SVLHEALHSHYTQKSLSLSPGK

Figure 7D

>scFv-Fc Side (SEQ ID NO: 75)

EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREQMTKNQVKLTCLVK
GFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVVFSCSVLHEALHSHYTQKSLSLSPGK

1 + 1 Fab-scFv-Fc Backbone 12**>Fab-Fc Side (SEQ ID NO: 76)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLG
TQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVAGFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVVFSCS
VMHEALHNHYTQKSLSLSPGK

>scFv-Fc Side (SEQ ID NO: 77)

ERKSSDKTHTCPRPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFKWYVDGVEVHNAKTKPRE
EQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREQMTKNQVKLTCLVK
GFYPSDIAVEWESNGQPENNYKTTPPVLDSDGSFFLYSKLTVDKSRWQQGNVVFSCSMHEALHNHYTQKSLSLSPGK

Figure 8A

2 + 1 Fab2-scFv-Fc Backbone 1**>Fab-Fc Side (SEQ ID NO: 78)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFCSS
VMHEALHNHYTQKSLSLSPGK

>Fab-scFv-Fc Side (SEQ ID NO: 79)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSSVMHEALHNHYTQKSLSLSPGK

2 + 1 Fab2-scFv-Fc Backbone 2**>Fab-Fc Side (SEQ ID NO: 80)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFCSS
VMHEALHNHYTQKSLSLSPGK

>Fab-scFv-Fc Side (SEQ ID NO: 81)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSSVMHEALHNHYTQKSLSLSPGK

2 + 1 Fab2-scFv-Fc Backbone 3**>Fab-Fc Side (SEQ ID NO: 82)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCEVSGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFCSS
VMHEALHNHYTQKSLSLSPGK

>Fab-scFv-Fc Side (SEQ ID NO: 83)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSSVMHEALHNHYTQKSLSLSPGK

2 + 1 Fab2-scFv-Fc Backbone 4**>Fab-Fc Side (SEQ ID NO: 84)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTENEVSLTCLVKGFYPSDIAVEWESDGQPENNYKTPPVLDSDGSFFLYSKLEVDKSRWEEQGDVFCSS
VMHEALHNHYTQKSLSLSPGK

Figure 8B

>Fab-scFv-Fc Side (SEQ ID NO: 85)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSDGSGFFLYSKLTVDKSRWQQGNVFSCSVMEALHNHYTQKSLSLSPGK

2 + 1 Fab2-scFv-Fc Backbone 5**>Fab-Fc Side (SEQ ID NO: 86)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSRDELTKNQVSLTCDVSGFYPSDIAVEWESDGGQPENNYKTTTPVLDSDGSGFFLYSKLTVDKSRWEQGDVFSCS
VMHEALHNHYTQKSLSLSPGK

>Fab-scFv-Fc Side (SEQ ID NO: 87)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSRDQLTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSDGSGFFLYSKLTVDKSRWQQGNVFSCSVMEALHNHYTQKSLSLSPGK

2 + 1 Fab2-scFv-Fc Backbone 6**>Fab-Fc Side (SEQ ID NO: 88)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYASTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGGQPENNYKTTTPVLDSDGSGFFLYSKLTVDKSRWEQGDVFSCS
VMHEALHNHYTQKSLSLSPGK

>Fab-scFv-Fc Side (SEQ ID NO: 89)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYASTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREQMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSDGSGFFLYSKLTVDKSRWQQGNVFSCSVMEALHNHYTQKSLSLSPGK

2 + 1 Fab2-scFv-Fc Backbone 7**>Fab-Fc Side (SEQ ID NO: 90)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYSSYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGGQPENNYKTTTPVLDSDGSGFFLYSKLTVDKSRWEQGDVFSCS
VMHEALHNHYTQKSLSLSPGK

>Fab-scFv-Fc Side (SEQ ID NO: 91)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYSSYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREQMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSDGSGFFLYSKLTVDKSRWQQGNVFSCSVMEALHNHYTQKSLSLSPGK

Figure 8C

2 + 1 Fab2-scFv-Fc Backbone 8**>Fab-Fc Side (SEQ ID NO: 92)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDGPENNYKTTTPVLDSGDSFFLYSKLTVDKSRWEQGDVFSCS
VLHEALHSHYTQKSLSLSPGK

>Fab-scFv-Fc Side (SEQ ID NO: 93)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREQMTKNQVCLTKLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSGDSFFLYSKLTVDKSRWQQGNVFSCSVLHEALHSHYTQKSLSLSPGK

2 + 1 Fab2-scFv-Fc Backbone 9**>Fab-Fc Side (SEQ ID NO: 94)**

ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLG
TQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPE
VKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVY
TLPPSREEMTKNQVSLTCDVAGFYPSDIAVEWESNGQPENNYKTTTPVLDSGDSFFLYSKLTVDKSRWQQGNVFSCS
VMHEALHNHYTQKSLSLSPGK

>Fab-scFv-Fc Side (SEQ ID NO: 95)

APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFKWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREQMTKNQVCLTKLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSGDSFFLYSKLTVDKSRWQQGNVFSCSVLHEALHNHYTQKSLSLSPGK

Figure 9

Constant Light Domain – Kappa (SEQ ID NO: 96)

RTVAAPSVFIFPPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNALQSGNSQESVTEQDSKDSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC

Constant Light Domain – Lambda (SEQ ID NO: 97)

GQPKAAPSVTLFPPSSEELQANKATLVCLISDFYPGAVTVAWKADSSPVKAGVETTTPSKQSNNKYAASSYLSLTPEQWKSHRSYSCQVTHEGSTVEKTVAPTECS

Figure 10A

CD3 High – [anti-CD3] H1.30 L1.47 scFv

	Sequence	SEQ ID NO:
scFv (VHVL)	<u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSK</u> <u>YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNFPGDSY</u> <u>VSWFAYWGQGTLLTVSS</u> / <u>GKPGSGKPGSGKPGSGKPGS</u> / <u>QAVVTQEP</u> <u>SLTVSPG</u> <u>GTVTLT</u> <u>CGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGV</u> <u>PARFSGSLL</u> <u>GGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTVL</u>	98
scFv (VLVH)	<u>QAVVTQEP</u> <u>SLTVSPGGT</u> <u>VTLT</u> <u>CGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGV</u> <u>PARFSGSLL</u> <u>GGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTV</u> <u>L</u> / <u>GKPGSGKPGSGKPGSGKPGS</u> / <u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFST</u> <u>YAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQM</u> <u>NSLRAEDTAVYYCVRHGNFPGDSYVSWFAYWGQGTLLTVSS</u>	99
Variable Heavy (vh) Domain	<u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSK</u> <u>YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNFPGDSY</u> <u>VSWFAYWGQGTLLTVSS</u>	100
vhCDR1	<u>TYAMN</u>	101
vhCDR2	<u>RIRSKYNNYATYYADSVKG</u>	102
vhCDR3	<u>HGNFPGDSYVSWFAY</u>	103
Variable Light (vl) Domain	<u>QAVVTQEP</u> <u>SLTVSPGGT</u> <u>VTLT</u> <u>CGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGV</u> <u>PARFSGSLL</u> <u>GGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTV</u> <u>L</u>	104
vlCDR1	<u>GSSTGAVTTSNYAN</u>	105
vlCDR2	<u>GTNKRAP</u>	106
vlCDR3	<u>ALWYSNHWV</u>	107
Linker	<u>GKPGSGKPGSGKPGSGKPGS</u>	108

Figure 10B

CD3 High-Int #1 – [anti-CD3] H1.32 L1.47 scFv

	Sequence	SEQ ID NO:
scFv (VHVL)	<u>EVQLVESGGGLVQPGGSLRLS</u> <u>CAASGFTTFSTYAMNWVRQAPGKGLEWVGRI</u> <u>RSK</u> <u>ANNYATYYADSVKGRFTISR</u> <u>DDSKNTLYLQMNSLRAEDTAVYYCVRHGNF</u> <u>GDSY</u> <u>VSWFAYWGQGLTVTVSS</u> / <u>GKPGSGKPGSGKPGSGKPGS</u> / <u>QAVVTQEP</u> <u>SLTVSPG</u> <u>GTVTLT</u> <u>CGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGVPARFSGS</u> <u>LL</u> <u>GGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTVL</u>	109
scFv (VLVH)	<u>QAVVTQEP</u> <u>SLTVSPGGT</u> <u>VTLT</u> <u>CGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGVPARFSGS</u> <u>LLGGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTV</u> <u>L</u> / <u>GKPGSGKPGSGKPGSGKPGS</u> / <u>EVQLVESGGGLVQPGGSLRLS</u> <u>CAASGFTTFST</u> <u>YAMNWVRQAPGKGLEWVGRI</u> <u>RSKANNYATYYADSVKGRFTISR</u> <u>DDSKNTLYLQM</u> <u>NSLRAEDTAVYYCVRHGNF</u> <u>GDSYVSWFAYWGQGLTVTVSS</u>	110
Variable Heavy (vh) Domain	<u>EVQLVESGGGLVQPGGSLRLS</u> <u>CAASGFTTFSTYAMNWVRQAPGKGLEWVGRI</u> <u>RSK</u> <u>ANNYATYYADSVKGRFTISR</u> <u>DDSKNTLYLQMNSLRAEDTAVYYCVRHGNF</u> <u>GDSY</u> <u>VSWFAYWGQGLTVTVSS</u>	111
vhCDR1	<u>TYAMN</u>	112
vhCDR2	<u>RIRSKANNYATYYADSVKG</u>	113
vhCDR3	<u>HGNFGDSYVSWFAY</u>	114
Variable Light (vl) Domain	<u>QAVVTQEP</u> <u>SLTVSPGGT</u> <u>VTLT</u> <u>CGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGVPARFSGS</u> <u>LLGGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTV</u> <u>L</u>	115
vlCDR1	<u>GSSTGAVTTSNYAN</u>	116
vlCDR2	<u>GTNKRAP</u>	117
vlCDR3	<u>ALWYSNHWV</u>	118
Linker	<u>GKPGSGKPGSGKPGSGKPGS</u>	119

Figure 10C

CD3 High-Int #2 – [anti-CD3] H1.89 L1.47 scFv

	Sequence	SEQ ID NO:
scFv (VHVL)	<u>EVQLVESGGGLVQPGGSLRLS</u> <u>CAASGFTTFSTYAMNWVRQAPGKGLEWVGRI</u> <u>RSK</u> <u>YNNYATYYADSVKGRFTISR</u> <u>DDSKNTLYLQMNSLRAEDTAVYYCVRHGNF</u> <u>GDEY</u> <u>VSWFAYWGQGLVTVSS</u> / <u>GKPGSGKPGSGKPGSGKPGS</u> / <u>QAVVTQEP</u> <u>SLTVSPG</u> <u>GTVTLT</u> <u>CGSSTGAVTTSNYANWVQKPGKSPRGLIGGT</u> <u>NKRAPGV</u> <u>PARFSGS</u> <u>LL</u> <u>GGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTVL</u>	120
scFv (VLVH)	<u>QAVVTQEP</u> <u>SLTVSPGGT</u> <u>VTLT</u> <u>CGSSTGAVTTSNYANWVQKPGKSPRGLIGGT</u> <u>N</u> <u>KRAPGV</u> <u>PARFSGS</u> <u>LLGGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTV</u> <u>L</u> / <u>GKPGSGKPGSGKPGSGKPGS</u> / <u>EVQLVESGGGLVQPGGSLRLS</u> <u>CAASGFTTFST</u> <u>YAMNWVRQAPGKGLEWVGRI</u> <u>RSKYNNYATYYADSVKGRFTISR</u> <u>DDSKNTLYLQM</u> <u>NSLRAEDTAVYYCVRHGNF</u> <u>GDEYVSWFAYWGQGLVTVSS</u>	121
Variable Heavy (vh) Domain	<u>EVQLVESGGGLVQPGGSLRLS</u> <u>CAASGFTTFSTYAMNWVRQAPGKGLEWVGRI</u> <u>RSK</u> <u>YNNYATYYADSVKGRFTISR</u> <u>DDSKNTLYLQMNSLRAEDTAVYYCVRHGNF</u> <u>GDEY</u> <u>VSWFAYWGQGLVTVSS</u>	122
vhCDR1	<u>TYAMN</u>	123
vhCDR2	<u>RIRSKYNNYATYYADSVKG</u>	124
vhCDR3	<u>HGNFGDEYVSWFAY</u>	125
Variable Light (vl) Domain	<u>QAVVTQEP</u> <u>SLTVSPGGT</u> <u>VTLT</u> <u>CGSSTGAVTTSNYANWVQKPGKSPRGLIGGT</u> <u>N</u> <u>KRAPGV</u> <u>PARFSGS</u> <u>LLGGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTV</u> <u>L</u>	126
vlCDR1	<u>GSSTGAVTTSNYAN</u>	127
vlCDR2	<u>GTNKRAP</u>	128
vlCDR3	<u>ALWYSNHWV</u>	129
Linker	<u>GKPGSGKPGSGKPGSGKPGS</u>	130

Figure 10D

CD3 High-Int #3 – [anti-CD3] H1.90 L1.47 scFv

	Sequence	SEQ ID NO:
scFv (VHVL)	<u>EVQLVESGGGLVQPGGSLRLS</u> <u>CAASGFTFSTYAMNWVRQAPGKGLEWVGRI</u> <u>RSK</u> <u>YNNYATYYADSVKGRFTISR</u> <u>DDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDPY</u> <u>VSWFAYWGQGLTVTVSS</u> / <u>GKPGSGKPGSGKPGSGKPGS</u> / <u>QAVVTQEPSLTVSPG</u> <u>GTVTLT</u> <u>CGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGVPARFSGSLL</u> <u>GGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTVL</u>	131
scFv (VLVH)	<u>QAVVTQEPSLTVSPGGTVTLT</u> <u>CGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTV</u> <u>L/GKPGSGKPGSGKPGSGKPGS</u> / <u>EVQLVESGGGLVQPGGSLRLS</u> <u>CAASGFTFST</u> <u>YAMNWVRQAPGKGLEWVGRI</u> <u>RSKYNNYATYYADSVKGRFTISR</u> <u>DDSKNTLYLQM</u> <u>NSLRAEDTAVYYCVRHGNFGDPYVSWFAYWGQGLTVTVSS</u>	132
Variable Heavy (vh) Domain	<u>EVQLVESGGGLVQPGGSLRLS</u> <u>CAASGFTFSTYAMNWVRQAPGKGLEWVGRI</u> <u>RSK</u> <u>YNNYATYYADSVKGRFTISR</u> <u>DDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDPY</u> <u>VSWFAYWGQGLTVTVSS</u>	133
vhCDR1	<u>TYAMN</u>	134
vhCDR2	<u>RIRSKYNNYATYYADSVKG</u>	135
vhCDR3	<u>HGNFGDPYVSWFAY</u>	136
Variable Light (vl) Domain	<u>QAVVTQEPSLTVSPGGTVTLT</u> <u>CGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTV</u> <u>L</u>	137
vlCDR1	<u>GSSTGAVTTSNYAN</u>	138
vlCDR2	<u>GTNKRAP</u>	139
vlCDR3	<u>ALWYSNHWV</u>	140
Linker	<u>GKPGSGKPGSGKPGSGKPGS</u>	141

Figure 10E

CD3-Intermediate – [anti-CD3] H1.33 L1.47 scFv

	Sequence	SEQ ID NO:
scFv (VHVL)	<u>EVQLVESGGGLVQPGGSLRLS</u> <u>CAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSK</u> <u>YNNYATYYADSVKGRFTISR</u> <u>DDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDSY</u> <u>VSWFDYWGQGT</u> <u>LVTVSS</u> / <u>GKPGSGKPGSGKPGSGKPGS</u> / <u>QAVVTQEP</u> <u>SLTVSPG</u> <u>GTVTLT</u> <u>CGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGV</u> <u>PARFSGSLL</u> <u>GGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTVL</u>	142
scFv (VLVH)	<u>QAVVTQEP</u> <u>SLTVSPGGT</u> <u>VTLT</u> <u>CGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGV</u> <u>PARFSGSLL</u> <u>GGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTV</u> <u>L</u> / <u>GKPGSGKPGSGKPGSGKPGS</u> / <u>EVQLVESGGGLVQPGGSLRLS</u> <u>CAASGFTTFST</u> <u>YAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISR</u> <u>DDSKNTLYLQM</u> <u>NSLRAEDTAVYYCVRHGNFGDSYVSWFDYWGQGT</u> <u>LVTVSS</u>	143
Variable Heavy (vh) Domain	<u>EVQLVESGGGLVQPGGSLRLS</u> <u>CAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSK</u> <u>YNNYATYYADSVKGRFTISR</u> <u>DDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDSY</u> <u>VSWFDYWGQGT</u> <u>LVTVSS</u>	144
vhCDR1	<u>TYAMN</u>	145
vhCDR2	<u>RIRSKYNNYATYYADSVKG</u>	146
vhCDR3	<u>HGNFGDSYVSWFDY</u>	147
Variable Light (vl) Domain	<u>QAVVTQEP</u> <u>SLTVSPGGT</u> <u>VTLT</u> <u>CGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGV</u> <u>PARFSGSLL</u> <u>GGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTV</u> <u>L</u>	148
vlCDR1	<u>GSSTGAVTTSNYAN</u>	149
vlCDR2	<u>GTNKRAP</u>	150
vlCDR3	<u>ALWYSNHWV</u>	151
Linker	<u>GKPGSGKPGSGKPGSGKPGS</u>	152

Figure 10F

CD3 Low – [anti-CD3] H1.31 L1.47 scFv

	Sequence	SEQ ID NO:
scFv (VHVL)	<u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMSWVRQAPGKGLEWVGRIRSK</u> <u>YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDSY</u> <u>VSWFAYWGQGT</u> <u>LVTVSS</u> / <u>GKPGSGKPGSGKPGSGKPGS</u> / <u>QAVVTQEPSLTVSPG</u> <u>GTVTLTGSSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLL</u> <u>GGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTVL</u>	153
scFv (VLVH)	<u>QAVVTQEPSLTVSPGGTVTLT</u> <u>CGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTV</u> <u>L</u> / <u>GKPGSGKPGSGKPGSGKPGS</u> / <u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFST</u> <u>YAMSWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQM</u> <u>NSLRAEDTAVYYCVRHGNFGDSYVSWFAYWGQGT</u> <u>LVTVSS</u>	154
Variable Heavy (vh) Domain	<u>EVQLVESGGGLVQPGGSLRLSCAASGFTTFSTYAMSWVRQAPGKGLEWVGRIRSK</u> <u>YNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVRHGNFGDSY</u> <u>VSWFAYWGQGT</u> <u>LVTVSS</u>	155
vhCDR1	<u>TYAMS</u>	156
vhCDR2	<u>RIRSKYNNYATYYADSVKG</u>	157
vhCDR3	<u>HGNFGDSYVSWFAY</u>	158
Variable Light (vl) Domain	<u>QAVVTQEPSLTVSPGGTVTLT</u> <u>CGSSTGAVTTSNYANWVQKPGKSPRGLIGGTN</u> <u>KRAPGVPARFSGSLLGGKAALTISGAQPEDEADYYCALWYSNHWVF</u> <u>GGGTKLTV</u> <u>L</u>	159
vlCDR1	<u>GSSTGAVTTSNYAN</u>	160
vlCDR2	<u>GTNKRAP</u>	161
vlCDR3	<u>ALWYSNHWV</u>	162
Linker	<u>GKPGSGKPGSGKPGSGKPGS</u>	163

Figure 11

Human CLDN6 sequence

>sp|P56747

MASAGMQILGVVLTLLGWVNGLVSCALPMWKVTAFIGNSIVVAQVWVEGLWMSCVVQSTGQMCKVYDSSLALPQDL
QAARALCVIALLLVALFGLLVYLAGAKCTTCVEEKDSKARLVLTSGIVFVISGVLTLPVCWTAHAIIRDFYNPLVAE
AQKREL GASLYLGWAASGLLLLGGGLLCCTCPSGGSQGPSHYMARYSTSAPAI SRGPSEYPTKNYV (SEQ ID
NO: 164)

Human CLDN6 sequence, N-terminal extracellular domain

>sp|P56747|29-81

MWKVTAFIGNSIVVAQVWVEGLWMSCVVQSTGQMCKVYDSSLALPQDLQAAR (SEQ ID NO: 165)

Human CLDN6 sequence, C-terminal extracellular domain

>sp|P56747|138-160

WTAHAIIRDFYNPLVAE AQKREL (SEQ ID NO: 166)

Mouse CLDN6 sequence

>sp|Q9Z262

MASTGLQILGIVLTLLGWVNALVSCALPMWKVTAFIGNSIVVAQMVEGLWMSCVVQSTGQMCKVYDSSLALPQDL
QAARALCVVTLIVLLGLLVYLAGAKCTTCVEDRNSKSRLVLISGII FVISGVLTLPVCWTAHSIIQDFYNPLVAD
AQKREL GASLYLGWAASGLLLLGGGLLCACSSGGTQGP RHYMACYSTSVPHSRGPSEYPTKNYV (SEQ ID NO:
167)

Mouse CLDN6 sequence, N-terminal extracellular domain

>sp|Q9Z262|29-81

MWKVTAFIGNSIVVAQMVEGLWMSCVVQSTGQMCKVYDSSLALPQDLQAAR (SEQ ID NO: 168)

Mouse CLDN6 sequence, C-terminal extracellular domain

>sp|Q9Z262|138-163

WTAHSIIQDFYNPLVADAQKREL GAS (SEQ ID NO: 169)

Macaca fascicularis CLDN6 sequence (predicted)

>tr|G7Q0B0

MASAGMQILGVVLTLLGWVNGLVSCALPMWKVTAFIGNSIVVAQVWVEGLWMSCVVQSTGQMCKVYDSSLALPQDL
QAARALCVIALLLVALFGLLVYLAGAKCTTCVEEKDSKARLVLTSGIVFVISGVLTLPVCWTAHAIIRDFYNPLVAE
AQKREL GASLYLGWAASGLLLLGGGLLCCTCPSGGSRGPSHYMARYSTSAPAI SRGPSEYPTKNYV (SEQ ID
NO: 170)

Macaca fascicularis CLDN6 sequence, N-terminal extracellular domain (predicted)

>tr|G7Q0B0|29-81

MWKVTAFIGNSIVVAQVWVEGLWMSCVVQSTGQMCKVYDSSLALPQDLQAAR (SEQ ID NO: 171)

Macaca fascicularis CLDN6 sequence, C-terminal extracellular domain (predicted)

>tr|G7Q0B0|138-160

WTAHAIIRDFYNPLVAE AQKREL (SEQ ID NO: 172)

```
===== EC loop
```

```
cldn6 MASAGMQILGWLTLLGWNGLVSCALP...KVTAFIGNSIIVAAQVVWEGLIHMSCVWQSTG 60
cldn9 MASTGLELLGMTLAVLGWLGTLVSCALP...KVTAFIGNSIIVAAQVVWEGLIHMSCVWQSTG 60
***::*:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*

cldn6 OMQCKVYDSLALLPODLQAARALCVIALLLVALFGLLVLAGAKCTTCVEEKDSKARLVLT 120
cldn9 OMQCKVYDSLALLPODLQAARALCVIALLLALLGLLVAITGAQCTTCVEDEGAKARIVLT 120
*****:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*

cldn6 SGIVFVISGVLTLPVCWTAHAITIDFYNPPLVAEADKKREL GASLYLGWAASGLLLLGGGL 180
cldn9 AGVILLLAGTILVLIPVCWTAHAITIDFYNPPLVAEADKKREL GASLYLGWAAAAALMLGGGL 180
*:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*:~::~*

cldn6 LCCTCPSGGSQGPPSHYMARYSTSAPAI SRGPSEYPTKNYV 220
cldn9 LCCTCPPPQVERPRGPRLGYSIPSR---SGASGLDKRDYV 217
***** : * ** : * * :~::~*
```

Figure 13

mC6-30[CLDN6]

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLQQSGPELVKPGASVKISCKTSGYTFTEYTMHWVKQSPGKSLEW IGGIDPNNGNTHYNQKFKGKATLTVDKSSSTAYMELRSLTSEDSAVY YCARIYYFGRLYFDFWGAGTTVTVSS	175
vhCDR1	EYTMH	176
vhCDR2	GIDPNNGNTHYNQKFKG	177
vhCDR3	IYYFGRLYFDF	178
Variable light (vl) domain	DIQMTQSSSSFSVSLGDRVITITCKASEDIYNRLAWYQQKPGNVPRLL ISGATSLETGVPSRFSGSGSGKDYTLSTSLQTEDVTYYCQQYWSS PLTFGGGKLEIK	179
vlCDR1	KASEDIYNRLA	180
vlCDR2	GATSLET	181
vlCDR3	QQYWSSPLT	182

>XENP34243 mC6-30mC6-30 Heavy Chain

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDSNNNGNTHYNQKFGQGHVTISVDKSI
TAYLQWSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVPSSSLGTQTYICNVNHKPSNTKVDKRVKVEPKSCDKHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVVFSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
183)

mC6-30 Light Chain

DIQMTQSPSSLSASVGDRTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTTIS
LQPEDIATYYCQQYWSGELTFGGGKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDSSTYSLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
184)

Figure 14

C6-30[CLDN6] H1

	sequence	SEQ ID NO:
Variable heavy (vh) domain	QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEW MGGIDPNNGNTHYNQKFQGRVTITVDKSASTAYMELSSLRSED TAVY YCARIYYFGRLYFDFWAGATLVTVSS	185
vhCDR1	EYTMH	186
vhCDR2	GIDPNNGNTHYNQKFQ	187
vhCDR3	IYYFGRLYFDF	188

C6-30[CLDN6] H2

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEW MGGIDPNNGNTHYNQKFQGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLVTVSS	189
vhCDR1	EYTMH	190
vhCDR2	GIDPNNGNTHYNQKFQ	191
vhCDR3	IYYFGRLYFDF	192

C6-30[CLDN6] L1

	sequence	SEQ ID NO:
Variable light (vl) domain	DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLL ISGATSLETGVPSRFSGSGSGKDYFTFTISSLQPEDIAITYYCQQYWSS PLTFGGGKVEIK	193
vlCDR1	QASEDIYNRLA	194
vlCDR2	GATSLET	195
vlCDR3	QQYWSSPLT	196

C6-30[CLDN6] L2

	sequence	SEQ ID NO:
Variable light (vl) domain	DIQMTQSPDSLAVSLGERATINCKASEDIYNRLAWYQQKPGQVPKLL ISGATSLETGVPSRFSGSGSGKDYFTFTISSLQAEDVAVYYCQQYWSS PLTFGGGKVEIK	197
vlCDR1	KASEDIYNRLA	198
vlCDR2	GATSLET	199
vlCDR3	QQYWSSPLT	200

Figure 15A

C6-30[CLDN6] H1.1

	sequence	SEQ ID NO:
Variable heavy (vh) domain	QVQLVQSGAEVKKPGASVKVSCKTSGYTFTTEYTMHWVRQAPGQSLEW MGGISPNNGNTHYNQKFQGRVTITVDKSASTAYMELSSLRSED TAVY YCARIYYFGRLYFDFWAGATLTVTVSS	201
vhCDR1	EYTMH	202
vhCDR2	GISPNNGNTHYNQKFQG	203
vhCDR3	IYYFGRLYFDF	204

C6-30[CLDN6] H1.2

	sequence	SEQ ID NO:
Variable heavy (vh) domain	QVQLVQSGAEVKKPGASVKVSCKTSGYTFTTEYTMHWVRQAPGQSLEW MGGINPNNGNTHYNQKFQGRVTITVDKSASTAYMELSSLRSED TAVY YCARIYYFGRLYFDFWAGATLTVTVSS	205
vhCDR1	EYTMH	206
vhCDR2	GINPNNGNTHYNQKFQG	207
vhCDR3	IYYFGRLYFDF	208

C6-30[CLDN6] H1.3

	sequence	SEQ ID NO:
Variable heavy (vh) domain	QVQLVQSGAEVKKPGASVKVSCKTSGYTFTTEYTMHWVRQAPGQSLEW MGGIDSNNGNTHYNQKFQGRVTITVDKSASTAYMELSSLRSED TAVY YCARIYYFGRLYFDFWAGATLTVTVSS	209
vhCDR1	EYTMH	210
vhCDR2	GIDSNNGNTHYNQKFQG	211
vhCDR3	IYYFGRLYFDF	212

C6-30[CLDN6] H1.4

	sequence	SEQ ID NO:
Variable heavy (vh) domain	QVQLVQSGAEVKKPGASVKVSCKTSGYTFTTEYTMHWVRQAPGQSLEW MGGIDANNNGNTHYNQKFQGRVTITVDKSASTAYMELSSLRSED TAVY YCARIYYFGRLYFDFWAGATLTVTVSS	213
vhCDR1	EYTMH	214
vhCDR2	GIDANNNGNTHYNQKFQG	215
vhCDR3	IYYFGRLYFDF	216

C6-30[CLDN6] H1.5

	sequence	SEQ ID NO:
Variable heavy (vh) domain	QVQLVQSGAEVKKPGASVKVSCKTSGYTFTTEYTMHWVRQAPGQSLEW MGGIDPQNGNTHYNQKFQGRVTITVDKSASTAYMELSSLRSED TAVY YCARIYYFGRLYFDFWAGATLTVTVSS	217
vhCDR1	EYTMH	218
vhCDR2	GIDPQNGNTHYNQKFQG	219
vhCDR3	IYYFGRLYFDF	220

Figure 15B

C6-30[CLDN6] H1.6

	sequence	SEQ ID NO:
Variable heavy (vh) domain	QVQLVQSGAEVKKPGASVKVSCKTSGYTFTTEYTMHWVRQAPGQSLEW MGGIDPNSGNTHYNQKFQGRVTITVDKSASTAYMELSSLRSED TAVY YCARIYYFGRLYFDFWAGATLTVVSS	221
vhCDR1	EYTMH	222
vhCDR2	GIDPNSGNTHYNQKFQG	223
vhCDR3	IYYFGRLYFDF	224

C6-30[CLDN6] H1.7

	sequence	SEQ ID NO:
Variable heavy (vh) domain	QVQLVQSGAEVKKPGASVKVSCKTSGYTFTTEYTMHWVRQAPGQSLEW MGGIDPNQGNTHYNQKFQGRVTITVDKSASTAYMELSSLRSED TAVY YCARIYYFGRLYFDFWAGATLTVVSS	225
vhCDR1	EYTMH	226
vhCDR2	GIDPNQGNTHYNQKFQG	227
vhCDR3	IYYFGRLYFDF	228

C6-30[CLDN6] H1.8

	sequence	SEQ ID NO:
Variable heavy (vh) domain	QVQLVQSGAEVKKPGASVKVSCKTSGYTFTTEYTMHWVRQAPGQSLEW MGGIDPNNNDNTHYNQKFQGRVTITVDKSASTAYMELSSLRSED TAVY YCARIYYFGRLYFDFWAGATLTVVSS	229
vhCDR1	EYTMH	230
vhCDR2	GIDPNNNDNTHYNQKFQG	231
vhCDR3	IYYFGRLYFDF	232

C6-30[CLDN6] H1.9

	sequence	SEQ ID NO:
Variable heavy (vh) domain	QVQLVQSGAEVKKPGASVKVSCKTSGYTFTTEYTMHWVRQAPGQSLEW MGGIDPNNANTHYNQKFQGRVTITVDKSASTAYMELSSLRSED TAVY YCARIYYFGRLYFDFWAGATLTVVSS	233
vhCDR1	EYTMH	234
vhCDR2	GIDPNNANTHYNQKFQG	235
vhCDR3	IYYFGRLYFDF	236

C6-30[CLDN6] H1.19

	sequence	SEQ ID NO:
Variable heavy (vh) domain	QVQLVQSGAEVKKPGASVKVSCKTSGYTFTTEYTMHWVRQAPGQSLEW MGGIDPNNGNTHYNQKFQGRVTITVDKSASTAYMELSSLRSED TATY YCARIYYFGRLYFDFWAGATLTVVSS	237
vhCDR1	EYTMH	238
vhCDR2	GIDPNNGNTHYNQKFQG	239
vhCDR3	IYYFGRLYFDF	240

Figure 15C

C6-30[CLDN6] H1.22

	sequence	SEQ ID NO:
Variable heavy (vh) domain	QVQLVQSGAEVKKPGASVKVSCRTSGYTFTEYTMHWVRQAPGQSLEW MGGIDPNNGNTHYNQKFQGRVTITVDKSASTAYMELSSLRSEDNAVY YCARILYFGRLYFDFWAGATLTVTVSS	241
vhCDR1	EYTMH	242
vhCDR2	GIDPNNGNTHYNQKFQG	243
vhCDR3	ILYFGRLYFDF	244

C6-30[CLDN6] H1.24

	sequence	SEQ ID NO:
Variable heavy (vh) domain	QVQLVQSGAEVKKPGASVKVSCRTSGYTFTEYTMHWVRQAPGQSLEW MGGIDPNNGNTHYNQKFQGRVTITVDKSASTAYMELSSLRSEDNAVY YCARIYYLGRLYFDFWAGATLTVTVSS	245
vhCDR1	EYTMH	246
vhCDR2	GIDPNNGNTHYNQKFQG	247
vhCDR3	IYYLGRLYFDF	248

C6-30[CLDN6] H2.1

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEW MGGISPNNNGNTHYNQKFQGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLTVTVSS	249
vhCDR1	EYTMH	250
vhCDR2	GISPNNGNTHYNQKFQG	251
vhCDR3	IYYFGRLYFDF	252

C6-30[CLDN6] H2.2

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEW MGGITPNNGNTHYNQKFQGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLTVTVSS	253
vhCDR1	EYTMH	254
vhCDR2	GLTPNNGNTHYNQKFQG	255
vhCDR3	IYYFGRLYFDF	256

C6-30[CLDN6] H2.3

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEW MGGIDSNNGNTHYNQKFQGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLTVTVSS	257
vhCDR1	EYTMH	258
vhCDR2	GIDSNNGNTHYNQKFQG	259
vhCDR3	IYYFGRLYFDF	260

Figure 15D

C6-30[CLDN6] H2.4

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEW MGGIDGNNGNTHYNQKFQGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLTVSS	261
vhCDR1	EYTMH	262
vhCDR2	GIDGNNGNTHYNQKFQG	263
vhCDR3	IYYFGRLYFDF	264

C6-30[CLDN6] H2.5

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEW MGGIDPQNGNTHYNQKFQGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLTVSS	265
vhCDR1	EYTMH	266
vhCDR2	GIDPQNGNTHYNQKFQG	267
vhCDR3	IYYFGRLYFDF	268

C6-30[CLDN6] H2.6

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEW MGGIDPNDGNTHYNQKFQGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLTVSS	269
vhCDR1	EYTMH	270
vhCDR2	GIDPNDGNTHYNQKFQG	271
vhCDR3	IYYFGRLYFDF	272

C6-30[CLDN6] H2.7

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEW MGGIDPNQGNTHYNQKFQGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLTVSS	273
vhCDR1	EYTMH	274
vhCDR2	GIDPNQGNTHYNQKFQG	275
vhCDR3	IYYFGRLYFDF	276

C6-30[CLDN6] H2.8

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEW MGGIDPNNDNTHYNQKFQGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLTVSS	277
vhCDR1	EYTMH	278
vhCDR2	GIDPNNDNTHYNQKFQG	279
vhCDR3	IYYFGRLYFDF	280

Figure 15E

C6-30[CLDN6] H2.9

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEW MGGIDPNNANTHYNQKFQGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLTVSS	281
vhCDR1	EYTMH	282
vhCDR2	GIDPNNANTHYNQKFQG	283
vhCDR3	IYYFGRLYFDF	284

C6-30[CLDN6] H2.11

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYSFTEYTMHWVRQMPGKSLEW MGGIDPNNGNTHYNQKFQGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLTVSS	285
vhCDR1	EYTMH	286
vhCDR2	GIDPNNGNTHYNQKFQG	287
vhCDR3	IYYFGRLYFDF	288

C6-30[CLDN6] H2.12

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYDFTEYTMHWVRQMPGKSLEW MGGIDPNNGNTHYNQKFQGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLTVSS	289
vhCDR1	EYTMH	290
vhCDR2	GIDPNNGNTHYNQKFQG	291
vhCDR3	IYYFGRLYFDF	292

C6-30[CLDN6] H2.71

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEW MGGIDLNNGNTHYNQKFQGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLTVSS	293
vhCDR1	EYTMH	294
vhCDR2	GIDLNNGNTHYNQKFQG	295
vhCDR3	IYYFGRLYFDF	296

C6-30[CLDN6] H2.75

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEW MGGIDPHNGNTHYNQKFQGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLTVSS	297
vhCDR1	EYTMH	298
vhCDR2	GIDPHNGNTHYNQKFQG	299
vhCDR3	IYYFGRLYFDF	300

Figure 15F

C6-30[CLDN6] H2.90

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEW MGGIDPNNLNTHYNQKFQGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLTVSS	301
vhCDR1	EYTMH	302
vhCDR2	GIDPNNLNTHYNQKFQG	303
vhCDR3	IYYFGRLYFDF	304

C6-30[CLDN6] H2.91

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEW MGGIDPNNFNTHYNQKFQGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLTVSS	305
vhCDR1	EYTMH	306
vhCDR2	GIDPNNFNTHYNQKFQG	307
vhCDR3	IYYFGRLYFDF	308

C6-30[CLDN6] H2.118

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEW MGGIDPNNGNTHYNQKKQGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLTVSS	309
vhCDR1	EYTMH	310
vhCDR2	GIDPNNGNTHYNQKKQG	311
vhCDR3	IYYFGRLYFDF	312

C6-30[CLDN6] H2.119

	sequence	SEQ ID NO:
Variable heavy (vh) domain	EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEW MGGIDPNNGNTHYNQKFEGHVTISVDKSISTAYLQWSSLKASDTAMY YCARIYYFGRLYFDFWAGATLTVSS	313
vhCDR1	EYTMH	314
vhCDR2	GIDPNNGNTHYNQKFEG	315
vhCDR3	IYYFGRLYFDF	316

C6-30[CLDN6] L1.1

	sequence	SEQ ID NO:
Variable light (vl) domain	DIQMTQSPSSLSASVGDRVTITCQASQDIYNRLAWYQQKPGKVPKLL ISGATSLETGVPSRFSGSGSGKDYFTFTISSLQPEDATYYCQQYWSS PLTFGGGKVEIK	317
vlCDR1	QASQDIYNRLA	318
vlCDR2	GATSLET	319
vlCDR3	QQYWSSPLT	320

Figure 15G

C6-30[CLDN6] L1.4

	sequence	SEQ ID NO:
Variable	DIQMTQSPSSLSASVGDRVITITCQASEDVYNRLAWYQQKPGKVPKLL	321
light (vl)	ISGATSLETIGVPSRFSGSGSGKDYTFTISSLPEDIATYYCQQYWSS	
domain	PLTFGGGKVEIK	
v1CDR1	QASEDVYNRLA	322
v1CDR2	GATSLET	323
v1CDR3	QQYWSSPLT	324

C6-30[CLDN6] L1.7

	sequence	SEQ ID NO:
Variable	DIQMTQSPSSLSASVGDRVITITCQASEDIYSRLAWYQQKPGKVPKLL	325
light (vl)	ISGATSLETIGVPSRFSGSGSGKDYTFTISSLPEDIATYYCQQYWSS	
domain	PLTFGGGKVEIK	
v1CDR1	QASEDIYSRLA	326
v1CDR2	GATSLET	327
v1CDR3	QQYWSSPLT	328

C6-30[CLDN6] L1.16

	sequence	SEQ ID NO:
Variable	DIQMTQSPSSLSASVGDRVITITCQASEDIYNRLAWYQQKPGKVPKLL	329
light (vl)	ISAATSLETIGVPSRFSGSGSGKDYTFTISSLPEDIATYYCQQYWSS	
domain	PLTFGGGKVEIK	
v1CDR1	QASEDIYNRLA	330
v1CDR2	AATSLET	331
v1CDR3	QQYWSSPLT	332

C6-30[CLDN6] L1.18

	sequence	SEQ ID NO:
Variable	DIQMTQSPSSLSASVGDRVITITCQASEDIYNRLAWYQQKPGKVPKLL	333
light (vl)	ISGTTTLETIGVPSRFSGSGSGKDYTFTISSLPEDIATYYCQQYWSS	
domain	PLTFGGGKVEIK	
v1CDR1	QASEDIYNRLA	334
v1CDR2	GTTTLET	335
v1CDR3	QQYWSSPLT	336

Figure 15H

C6-30[CLDN6] L1.19

	sequence	SEQ ID NO:
Variable light (v1) domain	DIQMTQSPSSLSASVGDRVITTCQASEDIYNRLAWYQQKPGKVPKLLISGASLETGVPSRFRSGSGSGKDYFTFTISSLQPEDATYYCQQYWSSPLTFGGGTKV EIK	337
v1CDR1	QASEDIYNRLA	338
v1CDR2	GASSLET	339
v1CDR3	QQYWSSPLT	340

C6-30[CLDN6] L1.21

	sequence	SEQ ID NO:
Variable light (v1) domain	DIQMTQSPSSLSASVGDRVITTCQASEDIYNRLAWYQQKPGKVPKLLISGATNLETGVPSRFRSGSGSGKDYFTFTISSLQPEDATYYCQQYWSSPLTFGGGTKVEIK	341
v1CDR1	QASEDIYNRLA	342
v1CDR2	GATNLET	343
v1CDR3	QQYWSSPLT	344

C6-30[CLDN6] L1.22

	sequence	SEQ ID NO:
Variable light (v1) domain	DIQMTQSPSSLSASVGDRVITTCQASEDIYNRLAWYQQKPGKVPKLLISGATQLETGVPSRFRSGSGSGKDYFTFTISSLQPEDATYYCQQYWSSPLTFGGGTKVEIK	345
v1CDR1	QASEDIYNRLA	346
v1CDR2	GATQLET	347
v1CDR3	QQYWSSPLT	348

C6-30[CLDN6] L1.23

	sequence	SEQ ID NO:
Variable light (v1) domain	DIQMTQSPSSLSASVGDRVITTCQASEDIYNRLAWYQQKPGKVPKLLISGATSRETGVPSRFRSGSGSGKDYFTFTISSLQPEDATYYCQQYWSSPLTFGGGTKVEIK	349
v1CDR1	QASEDIYNRLA	350
v1CDR2	GATSRET	351
v1CDR3	QQYWSSPLT	352

Figure 15I

C6-30[CLDN6] L1.27

	sequence	SEQ ID NO:
Variable light (v1) domain	DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLL ISGATSLES GVPSRFRSGSGSGKDYFT ISSLQPED IATYYCQQYWSS PLTFGGG TKVEIK	353
v1CDR1	QASEDIYNRLA	354
v1CDR2	GATSLES	355
v1CDR3	QQYWSSPLT	356

C6-30[CLDN6] L1.60

	sequence	SEQ ID NO:
Variable light (v1) domain	DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLL ISGATSLETGVPSRFRSGSGSGKDYFTISSLQPED IATYYCQQYWSS PLTFGGG TKVEIK	357
v1CDR1	QASEDIYNRLA	358
v1CDR2	GATSLET	359
v1CDR3	QQYWSSPLT	360

C6-30[CLDN6] L1.107

	sequence	SEQ ID NO:
Variable light (v1) domain	DIQMTQSPSSLSASVGDRVTITCQASEDIVNRLAWYQQKPGKVPKLL ISGATSLETGVPSRFRSGSGSGKDYFTISSLQPED IATYYCQQYWSS PLTFGGG TKVEIK	361
v1CDR1	QASEDIVNRLA	362
v1CDR2	GATSLET	363
v1CDR3	QQYWSSPLT	364

C6-30[CLDN6] L1.114

	sequence	SEQ ID NO:
Variable light (v1) domain	DIQMTQSPSSLSASVGDRVTITCQASEDIYLRLAWYQQKPGKVPKLL ISGATSLETGVPSRFRSGSGSGKDYFTISSLQPED IATYYCQQYWSS PLTFGGG TKVEIK	365
v1CDR1	QASEDIYLRLA	366
v1CDR2	GATSLET	367
v1CDR3	QQYWSSPLT	368

Figure 15J

C6-30[CLDN6] L1.187

	sequence	SEQ ID NO:
Variable light (v1) domain	DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLL ISGATSLETGVPSRFSGSGSGKDYFTISSLPEDIATYYCQQYWSA PLTFGGGKVEIK	369
v1CDR1	QASEDIYNRLA	370
v1CDR2	GATSLET	371
v1CDR3	QQYWSAPLT	372

C6-30[CLDN6] L1.189

	sequence	SEQ ID NO:
Variable light (v1) domain	DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLL ISGATSLETGVPSRFSGSGSGKDYFTISSLPEDIATYYCQQYWSG PLTFGGGKVEIK	373
v1CDR1	QASEDIYNRLA	374
v1CDR2	GATSLET	375
v1CDR3	QQYWSGPLT	376

Figure 16A

>XENP34218 C6-30[CLDN6]_H1_L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYDFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSLKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
377)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKKHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
378)

>XENP34219 C6-30[CLDN6]_H1_L2 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYDFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSLKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
379)

Light Chain - C6-30[CLDN6]_L2

DIQMTQSPDSLAVSLGERATINCKASEDIYNRLAWYQQKPGQVPKLLISGATSLETGVPSRFSGSGSGKDYTLTSS
LQAEDEVAVYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKKHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
380)

>XENP34220 C6-30[CLDN6]_H2_L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDSKIS
TAYLQWSSLKASDTAMYYCARIYYFGRLYDFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSLKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
381)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKKHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
382)

Figure 16B

>XENP34221 C6-30[CLDN6]_H2_L2_IgG1_PVA_/S267K**Heavy Chain - C6-30[CLDN6]_H2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
383)

Light Chain - C6-30[CLDN6]_L2

DIQMTQSPDLSAVSLGERATINCKASEDIYNRLAWYQQKPGQVPKLLISGATSLETGVPSRFSGSGSGKDYTLTISS
LQAEDVAVYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
384)

>XENP35044 C6-30[CLDN6]_H2_L1.1_IgG1_PVA_/S267K**Heavy Chain - C6-30[CLDN6]_H2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
385)

Light Chain - C6-30[CLDN6]_L1.1

DIQMTQSPSSLSASVGDRTTITCQASQDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSF (SEQ ID NO: 386)

>XENP35047 C6-30[CLDN6]_H2_L1.4_IgG1_PVA_/S267K**Heavy Chain - C6-30[CLDN6]_H2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
387)

Light Chain - C6-30[CLDN6]_L1.4

DIQMTQSPSSLSASVGDRTTITCQASEDVYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
388)

Figure 16C

>XENP35050 C6-30[CLDN6]_H2 L1.7 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDWAGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
389)

Light Chain - C6-30[CLDN6]_L1.7

DIQMTQSPSSLSASVGDRVTITCQASEDIYSRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
390)

>XENP35059 C6-30[CLDN6]_H2 L1.16 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDWAGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
391)

Light Chain - C6-30[CLDN6]_L1.16

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISAATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
392)

>XENP35061 C6-30[CLDN6]_H2 L1.18 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDWAGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
393)

Light Chain - C6-30[CLDN6]_L1.18

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGTTTLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
394)

Figure 16D

>XENP35062 C6-30[CLDN6]_H2 L1.19 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDLFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
395)

Light Chain - C6-30[CLDN6]_L1.19

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGASSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
396)

>XENP35064 C6-30[CLDN6]_H2 L1.21 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDLFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
397)

Light Chain - C6-30[CLDN6]_L1.21

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATNLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
398)

>XENP35065 C6-30[CLDN6]_H2 L1.22 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDLFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
399)

Light Chain - C6-30[CLDN6]_L1.22

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATQLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
400)

Figure 16E

>XENP35066 C6-30[CLDN6]_H2 L1.23 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDLFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
401)

Light Chain - C6-30[CLDN6]_L1.23

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSRETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
402)

>XENP35070 C6-30[CLDN6]_H2 L1.27 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDLFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
403)

Light Chain - C6-30[CLDN6]_L1.27

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLESQVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
404)

>XENP35085 C6-30[CLDN6]_H1.1 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.1**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGISPNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDLFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
405)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
406)

Figure 16F

>XENP35086 C6-30[CLDN6] H1.2 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.2**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGINPNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
407)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRTTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
408)

>XENP35087 C6-30[CLDN6] H1.3 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.3**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDSNNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
409)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRTTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
410)

>XENP35088 C6-30[CLDN6] H1.4 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.4**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDANNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
411)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRTTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
412)

Figure 16G

>XENP35089 C6-30[CLDN6] H1.5 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.5**

QVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPQNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDFWAGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
413)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVITITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
414)

>XENP35090 C6-30[CLDN6] H1.6 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.6**

QVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNSGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDFWAGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
415)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVITITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
416)

>XENP35091 C6-30[CLDN6] H1.7 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.7**

QVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPQNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDFWAGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
417)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVITITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
418)

Figure 16H

>XENP35092 C6-30[CLDN6]_H1.8 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.8**

QVQLVQSGAEVKKPGASVKVCKTSGYTFTTEYTMHWVRQAPGQSLEWMGGIDPNNDNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSED TAVYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
419)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPKGVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIAITYYCQQYWSSPLTFGGGTKVEIK/RTVAAPS FIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
420)

>XENP35093 C6-30[CLDN6]_H1.9 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.9**

QVQLVQSGAEVKKPGASVKVCKTSGYTFTTEYTMHWVRQAPGQSLEWMGGIDPNNANTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSED TAVYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
421)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPKGVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIAITYYCQQYWSSPLTFGGGTKVEIK/RTVAAPS FIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
422)

>XENP35094 C6-30[CLDN6]_H2.1 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.1**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTTEYTMHWVRQMPGKSLEWMGGISPNNGNTHYNQKFQGHVTISVDSKIS
TAYLQWSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
423)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPKGVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIAITYYCQQYWSSPLTFGGGTKVEIK/RTVAAPS FIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
424)

Figure 16l

>XENP35095 C6-30[CLDN6] H2.2 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGITPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
425)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
426)

>XENP35096 C6-30[CLDN6] H2.3 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.3**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDSNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
427)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
428)

>XENP35097 C6-30[CLDN6] H2.4 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.4**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDGNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
429)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
430)

Figure 16J

>XENP35098 C6-30[CLDN6] H2.5 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.5**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPQNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
431)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
432)

>XENP35099 C6-30[CLDN6] H2.6 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.6**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNDGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
433)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
434)

>XENP35100 C6-30[CLDN6] H2.7 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.7**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPQNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
435)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
436)

Figure 16K

>XENP35101 C6-30[CLDN6] H2.8 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.8**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNNDNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYDFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSLKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
437)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
438)

>XENP35102 C6-30[CLDN6] H2.9 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.9**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNANTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYDFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSLKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
439)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
440)

>XENP35865 C6-30[CLDN6] H2 L1.60 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYDFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSLKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
441)

Light Chain - C6-30[CLDN6]_L1.60

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYLWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
442)

Figure 16L

>XENP35883 C6-30[CLDN6] H1.19 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.19**

QVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNGNTHYNQKFGQGRVTITVDKSAS
TAYMELSSLRSEDTATYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
443)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVITITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
444)

>XENP35886 C6-30[CLDN6] H1.22 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.22**

QVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNGNTHYNQKFGQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARILYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
445)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVITITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
446)

>XENP35888 C6-30[CLDN6] H1.24 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.24**

QVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNGNTHYNQKFGQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYLGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
447)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVITITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
448)

Figure 16M

>XENP35890 C6-30[CLDN6] H2.11 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.11**

EVQLVQSGAEVKKPGESLRISCKTSGYSFTTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFEWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
449)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
450)

>XENP35891 C6-30[CLDN6] H2.12 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.12**

EVQLVQSGAEVKKPGESLRISCKTSGYDFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFEWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
451)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
452)

>XENP35979 C6-30[CLDN6] H2 L1.187 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFEWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
453)

Light Chain - C6-30[CLDN6]_L1.187

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSAPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
454)

Figure 16N

>XENP35981 C6-30[CLDN6]_H2 L1.189 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
455)

Light Chain - C6-30[CLDN6]_L1.189

DIQMTQSPSSLSASVGDRTTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSGLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
456)

>XENP35929 C6-30[CLDN6]_H2 L1.107 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
457)

Light Chain - C6-30[CLDN6]_L1.107

DIQMTQSPSSLSASVGDRTTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
458)

>XENP35936 C6-30[CLDN6]_H2 L1.114 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
459)

Light Chain - C6-30[CLDN6]_L1.114

DIQMTQSPSSLSASVGDRTTITCQASEDIYLRRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
460)

Figure 160

>XENP36021 C6-30[CLDN6] H2.71 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.71**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDLNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
461)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVITITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNIFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
462)

>XENP36022 C6-30[CLDN6] H2.72 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.72**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDANNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
463)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVITITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNIFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
464)

>XENP36025 C6-30[CLDN6] H2.75 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.75**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPHNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
465)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVITITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNIFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
466)

Figure 16P

>XENP36040 C6-30[CLDN6] H2.90 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.90**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNLNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
467)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
468)

>XENP36041 C6-30[CLDN6] H2.91 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.91**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNFNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
469)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
470)

>XENP36065 C6-30[CLDN6] H2.118 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.118**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKKQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
471)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
472)

Figure 16Q

>XENP36066 C6-30[CLDN6]_H2.119 L1 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.119**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFEQGHVTISVDKSI
TAYLQWSSSLKASDTAMYICARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
473)

Light Chain - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSFIFFPSDEQLKSGTASVVCLLNIFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
474)

>XENP36956 C6-30[CLDN6]_H1.22 L1.187 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.22**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNGNTHYNQKFEQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARILYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
475)

Light Chain - C6-30[CLDN6]_L1.187

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSAPLTFGGGTKVEIK/RTVAAPSFIFFPSDEQLKSGTASVVCLLNIFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
476)

>XENP36960 C6-30[CLDN6]_H1.22 L1.189 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.22**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNGNTHYNQKFEQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARILYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
477)

Light Chain - C6-30[CLDN6]_L1.189

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSGPLTFGGGTKVEIK/RTVAAPSFIFFPSDEQLKSGTASVVCLLNIFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
478)

Figure 16R

>XENP36968 C6-30[CLDN6]_H2.3 L1.189 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.3**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDSNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
479)

Light Chain - C6-30[CLDN6]_L1.189

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWWSGPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
480)

>XENP36963 C6-30[CLDN6]_H1.9 L1.189 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.9**

QVQLVQSGAEVKKPGASVKVSCCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDFNNANTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
481)

Light Chain - C6-30[CLDN6]_L1.189

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWWSGPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
482)

>XENP36964 C6-30[CLDN6]_H2.3 L1.187 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.3**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDSNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
483)

Light Chain - C6-30[CLDN6]_L1.187

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSAPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
484)

Figure 16S

>XENP36965 C6-30[CLDN6] H2.12 L1.187 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.12**

EVQLVQSGAEVKKPGESLRISCKTSGYDFTETMHWVRQMPGKSLLEWMGGIDPNNGNTHYNQKFQGHVTISVDSKIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
485)

Light Chain - C6-30[CLDN6]_L1.187

DIQMTQSPSSLSASVGDRVITITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSAPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
486)

>XENP36957 C6-30[CLDN6] H1.24 L1.187 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.24**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYLGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
487)

Light Chain - C6-30[CLDN6]_L1.187

DIQMTQSPSSLSASVGDRVITITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSAPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
488)

>XENP36958 C6-30[CLDN6] H1.19 L1.187 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.19**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTATYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
489)

Light Chain - C6-30[CLDN6]_L1.187

DIQMTQSPSSLSASVGDRVITITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSAPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
490)

Figure 16T

>XENP36959 C6-30[CLDN6] H1.9 L1.187 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.9**

QVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNANTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDFWAGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
491)

Light Chain - C6-30[CLDN6]_L1.187

DIQMTQSPSSLSASVGDRTTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSAPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
492)

>XENP36961 C6-30[CLDN6] H1.24 L1.189 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.24**

QVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYLGRLYFDFWAGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
493)

Light Chain - C6-30[CLDN6]_L1.189

DIQMTQSPSSLSASVGDRTTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSGPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
494)

>XENP36962 C6-30[CLDN6] H1.19 L1.189 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1.19**

QVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTATYYCARIYYFGRLYFDFWAGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
495)

Light Chain - C6-30[CLDN6]_L1.189

DIQMTQSPSSLSASVGDRTTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSGPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
496)

Figure 16U

>XENP36966 C6-30[CLDN6] H2.91 L1.187 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.91**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNFNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
497)

Light Chain - C6-30[CLDN6]_L1.187

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSAPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
498)

>XENP36967 C6-30[CLDN6] H2.118 L1.187 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.118**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
499)

Light Chain - C6-30[CLDN6]_L1.187

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSAPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
500)

>XENP36969 C6-30[CLDN6] H2.12 L1.189 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.12**

EVQLVQSGAEVKKPGESLRISCKTSGYDFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
501)

Light Chain - C6-30[CLDN6]_L1.189

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSGFLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
502)

Figure 16V

>XENP36970 C6-30[CLDN6]_H2.91 L1.189 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.91**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNFNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYDFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
503)

Light Chain - C6-30[CLDN6]_L1.189

DIQMTQSPSSLSASVGDRTITCQASEDIYNRLAWYQQKPKGVKPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIATYYCQQYWSGPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
504)

>XENP36972 C6-30[CLDN6]_H1 L1.187 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H1**

QVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDATAVYYCARIYYFGRLYDFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
505)

Light Chain - C6-30[CLDN6]_L1.187

DIQMTQSPSSLSASVGDRTITCQASEDIYNRLAWYQQKPKGVKPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIATYYCQQYWSAPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
506)

>XENP36971 C6-30[CLDN6]_H2.118 L1.189 IgG1 PVA /S267K**Heavy Chain - C6-30[CLDN6]_H2.118**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYDFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
507)

Light Chain - C6-30[CLDN6]_L1.189

DIQMTQSPSSLSASVGDRTITCQASEDIYNRLAWYQQKPKGVKPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIATYYCQQYWSGPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
508)

Figure 16W

>XENP36973 C6-30[CLDN6]_H1 L1.189 IgG1 PVA /S267K

Heavy Chain - C6-30[CLDN6]_H1

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDFNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSED TAVYYCARIYYFGRLYFD FFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWE
SNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
509)

Light Chain - C6-30[CLDN6]_L1.189

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIAITYCQQYWSGPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
510)

Figure 17A

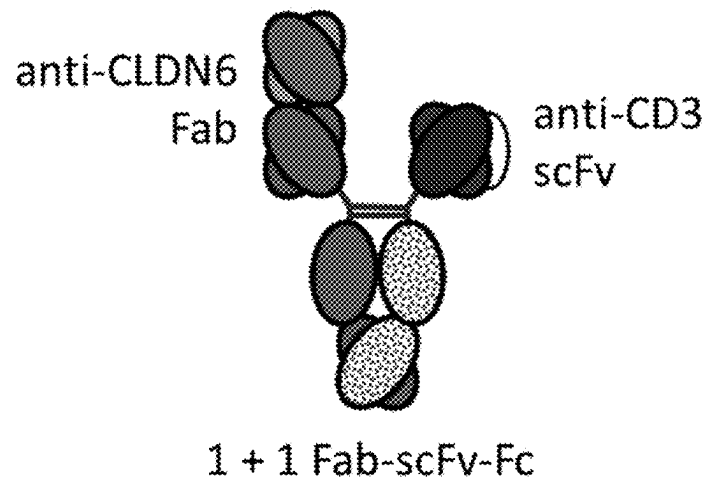


Figure 17B

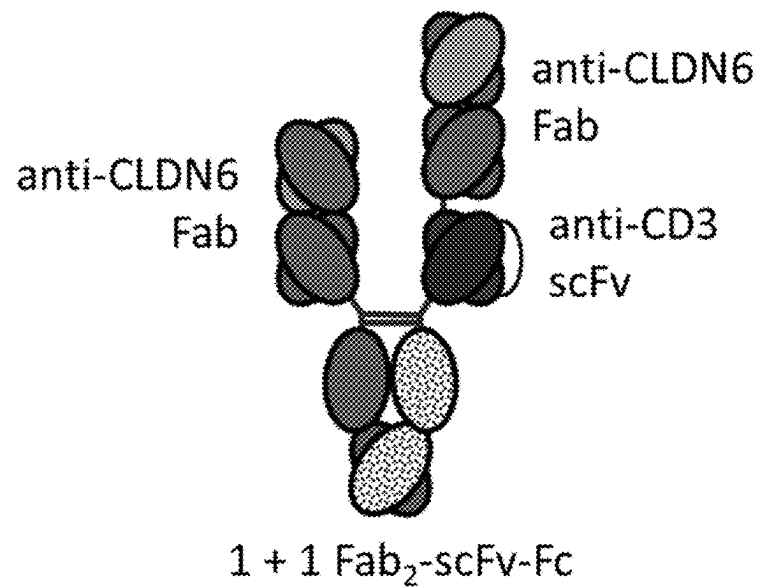


Figure 18

>XENP26863 mAb206-LCC[CLDN6]_H0L0_IgG1_PVA_/S267K**Heavy Chain - mAb206-LCC[CLDN6]_H0**

EVQLQQSGPELVKPGASMKISCKASGYSTGYTMNWVKQSHGKNLEWIGLINPYNGGTIYNQKFKGKATLTVDKSSS
TAYMELLSLTSEDSAVYYCARDYGFVLDYWGQGTTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEP
VTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCP
APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNG
QPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGNVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
511)

Light Chain - mAb206-LCC[CLDN6]_L0

QIVLTQSPAIMASAPGEKVTITCSASSSVSYLHWFAQKPGTSPKLVVYSTSNLPSGVPAREFGGSGSGTSYSLTISRM
EAEDAATYYCQQRSIYFPWTFGGGTKLEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
512)

**>XENP26849 mAb206-LCC[CLDN6]_H0L0_Fab-[ANTI-CD3]_H1.30_L1.47_scFv(GKPGS)4-IgG1_pl(-)
_Isosteric_A_/PVA_/S267K/L368D/K370S-Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q****Chain 1 - mAb206-LCC[CLDN6]_H0_IgG1_pl(-)_Isosteric_A_/PVA_/S267K/L368D/K370S**

EVQLQQSGPELVKPGASMKISCKASGYSTGYTMNWVKQSHGKNLEWIGLINPYNGGTIYNQKFKGKATLTVDKSSS
TAYMELLSLTSEDSAVYYCARDYGFVLDYWGQGTTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYFPEP
VTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCPPCP
APPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEYNSTYRVVSVLTV
LHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWESDG
QPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
513)

Chain 2 - [ANTI-CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q

EVQLVESGGGLVQPGGSLRLSCAASGFTFSTYAMNWVRQAPGKLEWVGRIKSKYNNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRAEDTAVYYCVRHGNEFGDSYVSWFAYWGQGTTLTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQE
PSLTVSPGGTVTLTCGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGAALTISGAQPED
EADYYCALWYSNHWVFGGGTKLTVL/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVKLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCFSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 514)

Chain 3 - mAb206-LCC[CLDN6]_L0

QIVLTQSPAIMASAPGEKVTITCSASSSVSYLHWFAQKPGTSPKLVVYSTSNLPSGVPAREFGGSGSGTSYSLTISRM
EAEDAATYYCQQRSIYFPWTFGGGTKLEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
515)

Figure 19

>XENP34229 C6-30[CLDN6] H1 L1 Fab-[ANTI-CD3] L1.47 H1.30 scFv(GKPGS)4-IgG1 PVA /S267K IgG1 pl(-)
Isosteric A PVA /S267K/L368D/K370S-Fc(216) IgG1 C220S/PVA /S267K/S364K/E357Q

Chain 1 - C6-30[CLDN6]_H1_IgG1_PVA_/S267K IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDFWAGAGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO:
516)

Chain 2 - [ANTI-CD3]_L1.47_H1.30_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q

QAVVTQEPSTLTVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGKKAALTI
SGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVYYCVR
HGNFGDSYVSWFAYWGQGTITVTVSS/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVCLTVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 517)

Chain 3 - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRTITCQASEDIYNRLAWYQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQYQWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
518)

>XENP34637 C6-30[CLDN6] H2 L1-[ANTI-CD3] H1.30 L1.47 scFv(GKPGS)4-IgG1 pl(-)
Isosteric A PVA /S267K/L368D/K370S-Fc(216) IgG1 C220S/PVA /S267K/S364K/E357Q

Chain 1 - C6-30[CLDN6]_H2_L1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLLEWMGGIDPNNGNTHYNQKFQGHVTISVDSKIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWAGAGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO:
519)

Chain 2 - [ANTI-CD3]_H1.30_L1.47_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q

EVQLVESGGGLVQPGGSLRLSCAASGFTFSTYAMNWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDS
KNTLYLQMNSLRAEDTAVYYCVRHGNFGDSYVSWFAYWGQGTITVTVSS/GKPGSGKPGSGKPGSGKPGS/QAVVTQE
PSLTVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGKKAALTISGAQPED
EADYYCALWYSNHWVFGGGTKLTVL/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVCLTVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 520)

Chain 3 - C6-30[CLDN6]_L1

DIQMTQSPSSLSASVGDRTITCQASEDIYNRLAWYQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQYQWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
521)

Figure 20

**>XENP35385 C6-30[CLDN6] H1 L1 Fab-[ANTI-CD3] L1.47 H1.32 scFv(GKPGS)4-IgG1 PVA /S267K IgG1 pl(-)
_Isosteric A PVA /S267K/L368D/K370S-Fc(216) IgG1 C220S/PVA /S267K/S364K/E357Q**

Chain 1 - C6-30[CLDN6] H1_IgG1_PVA_/S267K IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSED TAVYYCARIYYFGRLYFD FFWGAGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTQSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO:
522)

Chain 2 - [ANTI-CD3] L1.47 H1.32_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q

QAVVTQEPSTLTVSPGGTVTLTCGSSGTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFSGGKLTVL/GKPGSGKPGSGKPGSGKPGS/ EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNLSRAEDTAVYYCVR
HGNFGDSYVSWFAYWGQGTLLTVSS/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 523)

Chain 3 - C6-30[CLDN6] L1

DIQMTQSPSSLSASVGDRTITCQASEDIYNRLAWYQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQYQWSSPLTFGGGKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
524)

**>XENP35387 C6-30[CLDN6] H2 L1 Fab-[ANTI-CD3] L1.47 H1.32 scFv(GKPGS)4-IgG1 PVA /S267K IgG1 pl(-)
_Isosteric A PVA /S267K/L368D/K370S-Fc(216) IgG1 C220S/PVA /S267K/S364K/E357Q**

Chain 1 - C6-30[CLDN6] H2_IgG1_PVA_/S267K IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLLEWMGGIDPNNNGNTHYNQKFQGHVTISVDSKIS
TAYLQWSSLKASDTAMYYCARIYYFGRLYFD FFWGAGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTQSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO:
525)

Chain 2 - [ANTI-CD3] L1.47 H1.32_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q

QAVVTQEPSTLTVSPGGTVTLTCGSSGTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKAALTI
SGAQPEDEADYYCALWYSNHWVFSGGKLTVL/GKPGSGKPGSGKPGSGKPGS/ EVQLVESGGGLVQPGGSLRLSCA
ASGFTFSTYAMNWRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNLSRAEDTAVYYCVR
HGNFGDSYVSWFAYWGQGTLLTVSS/EPKSSDKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVK
HEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPR
EPQVYTLPPSREQMTKNQVLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWQQGN
VFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 526)

Chain 3 - C6-30[CLDN6] L1

DIQMTQSPSSLSASVGDRTITCQASEDIYNRLAWYQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQYQWSSPLTFGGGKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
527)

Figure 21A

>XENP34233 C6-30[CLDN6] H1 L1 Fab-C6-30[CLDN6] H1 L1 Fab (G4S)2 [ANTI-
CD3] L1.47 H1.30 scFv(GKPGS)4 (G4S)2 Fc(222) IgG1 PVA /S267K/S364K/E357Q-IgG1 pl(-
) Isosteric A PVA /S267K/L368D/K370S

Chain 1 - C6-30[CLDN6]_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO:
528)

Chain 2 - C6-30[CLDN6]_H1_Fab_(G4S)2_[ANTI-

CD3]_L1.47_H1.30_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCGGGGSGG
GGG/QAVVTQEPSTLVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKA
ALTISGAQPEDEADYYCALWYSNHWVFSGGKTCLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLR
LSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVY
YCVRHGNFQDSYVSWFAYWGQGLTVTVSS/GGGGSGGGGSKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT
ISKAKGQPREPQVYTLPPSREQMTKNQVKLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTV
DKSRWQQGNVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 529)

Chain 3 - C6-30[CLDN6]_L1_Light Chain

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTVKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSSLTLSKADYEKKHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
530)

Figure 21B

>XENP34638 C6-30[CLDN6]_H2_L1_Fab-C6-30[CLDN6]_H2_L1_Fab (G4S)2 [ANTI-CD3]_L1.47_H1.30_scFv(GKPGS)4_(G4S)2-IgG1_pl(-)_Isosteric A_PVA_/S267K/L368D/K370S-Fc(222)_IgG1_PVA_/S267K/S364K/E357Q

Chain 1 - C6-30[CLDN6]_H2_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSLKASDTAMYYCARIYYFGRLYFDFWAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
531)

Chain 2 - C6-30[CLDN6]_H2_(G4S)2_[ANTI-

CD3]_L1.47_H1.30_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNGNTHYNQKFQGHVTISVDKSIS
TAYLQWSSLKASDTAMYYCARIYYFGRLYFDFWAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCGGGGSGG
GGG/QAVVTQEPSTLTVSPGGTIVTLTCSSTQAVTTSNYANWVQQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGA
ALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLR
LSCAASGFTFTSYAMNHWVRQAPGKGLEWVGRIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVY
YCVRHGNFGDSYVSWFAYWGQGLTVTVSS/GGGGSGGGGSKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT
ISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTV
DKSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 532)

Chain 3 - C6-30[CLDN6]_L1_Light Chain

DIQMTQSPSSLSASVGRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQYQWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNFFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
533)

Figure 22A

>XENP35386 C6-30[CLDN6] H1 L1 Fab-C6-30[CLDN6] H1 L1 Fab (G4S)2 [ANTI-CD3] L1.47 H1.32 scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S-Fc(222) IgG1 PVA /S267K/S364K/E357Q

Chain 1 - C6-30[CLDN6]_H1_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSSLRSED TAVYYCARIYYFGRLYDFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO:
534)

Chain 2 - C6-30[CLDN6]_H1_L1_Fab_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSSLRSED TAVYYCARIYYFGRLYDFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCGGGGSGG
GGS/QAVVTQEPSTLTVSPGGTVTLTQGSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKA
ALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLR
LSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVY
YCVRHGNEFGDSYVSWFAYWGQGLTVTVSS/GGGGSGGGGSKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT
ISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTV
DKSRWQQGNVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 535)

Chain 3 - C6-30[CLDN6]_L1_Light Chain

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSFIFFPSDEQLKSGTASVVCLLNFFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKKHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
536)

>XENP35388 C6-30[CLDN6] H2 L1 Fab-C6-30[CLDN6] H2 L1 Fab (G4S)2 [ANTI-CD3] L1.47 H1.32 scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S-Fc(222) IgG1 PVA /S267K/S364K/E357Q

Chain 1 - C6-30[CLDN6]_H2_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNNGNTHYNQKFQGHVTISVDSKIS
TAYLQWSSLKASDTAMYYCARIYYFGRLYDFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO:
537)

Figure 22B

Chain 2 - C6-30[CLDN6]_H2_L1_Fab_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLLEWMGGIDPNNGNTHYNQKFQGHVTISVDSKIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDWAGAGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCGGGGSGG
GGG/QAVVTQEPSTLTVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKA
ALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLR
LSCAASGFTFSTYAMNWRQAPGKGLEWVGRIKSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVY
YCVRHGNGFGDSYVSWFAYWGQGTLLTVSS/GGGGSGGGGSKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKT
SKAKGQPREPQVYTLPPSREQMTKNQVQLTCLVKGFPYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVD
KSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 538)

Chain 3 - C6-30[CLDN6]_L1_Light Chain

DIQMTQSPSSLSASVGDRTVITCQASEDIYNRLAWYQQKPKGVKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSSPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
539)

**>XENP37227 C6-30[CLDN6]_H1.9_L1.187_Fab-C6-30[CLDN6]_H1.9_L1.187_Fab_(G4S)2_[ANTI-
CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2-IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S-
Fc(222)_IgG1_PVA_/S267K/S364K/E357Q****Chain 1 - C6-30[CLDN6]_H1.9_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNANTHYNQKFQGRVTITVDKSAS
TAYMELSSSLRSEDATAVYYCARIYYFGRLYFDWAGAGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEYNSTYRVVSV
LTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
540)

**Chain 2 - C6-30[CLDN6]_H1.9_(G4S)2_[ANTI-
CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNANTHYNQKFQGRVTITVDKSAS
TAYMELSSSLRSEDATAVYYCARIYYFGRLYFDWAGAGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCGGGGSGG
GGG/QAVVTQEPSTLTVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKA
ALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLR
LSCAASGFTFSTYAMNWRQAPGKGLEWVGRIKSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVY
YCVRHGNGFGDSYVSWFAYWGQGTLLTVSS/GGGGSGGGGSKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKT
SKAKGQPREPQVYTLPPSREQMTKNQVQLTCLVKGFPYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVD
KSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 541)

Chain 3 - C6-30[CLDN6]_L1.187_Light Chain

DIQMTQSPSSLSASVGDRTVITCQASEDIYNRLAWYQQKPKGVKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSAPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
542)

Figure 22C

>XENP37228 C6-30[CLDN6] H1.19 L1.187 Fab-C6-30[CLDN6] H1.19 L1.187 Fab (G4S)2 [ANTI-CD3] L1.47 H1.32_scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S-Fc(222) IgG1 PVA /S267K/S364K/E357Q

Chain 1 - C6-30[CLDN6]_H1.19_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNNGNTHYNQKFEQGRVTITVDKSAS
TAYMELSSLRSEDTATYYCARIYYFGRLYFDGAGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
543)

Chain 2 - C6-30[CLDN6]_H1.19_(G4S)2_[ANTI-CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNNGNTHYNQKFEQGRVTITVDKSAS
TAYMELSSLRSEDTATYYCARIYYFGRLYFDGAGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCGGGGSGG
GGG/QAVVTQEPSTLTVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGA
ALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLR
LSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVY
YCVRHGNFGDSYVSWFAYWGQGTITVTVSS/SGGGSGGGGSKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT
ISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVD
KSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 544)

Chain 3 - C6-30[CLDN6]_L1.187_Light Chain

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQYWSAPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
545)

>XENP37229 C6-30[CLDN6] H1.22 L1.187 Fab-C6-30[CLDN6] H1.22 L1.187 Fab (G4S)2 [ANTI-CD3] L1.47 H1.32_scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S-Fc(222) IgG1 PVA /S267K/S364K/E357Q

Chain 1 - C6-30[CLDN6]_H1.22_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNNGNTHYNQKFEQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDGAGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEEQGDVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
546)

Figure 22D

Chain 2 - C6-30[CLDN6]_H1.22_(G4S)2_[ANTI-**CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARILYFGRLYFD~~FW~~GAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCGGGGSGG
GGG/QAVVTQEP~~SL~~TVSPGGT~~VT~~LT~~CG~~SSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKA
ALTISGAQPEDEADYYCALWYSNHWVF~~GGG~~GKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLR
LSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRI~~SK~~ANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVY
YCVRHGNEFGDSYVSWFAYWGQGLTVTVSS/GGGGSGGGGSKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI
SKAKGQPREPQVYTLPPSREQMTKNQV~~KL~~TCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTV
D KSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 547)

Chain 3 - C6-30[CLDN6]_L1.187_Light Chain

DIQMTQSPSSLSASVGRVTITCQASEDIYNRLAWYQQKPKV~~PKLL~~ISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIA~~TY~~YCQYWSAPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVC~~LL~~NNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDS~~TY~~SLSTLTLSKADYEKKHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
548)

>XENP37230 C6-30[CLDN6]_H1.22_L1.189_Fab-C6-30[CLDN6]_H1.22_L1.189_Fab_(G4S)2_[ANTI-**CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2-IgG1_pl(-)_Isosteric A_PVA_/S267K/L368D/K370S-****Fc(222)_IgG1_PVA_/S267K/S364K/E357Q****Chain 1 - C6-30[CLDN6]_H1.22_IgG1_pl(-)_Isosteric A_PVA_/S267K/L368D/K370S**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARILYFGRLYFD~~FW~~GAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTI~~SK~~AKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO:
549)

Chain 2 - C6-30[CLDN6]_H1.22_(G4S)2_[ANTI-**CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARILYFGRLYFD~~FW~~GAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCGGGGSGG
GGG/QAVVTQEP~~SL~~TVSPGGT~~VT~~LT~~CG~~SSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKA
ALTISGAQPEDEADYYCALWYSNHWVF~~GGG~~GKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLR
LSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRI~~SK~~ANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVY
YCVRHGNEFGDSYVSWFAYWGQGLTVTVSS/GGGGSGGGGSKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI
SKAKGQPREPQVYTLPPSREQMTKNQV~~KL~~TCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTV
D KSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 550)

Chain 3 - C6-30[CLDN6]_L1.189_Light Chain

DIQMTQSPSSLSASVGRVTITCQASEDIYNRLAWYQQKPKV~~PKLL~~ISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIA~~TY~~YCQYWSG~~PL~~TFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVC~~LL~~NNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDS~~TY~~SLSTLTLSKADYEKKHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
551)

Figure 22E

>XENP37231 C6-30[CLDN6] H1.24 L1.187 Fab-C6-30[CLDN6] H1.24 L1.187 Fab (G4S)2 [ANTI-CD3] L1.47 H1.32 scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S-Fc(222) IgG1 PVA /S267K/S364K/E357Q

Chain 1 - C6-30[CLDN6] H1.24_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYLGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDFVSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO:
552)

Chain 2 - C6-30[CLDN6] H1.24_(G4S)2_[ANTI-

CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYLGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCGGGGSGG
GGG/QAVVTQEPSTLTVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGA
ALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLR
LSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRADTAVY
YCVRHGNGFSDSYVSWFAWVGQGLTVTVSS/GGGGSGGGGSKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT
ISKAKGQPREPQVYTLPPSREQMTKNQVLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVD
KSRWQQGNVFSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO: 553)

Chain 3 - C6-30[CLDN6]_L1.187_Light Chain

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSAPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNFFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYLSSTLTLSKADYEKKHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
554)

>XENP37232 C6-30[CLDN6] H1.24 L1.189 Fab-C6-30[CLDN6] H1.24 L1.189 Fab (G4S)2 [ANTI-CD3] L1.47 H1.32 scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S-Fc(222) IgG1 PVA /S267K/S364K/E357Q

Chain 1 - C6-30[CLDN6] H1.24_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYLGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDFVSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO:
555)

Figure 22F

Chain 2 - C6-30[CLDN6]_H1.24_(G4S)2_[ANTI-**CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q**

QVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTMHWVRQAPGQSLLEWMGGIDPNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYLGRLYFD~~FW~~GAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCGGGGSGG
GGG/QAVVTQEP~~SL~~TVSPGGTVTLT~~CG~~SSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKA
ALTISGAQPEDEADYYCALWYSNHWVF~~GGG~~GKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLR
LSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVY
YCVRHGNEFGDSYVSWFAYWGQGLTVTVSS/GGGGSGGGGSKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI
SKAKGQPREPQVYTLPPSREQMTKNQV~~KL~~TCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTV~~D~~
KSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 556)

Chain 3 - C6-30[CLDN6]_L1.189_Light Chain

DIQMTQSPSSLSASVGRVTITCQASEDIYNRLAWYQQKPKV~~PKLL~~ISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIA~~TY~~YCQ~~Q~~YWSG~~PL~~TFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVC~~LL~~NNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSS~~TL~~TL~~SK~~ADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
557)

>XENP37233 C6-30[CLDN6]_H2.91_L1.187_Fab-C6-30[CLDN6]_H2.91_L1.187_Fab (G4S)2_[ANTI-**CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2-IgG1_pl(-)_Isosteric A_PVA_/S267K/L368D/K370S-****Fc(222)_IgG1_PVA_/S267K/S364K/E357Q****Chain 1 - C6-30[CLDN6]_H2.91_IgG1_pl(-)_Isosteric A_PVA_/S267K/L368D/K370S**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNFNTHYNQKFQGHVTISVDK~~SI~~S
TAYLQWSS~~L~~KASDTAMYYCARIYYFGRLYFD~~FW~~GAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTI~~SK~~AKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTTTPVLDSDGSFFLYSKLTV~~DK~~SRWEQGDVFC~~SV~~MHEALHNHYTQKSLSLSPGK (SEQ ID NO:
558)

Chain 2 - C6-30[CLDN6]_H2.91_(G4S)2_[ANTI-**CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q**

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDPNNFNTHYNQKFQGHVTISVDK~~SI~~S
TAYLQWSS~~L~~KASDTAMYYCARIYYFGRLYFD~~FW~~GAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCGGGGSGG
GGG/QAVVTQEP~~SL~~TVSPGGTVTLT~~CG~~SSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKA
ALTISGAQPEDEADYYCALWYSNHWVF~~GGG~~GKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLR
LSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVY
YCVRHGNEFGDSYVSWFAYWGQGLTVTVSS/GGGGSGGGGSKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI
SKAKGQPREPQVYTLPPSREQMTKNQV~~KL~~TCLVKGFYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTV~~D~~
KSRWQQGNVFSCSVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 559)

Chain 3 - C6-30[CLDN6]_L1.187_Light Chain

DIQMTQSPSSLSASVGRVTITCQASEDIYNRLAWYQQKPKV~~PKLL~~ISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIA~~TY~~YCQ~~Q~~YWSA~~PL~~TFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVC~~LL~~NNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSS~~TL~~TL~~SK~~ADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
560)

Figure 22G

>XENP37547 C6-30[CLDN6] H2.91 L1.187 Fab-C6-30[CLDN6] H2.91 L1.187 Fab (G4S)2 [anti-CD3] L1.47 H1.32 scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S/M428L/N434S-Fc(222) IgG1 PVA /S267K/S364K/E357Q/M428L/N434S

Chain 1 - C6-30[CLDN6]_H2.91_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDFNNFNTHYNQKFQGHVTISVDSKIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDFVSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO:
561)

Chain 2 - C6-30[CLDN6]_H2.91_(G4S)2_[anti-

CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q/M428L/N434S

EVQLVQSGAEVKKPGESLRISCKTSGYTFTEYTMHWVRQMPGKSLEWMGGIDFNNFNTHYNQKFQGHVTISVDSKIS
TAYLQWSSSLKASDTAMYYCARIYYFGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCGGGGSGG
GGG/QAVVTQEPSTLTVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGA
ALTISGAQPEDEADYYCALWYSNHWYFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLR
LSCAASGFTTFSTYAMNWVRQAPGKLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVY
YCVRHGNEFGDSYVSWFAYWGQGTILTVTVSS/GGGGSGGGGSKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT
ISKAKGQPREPQVYTLPPSREQMTKNQVSLTCLVKGFIYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVD
KSRWQQGNVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 562)

Chain 3 - C6-30[CLDN6]_L1.187_Light Chain

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQYWSAPLTFGGGTKEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSTLTLSKADYEKKHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
563)

>XENP37545 C6-30[CLDN6] H1.24 L1.187 Fab-C6-30[CLDN6] H1.24 L1.187 Fab (G4S)2 [anti-CD3] L1.47 H1.32 scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S/M428L/N434S-Fc(222) IgG1 PVA /S267K/S364K/E357Q/M428L/N434S

Chain 1 - C6-30[CLDN6]_H1.24_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S

QVQLVQSGAEVKKPGASVKVCKTSGYTFTEYTMHWVRQAPGQSLWMMGGIDFNNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYLGRLYFDFWGAGTLVTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDFVSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO:
564)

Figure 22H

Chain 2 - C6-30[CLDN6]_H1.24_(G4S)2_[anti-**CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q/M428L/N434S**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNGNTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYLGRLYFDWAGAGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCGGGGSGG
GGG/QAVVTQEPSTLTVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKA
ALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLR
LSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVY
YCVRHGNFEGDSYVSWFAYWGQGTLLTVSS/SGGGSGGGGSKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI
SKAKGQPREPQVYTLPPSREQMTKNQVCLTCLVKGFPYSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVD
KSRWQQGNVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 565)

Chain 3 - C6-30[CLDN6]_L1.187_Light Chain

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPKGVKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSAPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
566)

**>XENP37541 C6-30[CLDN6]_H1.9_L1.187_Fab-C6-30[CLDN6]_H1.9_L1.187_Fab_(G4S)2_[anti-
CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2-IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S/M428L/N434S-
Fc(222)_IgG1_PVA_/S267K/S364K/E357Q/M428L/N434S****Chain 1 - C6-30[CLDN6]_H1.9_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNANTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDWAGAGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTI SKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO:
567)

Chain 2 - C6-30[CLDN6]_H1.9_(G4S)2_[anti-**CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q/M428L/N434S**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNANTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDWAGAGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCGGGGSGG
GGG/QAVVTQEPSTLTVSPGGTVTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKA
ALTISGAQPEDEADYYCALWYSNHWVFGGGTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLR
LSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVY
YCVRHGNFEGDSYVSWFAYWGQGTLLTVSS/SGGGSGGGGSKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKTI
SKAKGQPREPQVYTLPPSREQMTKNQVCLTCLVKGFPYSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVD
KSRWQQGNVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 568)

Chain 3 - C6-30[CLDN6]_L1.187_Light Chain

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPKGVKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDATYYCQQYWSAPLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCCLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
569)

Figure 23B

C6-11

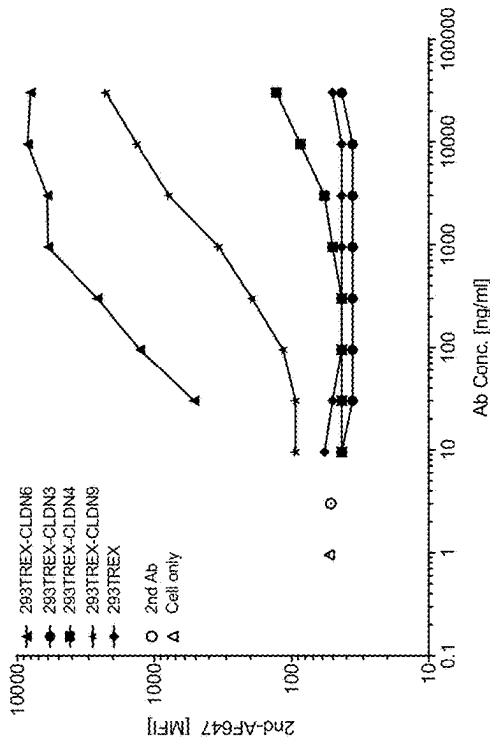


Figure 23D

C6-21

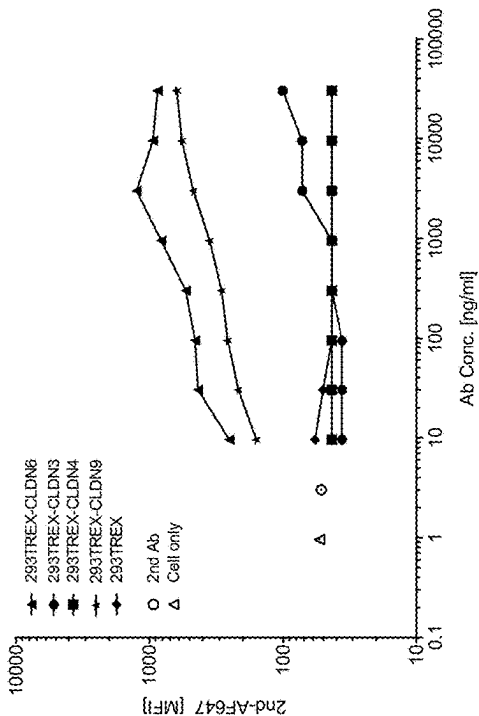


Figure 23A

C6-10

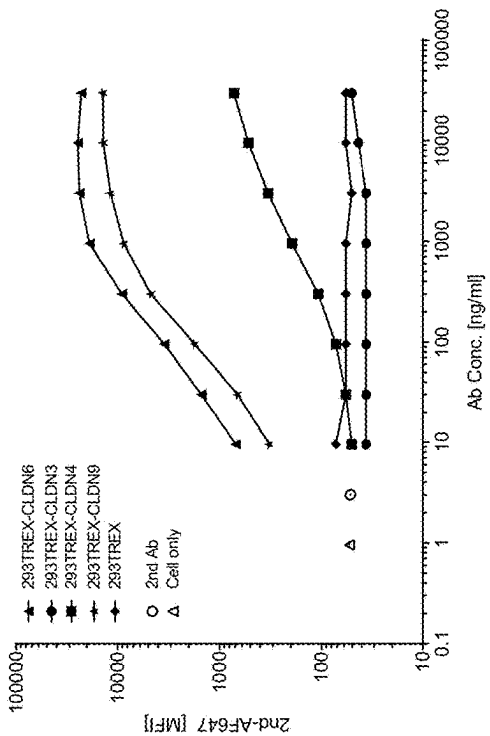


Figure 23C

C6-15

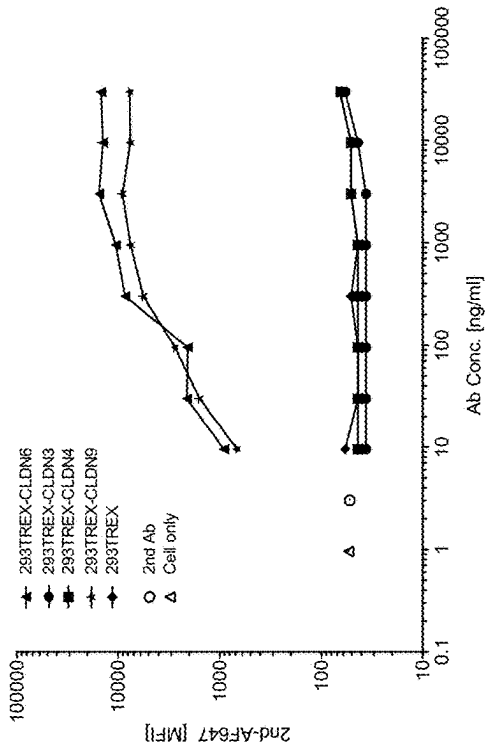


Figure 23F
C6-30

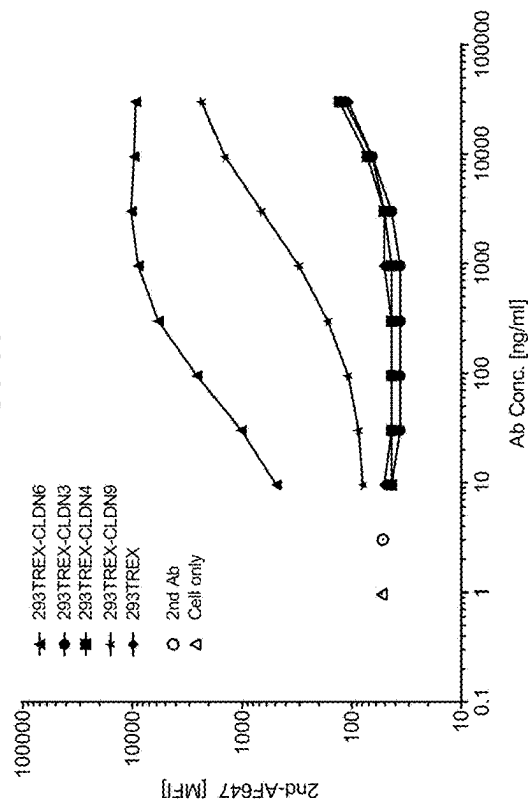
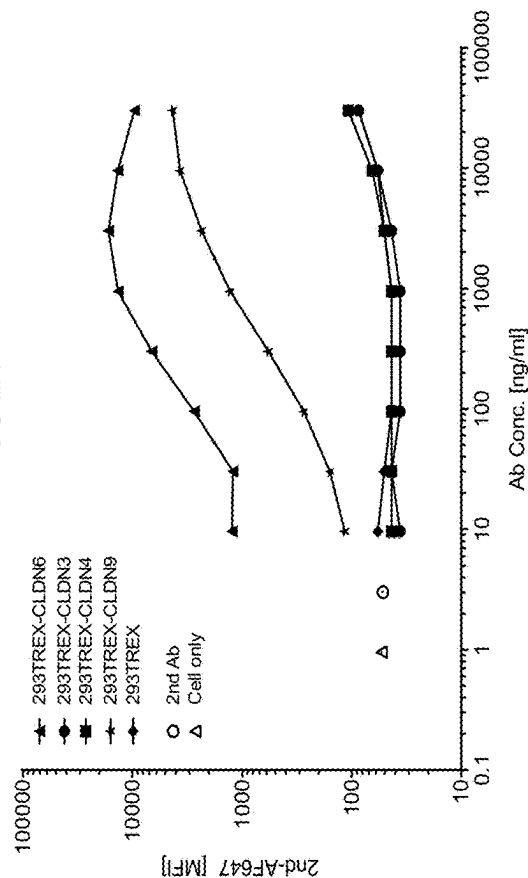
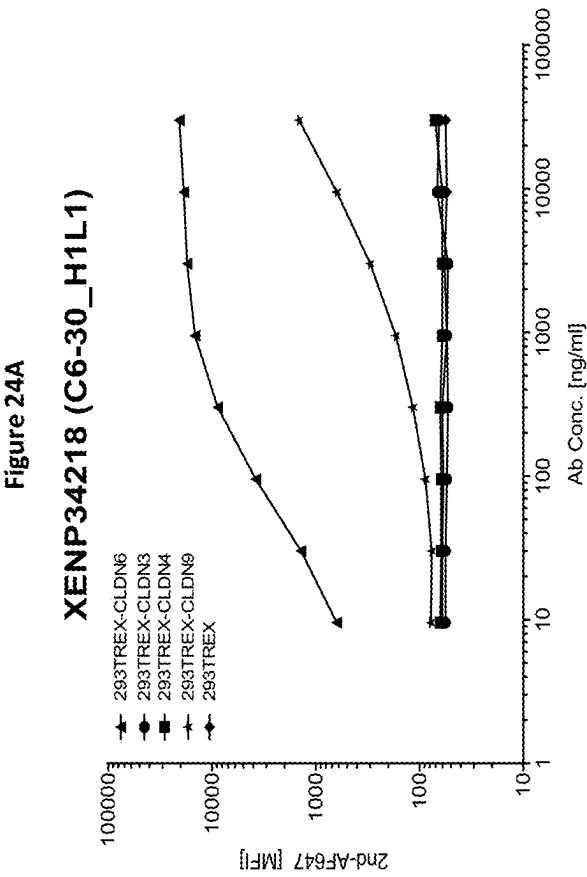
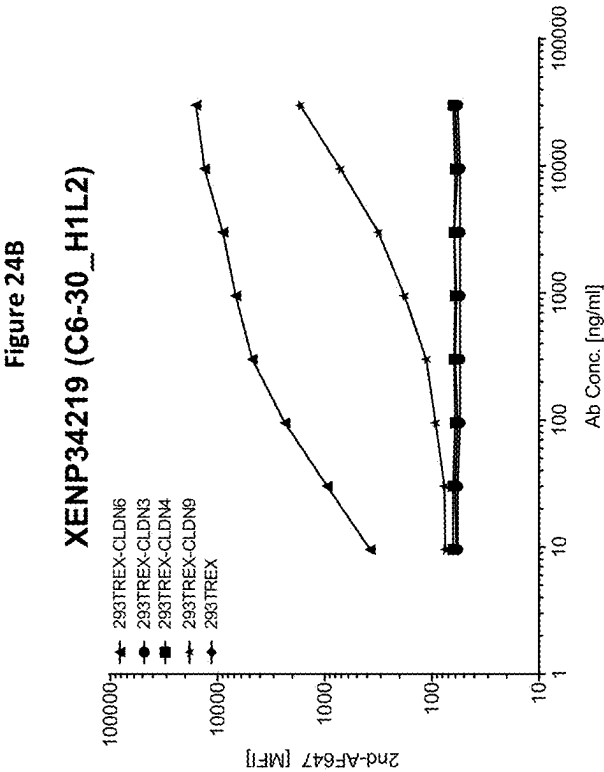


Figure 23E
C6-24





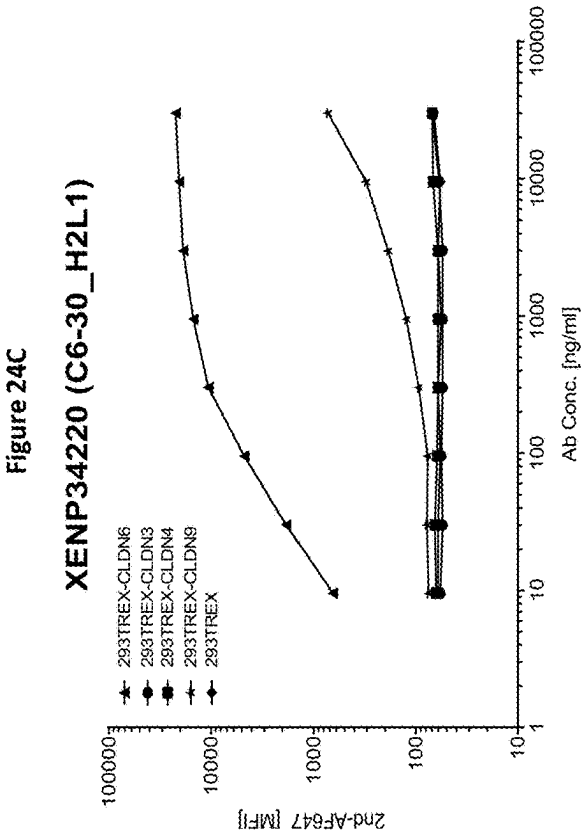
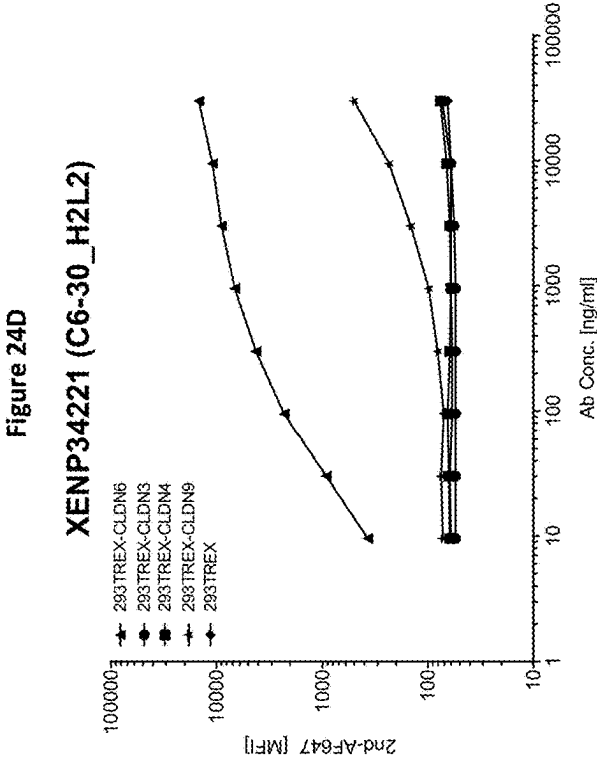


Figure 24E

C6-30 Murine Precursor

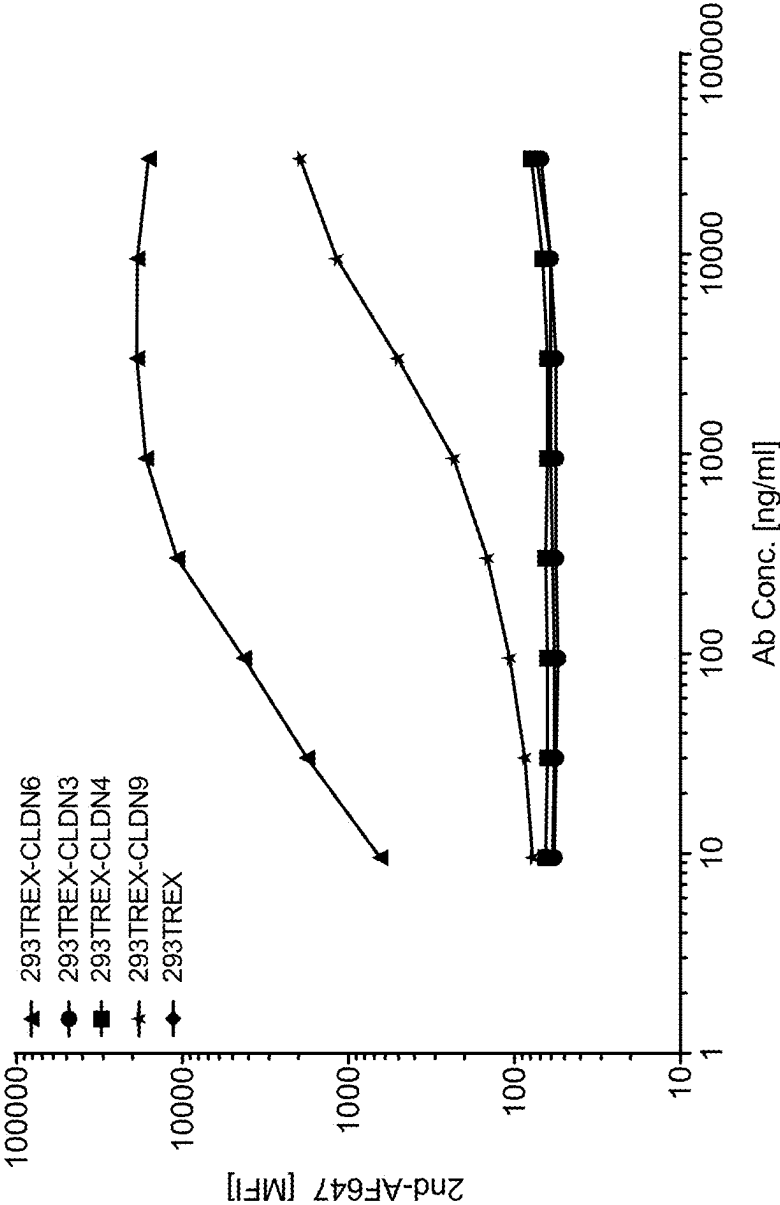


Figure 25A

XENP34208 (C6-11_H1L1)

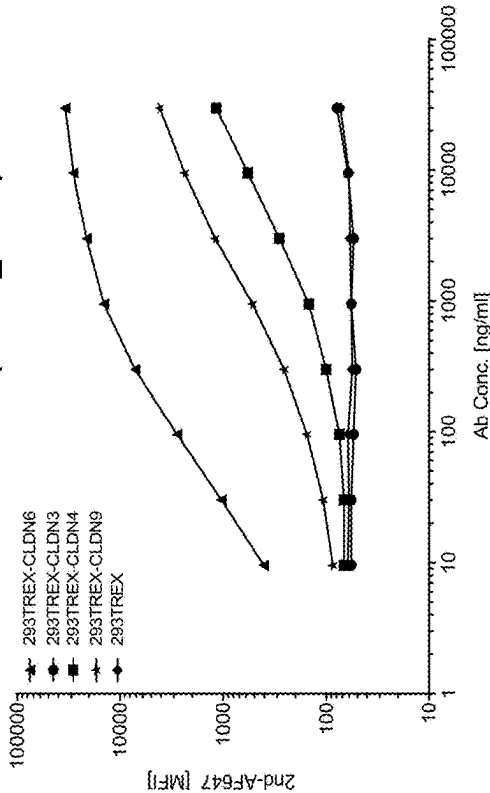


Figure 25B

XENP34209 (C6-11_H1L2)

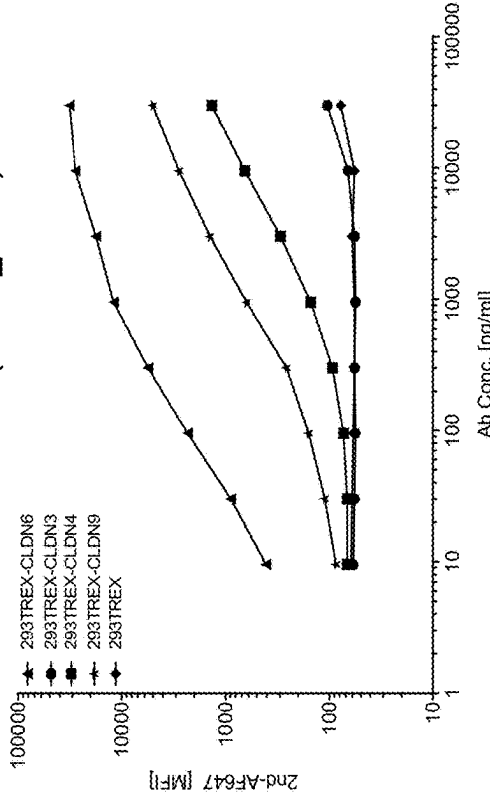


Figure 25C

XENP34212 (C6-11_H2L1)

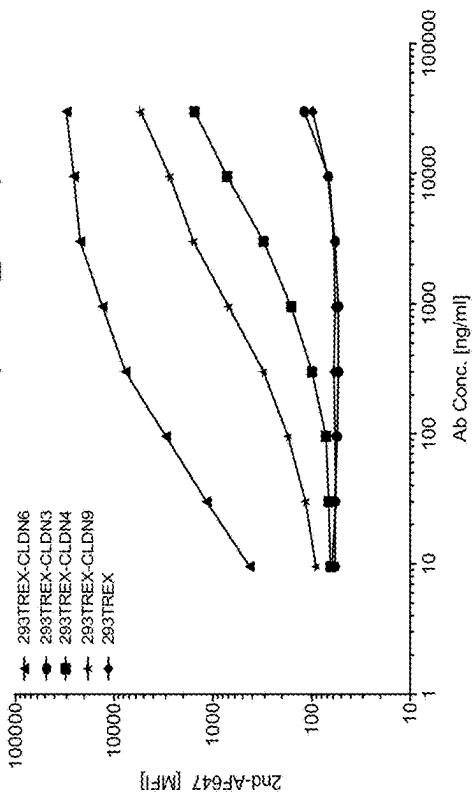


Figure 25D

XENP34213 (C6-11_H2L2)

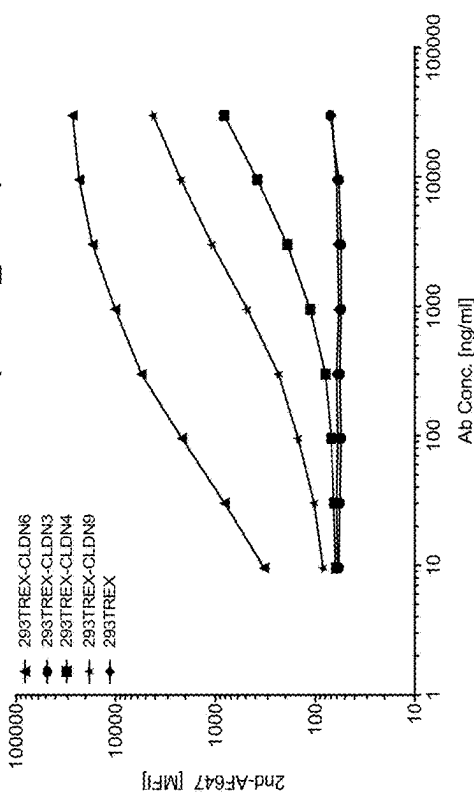


Figure 25E

C6-11 Murine Precursor

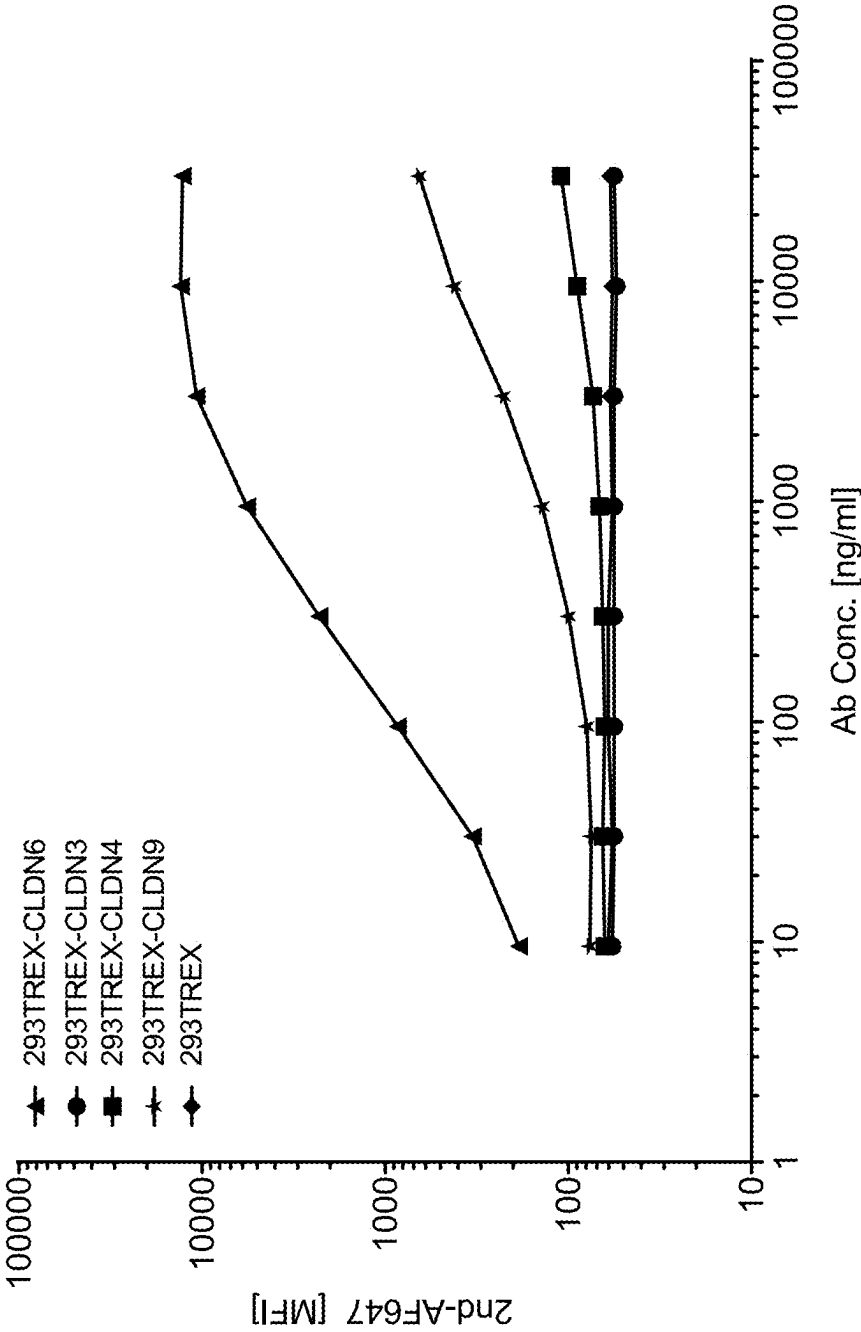


Figure 26A

XENP34214 (C6-24_H1L1)

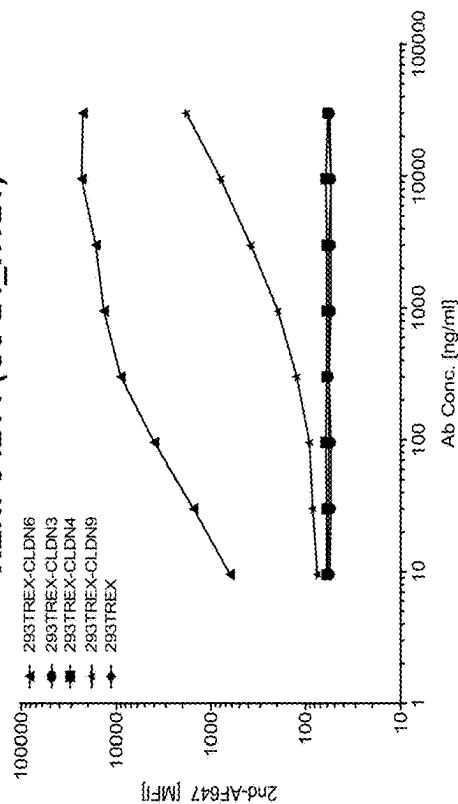


Figure 26B

XENP34215 (C6-24_H1L2)

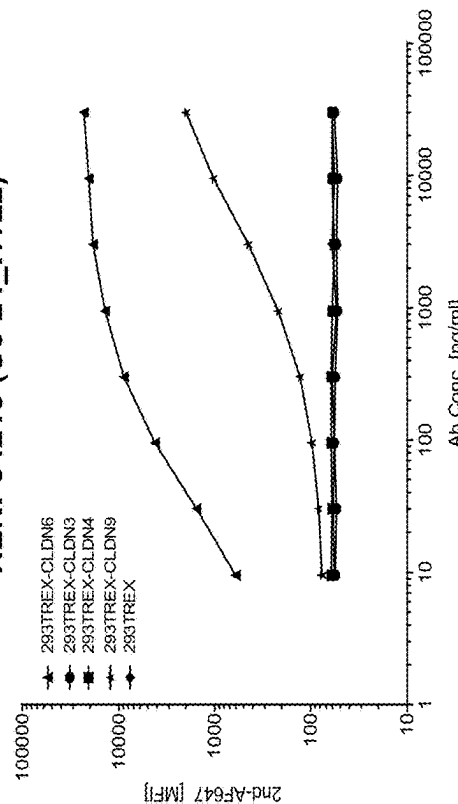


Figure 26C

XENP34216 (C6-24_H2L1)

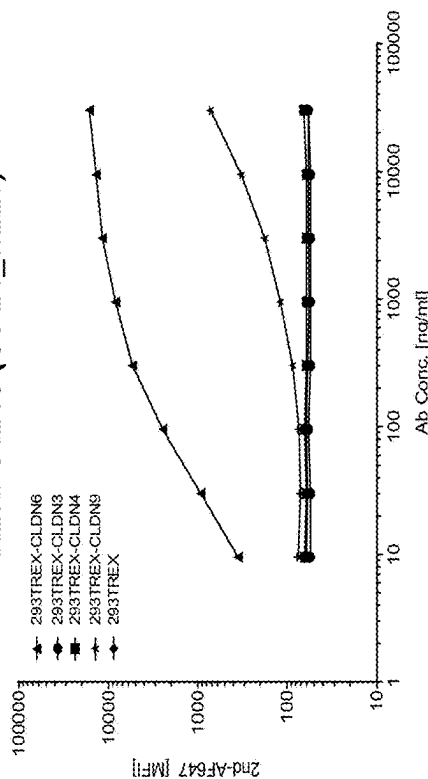


Figure 26D

XENP34217 (C6-24_H2L2)

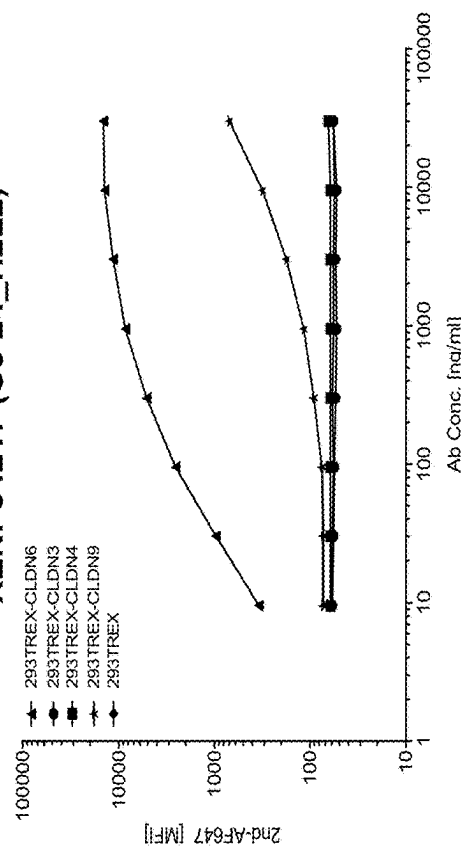


Figure 26E

C6-24 Murine Precursor

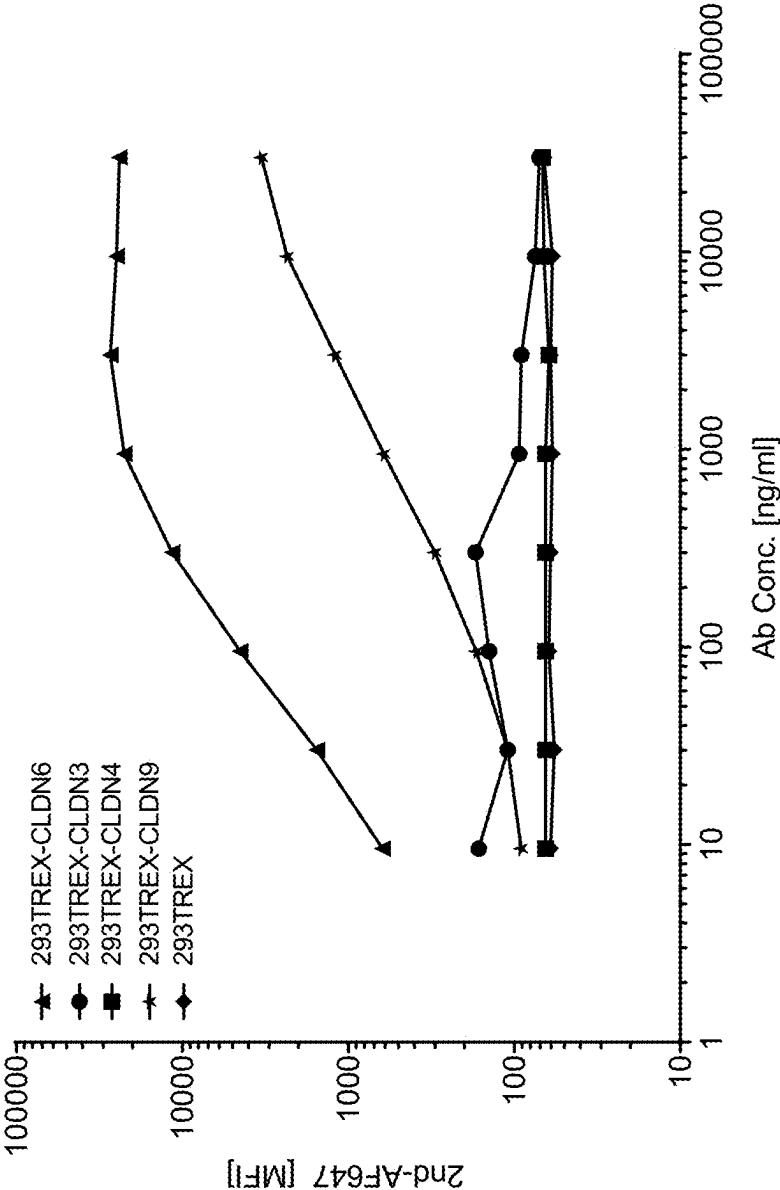


Figure 27

	<u>C6-30</u> <u>Variant</u>	<u>VH</u> <u>mutation</u> <u>(Kabat)</u>	<u>Estimated</u> <u>CLDN6</u> <u>EC50</u> <u>(ng/mL)</u>	<u>Estimated</u> <u>CLDN9</u> <u>EC50</u> <u>(ng/mL)</u>	<u>Estimated</u> <u>CLDN9/CLDN6</u> <u>EC50 ratio</u>	<u>CLDN9/CLDN6</u> <u>EC50 ratio</u> <u>normalized to</u> <u>30.1 H2L1</u>
XENP34220	H2L1	-	870.8	5821.0	6.7	1.0
XENP35085	H1.1_L1	D52S	2509.0	15452.5	6.2	0.9
XENP35086	H1.2_L1	D52N	1459.0	12133.9	8.3	1.2
XENP35087	H1.3_L1	P52aS	685.5	2818.4	4.1	0.6
XENP35088	H1.4_L1	P52aA	1124.0	5333.3	4.7	0.7
XENP35089	H1.5_L1	N53Q	1696.0	12560.3	7.4	1.1
XENP35091	H1.7_L1	N54Q	666.9	3357.4	5.0	0.8
XENP35092	H1.8_L1	G55D	2017.0	7261.1	3.6	0.5
XENP35093	H1.9_L1	G55A	1042.0	7430.2	7.1	1.1
XENP35094	H2.1_L1	D52S	1802.0	9817.5	5.4	0.8
XENP35095	H2.2_L1	D52T	2111.0	11912.4	5.6	0.8
XENP35096	H2.3_L1	P52aS	1001.0	5116.8	5.1	0.8
XENP35097	H2.4_L1	P52aG	1190.0	4446.3	3.7	0.6
XENP35098	H2.5_L1	N53Q	2204.0	7063.2	3.2	0.5
XENP35099	H2.6_L1	N54D	1628.0	13458.6	8.3	1.2
XENP35100	H2.7_L1	N54Q	800.6	3556.3	4.4	0.7
XENP35101	H2.8_L1	G55D	1190.0	9183.3	7.7	1.2

Figure 28A

XENP34228
C6-24_H1L1 x CD3 in 1 + 1 Fab-scFv-Fc

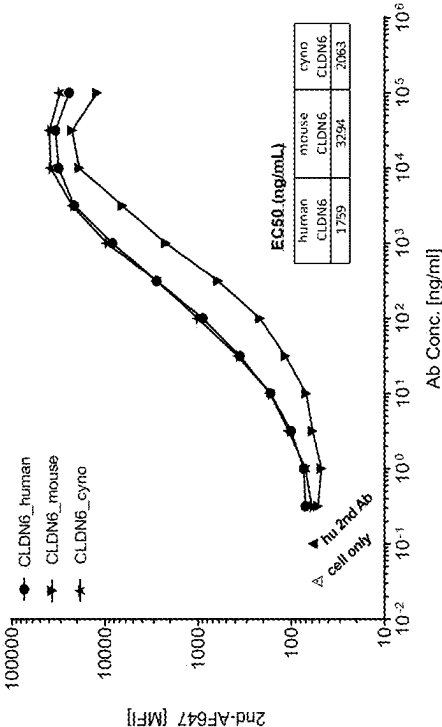


Figure 28C

XENP34229
C6-30_H1L1 x CD3 in 1 + 1 Fab-scFv-Fc

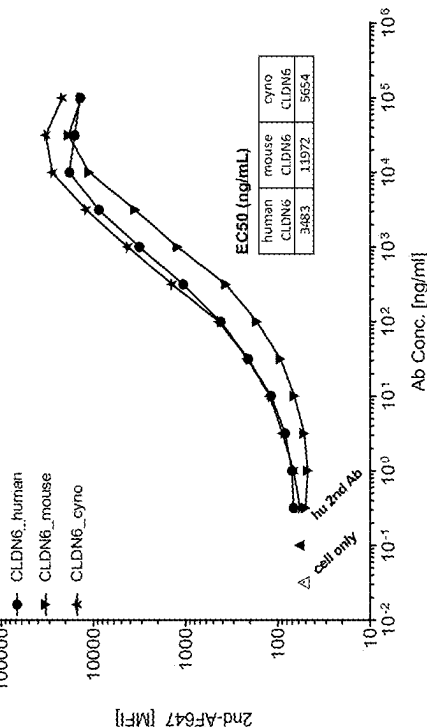


Figure 28B

XENP34232
C6-24_H1L1 x CD3 in 2 + 1 Fab₂-scFv-Fc

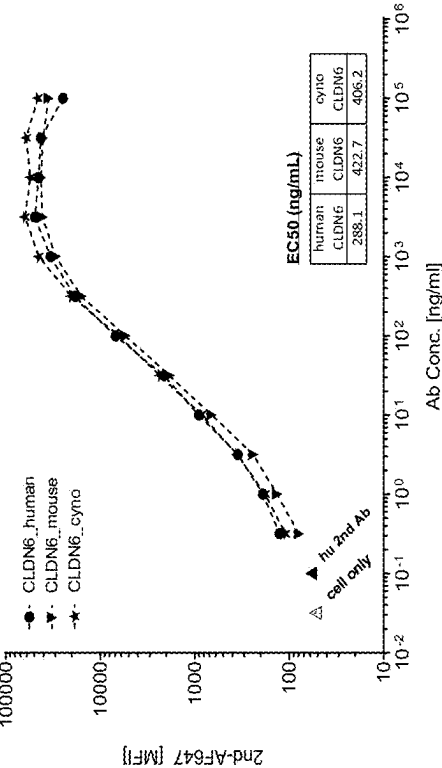


Figure 28D

XENP34233
C6-30_H1L1 x CD3 in 2 + 1 Fab₂-scFv-Fc

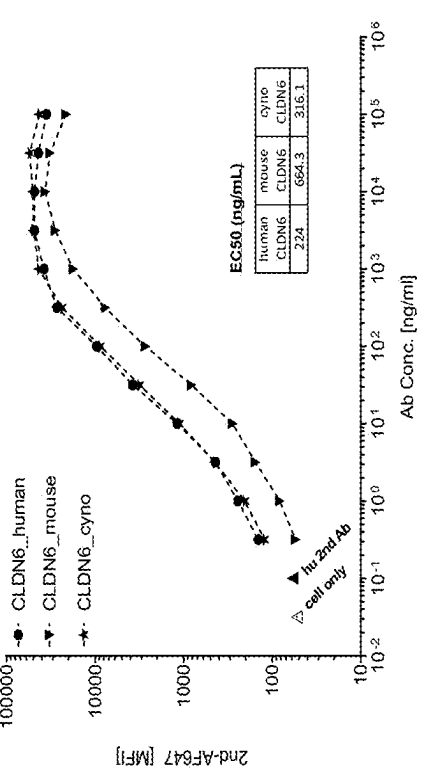


Figure 28E

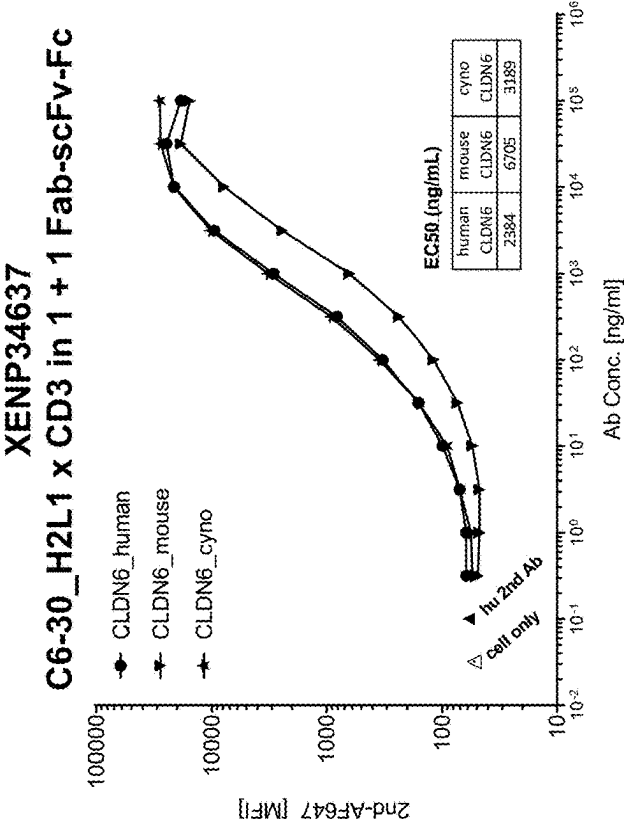


Figure 28F

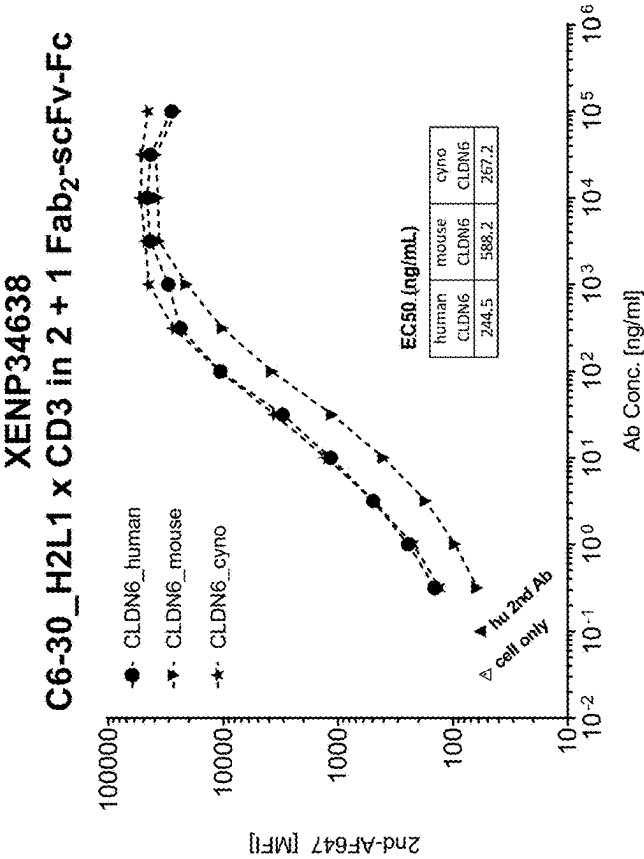


Figure 29A

XENP34228
C6-24_H1L1 x CD3 in 1 + 1 Fab-scFv-Fc

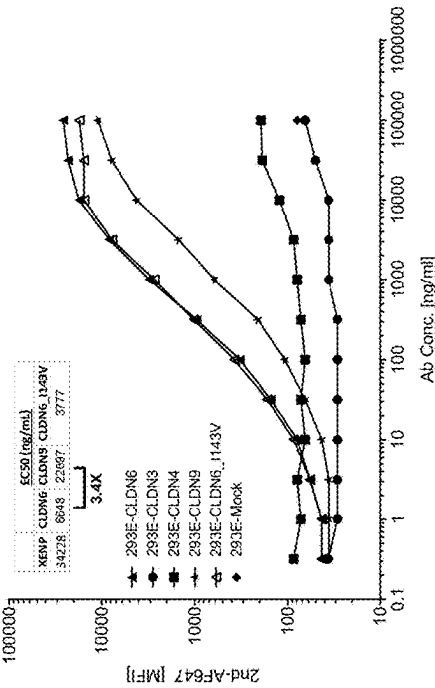


Figure 29B

XENP34232
C6-24_H1L1 x CD3 in 2 + 1 Fab₂-scFv-Fc

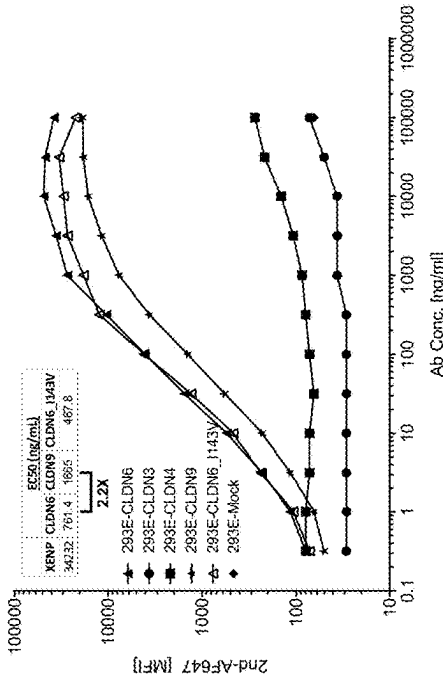


Figure 29C

XENP34229
C6-30_H1L1 x CD3 in 1 + 1 Fab-scFv-Fc

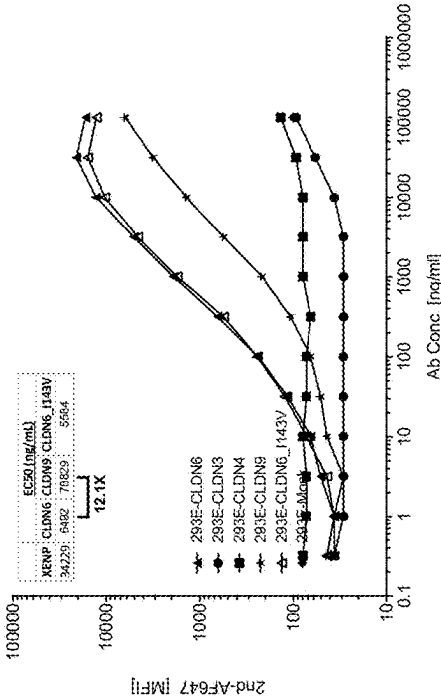


Figure 29D

XENP34233
C6-30_H1L1 x CD3 in 2 + 1 Fab₂-scFv-Fc

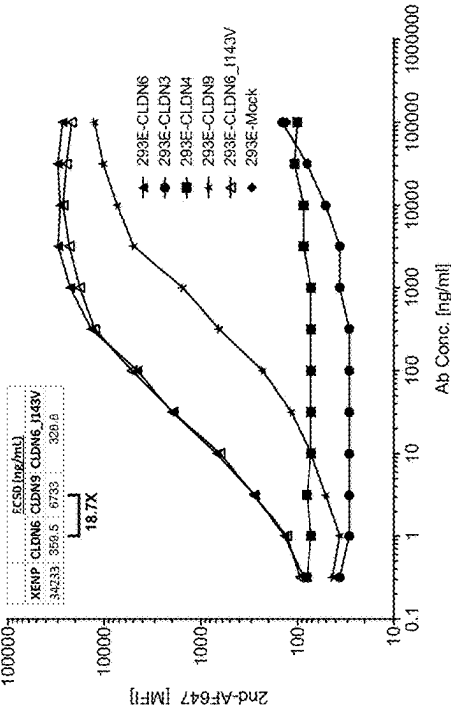


Figure 29E

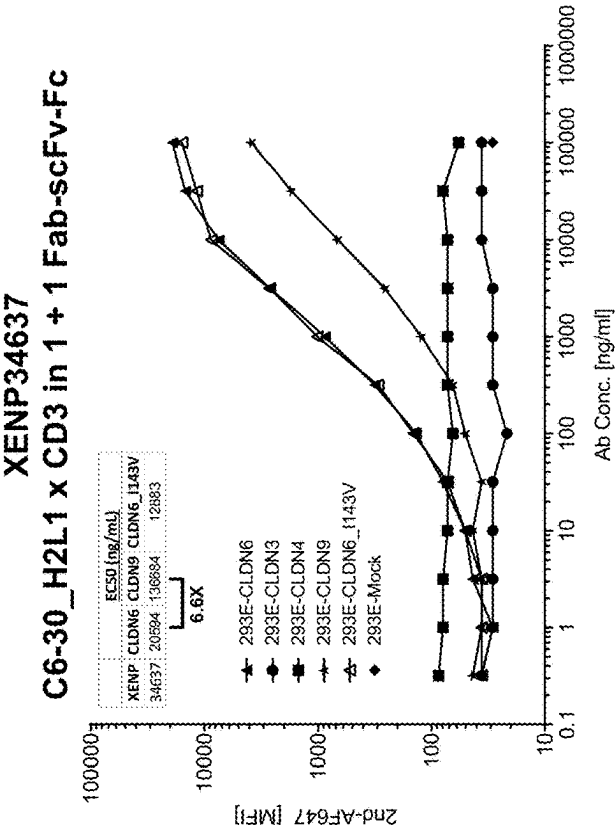


Figure 29F

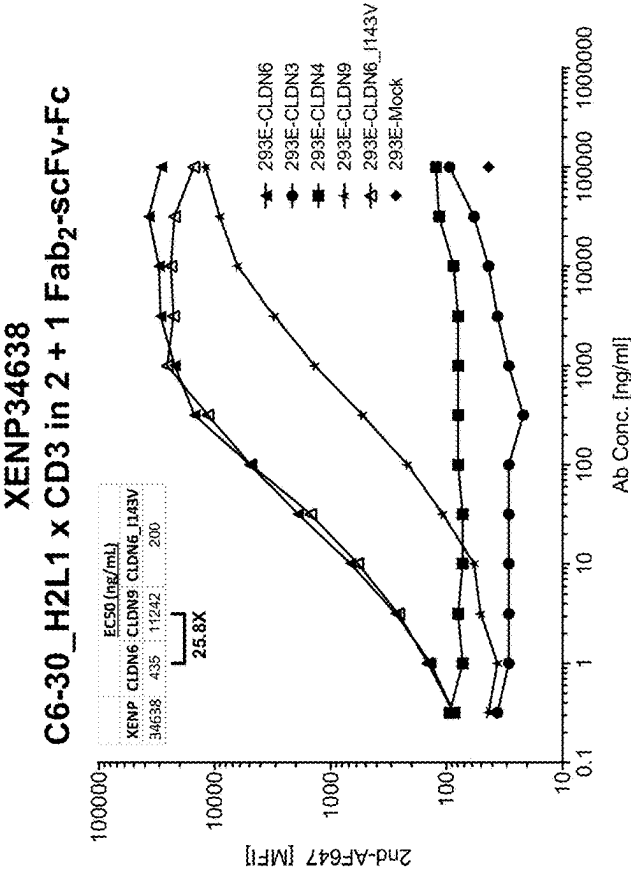


Figure 30

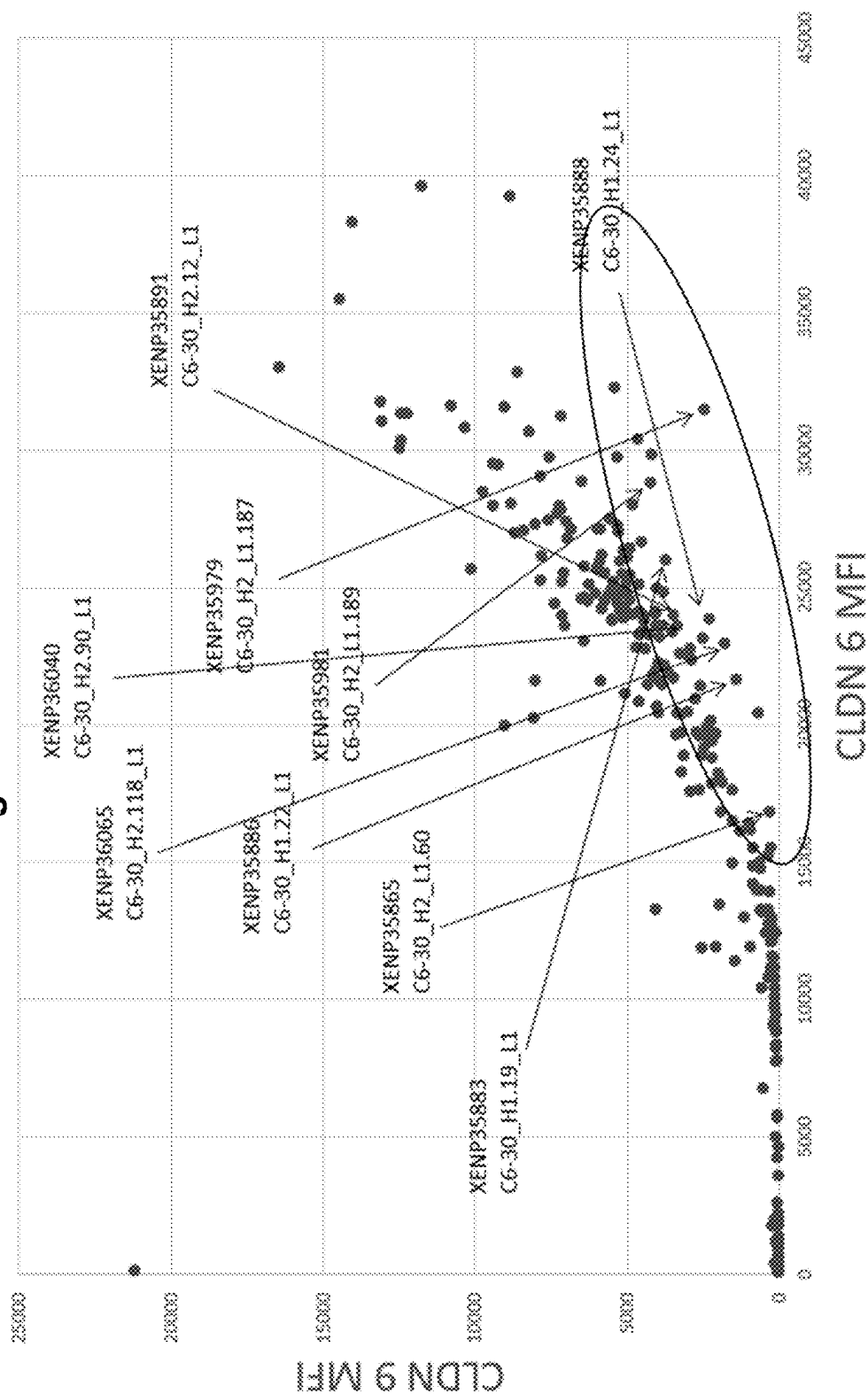


Figure 31A

A)	C6-30 Variant	EC50 (ng/ml)		CLDN9/6 ratio
		On CLDN6	On CLDN9	
XENP26863	COMPARATOR	273.8	7374	26.93
XENP34218	H1_L1	1241	54237	43.70
XENP34220	H2_L1	1632	43826	26.85
XENP35044	H2_L1.1	4338	NA	NA
XENP35047	H2_L1.4	795.2	17550	22.07
XENP35050	H2_L1.7	2218	99339	44.79
XENP35059	H2_L1.16	1276	32669	25.60
XENP35061	H2_L1.18	1248	56295	45.11
XENP35062	H2_L1.19	1225	86760	70.82
XENP35064	H2_L1.21	979.2	14147	14.45
XENP35065	H2_L1.22	1478	69069	46.73
XENP35066	H2_L1.23	1590	31803	20.00
XENP35070	H2_L1.27	1183	29684	25.09
XENP35087	H1.3_L1	1198	31760	26.51
XENP35088	H1.4_L1	1286	29242	22.74
XENP35090	H1.6_L1	1556	44628	28.68
XENP35091	H1.7_L1	1433	31427	21.93
XENP35092	H1.8_L1	1160	24998	21.55
XENP35093	H1.9_L1	978	53115	54.31
XENP35096	H2.3_L1	1241	110298	88.88
XENP35097	H2.4_L1	1103	72866	66.06
XENP35100	H2.7_L1	1139	63509	55.76
XENP35101	H2.8_L1	24783	1061	0.04
XENP35102	H2.9_L1	3526	1432	0.41

Figure 31B

B)	C6-30 Variant	EC50 (ng/ml)		CLDN9/6 ratio
		On CLDN6	On CLDN9	
XENP26863	COMPARATOR	73.97	1874	25.33
XENP34218	H1_L1	344.3	11556	33.56
XENP34220	H2_L1	390.8	11056	28.29
XENP35090	H1.6_L1	902.7	7062	7.82
XENP35865	H2_L1.60	685.7	123972	180.80
XENP35883	H1.19_L1	617.2	16767	27.17
XENP35886	H1.22_L1	941.1	24483	26.02
XENP35888	H1.24_L1	657.4	43820	66.66
XENP35890	H2.11_L1	408.1	34071	83.49
XENP35891	H2.12_L1	473.6	44135	93.19
XENP35929	H2_L1.107	791.8	99068	125.12
XENP35936	H2_L1.114	828.6	57436	69.32
XENP35979	H2_L1.187	323	29169	90.31
XENP35981	H2_L1.189	403.5	31700	78.56
XENP36021	H2.71_L1	902.4	49328	54.66
XENP36022	H2.72_L1	513.9	47177	91.80
XENP36025	H2.75_L1	1343	61089	45.49
XENP36040	H2.90_L1	705.5	66108	93.70
XENP36041	H2.91_L1	449.4	32521	72.37
XENP36065	H2.118_L1	783	491857	628.17
XENP36066	H2.119_L1	567	18031	31.80

Figure 32A

CLDN6

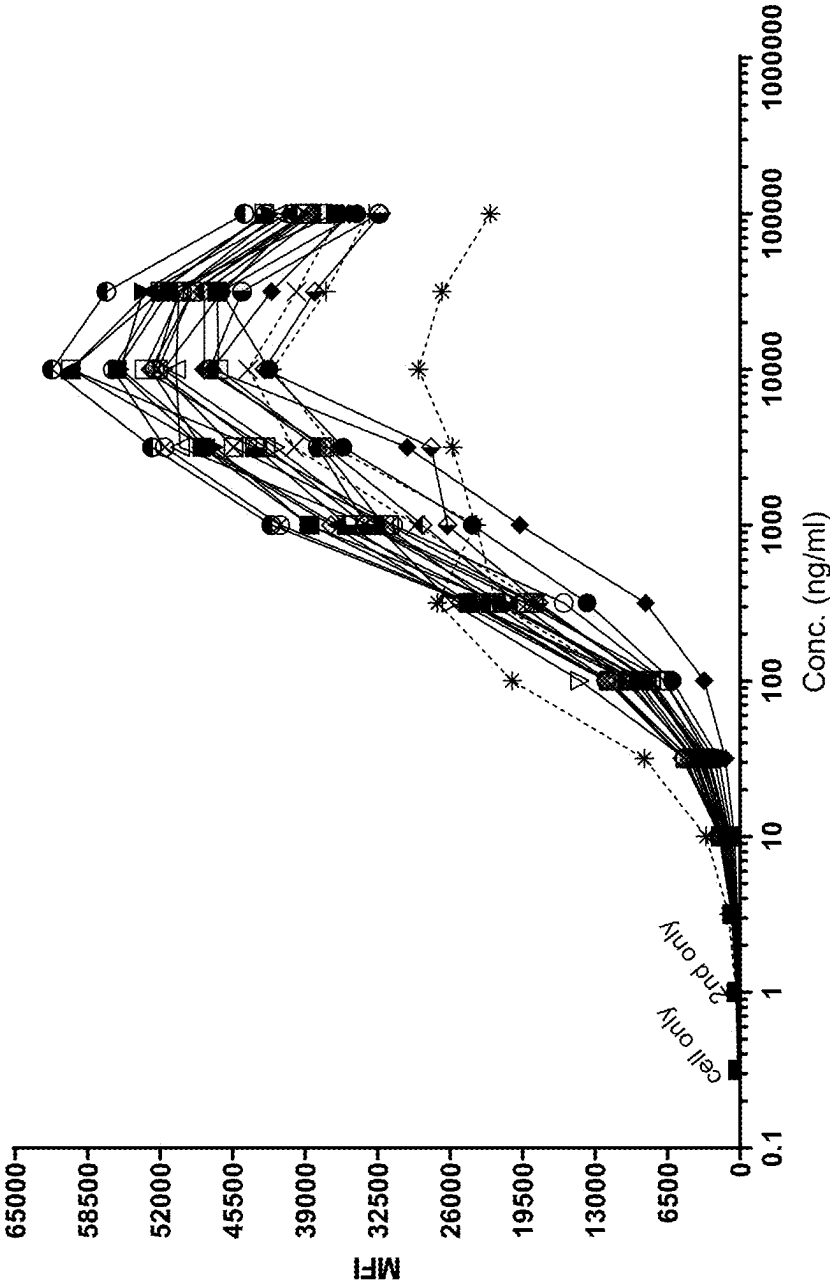


Figure 32B

CLDN9

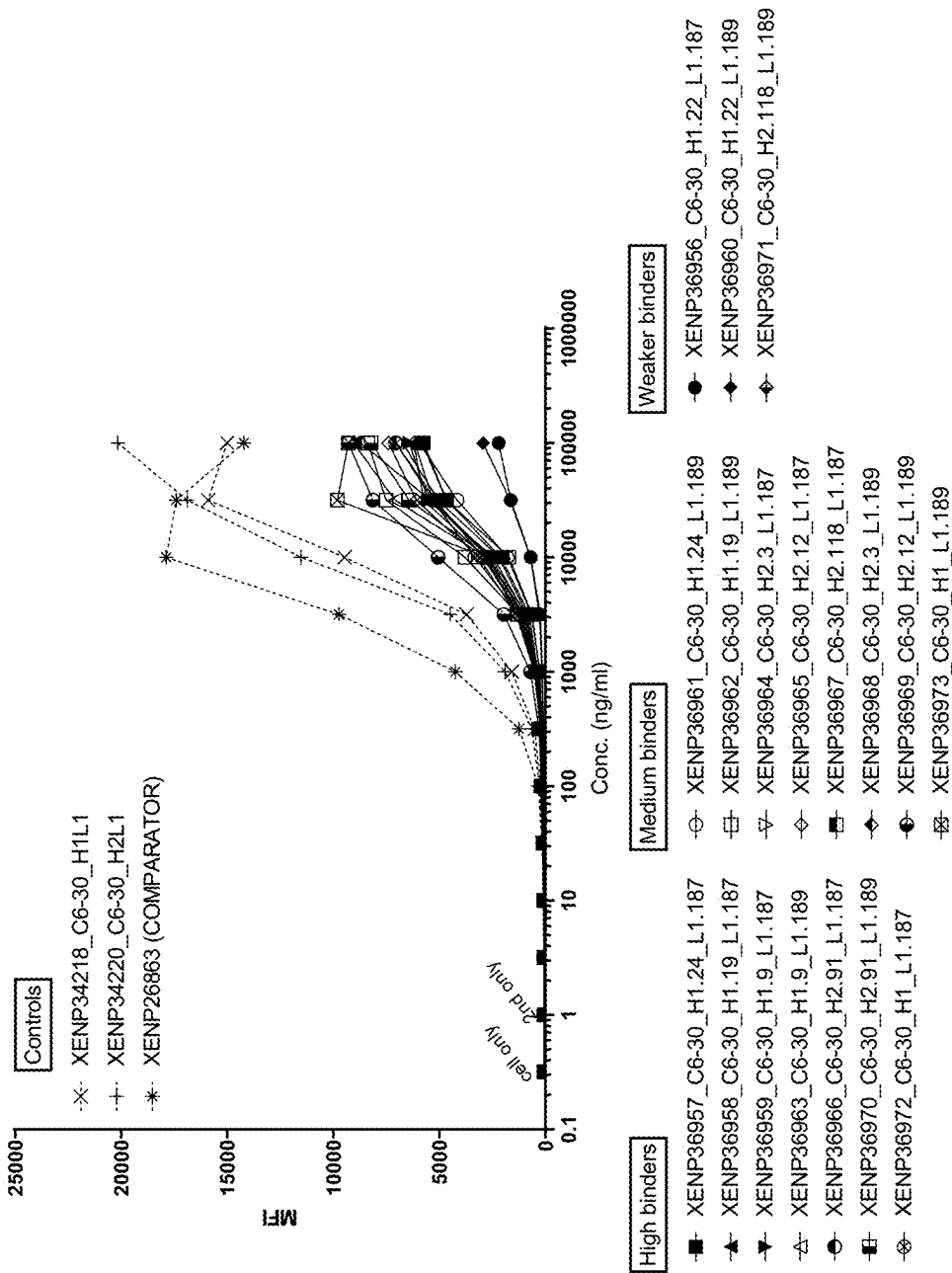


Figure 33

C6-30 Variants	CLDN6 EC50	CLDN9 EC50	CLDN9/CLDN6 EC50 Ratio	CLDN6 AUC	CLDN9 AUC	CLDN6/CLDN9 AUC Ratio
XENP34218_H1L1	373.2	8594	23.02787	3.88E+09	1.38E+09	2.802246
XENP34220_H2L1	294.7	10485	35.57855	3.62E+09	1.64E+09	2.20981
XENP26863 (COMPARATOR)	42.24	2146	50.80492	2.55E+09	1.58E+09	1.616028
XENP36956_H1.22_L1.187	658.3	29584	44.94	4.07E+09	1.61E+08	25.34196
XENP36957_H1.24_L1.187	305.3	21240	69.57091	4.63E+09	4.41E+08	10.49091
XENP36958_H1.19_L1.187	414.6	17235	41.57019	4.91E+09	4.92E+08	9.969486
XENP36959_H1.9_L1.187	392.2	18873	48.12086	4.92E+09	5.09E+08	9.67232
XENP36960_H1.22_L1.189	1072	60046	56.01306	3.93E+09	1.88E+08	20.88547
XENP36961_H1.24_L1.189	562.1	31368	55.80502	4.63E+09	4.2E+08	11.03723
XENP36962_H1.19_L1.189	450.1	15252	33.8858	4.6E+09	6.85E+08	6.710047
XENP36963_H1.9_L1.189	451.6	25916	57.38707	4.69E+09	6.73E+08	6.95726
XENP36964_H2.3_L1.187	259.5	26066	100.447	4.42E+09	4.94E+08	8.951393
XENP36965_H2.12_L1.187	279.1	16705	59.8531	4.57E+09	5.79E+08	7.881124
XENP36966_H2.91_L1.187	349.9	19604	56.02744	5.27E+09	5.24E+08	10.05439
XENP36967_H2.118_L1.187	313.6	26403	84.19324	4.29E+09	4.38E+08	9.796817
XENP36968_H2.3_L1.189	452	36292	80.29204	4.37E+09	5.86E+08	7.458967
XENP36969_H2.12_L1.189	295.1	11508	38.99695	4.02E+09	7.66E+08	5.251557
XENP36970_H2.91_L1.189	394.6	24490	62.06285	4.91E+09	6.15E+08	7.988906
XENP36971_H2.118_L1.189	378.8	14082	37.17529	3.6E+09	5.18E+08	6.948966
XENP36972_H1_L1.187	307.6	16192	52.63979	4.9E+09	5.34E+08	9.166706
XENP36973_H1_L1.189	466.5	18579	39.82637	4.55E+09	8.06E+08	5.640381

Figure 34A

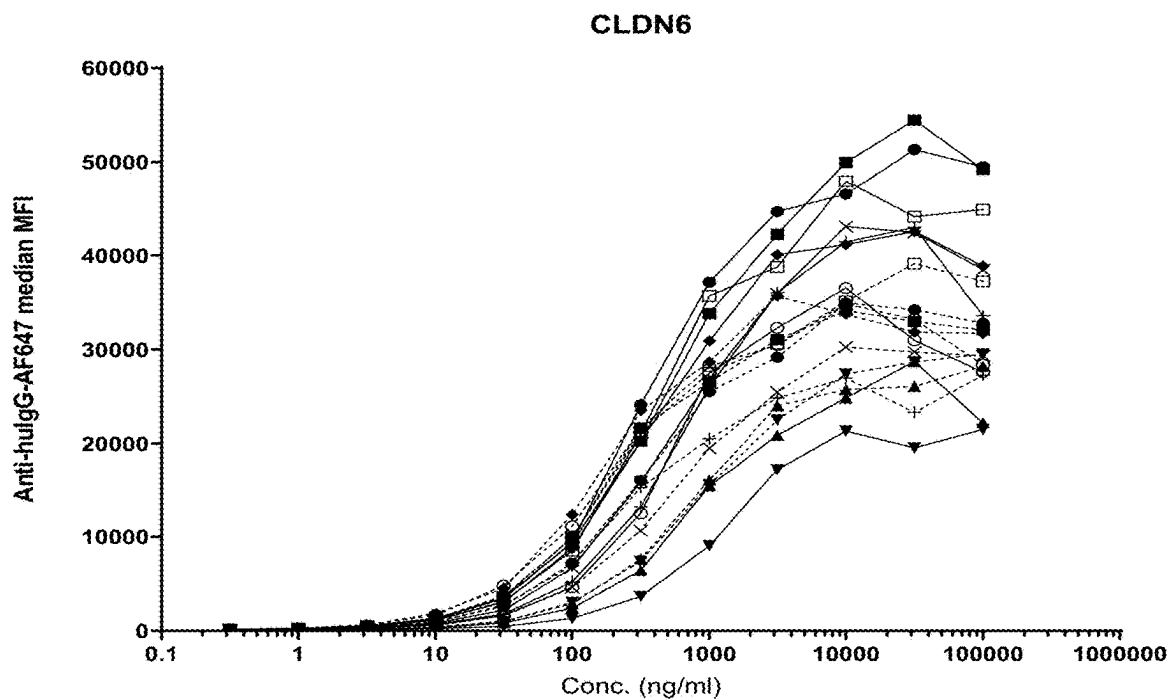


Figure 34B

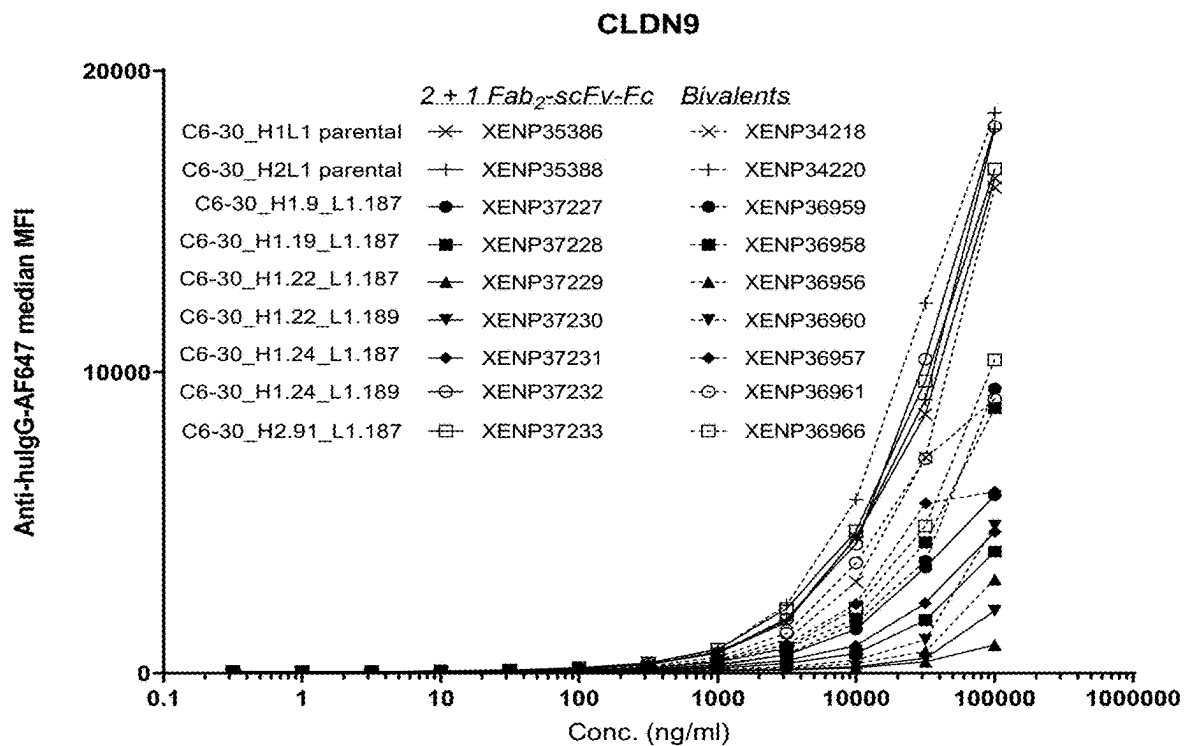


Figure 35B

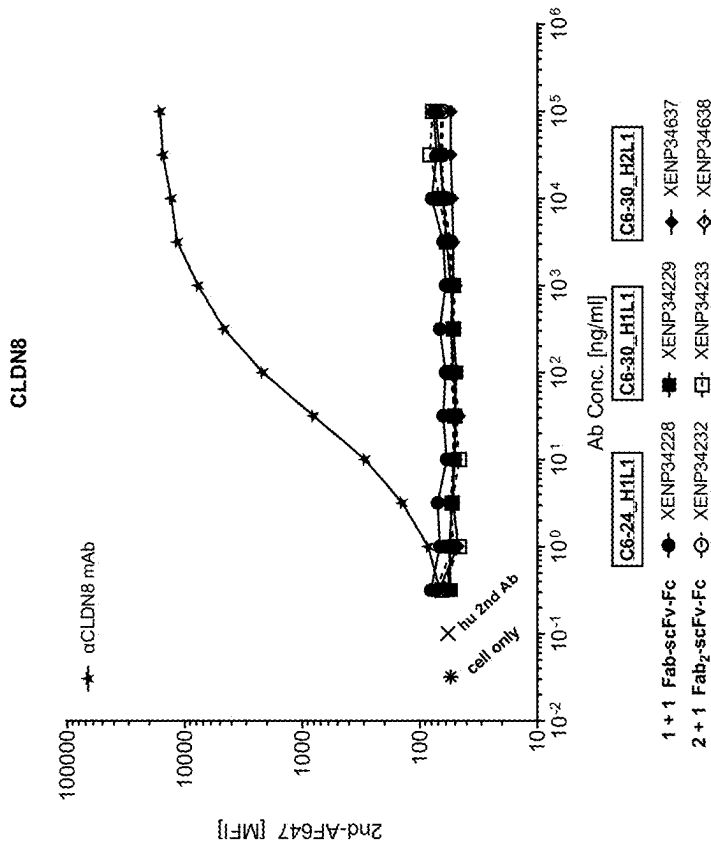


Figure 35A

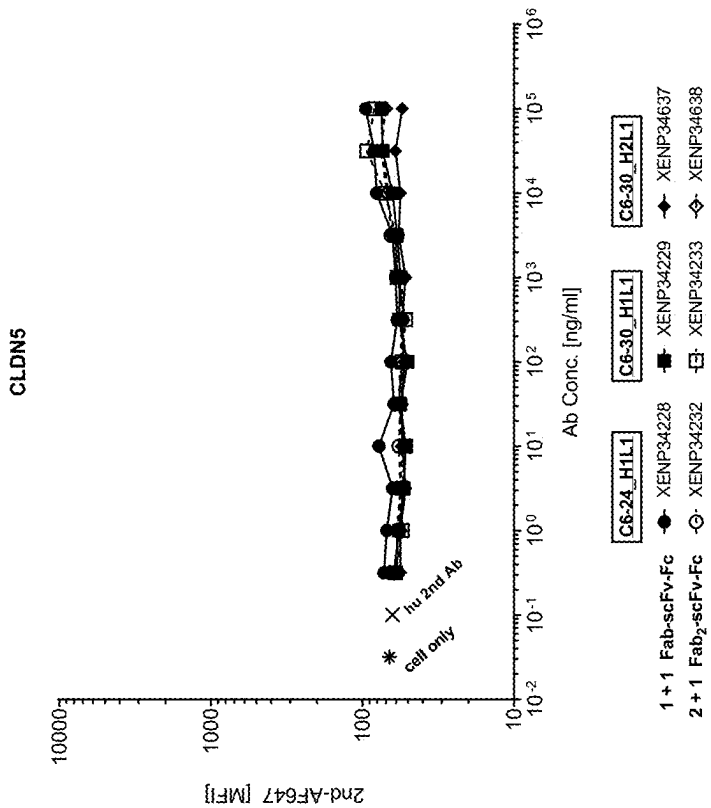


Figure 35C

CLDN17

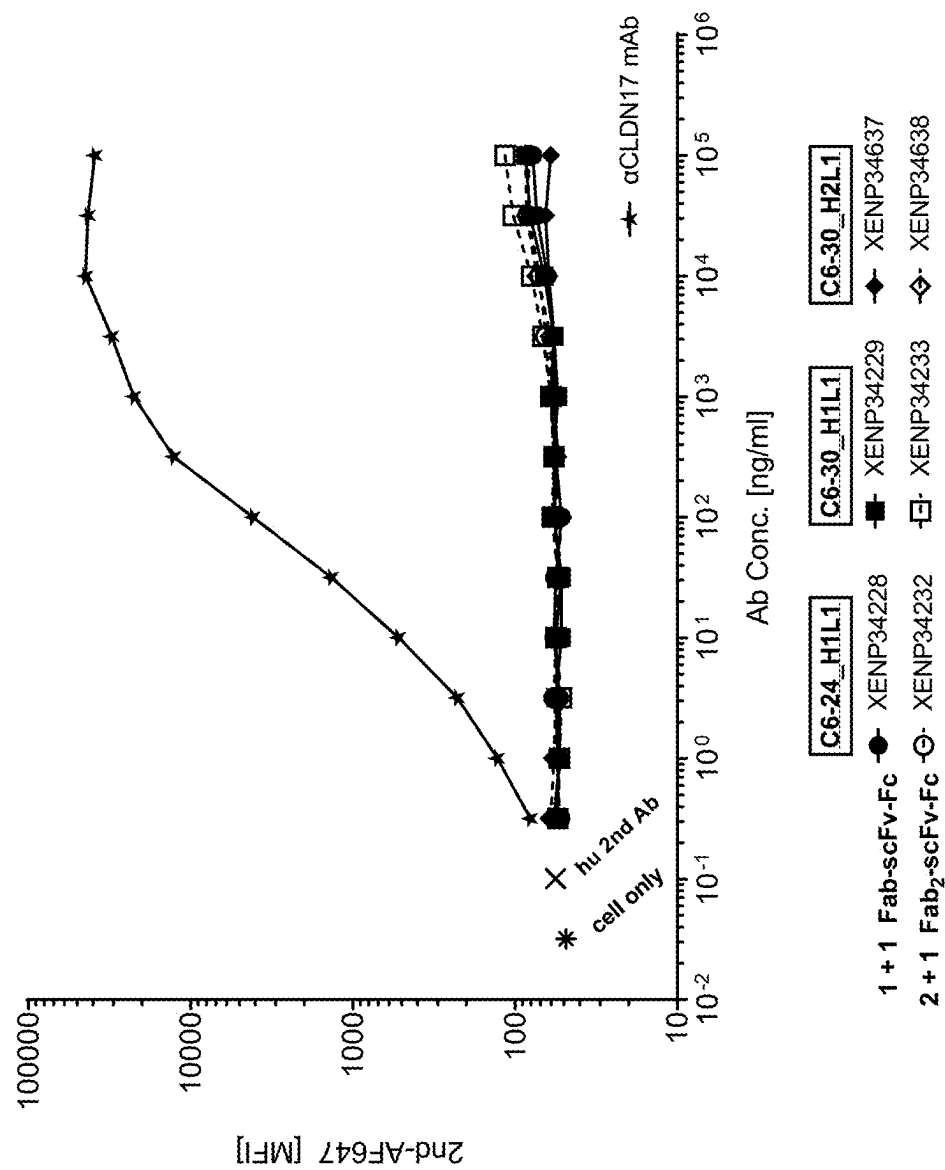
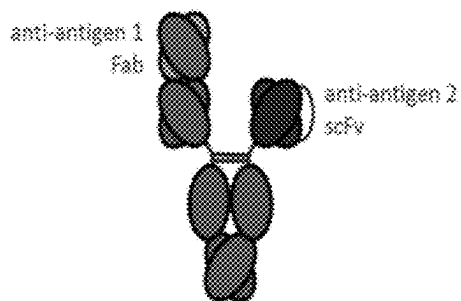


Figure 36A



Bottle Opener or
1+1 Fab-scFv-Fc

Figure 36B

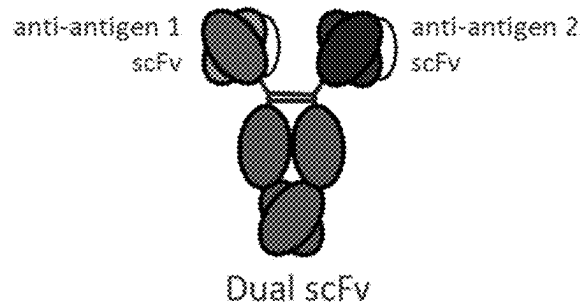
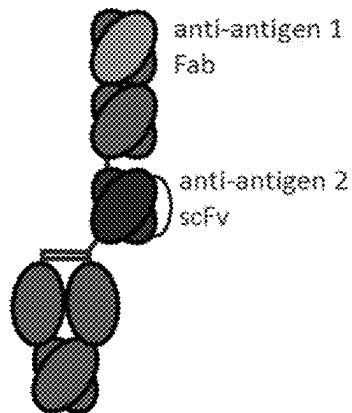
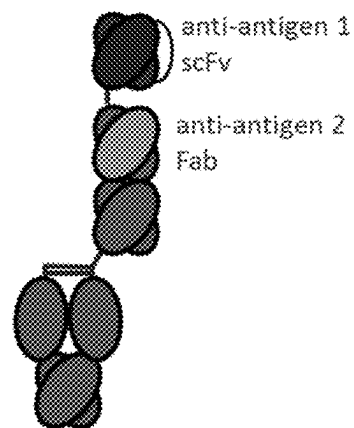


Figure 36C



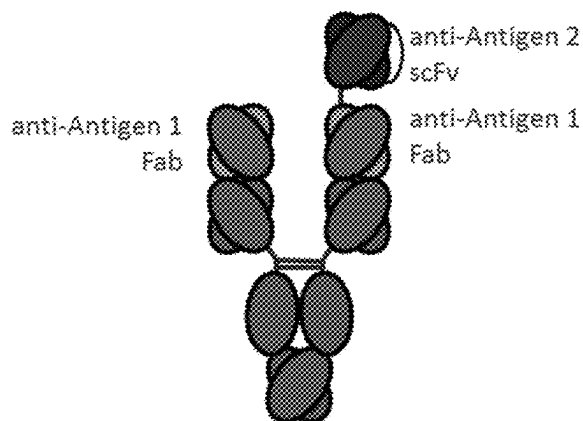
One-arm central-scFv

Figure 36D



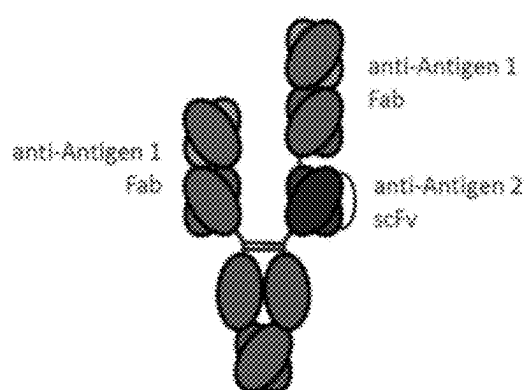
One-arm scFv-mAb

Figure 36E



scFv-mAb

Figure 36F



Central scFv or
2+1 Fab2-scFv-Fc

Figure 36G

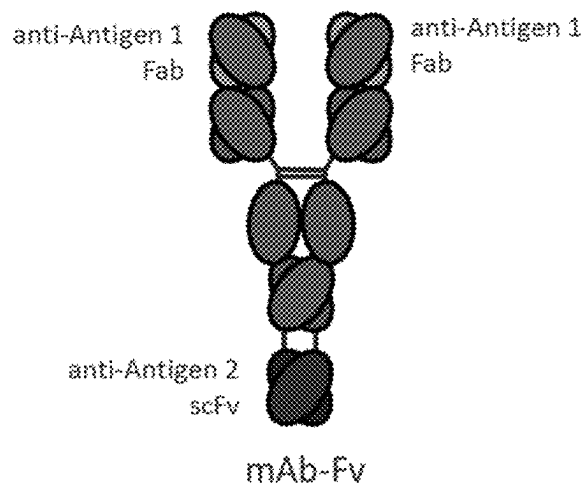


Figure 36H

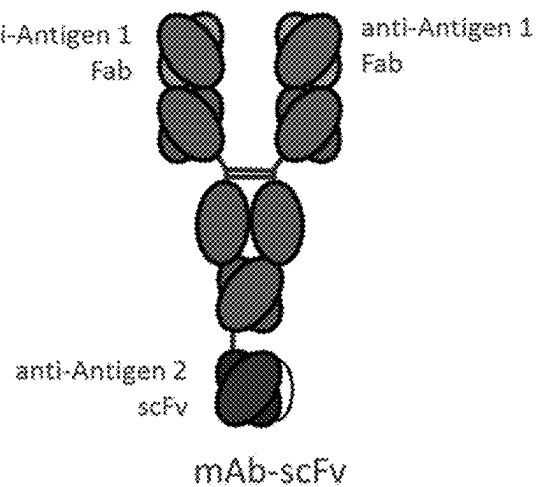


Figure 36I

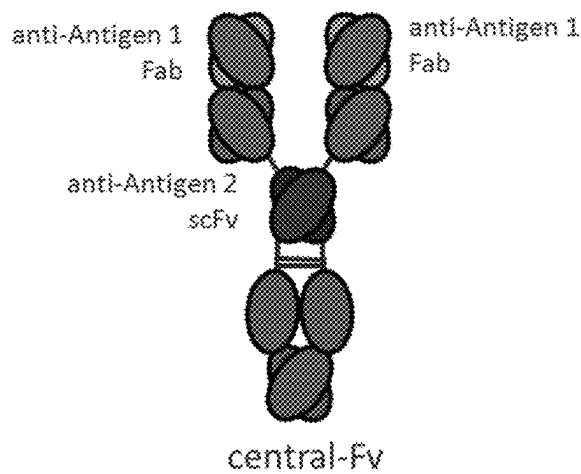


Figure 36J

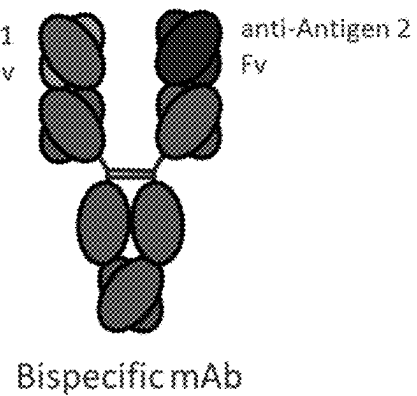
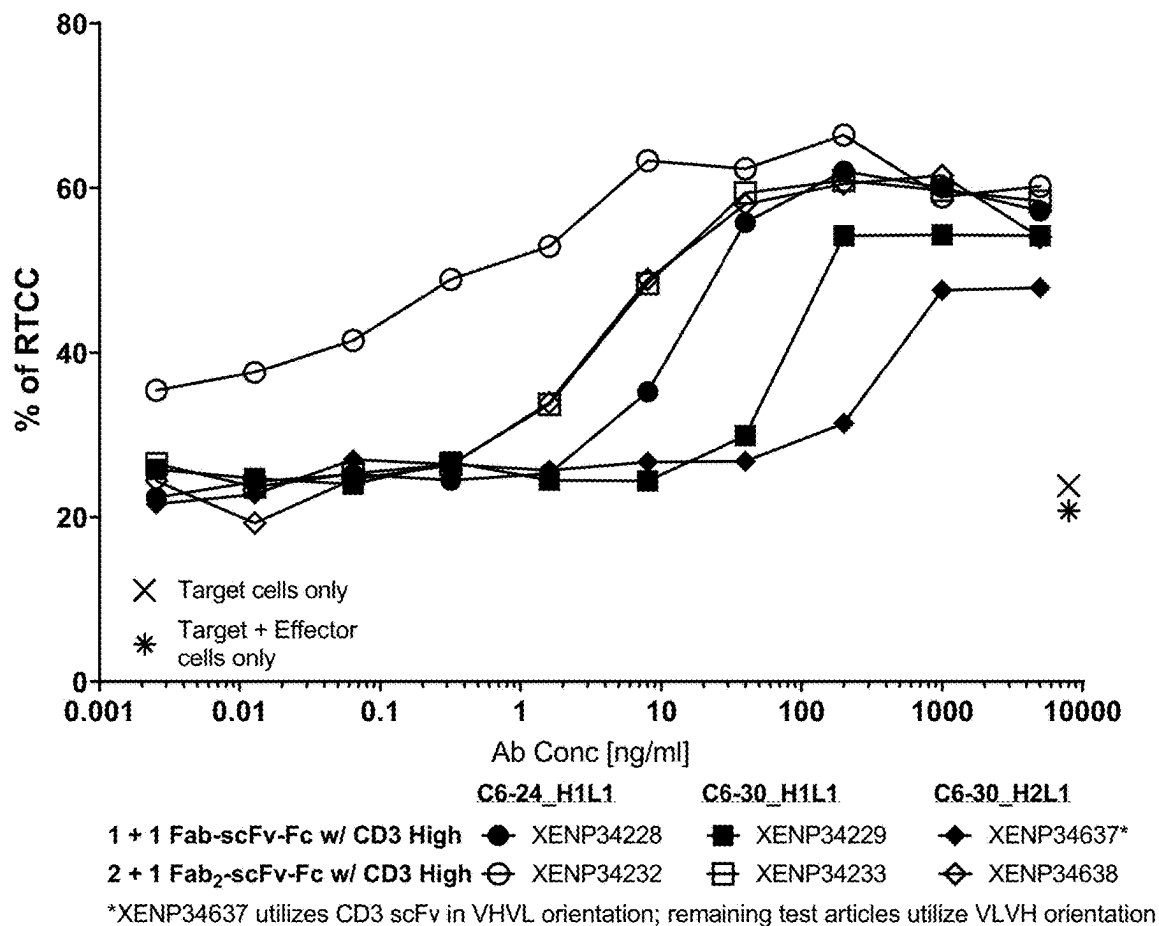


Figure 37



Sample ID	Format	FV	Humanization	PA-1 RTCC average EC50 (ng/mL)
XENP34229	1+1	C6-30	H1L1	75.6 ± 32.3
XENP34233	2+1	C6-30	H1L1	5.3 ± 1.5
XENP34637	1+1	C6-30	H2L1	205.5 ± 115.4
XENP34638	2+1	C6-30	H2L1	2.4 ± 1.2
XENP34228	1+1	C6-24	H1L1	11.4 ± 1.3
XENP34232	2+1	C6-24	H1L1	1.0 ± 0.9

Figure 38A

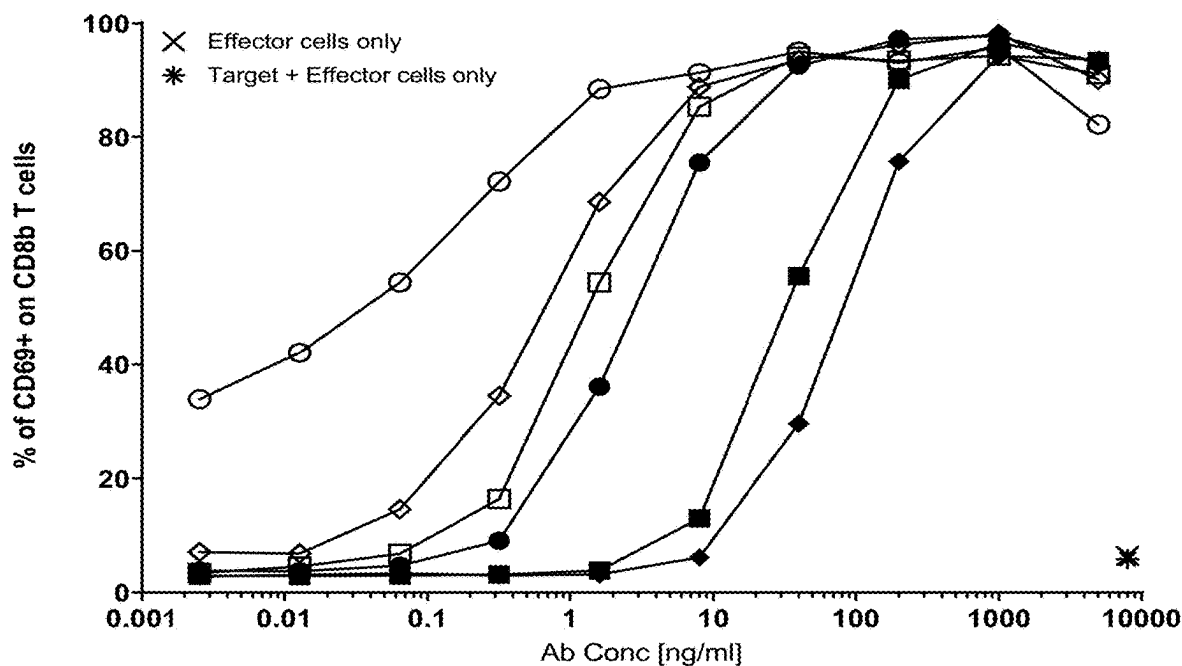
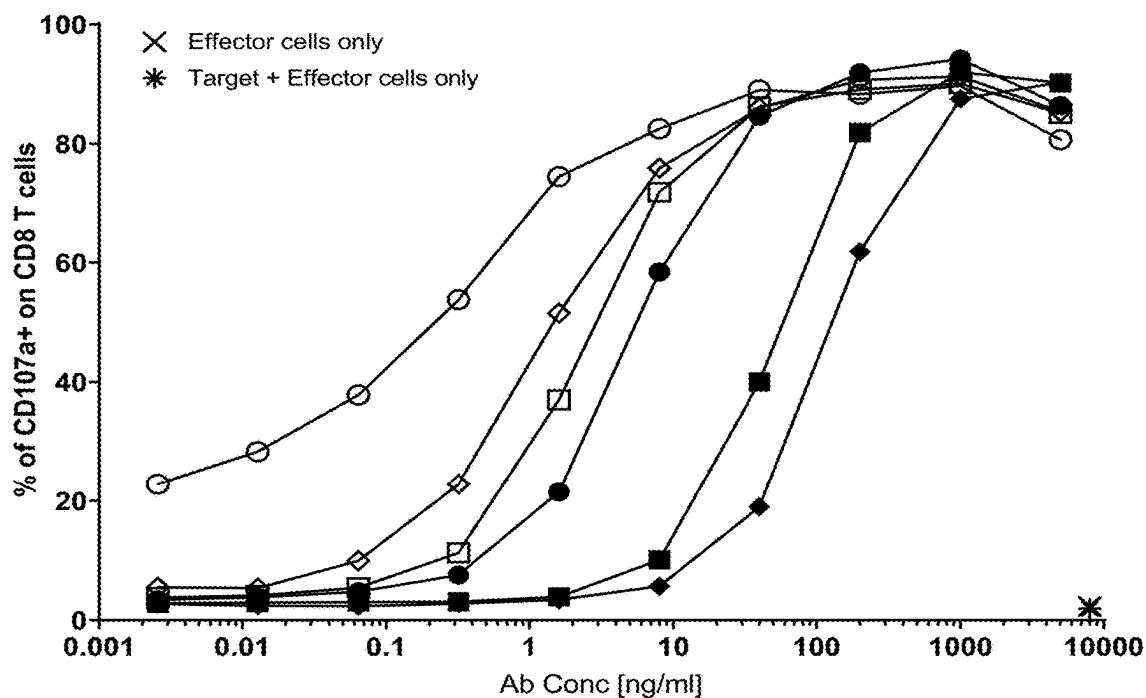


Figure 38B



1 + 1 Fab-scFv-Fc w/ CD3 High ● XENP34228 ■ XENP34229 ◆ XENP34637
 2 + 1 Fab₂-scFv-Fc w/ CD3 High ⊖ XENP34332 ⊞ XENP34233 ◇ XENP34638
 *XENP34637 utilizes CD3 scFv in VHVL orientation; remaining test articles utilize VLVH orientation

Figure 39A

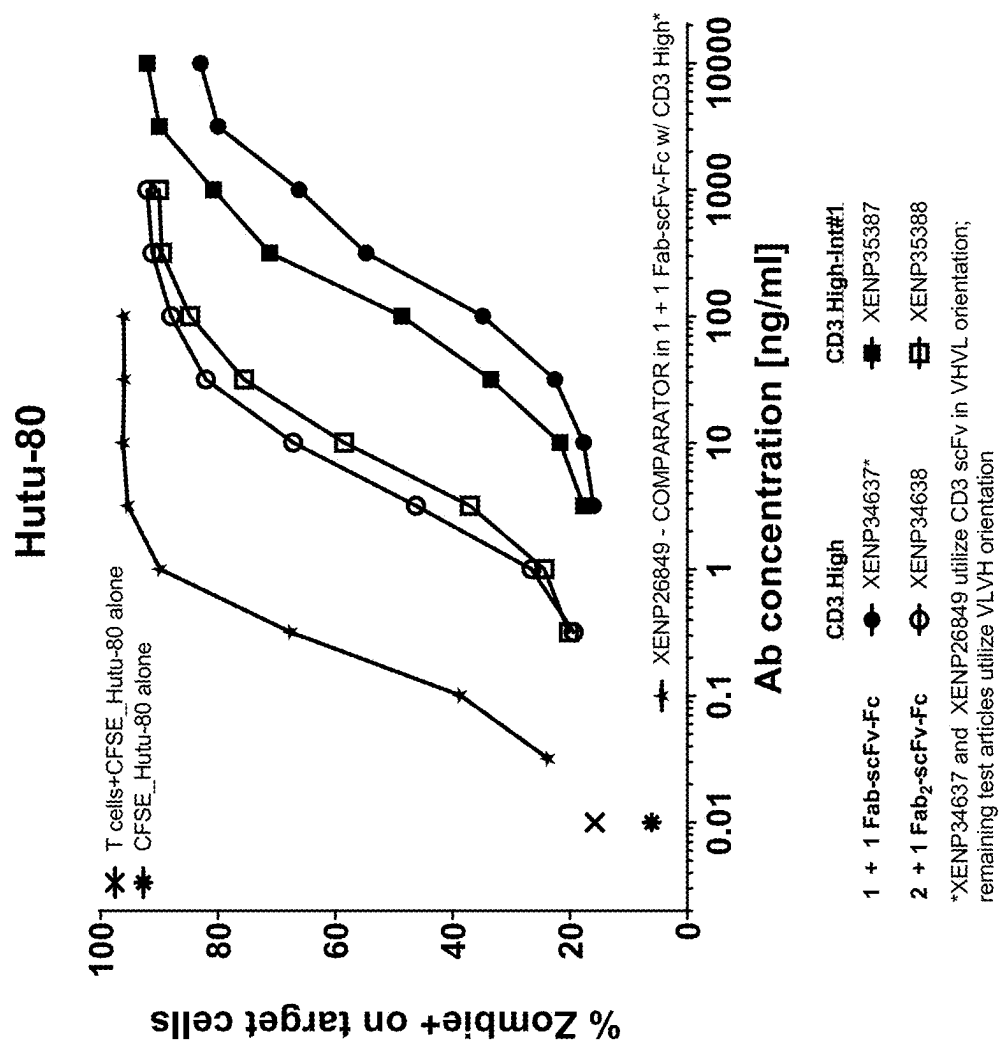
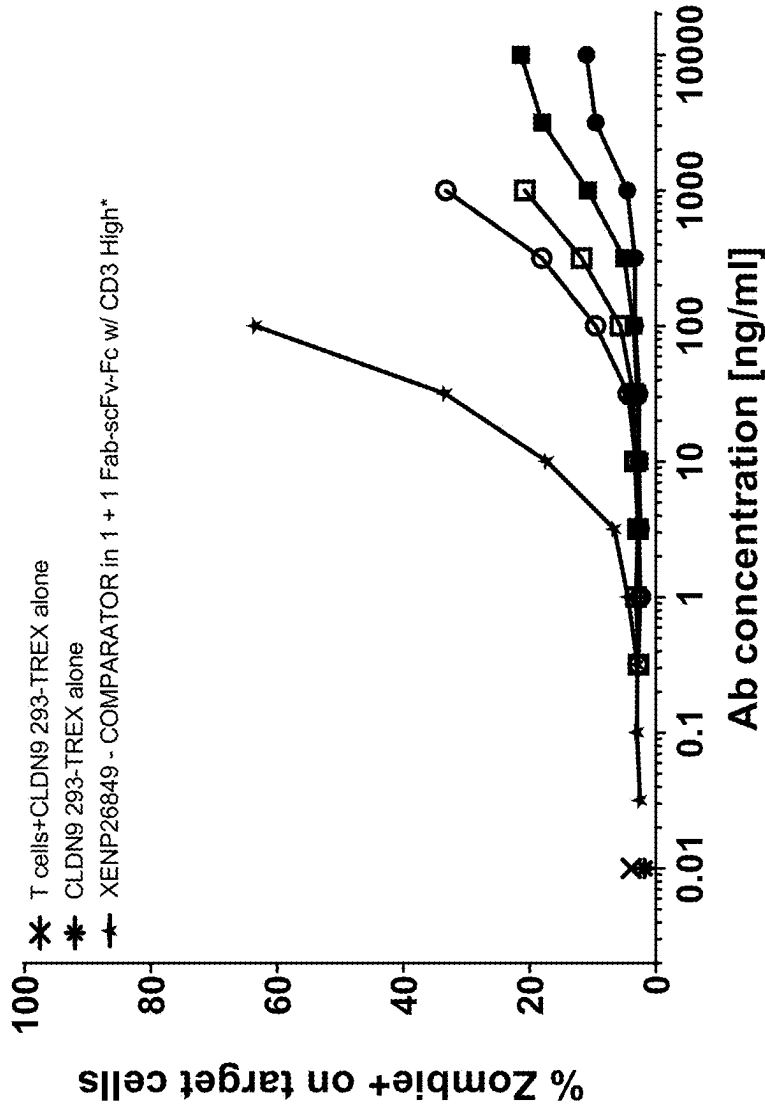


Figure 39B

CLDN9 293-TREX



*XENP34637 and XENP26849 utilize CD3 scFv in VHVL orientation; remaining test articles utilize VL VH orientation

Figure 40B

CLDN9 293-TREX

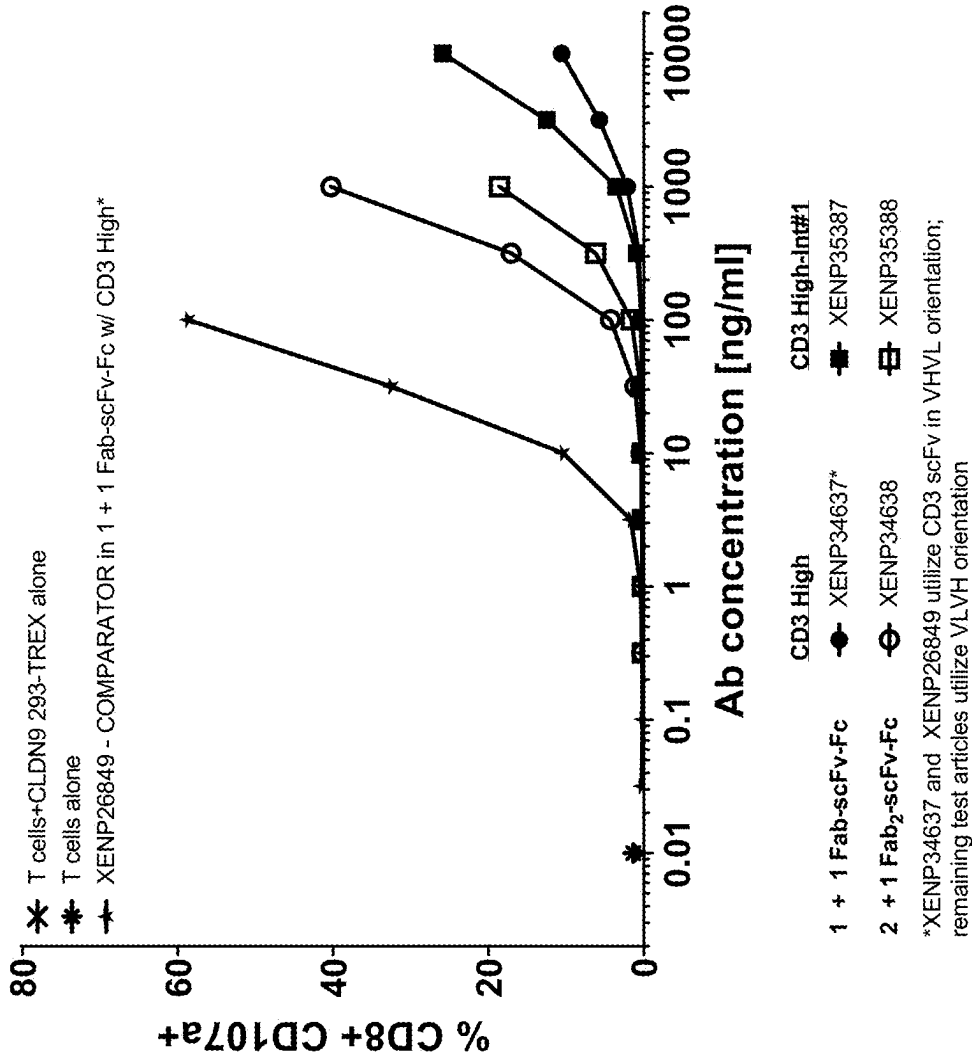


Figure 41B

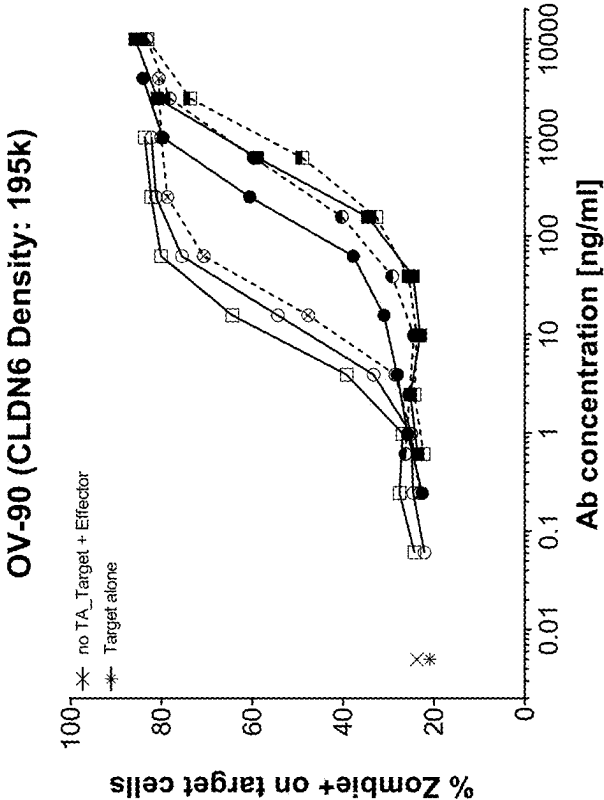


Figure 41A

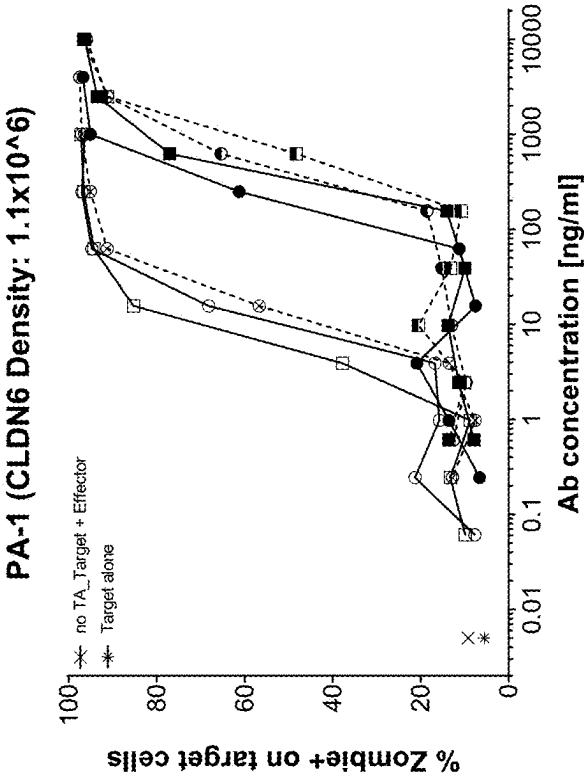
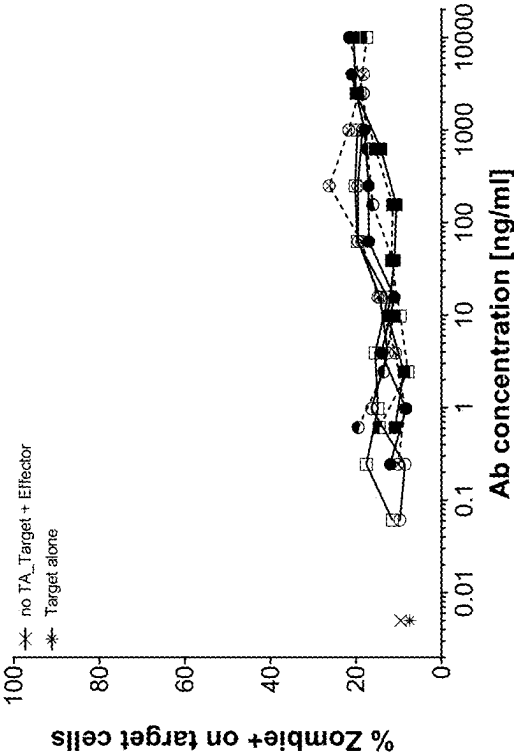


Figure 41C

NEC-8 (CLDN6 Density: 175k)



- | | C6-30_H1L1 | C6-30_H2L1 |
|---|-------------|--------------|
| 1 + 1 Fab-scFv-Fc w/ CD3 High | ● XENP34229 | ■ XENP34637* |
| 2 + 1 Fab ₂ -scFv-Fc w/ CD3 High | ○ XENP34233 | □ XENP34638 |
| 1 + 1 Fab-scFv-Fc w/ CD3 High-Int#1 | ● XENP35385 | ■ XENP35387 |
| 2 + 1 Fab ₂ -scFv-Fc w/ CD3 High-Int#1 | ○ XENP35386 | □ XENP35388 |

*XENP34637 utilizes CD3 scFv in VHVL orientation; remaining test articles utilize VL VH orientation

Figure 41D

COV-318 (CLDN6 Density: 11k)

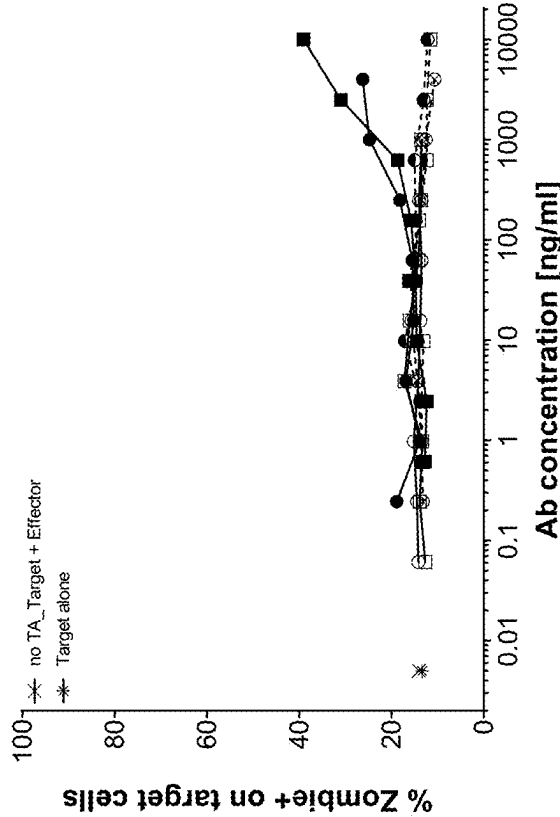


Figure 42

EC50_%Zombie+ on targets (ng/mL)		
	PA-1	OV-90
XENP34229	287	200.3
XENP34233	12.44	14.38
XENP34637	469.7	590.5
XENP34638	5.989	9.29
XENP26849	0.1084	0.299
XENP28817	1.352	5.173
XENP35385	570.1	472
XENP35386	14.26	19.19
XENP35387	1045	908.3

Figure 43B

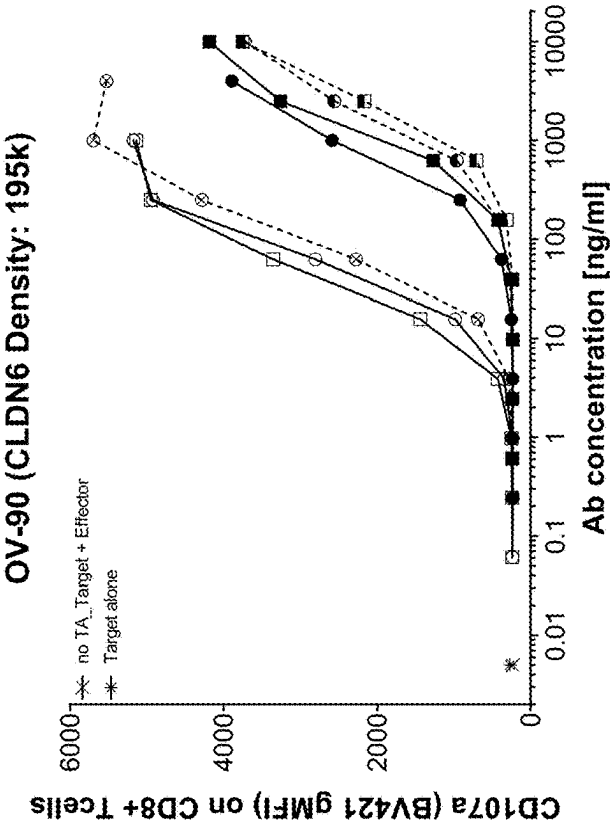


Figure 43A

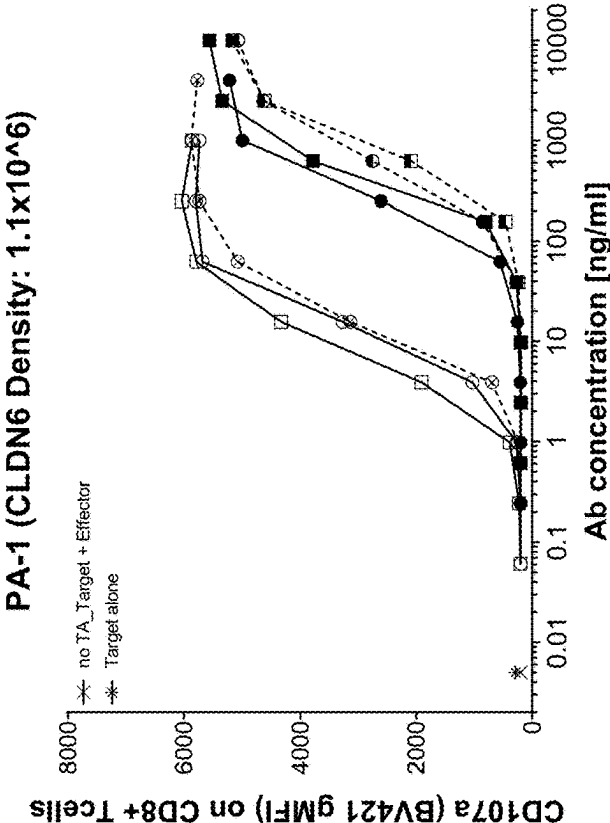


Figure 43C

NEC-8 (CLDN6 Density: 175k)

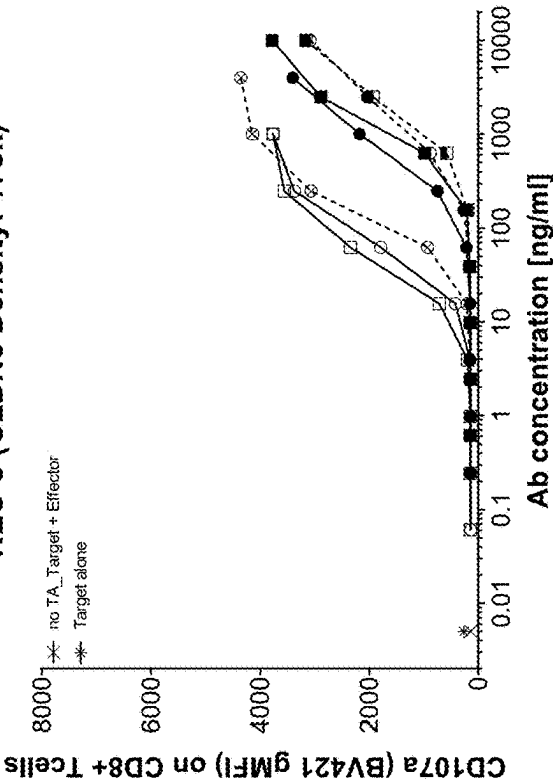
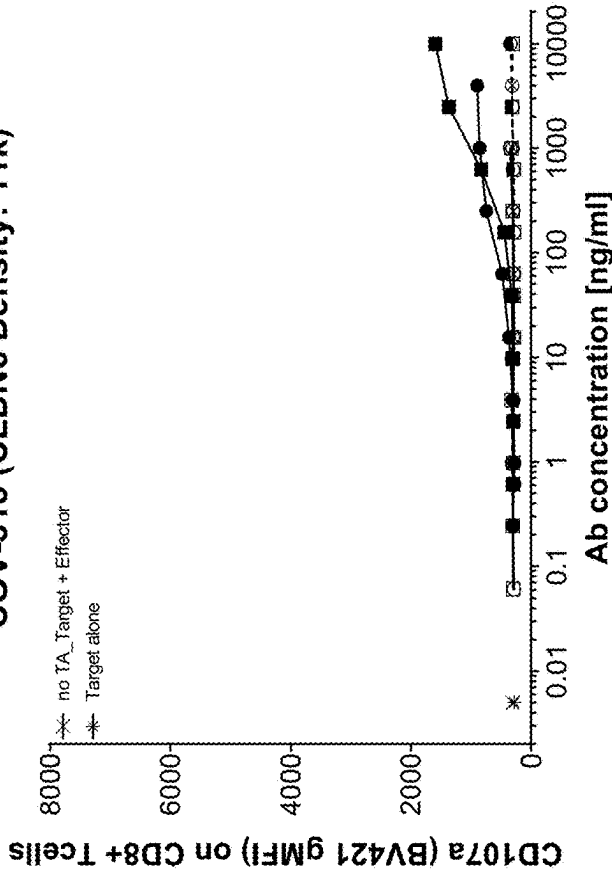


Figure 43D

COV-318 (CLDN6 Density: 11k)

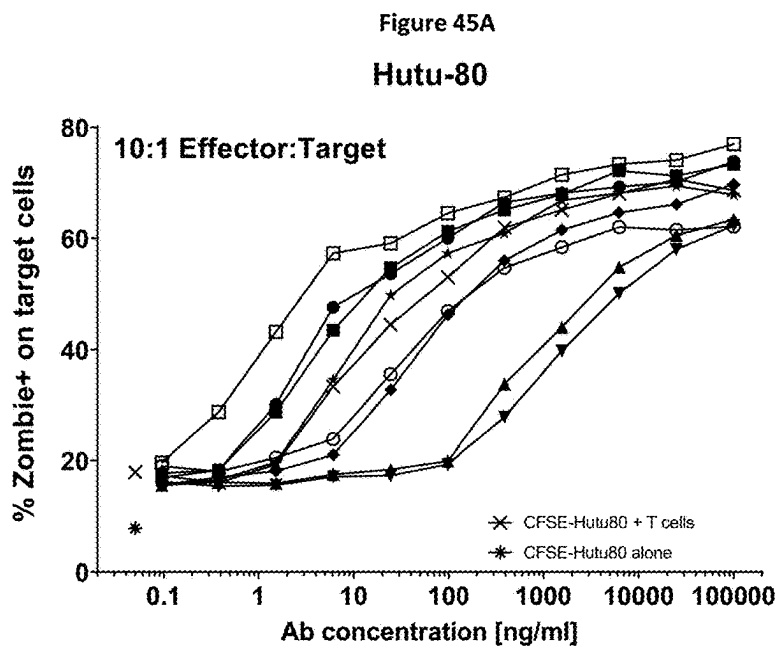


- | | | |
|---|-------------------|-------------------|
| | <u>C6.30_H1L1</u> | <u>C6.30_H2L1</u> |
| 1 + 1 Fab ₂ -scFv-Fc w/ CD3 High | ● XENP34229 | ■ XENP34637* |
| 2 + 1 Fab ₂ -scFv-Fc w/ CD3 High | ⊖ XENP34233 | ⊖ XENP34638 |
| 1 + 1 Fab ₂ -scFv-Fc w/ CD3 High-Int#1 | ● XENP35385 | ■ XENP35387 |
| 2 + 1 Fab ₂ -scFv-Fc w/ CD3 High-Int#1 | ⊖ XENP35386 | ⊖ XENP35388 |

*XENP34637 utilizes CD3 scFv in VHVL orientation, remaining test articles utilize VLVH orientation

Figure 44

EC50_CD107a on CD8+				
	PA-1	OV-90	NEC-8	COV-318
XENP34229	322.6	1097	1178	118
XENP34233	13.18	67.02	88.68	
XENP34637	512.9	1714	1909	1012
XENP34638	7.561	44.08	54.54	
XENP26849	0.2754	2.396	2.285	22.29
XENP28817	2.064	23.56	20.21	38.66
XENP35385	686.7	2444	2383	
XENP35386	14.72	103.9	173.2	
XENP35387	1079	4280	3613	



%Zombie+ on Hutu-80	
XENP	EC50 (ng/ml)
37227	5.4
37228	7
37229	857
37230	1426
37231	63
37232	44
37233	2.1
35386	21
35388	12

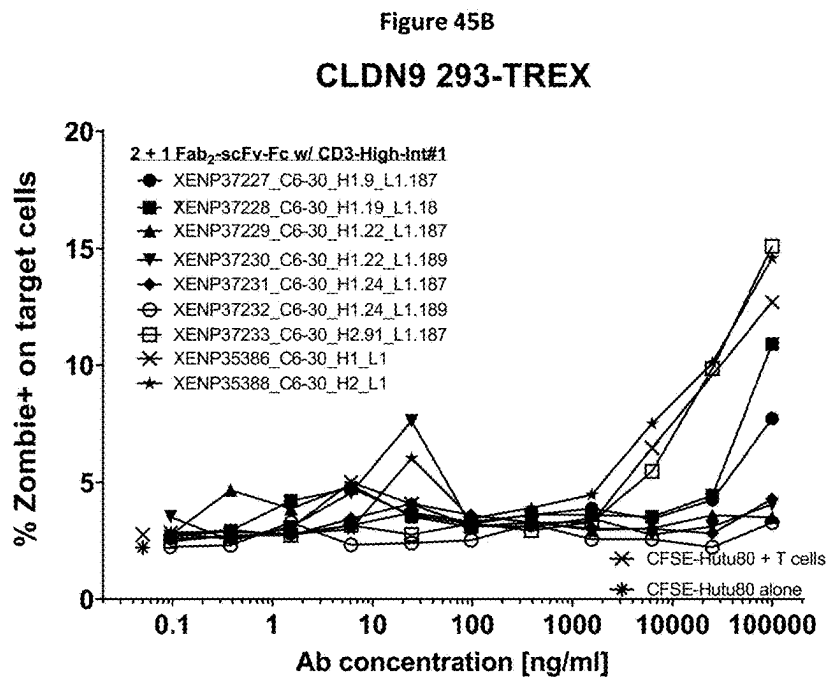
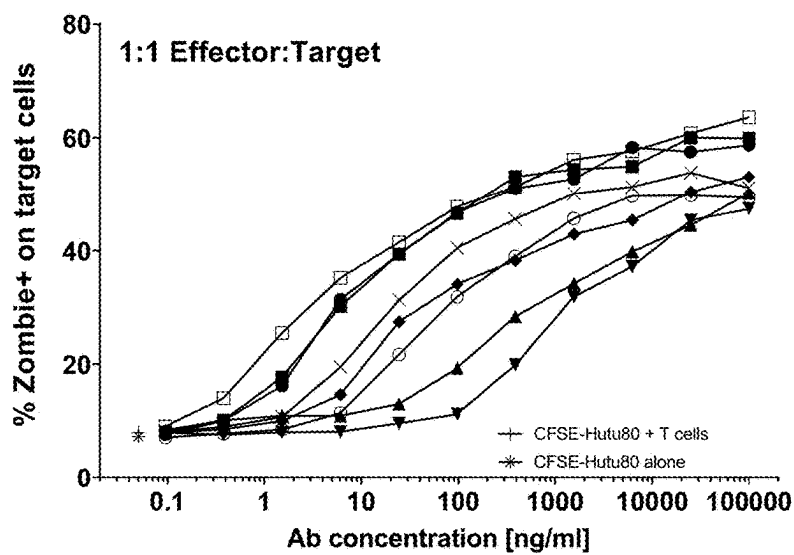


Figure 46A

Hutu-80



%Zombie+ on Hutu-80

XENP	EC50 (ng/ml)
37227	8.9
37228	10
37229	447
37230	1004
37231	38
37232	66
37233	7.2
35386	23

Figure 46B

CLDN9 293-TREX

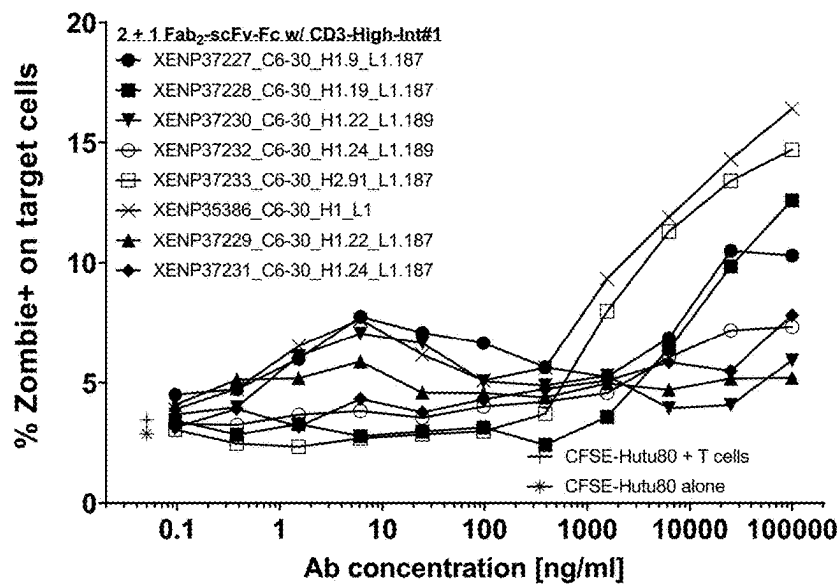


Figure 47A

Hutu-80

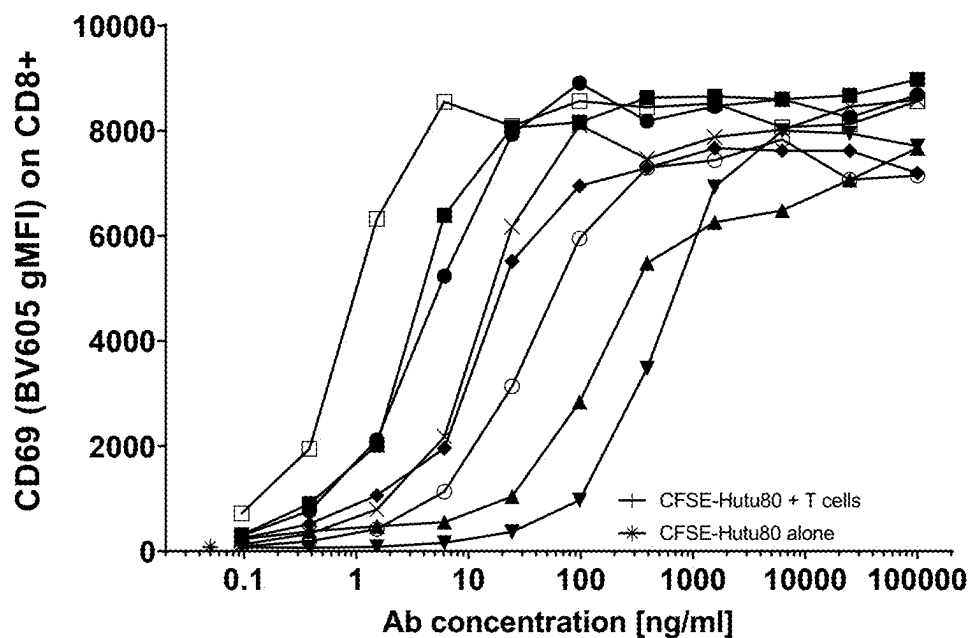


Figure 47B

CLDN9 293-TREX

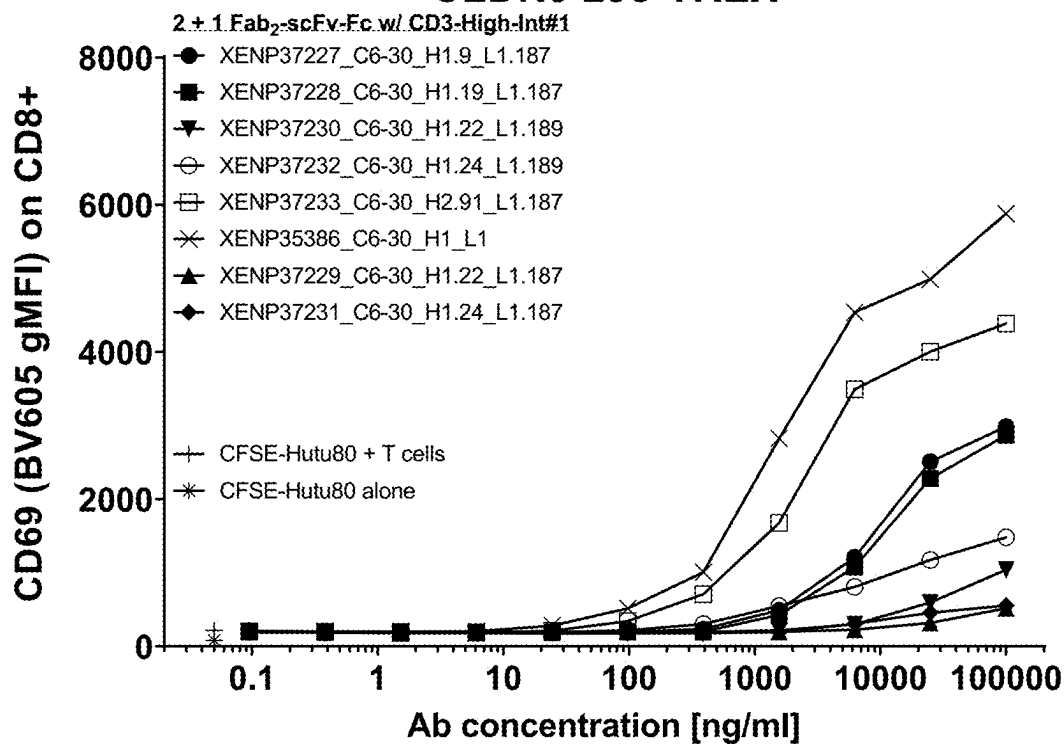


Figure 48A

Hutu-80

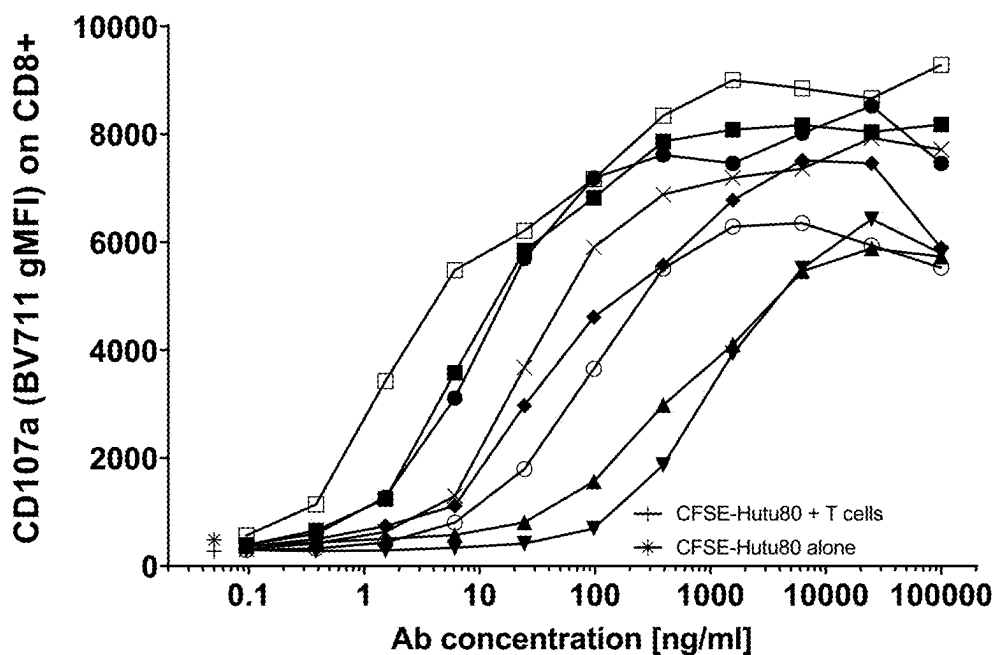


Figure 48B

CLDN9 293-TREX

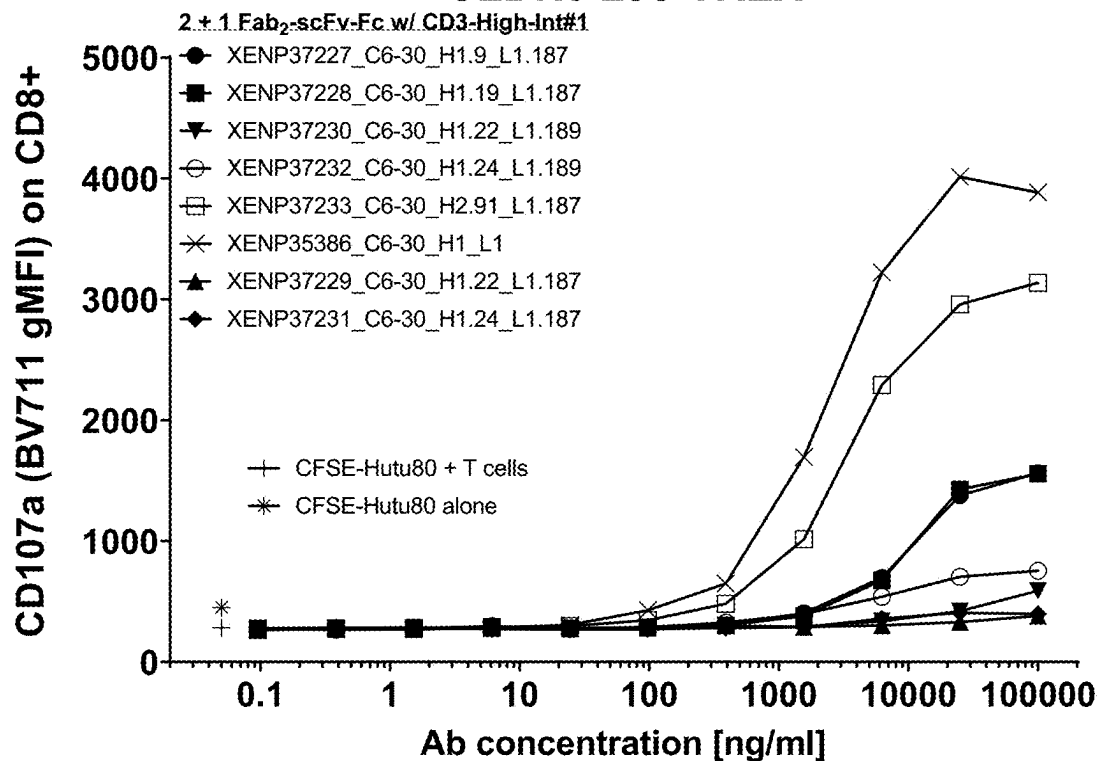


Figure 49A

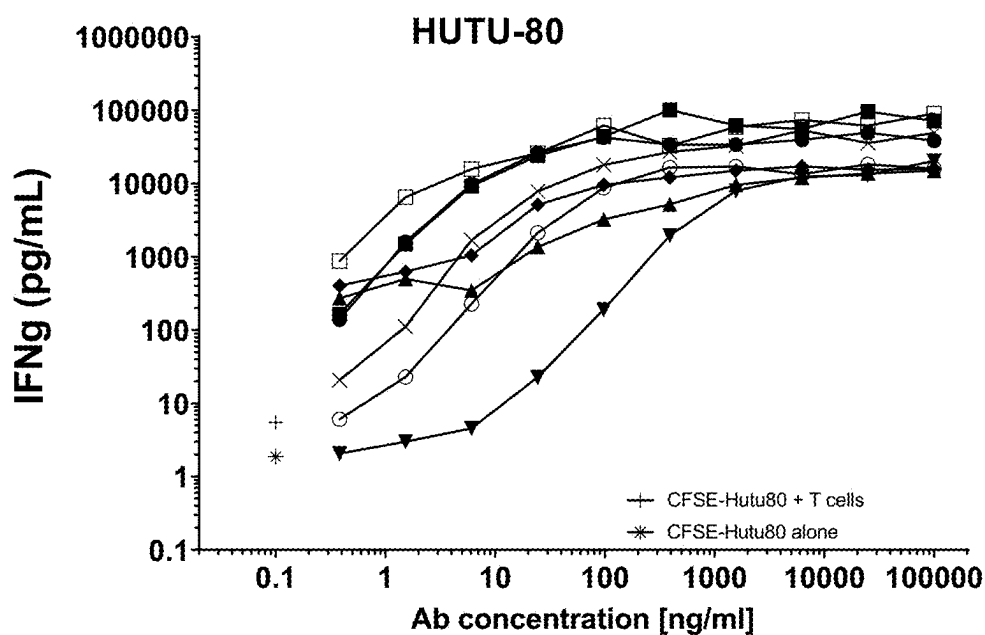


Figure 49B

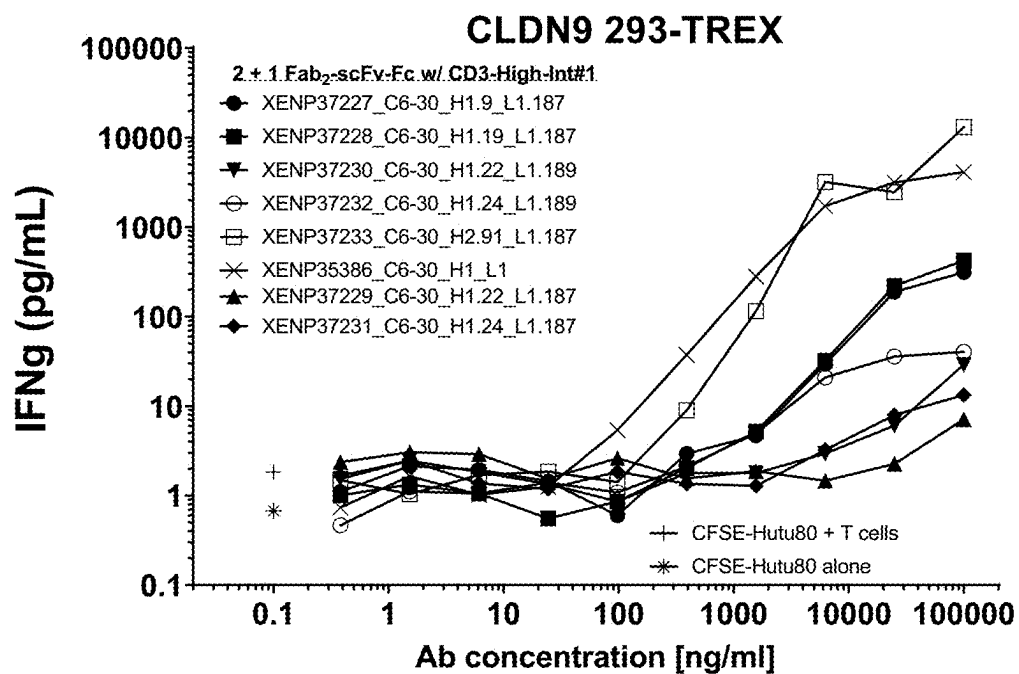


Figure 50A

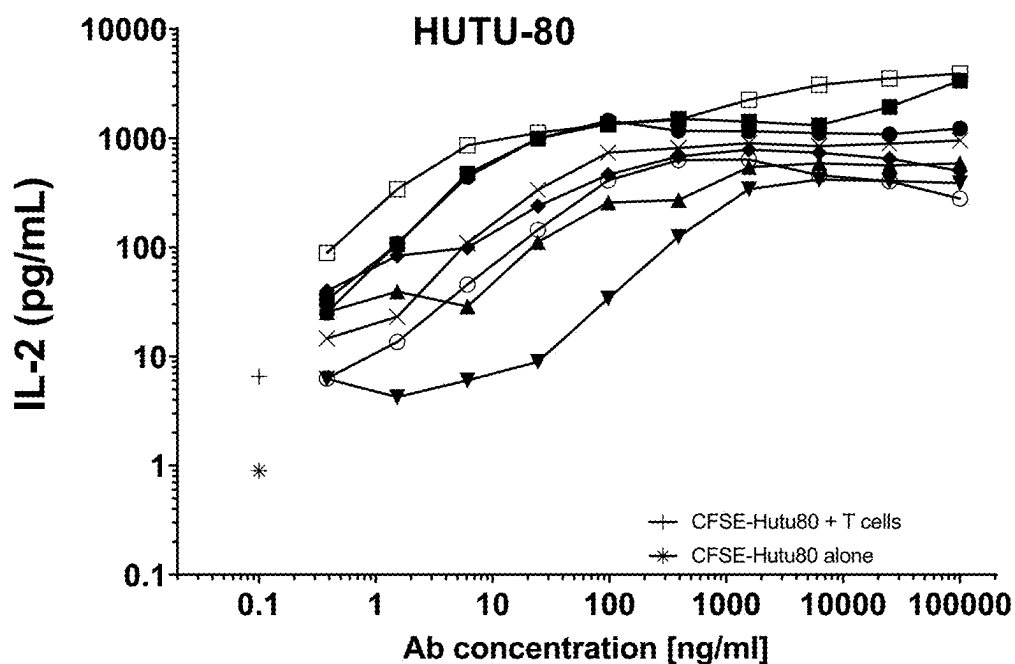


Figure 50B

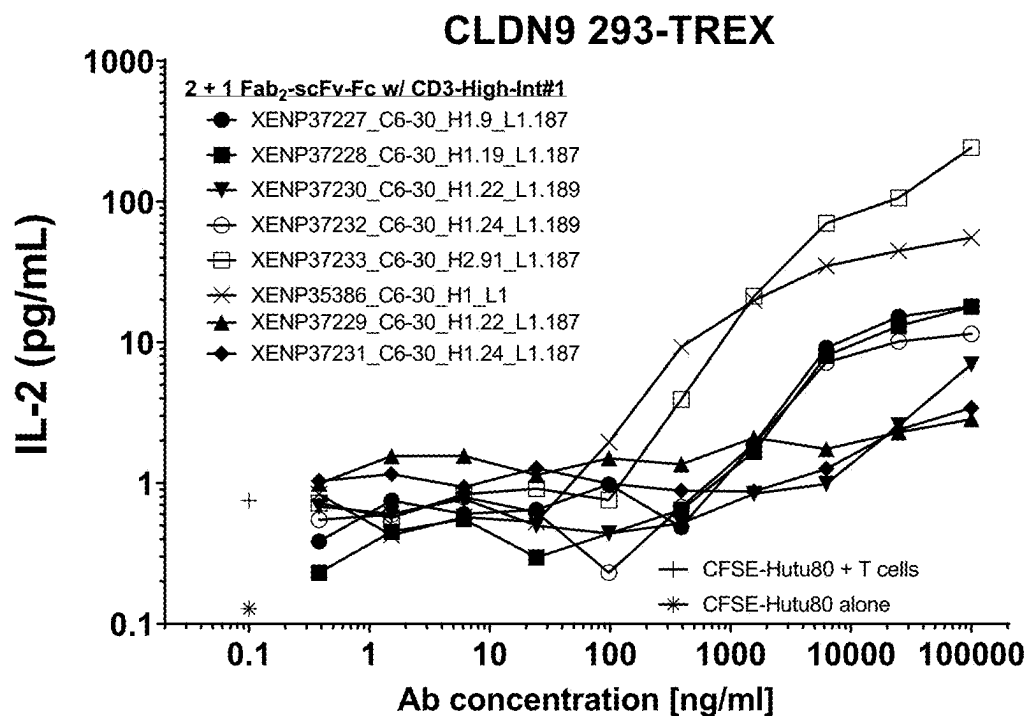


Figure 51A

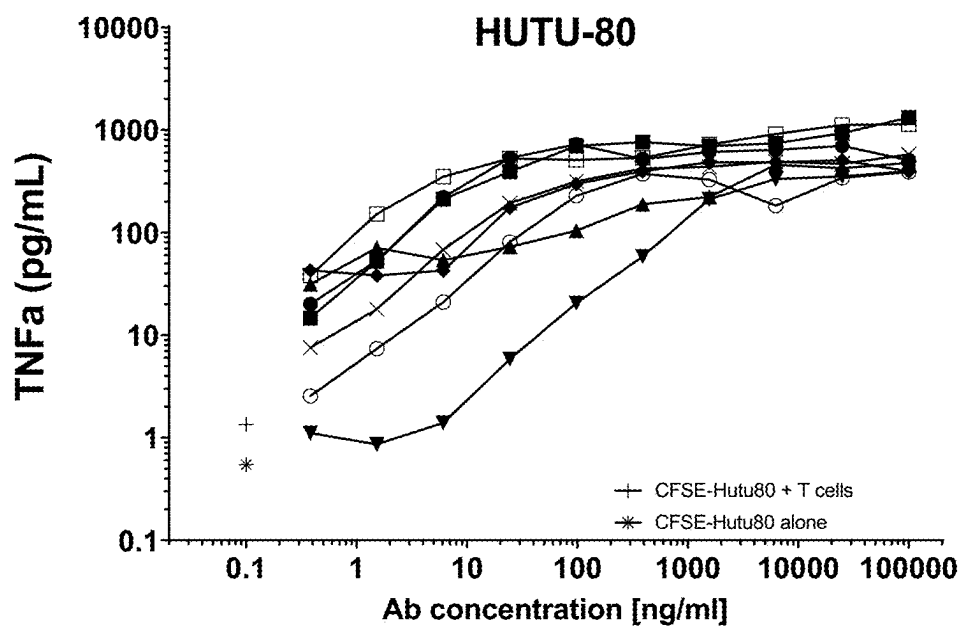


Figure 51B

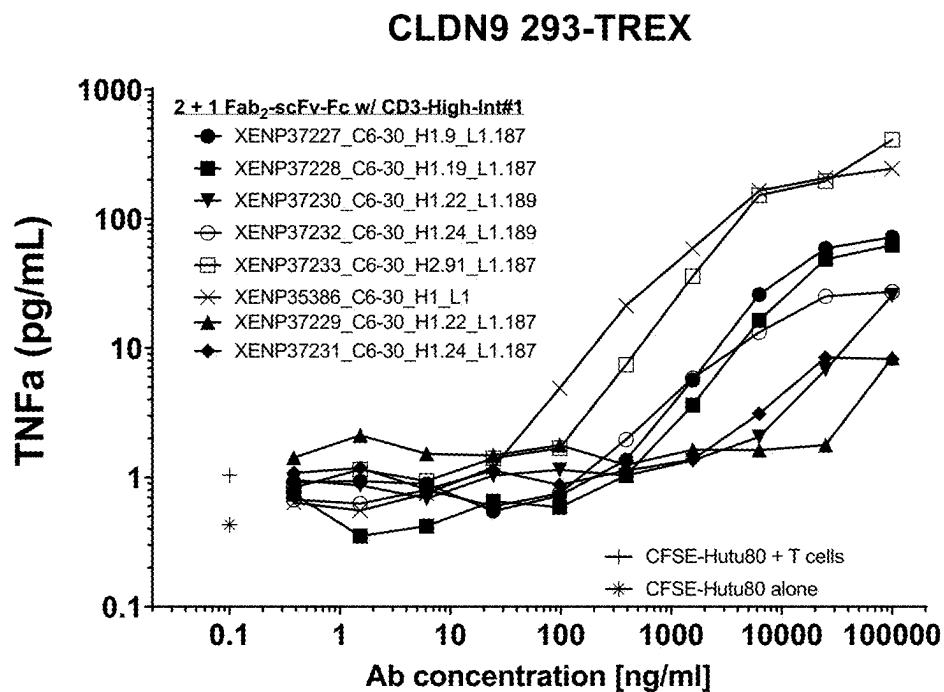


Figure 52A

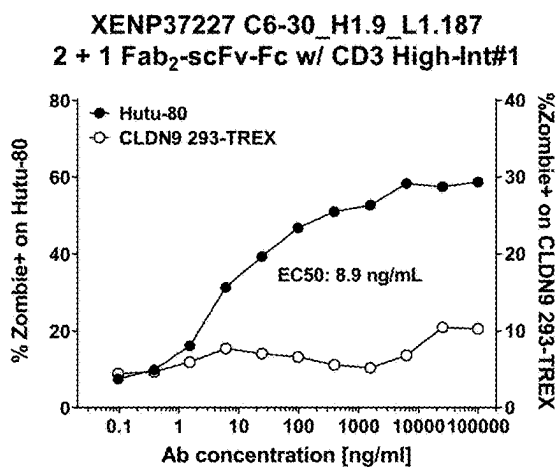


Figure 52B

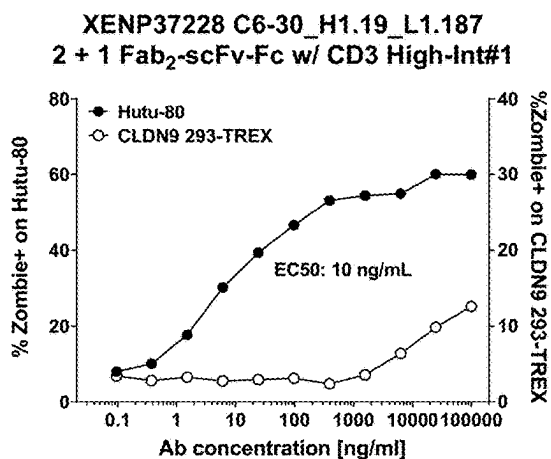


Figure 52C

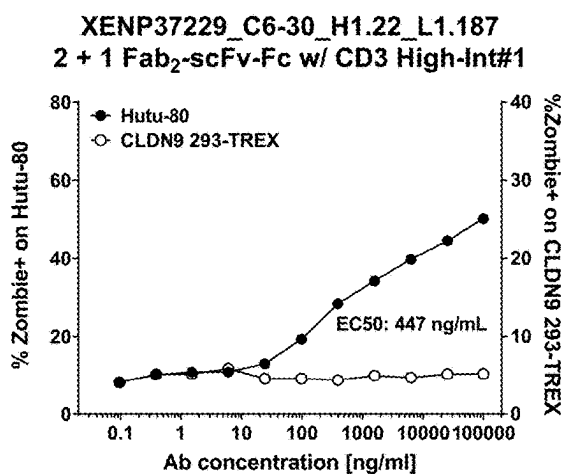


Figure 52D

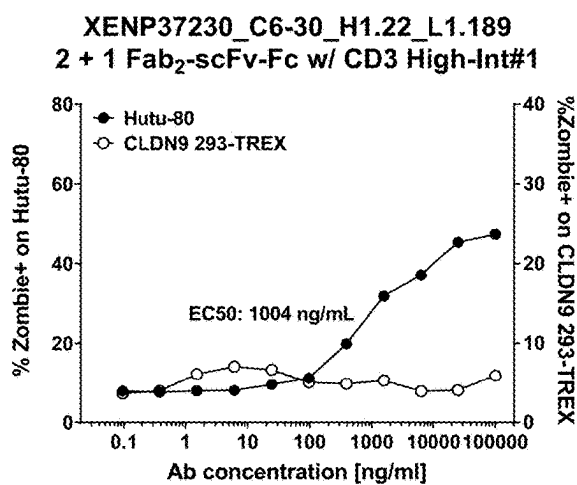


Figure 52E

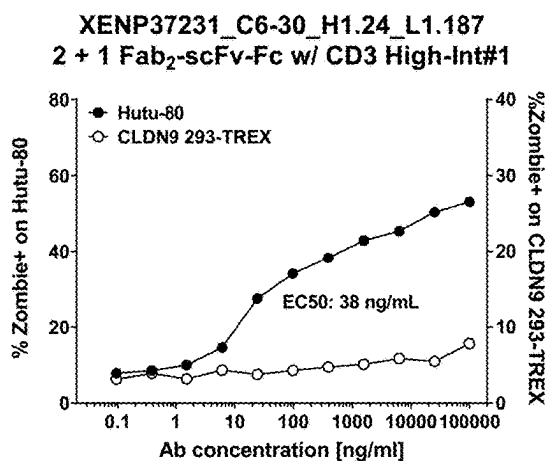


Figure 52F

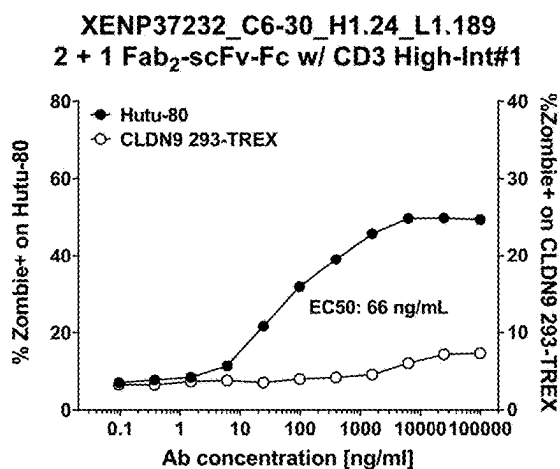


Figure 52G

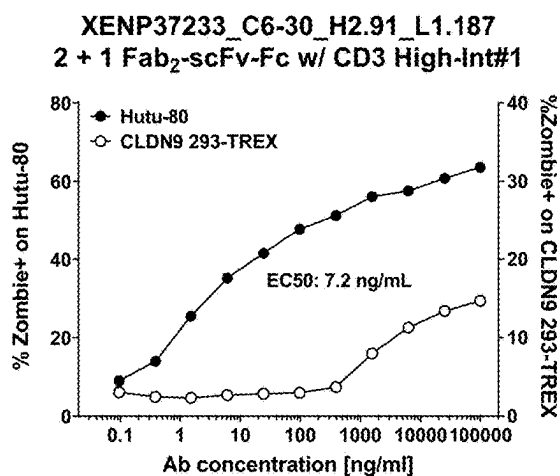


Figure 52H

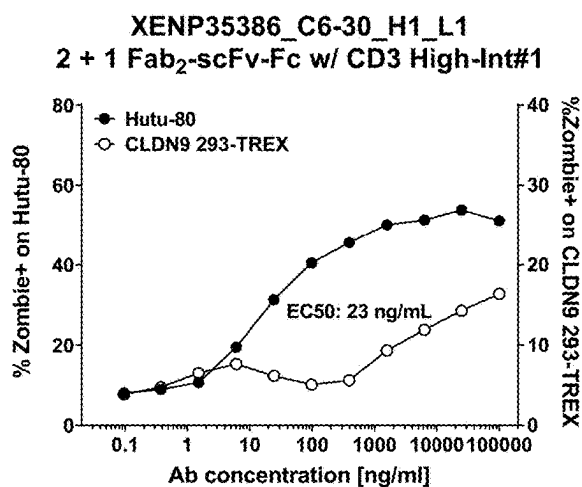


Figure 53B

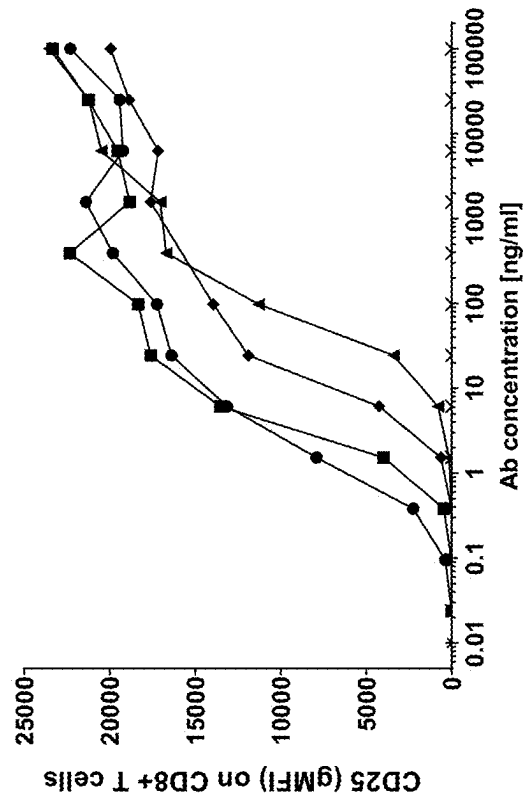


Figure 53A

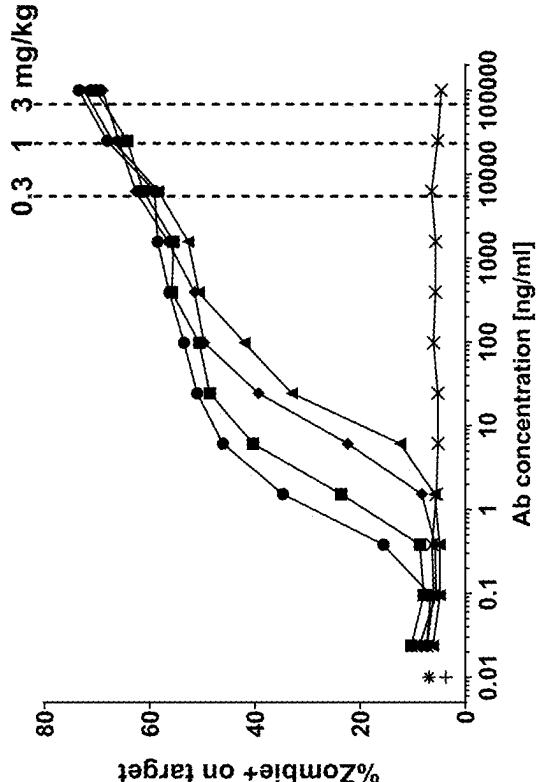


Figure 53C

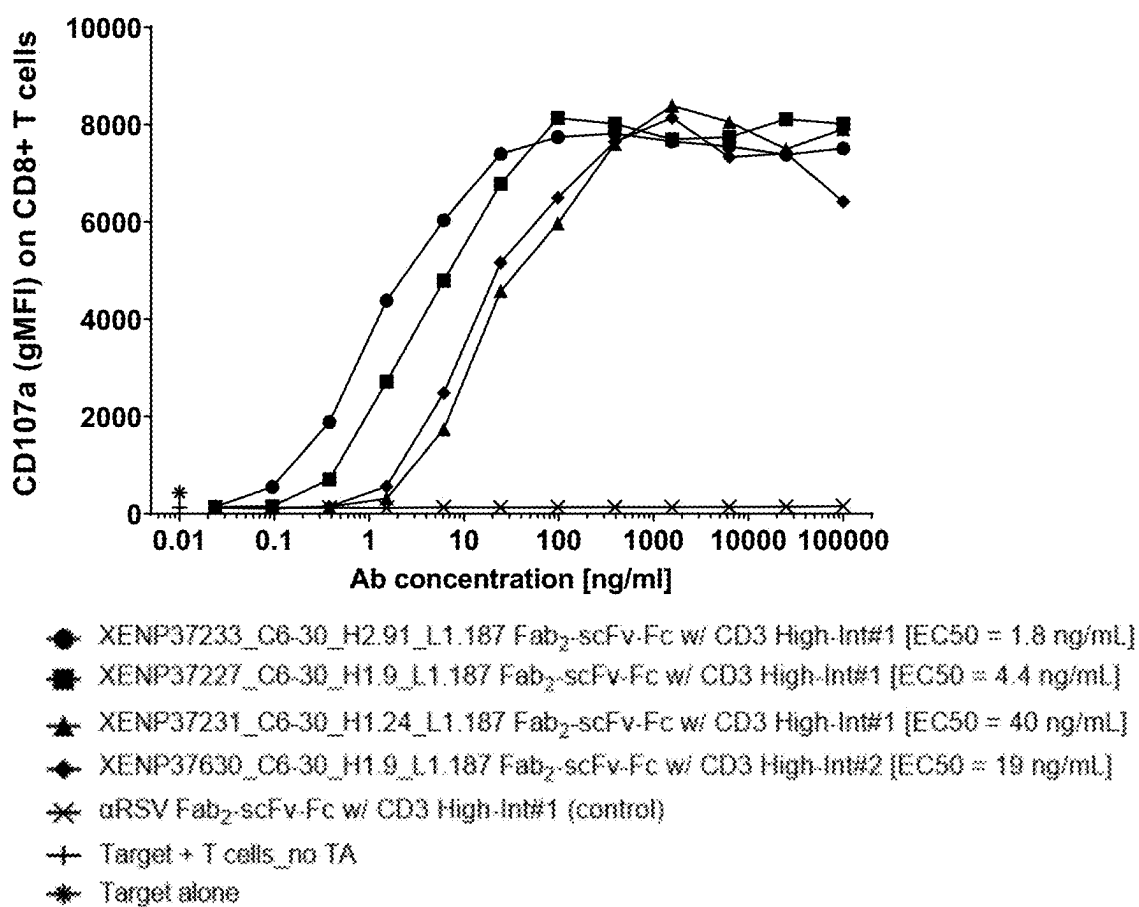


Figure 54

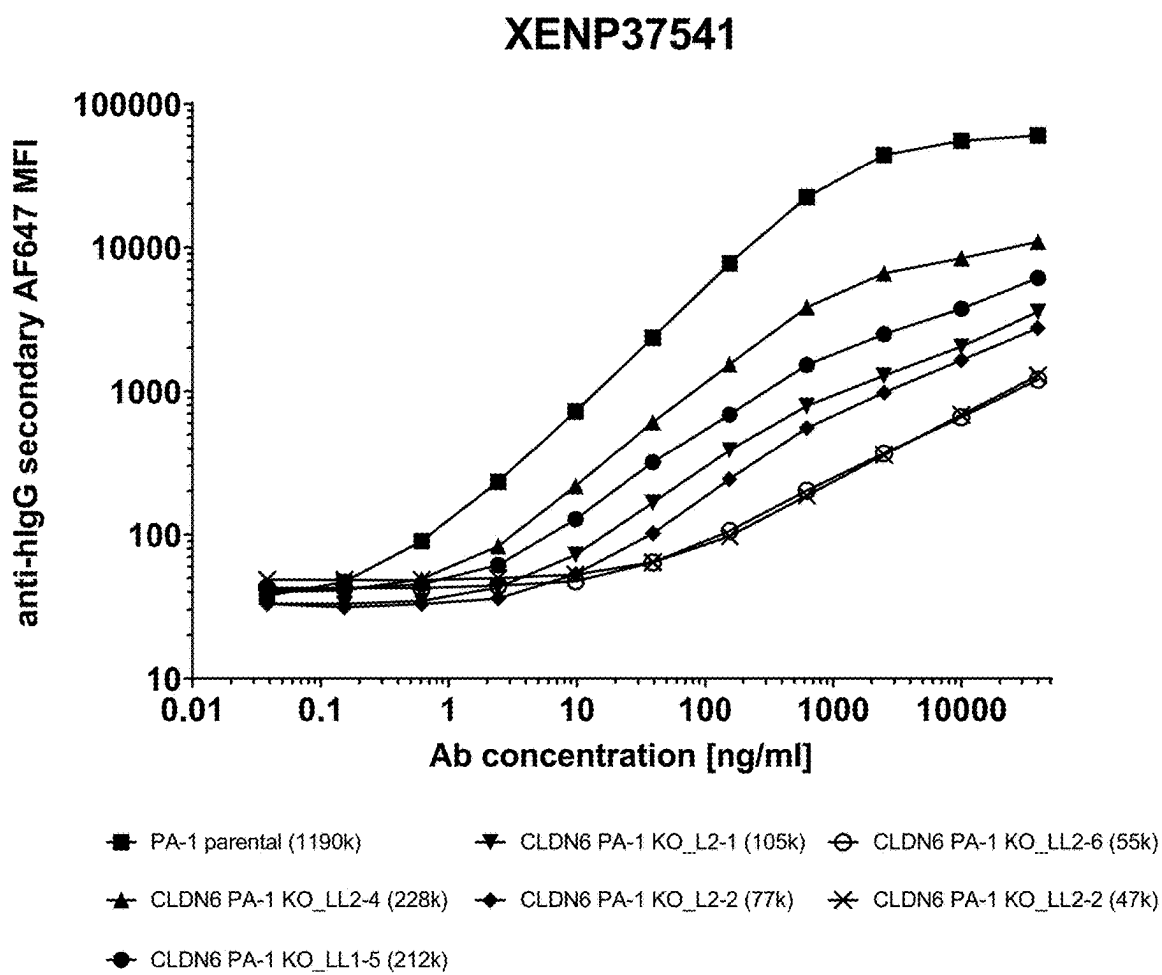
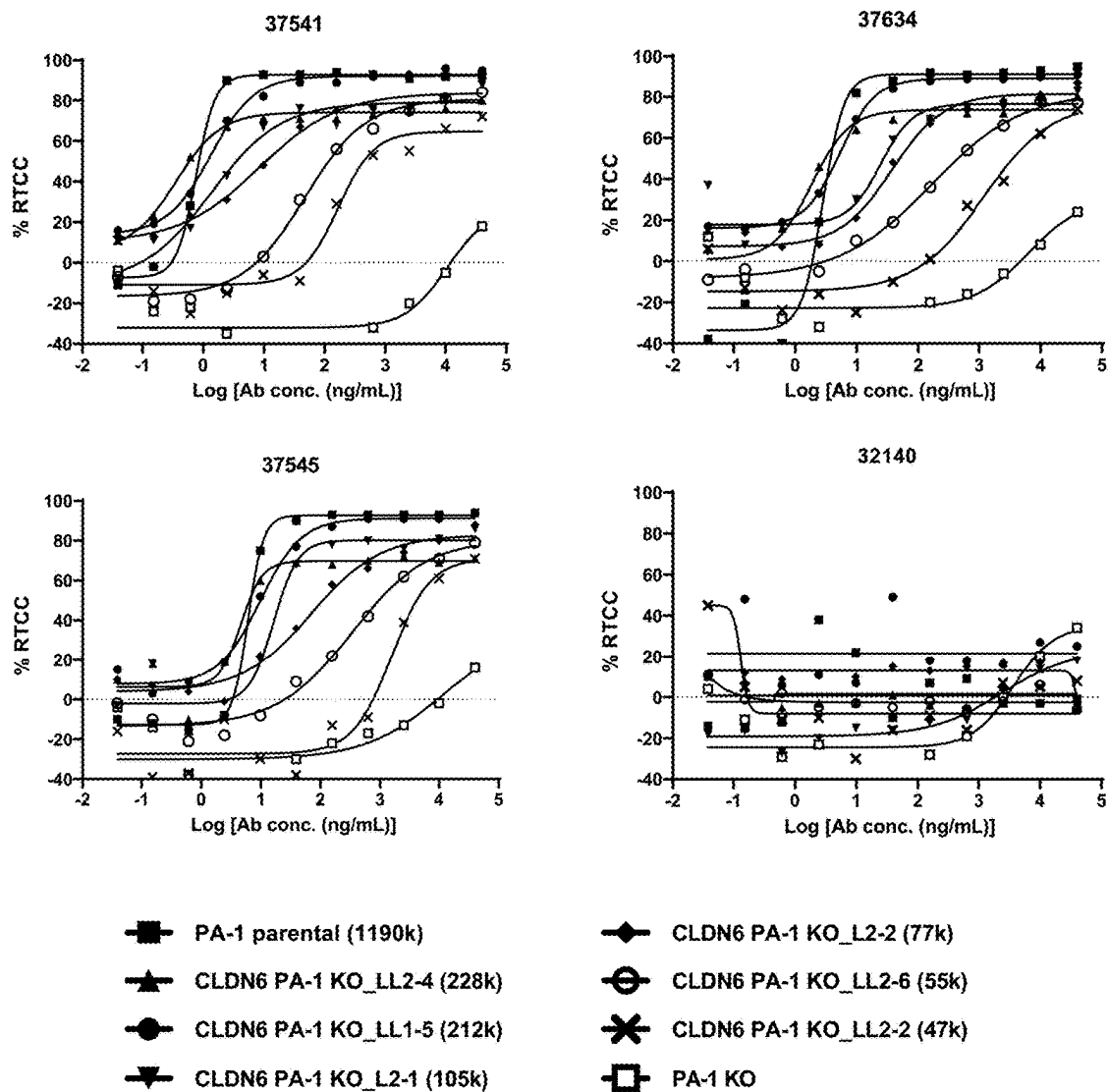


Figure 55



EC50	Parental (1190K)	LL2-4 (228K)	LL1-5 (212K)	L2-1 (105K)	L2-2 (77K)	LL2-6 (55K)	LL2-2 (47K)
<u>XENP37541</u>	0.76 ng/mL	0.35 ng/mL	1.3 ng/mL	1.4 ng/mL	8.3 ng/mL	47 ng/mL	170 ng/mL
<u>XENP37634</u>	2.8 ng/mL	1.8 ng/mL	5.3 ng/mL	24 ng/mL	37 ng/mL	160 ng/mL	1200 ng/mL
<u>XENP37545</u>	6.1 ng/mL	4.6 ng/mL	10 ng/mL	16 ng/mL	70 ng/mL	340 ng/mL	1600 ng/mL

Figure 56A

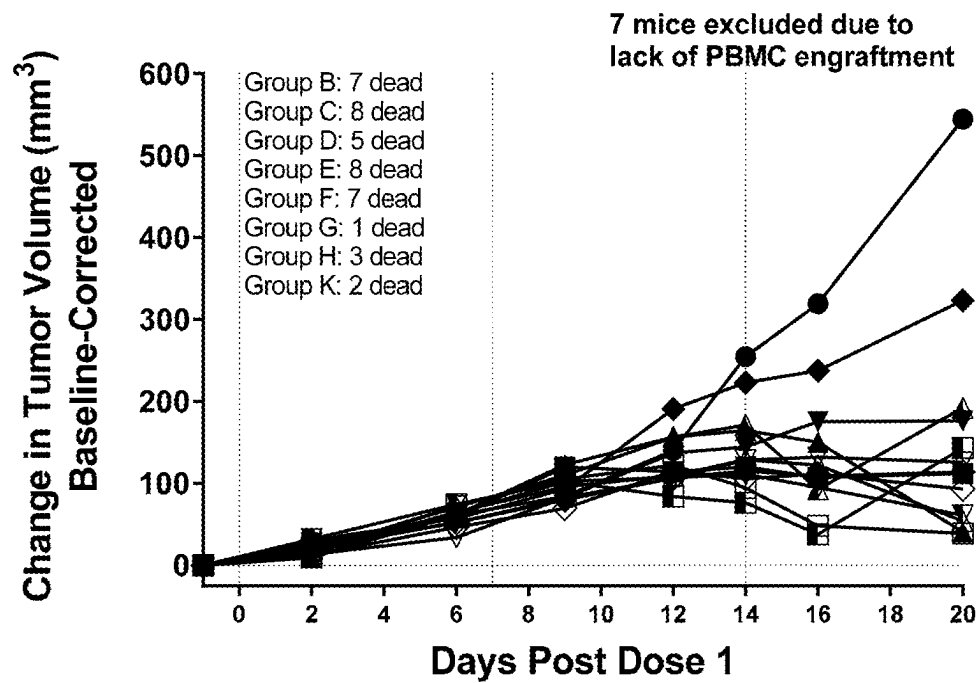


Figure 56B

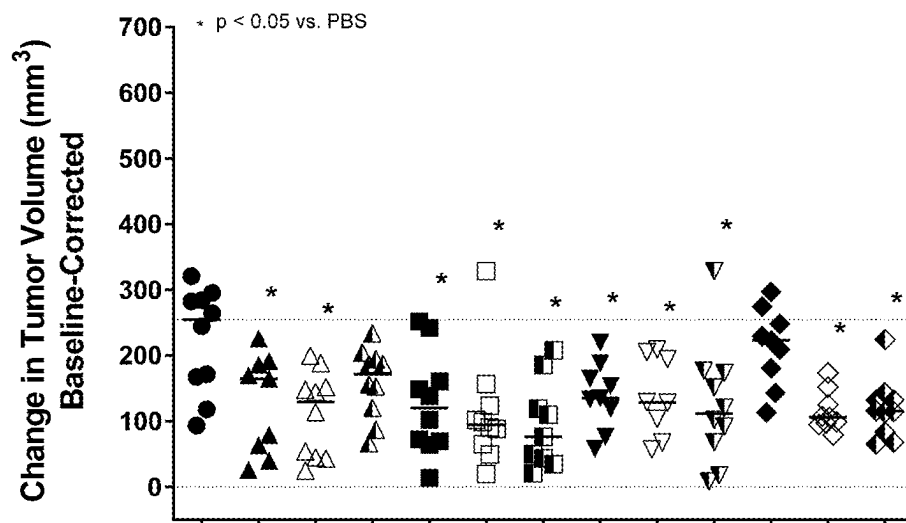


Figure 56C

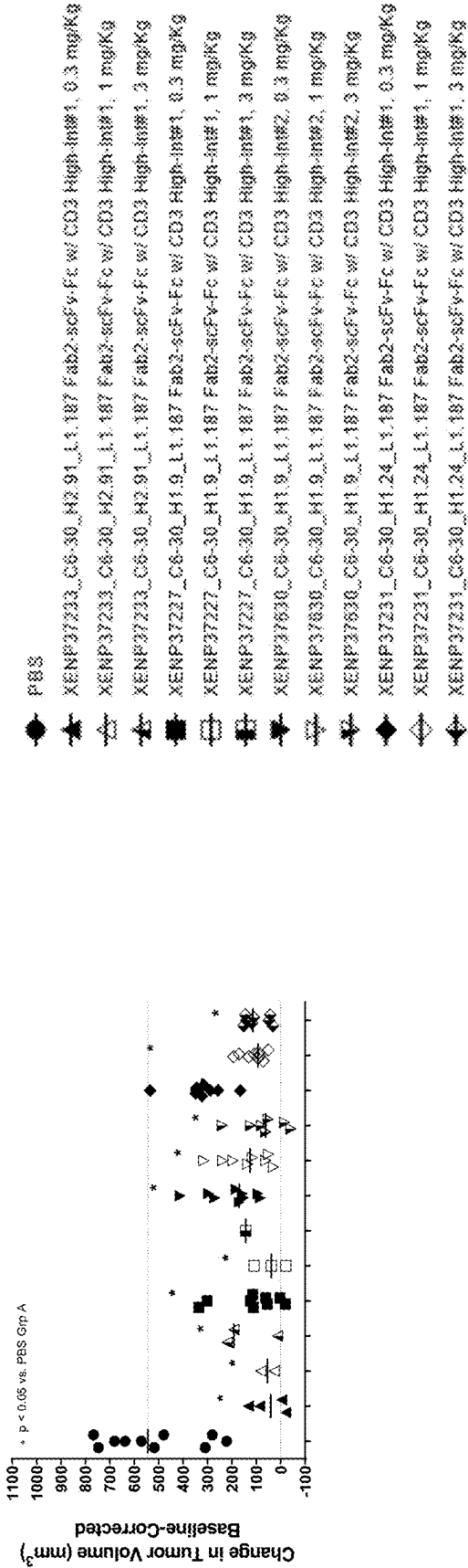


Figure 57A

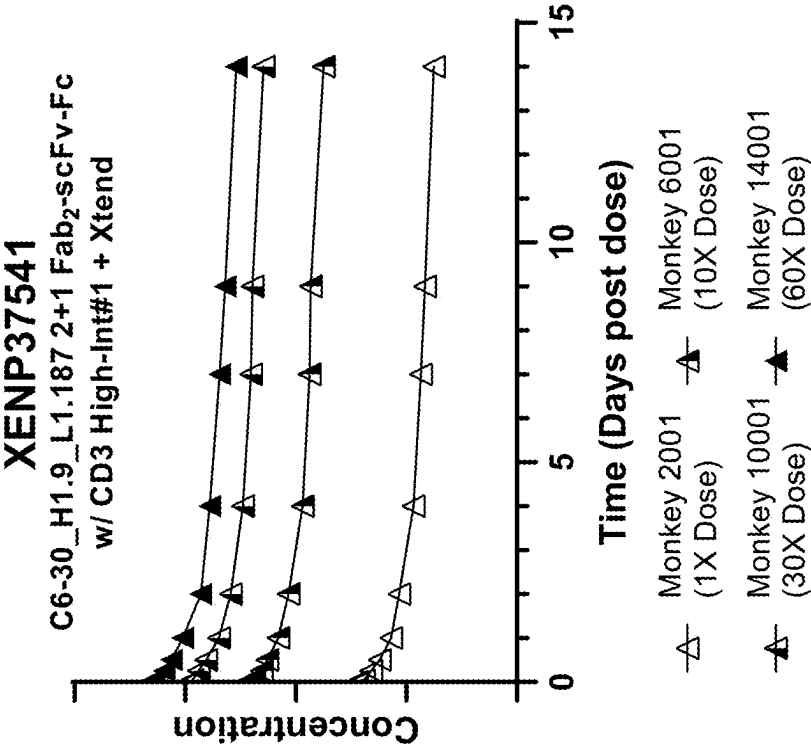


Figure 57B

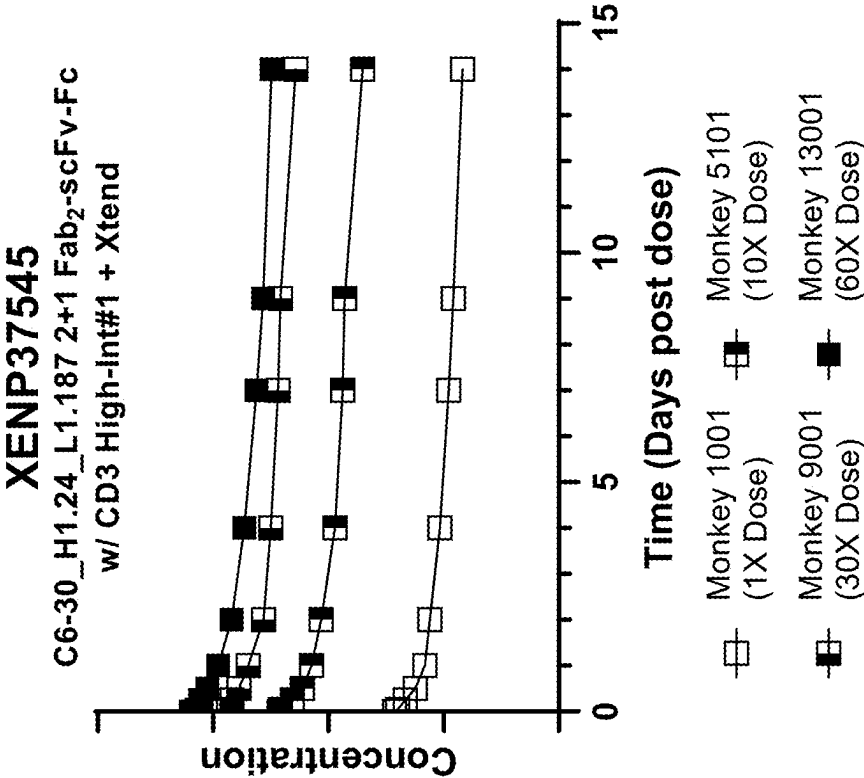


Figure 57C

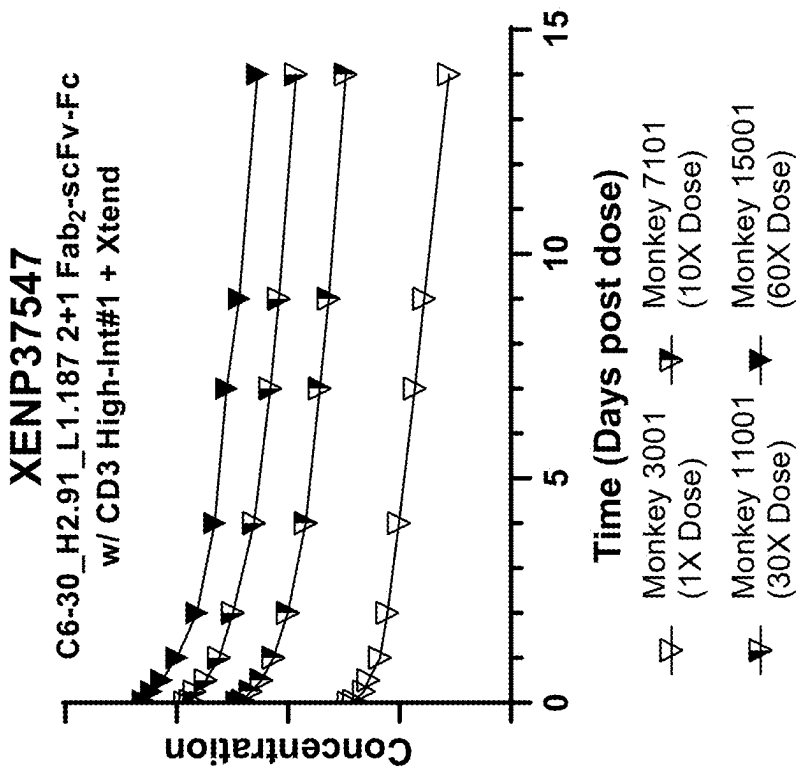


Figure 57D

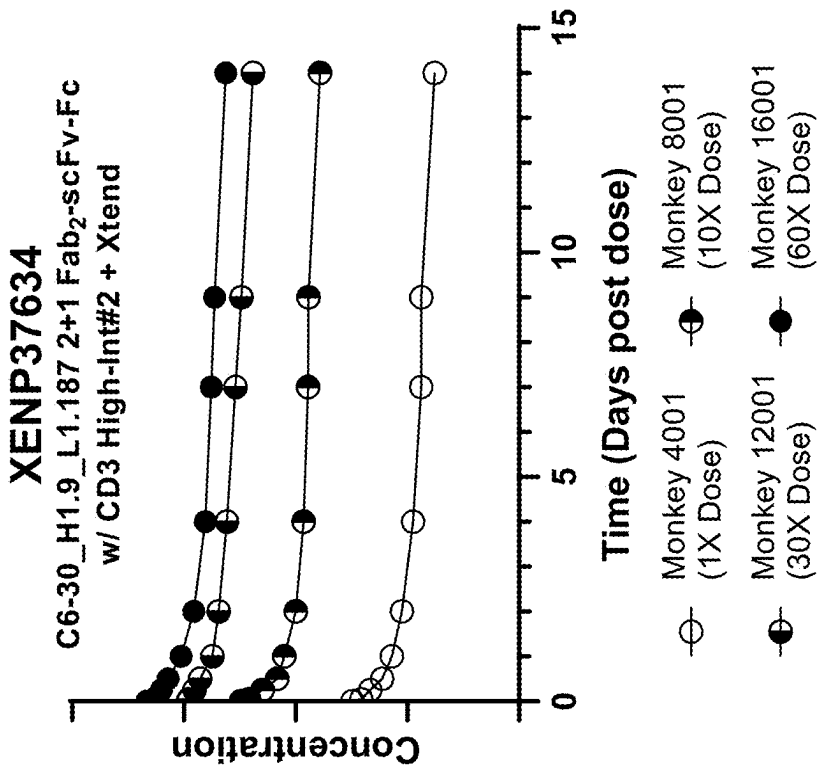
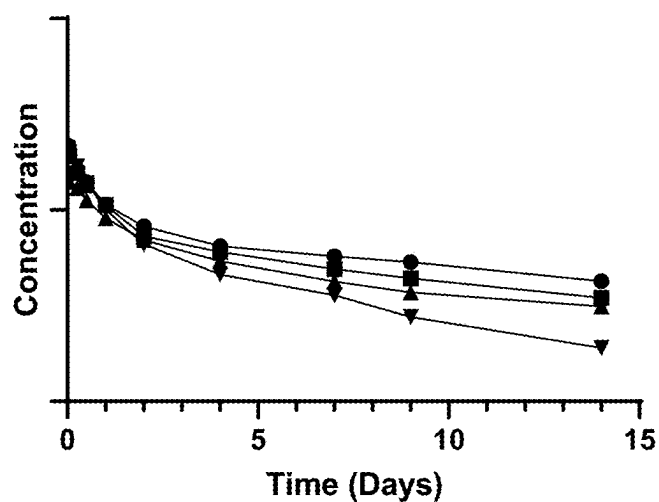


Figure 58

Blood concentration over time following high 60X dose



- ▲ XENP37541 C6-30_H1.9_L1.187 2+1 Fab₂-scFv-Fc w/ CD3 High-Int#1 + Xtend
- XENP37634 C6-30_H1.9_L1.187 2+1 Fab₂-scFv-Fc w/ CD3 High-Int#2 + Xtend
- XENP37545 C6-30_H1.24_L1.187 2+1 Fab₂-scFv-Fc w/ CD3 High-Int#1 + Xtend
- ▼ XENP37547 C6-30_H2.91_L1.187 2+1 Fab₂-scFv-Fc w/ CD3 High-Int#1 + Xtend

Figure 59A

>XENP37630 C6-30[CLDN6] H1.9 L1.187 Fab-C6-30[CLDN6] H1.9 L1.187 Fab (G4S)2 [anti-CD3] L1.47 H1.89 scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S -Fc(222) IgG1 PVA /S267K/S364K/E357Q

Chain 1 - C6-30[CLDN6]_H1.9_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNANTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDFWGAGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDFVSCSVMEALHNHYTQKSLSLSPGK (SEQ ID NO:
570)

Chain 2 - C6-30[CLDN6]_H1.9_(G4S)2_[anti-

CD3]_L1.47_H1.89_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNANTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDFWGAGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCGGGGSGG
GGG/QAVVTQEPSTLTVSPGGTTLTCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLGGA
ALTISGAQPEDEADYYCALWYSNHWVFGGCTKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGSLR
LSCAASGFTFTSTYAMNWVRQAPGKLEWVGRIIRSKYNNYATYYADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVY
YCVRHGMEFGDEYVSWFAYWGQGTLLTVSS/SGGGSGGGGSKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYKCKVSNKALPAPIEKT
ISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFYPSDIAVEWESNGQPENNYKTPPVLDSDGSFFLYSKLTVD
KSRWQQGNVFCSCVMHEALHNHYTQKSLSLSPGK (SEQ ID NO: 571)

Chain 3 - C6-30[CLDN6]_L1.187_Light Chain

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFSGSGSGKDYFTFTISS
LQPEDIAITYCQOYWSAFLTFGGGTKVEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKSTYSLSSTLTLSKADYEKKHYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
572)

>XENP37634 C6-30[CLDN6] H1.9 L1.187 Fab-C6-30[CLDN6] H1.9 L1.187 Fab (G4S)2 [anti-CD3] L1.47 H1.89 scFv(GKPGS)4 (G4S)2-IgG1 pl(-) Isosteric A PVA /S267K/L368D/K370S/M428L/N434S-Fc(222) IgG1 PVA /S267K/S364K/E357Q/M428L/N434S

Chain 1 - C6-30[CLDN6]_H1.9_IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S/M428L/N434S

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNANTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFDFWGAGTLTVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSDTKVDKKVEPKSCDKTHTCP
PCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVSV
LTVLHQDWLNGKEYKCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFYPSDIAVEWE
SDGQPENNYKTPPVLDSDGSFFLYSKLTVDKSRWEQGDFVSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO:
573)

Figure 59B

Chain 2 - C6-30[CLDN6]_H1.9_(G4S)2_[anti-**CD3]_L1.47_H1.89_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q/M428L/N434S**

QVQLVQSGAEVKKPGASVKVSCKTSGYTFTEYTMHWVRQAPGQSLEWMGGIDPNNANTHYNQKFQGRVTITVDKSAS
TAYMELSSLRSEDTAVYYCARIYYFGRLYFD~~FWGAGTL~~TVVSS/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDYF
PEPVTVSWNSGALTSGVHTFPAVLQSSGLYSLSSVTVPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCGGGGSGG
GGS/QAVVTQEPSTLVSPGGTVTLT~~CGSSTGAVTTS~~NYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGGKA
ALTISGAQPEDEADYYCALWYS~~NHWVF~~GGG~~TKLT~~VL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQP~~GGSLR~~
LSCAASGFTFSTYAMNWVRQAPGKGLEWVGRI~~RSKYNNYATYY~~ADSVKGRFTISRDDSKNTLYLQMNSLRAEDTAVY
YCVRHGNEFGDEYVSWFAYWGQ~~TLTV~~VSS/GGGGSGGGGSKTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRTPEV
TCVVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIEKTI
SKAKGQPREPQVYTLPPSREQMTKNQVKLTCLVKGFI~~PSDIA~~VEWESNGQPENNYKTTTPVLDSDGSFFLYSKLTVD
KSRWQQGNVFSCSVLHEALHSHYTQKSLSLSPGK (SEQ ID NO: 574)

Chain 3 - C6-30[CLDN6]_L1.187_Light Chain

DIQMTQSPSSLSASVGDRVTITCQASEDIYNRLAWYQQKPGKVPKLLISGATSLETGVPSRFRSGSGSGKDYFTFTISS
LQPEDIATYYCQQYWSAPLTFGGG~~TKVEIK~~/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
575)

Figure 60A

>XENP37217 Comp_CH02[CLDN6]_H0L0_Fab-

Comp_CH02[CLDN6]_H0L0_Fab_(G4S)2_SP34_L1.47_H1.32_scFv(GKPGS)4_(G4S)2-IgG1_pl(-

)_Isosteric_A_PVA_/S267K/L368D/K370S-Fc(222)_IgG1_PVA_/S267K/S364K/E357Q

Chain 1- Comp_CH02[CLDN6]_H0_Fab_-(G4S)2_SP34_L1.47_H1.32_scFv(GKPGS)4_-(G4S)2 -

Fc(222)_IgG1_PVA_/S267K/S364K/E357Q

EVQVEQSGLELVKPGASVKISCKTSGYTFTEYTMHWVKQSHGKSLEWIGGINPNNGNTRYNQKFKDKATLTVDKSSR
TAYMELHTLTSEDSAVYYCARCGDYDLFFFFAYWGQGLTVTVSA/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDY
FPEPVTISWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSC/GGGGS
GGGGS/QAVVTQEPSTLTVSPGGTVTLTCCSSTGAVTTSNYANWVQKPGKSPRGLIGGTNKRAPGVPARFSGSLLGG
KAALTISGAQPEDEADYYCALWYSNHWFVGGGTGKLTVL/GKPGSGKPGSGKPGSGKPGS/EVQLVESGGGLVQPGGS
LRLSCAASGFTTFSTYAMNWVRQAPGKGLEWVGRIRSKANNYATYYADSVKGPFTISRDDSKNTLYLQMNSLRAEDTA
VYYCVRHGNFGDSYVSWFAYWGQGLTVTVSS/GGGSGGGGS/KTHTCPPCPAPPVAGPSVFLFPPKPKDTLMISRT
PEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEQYNSTYRVVSVLTVLHQDWLNGKEYCKVSNKALPAPIE
KTISKAKGQPREPQVYTLPPSREQMTKQNQVLTCLVKGFIYPSDIAVEWESNGQPENNYKTTTPVLDSDGSFFLYSKL
TVDKSRWQQGNVFSCSVMHREALHNYTQKSLSLSPGK (SEQ ID NO: 576)

Chain 2- Comp_CH02[CLDN6]_H0_Fab-IgG1_pl(-)_Isosteric_A_PVA_/S267K/L368D/K370S

EVQVEQSGLELVKPGASVKISCKTSGYTFTEYTMHWVKQSHGKSLEWIGGINPNNGNTRYNQKFKDKATLTVDKSSR
TAYMELHTLTSEDSAVYYCARCGDYDLFFFFAYWGQGLTVTVSA/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDY
FPEPVTISWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTC
PPCPAPPVAGPSVFLFPPKPKDTLMISRTPEVTCVVDVKHEDPEVKFNWYVDGVEVHNAKTKPREEEYNSTYRVVS
VLTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCDVSGFIYPSDIAVEW
ESDQGPENNYKTTTPVLDSDGSFFLYSKLTVDKSRWEQGDVFSCSVMHREALHNYTQKSLSLSPGK (SEQ ID
NO: 577)

Chain 3- Comp_CH02[CLDN6]_L0_Light Chain

DIVMTQFQKFMSTSVGDRVSVTCASQNVGTNVAWYQKPGQSPEALISSASSRFSGVPDRFTGSGSGTDFTLTITN
VQSEDLADYFCQQYNSFPFTFGSGTELEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDSITYLSSTLTLSKADYEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
578)

>XENP38084 Comp_CH02[CLDN6]_H0_L0

Chain 1- Comp_CH02[CLDN6]_H0_Heavy Chain

EVQVEQSGLELVKPGASVKISCKTSGYTFTEYTMHWVKQSHGKSLEWIGGINPNNGNTRYNQKFKDKATLTVDKSSR
TAYMELHTLTSEDSAVYYCARCGDYDLFFFFAYWGQGLTVTVSA/ASTKGPSVFPLAPSSKSTSGGTAALGCLVKDY
FPEPVTISWNSGALTSGVHTFPAVLQSSGLYSLSSVTVTPSSSLGTQTYICNVNHKPSNTKVDKKVEPKSCDKTHTC
FPFCFAPFLPGGPKVFLFPPKPKDTLMISRTPEVTCVVDVSHEDPEVKFNWYVDGVEVHNAKTKPREEQYASTYRVV
SVLTVLHQDWLNGKEYCKVSNKALPAPIEKTISKAKGQPREPQVYTLPPSREEMTKNQVSLTCLVKGFIYPSDIAVE
WESNGQPENNYKTTTPPYLDSDGSFFLYSKLTVDKSRWQQGNVFSCSVMHREALHNYTQESLSLSP (SEQ ID NO:
579)

Figure 60B

Chain 2- Comp_CH02[CLDN6]_H0_L0

DIVMTQFQKFMSTSVGDRVSVTCKASQNVGTNVAWYQOKPGQSPEALISSASSRFSGVFDRFTGSGSGTDFTLTITN
VQSEDLADYFCQQYNSFPFTFGSGTELEIK/RTVAAPSVFIFPPSDEQLKSGTASVVCLLNNFYPREAKVQWKVDNA
LQSGNSQESVTEQDSKDYSLSSLTLSKADYEEKHKVYACEVTHQGLSSPVTKSFNRGEC (SEQ ID NO:
580)

Figure 61A

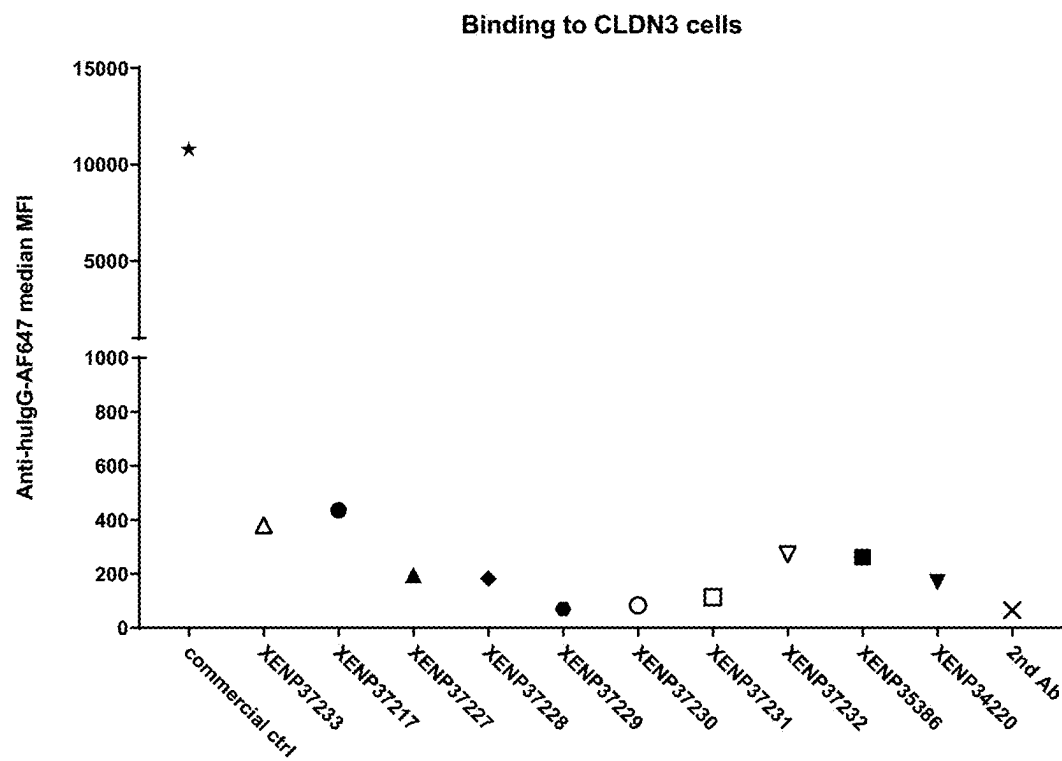


Figure 61B

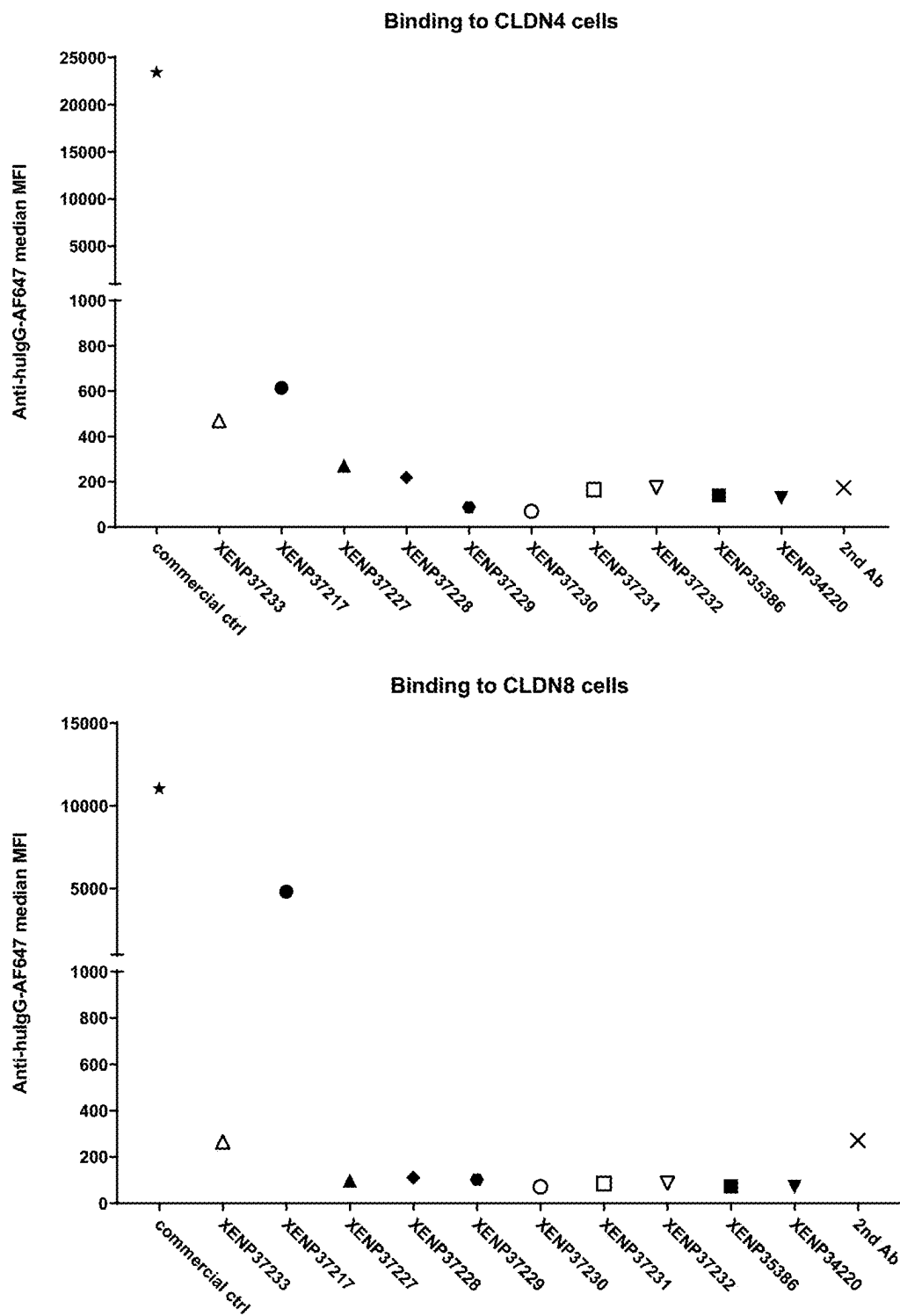
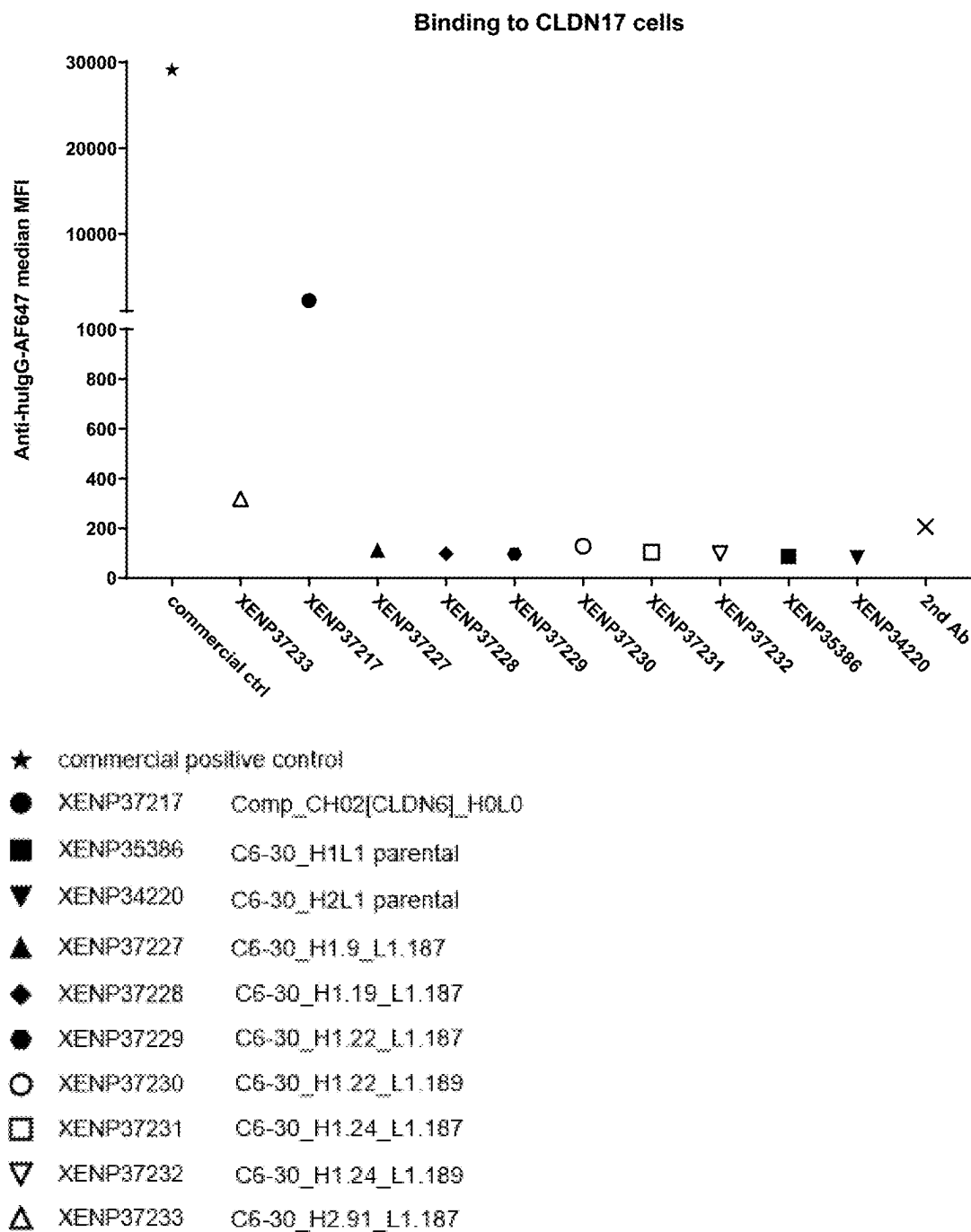


Figure 61C



1

**HETERODIMERIC ANTIBODIES THAT BIND
CD3 AND CLDN6****PRIORITY CLAIM**

This application is a continuation of U.S. patent application Ser. No. 18/348,244, filed Jul. 6, 2023, which is a continuation of U.S. patent application Ser. No. 17/690,702, filed Mar. 9, 2022, now U.S. Pat. No. 11,739,144, which claims the benefit of U.S. Provisional Patent Application No. 63/158,584, filed Mar. 9, 2021, which is hereby incorporated by reference in its entirety.

SEQUENCE LISTING

The instant application contains a Sequence Listing which has been submitted electronically in XML file format and is hereby incorporated by reference in its entirety. Said XML copy, created on Jun. 16, 2023, is named 067461-5281-US01.xml and is 1,071,543 bytes in size.

BACKGROUND

Antibody-based therapeutics have been used successfully to treat a variety of diseases, including cancer. An increasingly prevalent avenue being explored is the engineering of single immunoglobulin molecules that co-engage two different antigens. Such alternate antibody formats that engage two different antigens are often referred to as bispecific antibodies. Because the considerable diversity of the antibody variable region (Fv) makes it possible to produce an Fv that recognizes virtually any molecule, the typical approach to bispecific antibody generation is the introduction of new variable regions into the antibody.

A particularly useful approach for bispecific antibodies is to engineer a first binding domain which engages CD3 and a second binding domain which engages an antigen associated with or upregulated on cancer cells so that the bispecific antibody redirects CD3+ cells to destroy the cancer cells. Claudin 6 (CLDN6) has been determined to be upregulated in gastric, lung, and ovarian cancers. In view of this, it is believed that anti-CLDN6 antibodies are useful, for example, for localizing anti-tumor therapeutics (e.g., chemotherapeutic agents and T cells) to such CLDN6 expressing tumors. The present invention provides novel bispecific antibodies to CD3 and CLDN6 that are capable of localizing CD3+ effector T cells to CLDN6 expressing tumors.

SUMMARY

Provided herein are novel CLDN6 binding domains, and anti-CLDN6xanti-CD3 antibodies that include such CLDN6 binding domains. Also provided herein are methods of using such antibodies for the treatment of CLDN6-associated cancers.

In a first aspect, provided herein are compositions comprising a CLDN6 antigen binding domain. In some embodiments, the CLDN6 antigen binding domain comprises a set of 6 CDRs (vhCDR1, vhCDR2, vhCDR3, vlCDR1, vlCDR2 and vlCDR3) from a variable heavy domain (VH)/variable light domain (VL) pair selected from the group consisting: H1_L1, H1.1_L1, H1.2_L1, H1.3_L1, H1.4_L1, H1.5_L1, H1.6_L1, H1.7_L1, H1.8_L1, H1.9_L1, H1.19_L1, H1.22_L1, H1.24_L1, H2_L1, H2.1_L1, H2.2_L1, H2.3_L1, H2.4_L1, H2.5_L1, H2.6_L1, H2.7_L1, H2.8_L1, H2.9_L1, H2.11_L1, H2.12_L1, H2.71_L1, H2.75_L1, H2.90_L1, H2.91_L1, H2.118_L1, H2.119_L1, H1_L1.1,

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H1.1_L1.1, H1.2_L1.1, H1.3_L1.1, H1.4_L1.1, H1.5_L1.1, H1.6_L1.1, H1.7_L1.1, H1.8_L1.1, H1.9_L1.1, H1.19_L1.1, H1.22_L1.1, H1.24_L1.1, H2_L1.1, H2.1_L1.1, H2.2_L1.1, H2.3_L1.1, H2.4_L1.1, H2.5_L1.1, H2.6_L1.1, H2.7_L1.1, H2.8_L1.1, H2.9_L1.1, H2.11_L1.1, H2.12_L1.1, H2.71_L1.1, H2.75_L1.1, H2.90_L1.1, H2.91_L1.1, H2.118_L1.1, H2.119_L1.1, H1_L1.4, H1.1_L1.4, H1.2_L1.4, H1.3_L1.4, H1.4_L1.4, H1.5_L1.4, H1.6_L1.4, H1.7_L1.4, H1.8_L1.4, H1.9_L1.4, H1.19_L1.4, H1.22_L1.4, H1.24_L1.4, H2_L1.4, H2.1_L1.4, H2.2_L1.4, H2.3_L1.4, H2.4_L1.4, H2.5_L1.4, H2.6_L1.4, H2.7_L1.4, H2.8_L1.4, H2.9_L1.4, H2.11_L1.4, H2.12_L1.4, H2.71_L1.4, H2.75_L1.4, H2.90_L1.4, H2.91_L1.4, H2.118_L1.4, H2.119_L1.4, H1_L1.7, H1.1_L1.7, H1.2_L1.7, H1.3_L1.7, H1.4_L1.7, H1.5_L1.7, H1.6_L1.7, H1.7_L1.7, H1.8_L1.7, H1.9_L1.7, H1.19_L1.7, H1.22_L1.7, H1.24_L1.7, H2_L1.7, H2.1_L1.7, H2.2_L1.7, H2.3_L1.7, H2.4_L1.7, H2.5_L1.7, H2.6_L1.7, H2.7_L1.7, H2.8_L1.7, H2.9_L1.7, H2.11_L1.7, H2.12_L1.7, H2.71_L1.7, H2.75_L1.7, H2.90_L1.7, H2.91_L1.7, H2.118_L1.7, H2.119_L1.7, H1_L1.16, H1.1_L1.16, H1.2_L1.16, H1.3_L1.16, H1.4_L1.16, H1.5_L1.16, H1.6_L1.16, H1.7_L1.16, H1.8_L1.16, H1.9_L1.16, H1.19_L1.16, H1.22_L1.16, H1.24_L1.16, H2_L1.16, H2.1_L1.16, H2.2_L1.16, H2.3_L1.16, H2.4_L1.16, H2.5_L1.16, H2.6_L1.16, H2.7_L1.16, H2.8_L1.16, H2.9_L1.16, H2.11_L1.16, H2.12_L1.16, H2.71_L1.16, H2.75_L1.16, H2.90_L1.16, H2.91_L1.16, H2.118_L1.16, H2.119_L1.16, H1_L1.18, H1.1_L1.18, H1.2_L1.18, H1.3_L1.18, H1.4_L1.18, H1.5_L1.18, H1.6_L1.18, H1.7_L1.18, H1.8_L1.18, H1.9_L1.18, H1.19_L1.18, H1.22_L1.18, H1.24_L1.18, H2_L1.18, H2.1_L1.18, H2.2_L1.18, H2.3_L1.18, H2.4_L1.18, H2.5_L1.18, H2.6_L1.18, H2.7_L1.18, H2.8_L1.18, H2.9_L1.18, H2.11_L1.18, H2.12_L1.18, H2.71_L1.18, H2.75_L1.18, H2.90_L1.18, H2.91_L1.18, H2.118_L1.18, H2.119_L1.18, H1_L1.19, H1.1_L1.19, H1.2_L1.19, H1.3_L1.19, H1.4_L1.19, H1.5_L1.19, H1.6_L1.19, H1.7_L1.19, H1.8_L1.19, H1.9_L1.19, H1.19_L1.19, H1.22_L1.19, H1.24_L1.19, H2_L1.19, H2.1_L1.19, H2.2_L1.19, H2.3_L1.19, H2.4_L1.19, H2.5_L1.19, H2.6_L1.19, H2.7_L1.19, H2.8_L1.19, H2.9_L1.19, H2.11_L1.19, H2.12_L1.19, H2.71_L1.19, H2.75_L1.19, H2.90_L1.19, H2.91_L1.19, H2.118_L1.19, H2.119_L1.19, H1_L1.21, H1.1_L1.21, H1.2_L1.21, H1.3_L1.21, H1.4_L1.21, H1.5_L1.21, H1.6_L1.21, H1.7_L1.21, H1.8_L1.21, H1.9_L1.21, H1.19_L1.21, H1.22_L1.21, H1.24_L1.21, H2_L1.21, H2.1_L1.21, H2.2_L1.21, H2.3_L1.21, H2.4_L1.21, H2.5_L1.21, H2.6_L1.21, H2.7_L1.21, H2.8_L1.21, H2.9_L1.21, H2.11_L1.21, H2.12_L1.21, H2.71_L1.21, H2.75_L1.21, H2.90_L1.21, H2.91_L1.21, H2.118_L1.21, H2.119_L1.21, H1_L1.22, H1.1_L1.22, H1.2_L1.22, H1.3_L1.22, H1.4_L1.22, H1.5_L1.22, H1.6_L1.22, H1.7_L1.22, H1.8_L1.22, H1.9_L1.22, H1.19_L1.22, H1.22_L1.22, H1.24_L1.22, H2_L1.22, H2.1_L1.22, H2.2_L1.22, H2.3_L1.22, H2.4_L1.22, H2.5_L1.22, H2.6_L1.22, H2.7_L1.22, H2.8_L1.22, H2.9_L1.22, H2.11_L1.22, H2.12_L1.22, H2.71_L1.22, H2.75_L1.22, H2.90_L1.22, H2.91_L1.22, H2.118_L1.22, H2.119_L1.22, H1_L1.23, H1.1_L1.23, H1.2_L1.23, H1.3_L1.23, H1.4_L1.23, H1.5_L1.23, H1.6_L1.23, H1.7_L1.23, H1.8_L1.23, H1.9_L1.23, H1.19_L1.23, H1.22_L1.23, H1.24_L1.23, H2_L1.23, H2.1_L1.23, H2.2_L1.23, H2.3_L1.23, H2.4_L1.23, H2.5_L1.23, H2.6_L1.23, H2.7_L1.23, H2.8_L1.23, H2.9_L1.23, H2.11_L1.23, H2.12_L1.23, H2.71_L1.23, H2.75_L1.23, H2.90_L1.23, H2.91_L1.23,

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H2.118_L1.23, H2.119_L1.23, H1_L1.27, H1.1_L1.27, H1.2_L1.27, H1.3_L1.27, H1.4_L1.27, H1.5_L1.27, H1.6_L1.27, H1.7_L1.27, H1.8_L1.27, H1.9_L1.27, H1.19_L1.27, H1.22_L1.27, H1.24_L1.27, H2_L1.27, H2.1_L1.27, H2.2_L1.27, H2.3_L1.27, H2.4_L1.27, H2.5_L1.27, H2.6_L1.27, H2.7_L1.27, H2.8_L1.27, H2.9_L1.27, H2.11_L1.27, H2.12_L1.27, H2.71_L1.27, H2.75_L1.27, H2.90_L1.27, H2.91_L1.27, H2.118_L1.27, H2.119_L1.27, H1_L1.60, H1.1_L1.60, H1.2_L1.60, H1.3_L1.60, H1.4_L1.60, H1.5_L1.60, H1.6_L1.60, H1.7_L1.60, H1.8_L1.60, H1.9_L1.60, H1.19_L1.60, H1.22_L1.60, H1.24_L1.60, H2_L1.60, H2.1_L1.60, H2.2_L1.60, H2.3_L1.60, H2.4_L1.60, H2.5_L1.60, H2.6_L1.60, H2.7_L1.60, H2.8_L1.60, H2.9_L1.60, H2.11_L1.60, H2.12_L1.60, H2.71_L1.60, H2.75_L1.60, H2.90_L1.60, H2.91_L1.60, H2.118_L1.60, H2.119_L1.60, H1_L1.107, H1.1_L1.107, H1.2_L1.107, H1.3_L1.107, H1.4_L1.107, H1.5_L1.107, H1.6_L1.107, H1.7_L1.107, H1.8_L1.107, H1.9_L1.107, H1.19_L1.107, H1.22_L1.107, H1.24_L1.107, H2_L1.107, H2.1_L1.107, H2.2_L1.107, H2.3_L1.107, H2.4_L1.107, H2.5_L1.107, H2.6_L1.107, H2.7_L1.107, H2.8_L1.107, H2.9_L1.107, H2.11_L1.107, H2.12_L1.107, H2.71_L1.107, H2.75_L1.107, H2.90_L1.107, H2.91_L1.107, H2.118_L1.107, H2.119_L1.107, H1_L1.114, H1.1_L1.114, H1.2_L1.114, H1.3_L1.114, H1.4_L1.114, H1.5_L1.114, H1.6_L1.114, H1.7_L1.114, H1.8_L1.114, H1.9_L1.114, H1.19_L1.114, H1.22_L1.114, H1.24_L1.114, H2_L1.114, H2.1_L1.114, H2.2_L1.114, H2.3_L1.114, H2.4_L1.114, H2.5_L1.114, H2.6_L1.114, H2.7_L1.114, H2.8_L1.114, H2.9_L1.114, H2.11_L1.114, H2.12_L1.114, H2.71_L1.114, H2.75_L1.114, H2.90_L1.114, H2.91_L1.114, H2.118_L1.114, H2.119_L1.114, H1_L1.187, H1.1_L1.187, H1.2_L1.187, H1.3_L1.187, H1.4_L1.187, H1.5_L1.187, H1.6_L1.187, H1.7_L1.187, H1.8_L1.187, H1.9_L1.187, H1.19_L1.187, H1.22_L1.187, H1.24_L1.187, H2_L1.187, H2.1_L1.187, H2.2_L1.187, H2.3_L1.187, H2.4_L1.187, H2.5_L1.187, H2.6_L1.187, H2.7_L1.187, H2.8_L1.187, H2.9_L1.187, H2.11_L1.187, H2.12_L1.187, H2.71_L1.187, H2.75_L1.187, H2.90_L1.187, H2.91_L1.187, H2.118_L1.187, H2.119_L1.187, H1_L1.189, H1.1_L1.189, H1.2_L1.189, H1.3_L1.189, H1.4_L1.189, H1.5_L1.189, H1.6_L1.189, H1.7_L1.189, H1.8_L1.189, H1.9_L1.189, H1.19_L1.189, H1.22_L1.189, H1.24_L1.189, H2_L1.189, H2.1_L1.189, H2.2_L1.189, H2.3_L1.189, H2.4_L1.189, H2.5_L1.189, H2.6_L1.189, H2.7_L1.189, H2.8_L1.189, H2.9_L1.189, H2.11_L1.189, H2.12_L1.189, H2.71_L1.189, H2.75_L1.189, H2.90_L1.189, H2.91_L1.189, H2.118_L1.189, H2.119_L1.189, H1_L2, H1.1_L2, H1.2_L2, H1.3_L2, H1.4_L2, H1.5_L2, H1.6_L2, H1.7_L2, H1.8_L2, H1.9_L2, H1.19_L2, H1.22_L2, H1.24_L2, H2_L2, H2.1_L2, H2.2_L2, H2.3_L2, H2.4_L2, H2.5_L2, H2.6_L2, H2.7_L2, H2.8_L2, H2.9_L2, H2.11_L2, H2.12_L2, H2.71_L2, H2.75_L2, H2.90_L2, H2.91_L2, H2.118_L2 and H2.119_L2.

In some embodiments, the CLDN6 antigen binding domain comprises a VH/VL pair selected from the group consisting of: H1_L1, H1.1_L1, H1.2_L1, H1.3_L1, H1.4_L1, H1.5_L1, H1.6_L1, H1.7_L1, H1.8_L1, H1.9_L1, H1.19_L1, H1.22_L1, H1.24_L1, H2_L1, H2.1_L1, H2.2_L1, H2.3_L1, H2.4_L1, H2.5_L1, H2.6_L1, H2.7_L1, H2.8_L1, H2.9_L1, H2.11_L1, H2.12_L1, H2.71_L1, H2.75_L1, H2.90_L1, H2.91_L1, H2.118_L1, H2.119_L1, H1_L1.1, H1.1_L1.1, H1.2_L1.1, H1.3_L1.1, H1.4_L1.1, H1.5_L1.1, H1.6_L1.1, H1.7_L1.1, H1.8_L1.1, H1.9_L1.1, H1.19_L1.1, H1.22_L1.1, H1.24_L1.1, H2_L1.1,

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H2.1_L1.1, H2.2_L1.1, H2.3_L1.1, H2.4_L1.1, H2.5_L1.1, H2.6_L1.1, H2.7_L1.1, H2.8_L1.1, H2.9_L1.1, H2.11_L1.1, H2.12_L1.1, H2.71_L1.1, H2.75_L1.1, H2.90_L1.1, H2.91_L1.1, H2.118_L1.1, H2.119_L1.1, H1_L1.4, H1.1_L1.4, H1.2_L1.4, H1.3_L1.4, H1.4_L1.4, H1.5_L1.4, H1.6_L1.4, H1.7_L1.4, H1.8_L1.4, H1.9_L1.4, H1.19_L1.4, H1.22_L1.4, H1.24_L1.4, H2_L1.4, H2.1_L1.4, H2.2_L1.4, H2.3_L1.4, H2.4_L1.4, H2.5_L1.4, H2.6_L1.4, H2.7_L1.4, H2.8_L1.4, H2.9_L1.4, H2.11_L1.4, H2.12_L1.4, H2.71_L1.4, H2.75_L1.4, H2.90_L1.4, H2.91_L1.4, H2.118_L1.4, H2.119_L1.4, H1_L1.7, H1.1_L1.7, H1.2_L1.7, H1.3_L1.7, H1.4_L1.7, H1.5_L1.7, H1.6_L1.7, H1.7_L1.7, H1.8_L1.7, H1.9_L1.7, H1.19_L1.7, H1.22_L1.7, H1.24_L1.7, H2_L1.7, H2.1_L1.7, H2.2_L1.7, H2.3_L1.7, H2.4_L1.7, H2.5_L1.7, H2.6_L1.7, H2.7_L1.7, H2.8_L1.7, H2.9_L1.7, H2.11_L1.7, H2.12_L1.7, H2.71_L1.7, H2.75_L1.7, H2.90_L1.7, H2.91_L1.7, H2.118_L1.7, H2.119_L1.7, H1_L1.16, H1.1_L1.16, H1.2_L1.16, H1.3_L1.16, H1.4_L1.16, H1.5_L1.16, H1.6_L1.16, H1.7_L1.16, H1.8_L1.16, H1.9_L1.16, H1.19_L1.16, H1.22_L1.16, H1.24_L1.16, H2_L1.16, H2.1_L1.16, H2.2_L1.16, H2.3_L1.16, H2.4_L1.16, H2.5_L1.16, H2.6_L1.16, H2.7_L1.16, H2.8_L1.16, H2.9_L1.16, H2.11_L1.16, H2.12_L1.16, H2.71_L1.16, H2.75_L1.16, H2.90_L1.16, H2.91_L1.16, H2.118_L1.16, H2.119_L1.16, H1_L1.18, H1.1_L1.18, H1.2_L1.18, H1.3_L1.18, H1.4_L1.18, H1.5_L1.18, H1.6_L1.18, H1.7_L1.18, H1.8_L1.18, H1.9_L1.18, H1.19_L1.18, H1.22_L1.18, H1.24_L1.18, H2_L1.18, H2.1_L1.18, H2.2_L1.18, H2.3_L1.18, H2.4_L1.18, H2.5_L1.18, H2.6_L1.18, H2.7_L1.18, H2.8_L1.18, H2.9_L1.18, H2.11_L1.18, H2.12_L1.18, H2.71_L1.18, H2.75_L1.18, H2.90_L1.18, H2.91_L1.18, H2.118_L1.18, H2.119_L1.18, H1_L1.19, H1.1_L1.19, H1.2_L1.19, H1.3_L1.19, H1.4_L1.19, H1.5_L1.19, H1.6_L1.19, H1.7_L1.19, H1.8_L1.19, H1.9_L1.19, H1.19_L1.19, H1.22_L1.19, H1.24_L1.19, H2_L1.19, H2.1_L1.19, H2.2_L1.19, H2.3_L1.19, H2.4_L1.19, H2.5_L1.19, H2.6_L1.19, H2.7_L1.19, H2.8_L1.19, H2.9_L1.19, H2.11_L1.19, H2.12_L1.19, H2.71_L1.19, H2.75_L1.19, H2.90_L1.19, H2.91_L1.19, H2.118_L1.19, H2.119_L1.19, H1_L1.21, H1.1_L1.21, H1.2_L1.21, H1.3_L1.21, H1.4_L1.21, H1.5_L1.21, H1.6_L1.21, H1.7_L1.21, H1.8_L1.21, H1.9_L1.21, H1.19_L1.21, H1.22_L1.21, H1.24_L1.21, H2_L1.21, H2.1_L1.21, H2.2_L1.21, H2.3_L1.21, H2.4_L1.21, H2.5_L1.21, H2.6_L1.21, H2.7_L1.21, H2.8_L1.21, H2.9_L1.21, H2.11_L1.21, H2.12_L1.21, H2.71_L1.21, H2.75_L1.21, H2.90_L1.21, H2.91_L1.21, H2.118_L1.2, H2.119_L1.21, H1_L1.22, H1.1_L1.22, H1.2_L1.22, H1.3_L1.22, H1.4_L1.22, H1.5_L1.22, H1.6_L1.22, H1.7_L1.22, H1.8_L1.22, H1.9_L1.22, H1.19_L1.22, H1.22_L1.22, H1.24_L1.22, H2_L1.22, H2.1_L1.22, H2.2_L1.22, H2.3_L1.22, H2.4_L1.22, H2.5_L1.22, H2.6_L1.22, H2.7_L1.22, H2.8_L1.22, H2.9_L1.22, H2.11_L1.22, H2.12_L1.22, H2.71_L1.22, H2.75_L1.22, H2.90_L1.22, H2.91_L1.22, H2.118_L1.22, H2.119_L1.22, H1_L1.23, H1.1_L1.23, H1.2_L1.23, H1.3_L1.23, H1.4_L1.23, H1.5_L1.23, H1.6_L1.23, H1.7_L1.23, H1.8_L1.23, H1.9_L1.23, H1.19_L1.23, H1.22_L1.23, H1.24_L1.23, H2_L1.23, H2.1_L1.23, H2.2_L1.23, H2.3_L1.23, H2.4_L1.23, H2.5_L1.23, H2.6_L1.23, H2.7_L1.23, H2.8_L1.23, H2.9_L1.23, H2.11_L1.23, H2.12_L1.23, H2.71_L1.23, H2.75_L1.23, H2.90_L1.23, H2.91_L1.23, H2.118_L1.23, H2.119_L1.23, H1_L1.27, H1.1_L1.27, H1.2_L1.27, H1.3_L1.27, H1.4_L1.27, H1.5_L1.27, H1.6_L1.27, H1.7_L1.27, H1.8_L1.27, H1.9_L1.27,

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H1.19_L1.27, H1.22_L1.27, H1.24_L1.27, H2_L1.27, H2.1_L1.27, H2.2_L1.27, H2.3_L1.27, H2.4_L1.27, H2.5_L1.27, H2.6_L1.27, H2.7_L1.27, H2.8_L1.27, H2.9_L1.27, H2.11_L1.27, H2.12_L1.27, H2.71_L1.27, H2.75_L1.27, H2.90_L1.27, H2.91_L1.27, H2.118_L1.27, H2.119_L1.27, H1_L1.60, H1.1_L1.60, H1.2_L1.60, H1.3_L1.60, H1.4_L1.60, H1.5_L1.60, H1.6_L1.60, H1.7_L1.60, H1.8_L1.60, H1.9_L1.60, H1.19_L1.60, H1.22_L1.60, H1.24_L1.60, H2_L1.60, H2.1_L1.60, H2.2_L1.60, H2.3_L1.60, H2.4_L1.60, H2.5_L1.60, H2.6_L1.60, H2.7_L1.60, H2.8_L1.60, H2.9_L1.60, H2.11_L1.60, H2.12_L1.60, H2.71_L1.60, H2.75_L1.60, H2.90_L1.60, H2.91_L1.60, H2.118_L1.60, H2.119_L1.60, H1_L1.107, H1.1_L1.107, H1.2_L1.107, H1.3_L1.107, H1.4_L1.107, H1.5_L1.107, H1.6_L1.107, H1.7_L1.107, H1.8_L1.107, H1.9_L1.107, H1.19_L1.107, H1.22_L1.107, H1.24_L1.107, H2_L1.107, H2.1_L1.107, H2.2_L1.107, H2.3_L1.107, H2.4_L1.107, H2.5_L1.107, H2.6_L1.107, H2.7_L1.107, H2.8_L1.107, H2.9_L1.107, H2.11_L1.107, H2.12_L1.107, H2.71_L1.107, H2.75_L1.107, H2.90_L1.107, H2.91_L1.107, H2.118_L1.107, H2.119_L1.107, H1_L1.114, H1.1_L1.114, H1.2_L1.114, H1.3_L1.114, H1.4_L1.114, H1.5_L1.114, H1.6_L1.114, H1.7_L1.114, H1.8_L1.114, H1.9_L1.114, H1.19_L1.114, H1.22_L1.114, H1.24_L1.114, H2_L1.114, H2.1_L1.114, H2.2_L1.114, H2.3_L1.114, H2.4_L1.114, H2.5_L1.114, H2.6_L1.114, H2.7_L1.114, H2.8_L1.114, H2.9_L1.114, H2.11_L1.114, H2.12_L1.114, H2.71_L1.114, H2.75_L1.114, H2.90_L1.114, H2.91_L1.114, H2.118_L1.114, H2.119_L1.114, H1_L1.187, H1.1_L1.187, H1.2_L1.187, H1.3_L1.187, H1.4_L1.187, H1.5_L1.187, H1.6_L1.187, H1.7_L1.187, H1.8_L1.187, H1.9_L1.187, H1.19_L1.187, H1.22_L1.187, H1.24_L1.187, H2_L1.187, H2.1_L1.187, H2.2_L1.187, H2.3_L1.187, H2.4_L1.187, H2.5_L1.187, H2.6_L1.187, H2.7_L1.187, H2.8_L1.187, H2.9_L1.187, H2.11_L1.187, H2.12_L1.187, H2.71_L1.187, H2.75_L1.187, H2.90_L1.187, H2.91_L1.187, H2.118_L1.187, H2.119_L1.187, H1_L1.189, H1.1_L1.189, H1.2_L1.189, H1.3_L1.189, H1.4_L1.189, H1.5_L1.189, H1.6_L1.189, H1.7_L1.189, H1.8_L1.189, H1.9_L1.189, H1.19_L1.189, H1.22_L1.189, H1.24_L1.189, H2_L1.189, H2.1_L1.189, H2.2_L1.189, H2.3_L1.189, H2.4_L1.189, H2.5_L1.189, H2.6_L1.189, H2.7_L1.189, H2.8_L1.189, H2.9_L1.189, H2.11_L1.189, H2.12_L1.189, H2.71_L1.189, H2.75_L1.189, H2.90_L1.189, H2.91_L1.189, H2.118_L1.189, H2.119_L1.189, H1_L2, H1.1_L2, H1.2_L2, H1.3_L2, H1.4_L2, H1.5_L2, H1.6_L2, H1.7_L2, H1.8_L2, H1.9_L2, H1.19_L2, H1.22_L2, H1.24_L2, H2_L2, H2.1_L2, H2.2_L2, H2.3_L2, H2.4_L2, H2.5_L2, H2.6_L2, H2.7_L2, H2.8_L2, H2.9_L2, H2.11_L2, H2.12_L2, H2.71_L2, H2.75_L2, H2.90_L2, H2.91_L2, H2.118_L2 and H2.119_L2.

3. In Some Embodiments, the VH/VL Pair is Selected from the Group Consisting of H1.9_L1.187, H1.24_L1.187, H2.91_L1.187 and H1.9_L1.187.

In some embodiments, the composition is a monoclonal antibody.

In another aspect, provided herein is a heterodimeric antibody comprising: a) a first monomer, b) a second monomer, and c) a light chain. The first monomer comprises: i) an anti-CD3 scFv comprising a first variable light domain, an scFv linker and a first variable heavy domain; and ii) a first Fc domain, wherein the scFv is covalently attached to the N-terminus of the first Fc domain using a domain linker. The second monomer comprises a VH2-CH1-hinge-CH2-CH3 monomer, wherein VH is a second variable heavy domain

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and CH2-CH3 is a second Fc domain. The light chain comprises a second variable light domain. In this embodiment, the second variable heavy domain and the second variable light domain form a CLDN6 antigen binding domain (ABD).

In some embodiments, the CLDN6 binding domain comprises a set of 6 CDRs (vhCDR1, vhCDR2, vhCDR3, vlCDR1, vlCDR2 and vlCDR3) from a variable heavy domain/variable light domain pair selected from the group consisting of: H1_L1, H1.1_L1, H1.2_L1, H1.3_L1, H1.4_L1, H1.5_L1, H1.6_L1, H1.7_L1, H1.8_L1, H1.9_L1, H1.19_L1, H1.22_L1, H1.24_L1, H2_L1, H2.1_L1, H2.2_L1, H2.3_L1, H2.4_L1, H2.5_L1, H2.6_L1, H2.7_L1, H2.8_L1, H2.9_L1, H2.11_L1, H2.12_L1, H2.71_L1, H2.75_L1, H2.90_L1, H2.91_L1, H2.118_L1, H2.119_L1, H1_L1.1, H1.1_L1.1, H1.2_L1.1, H1.3_L1.1, H1.4_L1.1, H1.5_L1.1, H1.6_L1.1, H1.7_L1.1, H1.8_L1.1, H1.9_L1.1, H1.19_L1.1, H1.22_L1.1, H1.24_L1.1, H2_L1.1, H2.1_L1.1, H2.2_L1.1, H2.3_L1.1, H2.4_L1.1, H2.5_L1.1, H2.6_L1.1, H2.7_L1.1, H2.8_L1.1, H2.9_L1.1, H2.11_L1.1, H2.12_L1.1, H2.71_L1.1, H2.75_L1.1, H2.90_L1.1, H2.91_L1.1, H2.118_L1.1, H2.119_L1.1, H1_L1.4, H1.1_L1.4, H1.2_L1.4, H1.3_L1.4, H1.4_L1.4, H1.5_L1.4, H1.6_L1.4, H1.7_L1.4, H1.8_L1.4, H1.9_L1.4, H1.19_L1.4, H1.22_L1.4, H1.24_L1.4, H2_L1.4, H2.1_L1.4, H2.2_L1.4, H2.3_L1.4, H2.4_L1.4, H2.5_L1.4, H2.6_L1.4, H2.7_L1.4, H2.8_L1.4, H2.9_L1.4, H2.11_L1.4, H2.12_L1.4, H2.71_L1.4, H2.75_L1.4, H2.90_L1.4, H2.91_L1.4, H2.118_L1.4, H2.119_L1.4, H1_L1.7, H1.1_L1.7, H1.2_L1.7, H1.3_L1.7, H1.4_L1.7, H1.5_L1.7, H1.6_L1.7, H1.7_L1.7, H1.8_L1.7, H1.9_L1.7, H1.19_L1.7, H1.22_L1.7, H1.24_L1.7, H2_L1.7, H2.1_L1.7, H2.2_L1.7, H2.3_L1.7, H2.4_L1.7, H2.5_L1.7, H2.6_L1.7, H2.7_L1.7, H2.8_L1.7, H2.9_L1.7, H2.11_L1.7, H2.12_L1.7, H2.71_L1.7, H2.75_L1.7, H2.90_L1.7, H2.91_L1.7, H2.118_L1.7, H2.119_L1.7, H1_L1.16, H1.1_L1.16, H1.2_L1.16, H1.3_L1.16, H1.4_L1.16, H1.5_L1.16, H1.6_L1.16, H1.7_L1.16, H1.8_L1.16, H1.9_L1.16, H1.19_L1.16, H1.22_L1.16, H1.24_L1.16, H2_L1.16, H2.1_L1.16, H2.2_L1.16, H2.3_L1.16, H2.4_L1.16, H2.5_L1.16, H2.6_L1.16, H2.7_L1.16, H2.8_L1.16, H2.9_L1.16, H2.11_L1.16, H2.12_L1.16, H2.71_L1.16, H2.75_L1.16, H2.90_L1.16, H2.91_L1.16, H2.118_L1.16, H2.119_L1.16, H1_L1.18, H1.1_L1.18, H1.2_L1.18, H1.3_L1.18, H1.4_L1.18, H1.5_L1.18, H1.6_L1.18, H1.7_L1.18, H1.8_L1.18, H1.9_L1.18, H1.19_L1.18, H1.22_L1.18, H1.24_L1.18, H2_L1.18, H2.1_L1.18, H2.2_L1.18, H2.3_L1.18, H2.4_L1.18, H2.5_L1.18, H2.6_L1.18, H2.7_L1.18, H2.8_L1.18, H2.9_L1.18, H2.11_L1.18, H2.12_L1.18, H2.71_L1.18, H2.75_L1.18, H2.90_L1.18, H2.91_L1.18, H2.118_L1.18, H2.119_L1.18, H1_L1.19, H1.1_L1.19, H1.2_L1.19, H1.3_L1.19, H1.4_L1.19, H1.5_L1.19, H1.6_L1.19, H1.7_L1.19, H1.8_L1.19, H1.9_L1.19, H1.19_L1.19, H1.22_L1.19, H1.24_L1.19, H2_L1.19, H2.1_L1.19, H2.2_L1.19, H2.3_L1.19, H2.4_L1.19, H2.5_L1.19, H2.6_L1.19, H2.7_L1.19, H2.8_L1.19, H2.9_L1.19, H2.11_L1.19, H2.12_L1.19, H2.71_L1.19, H2.75_L1.19, H2.90_L1.19, H2.91_L1.19, H2.118_L1.19, H2.119_L1.19, H1_L1.21, H1.1_L1.21, H1.2_L1.21, H1.3_L1.21, H1.4_L1.21, H1.5_L1.21, H1.6_L1.21, H1.7_L1.21, H1.8_L1.21, H1.9_L1.21, H1.19_L1.21, H1.22_L1.21, H1.24_L1.21, H2_L1.21, H2.1_L1.21, H2.2_L1.21, H2.3_L1.21, H2.4_L1.21, H2.5_L1.21, H2.6_L1.21, H2.7_L1.21, H2.8_L1.21, H2.9_L1.21, H2.11_L1.21, H2.12_L1.21, H2.71_L1.21, H2.75_L1.21, H2.90_L1.21, H2.91_L1.21, H2.118_L1.2, H2.119_L1.21,

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H1_L1.22, H1.1_L1.22, H1.2_L1.22, H1.3_L1.22, H1.4_L1.22, H1.5_L1.22, H1.6_L1.22, H1.7_L1.22, H1.8_L1.22, H1.9_L1.22, H1.19_L1.22, H1.22_L1.22, H1.24_L1.22, H2_L1.22, H2.1_L1.22, H2.2_L1.22, H2.3_L1.22, H2.4_L1.22, H2.5_L1.22, H2.6_L1.22, H2.7_L1.22, H2.8_L1.22, H2.9_L1.22, H2.11_L1.22, H2.12_L1.22, H2.71_L1.22, H2.75_L1.22, H2.90_L1.22, H2.91_L1.22, H2.118_L1.22, H2.119_L1.22, H1_L1.23, H1.1_L1.23, H1.2_L1.23, H1.3_L1.23, H1.4_L1.23, H1.5_L1.23, H1.6_L1.23, H1.7_L1.23, H1.8_L1.23, H1.9_L1.23, H1.19_L1.23, H1.22_L1.23, H1.24_L1.23, H2_L1.23, H2.1_L1.23, H2.2_L1.23, H2.3_L1.23, H2.4_L1.23, H2.5_L1.23, H2.6_L1.23, H2.7_L1.23, H2.8_L1.23, H2.9_L1.23, H2.11_L1.23, H2.12_L1.23, H2.71_L1.23, H2.75_L1.23, H2.90_L1.23, H2.91_L1.23, H2.118_L1.23, H2.119_L1.23, H1_L1.27, H1.1_L1.27, H1.2_L1.27, H1.3_L1.27, H1.4_L1.27, H1.5_L1.27, H1.6_L1.27, H1.7_L1.27, H1.8_L1.27, H1.9_L1.27, H1.19_L1.27, H1.22_L1.27, H1.24_L1.27, H2_L1.27, H2.1_L1.27, H2.2_L1.27, H2.3_L1.27, H2.4_L1.27, H2.5_L1.27, H2.6_L1.27, H2.7_L1.27, H2.8_L1.27, H2.9_L1.27, H2.11_L1.27, H2.12_L1.27, H2.71_L1.27, H2.75_L1.27, H2.90_L1.27, H2.91_L1.27, H2.118_L1.27, H2.119_L1.27, H1_L1.60, H1.1_L1.60, H1.2_L1.60, H1.3_L1.60, H1.4_L1.60, H1.5_L1.60, H1.6_L1.60, H1.7_L1.60, H1.8_L1.60, H1.9_L1.60, H1.19_L1.60, H1.22_L1.60, H1.24_L1.60, H2_L1.60, H2.1_L1.60, H2.2_L1.60, H2.3_L1.60, H2.4_L1.60, H2.5_L1.60, H2.6_L1.60, H2.7_L1.60, H2.8_L1.60, H2.9_L1.60, H2.11_L1.60, H2.12_L1.60, H2.71_L1.60, H2.75_L1.60, H2.90_L1.60, H2.91_L1.60, H2.118_L1.60, H2.119_L1.60, H1_L1.107, H1.1_L1.107, H1.2_L1.107, H1.3_L1.107, H1.4_L1.107, H1.5_L1.107, H1.6_L1.107, H1.7_L1.107, H1.8_L1.107, H1.9_L1.107, H1.19_L1.107, H1.22_L1.107, H1.24_L1.107, H2_L1.107, H2.1_L1.107, H2.2_L1.107, H2.3_L1.107, H2.4_L1.107, H2.5_L1.107, H2.6_L1.107, H2.7_L1.107, H2.8_L1.107, H2.9_L1.107, H2.11_L1.107, H2.12_L1.107, H2.71_L1.107, H2.75_L1.107, H2.90_L1.107, H2.91_L1.107, H2.118_L1.107, H2.119_L1.107, H1_L1.114, H1.1_L1.114, H1.2_L1.114, H1.3_L1.114, H1.4_L1.114, H1.5_L1.114, H1.6_L1.114, H1.7_L1.114, H1.8_L1.114, H1.9_L1.114, H1.19_L1.114, H1.22_L1.114, H1.24_L1.114, H2_L1.114, H2.1_L1.114, H2.2_L1.114, H2.3_L1.114, H2.4_L1.114, H2.5_L1.114, H2.6_L1.114, H2.7_L1.114, H2.8_L1.114, H2.9_L1.114, H2.11_L1.114, H2.12_L1.114, H2.71_L1.114, H2.75_L1.114, H2.90_L1.114, H2.91_L1.114, H2.118_L1.114, H2.119_L1.114, H1_L1.187, H1.1_L1.187, H1.2_L1.187, H1.3_L1.187, H1.4_L1.187, H1.5_L1.187, H1.6_L1.187, H1.7_L1.187, H1.8_L1.187, H1.9_L1.187, H1.19_L1.187, H1.22_L1.187, H1.24_L1.187, H2_L1.187, H2.1_L1.187, H2.2_L1.187, H2.3_L1.187, H2.4_L1.187, H2.5_L1.187, H2.6_L1.187, H2.7_L1.187, H2.8_L1.187, H2.9_L1.187, H2.11_L1.187, H2.12_L1.187, H2.71_L1.187, H2.75_L1.187, H2.90_L1.187, H2.91_L1.187, H2.118_L1.187, H2.119_L1.187, H1_L1.189, H1.1_L1.189, H1.2_L1.189, H1.3_L1.189, H1.4_L1.189, H1.5_L1.189, H1.6_L1.189, H1.7_L1.189, H1.8_L1.189, H1.9_L1.189, H1.19_L1.189, H1.22_L1.189, H1.24_L1.189, H2_L1.189, H2.1_L1.189, H2.2_L1.189, H2.3_L1.189, H2.4_L1.189, H2.5_L1.189, H2.6_L1.189, H2.7_L1.189, H2.8_L1.189, H2.9_L1.189, H2.11_L1.189, H2.12_L1.189, H2.71_L1.189, H2.75_L1.189, H2.90_L1.189, H2.91_L1.189, H2.118_L1.189, H2.119_L1.189, H1_L2, H1.1_L2, H1.2_L2, H1.3_L2, H1.4_L2, H1.5_L2, H1.6_L2, H1.7_L2, H1.8_L2, H1.9_L2, H1.19_L2, H1.22_L2, H1.24_L2,

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H2_L2, H2.1_L2, H2.2_L2, H2.3_L2, H2.4_L2, H2.5_L2, H2.6_L2, H2.7_L2, H2.8_L2, H2.9_L2, H2.11_L2, H2.12_L2, H2.71_L2, H2.75_L2, H2.90_L2, H2.91_L2, H2.118_L2 and H2.119_L2.

5 In some embodiments, the CLDN6 binding domain comprises a VH/VL pair selected from the group consisting of: H1_L1, H1.1_L1, H1.2_L1, H1.3_L1, H1.4_L1, H1.5_L1, H1.6_L1, H1.7_L1, H1.8_L1, H1.9_L1, H1.19_L1, H1.22_L1, H1.24_L1, H2_L1, H2.1_L1, H2.2_L1, H2.3_L1, H2.4_L1, H2.5_L1, H2.6_L1, H2.7_L1, H2.8_L1, H2.9_L1, H2.11_L1, H2.12_L1, H2.71_L1, H2.75_L1, H2.90_L1, H2.91_L1, H2.118_L1, H2.119_L1, H1_L1.1, H1.1_L1.1, H1.2_L1.1, H1.3_L1.1, H1.4_L1.1, H1.5_L1.1, H1.6_L1.1, H1.7_L1.1, H1.8_L1.1, H1.9_L1.1, H1.19_L1.1, H1.22_L1.1, H1.24_L1.1, H2_L1.1, H2.1_L1.1, H2.2_L1.1, H2.3_L1.1, H2.4_L1.1, H2.5_L1.1, H2.6_L1.1, H2.7_L1.1, H2.8_L1.1, H2.9_L1.1, H2.11_L1.1, H2.12_L1.1, H2.71_L1.1, H2.75_L1.1, H2.90_L1.1, H2.91_L1.1, H2.118_L1.1, H2.119_L1.1, H1_L1.4, H1.1_L1.4, H1.2_L1.4, H1.3_L1.4, H1.4_L1.4, H1.5_L1.4, H1.6_L1.4, H1.7_L1.4, H1.8_L1.4, H1.9_L1.4, H1.19_L1.4, H1.22_L1.4, H1.24_L1.4, H2_L1.4, H2.1_L1.4, H2.2_L1.4, H2.3_L1.4, H2.4_L1.4, H2.5_L1.4, H2.6_L1.4, H2.7_L1.4, H2.8_L1.4, H2.9_L1.4, H2.11_L1.4, H2.12_L1.4, H2.71_L1.4, H2.75_L1.4, H2.90_L1.4, H2.91_L1.4, H2.118_L1.4, H2.119_L1.4, H1_L1.7, H1.1_L1.7, H1.2_L1.7, H1.3_L1.7, H1.4_L1.7, H1.5_L1.7, H1.6_L1.7, H1.7_L1.7, H1.8_L1.7, H1.9_L1.7, H1.19_L1.7, H1.22_L1.7, H1.24_L1.7, H2_L1.7, H2.1_L1.7, H2.2_L1.7, H2.3_L1.7, H2.4_L1.7, H2.5_L1.7, H2.6_L1.7, H2.7_L1.7, H2.8_L1.7, H2.9_L1.7, H2.11_L1.7, H2.12_L1.7, H2.71_L1.7, H2.75_L1.7, H2.90_L1.7, H2.91_L1.7, H2.118_L1.7, H2.119_L1.7, H1_L1.16, H1.1_L1.16, H1.2_L1.16, H1.3_L1.16, H1.4_L1.16, H1.5_L1.16, H1.6_L1.16, H1.7_L1.16, H1.8_L1.16, H1.9_L1.16, H1.19_L1.16, H1.22_L1.16, H1.24_L1.16, H2_L1.16, H2.1_L1.16, H2.2_L1.16, H2.3_L1.16, H2.4_L1.16, H2.5_L1.16, H2.6_L1.16, H2.7_L1.16, H2.8_L1.16, H2.9_L1.16, H2.11_L1.16, H2.12_L1.16, H2.71_L1.16, H2.75_L1.16, H2.90_L1.16, H2.91_L1.16, H2.118_L1.16, H2.119_L1.16, H1_L1.18, H1.1_L1.18, H1.2_L1.18, H1.3_L1.18, H1.4_L1.18, H1.5_L1.18, H1.6_L1.18, H1.7_L1.18, H1.8_L1.18, H1.9_L1.18, H1.19_L1.18, H1.22_L1.18, H1.24_L1.18, H2_L1.18, H2.1_L1.18, H2.2_L1.18, H2.3_L1.18, H2.4_L1.18, H2.5_L1.18, H2.6_L1.18, H2.7_L1.18, H2.8_L1.18, H2.9_L1.18, H2.11_L1.18, H2.12_L1.18, H2.71_L1.18, H2.75_L1.18, H2.90_L1.18, H2.91_L1.18, H2.118_L1.18, H2.119_L1.18, H1_L1.19, H1.1_L1.19, H1.2_L1.19, H1.3_L1.19, H1.4_L1.19, H1.5_L1.19, H1.6_L1.19, H1.7_L1.19, H1.8_L1.19, H1.9_L1.19, H1.19_L1.19, H1.22_L1.19, H1.24_L1.19, H2_L1.19, H2.1_L1.19, H2.2_L1.19, H2.3_L1.19, H2.4_L1.19, H2.5_L1.19, H2.6_L1.19, H2.7_L1.19, H2.8_L1.19, H2.9_L1.19, H2.11_L1.19, H2.12_L1.19, H2.71_L1.19, H2.75_L1.19, H2.90_L1.19, H2.91_L1.19, H2.118_L1.19, H2.119_L1.19, H1_L1.21, H1.1_L1.21, H1.2_L1.21, H1.3_L1.21, H1.4_L1.21, H1.5_L1.21, H1.6_L1.21, H1.7_L1.21, H1.8_L1.21, H1.9_L1.21, H1.19_L1.21, H1.22_L1.21, H1.24_L1.21, H2_L1.21, H2.1_L1.21, H2.2_L1.21, H2.3_L1.21, H2.4_L1.21, H2.5_L1.21, H2.6_L1.21, H2.7_L1.21, H2.8_L1.21, H2.9_L1.21, H2.11_L1.21, H2.12_L1.21, H2.71_L1.21, H2.75_L1.21, H2.90_L1.21, H2.91_L1.21, H2.118_L1.21, H2.119_L1.21, H1_L1.22, H1.1_L1.22, H1.2_L1.22, H1.3_L1.22, H1.4_L1.22, H1.5_L1.22, H1.6_L1.22, H1.7_L1.22, H1.8_L1.22, H1.9_L1.22, H1.19_L1.22, H1.22_L1.22,

H1.24_L1.22, H2_L1.22, H2.1_L1.22, H2.2_L1.22, H2.3_L1.22, H2.4_L1.22, H2.5_L1.22, H2.6_L1.22, H2.7_L1.22, H2.8_L1.22, H2.9_L1.22, H2.11_L1.22, H2.12_L1.22, H2.71_L1.22, H2.75_L1.22, H2.90_L1.22, H2.91_L1.22, H2.118_L1.22, H2.119_L1.22, H1_L1.23, H1.1_L1.23, H1.2_L1.23, H1.3_L1.23, H1.4_L1.23, H1.5_L1.23, H1.6_L1.23, H1.7_L1.23, H1.8_L1.23, H1.9_L1.23, H1.19_L1.23, H1.22_L1.23, H1.24_L1.23, H2_L1.23, H2.1_L1.23, H2.2_L1.23, H2.3_L1.23, H2.4_L1.23, H2.5_L1.23, H2.6_L1.23, H2.7_L1.23, H2.8_L1.23, H2.9_L1.23, H2.11_L1.23, H2.12_L1.23, H2.71_L1.23, H2.75_L1.23, H2.90_L1.23, H2.91_L1.23, H2.118_L1.23, H2.119_L1.23, H1_L1.27, H1.1_L1.27, H1.2_L1.27, H1.3_L1.27, H1.4_L1.27, H1.5_L1.27, H1.6_L1.27, H1.7_L1.27, H1.8_L1.27, H1.9_L1.27, H1.19_L1.27, H1.22_L1.27, H1.24_L1.27, H2_L1.27, H2.1_L1.27, H2.2_L1.27, H2.3_L1.27, H2.4_L1.27, H2.5_L1.27, H2.6_L1.27, H2.7_L1.27, H2.8_L1.27, H2.9_L1.27, H2.11_L1.27, H2.12_L1.27, H2.71_L1.27, H2.75_L1.27, H2.90_L1.27, H2.91_L1.27, H2.118_L1.27, H2.119_L1.27, H1_L1.60, H1.1_L1.60, H1.2_L1.60, H1.3_L1.60, H1.4_L1.60, H1.5_L1.60, H1.6_L1.60, H1.7_L1.60, H1.8_L1.60, H1.9_L1.60, H1.19_L1.60, H1.22_L1.60, H1.24_L1.60, H2_L1.60, H2.1_L1.60, H2.2_L1.60, H2.3_L1.60, H2.4_L1.60, H2.5_L1.60, H2.6_L1.60, H2.7_L1.60, H2.8_L1.60, H2.9_L1.60, H2.11_L1.60, H2.12_L1.60, H2.71_L1.60, H2.75_L1.60, H2.90_L1.60, H2.91_L1.60, H2.118_L1.60, H2.119_L1.60, H1_L1.107, H1.1_L1.107, H1.2_L1.107, H1.3_L1.107, H1.4_L1.107, H1.5_L1.107, H1.6_L1.107, H1.7_L1.107, H1.8_L1.107, H1.9_L1.107, H1.19_L1.107, H1.22_L1.107, H1.24_L1.107, H2_L1.107, H2.1_L1.107, H2.2_L1.107, H2.3_L1.107, H2.4_L1.107, H2.5_L1.107, H2.6_L1.107, H2.7_L1.107, H2.8_L1.107, H2.9_L1.107, H2.11_L1.107, H2.12_L1.107, H2.71_L1.107, H2.75_L1.107, H2.90_L1.107, H2.91_L1.107, H2.118_L1.107, H2.119_L1.107, H1_L1.114, H1.1_L1.114, H1.2_L1.114, H1.3_L1.114, H1.4_L1.114, H1.5_L1.114, H1.6_L1.114, H1.7_L1.114, H1.8_L1.114, H1.9_L1.114, H1.19_L1.114, H1.22_L1.114, H1.24_L1.114, H2_L1.114, H2.1_L1.114, H2.2_L1.114, H2.3_L1.114, H2.4_L1.114, H2.5_L1.114, H2.6_L1.114, H2.7_L1.114, H2.8_L1.114, H2.9_L1.114, H2.11_L1.114, H2.12_L1.114, H2.71_L1.114, H2.75_L1.114, H2.90_L1.114, H2.91_L1.114, H2.118_L1.114, H2.119_L1.114, H1_L1.187, H1.1_L1.187, H1.2_L1.187, H1.3_L1.187, H1.4_L1.187, H1.5_L1.187, H1.6_L1.187, H1.7_L1.187, H1.8_L1.187, H1.9_L1.187, H1.19_L1.187, H1.22_L1.187, H1.24_L1.187, H2_L1.187, H2.1_L1.187, H2.2_L1.187, H2.3_L1.187, H2.4_L1.187, H2.5_L1.187, H2.6_L1.187, H2.7_L1.187, H2.8_L1.187, H2.9_L1.187, H2.11_L1.187, H2.12_L1.187, H2.71_L1.187, H2.75_L1.187, H2.90_L1.187, H2.91_L1.187, H2.118_L1.187, H2.119_L1.187, H1_L1.189, H1.1_L1.189, H1.2_L1.189, H1.3_L1.189, H1.4_L1.189, H1.5_L1.189, H1.6_L1.189, H1.7_L1.189, H1.8_L1.189, H1.9_L1.189, H1.19_L1.189, H1.22_L1.189, H1.24_L1.189, H2_L1.189, H2.1_L1.189, H2.2_L1.189, H2.3_L1.189, H2.4_L1.189, H2.5_L1.189, H2.6_L1.189, H2.7_L1.189, H2.8_L1.189, H2.9_L1.189, H2.11_L1.189, H2.12_L1.189, H2.71_L1.189, H2.75_L1.189, H2.90_L1.189, H2.91_L1.189, H2.118_L1.189, H2.119_L1.189, H1_L2, H1.1_L2, H1.2_L2, H1.3_L2, H1.4_L2, H1.5_L2, H1.6_L2, H1.7_L2, H1.8_L2, H1.9_L2, H1.19_L2, H1.22_L2, H1.24_L2, H2_L2, H2.1_L2, H2.2_L2, H2.3_L2, H2.4_L2, H2.5_L2,

H2.6_L2, H2.7_L2, H2.8_L2, H2.9_L2, H2.11_L2, H2.12_L2, H2.71_L2, H2.75_L2, H2.90_L2, H2.91_L2, H2.118_L2 and H2.119_L2.

In certain embodiments, the CLDN6 binding domain comprises a VH/VL pair selected from the group consisting of: H1.9_L1.187, H1.24_L1.187, H2.91_L1.187 and H1.9_L1.187.

In some embodiments of the heterodimeric antibody, the anti-CD3 scFv comprises a VH/VL pair selected from the group consisting of H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31.

In some embodiments of the heterodimeric antibody, the scFv linker is a charged scFv linker.

In some embodiments, the first and second Fc domains are variant Fc domains. In some embodiments, the first and second Fc domains comprise a set of heterodimerization variants selected from the group consisting of those depicted in FIGS. 1A-1E. In some embodiments, the set of heterodimerization variants selected is from the group consisting of S364K/E357Q/L368D/K370S; S364K/L368D/K370S; S364K/L368E/K370S; D401K:T411E/K360E/Q362E; and T366W:T366S/L368A/Y407V, wherein numbering is according to EU numbering.

In some embodiments, the first and second monomers further comprise one or more ablation variants. In some embodiments, the one or more ablation variants are E233P/L234V/L235A/G236del/S267K, wherein numbering is according to EU numbering.

In exemplary embodiments, one of the first or second monomer comprises one or more pI variants. In some embodiments, the one or more pI variants are N208D/Q295E/N384D/Q418E/N421D, wherein numbering is according to EU numbering.

In exemplary embodiments of the heterodimeric antibody, the first monomer comprises amino acid variants S364K/E357Q/E233P/L234V/L235A/G236del/S267K, the second monomer comprises amino acid variants L368D/K370S/N208D/Q295E/N384D/Q418E/N421D/E233P/L234V/L235A/G236del/S267K, and wherein numbering is according to EU numbering.

In some embodiments, the first and second monomers each further comprise amino acid variants 428/434S, wherein numbering is according to EU numbering.

In another aspect, provided herein is a heterodimeric antibody comprising: a) a first monomer comprising, from N-terminal to C-terminal, a scFv-linker-CH2-CH3, wherein scFv is an anti-CD3 scFv and CH2-CH3 is a first Fc domain; b) a second monomer comprising, from N-terminal to C-terminal, a VH-CH1-hinge-CH2-CH3, wherein CH2-CH3 is a second Fc domain; and c) a light chain comprising VL-CL. The first variant Fc domain comprises amino acid variants S364K/E357Q, the second variant Fc domain comprises amino acid variants L368D/K370S, the first and second variant Fc domains each comprises amino acid variants E233P/L234V/L235A/G236del/S267K, and the CH1-hinge-CH2-CH3 of the second monomer comprises amino acid variants N208D/Q295E/N384D/Q418E/N421D (EU numbering). Further, the VH and VL form an CLDN6 binding domain comprising the variable heavy domain and the variable light domain, respectively, of an CLDN6 binding domain selected from H1.9_L1.187, H1.24_L1.187, H2.91_L1.187 and H1.9_L1.187; and the anti-CD3 scFv comprises the variable heavy domain and the variable light domain of a CD3 binding domain selected from H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47,

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H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31.

In some embodiments, the first and second variant Fc domains each further comprise amino acid variants 428/434S, wherein numbering is according to EU numbering.

In another aspect, provided herein is a heterodimeric antibody comprising: a) a first monomer, b) a second monomer, and c) a common light chain. The first monomer comprises, from N-terminal to C-terminal, a VH1-CH1-linker 1-scFv-linker 2-CH2-CH3, wherein VH1 is a first variable heavy domain, scFv is an anti-CD3 scFv, linker 1 and linker 2 are a first domain linker and second domain linker, respectively, and CH2-CH3 is a first Fc domain. The second monomer comprises, from N-terminal to C-terminal, a VH2-CH1-hinge-CH2-CH3, wherein VH2 is a second variable heavy domain and CH2-CH3 is a second Fc domain; and c) a common light chain comprising a variable light domain. The first variable heavy domain and the variable light domain form a first CLDN6 ABD, and the second variable heavy domain and the variable light domain form a second CLDN6 ABD.

In some embodiments, the first and second CLDN6 binding domains each comprise a set of 6 CDRs (vhCDR1, vhCDR2, vhCDR3, vlCDR1, vlCDR2 and vlCDR3) from a VH/VL pair selected from the group consisting of: H1_L1, H1.1_L1, H1.2_L1, H1.3_L1, H1.4_L1, H1.5_L1, H1.6_L1, H1.7_L1, H1.8_L1, H1.9_L1, H1.19_L1, H1.22_L1, H1.24_L1, H2_L1, H2.1_L1, H2.2_L1, H2.3_L1, H2.4_L1, H2.5_L1, H2.6_L1, H2.7_L1, H2.8_L1, H2.9_L1, H2.11_L1, H2.12_L1, H2.71_L1, H2.75_L1, H2.90_L1, H2.91_L1, H2.118_L1, H2.119_L1, H1_L1.1, H1.1_L1.1, H1.2_L1.1, H1.3_L1.1, H1.4_L1.1, H1.5_L1.1, H1.6_L1.1, H1.7_L1.1, H1.8_L1.1, H1.9_L1.1, H1.19_L1.1, H1.22_L1.1, H1.24_L1.1, H2_L1.1, H2.1_L1.1, H2.2_L1.1, H2.3_L1.1, H2.4_L1.1, H2.5_L1.1, H2.6_L1.1, H2.7_L1.1, H2.8_L1.1, H2.9_L1.1, H2.11_L1.1, H2.12_L1.1, H2.71_L1.1, H2.75_L1.1, H2.90_L1.1, H2.91_L1.1, H2.118_L1.1, H2.119_L1.1, H1_L1.4, H1.1_L1.4, H1.2_L1.4, H1.3_L1.4, H1.4_L1.4, H1.5_L1.4, H1.6_L1.4, H1.7_L1.4, H1.8_L1.4, H1.9_L1.4, H1.19_L1.4, H1.22_L1.4, H1.24_L1.4, H2_L1.4, H2.1_L1.4, H2.2_L1.4, H2.3_L1.4, H2.4_L1.4, H2.5_L1.4, H2.6_L1.4, H2.7_L1.4, H2.8_L1.4, H2.9_L1.4, H2.11_L1.4, H2.12_L1.4, H2.71_L1.4, H2.75_L1.4, H2.90_L1.4, H2.91_L1.4, H2.118_L1.4, H2.119_L1.4, H1_L1.7, H1.1_L1.7, H1.2_L1.7, H1.3_L1.7, H1.4_L1.7, H1.5_L1.7, H1.6_L1.7, H1.7_L1.7, H1.8_L1.7, H1.9_L1.7, H1.19_L1.7, H1.22_L1.7, H1.24_L1.7, H2_L1.7, H2.1_L1.7, H2.2_L1.7, H2.3_L1.7, H2.4_L1.7, H2.5_L1.7, H2.6_L1.7, H2.7_L1.7, H2.8_L1.7, H2.9_L1.7, H2.11_L1.7, H2.12_L1.7, H2.71_L1.7, H2.75_L1.7, H2.90_L1.7, H2.91_L1.7, H2.118_L1.7, H2.119_L1.7, H1_L1.16, H1.1_L1.16, H1.2_L1.16, H1.3_L1.16, H1.4_L1.16, H1.5_L1.16, H1.6_L1.16, H1.7_L1.16, H1.8_L1.16, H1.9_L1.16, H1.19_L1.16, H1.22_L1.16, H1.24_L1.16, H2_L1.16, H2.1_L1.16, H2.2_L1.16, H2.3_L1.16, H2.4_L1.16, H2.5_L1.16, H2.6_L1.16, H2.7_L1.16, H2.8_L1.16, H2.9_L1.16, H2.11_L1.16, H2.12_L1.16, H2.71_L1.16, H2.75_L1.16, H2.90_L1.16, H2.91_L1.16, H2.118_L1.16, H2.119_L1.16, H1_L1.18, H1.1_L1.18, H1.2_L1.18, H1.3_L1.18, H1.4_L1.18, H1.5_L1.18, H1.6_L1.18, H1.7_L1.18, H1.8_L1.18, H1.9_L1.18, H1.19_L1.18, H1.22_L1.18, H1.24_L1.18, H2_L1.18, H2.1_L1.18, H2.2_L1.18, H2.3_L1.18, H2.4_L1.18, H2.5_L1.18, H2.6_L1.18, H2.7_L1.18, H2.8_L1.18, H2.9_L1.18, H2.11_L1.18, H2.12_L1.18, H2.71_L1.18, H2.75_L1.18,

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H2.90_L1.18, H2.91_L1.18, H2.118_L1.18, H2.119_L1.18, H1_L1.19, H1.1_L1.19, H1.2_L1.19, H1.3_L1.19, H1.4_L1.19, H1.5_L1.19, H1.6_L1.19, H1.7_L1.19, H1.8_L1.19, H1.9_L1.19, H1.19_L1.19, H1.22_L1.19, H1.24_L1.19, H2_L1.19, H2.1_L1.19, H2.2_L1.19, H2.3_L1.19, H2.4_L1.19, H2.5_L1.19, H2.6_L1.19, H2.7_L1.19, H2.8_L1.19, H2.9_L1.19, H2.11_L1.19, H2.12_L1.19, H2.71_L1.19, H2.75_L1.19, H2.90_L1.19, H2.91_L1.19, H2.118_L1.19, H2.119_L1.19, H1_L1.21, H1.1_L1.21, H1.2_L1.21, H1.3_L1.21, H1.4_L1.21, H1.5_L1.21, H1.6_L1.21, H1.7_L1.21, H1.8_L1.21, H1.9_L1.21, H1.19_L1.21, H1.22_L1.21, H1.24_L1.21, H2_L1.21, H2.1_L1.21, H2.2_L1.21, H2.3_L1.21, H2.4_L1.21, H2.5_L1.21, H2.6_L1.21, H2.7_L1.21, H2.8_L1.21, H2.9_L1.21, H2.11_L1.21, H2.12_L1.21, H2.71_L1.21, H2.75_L1.21, H2.90_L1.21, H2.91_L1.21, H2.118_L1.2, H2.119_L1.21, H1_L1.22, H1.1_L1.22, H1.2_L1.22, H1.3_L1.22, H1.4_L1.22, H1.5_L1.22, H1.6_L1.22, H1.7_L1.22, H1.8_L1.22, H1.9_L1.22, H1.19_L1.22, H1.22_L1.22, H1.24_L1.22, H2_L1.22, H2.1_L1.22, H2.2_L1.22, H2.3_L1.22, H2.4_L1.22, H2.5_L1.22, H2.6_L1.22, H2.7_L1.22, H2.8_L1.22, H2.9_L1.22, H2.11_L1.22, H2.12_L1.22, H2.71_L1.22, H2.75_L1.22, H2.90_L1.22, H2.91_L1.22, H2.118_L1.22, H2.119_L1.22, H1_L1.23, H1.1_L1.23, H1.2_L1.23, H1.3_L1.23, H1.4_L1.23, H1.5_L1.23, H1.6_L1.23, H1.7_L1.23, H1.8_L1.23, H1.9_L1.23, H1.19_L1.23, H1.22_L1.23, H1.24_L1.23, H2_L1.23, H2.1_L1.23, H2.2_L1.23, H2.3_L1.23, H2.4_L1.23, H2.5_L1.23, H2.6_L1.23, H2.7_L1.23, H2.8_L1.23, H2.9_L1.23, H2.11_L1.23, H2.12_L1.23, H2.71_L1.23, H2.75_L1.23, H2.90_L1.23, H2.91_L1.23, H2.118_L1.23, H2.119_L1.23, H1_L1.27, H1.1_L1.27, H1.2_L1.27, H1.3_L1.27, H1.4_L1.27, H1.5_L1.27, H1.6_L1.27, H1.7_L1.27, H1.8_L1.27, H1.9_L1.27, H1.19_L1.27, H1.22_L1.27, H1.24_L1.27, H2_L1.27, H2.1_L1.27, H2.2_L1.27, H2.3_L1.27, H2.4_L1.27, H2.5_L1.27, H2.6_L1.27, H2.7_L1.27, H2.8_L1.27, H2.9_L1.27, H2.11_L1.27, H2.12_L1.27, H2.71_L1.27, H2.75_L1.27, H2.90_L1.27, H2.91_L1.27, H2.118_L1.27, H2.119_L1.27, H1_L1.60, H1.1_L1.60, H1.2_L1.60, H1.3_L1.60, H1.4_L1.60, H1.5_L1.60, H1.6_L1.60, H1.7_L1.60, H1.8_L1.60, H1.9_L1.60, H1.19_L1.60, H1.22_L1.60, H1.24_L1.60, H2_L1.60, H2.1_L1.60, H2.2_L1.60, H2.3_L1.60, H2.4_L1.60, H2.5_L1.60, H2.6_L1.60, H2.7_L1.60, H2.8_L1.60, H2.9_L1.60, H2.11_L1.60, H2.12_L1.60, H2.71_L1.60, H2.75_L1.60, H2.90_L1.60, H2.91_L1.60, H2.118_L1.60, H2.119_L1.60, H1_L1.107, H1.1_L1.107, H1.2_L1.107, H1.3_L1.107, H1.4_L1.107, H1.5_L1.107, H1.6_L1.107, H1.7_L1.107, H1.8_L1.107, H1.9_L1.107, H1.19_L1.107, H1.22_L1.107, H1.24_L1.107, H2_L1.107, H2.1_L1.107, H2.2_L1.107, H2.3_L1.107, H2.4_L1.107, H2.5_L1.107, H2.6_L1.107, H2.7_L1.107, H2.8_L1.107, H2.9_L1.107, H2.11_L1.107, H2.12_L1.107, H2.71_L1.107, H2.75_L1.107, H2.90_L1.107, H2.91_L1.107, H2.118_L1.107, H2.119_L1.107, H1_L1.114, H1.1_L1.114, H1.2_L1.114, H1.3_L1.114, H1.4_L1.114, H1.5_L1.114, H1.6_L1.114, H1.7_L1.114, H1.8_L1.114, H1.9_L1.114, H1.19_L1.114, H1.22_L1.114, H1.24_L1.114, H2_L1.114, H2.1_L1.114, H2.2_L1.114, H2.3_L1.114, H2.4_L1.114, H2.5_L1.114, H2.6_L1.114, H2.7_L1.114, H2.8_L1.114, H2.9_L1.114, H2.11_L1.114, H2.12_L1.114, H2.71_L1.114, H2.75_L1.114, H2.90_L1.114, H2.91_L1.114, H2.118_L1.114, H2.119_L1.114, H1_L1.187, H1.1_L1.187, H1.2_L1.187, H1.3_L1.187, H1.4_L1.187, H1.5_L1.187, H1.6_L1.187, H1.7_L1.187, H1.8_L1.187, H1.9_L1.187, H1.19_L1.187,

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H1.22_L1.187, H1.24_L1.187, H2_L1.187, H2.1_L1.187, H2.2_L1.187, H2.3_L1.187, H2.4_L1.187, H2.5_L1.187, H2.6_L1.187, H2.7_L1.187, H2.8_L1.187, H2.9_L1.187, H2.11_L1.187, H2.12_L1.187, H2.71_L1.187, H2.75_L1.187, H2.90_L1.187, H2.91_L1.187, H2.118_L1.187, H2.119_L1.187, H1_L1.189, H1.1_L1.189, H1.2_L1.189, H1.3_L1.189, H1.4_L1.189, H1.5_L1.189, H1.6_L1.189, H1.7_L1.189, H1.8_L1.189, H1.9_L1.189, H1.19_L1.189, H1.22_L1.189, H1.24_L1.189, H2_L1.189, H2.1_L1.189, H2.2_L1.189, H2.3_L1.189, H2.4_L1.189, H2.5_L1.189, H2.6_L1.189, H2.7_L1.189, H2.8_L1.189, H2.9_L1.189, H2.11_L1.189, H2.12_L1.189, H2.71_L1.189, H2.75_L1.189, H2.90_L1.189, H2.91_L1.189, H2.118_L1.189, H2.119_L1.189, H1_L2, H1.1_L2, H1.2_L2, H1.3_L2, H1.4_L2, H1.5_L2, H1.6_L2, H1.7_L2, H1.8_L2, H1.9_L2, H1.19_L2, H1.22_L2, H1.24_L2, H2_L2, H2.1_L2, H2.2_L2, H2.3_L2, H2.4_L2, H2.5_L2, H2.6_L2, H2.7_L2, H2.8_L2, H2.9_L2, H2.11_L2, H2.12_L2, H2.71_L2, H2.75_L2, H2.90_L2, H2.91_L2, H2.118_L2 and H2.119_L2.

In exemplary embodiments, each of the first and second CLDN6 binding domains comprise a VH/VL pair selected from the group consisting of: H1_L1, H1.1_L1, H1.2_L1, H1.3_L1, H1.4_L1, H1.5_L1, H1.6_L1, H1.7_L1, H1.8_L1, H1.9_L1, H1.19_L1, H1.22_L1, H1.24_L1, H2_L1, H2.1_L1, H2.2_L1, H2.3_L1, H2.4_L1, H2.5_L1, H2.6_L1, H2.7_L1, H2.8_L1, H2.9_L1, H2.11_L1, H2.12_L1, H2.71_L1, H2.75_L1, H2.90_L1, H2.91_L1, H2.118_L1, H2.119_L1, H1_L1.1, H1.1_L1.1, H1.2_L1.1, H1.3_L1.1, H1.4_L1.1, H1.5_L1.1, H1.6_L1.1, H1.7_L1.1, H1.8_L1.1, H1.9_L1.1, H1.19_L1.1, H1.22_L1.1, H1.24_L1.1, H2_L1.1, H2.1_L1.1, H2.2_L1.1, H2.3_L1.1, H2.4_L1.1, H2.5_L1.1, H2.6_L1.1, H2.7_L1.1, H2.8_L1.1, H2.9_L1.1, H2.11_L1.1, H2.12_L1.1, H2.71_L1.1, H2.75_L1.1, H2.90_L1.1, H2.91_L1.1, H2.118_L1.1, H2.119_L1.1, H1_L1.4, H1.1_L1.4, H1.2_L1.4, H1.3_L1.4, H1.4_L1.4, H1.5_L1.4, H1.6_L1.4, H1.7_L1.4, H1.8_L1.4, H1.9_L1.4, H1.19_L1.4, H1.22_L1.4, H1.24_L1.4, H2_L1.4, H2.1_L1.4, H2.2_L1.4, H2.3_L1.4, H2.4_L1.4, H2.5_L1.4, H2.6_L1.4, H2.7_L1.4, H2.8_L1.4, H2.9_L1.4, H2.11_L1.4, H2.12_L1.4, H2.71_L1.4, H2.75_L1.4, H2.90_L1.4, H2.91_L1.4, H2.118_L1.4, H2.119_L1.4, H1_L1.7, H1.1_L1.7, H1.2_L1.7, H1.3_L1.7, H1.4_L1.7, H1.5_L1.7, H1.6_L1.7, H1.7_L1.7, H1.8_L1.7, H1.9_L1.7, H1.19_L1.7, H1.22_L1.7, H1.24_L1.7, H2_L1.7, H2.1_L1.7, H2.2_L1.7, H2.3_L1.7, H2.4_L1.7, H2.5_L1.7, H2.6_L1.7, H2.7_L1.7, H2.8_L1.7, H2.9_L1.7, H2.11_L1.7, H2.12_L1.7, H2.71_L1.7, H2.75_L1.7, H2.90_L1.7, H2.91_L1.7, H2.118_L1.7, H2.119_L1.7, H1_L1.16, H1.1_L1.16, H1.2_L1.16, H1.3_L1.16, H1.4_L1.16, H1.5_L1.16, H1.6_L1.16, H1.7_L1.16, H1.8_L1.16, H1.9_L1.16, H1.19_L1.16, H1.22_L1.16, H1.24_L1.16, H2_L1.16, H2.1_L1.16, H2.2_L1.16, H2.3_L1.16, H2.4_L1.16, H2.5_L1.16, H2.6_L1.16, H2.7_L1.16, H2.8_L1.16, H2.9_L1.16, H2.11_L1.16, H2.12_L1.16, H2.71_L1.16, H2.75_L1.16, H2.90_L1.16, H2.91_L1.16, H2.118_L1.16, H2.119_L1.16, H1_L1.18, H1.1_L1.18, H1.2_L1.18, H1.3_L1.18, H1.4_L1.18, H1.5_L1.18, H1.6_L1.18, H1.7_L1.18, H1.8_L1.18, H1.9_L1.18, H1.19_L1.18, H1.22_L1.18, H1.24_L1.18, H2_L1.18, H2.1_L1.18, H2.2_L1.18, H2.3_L1.18, H2.4_L1.18, H2.5_L1.18, H2.6_L1.18, H2.7_L1.18, H2.8_L1.18, H2.9_L1.18, H2.11_L1.18, H2.12_L1.18, H2.71_L1.18, H2.75_L1.18, H2.90_L1.18, H2.91_L1.18, H2.118_L1.18, H2.119_L1.18, H1_L1.19, H1.1_L1.19, H1.2_L1.19, H1.3_L1.19, H1.4_L1.19, H1.5_L1.19,

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H1.6_L1.19, H1.7_L1.19, H1.8_L1.19, H1.9_L1.19, H1.19_L1.19, H1.22_L1.19, H1.24_L1.19, H2_L1.19, H2.1_L1.19, H2.2_L1.19, H2.3_L1.19, H2.4_L1.19, H2.5_L1.19, H2.6_L1.19, H2.7_L1.19, H2.8_L1.19, H2.9_L1.19, H2.11_L1.19, H2.12_L1.19, H2.71_L1.19, H2.75_L1.19, H2.90_L1.19, H2.91_L1.19, H2.118_L1.19, H2.119_L1.19, H1_L1.21, H1.1_L1.21, H1.2_L1.21, H1.3_L1.21, H1.4_L1.21, H1.5_L1.21, H1.6_L1.21, H1.7_L1.21, H1.8_L1.21, H1.9_L1.21, H1.19_L1.21, H1.22_L1.21, H1.24_L1.21, H2_L1.21, H2.1_L1.21, H2.2_L1.21, H2.3_L1.21, H2.4_L1.21, H2.5_L1.21, H2.6_L1.21, H2.7_L1.21, H2.8_L1.21, H2.9_L1.21, H2.11_L1.21, H2.12_L1.21, H2.71_L1.21, H2.75_L1.21, H2.90_L1.21, H2.91_L1.21, H2.118_L1.2, H2.119_L1.21, H1_L1.22, H1.1_L1.22, H1.2_L1.22, H1.3_L1.22, H1.4_L1.22, H1.5_L1.22, H1.6_L1.22, H1.7_L1.22, H1.8_L1.22, H1.9_L1.22, H1.19_L1.22, H1.22_L1.22, H1.24_L1.22, H2_L1.22, H2.1_L1.22, H2.2_L1.22, H2.3_L1.22, H2.4_L1.22, H2.5_L1.22, H2.6_L1.22, H2.7_L1.22, H2.8_L1.22, H2.9_L1.22, H2.11_L1.22, H2.12_L1.22, H2.71_L1.22, H2.75_L1.22, H2.90_L1.22, H2.91_L1.22, H2.118_L1.22, H2.119_L1.22, H2.118_L1.22, H2.119_L1.22, H1_L1.23, H1.1_L1.23, H1.2_L1.23, H1.3_L1.23, H1.4_L1.23, H1.5_L1.23, H1.6_L1.23, H1.7_L1.23, H1.8_L1.23, H1.9_L1.23, H1.19_L1.23, H1.22_L1.23, H1.24_L1.23, H2_L1.23, H2.1_L1.23, H2.2_L1.23, H2.3_L1.23, H2.4_L1.23, H2.5_L1.23, H2.6_L1.23, H2.7_L1.23, H2.8_L1.23, H2.9_L1.23, H2.11_L1.23, H2.12_L1.23, H2.71_L1.23, H2.75_L1.23, H2.90_L1.23, H2.91_L1.23, H2.118_L1.23, H2.119_L1.23, H1_L1.27, H1.1_L1.27, H1.2_L1.27, H1.3_L1.27, H1.4_L1.27, H1.5_L1.27, H1.6_L1.27, H1.7_L1.27, H1.8_L1.27, H1.9_L1.27, H1.19_L1.27, H1.22_L1.27, H1.24_L1.27, H2_L1.27, H2.1_L1.27, H2.2_L1.27, H2.3_L1.27, H2.4_L1.27, H2.5_L1.27, H2.6_L1.27, H2.7_L1.27, H2.8_L1.27, H2.9_L1.27, H2.11_L1.27, H2.12_L1.27, H2.71_L1.27, H2.75_L1.27, H2.90_L1.27, H2.91_L1.27, H2.118_L1.27, H2.119_L1.27, H1_L1.60, H1.1_L1.60, H1.2_L1.60, H1.3_L1.60, H1.4_L1.60, H1.5_L1.60, H1.6_L1.60, H1.7_L1.60, H1.8_L1.60, H1.9_L1.60, H1.19_L1.60, H1.22_L1.60, H1.24_L1.60, H2_L1.60, H2.1_L1.60, H2.2_L1.60, H2.3_L1.60, H2.4_L1.60, H2.5_L1.60, H2.6_L1.60, H2.7_L1.60, H2.8_L1.60, H2.9_L1.60, H2.11_L1.60, H2.12_L1.60, H2.71_L1.60, H2.75_L1.60, H2.90_L1.60, H2.91_L1.60, H2.118_L1.60, H2.119_L1.60, H1_L1.107, H1.1_L1.107, H1.2_L1.107, H1.3_L1.107, H1.4_L1.107, H1.5_L1.107, H1.6_L1.107, H1.7_L1.107, H1.8_L1.107, H1.9_L1.107, H1.19_L1.107, H1.22_L1.107, H1.24_L1.107, H2_L1.107, H2.1_L1.107, H2.2_L1.107, H2.3_L1.107, H2.4_L1.107, H2.5_L1.107, H2.6_L1.107, H2.7_L1.107, H2.8_L1.107, H2.9_L1.107, H2.11_L1.107, H2.12_L1.107, H2.71_L1.107, H2.75_L1.107, H2.90_L1.107, H2.91_L1.107, H2.118_L1.107, H2.119_L1.107, H1_L1.114, H1.1_L1.114, H1.2_L1.114, H1.3_L1.114, H1.4_L1.114, H1.5_L1.114, H1.6_L1.114, H1.7_L1.114, H1.8_L1.114, H1.9_L1.114, H1.19_L1.114, H1.22_L1.114, H1.24_L1.114, H2_L1.114, H2.1_L1.114, H2.2_L1.114, H2.3_L1.114, H2.4_L1.114, H2.5_L1.114, H2.6_L1.114, H2.7_L1.114, H2.8_L1.114, H2.9_L1.114, H2.11_L1.114, H2.12_L1.114, H2.71_L1.114, H2.75_L1.114, H2.90_L1.114, H2.91_L1.114, H2.118_L1.114, H2.119_L1.114, H1_L1.187, H1.1_L1.187, H1.2_L1.187, H1.3_L1.187, H1.4_L1.187, H1.5_L1.187, H1.6_L1.187, H1.7_L1.187, H1.8_L1.187, H1.9_L1.187, H1.19_L1.187, H1.22_L1.187, H1.24_L1.187, H2_L1.187, H2.1_L1.187, H2.2_L1.187, H2.3_L1.187, H2.4_L1.187, H2.5_L1.187, H2.6_L1.187,

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H2.7_L1.187, H2.8_L1.187, H2.9_L1.187, H2.11_L1.187, H2.12_L1.187, H2.71_L1.187, H2.75_L1.187, H2.90_L1.187, H2.91_L1.187, H2.118_L1.187, H2.119_L1.187, H1_L1.189, H1.1_L1.189, H1.2_L1.189, H1.3_L1.189, H1.4_L1.189, H1.5_L1.189, H1.6_L1.189, H1.7_L1.189, H1.8_L1.189, H1.9_L1.189, H1.19_L1.189, H1.22_L1.189, H1.24_L1.189, H2_L1.189, H2.1_L1.189, H2.2_L1.189, H2.3_L1.189, H2.4_L1.189, H2.5_L1.189, H2.6_L1.189, H2.7_L1.189, H2.8_L1.189, H2.9_L1.189, H2.11_L1.189, H2.12_L1.189, H2.71_L1.189, H2.75_L1.189, H2.90_L1.189, H2.91_L1.189, H2.118_L1.189, H2.119_L1.189, H1_L2, H1.1_L2, H1.2_L2, H1.3_L2, H1.4_L2, H1.5_L2, H1.6_L2, H1.7_L2, H1.8_L2, H1.9_L2, H1.19_L2, H1.22_L2, H1.24_L2, H2_L2, H2.1_L2, H2.2_L2, H2.3_L2, H2.4_L2, H2.5_L2, H2.6_L2, H2.7_L2, H2.8_L2, H2.9_L2, H2.11_L2, H2.12_L2, H2.71_L2, H2.75_L2, H2.90_L2, H2.91_L2, H2.118_L2 and H2.119_L2.

In some embodiments, the VH/VL pairs are selected from the group consisting of H1.9_L1.187, H1.24_L1.187, H2.91_L1.187 and H1.9_L1.187.

In some embodiments, the scFv comprises a set of 6 CDRs (vhCDR1, vhCDR2, vhCDR3, vlCDR1, vlCDR2 and vlCDR3) from VH/VL pairs selected from the group consisting of: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31. In exemplary embodiments, the scFv comprises the variable heavy domain and variable light domain of any of the following CD3 binding domains: H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31.

In some embodiments of the heterodimeric antibody, the scFv linker is a charged scFv linker. In some embodiments, the scFv linker is a charged scFv linker having the amino acid sequence (GKPGS)₄ (SEQ ID NO: 1).

In some embodiments, the first and second Fc domains are variant Fc domains. In some embodiments, the first and second Fc domains comprise a set of heterodimerization variants selected from the group consisting of those depicted in FIG. 1A-1E. In some embodiments, the set of heterodimerization variants selected is from the group consisting of S364K/E357Q/L368D/K370S; S364K/L368D/K370S; S364K/L368E/K370S; D401K/T411E/K360E/Q362E; and T366W/T366S/L368A/Y407V, wherein numbering is according to EU numbering.

In some embodiments, the first and second monomers further comprise one or more ablation variants. In some embodiments, the one or more ablation variants are E233P/L234V/L235A/G236del/S267K, wherein numbering is according to EU numbering.

In exemplary embodiments, one of the first or second monomer comprises one or more pI variants. In some embodiments, the one or more pI variants are N208D/Q295E/N384D/Q418E/N421D, wherein numbering is according to EU numbering.

In exemplary embodiments of the heterodimeric antibody, the first monomer comprises amino acid variants S364K/E357Q/E233P/L234V/L235A/G236del/S267K, the second monomer comprises amino acid variants L368D/K370S/N208D/Q295E/N384D/Q418E/N421D/E233P/L234V/L235A/G236del/S267K, and wherein numbering is according to EU numbering.

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In some embodiments, the first and second monomers each further comprise amino acid variants 428/434S, wherein numbering is according to EU numbering.

In another aspect, provided herein is heterodimeric antibody comprising: a) a first monomer, b) a second monomer, and c) a common light chain. The first monomer comprises, from N-terminal to C-terminal, a VH1-CH1-linker 1-scFv-linker 2-CH2-CH3, wherein scFv is an anti-CD3 scFv and CH2-CH3 is a first Fc domain. The second monomer comprises, from N-terminal to C-terminal, a VH1-CH1-hinge-CH2-CH3, wherein CH2-CH3 is a second Fc domain. The common light chain comprising VL-CL. The first variant Fc domain comprises amino acid variants S364K/E357Q, the second variant Fc domain comprises amino acid variants L368D/K370S, the first and second variant Fc domains each comprises amino acid variants E233P/L234V/L235A/G236del/S267K, and the CH1-hinge-CH2-CH3 of the second monomer comprises amino acid variants N208D/Q295E/N384D/Q418E/N421D (EU numbering). The VH and VL comprise the variable heavy domain and the variable light domain of a CLDN6 ABD selected from H1.9_L1.187, H1.24_L1.187, H2.91_L1.187 and H1.9_L1.187; and the anti-CD3 scFv comprises the variable heavy domain and the variable light domain of a CD3 binding domain selected from H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.32, L1.47_H1.33, and L1.47_H1.31.

Also provided herein are nucleic acid compositions comprising nucleic acids encoding the antibodies described herein, expression vector compositions that include such nucleic acids, host cells for making the antibodies that comprise the expression vector compositions, and methods of making the antibodies.

BRIEF DESCRIPTION OF THE DRAWINGS

FIGS. 1A-1E depict useful pairs of Fc heterodimerization variant sets (including skew and pI variants) that lead to Fc heterodimerization. There are variants for which there are no corresponding "monomer 2" variants; these are pI variants which can be used alone on either monomer.

FIG. 2 depicts a list of isosteric variant antibody constant regions and their respective substitutions. pI₍₋₎ indicates lower pI variants, while pI₍₊₎ indicates higher pI variants. These can be optionally and independently combined with other heterodimerization variants of the inventions (and other variant types as well, as outlined herein.)

FIG. 3 depicts useful ablation variants that ablate FcγR binding (sometimes referred to as "knock outs" or "KO" variants). Generally, ablation variants are found on both monomers, although in some cases they may be on only one monomer.

FIG. 4 depicts particularly useful embodiments of "non-Fv" components of the invention.

FIG. 5 depicts a number of charged scFv linkers that find use in increasing or decreasing the pI of the subject heterodimeric bsAbs that utilize one or more scFv as a component, as described herein. The (+H) positive linker finds particular use herein, particularly with anti-CD3 V_L and V_H sequences shown herein. A single prior art scFv linker with a single charge is referenced as "Whitlow", from Whitlow et al., Protein Engineering 6(8):989-995 (1993). It should be noted that this linker was used for reducing aggregation and enhancing proteolytic stability in scFvs. Such charged scFv linkers can be used in any of the subject antibody formats.

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disclosed herein that include scFvs (e.g., 1+1 Fab-scFv-Fc and 2+1 Fab₂-scFv-Fc formats).

FIG. 6 depicts a number of exemplary domain linkers. In some embodiments, these linkers find use linking a single-chain Fv to an Fc chain. In some embodiments, these linkers may be combined. For example, a GGGGS linker (SEQ ID NO: 5) may be combined with a “half hinge” linker.

FIG. 7A-7D depicts the sequences of several useful 1+1 Fab-scFv-Fc bispecific antibody format heavy chain backbones based on human IgG1, without the Fv sequences (e.g. the scFv and the VH for the Fab side). Backbone 1 is based on human IgG1 (356E/358M allotype), and includes the S364K/E357Q:L368D/K370S skew variants, C220S on the chain with the S364K/E357Q skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 2 is based on human IgG1 (356E/358M allotype), and includes S364K:L368D/K370S skew variants, C220S on the chain with the S364K skew variant, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants, and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 3 is based on human IgG1 (356E/358M allotype), and includes S364K:L368E/K370S skew variants, C220S on the chain with the S364K skew variant, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368E/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 4 is based on human IgG1 (356E/358M allotype), and includes D401K:K360E/Q362E/T411E skew variants, C220S on the chain with the D401K skew variant, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with K360E/Q362E/T411E skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 5 is based on human IgG1 (356D/358L allotype), and includes S364K/E357Q:L368D/K370S skew variants, C220S on the chain with the S364K/E357Q skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 6 is based on human IgG1 (356E/358M allotype), and includes S364K/E357Q:L368D/K370S skew variants, C220S on the chain with the S364K/E357Q skew variants, N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains, as well as an N297A variant on both chains. Backbone 7 is identical to 6 except the mutation is N297S. Backbone 8 is based on human IgG4, and includes the S364K/E357Q:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants, as well as a S228P (EU numbering, this is S241P in Kabat) variant on both chains that ablates Fab arm exchange as is known in the art. Backbone 9 is based on human IgG2, and includes the S364K/E357Q:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants. Backbone 10 is based on human IgG2, and includes the S364K/E357Q:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants as well as a S267K variant on both chains. Backbone 11 is identical to backbone 1, except it includes M428L/N434S Xtend mutations. Backbone 12 is based on human IgG1 (356E/358M allotype), and includes S364K/E357Q:L368D/K370S skew variants, C220S and the

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P217R/P229R/N276K pI variants on the chain with S364K/E357Q skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Included within each of these backbones are sequences that are 90, 95, 98 and 99% identical (as defined herein) to the recited sequences, and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 additional amino acid substitutions (as compared to the “parent” of the Figure, which, as will be appreciated by those in the art, already contain a number of amino acid modifications as compared to the parental human IgG1 (or IgG2 or IgG4, depending on the backbone). That is, the recited backbones may contain additional amino acid modifications (generally amino acid substitutions) in addition to the skew, pI and ablation variants contained within the backbones of this figure.

FIGS. 8A-8C depict the sequences of several useful 2+1 Fab₂-scFv-Fc bispecific antibody format heavy chain backbones based on human IgG1, without the Fv sequences (e.g. the scFv and the VH for the Fab side). Backbone 1 is based on human IgG1 (356E/358M allotype), and includes the S364K/E357Q:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 2 is based on human IgG1 (356E/358M allotype), and includes S364K:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants, and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 3 is based on human IgG1 (356E/358M allotype), and includes S364K:L368E/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368E/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 4 is based on human IgG1 (356E/358M allotype), and includes D401K:K360E/Q362E/T411E skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with K360E/Q362E/T411E skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 5 is based on human IgG1 (356D/358L allotype), and includes S364K/E357Q:L368D/K370S skew variants, the N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Backbone 6 is based on human IgG1 (356E/358M allotype), and includes S364K/E357Q:L368D/K370S skew variants, N208D/Q295E/N384D/Q418E/N421D pI variants on the chain with L368D/K370S skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains, as well as an N297A variant on both chains. Backbone 7 is identical to 6 except the mutation is N297S. Backbone 8 is identical to backbone 1, except it includes M428L/N434S Xtend mutations. Backbone 9 is based on human IgG1 (356E/358M allotype), and includes S364K/E357Q:L368D/K370S skew variants, the P217R/P229R/N276K pI variants on the chain with S364K/E357Q skew variants and the E233P/L234V/L235A/G236del/S267K ablation variants on both chains. Included within each of these backbones are sequences that are 90, 95, 98 and 99% identical (as defined herein) to the recited sequences, and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 additional amino acid substitutions (as compared to the “parent” of the Figure, which, as will be appreciated by those in the art, already contain a number of amino acid modifications as compared to the parental human IgG1 (or IgG2 or IgG4, depending on the backbone). That is, the recited backbones may contain additional amino acid modi-

fications (generally amino acid substitutions) in addition to the skew, pI and ablation variants contained within the backbones of this figure.

FIG. 9 depicts the sequences of several useful constant light domain backbones based on human IgG1, without the Fv sequences (e.g. the scFv or the Fab). Included herein are constant light backbone sequences that are 90, 95, 98 and 99% identical (as defined herein) to the recited sequences, and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 additional amino acid modifications.

FIG. 10A-10F depicts sequences for exemplary anti-CD3 scFvs suitable for use in the bispecific antibodies of the invention. The CDRs are underlined, the scFv linker is double underlined (in the sequences, the scFv linker is a positively charged scFv (GKPGS)₄ linker (SEQ ID NO: 1), although as will be appreciated by those in the art, this linker can be replaced by other linkers, including uncharged or negatively charged linkers, some of which are depicted in FIG. 5), and the slashes indicate the border(s) of the variable domains. In addition, the naming convention illustrates the orientation of the scFv from N- to C-terminus. As noted herein and is true for every sequence herein containing CDRs, the exact identification of the CDR locations may be slightly different depending on the numbering used as is shown in Table 2, and thus included herein are not only the CDRs that are underlined but also CDRs included within the V_H and V_L domains using other numbering systems. Furthermore, as for all the sequences in the Figures, these V_H and V_L sequences can be used either in a scFv format or in a Fab format.

FIG. 11 depicts the antigen sequences for a number of antigens of use in the invention, including both human and cyano, to facilitate the development of antigen binding domains that bind to both for ease of clinical development.

FIG. 12 depicts alignment of human CLDN6 and CLDN9 sequences and highlights the 3 differences in their respective extracellular loops. FIG. 12 discloses SEQ ID NOS 173-174, respectively, in order of appearance.

FIG. 13 depicts the variable heavy and variable light chain sequences for exemplary murine CLDN6 binding domain referred to herein as mC6-30 as well as sequences for XENP34243, a bivalent IgG1 mAb based on mC6-30 and with E233P/L234V/L235A/G236del/S267K ablation variant. CDRs are underlined and slashes indicate the border(s) between the variable regions and constant domain. As noted herein and is true for every sequence herein containing CDRs, the exact identification of the CDR locations may be slightly different depending on the numbering used as is shown in Table 2, and thus included herein are not only the CDRs that are underlined but also CDRs included within the V_H and V_L domains using other numbering systems. Furthermore, as for all the sequences in the Figures, these V_H and V_L sequences can be used either in a scFv format or in a Fab format.

FIG. 14 depicts the variable heavy and variable light chain sequences for humanized C6-30 variants. CDRs are underlined and slashes indicate the border(s) between the variable regions and constant domain. As noted herein and is true for every sequence herein containing CDRs, the exact identification of the CDR locations may be slightly different depending on the numbering used as is shown in Table 2, and thus included herein are not only the CDRs that are underlined but also CDRs included within the V_H and V_L domains using other numbering systems. Further, as for all the sequences in the Figures, these V_H and V_L sequences can be used either in a scFv format or in a Fab format. Furthermore, each of the variable heavy domains depicted herein

can be paired with any other α CLDN6 variable light domain; and each of the variable light domains depicted herein can be paired with any other α CLDN6 variable heavy domain.

FIG. 15A-15J depicts the variable heavy and variable light chain sequences for C6-30 variants engineered for reduced degradation (e.g. aspartic acid isomerization and deamidation) liability, enhanced selectivity for CLDN6, and/or modulated CLDN6 binding affinity. CDRs are underlined and slashes indicate the border(s) between the variable regions and constant domain. As noted herein and is true for every sequence herein containing CDRs, the exact identification of the CDR locations may be slightly different depending on the numbering used as is shown in Table 2, and thus included herein are not only the CDRs that are underlined but also CDRs included within the V_H and V_L domains using other numbering systems. Further, as for all the sequences in the Figures, these V_H and V_L sequences can be used either in a scFv format or in a Fab format. Furthermore, each of the variable heavy domains depicted herein can be paired with any other α CLDN6 variable light domain; and each of the variable light domains depicted herein can be paired with any other α CLDN6 variable heavy domain.

FIGS. 16A-16W depict illustrative C6-30[CLDN6] variants (parental humanized variants and further engineered variants) formatted as bivalent anti-CLDN6 mAb and IgG1 backbone with E233P/L234V/L235A/G236del/S267K ablation variant. CDRs are underlined and slashes indicate the border(s) between the variable regions and constant domain. As noted herein and is true for every sequence herein containing CDRs, the exact identification of the CDR locations may be slightly different depending on the numbering used as is shown in Table 2, and thus included herein are not only the CDRs that are underlined but also CDRs included within the V_H and V_L domains using other numbering systems. Furthermore, as for all the sequences in the Figures, these V_H and V_L sequences can be used either in a scFv format or in a Fab format.

FIGS. 17A and 17B depict a couple of formats of the present invention. FIG. 17A depicts the "1+1 Fab-scFv-Fc" format, with a first Fab arm binding CLDN6 and a second scFv arm binding CD3. FIG. 17B depicts the "2+1 Fab₂-scFv-Fc" format, with a first Fab arm binding CLDN6 and a second Fab-scFv arm, wherein the Fab binds CLDN6 and the scFv binds CD3.

FIG. 18 depicts sequences of a comparator CLDN6 binding domain (sequences as disclosed in U.S. Pat. No. 10,233,253) formatted as a bivalent anti-CLDN6 IgG1 mAb with E233P/L234V/L235A/G236del/S267K ablation variant (XENP26863), and formatted as a 1+1 Fab-scFv-Fc bsAb with CD3 High (XENP26849).

FIG. 19 depicts the sequences for illustrative α CLDN6 $\times$ α CD3 bsAbs in the 1+1 Fab-scFv-Fc format and comprising a CD3 High scFv (H1.30_L1.47 in either the VHV or the VLHV orientation). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α CLDN6 $\times$ α CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum

FIG. 20 depicts the sequences for illustrative α CLDN6 $\times$ α CD3 bsAbs in the 1+1 Fab-scFv-Fc format and comprising a CD3 High-Int #1 scFv (H1.32_L1.47 in either the VHVL orientation or the VLVH orientation). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α CLDN6 $\times$ α CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum

FIGS. 21A and 21B depict the sequences for illustrative α CLDN6 $\times$ α CD3 bsAbs in the 2+1 Fab₂-scFv-Fc format and comprising a CD3 High scFv (H1.30_L1.47 in either the VHVL orientation or the VLVH orientation). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α CLDN6 $\times$ α CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

FIGS. 22A-22H depict the sequences for illustrative α CLDN6 $\times$ α CD3 bsAbs in the 2+1 Fab₂-scFv-Fc format and comprising a CD3 High-Int #1 scFv (H1.32_L1.47 in either the VHVL orientation or the VLVH orientation). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α CLDN6 $\times$ α CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

FIGS. 23A-23F depicts binding of six murine anti-human CLDN6 bivalent mAbs designated A) C6-10, B) C6-11, C) C6-15, D) C6-21, E) C6-24, and F) C6-30 to HEK293-Trex cells that have been transfected to stably express human CLDN6, CLDN3, CLDN4, or CLDN9 as well as parental HEK293-Trex cells. Clones C6-11, C6-24, and C6-30 showed selectivity for CLDN6 over CLDN9.

FIGS. 24A-24E depicts binding of humanized variants of C6-30 in different humanization frameworks—(A) H1L1, (B) H1L2, (C) H2L1, (D) H2L2, and (E) the original mouse mAb to HEK293-Trex cells that have been transfected to stably express human CLDN6, CLDN3, CLDN4, or CLDN9 as well as parental HEK293-Trex cells. Clones C6-11, C6-24, and C6-30 showed selectivity for CLDN6 over CLDN9. Each of the humanized variants of C6-30 retained selectivity for CLDN6 over CLDN9 (as well as over CLDN3 and CLDN4).

FIGS. 25A-25E depicts binding of humanized variants of C6-11 in different humanization frameworks—(A) H1L1, (B) H1L2, (C) H2L1, (D) H2L2, and (E) the original mouse mAb to HEK293-Trex cells that have been transfected to stably express human CLDN6, CLDN3, CLDN4, or CLDN9 as well as parental HEK293-Trex cells. Clones C6-11, C6-24, and C6-30 showed selectivity for CLDN6 over

CLDN9. Each of the humanized variants of C6-30 retained selectivity for CLDN6 over CLDN9 (as well as over CLDN3 and CLDN4).

FIGS. 26A-26E depicts binding of humanized variants of C6-24 in different humanization frameworks—(A) H1L1, (B) H1L2, (C) H2L1, (D) H2L2, and (E) the original mouse mAb to HEK293-Trex cells that have been transfected to stably express human CLDN6, CLDN3, CLDN4, or CLDN9 as well as parental HEK293-Trex cells. Clones C6-11, C6-24, and C6-30 showed selectivity for CLDN6 over CLDN9. Each of the humanized variants of C6-30 retained selectivity for CLDN6 over CLDN9 (as well as over CLDN3 and CLDN4).

FIG. 27 depicts variants of humanized C6-30 engineered with substitutions in the variable heavy domain to remove degradation (e.g. deamidation and aspartic acid isomerization) liabilities and their binding potency to CLDN6 and CLDN9. XENP35093 (having C6-30 H1.9_L1) maintained selectivity with minimal loss to CLDN6 binding.

FIGS. 28A-28F depicts binding of A) XENP34228 (C6-24_H1L1 $\times$ CD3 bsAb in the 1+1 Fab-scFv-Fc format), B) XENP34232 (C6-24_H1L1 $\times$ CD3 bsAb in the 2+1 Fab₂-scFv-Fc format), C) XENP34229 (C6-30_H1L1 $\times$ CD3 bsAb in the 1+1 Fab-scFv-Fc format), D) XENP34233 (C6-30_H1L1 $\times$ CD3 bsAb in the 2+1 Fab₂-scFv-Fc format), E) XENP34637 (C6-30_H2L1 $\times$ CD3 bsAb in the 1+1 Fab-scFv-Fc format), and F) XENP34638 (C6-30_H2L1 $\times$ CD3 bsAb in the 2+1 Fab₂-scFv-Fc format) to HEK293E cells expressing human, cynomolgus, and mouse CLDN6. Each of the variants were cross-reactive for human, cynomolgus, and mouse CLDN6 irrespective of bispecific format.

FIGS. 29A-29F depicts binding of A) XENP34228 (C6-24_H1L1 $\times$ CD3 bsAb in the 1+1 Fab-scFv-Fc format), B) XENP34232 (C6-24_H1L1 $\times$ CD3 bsAb in the 2+1 Fab₂-scFv-Fc format), C) XENP34229 (C6-30_H1L1 $\times$ CD3 bsAb in the 1+1 Fab-scFv-Fc format), D) XENP34233 (C6-30_H1L1 $\times$ CD3 bsAb in the 2+1 Fab₂-scFv-Fc format), E) XENP34637 (C6-30_H2L1 $\times$ CD3 bsAb in the 1+1 Fab-scFv-Fc format), and F) XENP34638 (C6-30_H2L1 $\times$ CD3 bsAb in the 2+1 Fab₂-scFv-Fc format) to HEK293E cells expressing CLDN6, CLDN6 I143V isotype, CLDN9, CLDN3, and CLDN4. Each of the humanized C6-30 variants demonstrated enhanced selectivity in the 2+1 Fab₂-scFv-Fc format, in contrast to the humanized C6-24 variant which demonstrated similar selectivity in the 1+1 Fab-scFv-Fc format and the 2+1 Fab₂-scFv-Fc format.

FIG. 30 depicts the binding of variants from a single point mutation library of humanized C6-30 to 293 cells expressing CLDN6 or CLDN9. Each point on the plot represents a test article which was screened for binding at a single dose (30 ug/ml) to CLDN6 and CLDN9 and then analyzed with flow cytometry to obtain an MFI value. Test articles that skewed toward higher CLDN6 MFI values and lower CLDN9 MFI values, such as those circled on the scatter plot, were selected for further development.

FIGS. 31A and 31B depict EC50 of binding of illustrative C6-30 variants (in the context of bivalent mAbs) from the library combining the favorite variable heavy domain and variable light domain variants to HEK293E cells expressing CLDN6 or CLDN9 in A) a first and B) a second experiment. XENP26863 is a bivalent mAb based on a comparator CLDN6 binding domain. It should be noted that binding data from the experiments cannot be compared head-to-head as antigen density on the transfected cells vary between experiments. Nonetheless, several variants were enhanced in selectivity in comparison to the parental humanized clone (i.e.

XENP34218_H1L1 and XENP34220_H2L1) and in comparison to comparator XENP26863.

FIGS. 32A-32B depicts binding of illustrative C6-30 variants from library combining favorite variable heavy domain and variable light domain variants to HEK293E cells expressing A) CLDN6 or B) CLDN9.

FIG. 33 depicts EC50 of binding of illustrative C6-30 variants (in the context of bivalent mAbs) from the library combining the favorite variable heavy domain and variable light domain variants to HEK293E cells expressing CLDN6 or CLDN9. XENP26863 is a bivalent mAb based on a combination variants were enhanced in selectivity in comparison to the parental humanized clone (i.e. XENP34218_H1L1 and XENP34220_H2L1). 10 of the 16 combination variants demonstrated enhanced selectivity in comparison to comparator XENP26863.

FIGS. 34A-34B depicts binding of select C6-30 variants formatted as bivalent monospecific mAbs and 2+1 Fab₂-scFv-Fc bsAbs to HEK293E cells expressing A) CLDN6 or B) CLDN9.

FIGS. 35A-35C depicts binding of C6-24_H1L1, C6-30_H1L1, and C6-30_H2L1 in 1+1 Fab-scFv-Fc and 2+1 Fab₂-scFv-Fc to cells transfected to express human A) CLDN5, B) CLDN8, and C) CLDN17. None of the clones were cross-reactive for the additional claudins investigated.

FIGS. 36A to 36J depict several formats of the present invention. The first is the 1+1 Fab-scFv-Fc format, with a first and a second anti-antigen binding domain. Additionally, mAb-Fv, mAb-scFv, Central-scFv, Central-Fv, one-armed central-scFv, one scFv-mAb, scFv-mAb and a dual scFv format are all shown. For all of the scFv domains depicted, they can be either N- to C-terminus variable heavy-(optional linker)-variable light, or the opposite. In addition, for the one-armed scFv-mAb, the scFv can be attached either to the N-terminus of a heavy chain monomer or to the N-terminus of the light chain.

FIG. 37 depicts induction of RTCC on PA-1 (CLDN6^{high}) cells by C6-24_H1L1, C6-30_H1L1, and C6-30_H2L1 in 1+1 Fab-scFv-Fc and 2+1 Fab₂-scFv-Fc bsAbs (CD3 High). The 2+1 Fab₂-scFv-Fc constructs showed 20-100 fold lower EC50 in comparison to 1+1 constructs. Additionally, potency shifts between the 2+1 vs. 1+1 construct were ~half-log more for C6-30_H2L1 in comparison to C6-30_H1L1.

FIGS. 38A-38B depicts activation of CD8 T cells in the presence of PA-1 (CLDN6^{high}) cells by C6-24_H1L1, C6-30_H1L1, and C6-30_H2L1 in 1+1 Fab-scFv-Fc and 2+1 Fab₂-scFv-Fc bsAbs with CD3 High as indicated by A) percentage CD8 T cells expressing CD69 and B) percentage of CD8 T cells expressing CD107a.

FIGS. 39A-39B depicts induction of RTCC on A) HUTU-80 (CLDN6^{high}) and B) CLDN293-Trex stably transfected to express CLDN9 (CLDN9^{high}) by C6-30 formatted as 1+1 Fab-scFv-Fc or 2+1 Fab₂-scFv-Fc bsAbs with CD3 High or CD3 High-Int #1. The 2+1 format enabled much more potent RTCC activity on CLDN6⁺ cells in comparison to the 1+1 format. The bsAbs based on C6-30 demonstrated much weaker induction of RTCC on CLDN9⁺ cells in comparison to comparator bsAb XENP26849. 2+1 Fab₂-scFv-Fc bsAb having the lower affinity CD3_High-Int #1 induced RTCC on off-target CLDN9⁺ cells less potently than 2+1 bsAbs having high affinity CD3_High.

FIGS. 40A-40B depicts activation of CD8 T cells in the presence of A) HUTU-80 (CLDN6^{high}) and B) CLDN293-Trex stably transfected to express CLDN9 (CLDN9^{high}) by

C6-30 formatted as 1+1 Fab-scFv-Fc or 2+1 Fab₂-scFv-Fc bsAbs with CD3 High or CD3 High-Int #1 (as indicated by CD107a expression).

FIGS. 41A-41D depicts induction of RTCC on A) PA-1 (1.1×10⁶ CLDN6 density), B) OV-90 (195K CLDN6 density), C) NEC-8 (175K CLDN6 density), and D) COV-318 (11K CLDN6 density) cells by C6-30_H1L1 or C6-30_H2L1 in 1+1 Fab-scFv-Fc or 2+1 Fab₂-scFv-Fc bsAbs with CD3 High or CD3 High-Int #1. Cell killing activity correlates with CLDN6 antigen density and affinity of the CD3 binding domain.

FIG. 42 depicts EC50 of RTCC induction on PA-1 (1.1×10⁶ CLDN6 density) and B) OV-90 (195K CLDN6 density) by C6-30_H1L1 or C6-30_H2L1 in 1+1 Fab-scFv-Fc or 2+1 Fab₂-scFv-Fc bsAbs with CD3 High or CD3 High-Int #1.

FIGS. 43A-43D depicts activation of CD8 T cells in the presence of A) PA-1 (1.1×10⁶ CLDN6 density), B) OV-90 (195K CLDN6 density), C) NEC-8 (175K CLDN6 density), and D) COV-318 (11K CLDN6 density) cells by C6-30_H1L1 or C6-30_H2L1 in 1+1 Fab-scFv-Fc or 2+1 Fab₂-scFv-Fc bsAbs with CD3 High or CD3 High-Int #1 (as indicated by CD107a expression). T cell activation correlates with CLDN6 antigen density and affinity of the CD3 binding domain.

FIG. 44 depicts EC50 of CD8 T cell activation in the presence of A) PA-1 (1.1×10⁶ CLDN6 density), B) OV-90 (195K CLDN6 density), C) NEC-8 (175K CLDN6 density), and D) COV-318 (11K CLDN6 density) cells by C6-30_H1L1 or C6-30_H2L1 in 1+1 Fab-scFv-Fc or 2+1 Fab₂-scFv-Fc bsAbs with CD3 High or CD3 High-Int #1 (as indicated by CD107a expression).

FIGS. 45A-45B depicts induction of RTCC on A) HUTU-80 (CLDN6^{high}) and B) CLDN293-Trex stably transfected to express CLDN9 (CLDN9^{high}) by C6-30 variants formatted as 2+1 Fab₂-scFv-Fc bsAbs with CD3 High-Int #1 at a 10:1 effector:target ratio. The bsAbs having selectivity-engineered CLDN6 binding domains demonstrated modulated activity on CLDN6⁺ cells and/or modulated activity on CLDN9⁺ cells in comparison to bsAbs having the parental C6-30_H1L1 and C6-30_H2L1. XENP37233 demonstrated similar activity on CLDN9⁺ cells but significantly enhanced activity on CLDN6⁺ cells in comparison to the parental clones. XENP37227 demonstrated reduced activity on CLDN9⁺ cells and enhanced activity on CLDN6⁺ cells (albeit, less enhanced in comparison to XENP37233) in comparison to the parental clones. XENP37231 demonstrated little to no activity on CLDN9⁺ cells but slightly reduced activity on CLDN6⁺ cells in comparison to the parental clones.

FIGS. 46A-46B depicts induction of RTCC on A) HUTU-80 (CLDN6^{high}) and B) CLDN293-Trex stably transfected to express CLDN9 (CLDN9^{high}) by C6-30 variants formatted as 2+1 Fab₂-scFv-Fc bsAbs with CD3 High-Int #1 at a 1:1 effector:target ratio.

FIGS. 47A-47B depicts activation of CD8 T cells in the presence of A) HUTU-80 (CLDN6^{high}) and B) CLDN293-Trex stably transfected to express CLDN9 (CLDN9^{high}) by C6-30 formatted as 1+1 Fab-scFv-Fc or 2+1 Fab₂-scFv-Fc bsAbs with CD3 High or CD3 High-Int #1 (as indicated by CD69 expression).

FIGS. 48A-48B depicts activation of CD8 T cells in the presence of A) HUTU-80 (CLDN6^{high}) and B) CLDN293-Trex stably transfected to express CLDN9 (CLDN9^{high}) by C6-30 formatted as 1+1 Fab-scFv-Fc or 2+1 Fab₂-scFv-Fc bsAbs with CD3 High or CD3 High-Int #1 (as indicated by CD107a expression).

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FIGS. 49A-49B depicts IFN γ secretion by T cells in the presence of A) HUTU-80 (CLDN6^{high}) and B) CLDN293-Trex stably transfected to express CLDN9 (CLDN9^{high}) by C6-30 formatted as 1+1 Fab-scFv-Fc or 2+1 Fab₂-scFv-Fc bsAbs with CD3 High or CD3 High-Int #1 (as indicated by IFN γ secretion).

FIGS. 50A-50B depicts IL-2 secretion by T cells in the presence of A) HUTU-80 (CLDN6^{high}) and B) CLDN293-Trex stably transfected to express CLDN9 (CLDN9^{high}) by C6-30 formatted as 1+1 Fab-scFv-Fc or 2+1 Fab₂-scFv-Fc bsAbs with CD3 High or CD3 High-Int #1 (as indicated by IL-2 secretion).

FIGS. 51A-51B depicts TNF α secretion by T cells in the presence of A) HUTU-80 (CLDN6^{high}) and B) CLDN293-Trex stably transfected to express CLDN9 (CLDN9^{high}) by C6-30 formatted as 1+1 Fab-scFv-Fc or 2+1 Fab₂-scFv-Fc bsAbs with CD3 High or CD3 High-Int #1 (as indicated by TNF α secretion).

FIGS. 52A-52H depict overlay of induction of RTCC on HUTU-80 (CLDN6^{high}) and CLDN293-Trex stably transfected to express CLDN9 (CLDN9^{high}) by A) XENP37227 (C6-30_H1.9_L1.187), B) XENP37228 (C6-30_H1.19_L1.187), C) XENP37229 (C6-30_H1.22_L1.187), D) XENP37230 (C6-30_H1.22_L1.189), E) XENP37231 (C6-30_H1.24_L1.187), F) XENP37232 (C6-30_H1.24_L1.189), G) XENP37233 (C6-30_H2.91_L1.187), and H) XENP35386 (C6-30_H1L1) each of which are bispecific antibodies in the 2+1 Fab₂-scFv-Fc format with CD3 High-Int #1. Each bsAb can be dosed at high concentrations to achieve efficacious killing of CLDN6⁺ cells while avoiding killing of CLDN9⁺ cells.

FIGS. 53A-53C depicts A) induction of RTCC (including EC50 values) on PA-1 (CLDN6^{high}) cells by XENP37233, XENP37227, XENP37231, and XENP37630 at a 1:1 effector:target ratio, as well a CD8⁺ T cell activation as indicated by B) CD25 expression and C) CD107a expression.

FIG. 54 depicts the ability of XENP37541 to bind cells expressing CLDN6 in a density dependent manner. PA-1 cells were engineered to stably express a range of CLDN6 antigen densities. XENP37541 was incubated with cells at a range of doses. Cells were washed, stained with a secondary antibody, and washed again prior to being analyzed using flow cytometry. XENP37541 was able to bind cells with lower levels of CLDN6 expression, even down to 47 k CLDN6 antigens per cell.

FIG. 55 depicts the induction of RTCC on PA-1 cells engineered to express a range of CLDN6 antigen levels, which are annotated in the EC50 chart in the figure. A 1:1 effector:target cell ratio and 72 hour incubation time was used before measuring RTCC. The results show that XENP37541, XENP37634, and XENP37545 were all able to induce RTCC in a CLDN6 density dependent manner, while negative control RSV $\times$ CD3 bsAb XENP32140 was not. XENP37541 displayed the strongest potency and lowest EC50 values across all different levels of CLDN6 expression.

FIGS. 56A-56C depict the change in tumor volume (as determined by caliper measurements) A) over time, B) by Day 14 and C) by Day 20 (after 1st dose on Day 0) in PA-1 and huPBM-engrafted NSG mice dosed with PBS and XENP37227, XENP37630, or XENP37231 at 0.3, 1.0, or 3.0 mg/kg* indicates p<0.05 vs. PBS group (statistical analysis performed using Mann-Whitney test on baseline corrected tumor volume). By Day 14, each of the bispecific antibodies enhanced anti-tumor activity in comparison to PBS control.

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FIGS. 57A-57D depicts dose response observed in pharmacokinetics for each of A) XENP37541, B) XENP37545, C) XENP37547, and D) XENP37634.

FIG. 58 depicts overlay of blood concentration of for each of XENP37541, XENP37545, XENP37547, and XENP37634 following 60 $\times$ dose.

FIGS. 59A and 59B depict the sequences for illustrative α CLDN6 $\times$ α CD3 bsAbs in the 2+1 Fab₂-scFv-Fc format and comprising a CD3 High-Int #2 scFv (H1.89_L1.47 in either the VHVL orientation or the VLHV orientation). CDRs are underlined and slashes indicate the border(s) between the variable regions and other chain components (e.g. constant region and domain linkers). It should be noted that the α CLDN6 $\times$ α CD3 bsAbs can utilize variable region, Fc region, and constant domain sequences that are 90, 95, 98 and 99% identical (as defined herein), and/or contain from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid substitutions. In addition, each sequence outlined herein can include or exclude the M428L/N434S variants in one or preferably both Fc domains, which results in longer half-life in serum.

FIGS. 60A and 60B depict sequences comprising a comparator CLDN6 binding domain Comp_CHO02. XENP37217 comprises this comparator CLDN6 binding domain in the 2+1 Fab₂-scFv-Fc format, while XENP38084 comprises the binding domain in the bivalent antibody format.

FIGS. 61A-61C depict the binding of C6-30 variants, as well as comparator CLDN6 $\times$ CD3 bsAb XENP37217, to other claudin family members CLDN3, CLDN4, CLDN8 and CLDN17. 293 cells were transiently transfected to express CLDN3, CLDN4, CLDN8, or CLDN17. Cells were plated and each test article was added to each cell line at a 100 μ g/ml dose. After 1 hour incubation at 4 $^{\circ}$ C., cells were washed and a secondary AF647 antibody was added. Cells were incubated at 4 $^{\circ}$ C. for another hour, followed by additional washing and then analysis by flow cytometry. Comparator molecule XENP37217 showed significantly higher binding to CLDN8 and CLDN17 than any of the C6-30 variants.

DETAILED DESCRIPTION

The present invention provides heterodimeric bispecific antibodies that bind to human CD3 ϵ and human CLDN6.

I. Overview

Anti-bispecific antibodies that co-engage CD3 and a tumor antigen target are used to redirect T cells to attack and lyse targeted tumor cells. Examples include the BiTE $^{\circledR}$ and DART formats, which monovalently engage CD3 and a tumor antigen. While the CD3-targeting approach has shown considerable promise, a common side effect of such therapies is the associated production of cytokines, often leading to toxic cytokine release syndrome. Because the anti-CD3 binding domain of the bispecific antibody engages all T cells, the high cytokine-producing CD4 T cell subset is recruited. Moreover, the CD4 T cell subset includes regulatory T cells, whose recruitment and expansion can potentially lead to immune suppression and have a negative impact on long-term tumor suppression. In addition, these formats do not contain Fc domains and show very short serum half-lives in patients.

Provided herein are novel anti-CD3 $\times$ anti-CLDN6 (also referred to as anti-CLDN6 $\times$ anti-CD3, α CD3 $\times$ α CLDN6, α CLDN6 $\times$ α CD3 or sometimes just CLDN6 $\times$ CD3) heterodimeric bispecific antibodies and methods of using such

antibodies for the treatment of cancers. In particular, provided herein are anti-CD3, anti-CLDN6 bispecific antibodies in a variety of formats. These bispecific antibodies are useful for the treatment of cancers, particularly those with increased CLDN6 expression such as renal cell carcinoma. Such antibodies are used to direct CD3+ effector T cells to CLDN6+ tumors, thereby allowing the CD3+ effector T cells to attack and lyse the CLDN6+ tumors.

Additionally, the bispecific antibodies of the invention additionally distinguish between CLDN6 and other members of the CLDN family. While CLDN6 is expressed on cancers, CLDN9 is more highly expressed in healthy tissues (e.g. the cervix and the esophagus), so cross-reactivity of CLDN6 therapeutic with CLDN9 could lead to off-target toxicity. However, CLDN6 is 96% identical to CLDN9, differing by only 3 residues in their extracellular loops (as depicted in FIG. 12); therefore, it is a significant challenge to develop an antibody capable of binding CLDN6 selectively over CLDN9, as well as selectivity of CLDN6 antigen binding domains (ABDs) that bind CLDN6 preferentially over other members of the CLDN family, including CLDN3 and CLDN4. Thus, included in the present invention are antibodies and/or antigen binding domains that bind CLDN6 preferentially over CLDN9 as well as these ABDs in anti-CLDN6×anti-CD3 bispecific antibodies.

Additionally, in some embodiments, the disclosure provides bispecific antibodies that have different binding affinities to human CD3 that can alter or reduce the potential side effects of anti-CD3 therapy. That is, in some embodiments the antibodies described herein provide antibody constructs comprising anti-CD3 antigen binding domains that are “strong” or “high affinity” binders to CD3 (e.g. one example are heavy and light variable domains depicted as H1.30_L1.47 (optionally including a charged linker as appropriate)) and also bind to CLDN6. In other embodiments, the antibodies described herein provide antibody constructs comprising anti-CD3 antigen binding domains that are “lite” or “lower affinity” binders to CD3. Additional embodiments provides antibody constructs comprising anti-CD3 antigen binding domains that have intermediate or “medium” affinity to CD3 that also bind to CLDN6. While a very large number of anti-CD3 antigen binding domains (ABDs) can be used, particularly useful embodiments use 6 different anti-CD3 ABDs, although they can be used in two scFv orientations as discussed herein. Affinity is generally measured using a Biacore assay.

It should be appreciated that the “high, medium, low” anti-CD3 sequences provided herein can be used in a variety of heterodimerization formats as discussed herein. In general, due to the potential side effects of T cell recruitment, exemplary embodiments utilize formats that only bind CD3 monovalently, such as depicted in FIGS. 17A and 17B, and in the formats depicted herein, it is the CD3 ABD that is a scFv as more fully described herein. In contrast, the subject bispecific antibodies can bind CLDN6 either monovalently (e.g. FIG. 17A) or bivalently (e.g. FIG. 17B).

Provided herein are compositions that include CLDN6 binding domains, including antibodies with such CLDN6 binding domains (e.g., CLDN6×CD3 bispecific antibodies). Subject antibodies that include such CLDN6 binding domains advantageously elicit a range of different immune responses, depending on the particular CLDN6 binding domain used. For example, the subject antibodies exhibit differences in selectivity for cells with different CLDN6 expression, potencies for CLDN6 expressing cells, ability to elicit cytokine release, and sensitivity to soluble CLDN6.

Such CLDN6 binding domains and related antibodies find use, for example, in the treatment of CLDN6-associated cancers.

Accordingly, in one aspect, provided herein are heterodimeric antibodies that bind to two different antigens, e.g. the antibodies are “bispecific”, in that they bind two different target antigens, generally CLDN6 and CD3 as described herein. These heterodimeric antibodies can bind these target antigens either monovalently (e.g. there is a single antigen binding domain such as a variable heavy and variable light domain pair) or bivalently (there are two antigen binding domains that each independently bind the antigen). In some embodiments, the heterodimeric antibody provided herein includes one CD3 binding domain and one CLDN6 binding domain (e.g., heterodimeric antibodies in the “1+1 Fab-scFv-Fc” format described herein). In other embodiments, the heterodimeric antibody provided herein includes one CD3 binding domain and two CLDN6 binding domains (e.g., heterodimeric antibodies in the “2+1 Fab₂-scFv-Fc” formats described herein). The heterodimeric antibodies provided herein are based on the use different monomers which contain amino acid substitutions that “skew” formation of heterodimers over homodimers, as is more fully outlined below, coupled with “pI variants” that allow simple purification of the heterodimers away from the homodimers, as is similarly outlined below. The heterodimeric bispecific antibodies provided generally rely on the use of engineered or variant Fc domains that can self-assemble in production cells to produce heterodimeric proteins, and methods to generate and purify such heterodimeric proteins.

II. Nomenclature

The antibodies provided herein are listed in several different formats. In some instances, each monomer of a particular antibody is given a unique “XENP” number, although as will be appreciated in the art, a longer sequence might contain a shorter one. For example, a “scFv-Fc” monomer of a 1+1 Fab-scFv-Fc format antibody may have a first XENP number, while the scFv domain itself will have a different XENP number. Some molecules have three polypeptides, so the XENP number, with the components, is used as a name. Thus, the molecule XENP37630, which is in 2+1 Fab₂-scFv-Fc format, comprises three sequences (see FIG. 59A) a “Fab-Fc Heavy Chain” monomer (“Chain 1”); 2) a “Fab-scFv-Fc Heavy Chain” monomer (“Chain 2”); and 3) a “Light Chain” monomer (“Chain 3”) or equivalents, although one of skill in the art would be able to identify these easily through sequence alignment. These XENP numbers are in the sequence listing as well as identifiers, and used in the Figures. In addition, one molecule, comprising the three components, gives rise to multiple sequence identifiers. For example, the listing of the Fab includes, the full heavy chain sequence, the variable heavy domain sequence and the three CDRs of the variable heavy domain sequence, the full light chain sequence, a variable light domain sequence and the three CDRs of the variable light domain sequence. A Fab-scFv-Fc monomer includes a full length sequence, a variable heavy domain sequence, 3 heavy CDR sequences, and an scFv sequence (include scFv variable heavy domain sequence, scFv variable light domain sequence and scFv linker). Note that some molecules herein with a scFv domain use a single charged scFv linker (+H), although others can be used. In addition, the naming nomenclature of particular antigen binding domains (e.g., CLDN6 and CD3 binding domains) use a “Hx.xx_Ly.yy” type of format, with the numbers being unique identifiers to particular variable chain

sequences. Thus, an Fv domain of the antigen binding domain is "H1 L1", which indicates that the variable heavy domain, H1, was combined with the light domain L1. In the case that these sequences are used as scFvs, the designation "H1 L1", indicates that the variable heavy domain, H1 is combined with the light domain, L1, and is in VH-linker-VL orientation, from N- to C-terminus. This molecule with the identical sequences of the heavy and light variable domains but in the reverse order (VL-linker-VH orientation, from N- to C-terminus) would be designated "L111.1". Similarly, different constructs may "mix and match" the heavy and light chains as will be evident from the sequence listing and the figures.

Additionally, the bispecific antibodies of the invention are referred to herein as "anti-CD3×anti-CLDN6", "αCD3×αCLDN6", "αCLDN6×αCD3" or sometimes just "CLDN6×CD3". The order of the antigens is not determinative as will be discussed below, although the majority of the formats that utilize as scFv have the an anti-CD3 ABD as the scFv.

III. Definitions

In order that the application may be more completely understood, several definitions are set forth below. Such definitions are meant to encompass grammatical equivalents.

By "CLDN6" herein is meant a protein belonging to the claudin family. CLDN6 sequences are depicted, for example, in FIG. 11. The ABDs of the invention bind to human CLDN6.

By "ablation" herein is meant a decrease or removal of activity. Thus for example, "ablating FcγR binding" means the Fc region amino acid variant has less than 50% starting binding as compared to an Fc region not containing the specific variant, with more than 70-80-90-95-98% loss of activity being preferred, and in general, with the activity being below the level of detectable binding in a Biacore, SPR or BLI assay. Of particular use in the ablation of FcγR binding are those shown in FIG. 3, which generally are added to both monomers.

By "ADCC" or "antibody dependent cell-mediated cytotoxicity" as used herein is meant the cell-mediated reaction wherein nonspecific cytotoxic cells that express FcγRs recognize bound antibody on a target cell and subsequently cause lysis of the target cell. ADCC is correlated with binding to FcγRIIIa; increased binding to FcγRIIIa leads to an increase in ADCC activity.

By "ADCP" or antibody dependent cell-mediated phagocytosis as used herein is meant the cell-mediated reaction wherein nonspecific phagocytic cells that express FcγRs recognize bound antibody on a target cell and subsequently cause phagocytosis of the target cell.

As used herein, term "antibody" is used generally. Antibodies described herein can take on a number of formats as described herein, including traditional antibodies as well as antibody derivatives, fragments and mimetics, including a number of bispecific formats described herein.

Traditional immunoglobulin (Ig) antibodies are "Y" shaped tetramers. Each tetramer is typically composed of two identical pairs of polypeptide chains, each pair having one "light chain" monomer (typically having a molecular weight of about 25 kDa) and one "heavy chain" monomer (typically having a molecular weight of about 50-70 kDa).

Other useful antibody formats include, but are not limited to, the 1+1 Fab-scFv-Fc format and 2+1 Fab-scFv-Fc antibody formats described herein, as well as "mAb-Fv," "mAb-

scFv," "central-Fv", "one-armed scFv-mAb," "scFv-mAb," "dual scFv," and "trident" format antibodies, as discussed below.

Antibody heavy chains typically include a variable heavy (VH) domain, which includes vHCDR1-3, and an Fc domain, which includes a CH2-CH3 monomer. In some embodiments, antibody heavy chains include a hinge and CH1 domain. Traditional antibody heavy chains are monomers that are organized, from N- to C-terminus: VH-CH1-hinge-CH2-CH3. The CH1-hinge-CH2-CH3 is collectively referred to as the heavy chain "constant domain" or "constant region" of the antibody, of which there are five different categories or "isotypes": IgA, IgD, IgG, IgE and IgM. Thus, "isotype" as used herein is meant any of the subclasses of immunoglobulins defined by the chemical and antigenic characteristics of their constant regions. It should be understood that therapeutic antibodies can also comprise hybrids of isotypes and/or subclasses. For example, as shown in US Publication 2009/0163699, incorporated by reference, the antibodies described herein include the use of human IgG1/G2 hybrids.

In some embodiments, the antibodies provided herein include IgG isotype constant domains, which has several subclasses, including, but not limited to IgG1, IgG2, IgG3, and IgG4. In the IgG subclass of immunoglobulins, there are several immunoglobulin domains in the heavy chain. By "immunoglobulin (Ig) domain" herein is meant a region of an immunoglobulin having a distinct tertiary structure. Of interest in the antibodies described herein are the heavy chain domains, including, the constant heavy (CH) domains and the hinge domains. In the context of IgG antibodies, the IgG isotypes each have three CH regions. Accordingly, "CH" domains in the context of IgG are as follows: "CH1" refers to positions 118-220 according to the EU index as in Kabat. "CH2" refers to positions 237-340 according to the EU index as in Kabat, and "CH3" refers to positions 341-447 according to the EU index as in Kabat. As shown herein and described below, the pI variants can be in one or more of the CH regions, as well as the hinge region, discussed below.

It should be noted that IgG1 has different allotypes with polymorphisms at 356 (D or E) and 358 (L or M). The sequences depicted herein use the 356D/358M allotype, however the other allotype is included herein. That is, any sequence inclusive of an IgG1 Fc domain included herein can have 356E/358L replacing the 356D/358M allotype. It should be understood that therapeutic antibodies can also comprise hybrids of isotypes and/or subclasses. For example, as shown in US Publication 2009/0163699, incorporated by reference, the present antibodies, in some embodiments, include IgG1/IgG2 hybrids.

By "Fc" or "Fc region" or "Fc domain" as used herein is meant the polypeptide comprising the constant region of an antibody, in some instances, excluding all of the first constant region immunoglobulin domain (e.g., CH1) or a portion thereof, and in some cases, optionally including all or part of the hinge. For IgG, the Fc domain comprises immunoglobulin domains CH2 and CH3 (C72 and C73), and optionally all or a portion of the hinge region between CH1 (C71) and CH2 (C72). Thus, in some cases, the Fc domain includes, from N- to C-terminal, CH2-CH3 and hinge-CH2-CH3. In some embodiments, the Fc domain is that from human IgG1, IgG2, IgG3 or IgG4, with human IgG1 hinge-CH2-CH3 and IgG4 hinge-CH2-CH3 finding particular use in many embodiments. Additionally, in the case of human IgG1 Fc domains, frequently the hinge includes a C220S amino acid substitution. Furthermore, in the case of human

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IgG4 Fc domains, frequently the hinge includes a S228P amino acid substitution. Although the boundaries of the Fc region may vary, the human IgG heavy chain Fc region is usually defined to include residues E216, C226, or A231 to its carboxyl-terminal, wherein the numbering is according to the EU index as in Kabat. In some embodiments, as is more fully described below, amino acid modifications are made to the Fc region, for example to alter binding to one or more FcγR or to the FcRn.

By “heavy chain constant region” herein is meant the CH1-hinge-CH2-CH3 portion of an antibody (or fragments thereof), excluding the variable heavy domain; in EU numbering of human IgG1 this is amino acids 118-447. By “heavy chain constant region fragment” herein is meant a heavy chain constant region that contains fewer amino acids from either or both of the N- and C-termini but still retains the ability to form a dimer with another heavy chain constant region.

Another type of Ig domain of the heavy chain is the hinge region. By “hinge” or “hinge region” or “antibody hinge region” or “hinge domain” herein is meant the flexible polypeptide comprising the amino acids between the first and second constant domains of an antibody. Structurally, the IgG CH1 domain ends at EU position 215, and the IgG CH2 domain begins at residue EU position 231. Thus for IgG the antibody hinge is herein defined to include positions 216 (E216 in IgG1) to 230 (p230 in IgG1), wherein the numbering is according to the EU index as in Kabat. In some cases, a “hinge fragment” is used, which contains fewer amino acids at either or both of the N- and C-termini of the hinge domain. As noted herein, pl variants can be made in the hinge region as well. Many of the antibodies herein have at least one the cysteines at position 220 according to EU numbering (hinge region) replaced by a serine. Generally, this modification is on the “scFv monomer” side for most of the sequences depicted herein, although it can also be on the “Fab monomer” side, or both, to reduce disulfide formation. Specifically included within the sequences herein are one or both of these cysteines replaced (C220S).

As will be appreciated by those in the art, the exact numbering and placement of the heavy constant region domains can be different among different numbering systems. A useful comparison of heavy constant region numbering according to EU and Kabat is as below, see Edelman et al., 1969, *Proc Natl Acad Sci USA* 63:78-85 and Kabat et al., 1991, *Sequences of Proteins of Immunological Interest*, 5th Ed., United States Public Health Service, National Institutes of Health, Bethesda, entirely incorporated by reference.

TABLE 1

	EU Numbering	Kabat Numbering
CH1	118-215	114-223
Hinge	216-230	226-243

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TABLE 1-continued

	EU Numbering	Kabat Numbering
CH2	231-340	244-360
CH3	341-447	361-478

The antibody light chain generally comprises two domains: the variable light domain (VL), which includes light chain CDRs vLCDR1-3, and a constant light chain region (often referred to as CL or Cx). The antibody light chain is typically organized from N- to C-terminus: VL-CL.

By “antigen binding domain” or “ABD” herein is meant a set of six Complementary Determining Regions (CDRs) that, when present as part of a polypeptide sequence, specifically binds a target antigen (e.g., CLDN6 or CD3) as discussed herein. As is known in the art, these CDRs are generally present as a first set of variable heavy CDRs (vhCDRs or VHCDRs) and a second set of variable light CDRs (vLCDRs or VLCDRs), each comprising three CDRs: vhCDR1, vhCDR2, vhCDR3 variable heavy CDRs and vLCDR1, vLCDR2 and vLCDR3 variable light CDRs. The CDRs are present in the variable heavy domain (vhCDR1-3) and variable light domain (vLCDR1-3). The variable heavy domain and variable light domain from an Fv region.

The antibodies described herein provide a large number of different CDR sets. In this case, a “full CDR set” comprises the three variable light and three variable heavy CDRs, e.g., a vLCDR1, vLCDR2, vLCDR3, vhCDR1, vhCDR2 and vhCDR3. These can be part of a larger variable light or variable heavy domain, respectfully. In addition, as more fully outlined herein, the variable heavy and variable light domains can be on separate polypeptide chains, when a heavy and light chain is used (for example when Fabs are used), or on a single polypeptide chain in the case of scFv sequences.

As will be appreciated by those in the art, the exact numbering and placement of the CDRs can be different among different numbering systems. However, it should be understood that the disclosure of a variable heavy and/or variable light sequence includes the disclosure of the associated (inherent) CDRs. Accordingly, the disclosure of each variable heavy region is a disclosure of the vhCDRs (e.g., vhCDR1, vhCDR2 and vhCDR3) and the disclosure of each variable light region is a disclosure of the vLCDRs (e.g., vLCDR1, vLCDR2 and vLCDR3). A useful comparison of CDR numbering is as below, see Lafranc et al., *Dev. Comp. Immunol.* 27(1):55-77 (2003):

TABLE 2

	Kabat + Chothia	IMGT	Kabat	AbM	Chothia	Contact	Xencor
vhCDR1	26-35	27-38	31-35	26-35	26-32	30-35	27-35
vhCDR2	50-65	56-65	50-65	50-58	52-56	47-58	54-61
vhCDR3	95-102	105-117	95-102	95-102	95-102	93-101	103-116
vLCDR1	24-34	27-38	24-34	24-34	24-34	30-36	27-38
vLCDR2	50-56	56-65	50-56	50-56	50-56	46-55	56-62
vLCDR3	89-97	105-117	89-97	89-97	89-97	89-96	97-105

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Throughout the present specification, the Kabat numbering system is generally used when referring to a residue in the variable domain (approximately, residues 1-107 of the light chain variable region and residues 1-113 of the heavy chain variable region) and the EU numbering system for Fc regions (e.g., Kabat et al., *supra* (1991)).

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The CDRs contribute to the formation of the antigen-binding, or more specifically, epitope binding site of the antigen binding domains and antibodies. "Epitope" refers to a determinant that interacts with a specific antigen binding site in the variable region of an antibody molecule known as a paratope. Epitopes are groupings of molecules such as amino acids or sugar side chains and usually have specific structural characteristics, as well as specific charge characteristics. A single antigen may have more than one epitope.

The epitope may comprise amino acid residues directly involved in the binding (also called immunodominant component of the epitope) and other amino acid residues, which are not directly involved in the binding, such as amino acid residues which are effectively blocked by the specifically antigen binding peptide; in other words, the amino acid residue is within the footprint of the specifically antigen binding peptide.

Epitopes may be either conformational or linear. A conformational epitope is produced by spatially juxtaposed amino acids from different segments of the linear polypeptide chain. A linear epitope is one produced by adjacent amino acid residues in a polypeptide chain. Conformational and nonconformational epitopes may be distinguished in that the binding to the former but not the latter is lost in the presence of denaturing solvents.

An epitope typically includes at least 3, and more usually, at least 5 or 8-10 amino acids in a unique spatial conformation. Antibodies that recognize the same epitope can be verified in a simple immunoassay showing the ability of one antibody to block the binding of another antibody to a target antigen, for example "binning." As outlined below, the disclosure not only includes the enumerated antigen binding domains and antibodies herein, but those that compete for binding with the epitopes bound by the enumerated antigen binding domains.

In some embodiments, the six CDRs of the antigen binding domain are contributed by a variable heavy and a variable light domain. In a "Fab" format, the set of 6 CDRs are contributed by two different polypeptide sequences, the variable heavy domain (vh or VH; containing the vhCDR1, vhCDR2 and vhCDR3) and the variable light domain (vl or VL; containing the vlCDR1, vlCDR2 and vlCDR3), with the C-terminus of the vh domain being attached to the N-terminus of the CH1 domain of the heavy chain and the C-terminus of the vl domain being attached to the N-terminus of the constant light domain (and thus forming the light chain). In a scFv format, the vh and vl domains are covalently attached, generally through the use of a linker (a "scFv linker") as outlined herein, into a single polypeptide sequence, which can be either (starting from the N-terminus) vh-linker-vl or vl-linker-vh, with the former being generally preferred (including optional domain linkers on each side, depending on the format used (e.g., from FIG. 36). In general, the C-terminus of the scFv domain is attached to the N-terminus of the hinge in the second monomer.

By "variable region" or "variable domain" as used herein is meant the region of an immunoglobulin that comprises one or more Ig domains substantially encoded by any of the Vx, VW, and/or VH genes that make up the kappa, lambda, and heavy chain immunoglobulin genetic loci respectively, and contains the CDRs that confer antigen specificity. Thus, a "variable heavy domain" pairs with a "variable light domain" to form an antigen binding domain ("ABD"). In addition, each variable domain comprises three hypervariable regions ("complementary determining regions," "CDRs") (VHCDR1, VHCDR2 and VHCDR3 for the variable heavy domain and VLCDR1, VLCDR2 and VLCDR3

for the variable light domain) and four framework (FR) regions, arranged from amino-terminus to carboxy-terminus in the following order: FR1-CDR1-FR2-CDR2-FR3-CDR3-FR4. The hypervariable region generally encompasses amino acid residues from about amino acid residues 24-34 (LCDR1; "L" denotes light chain), 50-56 (LCDR2) and 89-97 (LCDR3) in the light chain variable region and around about 31-35B (HCDR1; "H" denotes heavy chain), 50-65 (HCDR2), and 95-102 (HCDR3) in the heavy chain variable region; Kabat et al., SEQUENCES OF PROTEINS OF IMMUNOLOGICAL INTEREST, 5th Ed. Public Health Service, National Institutes of Health, Bethesda, Md. (1991) and/or those residues forming a hypervariable loop (e.g. residues 26-32 (LCDR1), 50-52 (LCDR2) and 91-96 (LCDR3) in the light chain variable region and 26-32 (HCDR1), 53-55 (HCDR2) and 96-101 (HCDR3) in the heavy chain variable region; Chothia and Lesk (1987) J. Mol. Biol. 196:901-917. Specific CDRs of the invention are described in Table 2.

By "Fab" or "Fab region" as used herein is meant the polypeptide that comprises the VH, CH1, VL, and CL immunoglobulin domains, generally on two different polypeptide chains (e.g. VH-CH1 on one chain and VL-CL on the other). Fab may refer to this region in isolation, or this region in the context of a bispecific antibody described herein. In the context of a Fab, the Fab comprises an Fv region in addition to the CH1 and CL domains.

By "Fv" or "Fv fragment" or "Fv region" as used herein is meant a polypeptide that comprises the VL and VH domains of an ABD. Fv regions can be formatted as both Fabs (as discussed above, generally two different polypeptides that also include the constant regions as outlined above) and scFvs, where the VL and VH domains are combined (generally with a linker as discussed herein) to form an scFv.

By "single chain Fv" or "scFv" herein is meant a variable heavy domain covalently attached to a variable light domain, generally using a scFv linker as discussed herein, to form a scFv or scFv domain. A scFv domain can be in either orientation from N- to C-terminus (VH-linker-VL or VL-linker-VH). In the sequences depicted in the sequence listing and in the figures, the order of the VH and VL domain is indicated in the name, e.g. H.X₁L.Y means N- to C-terminal is VH-linker-VL, and L.Y₁H.X is VL-linker-VH.

Some embodiments of the subject antibodies provided herein comprise at least one scFv domain, which, while not naturally occurring, generally includes a variable heavy domain and a variable light domain, linked together by a scFv linker. As outlined herein, while the scFv domain is generally from N- to C-terminus oriented as VH-scFv linker-VL, this can be reversed for any of the scFv domains (or those constructed using vh and vl sequences from Fabs), to VL-scFv linker-VH, with optional linkers at one or both ends depending on the format.

By "modification" herein is meant an amino acid substitution, insertion, and/or deletion in a polypeptide sequence or an alteration to a moiety chemically linked to a protein. For example, a modification may be an altered carbohydrate or PEG structure attached to a protein. By "amino acid modification" herein is meant an amino acid substitution, insertion, and/or deletion in a polypeptide sequence. For clarity, unless otherwise noted, the amino acid modification is always to an amino acid coded for by DNA, e.g. the 20 amino acids that have codons in DNA and RNA.

By "amino acid substitution" or "substitution" herein is meant the replacement of an amino acid at a particular position in a parent polypeptide sequence with a different

amino acid. In particular, in some embodiments, the substitution is to an amino acid that is not naturally occurring at the particular position, either not naturally occurring within the organism or in any organism. For example, the substitution E272Y refers to a variant polypeptide, in this case an Fc variant, in which the glutamic acid at position 272 is replaced with tyrosine. For clarity, a protein which has been engineered to change the nucleic acid coding sequence but not change the starting amino acid (for example exchanging CGG (encoding arginine) to CGA (still encoding arginine) to increase host organism expression levels) is not an "amino acid substitution"; that is, despite the creation of a new gene encoding the same protein, if the protein has the same amino acid at the particular position that it started with, it is not an amino acid substitution.

By "amino acid insertion" or "insertion" as used herein is meant the addition of an amino acid sequence at a particular position in a parent polypeptide sequence. For example, -233E or 233E designates an insertion of glutamic acid after position 233 and before position 234. Additionally, -233ADE or A233ADE designates an insertion of AlaAsp-Glu after position 233 and before position 234.

By "amino acid deletion" or "deletion" as used herein is meant the removal of an amino acid sequence at a particular position in a parent polypeptide sequence. For example, E233- or E233 #, E233() or E233del designates a deletion of glutamic acid at position 233. Additionally, EDA233- or EDA233 # designates a deletion of the sequence GluAspAla that begins at position 233.

By "variant protein" or "protein variant", or "variant" as used herein is meant a protein that differs from that of a parent protein by virtue of at least one amino acid modification. The protein variant has at least one amino acid modification compared to the parent protein, yet not so many that the variant protein will not align with the parental protein using an alignment program such as that described below. In general, variant proteins (such as variant Fc domains, etc., outlined herein, are generally at least 75, 80, 85, 90, 91, 92, 93, 94, 95, 96, 97, 98 or 99% identical to the parent protein, using the alignment programs described below, such as BLAST. "Variant" as used herein also refers to particular amino acid modifications that confer particular function (e.g., a "heterodimerization variant," "pI variant," "ablation variant," etc.).

As described below, in some embodiments the parent polypeptide, for example an Fc parent polypeptide, is a human wild-type sequence, such as the heavy constant domain or Fc region from IgG1, IgG2, IgG3 or IgG4, although human sequences with variants can also serve as "parent polypeptides", for example the IgG1/2 hybrid of US Publication 2006/0134105 can be included. The protein variant sequence herein will preferably possess at least about 80% identity with a parent protein sequence, and most preferably at least about 90% identity, more preferably at least about 95-98-99% identity. Accordingly, by "antibody variant" or "variant antibody" as used herein is meant an antibody that differs from a parent antibody by virtue of at least one amino acid modification, "IgG variant" or "variant IgG" as used herein is meant an antibody that differs from a parent IgG (again, in many cases, from a human IgG sequence) by virtue of at least one amino acid modification, and "immunoglobulin variant" or "variant immunoglobulin" as used herein is meant an immunoglobulin sequence that differs from that of a parent immunoglobulin sequence by virtue of at least one amino acid modification. "Fc variant" or "variant Fc" as used herein is meant a protein comprising

an amino acid modification in an Fc domain as compared to an Fc domain of human IgG1, IgG2 or IgG4.

"Fc variant" or "variant Fc" as used herein is meant a protein comprising an amino acid modification in an Fc domain. The modification can be an addition, deletion, or substitution. The Fc variants are defined according to the amino acid modifications that compose them. Thus, for example, N434S or 434S is an Fc variant with the substitution for serine at position 434 relative to the parent Fc polypeptide, wherein the numbering is according to the EU index. Likewise, M428L/N434S defines an Fc variant with the substitutions M428L and N434S relative to the parent Fc polypeptide. The identity of the WT amino acid may be unspecified, in which case the aforementioned variant is referred to as 428L/434S. It is noted that the order in which substitutions are provided is arbitrary, that is to say that, for example, 428L/434S is the same Fc variant as 434S/428L, and so on. For all positions discussed herein that relate to antibodies or derivatives and fragments thereof (e.g., Fc domains), unless otherwise noted, amino acid position numbering is according to the EU index. The "EU index" or "EU index as in Kabat" or "EU numbering" scheme refers to the numbering of the EU antibody (Edelman et al., 1969, Proc Natl Acad Sci USA 63:78-85, hereby entirely incorporated by reference).

In general, variant Fc domains have at least about 80, 85, 90, 95, 97, 98 or 99 percent identity to the corresponding parental human IgG Fc domain (using the identity algorithms discussed below, with one embodiment utilizing the BLAST algorithm as is known in the art, using default parameters). Alternatively, the variant Fc domains can have from 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 amino acid modifications as compared to the parental Fc domain. Alternatively, the variant Fc domains can have up to 1, 2, 3, 4, 5, 6, 7, 8, 9, 10, 11, 12, 13, 14, 15, 16, 11, 12, 13, 14, 15, 16, 17, 18, 19 or 20 amino acid modifications as compared to the parental Fc domain. Additionally, as discussed herein, the variant Fc domains described herein still retain the ability to form a dimer with another Fc domain as measured using known techniques as described herein, such as non-denaturing gel electrophoresis.

By "protein" herein is meant at least two covalently attached amino acids, which includes proteins, polypeptides, oligopeptides and peptides. In addition, polypeptides that make up the antibodies described herein may include synthetic derivatization of one or more side chains or termini, glycosylation, PEGylation, circular permutation, cyclization, linkers to other molecules, fusion to proteins or protein domains, and addition of peptide tags or labels.

By "residue" as used herein is meant a position in a protein and its associated amino acid identity. For example, Asparagine 297 (also referred to as Asn297 or N297) is a residue at position 297 in the human antibody IgG1.

By "IgG subclass modification" or "isotype modification" as used herein is meant an amino acid modification that converts one amino acid of one IgG isotype to the corresponding amino acid in a different, aligned IgG isotype. For example, because IgG1 comprises a tyrosine and IgG2 a phenylalanine at EU position 296, a F296Y substitution in IgG2 is considered an IgG subclass modification.

By "non-naturally occurring modification" as used herein is meant an amino acid modification that is not isotypic. For example, because none of the human IgGs comprise a serine at position 434, the substitution 434S in IgG1, IgG2, IgG3, or IgG4 (or hybrids thereof) is considered a non-naturally occurring modification.

By “amino acid” and “amino acid identity” as used herein is meant one of the 20 naturally occurring amino acids that are coded for by DNA and RNA.

By “effector function” as used herein is meant a biochemical event that results from the interaction of an antibody Fc region with an Fc receptor or ligand. Effector functions include but are not limited to ADCC, ADCP, and CDC.

By “IgG Fc ligand” as used herein is meant a molecule, preferably a polypeptide, from any organism that binds to the Fc region of an IgG antibody to form an Fc/Fc ligand complex. Fc ligands include but are not limited to FcγRI, FcγRII, FcγRIII, FcRn, C1q, C3, mannan binding lectin, mannose receptor, staphylococcal protein A, streptococcal protein G, and viral FcγR. Fc ligands also include Fc receptor homologs (FcRH), which are a family of Fc receptors that are homologous to the FcγRs (Davis et al., 2002, Immunological Reviews 190:123-136, entirely incorporated by reference). Fc ligands may include undiscovered molecules that bind Fc. Particular IgG Fc ligands are FcRn and Fc gamma receptors. By “Fc ligand” as used herein is meant a molecule, preferably a polypeptide, from any organism that binds to the Fc region of an antibody to form an Fc/Fc ligand complex.

By “Fc gamma receptor”, “FcγR” or “FcγR” as used herein is meant any member of the family of proteins that bind the IgG antibody Fc region and is encoded by an FcγR gene. In humans this family includes but is not limited to FcγRI (CD64), including isoforms FcγRIa, FcγRIb, and FcγRIc; FcγRII (CD32), including isoforms FcγRIIa (including allotypes H131 and R131), FcγRIIb (including FcγRIIb-1 and FcγRIIb-2), and FcγRIIc; and FcγRIII (CD16), including isoforms FcγRIIIa (including allotypes V158 and F158) and FcγRIIIb (including allotypes FcγRIIIb-NA1 and FcγRIIIb-NA2) (Jeffers et al., 2002, Immunol Lett 82:57-65, entirely incorporated by reference), as well as any undiscovered human FcγRs or FcγR isoforms or allotypes. An FcγR may be from any organism, including but not limited to humans, mice, rats, rabbits, and monkeys. Mouse FcγRs include but are not limited to FcγRI (CD64), FcγRII (CD32), FcγRIII (CD16), and FcγRIII-2 (CD16-2), as well as any undiscovered mouse FcγRs or FcγR isoforms or allotypes.

By “FcRn” or “neonatal Fc Receptor” as used herein is meant a protein that binds the IgG antibody Fc region and is encoded at least in part by an FcRn gene. The FcRn may be from any organism, including but not limited to humans, mice, rats, rabbits, and monkeys. As is known in the art, the functional FcRn protein comprises two polypeptides, often referred to as the heavy chain and light chain. The light chain is beta-2-microglobulin and the heavy chain is encoded by the FcRn gene. Unless otherwise noted herein, FcRn or an FcRn protein refers to the complex of FcRn heavy chain with beta-2-microglobulin. A variety of FcRn variants used to increase binding to the FcRn receptor, and in some cases, to increase serum half-life. An “FcRn variant” is one that increases binding to the FcRn receptor, and suitable FcRn variants are shown below.

By “parent polypeptide” as used herein is meant a starting polypeptide that is subsequently modified to generate a variant. The parent polypeptide may be a naturally occurring polypeptide, or a variant or engineered version of a naturally occurring polypeptide. Accordingly, by “parent immunoglobulin” as used herein is meant an unmodified immunoglobulin polypeptide that is modified to generate a variant, and by “parent antibody” as used herein is meant an unmodified antibody that is modified to generate a variant antibody. It should be noted that “parent antibody” includes known

commercial, recombinantly produced antibodies as outlined below. In this context, a “parent Fc domain” will be relative to the recited variant; thus, a “variant human IgG1 Fc domain” is compared to the parent Fc domain of human IgG1, a “variant human IgG4 Fc domain” is compared to the parent Fc domain human IgG4, etc.

By “position” as used herein is meant a location in the sequence of a protein. Positions may be numbered sequentially, or according to an established format, for example the EU index for antibody numbering.

By “target antigen” as used herein is meant the molecule that is bound specifically by the antigen binding domain comprising the variable regions of a given antibody.

By “strandedness” in the context of the monomers of the heterodimeric antibodies described herein is meant that, similar to the two strands of DNA that “match”, heterodimerization variants are incorporated into each monomer so as to preserve the ability to “match” to form heterodimers. For example, if some pI variants are engineered into monomer A (e.g. making the pI higher) then steric variants that are “charge pairs” that can be utilized as well do not interfere with the pI variants, e.g. the charge variants that make a pI higher are put on the same “strand” or “monomer” to preserve both functionalities. Similarly, for “skew” variants that come in pairs of a set as more fully outlined below, the skilled artisan will consider pI in deciding into which strand or monomer one set of the pair will go, such that pI separation is maximized using the pI of the skews as well.

By “target cell” as used herein is meant a cell that expresses a target antigen.

By “host cell” in the context of producing a bispecific antibody according to the antibodies described herein is meant a cell that contains the exogenous nucleic acids encoding the components of the bispecific antibody and is capable of expressing the bispecific antibody under suitable conditions. Suitable host cells are discussed below.

By “wild type or WT” herein is meant an amino acid sequence or a nucleotide sequence that is found in nature, including allelic variations. A WT protein has an amino acid sequence or a nucleotide sequence that has not been intentionally modified.

Provided herein are a number of antibody domains that have sequence identity to human antibody domains. Sequence identity between two similar sequences (e.g., antibody variable domains) can be measured by algorithms such as that of Smith, T. F. & Waterman, M. S. (1981) “Comparison Of Biosequences,” Adv. Appl. Math. 2:482 [local homology algorithm]; Needleman, S. B. & Wunsch, C. D. (1970) “A General Method Applicable To The Search For Similarities In The Amino Acid Sequence Of Two Proteins,” J. Mol. Biol. 48:443 [homology alignment algorithm]; Pearson, W. R. & Lipman, D. J. (1988) “Improved Tools For Biological Sequence Comparison,” Proc. Natl. Acad. Sci. (U.S.A.) 85:2444 [search for similarity method]; or Altschul, S. F. et al, (1990) “Basic Local Alignment Search Tool,” J. Mol. Biol. 215:403-10, the “BLAST” algorithm, see <https://blast.ncbi.nlm.nih.gov/Blast.cgi>. When using any of the aforementioned algorithms, the default parameters (for Window length, gap penalty, etc.) are used. In one embodiment, sequence identity is done using the BLAST algorithm, using default parameters.

The antibodies described herein are generally isolated or recombinant. “Isolated,” when used to describe the various polypeptides disclosed herein, means a polypeptide that has been identified and separated and/or recovered from a cell or cell culture from which it was expressed. Ordinarily, an isolated polypeptide will be prepared by at least one puri-

fication step. An “isolated antibody,” refers to an antibody which is substantially free of other antibodies having different antigenic specificities. “Recombinant” means the antibodies are generated using recombinant nucleic acid techniques in exogenous host cells, and they can be isolated as well.

“Specific binding” or “specifically binds to” or is “specific for” a particular antigen or an epitope means binding that is measurably different from a non-specific interaction. Specific binding can be measured, for example, by determining binding of a molecule compared to binding of a control molecule, which generally is a molecule of similar structure that does not have binding activity. For example, specific binding can be determined by competition with a control molecule that is similar to the target.

Specific binding for a particular antigen or an epitope can be exhibited, for example, by an antibody having a KD for an antigen or epitope of at least about 10^{-4} M, at least about 10^{-5} M, at least about 10^{-6} M, at least about 10^{-7} M, at least about 108 M, at least about 10^{-9} M, alternatively at least about 10^{-10} M, at least about 10^{-11} M, at least about 10^{-12} M, or greater, where KD refers to a dissociation rate of a particular antibody-antigen interaction. Typically, an antibody that specifically binds an antigen will have a KD that is 20-, 50-, 100-, 500-, 1000-, 5,000-, 10,000- or more times greater for a control molecule relative to the antigen or epitope.

Also, specific binding for a particular antigen or an epitope can be exhibited, for example, by an antibody having a KA or Ka for an antigen or epitope of at least 20-, 50-, 100-, 500-, 1000-, 5,000-, 10,000- or more times greater for the epitope relative to a control, where KA or Ka refers to an association rate of a particular antibody-antigen interaction. Binding affinity is generally measured using a Biacore, SPR or BLI assay.

IV. Antibodies of the Invention

The present invention provides antibodies, including monoclonal antibodies and bispecific antibodies, that bind to human CLDN6 (it should be noted that many, if not most, of the exemplified antibodies also bind to cyano CLDN6 for ease of pre-clinical testing, but this is not required in all embodiments). In particularly, bispecific antibodies are provided that bind CD3 and CLDN6 that make take on a variety of formats as more fully described below.

Of particular interest are anti-CLDN6 antigen binding domains (and antibodies containing them) that preferentially bind CLDN6 over CLDN9.

1. Antibodies

The antibodies provided herein include different antibody domains as is more fully described below. As described herein and known in the art, the antibodies described herein include different domains within the heavy and light chains, which can be overlapping as well. These domains include, but are not limited to, the Fc domain, the CH1 domain, the CH2 domain, the CH3 domain, the hinge domain, the heavy constant domain (CH1-hinge-Fc domain or CH1-hinge-CH2-CH3), the variable heavy domain, the variable light domain, the light constant domain, Fab domains and scFv domains.

In particular, the formats depicted in FIGS. 17 and 36 are usually referred to as “heterodimeric antibodies”, meaning that the protein has at least two associated Fc sequences self-assembled into a heterodimeric Fc domain and at least two Fv regions, whether as Fabs or as scFvs.

a. Chimeric and Humanized Antibodies

In certain embodiments, the antibodies described herein comprise a heavy chain variable region from a particular germline heavy chain immunoglobulin gene and/or a light chain variable region from a particular germline light chain immunoglobulin gene. For example, such antibodies may comprise or consist of a human antibody comprising heavy or light chain variable regions that are “the product of” or “derived from” a particular germline sequence. A human antibody that is “the product of” or “derived from” a human germline immunoglobulin sequence can be identified as such by comparing the amino acid sequence of the human antibody to the amino acid sequences of human germline immunoglobulins and selecting the human germline immunoglobulin sequence that is closest in sequence (i.e., greatest % identity) to the sequence of the human antibody (using the methods outlined herein). A human antibody that is “the product of” or “derived from” a particular human germline immunoglobulin sequence may contain amino acid differences as compared to the germline sequence, due to, for example, naturally-occurring somatic mutations or intentional introduction of site-directed mutation. However, a humanized antibody typically is at least 90% identical in amino acids sequence to an amino acid sequence encoded by a human germline immunoglobulin gene and contains amino acid residues that identify the antibody as being derived from human sequences when compared to the germline immunoglobulin amino acid sequences of other species (e.g., murine germline sequences). In certain cases, a humanized antibody may be at least 95, 96, 97, 98 or 99%, or even at least 96%, 97%, 98%, or 99% identical in amino acid sequence to the amino acid sequence encoded by the germline immunoglobulin gene. Typically, a humanized antibody derived from a particular human germline sequence will display no more than 10-20 amino acid differences from the amino acid sequence encoded by the human germline immunoglobulin gene (prior to the introduction of any skew, pI and ablation variants herein; that is, the number of variants is generally low, prior to the introduction of the variants described herein). In certain cases, the humanized antibody may display no more than 5, or even no more than 4, 3, 2, or 1 amino acid difference from the amino acid sequence encoded by the germline immunoglobulin gene (again, prior to the introduction of any skew, pI and ablation variants herein; that is, the number of variants is generally low, prior to the introduction of the variants described herein). In some embodiments, the amino acid differences are in one or more of the 6 CDRs. In some embodiments, the amino acid differences are in a VH and/or VL framework region.

In one embodiment, the parent antibody has been affinity matured, as is known in the art. Structure-based methods may be employed for humanization and affinity maturation, for example as described in U.S. Ser. No. 11/004,590. Selection based methods may be employed to humanize and/or affinity mature antibody variable regions, including but not limited to methods described in Wu et al., 1999, J. Mol. Biol. 294:151-162; Baca et al., 1997, J. Biol. Chem. 272(16):10678-10684; Rosok et al., 1996, J. Biol. Chem. 271(37): 22611-22618; Rader et al., 1998, Proc. Natl. Acad. Sci. USA 95: 8910-8915; Krauss et al., 2003, Protein Engineering 16(10):753-759, all entirely incorporated by reference. Other humanization methods may involve the grafting of only parts of the CDRs, including but not limited to methods described in U.S. Ser. No. 09/810,510; Tan et al., 2002, J. Immunol. 169:1119-1125; De Pascalis et al., 2002, J. Immunol. 169:3076-3084, all entirely incorporated by reference.

2. Heterodimeric Antibodies

In exemplary embodiments, the bispecific antibodies provided herein are heterodimeric bispecific antibodies that include two variant Fc domain sequences. Such variant Fc domains include amino acid modifications to facilitate the self-assembly and/or purification of the heterodimeric antibodies.

An ongoing problem in antibody technologies is the desire for “bispecific” antibodies that bind to two different antigens simultaneously, in general thus allowing the different antigens to be brought into proximity and resulting in new functionalities and new therapies. In general, these antibodies are made by including genes for each heavy and light chain into the host cells. This generally results in the formation of the desired heterodimer (A-B), as well as the two homodimers (A-A and B-B (not including the light chain heterodimeric issues)). However, a major obstacle in the formation of bispecific antibodies is the difficulty in biasing the formation of the desired heterodimeric antibody over the formation of the homodimers and/or purifying the heterodimeric antibody away from the homodimers.

There are a number of mechanisms that can be used to generate the subject heterodimeric antibodies. In addition, as will be appreciated by those in the art, these different mechanisms can be combined to ensure high heterodimerization. Amino acid modifications that facilitate the production and purification of heterodimers are collectively referred to generally as “heterodimerization variants.” As discussed below, heterodimerization variants include “skew” variants (e.g., the “knobs and holes” and the “charge pairs” variants described below) as well as “pI variants,” which allow purification of heterodimers from homodimers. As is generally described in U.S. Pat. No. 9,605,084, hereby incorporated by reference in its entirety and specifically as below for the discussion of heterodimerization variants, useful mechanisms for heterodimerization include “knobs and holes” (“KIH”) as described in U.S. Pat. No. 9,605,084, “electrostatic steering” or “charge pairs” as described in U.S. Pat. No. 9,605,084, pI variants as described in U.S. Pat. No. 9,605,084, and general additional Fc variants as outlined in U.S. Pat. No. 9,605,084 and below.

Heterodimerization variants that are useful for the formation and purification of the subject heterodimeric antibody (e.g., bispecific antibodies) are further discussed in detailed below.

a. Skew Variants

In some embodiments, the heterodimeric antibody includes skew variants which are one or more amino acid modifications in a first Fc domain (A) and/or a second Fc domain (B) that favor the formation of Fc heterodimers (Fc dimers) that include the first and the second Fc domain; (A-B) over Fc homodimers (Fc dimers that include two of the first Fc domain or two of the second Fc domain; A-A or B-B). Suitable skew variants are included in the FIG. 29 of US Publ. App. No. 2016/0355608, hereby incorporated by reference in its entirety and specifically for its disclosure of skew variants, as well as in FIG. 1.

Thus, suitable Fc heterodimerization variant pairs that will permit the formation of heterodimeric Fc regions are shown in FIG. 1. Thus a first Fc domain has first Fc heterodimerization variants and the second Fc domain has second Fc heterodimerization variants selected from the pairs in FIG. 1.

One mechanism is generally referred to in the art as “knobs and holes”, referring to amino acid engineering that creates steric influences to favor heterodimeric formation and disfavor homodimeric formation can also optionally be

used; this is sometimes referred to as “knobs and holes”, as described in U.S. Ser. No. 61/596,846, Ridgway et al., Protein Engineering 9(7):617 (1996); Atwell et al., J. Mol. Biol. 1997 270:26; U.S. Pat. No. 8,216,805, all of which are hereby incorporated by reference in their entirety. The Figures identify a number of “monomer A—monomer B” pairs that rely on “knobs and holes”. In addition, as described in Merchant et al., Nature Biotech. 16:677 (1998), these “knobs and hole” mutations can be combined with disulfide bonds to skew formation to heterodimerization.

An additional mechanism that finds use in the generation of heterodimers is sometimes referred to as “electrostatic steering” as described in Gunasekaran et al., J. Biol. Chem. 285(25):19637 (2010), hereby incorporated by reference in its entirety. This is sometimes referred to herein as “charge pairs”. In this embodiment, electrostatics are used to skew the formation towards heterodimerization. As those in the art will appreciate, these may also have an effect on pI, and thus on purification, and thus could in some cases also be considered pI variants. However, as these were generated to force heterodimerization and were not used as purification tools, they are classified as “steric variants”. These include, but are not limited to, D221E/P228E/L368E paired with D221R/P228R/K409R (e.g. these are “monomer corresponding sets) and C220E/P228E/368E paired with C220R/E224R/P228R/K409R.

In some embodiments, the skew variants advantageously and simultaneously favor heterodimerization based on both the “knobs and holes” mechanism as well as the “electrostatic steering” mechanism. In some embodiments, the heterodimeric antibody includes one or more sets of such heterodimerization skew variants. These variants come in “pairs” of “sets”. That is, one set of the pair is incorporated into the first monomer and the other set of the pair is incorporated into the second monomer. It should be noted that these sets do not necessarily behave as “knobs in holes” variants, with a one-to-one correspondence between a residue on one monomer and a residue on the other. That is, these pairs of sets may instead form an interface between the two monomers that encourages heterodimer formation and discourages homodimer formation, allowing the percentage of heterodimers that spontaneously form under biological conditions to be over 90%, rather than the expected 50% (25% homodimer A/A:50% heterodimer A/B:25% homodimer B/B). Exemplary heterodimerization “skew” variants are depicted in FIG. 1. Such “skew” variants include, but are not limited to: S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L; K370S:S364K/E357Q (EU numbering).

In exemplary embodiments, the heterodimeric antibody includes Fc heterodimerization variants as sets: S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L; K370S:S364K/E357Q; or a T366S/L368A/Y407V:T366W (optionally including a bridging disulfide, T366S/L368A/Y407V/Y349C:T366W/S354C) are all “skew” variant amino acid substitution sets of Fc heterodimerization variants. In an exemplary embodiment, the heterodimeric antibody includes a “S364K/E357Q:L368D/K370S” amino acid substitution set. In terms of nomenclature, the pair “S364K/E357Q:L368D/K370S” means that one of the monomers includes an Fc domain that includes the amino acid substitutions S364K and E357Q and the other monomer includes an Fc domain that includes the amino acid substitutions L368D and K370S; as above, the “strandedness” of these pairs depends on the starting pI.

In some embodiments, the skew variants provided herein can be optionally and independently incorporated with any other modifications, including, but not limited to, other skew variants (see, e.g., in FIG. 37 of US Publ. App. No. 2012/0149876, herein incorporated by reference, particularly for its disclosure of skew variants), pI variants, isotypic variants, FcRn variants, ablation variants, etc. into one or both of the first and second Fc domains of the heterodimeric antibody. Further, individual modifications can also independently and optionally be included or excluded from the subject the heterodimeric antibody.

Additional monomer A and monomer B variants that can be combined with other variants, optionally and independently in any amount, such as pI variants outlined herein or other steric variants that are shown in FIG. 37 of US 2012/0149876, the figure and legend and SEQ ID NOs of which are incorporated expressly by reference herein.

In some embodiments, the steric variants outlined herein can be optionally and independently incorporated with any pI variant (or other variants such as Fc variants, FcRn variants, etc.) into one or both monomers, and can be independently and optionally included or excluded from the proteins of the antibodies described herein.

A list of suitable skew variants is found in FIG. 1. Of particular use in many embodiments are the pairs of sets including, but not limited to, S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L and K370S:S364K/E357Q. In terms of nomenclature, the pair “S364K/E357Q:L368D/K370S” means that one of the monomers has the double variant set S364K/E357Q and the other has the double variant set L368D/K370S.

b. pI (Isoelectric point) Variants for Heterodimers

In some embodiments, the heterodimeric antibody includes purification variants that advantageously allow for the separation of heterodimeric antibody (e.g., anti-CLDN6xanti-CD3 bispecific antibody) from homodimeric proteins.

There are several basic mechanisms that can lead to ease of purifying heterodimeric antibodies. For example, modifications to one or both of the antibody heavy chain monomers A and B such that each monomer has a different pI allows for the isoelectric purification of heterodimeric A-B antibody from monomeric A-A and B-B proteins. Alternatively, some scaffold formats, such as the “1+1 Fab-scFv-Fc” format and the “2+1 Fab₂-scFv-Fc” format, also allows separation on the basis of size. As described above, it is also possible to “skew” the formation of heterodimers over homodimers using skew variants. Thus, a combination of heterodimerization skew variants and pI variants find particular use in the heterodimeric antibodies provided herein.

Additionally, as more fully outlined below, depending on the format of the heterodimeric antibody, pI variants either contained within the constant region and/or Fc domains of a monomer, and/or domain linkers can be used. In some embodiments, the heterodimeric antibody includes additional modifications for alternative functionalities that can also create pI changes, such as Fc, FcRn and KO variants.

In some embodiments, the subject heterodimeric antibodies provided herein include at least one monomer with one or more modifications that alter the pI of the monomer (i.e., a “pI variant”). In general, as will be appreciated by those in the art, there are two general categories of pI variants: those that increase the pI of the protein (basic changes) and those that decrease the pI of the protein (acidic changes). As described herein, all combinations of these variants can be done: one monomer may be wild type, or a variant that does

not display a significantly different pI from wild-type, and the other can be either more basic or more acidic. Alternatively, each monomer is changed, one to more basic and one to more acidic.

Depending on the format of the heterodimer antibody, pI variants can be either contained within the constant and/or Fc domains of a monomer, or charged linkers, either domain linkers or scFv linkers, can be used. That is, antibody formats that utilize scFv(s) such as “1+1 Fab-scFv-Fc”, format can include charged scFv linkers (either positive or negative), that give a further pI boost for purification purposes. As will be appreciated by those in the art, some 1+1 Fab-scFv-Fc formats are useful with just charged scFv linkers and no additional pI adjustments, although the antibodies described herein do provide pI variants that are on one or both of the monomers, and/or charged domain linkers as well. In addition, additional amino acid engineering for alternative functionalities may also confer pI changes, such as Fc, FcRn and KO variants.

In subject heterodimeric antibodies that utilizes pI as a separation mechanism to allow the purification of heterodimeric proteins, amino acid variants are introduced into one or both of the monomer polypeptides. That is, the pI of one of the monomers (referred to herein for simplicity as “monomer A”) can be engineered away from monomer B, or both monomer A and B change be changed, with the pI of monomer A increasing and the pI of monomer B decreasing. As is outlined more fully below, the pI changes of either or both monomers can be done by removing or adding a charged residue (e.g., a neutral amino acid is replaced by a positively or negatively charged amino acid residue, e.g., glycine to glutamic acid), changing a charged residue from positive or negative to the opposite charge (aspartic acid to lysine) or changing a charged residue to a neutral residue (e.g., loss of a charge; lysine to serine.). A number of these variants are shown in the FIGS. 1 and 2.

Thus, in some embodiments, the subject heterodimeric antibody includes amino acid modifications in the constant regions that alter the isoelectric point (pI) of at least one, if not both, of the monomers of a dimeric protein to form “pI antibodies”) by incorporating amino acid substitutions (“pI variants” or “pI substitutions”) into one or both of the monomers. As shown herein, the separation of the heterodimers from the two homodimers can be accomplished if the pIs of the two monomers differ by as little as 0.1 pH unit, with 0.2, 0.3, 0.4 and 0.5 or greater all finding use in the antibodies described herein.

As will be appreciated by those in the art, the number of pI variants to be included on each or both monomer(s) to get good separation will depend in part on the starting pI of the components, for example in the 1+1 Fab-scFv-Fc and 2+1 Fab₂-scFv-Fc formats, the starting pI of the scFv and Fab(s) of interest. That is, to determine which monomer to engineer or in which “direction” (e.g., more positive or more negative), the Fv sequences of the two target antigens are calculated and a decision is made from there. As is known in the art, different Fvs will have different starting pIs which are exploited in the antibodies described herein. In general, as outlined herein, the pIs are engineered to result in a total pI difference of each monomer of at least about 0.1 logs, with 0.2 to 0.5 being preferred as outlined herein.

In the case where pI variants are used to achieve heterodimerization, by using the constant region(s) of the heavy chain(s), a more modular approach to designing and purifying bispecific proteins, including antibodies, is provided. Thus, in some embodiments, heterodimerization variants (including skew and pI heterodimerization variants) are not

included in the variable regions, such that each individual antibody must be engineered. In addition, in some embodiments, the possibility of immunogenicity resulting from the pI variants is significantly reduced by importing pI variants from different IgG isotypes such that pI is changed without introducing significant immunogenicity. Thus, an additional problem to be solved is the elucidation of low pI constant domains with high human sequence content, e.g., the minimization or avoidance of non-human residues at any particular position. Alternatively or in addition to isotopic substitutions, the possibility of immunogenicity resulting from the pI variants is significantly reduced by utilizing isosteric substitutions (e.g. Asn to Asp; and Gln to Glu).

As discussed below, a side benefit that can occur with this pI engineering is also the extension of serum half-life and increased FcRn binding. That is, as described in US Publ. App. No. US 2012/0028304 (incorporated by reference in its entirety), lowering the pI of antibody constant domains (including those found in antibodies and Fc fusions) can lead to longer serum retention in vivo. These pI variants for increased serum half-life also facilitate pI changes for purification.

In addition, it should be noted that the pI variants give an additional benefit for the analytics and quality control process of bispecific antibodies, as the ability to either eliminate, minimize and distinguish when homodimers are present is significant. Similarly, the ability to reliably test the reproducibility of the heterodimeric antibody production is important.

In general, embodiments of particular use rely on sets of variants that include skew variants, which encourage heterodimerization formation over homodimerization formation, coupled with pI variants, which increase the pI difference between the two monomers to facilitate purification of heterodimers away from homodimers.

Exemplary combinations of pI variants are shown in FIGS. 4 and 5, and FIG. 30 of US Publ. App. No. 2016/0355608, all of which are herein incorporated by reference in its entirety and specifically for the disclosure of pI variants. Preferred combinations of pI variants are shown in FIGS. 1 and 2. As outlined herein and shown in the figures, these changes are shown relative to IgG1, but all isotypes can be altered this way, as well as isotype hybrids. In the case where the heavy chain constant domain is from IgG2-4, R133E and R133Q can also be used.

In one embodiment, a preferred combination of pI variants has one monomer (the negative Fab side) comprising 208D/295E/384D/418E/421D variants (N208D/Q295E/N384D/Q418E/N421D when relative to human IgG1) and a second monomer (the positive scFv side) comprising a positively charged scFv linker, including (GKPGS)₄ (SEQ ID NO: 1). However, as will be appreciated by those in the art, the first monomer includes a CH1 domain, including position 208. Accordingly, in constructs that do not include a CH1 domain (for example for antibodies that do not utilize a CH1 domain on one of the domains), a preferred negative pI variant Fc set includes 295E/384D/418E/421D variants (Q295E/N384D/Q418E/N421D when relative to human IgG1).

Accordingly, in some embodiments, one monomer has a set of substitutions from FIG. 2 and the other monomer has a charged linker (either in the form of a charged scFv linker because that monomer comprises an scFv or a charged domain linker, as the format dictates, which can be selected from those depicted in FIG. 5).

In some embodiments, modifications are made in the hinge of the Fc domain, including positions 216, 217, 218,

219, 220, 221, 222, 223, 224, 225, 226, 227, 228, 229, and 230 based on EU numbering. Thus, pI mutations and particularly substitutions can be made in one or more of positions 216-230, with 1, 2, 3, 4 or 5 mutations finding use.

Again, all possible combinations are contemplated, alone or with other pI variants in other domains.

Specific substitutions that find use in lowering the pI of hinge domains include, but are not limited to, a deletion at position 221, a non-native valine or threonine at position 222, a deletion at position 223, a non-native glutamic acid at position 224, a deletion at position 225, a deletion at position 235 and a deletion or a non-native alanine at position 236. In some cases, only pI substitutions are done in the hinge domain, and in others, these substitution(s) are added to other pI variants in other domains in any combination.

In some embodiments, mutations can be made in the CH2 region, including positions 233, 234, 235, 236, 274, 296, 300, 309, 320, 322, 326, 327, 334 and 339, based on EU numbering. It should be noted that changes in 233-236 can be made to increase effector function (along with 327A) in the IgG2 backbone. Again, all possible combinations of these 14 positions can be made; e.g., may include a variant Fc domain with 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 CH2 pI substitutions.

Specific substitutions that find use in lowering the pI of CH2 domains include, but are not limited to, a non-native glutamine or glutamic acid at position 274, a non-native phenylalanine at position 296, a non-native phenylalanine at position 300, a non-native valine at position 309, a non-native glutamic acid at position 320, a non-native glutamic acid at position 322, a non-native glutamic acid at position 326, a non-native glycine at position 327, a non-native glutamic acid at position 334, a non-native threonine at position 339, and all possible combinations within CH2 and with other domains.

In this embodiment, the modifications can be independently and optionally selected from position 355, 359, 362, 384, 389, 392, 397, 418, 419, 444 and 447 (EU numbering) of the CH3 region. Specific substitutions that find use in lowering the pI of CH3 domains include, but are not limited to, a non-native glutamine or glutamic acid at position 355, a non-native serine at position 384, a non-native asparagine or glutamic acid at position 392, a non-native methionine at position 397, a non-native glutamic acid at position 419, a non-native glutamic acid at position 359, a non-native glutamic acid at position 362, a non-native glutamic acid at position 389, a non-native glutamic acid at position 418, a non-native glutamic acid at position 444, and a deletion or non-native aspartic acid at position 447.

In general, as will be appreciated by those in the art, there are two general categories of pI variants: those that increase the pI of the protein (basic changes) and those that decrease the pI of the protein (acidic changes). As described herein, all combinations of these variants can be done: one monomer may be wild type, or a variant that does not display a significantly different pI from wild-type, and the other can be either more basic or more acidic. Alternatively, each monomer is changed, one to more basic and one to more acidic.

Preferred combinations of pI variants are shown in FIG. 2. As outlined herein and shown in the figures, these changes are shown relative to IgG1, but all isotypes can be altered this way, as well as isotype hybrids. In the case where the heavy chain constant domain is from IgG2-4, R133E and R133Q can also be used.

In one embodiment, for example in the FIG. 36 formats, a preferred combination of pI variants has one monomer (the

negative Fab side) comprising 208D/295E/384D/418E/421D variants (N208D/Q295E/N384D/Q418E/N421D when relative to human IgG1) and a second monomer (the positive scFv side) comprising a positively charged scFv linker, including (GKPGS)₄ (SEQ ID NO: 1). However, as will be appreciated by those in the art, the first monomer includes a CH1 domain, including position 208. Accordingly, in constructs that do not include a CH1 domain (for example for antibodies that do not utilize a CH1 domain on one of the domains, for example in a dual scFv format or a “one-armed” format such as those depicted in FIG. 36B, C or D), a preferred negative pI variant Fc set includes 295E/384D/418E/421D variants (Q295E/N384D/Q418E/N421D when relative to human IgG1).

Accordingly, in some embodiments, one monomer has a set of substitutions from FIG. 4 and the other monomer has a charged linker (either in the form of a charged scFv linker because that monomer comprises an scFv or a charged domain linker, as the format dictates, which can be selected from those depicted in FIG. 5).

c. Isotypic Variants

In addition, many embodiments of the antibodies described herein rely on the “importation” of pI amino acids at particular positions from one IgG isotype into another, thus reducing or eliminating the possibility of unwanted immunogenicity being introduced into the variants. A number of these are shown in FIG. 21 of US Publ. 2014/0370013, hereby incorporated by reference. That is, IgG1 is a common isotype for therapeutic antibodies for a variety of reasons, including high effector function. However, the heavy constant region of IgG1 has a higher pI than that of IgG2 (8.10 versus 7.31). By introducing IgG2 residues at particular positions into the IgG1 backbone, the pI of the resulting monomer is lowered (or increased) and additionally exhibits longer serum half-life. For example, IgG1 has a glycine (pI 5.97) at position 137, and IgG2 has a glutamic acid (pI 3.22); importing the glutamic acid will affect the pI of the resulting protein. As is described below, a number of amino acid substitutions are generally required to significantly affect the pI of the variant antibody. However, it should be noted as discussed below that even changes in IgG2 molecules allow for increased serum half-life.

In other embodiments, non-isotypic amino acid changes are made, either to reduce the overall charge state of the resulting protein (e.g. by changing a higher pI amino acid to a lower pI amino acid), or to allow accommodations in structure for stability, etc. as is further described below.

In addition, by pI engineering both the heavy and light constant domains, significant changes in each monomer of the heterodimer can be seen. As discussed herein, having the pIs of the two monomers differ by at least 0.5 can allow separation by ion exchange chromatography or isoelectric focusing, or other methods sensitive to isoelectric point.

d. Calculating pI

The pI of each monomer can depend on the pI of the variant heavy chain constant domain and the pI of the total monomer, including the variant heavy chain constant domain and the fusion partner. Thus, in some embodiments, the change in pI is calculated on the basis of the variant heavy chain constant domain, using the chart in the FIG. 19 of US Pub. 2014/0370013. As discussed herein, which monomer to engineer is generally decided by the inherent pI of the Fv and scaffold regions. Alternatively, the pI of each monomer can be compared.

e. pI Variants that Also Confer Better FcRn In Vivo Binding
In the case where the pI variant decreases the pI of the monomer, they can have the added benefit of improving serum retention in vivo.

Although still under examination, Fc regions are believed to have longer half-lives in vivo, because binding to FcRn at pH 6 in an endosome sequesters the Fc (Ghetie and Ward, 1997 Immunol Today. 18(12): 592-598, entirely incorporated by reference). The endosomal compartment then recycles the Fc to the cell surface. Once the compartment opens to the extracellular space, the higher pH, ~7.4, induces the release of Fc back into the blood. In mice, Dall’Acqua et al. showed that Fc mutants with increased FcRn binding at pH 6 and pH 7.4 actually had reduced serum concentrations and the same half-life as wild-type Fc (Dall’Acqua et al. 2002, J. Immunol. 169:5171-5180, entirely incorporated by reference). The increased affinity of Fc for FcRn at pH 7.4 is thought to forbid the release of the Fc back into the blood. Therefore, the Fc mutations that will increase Fc’s half-life in vivo will ideally increase FcRn binding at the lower pH while still allowing release of Fc at higher pH. The amino acid histidine changes its charge state in the pH range of 6.0 to 7.4. Therefore, it is not surprising to find His residues at important positions in the Fc/FcRn complex.

Recently it has been suggested that antibodies with variable regions that have lower isoelectric points may also have longer serum half-lives (Igawa et al., 2010 PEDS. 23(5): 385-392, entirely incorporated by reference). However, the mechanism of this is still poorly understood. Moreover, variable regions differ from antibody to antibody. Constant region variants with reduced pI and extended half-life would provide a more modular approach to improving the pharmacokinetic properties of antibodies, as described herein.
f. Additional Fc Variants for Additional Functionality

In addition to the heterodimerization variants discussed above, there are a number of useful Fc amino acid modifications that can be made for a variety of reasons, including, but not limited to, altering binding to one or more FcγR receptors, altered binding to FcRn receptors, etc., as discussed below.

Accordingly, the antibodies provided herein (heterodimeric, as well as homodimeric) can include such amino acid modifications with or without the heterodimerization variants outlined herein (e.g., the pI variants and steric variants). Each set of variants can be independently and optionally included or excluded from any particular heterodimeric protein.

(i) FcγR Variants

Accordingly, there are a number of useful Fc substitutions that can be made to alter binding to one or more of the FcγR receptors. In certain embodiments, the subject antibody includes modifications that alter the binding to one or more FcγR receptors (i.e., “FcγR variants”). Substitutions that result in increased binding as well as decreased binding can be useful. For example, it is known that increased binding to FcγRIIIa generally results in increased ADCC (antibody dependent cell-mediated cytotoxicity; the cell-mediated reaction wherein nonspecific cytotoxic cells that express FcγRs recognize bound antibody on a target cell and subsequently cause lysis of the target cell). Similarly, decreased binding to FcγRIIb (an inhibitory receptor) can be beneficial as well in some circumstances. Amino acid substitutions that find use in the antibodies described herein include those listed in U.S. Pat. No. 8,188,321 (particularly FIG. 41) and U.S. Pat. No. 8,084,582, and US Publ. App. Nos. 20060235208 and 20070148170, all of which are expressly incorporated herein by reference in their entirety and specifically for the variants disclosed therein. Particular variants

that find use include, but are not limited to, 236A, 239D, 239E, 332E, 332D, 239D/332E, 267D, 267E, 328F, 267E/328F, 236A/332E, 239D/332E/330Y, 239D/332E/330L, 243A, 243L, 264A, 264V and 299T.

In addition, there are additional Fc substitutions that find use in increased binding to the FcRn receptor and increased serum half-life, as specifically disclosed in U.S. Ser. No. 12/341,769, hereby incorporated by reference in its entirety, including, but not limited to, 434S, 434A, 428L, 308F, 259I, 428L/434S, 428L/434A, 259I/308F, 436I/428L, 436I or V/434S, 436V/428L and 259I/308F/428L. Such modification may be included in one or both Fc domains of the subject antibody.

(ii) Ablation Variants

Similarly, another category of functional variants are “FcγR ablation variants” or “Fc knock out (FcKO or KO)” variants. In these embodiments, for some therapeutic applications, it is desirable to reduce or remove the normal binding of the Fc domain to one or more or all of the Fcγ receptors (e.g. FcγR1, FcγRIIa, FcγRIIb, FcγRIIIa, etc.) to avoid additional mechanisms of action. That is, for example, in many embodiments, particularly in the use of bispecific antibodies that bind CD3 monovalently it is generally desirable to ablate FcγRIIIa binding to eliminate or significantly reduce ADCC activity. Wherein one of the Fc domains comprises one or more Fcγ receptor ablation variants. These ablation variants are depicted in FIG. 3, and each can be independently and optionally included or excluded, with preferred aspects utilizing ablation variants selected from the group consisting of G236R/L328R, E233P/L234V/L235A/G236del/S239K, E233P/L234V/L235A/G236del/S267K, E233P/L234V/L235A/G236del/S239K/A327G, E233P/L234V/L235A/G236del/S267K/A327G and E233P/L234V/L235A/G236del. It should be noted that the ablation variants referenced herein ablate FcγR binding but generally not FcRn binding.

As is known in the art, the Fc domain of human IgG1 has the highest binding to the Fcγ receptors, and thus ablation variants can be used when the constant domain (or Fc domain) in the backbone of the heterodimeric antibody is IgG1. Alternatively, or in addition to ablation variants in an IgG1 background, mutations at the glycosylation position 297 (generally to A or S) can significantly ablate binding to FcγRIIIa, for example. Human IgG2 and IgG4 have naturally reduced binding to the Fcγ receptors, and thus those backbones can be used with or without the ablation variants.

B. Combination of Heterodimeric and Fc Variants

As will be appreciated by those in the art, all of the recited heterodimerization variants (including skew and/or pI variants) can be optionally and independently combined in any way, as long as they retain their “strandedness” or “monomer partition”. In some embodiments, the heterodimeric antibodies provided herein include the combination of heterodimerization skew variants, isosteric pI substitutions and FcKO variants as depicted in FIG. 4. In addition, all of these variants can be combined into any of the heterodimerization formats.

In the case of pI variants, while embodiments finding particular use are shown in the Figures, other combinations can be generated, following the basic rule of altering the pI difference between two monomers to facilitate purification.

In addition, any of the heterodimerization variants, skew and pI, are also independently and optionally combined with Fc ablation variants, Fc variants, FcRn variants, as generally outlined herein.

Exemplary combination of variants that are included in some embodiments of the heterodimeric 1+1 Fab-scFv-Fc

and 2+1 Fab₂-scFv-Fc format antibodies are included in FIG. 4. In certain embodiments, the antibody is a heterodimeric 1+1 Fab-scFv-Fc or 2+1 Fab₂-scFv-Fc format antibody as shown in FIGS. 17A and 17B.

C. Anti-CLDN6×Anti-CD3 Bispecific Antibodies

In another aspect, provided herein are anti-CLDN6×anti-CD3 (also referred to herein as “αCLDN6×αCD3”) bispecific antibodies. Such antibodies include at least one CLDN6 binding domain and at least one CD3 binding domain. In some embodiments, bispecific αCLDN6×αCD3 provided herein immune responses selectively in tumor sites that express CLDN6.

Note that unless specified herein, the order of the antigen list in the name does not confer structure; that is a CLDN6×CD3 1+1 Fab-scFv-Fc antibody can have the scFv bind to CLDN6 or CD3, although in some cases, the order specifies structure as indicated.

As is more fully outlined herein, these combinations of ABDs can be in a variety of formats, as outlined below, generally in combinations where one ABD is in a Fab format and the other is in an scFv format. Exemplary formats that are used in the bispecific antibodies provided herein include the 1+1 Fab-scFv-Fc and 2+1 Fab₂-scFv-Fv formats (see, e.g., FIGS. 17A and 17B). Other useful antibody formats include, but are not limited to, “mAb-Fv,” “mAb-scFv,” “central-Fv,” “one-armed scFv-mAb,” “scFv-mAb,” “dual scFv,” and “trident” format antibodies, as depicted in FIG. 36 and more fully described below.

In addition, in general, one of the ABDs comprises a scFv as outlined herein, in an orientation from N- to C-terminus of VH-scFv linker-VL or VL-scFv linker-VH. One or both of the other ABDs, according to the format, generally is a Fab, comprising a VH domain on one protein chain (generally as a component of a heavy chain) and a VL on another protein chain (generally as a component of a light chain).

As will be appreciated by those in the art, any set of 6 CDRs or VH and VL domains can be in the scFv format or in the Fab format, which is then added to the heavy and light constant domains, where the heavy constant domains comprise variants (including within the CH1 domain as well as the Fc domain). The scFv sequences contained in the sequence listing utilize a particular charged linker, but as outlined herein, uncharged or other charged linkers can be used, including those depicted in FIG. 5.

In addition, as discussed above, the numbering used in the Sequence Listing for the identification of the CDRs is Kabat, however, different numbering can be used, which will change the amino acid sequences of the CDRs as shown in Table 2.

For all of the variable heavy and light domains listed herein, further variants can be made. As outlined herein, in some embodiments the set of 6 CDRs can have from 0, 1, 2, 3, 4 or 5 amino acid modifications (with amino acid substitutions finding particular use), as well as changes in the framework regions of the variable heavy and light domains, as long as the frameworks (excluding the CDRs) retain at least about 80, 85, 90, 95 or 99% identity to a human germline sequence selected from those listed in FIG. 1 of U.S. Pat. No. 7,657,380, which Figure and Legend is incorporated by reference in its entirety herein. Thus, for example, the identical CDRs as described herein can be combined with different framework sequences from human germline sequences, as long as the framework regions retain at least 80, 85, 90, 95 or 99% identity to a human germline sequence selected from those listed in FIG. 1 of U.S. Pat. No. 7,657,380. Alternatively, the CDRs can have amino acid modifications (e.g., from 1, 2, 3, 4 or 5 amino acid modi-

fications in the set of CDRs (that is, the CDRs can be modified as long as the total number of changes in the set of 6 CDRs is less than 6 amino acid modifications, with any combination of CDRs being changed; e.g., there may be one change in vCDR1, two in vhCDR2, none in vhCDR3, etc.)), as well as having framework region changes, as long as the framework regions retain at least 80, 85 90, 95 or 99% identity to a human germline sequence selected from those listed in FIG. 1 of U.S. Pat. No. 7,657,380.

As discussed herein, the subject heterodimeric antibodies include two antigen binding domains (ABDs), each of which bind to CLDN6 or CD3. As outlined herein, these heterodimeric antibodies can be bispecific and bivalent (each antigen is bound by a single ABD, for example, in the format depicted in FIG. 17A), or bispecific and trivalent (one antigen is bound by a single ABD and the other is bound by two ABDs, for example as depicted in FIG. 17B).

In addition, in general, one of the ABDs comprises a scFv as outlined herein, in an orientation from N- to C-terminus of VH-scFv linker-VL or VL-scFv linker-VH. One or both of the other ABDs, according to the format, generally is a Fab, comprising a VH domain on one protein chain (generally as a component of a heavy chain) and a VL on another protein chain (generally as a component of a light chain).

The disclosure provides a number of ABDs as outlined below. As will be appreciated by those in the art, any set of 6 CDRs or VH and VL domains can be in the scFv format or in the Fab format, which is then added to the heavy and light constant domains, where the heavy constant domains comprise variants (including within the CH1 domain as well as the Fc domain). The scFv sequences contained in the sequence listing utilize a particular charged linker, but as outlined herein, uncharged or other charged linkers can be used, including those depicted in FIG. 5.

In addition, as discussed above, the numbering used in the Sequence Listing for the identification of the CDRs is Kabat, however, different numbering can be used, which will change the amino acid sequences of the CDRs as shown in Table 2.

For all of the variable heavy and light domains listed herein, further variants can be made. As outlined herein, in some embodiments the set of 6 CDRs can have from 0, 1, 2, 3, 4 or 5 amino acid modifications (with amino acid substitutions finding particular use), as well as changes in the framework regions of the variable heavy and light domains, as long as the frameworks (excluding the CDRs) retain at least about 80, 85 90, 95 or 99% identity to a human germline sequence selected from those listed in FIG. 1 of U.S. Pat. No. 7,657,380, which Figure and Legend is incorporated by reference in its entirety herein. Thus, for example, the identical CDRs as described herein can be combined with different framework sequences from human germline sequences, as long as the framework regions retain at least 80, 85 90, 95 or 99% identity to a human germline sequence selected from those listed in FIG. 1 of U.S. Pat. No. 7,657,380. Alternatively, the CDRs can have amino acid modifications (e.g. from 1, 2, 3, 4 or 5 amino acid modifications in the set of CDRs (that is, the CDRs can be modified as long as the total number of changes in the set of 6 CDRs is less than 6 amino acid modifications, with any combination of CDRs being changed; e.g. there may be one change in vCDR1, two in VHCDR2, none in VHCDR3, etc.)), as well as having framework region changes, as long as the framework regions retain at least 80, 85 90, 95 or 99% identity to a human germline sequence selected from those listed in FIG. 1 of U.S. Pat. No. 7,657,380.

1. CLDN6 Antigen Binding Domains

Herein is provided monoclonal and bispecific antibodies (e.g., the anti-CLDN6×anti-CD3 antibodies provided herein), and fusion proteins that contain antigen binding domains that bind to human CLDN6. Suitable sets of 6 CDRs and/or VH and VL domains are depicted in FIGS. 13, 14, 15 and 18. In some embodiments, the heterodimeric antibody is a 1+1 Fab-scFv-Fc or 2+1 Fab₂-scFv-Fv format antibody (see, e.g., FIGS. 17A and 17B) although any of the formats outlined below can be utilized.

In some embodiments, the CLDN6 ABD has a set of vhCDRs selected from the vhCDR1, vhCDR2 and vhCDR3 sequences from a VH selected from the group consisting of H1, H1.1, H1.2, H1.3, H1.4, H1.5, H1.6, H1.7, H1.8, H1.9, H1.19, H1.22, H1.24, H2, H2.1, H2.2, H2.3, H2.4, H2.5, H2.6, H2.7, H2.8, H2.9, H2.11, H2.12, H2.71, H2.75, H2.90, H2.91, H2.118 and H2.119, see FIGS. 14 and 15.

In some embodiments, the VH domain of the CLDN6 ABD is selected from the group consisting of H1, H1.1, H1.2, H1.3, H1.4, H1.5, H1.6, H1.7, H1.8, H1.9, H1.19, H1.22, H1.24, H2, H2.1, H2.2, H2.3, H2.4, H2.5, H2.6, H2.7, H2.8, H2.9, H2.11, H2.12, H2.71, H2.75, H2.90, H2.91, H2.118 and H2.119, see FIGS. 14 and 15.

In some embodiments, the CLDN6 ABD has a set of vlCDRs selected from the vlCDR1, vlCDR2 and vlCDR3 sequences from a VL selected from the group consisting of L1, L1.1, L1.4, L1.7, L1.16, L1.18, L1.19, L1.21, L1.22, L1.23, L1.27, L1.60, L1.107, L1.114, L1.187, L1.189 and L2, see FIGS. 14 and 15.

In some embodiments, the VL of the CLDN6 ABD is selected from the group consisting of L1, L1.1, L1.4, L1.7, L1.16, L1.7, L1.16, L1.18, L1.19, L1.21, L1.22, L1.23, L1.27, L1.60, L1.107, L1.114, L1.187, L1.189 and L2, see FIGS. 14 and 15.

Accordingly, included herein are CLDN6 ABDs that have a set of 6 CDRs (vhCDR1, vhCDR2, vhCDR3, vlCDR1, vlCDR2 and vlCDR3) from VH/VL pairs selected from the group consisting of: H1_L1, H1.1_L1, H1.2_L1, H1.3_L1, H1.4_L1, H1.5_L1, H1.6_L1, H1.7_L1, H1.8_L1, H1.9_L1, H1.19_L1, H1.22_L1, H1.24_L1, H2_L1, H2.1_L1, H2.2_L1, H2.3_L1, H2.4_L1, H2.5_L1, H2.6_L1, H2.7_L1, H2.8_L1, H2.9_L1, H2.11_L1, H2.12_L1, H2.71_L1, H2.75_L1, H2.90_L1, H2.91_L1, H2.118_L1, H2.119_L1, H1_L1.1, H1.1_L1.1, H1.2_L1.1, H1.3_L1.1, H1.4_L1.1, H1.5_L1.1, H1.6_L1.1, H1.7_L1.1, H1.8_L1.1, H1.9_L1.1, H1.19_L1.1, H1.22_L1.1, H1.24_L1.1, H2_L1.1, H2.1_L1.1, H2.2_L1.1, H2.3_L1.1, H2.4_L1.1, H2.5_L1.1, H2.6_L1.1, H2.7_L1.1, H2.8_L1.1, H2.9_L1.1, H2.11_L1.1, H2.12_L1.1, H2.71_L1.1, H2.75_L1.1, H2.90_L1.1, H2.91_L1.1, H2.118_L1.1, H2.119_L1.1, H1_L1.4, H1.1_L1.4, H1.2_L1.4, H1.3_L1.4, H1.4_L1.4, H1.5_L1.4, H1.6_L1.4, H1.7_L1.4, H1.8_L1.4, H1.9_L1.4, H1.19_L1.4, H1.22_L1.4, H1.24_L1.4, H2_L1.4, H2.1_L1.4, H2.2_L1.4, H2.3_L1.4, H2.4_L1.4, H2.5_L1.4, H2.6_L1.4, H2.7_L1.4, H2.8_L1.4, H2.9_L1.4, H2.11_L1.4, H2.12_L1.4, H2.71_L1.4, H2.75_L1.4, H2.90_L1.4, H2.91_L1.4, H2.118_L1.4, H2.119_L1.4, H1_L1.7, H1.1_L1.7, H1.2_L1.7, H1.3_L1.7, H1.4_L1.7, H1.5_L1.7, H1.6_L1.7, H1.7_L1.7, H1.8_L1.7, H1.9_L1.7, H1.19_L1.7, H1.22_L1.7, H1.24_L1.7, H2_L1.7, H2.1_L1.7, H2.2_L1.7, H2.3_L1.7, H2.4_L1.7, H2.5_L1.7, H2.6_L1.7, H2.7_L1.7, H2.8_L1.7, H2.9_L1.7, H2.11_L1.7, H2.12_L1.7, H2.71_L1.7, H2.75_L1.7, H2.90_L1.7, H2.91_L1.7, H2.118_L1.7, H2.119_L1.7, H1_L1.16, H1.1_L1.16, H1.2_L1.16, H1.3_L1.16, H1.4_L1.16, H1.5_L1.16, H1.6_L1.16, H1.7_L1.16, H1.8_L1.16, H1.9_L1.16, H1.19_L1.16, H1.22_L1.16, H1.24_L1.16, H2_L1.16, H2.1_L1.16, H2.2_L1.16,

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H2.3_L1.16, H2.4_L1.16, H2.5_L1.16, H2.6_L1.16,
 H2.7_L1.16, H2.8_L1.16, H2.9_L1.16, H2.11_L1.16,
 H2.12_L1.16, H2.71_L1.16, H2.75_L1.16, H2.90_L1.16,
 H2.91_L1.16, H2.118_L1.16, H2.119_L1.16, H1_L1.18,
 H1.1_L1.18, H1.2_L1.18, H1.3_L1.18, H1.4_L1.18,
 H1.5_L1.18, H1.6_L1.18, H1.7_L1.18, H1.8_L1.18,
 H1.9_L1.18, H1.19_L1.18, H1.22_L1.18, H1.24_L1.18,
 H2_L1.18, H2.1_L1.18, H2.2_L1.18, H2.3_L1.18,
 H2.4_L1.18, H2.5_L1.18, H2.6_L1.18, H2.7_L1.18,
 H2.8_L1.18, H2.9_L1.18, H2.11_L1.18, H2.12_L1.18,
 H2.71_L1.18, H2.75_L1.18, H2.90_L1.18, H2.91_L1.18,
 H2.118_L1.18, H2.119_L1.18, H1_L1.19, H1.1_L1.19,
 H1.2_L1.19, H1.3_L1.19, H1.4_L1.19, H1.5_L1.19,
 H1.6_L1.19, H1.7_L1.19, H1.8_L1.19, H1.9_L1.19,
 H1.19_L1.19, H1.22_L1.19, H1.24_L1.19, H2_L1.19,
 H2.1_L1.19, H2.2_L1.19, H2.3_L1.19, H2.4_L1.19,
 H2.5_L1.19, H2.6_L1.19, H2.7_L1.19, H2.8_L1.19,
 H2.9_L1.19, H2.11_L1.19, H2.12_L1.19, H2.71_L1.19,
 H2.75_L1.19, H2.90_L1.19, H2.91_L1.19, H2.118_L1.19,
 H2.119_L1.19, H1_L1.21, H1.1_L1.21, H1.2_L1.21,
 H1.3_L1.21, H1.4_L1.21, H1.5_L1.21, H1.6_L1.21,
 H1.7_L1.21, H1.8_L1.21, H1.9_L1.21, H1.19_L1.21,
 H1.22_L1.21, H1.24_L1.21, H2_L1.21, H2.1_L1.21,
 H2.2_L1.21, H2.3_L1.21, H2.4_L1.21, H2.5_L1.21,
 H2.6_L1.21, H2.7_L1.21, H2.8_L1.21, H2.9_L1.21,
 H2.11_L1.21, H2.12_L1.21, H2.71_L1.21, H2.75_L1.21,
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 H1.4_L1.22, H1.5_L1.22, H1.6_L1.22, H1.7_L1.22,
 H1.8_L1.22, H1.9_L1.22, H1.19_L1.22, H1.22_L1.22,
 H1.24_L1.22, H2_L1.22, H2.1_L1.22, H2.2_L1.22,
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 H2.12_L1.22, H2.71_L1.22, H2.75_L1.22, H2.90_L1.22,
 H2.91_L1.22, H2.118_L1.22, H2.119_L1.22, H1_L1.23,
 H1.1_L1.23, H1.2_L1.23, H1.3_L1.23, H1.4_L1.23,
 H1.5_L1.23, H1.6_L1.23, H1.7_L1.23, H1.8_L1.23,
 H1.9_L1.23, H1.19_L1.23, H1.22_L1.23, H1.24_L1.23,
 H2_L1.23, H2.1_L1.23, H2.2_L1.23, H2.3_L1.23,
 H2.4_L1.23, H2.5_L1.23, H2.6_L1.23, H2.7_L1.23,
 H2.8_L1.23, H2.9_L1.23, H2.11_L1.23, H2.12_L1.23,
 H2.71_L1.23, H2.75_L1.23, H2.90_L1.23, H2.91_L1.23,
 H2.118_L1.23, H2.119_L1.23, H1_L1.27, H1.1_L1.27,
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 H1.6_L1.27, H1.7_L1.27, H1.8_L1.27, H1.9_L1.27,
 H1.19_L1.27, H1.22_L1.27, H1.24_L1.27, H2_L1.27,
 H2.1_L1.27, H2.2_L1.27, H2.3_L1.27, H2.4_L1.27,
 H2.5_L1.27, H2.6_L1.27, H2.7_L1.27, H2.8_L1.27,
 H2.9_L1.27, H2.11_L1.27, H2.12_L1.27, H2.71_L1.27,
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 H1.7_L1.60, H1.8_L1.60, H1.9_L1.60, H1.19_L1.60,
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 H2.6_L1.60, H2.7_L1.60, H2.8_L1.60, H2.9_L1.60,
 H2.11_L1.60, H2.12_L1.60, H2.71_L1.60, H2.75_L1.60,
 H2.90_L1.60, H2.91_L1.60, H2.118_L1.60, H2.119_L1.60,
 H1_L1.107, H1.1_L1.107, H1.2_L1.107, H1.3_L1.107,
 H1.4_L1.107, H1.5_L1.107, H1.6_L1.107, H1.7_L1.107,
 H1.8_L1.107, H1.9_L1.107, H1.19_L1.107, H1.22_L1.107,
 H1.24_L1.107, H2_L1.107, H2.1_L1.107, H2.2_L1.107,
 H2.3_L1.107, H2.4_L1.107, H2.5_L1.107, H2.6_L1.107,
 H2.7_L1.107, H2.8_L1.107, H2.9_L1.107, H2.11_L1.107,
 H2.12_L1.107, H2.71_L1.107, H2.75_L1.107,
 H2.90_L1.107, H2.91_L1.107, H2.118_L1.107,
 H2.119_L1.107, H1_L1.114, H1.1_L1.114, H1.2_L1.114,

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H1.3_L1.114, H1.4_L1.114, H1.5_L1.114, H1.6_L1.114,
 H1.7_L1.114, H1.8_L1.114, H1.9_L1.114, H1.19_L1.114,
 H1.22_L1.114, H1.24_L1.114, H2_L1.114, H2.1_L1.114,
 H2.2_L1.114, H2.3_L1.114, H2.4_L1.114, H2.5_L1.114,
 H2.6_L1.114, H2.7_L1.114, H2.8_L1.114, H2.9_L1.114,
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 H2.75_L1.114, H2.90_L1.114, H2.91_L1.114,
 H2.118_L1.114, H2.119_L1.114, H1_L1.187,
 H1.1_L1.187, H1.2_L1.187, H1.3_L1.187, H1.4_L1.187,
 H1.5_L1.187, H1.6_L1.187, H1.7_L1.187, H1.8_L1.187,
 H1.9_L1.187, H1.19_L1.187, H1.22_L1.187,
 H1.24_L1.187, H2_L1.187, H2.1_L1.187, H2.2_L1.187,
 H2.3_L1.187, H2.4_L1.187, H2.5_L1.187, H2.6_L1.187,
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 H2.2_L1.189, H2.3_L1.189, H2.4_L1.189, H2.5_L1.189,
 H2.6_L1.189, H2.7_L1.189, H2.8_L1.189, H2.9_L1.189,
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 H2.75_L1.189, H2.90_L1.189, H2.91_L1.189,
 H2.118_L1.189, H2.119_L1.189, H1_L2, H1.1_L2,
 H1.2_L2, H1.3_L2, H1.4_L2, H1.5_L2, H1.6_L2, H1.7_L2,
 H1.8_L2, H1.9_L2, H1.19_L2, H1.22_L2, H1.24_L2,
 H2_L2, H2.1_L2, H2.2_L2, H2.3_L2, H2.4_L2, H2.5_L2,
 H2.6_L2, H2.7_L2, H2.8_L2, H2.9_L2, H2.11_L2,
 H2.12_L2, H2.71_L2, H2.75_L2, H2.90_L2, H2.91_L2,
 H2.118_L2 and H2.119_L2.
 Additionally, included herein are CLDN6 ABDs that have
 VH/VL pairs selected from the group consisting of: H1_L1,
 H1.1_L1, H1.2_L1, H1.3_L1, H1.4_L1, H1.5_L1, H1.6_L1,
 H1.7_L1, H1.8_L1, H1.9_L1, H1.19_L1, H1.22_L1,
 H1.24_L1, H2_L1, H2.1_L1, H2.2_L1, H2.3_L1, H2.4_L1,
 H2.5_L1, H2.6_L1, H2.7_L1, H2.8_L1, H2.9_L1,
 H2.11_L1, H2.12_L1, H2.71_L1, H2.75_L1, H2.90_L1,
 H2.91_L1, H2.118_L1, H2.119_L1, H1_L1.1, H1.1_L1.1,
 H1.2_L1.1, H1.3_L1.1, H1.4_L1.1, H1.5_L1.1, H1.6_L1.1,
 H1.7_L1.1, H1.8_L1.1, H1.9_L1.1, H1.19_L1.1,
 H1.22_L1.1, H1.24_L1.1, H2_L1.1, H2.1_L1.1, H2.2_L1.1,
 H2.3_L1.1, H2.4_L1.1, H2.5_L1.1, H2.6_L1.1, H2.7_L1.1,
 H2.8_L1.1, H2.9_L1.1, H2.11_L1.1, H2.12_L1.1,
 H2.71_L1.1, H2.75_L1.1, H2.90_L1.1, H2.91_L1.1,
 H2.118_L1.1, H2.119_L1.1, H1_L1.4, H1.1_L1.4,
 H1.2_L1.4, H1.3_L1.4, H1.4_L1.4, H1.5_L1.4, H1.6_L1.4,
 H1.7_L1.4, H1.8_L1.4, H1.9_L1.4, H1.19_L1.4,
 H1.22_L1.4, H1.24_L1.4, H2_L1.4, H2.1_L1.4, H2.2_L1.4,
 H2.3_L1.4, H2.4_L1.4, H2.5_L1.4, H2.6_L1.4, H2.7_L1.4,
 H2.8_L1.4, H2.9_L1.4, H2.11_L1.4, H2.12_L1.4,
 H2.71_L1.4, H2.75_L1.4, H2.90_L1.4, H2.91_L1.4,
 H2.118_L1.4, H2.119_L1.4, H1_L1.7, H1.1_L1.7,
 H1.2_L1.7, H1.3_L1.7, H1.4_L1.7, H1.5_L1.7, H1.6_L1.7,
 H1.7_L1.7, H1.8_L1.7, H1.9_L1.7, H1.19_L1.7,
 H1.22_L1.7, H1.24_L1.7, H2_L1.7, H2.1_L1.7, H2.2_L1.7,
 H2.3_L1.7, H2.4_L1.7, H2.5_L1.7, H2.6_L1.7, H2.7_L1.7,
 H2.8_L1.7, H2.9_L1.7, H2.11_L1.7, H2.12_L1.7,
 H2.71_L1.7, H2.75_L1.7, H2.90_L1.7, H2.91_L1.7,
 H2.118_L1.7, H2.119_L1.7, H1_L1.16, H1.1_L1.16,
 H1.2_L1.16, H1.3_L1.16, H1.4_L1.16, H1.5_L1.16,
 H1.6_L1.16, H1.7_L1.16, H1.8_L1.16, H1.9_L1.16,
 H1.19_L1.16, H1.22_L1.16, H1.24_L1.16, H2_L1.16,
 H2.1_L1.16, H2.2_L1.16, H2.3_L1.16, H2.4_L1.16,
 H2.5_L1.16, H2.6_L1.16, H2.7_L1.16, H2.8_L1.16,
 H2.9_L1.16, H2.11_L1.16, H2.12_L1.16, H2.71_L1.16,
 H2.75_L1.16, H2.90_L1.16, H2.91_L1.16, H2.118_L1.16,

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H2.119_L1.16, H1_L1.18, H1.1_L1.18, H1.2_L1.18, H1.3_L1.18, H1.4_L1.18, H1.5_L1.18, H1.6_L1.18, H1.7_L1.18, H1.8_L1.18, H1.9_L1.18, H1.19_L1.18, H1.22_L1.18, H1.24_L1.18, H2_L1.18, H2.1_L1.18, H2.2_L1.18, H2.3_L1.18, H2.4_L1.18, H2.5_L1.18, H2.6_L1.18, H2.7_L1.18, H2.8_L1.18, H2.9_L1.18, H2.11_L1.18, H2.12_L1.18, H2.71_L1.18, H2.75_L1.18, H2.90_L1.18, H2.91_L1.18, H2.118_L1.18, H2.119_L1.18, H1_L1.19, H1.1_L1.19, H1.2_L1.19, H1.3_L1.19, H1.4_L1.19, H1.5_L1.19, H1.6_L1.19, H1.7_L1.19, H1.8_L1.19, H1.9_L1.19, H1.19_L1.19, H1.22_L1.19, H1.24_L1.19, H2_L1.19, H2.1_L1.19, H2.2_L1.19, H2.3_L1.19, H2.4_L1.19, H2.5_L1.19, H2.6_L1.19, H2.7_L1.19, H2.8_L1.19, H2.9_L1.19, H2.11_L1.19, H2.12_L1.19, H2.71_L1.19, H2.75_L1.19, H2.90_L1.19, H2.91_L1.19, H2.118_L1.19, H2.119_L1.19, H1_L1.21, H1.1_L1.21, H1.2_L1.21, H1.3_L1.21, H1.4_L1.21, H1.5_L1.21, H1.6_L1.21, H1.7_L1.21, H1.8_L1.21, H1.9_L1.21, H1.19_L1.21, H1.22_L1.21, H1.24_L1.21, H2_L1.21, H2.1_L1.21, H2.2_L1.21, H2.3_L1.21, H2.4_L1.21, H2.5_L1.21, H2.6_L1.21, H2.7_L1.21, H2.8_L1.21, H2.9_L1.21, H2.11_L1.21, H2.12_L1.21, H2.71_L1.21, H2.75_L1.21, H2.90_L1.21, H2.91_L1.21, H2.118_L1.2, H2.119_L1.21, H1_L1.22, H1.1_L1.22, H1.2_L1.22, H1.3_L1.22, H1.4_L1.22, H1.5_L1.22, H1.6_L1.22, H1.7_L1.22, H1.8_L1.22, H1.9_L1.22, H1.19_L1.22, H1.22_L1.22, H1.24_L1.22, H2_L1.22, H2.1_L1.22, H2.2_L1.22, H2.3_L1.22, H2.4_L1.22, H2.5_L1.22, H2.6_L1.22, H2.7_L1.22, H2.8_L1.22, H2.9_L1.22, H2.11_L1.22, H2.12_L1.22, H2.71_L1.22, H2.75_L1.22, H2.90_L1.22, H2.91_L1.22, H2.118_L1.22, H2.119_L1.22, H1_L1.23, H1.1_L1.23, H1.2_L1.23, H1.3_L1.23, H1.4_L1.23, H1.5_L1.23, H1.6_L1.23, H1.7_L1.23, H1.8_L1.23, H1.9_L1.23, H1.19_L1.23, H1.22_L1.23, H1.24_L1.23, H2_L1.23, H2.1_L1.23, H2.2_L1.23, H2.3_L1.23, H2.4_L1.23, H2.5_L1.23, H2.6_L1.23, H2.7_L1.23, H2.8_L1.23, H2.9_L1.23, H2.11_L1.23, H2.12_L1.23, H2.71_L1.23, H2.75_L1.23, H2.90_L1.23, H2.91_L1.23, H2.118_L1.23, H2.119_L1.23, H1_L1.27, H1.1_L1.27, H1.2_L1.27, H1.3_L1.27, H1.4_L1.27, H1.5_L1.27, H1.6_L1.27, H1.7_L1.27, H1.8_L1.27, H1.9_L1.27, H1.19_L1.27, H1.22_L1.27, H1.24_L1.27, H2_L1.27, H2.1_L1.27, H2.2_L1.27, H2.3_L1.27, H2.4_L1.27, H2.5_L1.27, H2.6_L1.27, H2.7_L1.27, H2.8_L1.27, H2.9_L1.27, H2.11_L1.27, H2.12_L1.27, H2.71_L1.27, H2.75_L1.27, H2.90_L1.27, H2.91_L1.27, H2.118_L1.27, H2.119_L1.27, H1_L1.60, H1.1_L1.60, H1.2_L1.60, H1.3_L1.60, H1.4_L1.60, H1.5_L1.60, H1.6_L1.60, H1.7_L1.60, H1.8_L1.60, H1.9_L1.60, H1.19_L1.60, H1.22_L1.60, H1.24_L1.60, H2_L1.60, H2.1_L1.60, H2.2_L1.60, H2.3_L1.60, H2.4_L1.60, H2.5_L1.60, H2.6_L1.60, H2.7_L1.60, H2.8_L1.60, H2.9_L1.60, H2.11_L1.60, H2.12_L1.60, H2.71_L1.60, H2.75_L1.60, H2.90_L1.60, H2.91_L1.60, H2.118_L1.60, H2.119_L1.60, H1_L1.107, H1.1_L1.107, H1.2_L1.107, H1.3_L1.107, H1.4_L1.107, H1.5_L1.107, H1.6_L1.107, H1.7_L1.107, H1.8_L1.107, H1.9_L1.107, H1.19_L1.107, H1.22_L1.107, H1.24_L1.107, H2_L1.107, H2.1_L1.107, H2.2_L1.107, H2.3_L1.107, H2.4_L1.107, H2.5_L1.107, H2.6_L1.107, H2.7_L1.107, H2.8_L1.107, H2.9_L1.107, H2.11_L1.107, H2.12_L1.107, H2.71_L1.107, H2.75_L1.107, H2.90_L1.107, H2.91_L1.107, H2.118_L1.107, H2.119_L1.107, H1_L1.114, H1.1_L1.114, H1.2_L1.114, H1.3_L1.114, H1.4_L1.114, H1.5_L1.114, H1.6_L1.114, H1.7_L1.114, H1.8_L1.114, H1.9_L1.114, H1.19_L1.114, H1.22_L1.114, H1.24_L1.114, H2_L1.114, H2.1_L1.114, H2.2_L1.114,

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H2.3_L1.114, H2.4_L1.114, H2.5_L1.114, H2.6_L1.114, H2.7_L1.114, H2.8_L1.114, H2.9_L1.114, H2.11_L1.114, H2.12_L1.114, H2.71_L1.114, H2.75_L1.114, H2.90_L1.114, H2.91_L1.114, H2.118_L1.114, H2.119_L1.114, H1_L1.187, H1.1_L1.187, H1.2_L1.187, H1.3_L1.187, H1.4_L1.187, H1.5_L1.187, H1.6_L1.187, H1.7_L1.187, H1.8_L1.187, H1.9_L1.187, H1.19_L1.187, H1.22_L1.187, H1.24_L1.187, H2_L1.187, H2.1_L1.187, H2.2_L1.187, H2.3_L1.187, H2.4_L1.187, H2.5_L1.187, H2.6_L1.187, H2.7_L1.187, H2.8_L1.187, H2.9_L1.187, H2.11_L1.187, H2.12_L1.187, H2.71_L1.187, H2.75_L1.187, H2.90_L1.187, H2.91_L1.187, H2.118_L1.187, H2.119_L1.187, H1_L1.189, H1.1_L1.189, H1.2_L1.189, H1.3_L1.189, H1.4_L1.189, H1.5_L1.189, H1.6_L1.189, H1.7_L1.189, H1.8_L1.189, H1.9_L1.189, H1.19_L1.189, H1.22_L1.189, H1.24_L1.189, H2_L1.189, H2.1_L1.189, H2.2_L1.189, H2.3_L1.189, H2.4_L1.189, H2.5_L1.189, H2.6_L1.189, H2.7_L1.189, H2.8_L1.189, H2.9_L1.189, H2.11_L1.189, H2.12_L1.189, H2.71_L1.189, H2.75_L1.189, H2.90_L1.189, H2.91_L1.189, H2.118_L1.189, H2.119_L1.189, H1_L2, H1.1_L2, H1.2_L2, H1.3_L2, H1.4_L2, H1.5_L2, H1.6_L2, H1.7_L2, H1.8_L2, H1.9_L2, H1.19_L2, H1.22_L2, H1.24_L2, H2_L2, H2.1_L2, H2.2_L2, H2.3_L2, H2.4_L2, H2.5_L2, H2.6_L2, H2.7_L2, H2.8_L2, H2.9_L2, H2.11_L2, H2.12_L2, H2.71_L2, H2.75_L2, H2.90_L2, H2.91_L2, H2.118_L2 and H2.119_L2.

In particular embodiments, the VH/VL pairs are selected from the group consisting of H1.9_L1.187, H1.24_L1.187, H2.91_L1.187 and H1.9_L1.187.

In particular embodiments, the VH/VL pairs are Fabs and are selected from the group consisting of H1.9_L1.187, H1.24_L1.187, H2.91_L1.187 and H1.9_L1.187.

As will be appreciated by those in the art, suitable CLDN6 binding domains can comprise a set of 6 CDRs as depicted in the Figures, either as they are underlined or, in the case where a different numbering scheme is used as described herein and as shown in Table 2, as the CDRs that are identified using other alignments within the VH and VL sequences of those depicted in FIGS. 13, 14 and 15. Suitable ABDs can also include the entire VH and VL sequences as depicted in these sequences and Figures, used as scFvs or as Fabs. In many of the embodiments herein that contain an Fv to CLDN6, it is the Fab monomer that binds CLDN6.

In addition to the parental CDR sets disclosed in the figures and sequence listing that form an ABD to CLDN6, provided herein are variant CLDN6 ABDs having CDRs that include at least one modification of the CLDN6 ABD CDRs disclosed herein (e.g., (FIGS. 13-15 and 18 and the sequence listing). In one embodiment, the CLDN6 ABD of the subject heterodimeric antibody includes a set of 6 CDRs with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 amino acid modifications as compared to the 6 CDRs of a CLDN6 binding domain VH/VL pair as described herein, including the figures and sequence listing. In exemplary embodiments, the CLDN6 ABD of the subject heterodimeric antibody includes a set of 6 CDRs with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 amino acid modifications as compared to the 6 CDRs of one of the following CLDN6 binding domain VH/VL pairs: H1.9_L1.187, H1.24_L1.187, H2.91_L1.187 and H1.9_L1.187. In certain embodiments, the CLDN6 ABD of the subject antibody is capable of binding to CLDN6, as measured at least one of a Biacore, surface plasmon resonance (SPR), BLI (biolayer interferometry, e.g., Octet assay) assay, and/or flow cytometry, with the latter finding particular use in many embodiments. In particular embodiments, the CLDN6 ABD is capable of

binding human CLDN6 (see FIG. 11). In some cases, each variant CDR has no more than 1 or 2 amino acid changes, with no more than 1 per CDR being particularly useful.

In some embodiments, the CLDN6 ABD of the subject antibody includes 6 CDRs that are at least 90, 95, 97, 98 or 99% identical to the 6 CDRs of a CLDN6 ABD as described herein, including the figures and sequence listing. In exemplary embodiments, the CLDN6 ABD of the subject antibody includes 6 CDRs that are at least 90, 95, 97, 98 or 99% identical to the 6 CDRs of one of the following CLDN6 binding domain VH/VL pairs: H1.9_L1.187, H1.24_L1.187, H2.91_L1.187 and H1.9_L1.187. In certain embodiments, the CLDN6 ABD of the subject antibody is capable of binding to CLDN6, as measured at least one of a Biacore, surface plasmon resonance (SPR), BLI (biolayer interferometry, e.g., Octet assay) assay, and/or flow cytometry, with the latter finding particular use in many embodiments. In particular embodiments, the CLDN6 ABD is capable of binding human CLDN6 antigen (see FIG. 11).

In another exemplary embodiment, the CLDN6 ABD of the subject antibody includes the variable heavy (VH) domain and variable light (VL) domain of any one of the CLDN6 binding domain VH/VL pairs described herein, including the figures and sequence listing.

In some embodiments, the subject antibody includes a CLDN6 ABD that includes a variable heavy domain and/or a variable light domain that are variants of a CLDN6 ABD VH and VL domain disclosed herein. In one embodiment, the variant VH domain and/or VL domain has from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid changes from a VH and/or VL domain of a CLDN6 ABD described herein, including the figures and sequence listing. In exemplary embodiments, the variant VH domain and/or VL domain has from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid changes from a VH and/or VL domain of one of the following CLDN6 binding domain VH/VL pairs: H1.9_L1.187, H1.24_L1.187, H2.91_L1.187 and H1.9_L1.187. In some embodiments, the changes are in a VH domain depicted in FIGS. 13-15 and 18. In some embodiments, the changes are in a VL domain are depicted in FIGS. 13-15 and 18. In some embodiments, one or more amino acid changes are in the VH and/or VL framework regions (FR1, FR2, FR3, and/or FR4). In some embodiments, one or more amino acid changes are in one or more CDRs. In certain embodiments, the CLDN6 ABD of the subject antibody is capable of binding to CLDN6, as measured at least one of a Biacore, surface plasmon resonance (SPR), BLI (biolayer interferometry, e.g., Octet assay) assay, and/or flow cytometry, with the latter finding particular use in many embodiments. In particular embodiments, the CLDN6 ABD is capable of binding human CLDN6 antigen (see FIG. 11).

In one embodiment, the variant VH and/or VL domain is at least 90, 95, 97, 98 or 99% identical to the VH and/or VL of a CLDN6 ABD as described herein, including the figures and sequence listing. In exemplary embodiments, the variant VH and/or VL domain is at least 90, 95, 97, 98 or 99% identical to the VH and/or VL of one of the following CLDN6 binding domain VH/VL pairs: H1.9_L1.187, H1.24_L1.187, H2.91_L1.187 and H1.9_L1.187. In some embodiments, the CLDN6 ABD includes a VH that is at least 90, 95, 97, 98 or 99% identical to VH domain depicted in FIGS. 13-15 and 18. In some embodiments, the CLDN6 ABD includes a VL that is at least 90, 95, 97, 98 or 99% identical to VL domain depicted in FIGS. 13-15 and 18. In some embodiments, the CLDN6 ABD includes a VH and a VL that is at least 90, 95, 97, 98 or 99% identical to a VH

domain and a VL domain depicted in FIGS. 13-15 and 18. In certain embodiments, the CLDN6 ABD of the subject antibody is capable of binding to CLDN6, as measured at least one of a Biacore, surface plasmon resonance (SPR), BLI (biolayer interferometry, e.g., Octet assay) assay, and/or flow cytometry, with the latter finding particular use in many embodiments. In particular embodiments, the CLDN6 ABD is capable of binding human CLDN6 antigen (see FIG. 11).

2. CD3 Antigen Binding Domains
The heterodimeric bispecific of the invention (e.g., anti-CLDN6xanti-CD3 antibodies) also include an ABD that binds to human epsilon CD3 (CD3ε).

Suitable sets of 6 CDRs and/or VH and VL domains, as well as scFv sequences, are depicted in FIG. 10. CD3 binding domain sequences that are of particular use include, but are not limited to, anti-CD3 H1.30_L1.47, anti-CD3 H1.32_L1.47, anti-CD3 H1.89_L1.47, anti-CD3 H1.90_L1.47, anti-CD3 H1.33_L1.47, anti-CD3 H1.31_L1.47, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.32, anti-CD3 L1.47_H1.89, anti-CD3 L1.47_H1.90, anti-CD3 L1.47_H1.33, and anti-CD3 L1.47_H1.31 as depicted in FIG. 10. As shown in FIG. 10, when the anti-CD3 ABD is a scFv domain, the VH and VL domains can be in either orientation.

As will be appreciated by those in the art, suitable CD3 binding domains can comprise a set of 6 CDRs as depicted in FIG. 10, either as they are underlined or, in the case where a different numbering scheme is used as described herein and as shown in Table 2, as the CDRs that are identified using other alignments within the VH and VL sequences of those depicted in FIG. 10A-10F. Suitable ABDs can also include the entire VH and VL sequences as depicted in these sequences and Figures, used as scFvs or as Fabs. In many of the embodiments herein that contain an Fv to CD3, it is the scFv monomer that binds CD3.

In addition to the parental CDR sets disclosed in the figures and sequence listing that form an ABD to CD3, provided herein are variant CD3 ABDs having CDRs that include at least one modification of the CD3 ABD CDRs disclosed herein (e.g., (FIG. 10 and the sequence listing). In one embodiment, the CD3 ABD of the subject heterodimeric antibody (e.g., anti-CLDN6xanti-CD3 antibody) includes a set of 6 CDRs with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 amino acid modifications as compared to the 6 CDRs of a CD3 ABD as described herein, including the figures and sequence listing. In exemplary embodiments, the CD3 ABD of the subject heterodimeric antibody includes a set of 6 CDRs with 1, 2, 3, 4, 5, 6, 7, 8, 9, 10 amino acid modifications as compared to the 6 CDRs of one of the following CD3 binding domains: anti-CD3 H1.30_L1.47, anti-CD3 H1.32_L1.47, anti-CD3 H1.89_L1.47, anti-CD3 H1.90_L1.47, anti-CD3 H1.33_L1.47, anti-CD3 H1.31_L1.47, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.32, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.32, anti-CD3 L1.47_H1.89, anti-CD3 L1.47_H1.90, anti-CD3 L1.47_H1.33, and anti-CD3 L1.47_H1.31 (FIG. 10). In certain embodiments, the CD3 ABD of the subject antibody is capable of binding CD3 antigen, as measured by at least one of a Biacore, surface plasmon resonance (SPR), flow cytometry, and/or BLI (biolayer interferometry, e.g., Octet assay) assay, with the latter finding particular use in many embodiments. In particular embodiments, the CD3ABD is capable of binding human CD3.

In some embodiments, the CD3 ABD of the subject antibody includes 6 CDRs that are at least 90, 95, 97, 98 or 99% identical to the 6 CDRs of a CD3 ABD as described

herein, including the figures and sequence listing. In exemplary embodiments, the CD3 ABD of the subject antibody includes 6 CDRs that are at least 90, 95, 97, 98 or 99% identical to the 6 CDRs of one of the following CD3 binding domains: anti-CD3 H1.30_L1.47, anti-CD3 H1.32_L1.47, anti-CD3 H1.89_L1.47, anti-CD3 H1.90_L1.47, anti-CD3 H1.33_L1.47, anti-CD3 H1.31_L1.47, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.32, anti-CD3 L1.47_H1.89, anti-CD3 L1.47_H1.90, anti-CD3 L1.47_H1.33, and anti-CD3 L1.47_H1.31 (FIG. 10). In certain embodiments, the CD3 ABD is capable of binding to the CD3, as measured by at least one of a Biacore, surface plasmon resonance (SPR), flow cytometry, and/or BLI (biolayer interferometry, e.g., Octet assay) assay, with the latter finding particular use in many embodiments. In particular embodiments, the CD3 ABD is capable of binding human CD3 antigen.

In another exemplary embodiment, the CD3 ABD of the subject antibody includes the variable heavy (VH) domain and variable light (VL) domain of any one of the CD3 binding domains described herein, including the figures and sequence listing.

In some embodiments, the subject antibody includes a CD3 ABD that includes a variable heavy domain and/or a variable light domain that are variants of a CD3 ABD VH and VL domain disclosed herein. In one embodiment, the variant VH domain and/or VL domain has from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid changes from a VH and/or VL domain of a CD3 ABD described herein, including the figures and sequence listing. In exemplary embodiments, the variant VH domain and/or VL domain has from 1, 2, 3, 4, 5, 6, 7, 8, 9 or 10 amino acid changes from a VH and/or VL domain of one of the following CD3 binding domains: anti-CD3 H1.30_L1.47, anti-CD3 H1.32_L1.47, anti-CD3 H1.89_L1.47, anti-CD3 H1.90_L1.47, anti-CD3 H1.33_L1.47, anti-CD3 H1.31_L1.47, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.32, anti-CD3 L1.47_H1.89, anti-CD3 L1.47_H1.90, anti-CD3 L1.47_H1.33, and anti-CD3 L1.47_H1.31 (FIG. 10). In some embodiments, the changes are in a VH domain depicted in FIG. 10. In some embodiments, the changes are in a VL domain depicted in FIG. 10. In some embodiments, the changes are in a VH and VL domain depicted in FIG. 10. In some embodiments, one or more amino acid changes are in the VH and/or VL framework regions (FR1, FR2, FR3, and/or FR4). In some embodiments, one or more amino acid changes are in one or more CDRs. In certain embodiments, the CD3 ABD of the subject antibody is capable of binding to CD3, as measured at least one of a Biacore, surface plasmon resonance (SPR), flow cytometry, and/or BLI (biolayer interferometry, e.g., Octet assay) assay, with the latter finding particular use in many embodiments. In particular embodiments, the CD3 ABD is capable of binding human CD3 antigen.

In one embodiment, the variant VH and/or VL domain is at least 90, 95, 97, 98 or 99% identical to the VH and/or VL of a CD3 ABD as described herein, including the figures and sequence listing. In exemplary embodiments, the variant VH and/or VL domain is at least 90, 95, 97, 98 or 99% identical to the VH and/or VL of one of the following CD3 binding domains: anti-CD3 H1.30_L1.47, anti-CD3 H1.32_L1.47, anti-CD3 H1.89_L1.47, anti-CD3 H1.90_L1.47, anti-CD3 H1.33_L1.47, anti-CD3 H1.31_L1.47, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.32, anti-CD3 L1.47_H1.89, anti-CD3 L1.47_H1.90, anti-CD3 L1.47_H1.33, and anti-CD3 L1.47_H1.31 (FIG. 10). In some embodiments, the CD3

ABD includes a VH that is at least 90, 95, 97, 98 or 99% identical to VH domain depicted in FIG. 10. In some embodiments, the CD3 ABD includes a VL that is at least 90, 95, 97, 98 or 99% identical to VL domain depicted in FIG. 10. In some embodiments, the CD3 ABD includes a VH and a VL that is at least 90, 95, 97, 98 or 99% identical to a VH domain and a VL domain depicted in FIG. 10. In certain embodiments, the CD3 ABD is capable of binding to CD3, as measured by at least one of a Biacore, surface plasmon resonance (SPR), flow cytometry, and/or BLI (biolayer interferometry, e.g., Octet assay) assay, with the latter finding particular use in many embodiments. In particular embodiments, the CD3 ABD is capable of binding human CD3 antigen.

In addition to the α CD3 ABDs of FIG. 10, additional ABDs of use in the invention include those depicted in FIGS. 14 and 15 of WO2014/145806, hereby expressly incorporated herein in their entirety including the Figures and Legends therein.

3. Linkers

As shown herein, there are a number of suitable linkers (for use as either domain linkers or scFv linkers) that can be used to covalently attach the recited domains (e.g., scFvs, Fabs, Fc domains, etc.), including traditional peptide bonds, generated by recombinant techniques. Exemplary linkers to attach domains of the subject antibody to each other are depicted in FIG. 6. In some embodiments, the linker peptide may predominantly include the following amino acid residues: Gly, Ser, Ala, or Thr. The linker peptide should have a length that is adequate to link two molecules in such a way that they assume the correct conformation relative to one another so that they retain the desired activity. In one embodiment, the linker is from about 1 to 50 amino acids in length, preferably about 1 to 30 amino acids in length. In one embodiment, linkers of 1 to 20 amino acids in length may be used, with from about 5 to about 10 amino acids finding use in some embodiments. Useful linkers include glycine-serine polymers, including for example (GS)_n, (GSGGS)_n (SEQ ID NO: 3), (GGGGS)_n (SEQ ID NO: 2), and (GGGS)_n (SEQ ID NO: 4), where n is an integer of at least one (and generally from 3 to 4), glycine-alanine polymers, alanine-serine polymers, and other flexible linkers, some of which are shown in FIG. 5 and FIG. 6. Alternatively, a variety of nonproteinaceous polymers, including but not limited to polyethylene glycol (PEG), polypropylene glycol, polyoxyalkylenes, or copolymers of polyethylene glycol and polypropylene glycol, may find use as linkers.

Other linker sequences may include any sequence of any length of CL/CH1 domain but not all residues of CL/CH1 domain; for example the first 5-12 amino acid residues of the CL/CH1 domains. Linkers can be derived from immunoglobulin light chain, for example C κ or C λ . Linkers can be derived from immunoglobulin heavy chains of any isotype, including for example C γ 1, C γ 2, C γ 3, C γ 4, C α 1, C α 2, C δ , C ϵ , and C μ . Linker sequences may also be derived from other proteins such as Ig-like proteins (e.g. TCR, FcR, KIR), hinge region-derived sequences, and other natural sequences from other proteins.

In some embodiments, the linker is a "domain linker", used to link any two domains as outlined herein together. For example, in FIG. 17B, there may be a domain linker that attaches the C-terminus of the CH1 domain of the Fab to the N-terminus of the scFv, with another optional domain linker attaching the C-terminus of the scFv to the CH2 domain (although in many embodiments the hinge is used as this domain linker). While any suitable linker can be used, many embodiments utilize a glycine-serine polymer as the domain

linker, including for example (GS)_n, (GSGGS)_n (SEQ ID NO: 3), (GGGGGS)_n (SEQ ID NO: 2), and (GGGS)_n (SEQ ID NO: 4), where *n* is an integer of at least one (and generally from 3 to 4 to 5) as well as any peptide sequence that allows for recombinant attachment of the two domains with sufficient length and flexibility to allow each domain to retain its biological function. In some cases, and with attention being paid to “strandedness”, as outlined below, charged domain linkers, as used in some embodiments of scFv linkers can be used. Exemplary useful domain linkers are depicted in FIG. 6.

With particular reference to the domain linker used to attach the scFv domain to the Fc domain in the “2+1” format, there are several domain linkers that find particular use, including “full hinge C220S variant,” “flex half hinge,” “charged half hinge 1,” and “charged half hinge 2” as shown in FIG. 6.

In some embodiments, the linker is a “scFv linker”, used to covalently attach the VH and VL domains as discussed herein. In many cases, the scFv linker is a charged scFv linker, a number of which are shown in FIG. 5. Accordingly, in some embodiments, the antibodies described herein further provide charged scFv linkers, to facilitate the separation in pI between a first and a second monomer. That is, by incorporating a charged scFv linker, either positive or negative (or both, in the case of scaffolds that use scFvs on different monomers), this allows the monomer comprising the charged linker to alter the pI without making further changes in the Fc domains. These charged linkers can be substituted into any scFv containing standard linkers. Again, as will be appreciated by those in the art, charged scFv linkers are used on the correct “strand” or monomer, according to the desired changes in pI. For example, as discussed herein, to make 1+1 Fab-scFv-Fc format heterodimeric antibody, the original pI of the Fv region for each of the desired antigen binding domains are calculated, and one is chosen to make an scFv, and depending on the pI, either positive or negative linkers are chosen.

Charged domain linkers can also be used to increase the pI separation of the monomers of the antibodies described herein as well, and thus those included in FIG. 5 can be used in any embodiment herein where a linker is utilized.

D. Useful Formats of the Invention

As will be appreciated by those in the art and discussed more fully below, the heterodimeric bispecific antibodies provided herein can take on a wide variety of configurations, as are generally depicted in FIG. 17 as well as FIG. 36. Some figures depict “single ended” configurations, where there is one type of specificity on one “arm” of the molecule and a different specificity on the other “arm”. Other figures depict “dual ended” configurations, where there is at least one type of specificity at the “top” of the molecule and one or more different specificities at the “bottom” of the molecule. Thus, in some embodiments, the antibodies described herein are directed to novel immunoglobulin compositions that co-engage a different first and a second antigen.

As will be appreciated by those in the art, the heterodimeric formats of the antibodies described herein can have different valencies as well as be bispecific. That is, heterodimeric antibodies of the antibodies described herein can be bivalent and bispecific, wherein one target tumor antigen (e.g. CD3) is bound by one binding domain and the other target tumor antigen (e.g. CLDN6) is bound by a second binding domain. The heterodimeric antibodies can also be trivalent and bispecific, wherein the first antigen is bound by two binding domains and the second antigen by a second binding domain. As is outlined herein, when CD3 is one of

the target antigens, it is preferable that the CD3 is bound only monovalently, to reduce potential side effects.

The antibodies described herein utilize anti-CD3 antigen binding domains in combination with anti-CLDN6 binding domains. As will be appreciated by those in the art, any collection of anti-CD3 CDRs, anti-CD3 variable light and variable heavy domains, Fabs and scFvs as described herein, and depicted in any of the Figures can be used. Similarly, any of the anti-CLDN6 antigen binding domains can be used, whether CDRs, variable light and variable heavy domains, Fabs and scFvs as described herein, and depicted in any of the Figures can be used, optionally and independently combined in any combination.

1. 1+1 Fab-scFv-Fc Format

One heterodimeric scaffold that finds particular use in the antibodies described herein is the “1+1 Fab-scFv-Fc” or “bottle-opener” format as shown in FIG. 17A with an exemplary combination of a CD3 binding domain and a tumor target antigen (CLDN6) binding domain. In this embodiment, one heavy chain monomer of the antibody contains a single chain Fv (“scFv”, as defined below) and an Fc domain. The scFv includes a variable heavy domain (VH1) and a variable light domain (VL1), wherein the VH1 is attached to the VL1 using an scFv linker that can be charged (see, e.g., FIG. 5). The scFv is attached to the heavy chain using a domain linker (see, e.g., FIG. 6). The other heavy chain monomer is a “regular” heavy chain (VH-CH1-hinge-CH2-CH3). The 1+1 Fab-scFv-Fc also includes a light chain that interacts with the VH-CH1 to form a Fab. This structure is sometimes referred to herein as the “bottle-opener” format, due to a rough visual similarity to a bottle-opener. The two heavy chain monomers are brought together by the use of amino acid variants (e.g., heterodimerization variants, discussed above) in the constant regions (e.g., the Fc domain, the CH1 domain and/or the hinge region) that promote the formation of heterodimeric antibodies as is described more fully below.

There are several distinct advantages to the present “1+1 Fab-scFv-Fc” format. As is known in the art, antibody analogs relying on two scFv constructs often have stability and aggregation problems, which can be alleviated in the antibodies described herein by the addition of a “regular” heavy and light chain pairing. In addition, as opposed to formats that rely on two heavy chains and two light chains, there is no issue with the incorrect pairing of heavy and light chains (e.g. heavy 1 pairing with light 2, etc.).

Many of the embodiments outlined herein rely in general on the 1+1 Fab-scFv-Fc or “bottle opener” format antibody that comprises a first monomer comprising an scFv, comprising a variable heavy and a variable light domain, covalently attached using an scFv linker (charged, in many but not all instances), where the scFv is covalently attached to the N-terminus of a first Fc domain usually through a domain linker. The domain linker can be either charged or uncharged and exogenous or endogenous (e.g., all or part of the native hinge domain). Any suitable linker can be used to attach the scFv to the N-terminus of the first Fc domain. In some embodiments, the domain linker is chosen from the domain linkers in FIG. 6. The second monomer of the 1+1 Fab-scFv-Fc format or “bottle opener” format is a heavy chain, and the composition further comprises a light chain.

In general, in many preferred embodiments, the scFv is the domain that binds to the CD3, and the Fab forms an CLDN6 binding domain. An exemplary anti-CLDN6×anti-CD3 bispecific antibody in the 1+1 Fab-scFv-Fc format is

depicted in FIG. 17A. Exemplary anti-CLDN6xanti-CD3 bispecific antibodies in the 1+1 Fab-scFv-Fc format are depicted in FIGS. 19 and 20.

In addition, the Fe domains of the antibodies described herein generally include skew variants (e.g. a set of amino acid substitutions as shown in FIG. 1, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E: D401K; L368D/K370S:S364K/E357L; K370S:S364K/E357Q; T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in FIG. 3), optionally charged scFv linkers (including those shown in FIG. 5) and the heavy chain comprises pl variants (including those shown in FIG. 2).

In certain embodiments, the 1+1 Fab-scFv-Fc scaffold format includes a first monomer that includes a scFv-domain linker-CH2-CH3 monomer, a second monomer that includes a first variable heavy domain-CH1-hinge-CH2-CH3 monomer and a third monomer that includes a first variable light domain. In some embodiments, the CH2-CH3 of the first monomer is a first variant Fc domain and the CH2-CH3 of the second monomer is a second variant Fc domain. In some embodiments, the scFv includes a scFv variable heavy domain and a scFv variable light domain that form a CD3 binding moiety. In certain embodiments, the scFv variable heavy domain and scFv variable light domain are covalently attached using an scFv linker (charged, in many but not all instances. See, e.g., FIG. 5). In some embodiments, the first variable heavy domain and first variable light domain form a CLDN6 binding domain.

In some embodiments, the 1+1 Fab-scFv-Fc format includes skew variants, pl variants, and ablation variants. Accordingly, some embodiments include 1+1 Fab-scFv-Fc formats that comprise: a) a first monomer (the “scFv monomer”) that comprises a charged scFv linker (with the +H sequence of FIG. 5 being preferred in some embodiments), the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and an scFv that binds to CD3 as outlined herein; b) a second monomer (the “Fab monomer”) that comprises the skew variants L368D/K370S, the pl variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain; and c) a light chain that includes a variable light domain light domain (VL) and a constant light domain (CL), wherein numbering is according to EU numbering. The variable heavy domain and variable light domain make up an CLDN6 binding moiety.

Any suitable CD3 ABD can be included in the 1+1 Fab-scFv-Fc format antibody, included those provided herein. CD3 binding domain sequences finding particular use in these embodiments include, but are not limited to, H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 or a variant thereof, as well as those depicted in FIG. 10 and those depicted in FIGS. 14 and 15 of WO2014/145806, hereby incorporated by reference including the Legends.

Any suitable CLDN6 ABD can be included in the 1+1 Fab-scFv-Fc format antibody, included those provided herein. CLDN6 ABDs that are of particular use in these embodiments include, but are not limited to, VH and VL domains selected from have VH/VL pairs selected from the group consisting of: H1_L1, H1.1_L1, H1.2_L1, H1.3_L1,

H1.4_L1, H1.5_L1, H1.6_L1, H1.7_L1, H1.8_L1, H1.9_L1, H1.19_L1, H1.22_L1, H1.24_L1, H2_L1, H2.1_L1, H2.2_L1, H2.3_L1, H2.4_L1, H2.5_L1, H2.6_L1, H2.7_L1, H2.8_L1, H2.9_L1, H2.11_L1, H2.12_L1, H2.71_L1, H2.75_L1, H2.90_L1, H2.91_L1, H2.118_L1, H2.119_L1, H1_L1.1, H1.1_L1.1, H1.2_L1.1, H1.3_L1.1, H1.4_L1.1, H1.5_L1.1, H1.6_L1.1, H1.7_L1.1, H1.8_L1.1, H1.9_L1.1, H1.19_L1.1, H1.22_L1.1, H1.24_L1.1, H2_L1.1, H2.1_L1.1, H2.2_L1.1, H2.3_L1.1, H2.4_L1.1, H2.5_L1.1, H2.6_L1.1, H2.7_L1.1, H2.8_L1.1, H2.9_L1.1, H2.11_L1.1, H2.12_L1.1, H2.71_L1.1, H2.75_L1.1, H2.90_L1.1, H2.91_L1.1, H2.118_L1.1, H2.119_L1.1, H1_L1.4, H1.1_L1.4, H1.2_L1.4, H1.3_L1.4, H1.4_L1.4, H1.5_L1.4, H1.6_L1.4, H1.7_L1.4, H1.8_L1.4, H1.9_L1.4, H1.19_L1.4, H1.22_L1.4, H1.24_L1.4, H2_L1.4, H2.1_L1.4, H2.2_L1.4, H2.3_L1.4, H2.4_L1.4, H2.5_L1.4, H2.6_L1.4, H2.7_L1.4, H2.8_L1.4, H2.9_L1.4, H2.11_L1.4, H2.12_L1.4, H2.71_L1.4, H2.75_L1.4, H2.90_L1.4, H2.91_L1.4, H2.118_L1.4, H2.119_L1.4, H1_L1.7, H1.1_L1.7, H1.2_L1.7, H1.3_L1.7, H1.4_L1.7, H1.5_L1.7, H1.6_L1.7, H1.7_L1.7, H1.8_L1.7, H1.9_L1.7, H1.19_L1.7, H1.22_L1.7, H1.24_L1.7, H2_L1.7, H2.1_L1.7, H2.2_L1.7, H2.3_L1.7, H2.4_L1.7, H2.5_L1.7, H2.6_L1.7, H2.7_L1.7, H2.8_L1.7, H2.9_L1.7, H2.11_L1.7, H2.12_L1.7, H2.71_L1.7, H2.75_L1.7, H2.90_L1.7, H2.91_L1.7, H2.118_L1.7, H2.119_L1.7, H1_L1.16, H1.1_L1.16, H1.2_L1.16, H1.3_L1.16, H1.4_L1.16, H1.5_L1.16, H1.6_L1.16, H1.7_L1.16, H1.8_L1.16, H1.9_L1.16, H1.19_L1.16, H1.22_L1.16, H1.24_L1.16, H2_L1.16, H2.1_L1.16, H2.2_L1.16, H2.3_L1.16, H2.4_L1.16, H2.5_L1.16, H2.6_L1.16, H2.7_L1.16, H2.8_L1.16, H2.9_L1.16, H2.11_L1.16, H2.12_L1.16, H2.71_L1.16, H2.75_L1.16, H2.90_L1.16, H2.91_L1.16, H2.118_L1.16, H2.119_L1.16, H1_L1.18, H1.1_L1.18, H1.2_L1.18, H1.3_L1.18, H1.4_L1.18, H1.5_L1.18, H1.6_L1.18, H1.7_L1.18, H1.8_L1.18, H1.9_L1.18, H1.19_L1.18, H1.22_L1.18, H1.24_L1.18, H2_L1.18, H2.1_L1.18, H2.2_L1.18, H2.3_L1.18, H2.4_L1.18, H2.5_L1.18, H2.6_L1.18, H2.7_L1.18, H2.8_L1.18, H2.9_L1.18, H2.11_L1.18, H2.12_L1.18, H2.71_L1.18, H2.75_L1.18, H2.90_L1.18, H2.91_L1.18, H2.118_L1.18, H2.119_L1.18, H1_L1.19, H1.1_L1.19, H1.2_L1.19, H1.3_L1.19, H1.4_L1.19, H1.5_L1.19, H1.6_L1.19, H1.7_L1.19, H1.8_L1.19, H1.9_L1.19, H1.19_L1.19, H1.22_L1.19, H1.24_L1.19, H2_L1.19, H2.1_L1.19, H2.2_L1.19, H2.3_L1.19, H2.4_L1.19, H2.5_L1.19, H2.6_L1.19, H2.7_L1.19, H2.8_L1.19, H2.9_L1.19, H2.11_L1.19, H2.12_L1.19, H2.71_L1.19, H2.75_L1.19, H2.90_L1.19, H2.91_L1.19, H2.118_L1.19, H2.119_L1.19, H1_L1.21, H1.1_L1.21, H1.2_L1.21, H1.3_L1.21, H1.4_L1.21, H1.5_L1.21, H1.6_L1.21, H1.7_L1.21, H1.8_L1.21, H1.9_L1.21, H1.19_L1.21, H1.22_L1.21, H1.24_L1.21, H2_L1.21, H2.1_L1.21, H2.2_L1.21, H2.3_L1.21, H2.4_L1.21, H2.5_L1.21, H2.6_L1.21, H2.7_L1.21, H2.8_L1.21, H2.9_L1.21, H2.11_L1.21, H2.12_L1.21, H2.71_L1.21, H2.75_L1.21, H2.90_L1.21, H2.91_L1.21, H2.118_L1.21, H2.119_L1.21, H1_L1.22, H1.1_L1.22, H1.2_L1.22, H1.3_L1.22, H1.4_L1.22, H1.5_L1.22, H1.6_L1.22, H1.7_L1.22, H1.8_L1.22, H1.9_L1.22, H1.19_L1.22, H1.22_L1.22, H1.24_L1.22, H2_L1.22, H2.1_L1.22, H2.2_L1.22, H2.3_L1.22, H2.4_L1.22, H2.5_L1.22, H2.6_L1.22, H2.7_L1.22, H2.8_L1.22, H2.9_L1.22, H2.11_L1.22, H2.12_L1.22, H2.71_L1.22, H2.75_L1.22, H2.90_L1.22, H2.91_L1.22, H2.118_L1.22, H2.119_L1.22, H1_L1.23, H1.1_L1.23, H1.2_L1.23, H1.3_L1.23, H1.4_L1.23, H1.5_L1.23, H1.6_L1.23, H1.7_L1.23, H1.8_L1.23,

H1.9_L1.23, H1.19_L1.23, H1.22_L1.23, H1.24_L1.23, H2_L1.23, H2.1_L1.23, H2.2_L1.23, H2.3_L1.23, H2.4_L1.23, H2.5_L1.23, H2.6_L1.23, H2.7_L1.23, H2.8_L1.23, H2.9_L1.23, H2.11_L1.23, H2.12_L1.23, H2.71_L1.23, H2.75_L1.23, H2.90_L1.23, H2.91_L1.23, H2.118_L1.23, H2.119_L1.23, H1_L1.27, H1.1_L1.27, H1.2_L1.27, H1.3_L1.27, H1.4_L1.27, H1.5_L1.27, H1.6_L1.27, H1.7_L1.27, H1.8_L1.27, H1.9_L1.27, H1.19_L1.27, H1.22_L1.27, H1.24_L1.27, H2_L1.27, H2.1_L1.27, H2.2_L1.27, H2.3_L1.27, H2.4_L1.27, H2.5_L1.27, H2.6_L1.27, H2.7_L1.27, H2.8_L1.27, H2.9_L1.27, H2.11_L1.27, H2.12_L1.27, H2.71_L1.27, H2.75_L1.27, H2.90_L1.27, H2.91_L1.27, H2.118_L1.27, H2.119_L1.27, H1_L1.60, H1.1_L1.60, H1.2_L1.60, H1.3_L1.60, H1.4_L1.60, H1.5_L1.60, H1.6_L1.60, H1.7_L1.60, H1.8_L1.60, H1.9_L1.60, H1.19_L1.60, H1.22_L1.60, H1.24_L1.60, H2_L1.60, H2.1_L1.60, H2.2_L1.60, H2.3_L1.60, H2.4_L1.60, H2.5_L1.60, H2.6_L1.60, H2.7_L1.60, H2.8_L1.60, H2.9_L1.60, H2.11_L1.60, H2.12_L1.60, H2.71_L1.60, H2.75_L1.60, H2.90_L1.60, H2.91_L1.60, H2.118_L1.60, H2.119_L1.60, H1_L1.107, H1.1_L1.107, H1.2_L1.107, H1.3_L1.107, H1.4_L1.107, H1.5_L1.107, H1.6_L1.107, H1.7_L1.107, H1.8_L1.107, H1.9_L1.107, H1.19_L1.107, H1.22_L1.107, H1.24_L1.107, H2_L1.107, H2.1_L1.107, H2.2_L1.107, H2.3_L1.107, H2.4_L1.107, H2.5_L1.107, H2.6_L1.107, H2.7_L1.107, H2.8_L1.107, H2.9_L1.107, H2.11_L1.107, H2.12_L1.107, H2.71_L1.107, H2.75_L1.107, H2.90_L1.107, H2.91_L1.107, H2.118_L1.107, H2.119_L1.107, H1_L1.114, H1.1_L1.114, H1.2_L1.114, H1.3_L1.114, H1.4_L1.114, H1.5_L1.114, H1.6_L1.114, H1.7_L1.114, H1.8_L1.114, H1.9_L1.114, H1.19_L1.114, H1.22_L1.114, H1.24_L1.114, H2_L1.114, H2.1_L1.114, H2.2_L1.114, H2.3_L1.114, H2.4_L1.114, H2.5_L1.114, H2.6_L1.114, H2.7_L1.114, H2.8_L1.114, H2.9_L1.114, H2.11_L1.114, H2.12_L1.114, H2.71_L1.114, H2.75_L1.114, H2.90_L1.114, H2.91_L1.114, H2.118_L1.114, H2.119_L1.114, H1_L1.187, H1.1_L1.187, H1.2_L1.187, H1.3_L1.187, H1.4_L1.187, H1.5_L1.187, H1.6_L1.187, H1.7_L1.187, H1.8_L1.187, H1.9_L1.187, H1.19_L1.187, H1.22_L1.187, H1.24_L1.187, H2_L1.187, H2.1_L1.187, H2.2_L1.187, H2.3_L1.187, H2.4_L1.187, H2.5_L1.187, H2.6_L1.187, H2.7_L1.187, H2.8_L1.187, H2.9_L1.187, H2.11_L1.187, H2.12_L1.187, H2.71_L1.187, H2.75_L1.187, H2.90_L1.187, H2.91_L1.187, H2.118_L1.187, H2.119_L1.187, H1_L1.189, H1.1_L1.189, H1.2_L1.189, H1.3_L1.189, H1.4_L1.189, H1.5_L1.189, H1.6_L1.189, H1.7_L1.189, H1.8_L1.189, H1.9_L1.189, H1.19_L1.189, H1.22_L1.189, H1.24_L1.189, H2_L1.189, H2.1_L1.189, H2.2_L1.189, H2.3_L1.189, H2.4_L1.189, H2.5_L1.189, H2.6_L1.189, H2.7_L1.189, H2.8_L1.189, H2.9_L1.189, H2.11_L1.189, H2.12_L1.189, H2.71_L1.189, H2.75_L1.189, H2.90_L1.189, H2.91_L1.189, H2.118_L1.189, H2.119_L1.189, H1_L2, H1.1_L2, H1.2_L2, H1.3_L2, H1.4_L2, H1.5_L2, H1.6_L2, H1.7_L2, H1.8_L2, H1.9_L2, H1.19_L2, H1.22_L2, H1.24_L2, H2_L2, H2.1_L2, H2.2_L2, H2.3_L2, H2.4_L2, H2.5_L2, H2.6_L2, H2.7_L2, H2.8_L2, H2.9_L2, H2.11_L2, H2.12_L2, H2.71_L2, H2.75_L2, H2.90_L2, H2.91_L2, H2.118_L2 and H2.119_L2 or a variant thereof.

In particular embodiments, the α CLDN6 ABD VH/VL pairs are selected from the group consisting of H1.9_L1.187, H1.24_L1.187, H2.91_L1.187 and H1.9_L1.187 or a variant thereof.

In some embodiments, the 1+1 Fab-scFv-Fc format includes skew variants, pl variants, ablation variants and

FcRn variants. Accordingly, some embodiments include 1+1 Fab-scFv-Fc formats that comprise: a) a first monomer (the “scFv monomer”) that comprises a charged scFv linker (with the +H sequence of FIG. 6 being preferred in some embodiments), the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and an scFv that binds to CD3 as outlined herein; b) a second monomer (the “Fab monomer”) that comprises the skew variants L368D/K370S, the pl variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S, and a variable heavy domain; and c) a light chain that includes a variable light domain (VL) and a constant light domain (CL), wherein numbering is according to EU numbering. The variable heavy domain and variable light domain make up a CLDN6 binding domain. CD3 binding domain sequences finding particular use in these embodiments include, but are not limited to, H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 or a variant thereof, as well as those depicted in FIG. 10. CLDN6 binding domain sequences that are of particular use in these embodiments include, but are not limited to, the α CLDN6 ABD VH_VL pairs are selected from the group consisting of H1.9_L1.187, H1.24_L1.187, H2.91_L1.187 and H1.9_L1.187 or a variant thereof.

Particularly useful CLDN6 and CD3 sequence combinations for use with the 1+1 format antibody include, for example, are disclosed in FIGS. 19 and 20.

FIGS. 7A-7D show some exemplary Fc domain sequences that are useful in the 1+1 Fab-scFv-Fc format antibodies. The “monomer 1” sequences depicted in FIGS. 7A-7D typically refer to the Fc domain of the “Fab-Fc heavy chain” and the “monomer 2” sequences refer to the Fc domain of the “scFv-Fc heavy chain.” Further, FIG. 9 provides useful CL sequences that can be used with this format.

In some embodiments, any of the VH and VL sequences depicted herein (including all VH and VL sequences depicted in the Figures and Sequence Listings, including those directed to CLDN6) can be added to the bottle opener backbone formats of FIG. 7A-7D as the “Fab side”, using any of the anti-CD3 scFv sequences shown in the Figures and Sequence Listings.

For bottle opener backbone 1 from FIG. 7A, (optionally including the 428L/434S variants), CD binding domain sequences finding particular use in these embodiments include, but are not limited to, CD3 binding domain anti-CD3 H1.30_L1.47, anti-CD3 H1.32_L1.47, anti-CD3 H1.89_L1.47, anti-CD3 H1.90_L1.47, anti-CD3 H1.33_L1.47 and anti-CD3 H1.31_L1.47 attached as the scFv side of the backbones shown in FIG. 7.

2. mAb-Fv

One heterodimeric scaffold that finds particular use in the antibodies described herein is the mAb-Fv format (FIG. 36G). In this embodiment, the format relies on the use of a C-terminal attachment of an “extra” variable heavy domain to one monomer and the C-terminal attachment of an “extra” variable light domain to the other monomer, thus forming a third antigen binding domain, wherein the Fab portions of the two monomers bind a CLDN6 and the “extra” scFv domain binds CD3.

In this embodiment, the first monomer comprises a first heavy chain, comprising a first variable heavy domain and a first constant heavy domain comprising a first Fc domain, with a first variable light domain covalently attached to the

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C-terminus of the first Fc domain using a domain linker (VH1-CH1-hinge-CH2-CH3-[optional linker]-VL2). The second monomer comprises a second variable heavy domain of the second constant heavy domain comprising a second Fc domain, and a third variable heavy domain covalently attached to the C-terminus of the second Fc domain using a domain linker (vh1-CH1-hinge-CH2-CH3-[optional linker]-VH2). The two C-terminally attached variable domains make up a Fv that binds CD3 (as it is less preferred to have bivalent CD3 binding). This embodiment further utilizes a common light chain comprising a variable light domain and a constant light domain that associates with the heavy chains to form two identical Fabs that bind a CLDN6. As for many of the embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

The antibodies described herein provide mAb-Fv formats where the CD3 binding domain sequences are as shown in FIG. 10 or a variant thereof. The antibodies described herein provide mAb-Fv formats wherein the CLDN6 binding domain sequences are as shown in FIGS. 13, 14 and 15 or a variant thereof.

In addition, the Fc domains of the mAb-Fv format comprise skew variants (e.g. a set of amino acid substitutions as shown in FIG. 1, with particularly useful skew variants being selected from the group consisting of S364K/E357Q: L368D/K370S; L368D/K370S:S364K; L368E/K370S: S364K; T411T/E360E/Q362E:D401K; L368D/K370S: S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in FIG. 3), optionally charged scFv linkers (including those shown in FIG. 5) and the heavy chain comprises pI variants (including those shown in FIG. 2).

In some embodiments, the mAb-Fv format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments include mAb-Fv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a first variable heavy domain that, with the first variable light domain of the light chain, makes up an Fv that binds to CLDN6, and a second variable heavy domain; b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and a first variable heavy domain that, with the first variable light domain, makes up the Fv that binds to CLDN6 as outlined herein, and a second variable light chain, that together with the second variable heavy domain forms an Fv (ABD) that binds to CD3; and c) a light chain comprising a first variable light domain and a constant light domain.

In some embodiments, the mAb-Fv format includes skew variants, pI variants, ablation variants and FcRn variants. Accordingly, some embodiments include mAb-Fv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a first variable heavy domain that, with the first variable light domain of the light chain, makes up an Fv that binds to CLDN6, and a second variable heavy domain; b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a first variable heavy domain that, with the first variable light domain, makes up the Fv that binds to CLDN6 as outlined herein, and a second

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variable light chain, that together with the second variable heavy domain of the first monomer forms an Fv (ABD) that binds CD3; and c) a light chain comprising a first variable light domain and a constant light domain.

3. mAb-scFv

One heterodimeric scaffold that finds particular use in the antibodies described herein is the mAb-scFv format (FIG. 36H). In this embodiment, the format relies on the use of a C-terminal attachment of a scFv to one of the monomers, thus forming a third antigen binding domain, wherein the Fab portions of the two monomers bind CLDN6 and the "extra" scFv domain binds CD3. Thus, the first monomer comprises a first heavy chain (comprising a variable heavy domain and a constant domain), with a C-terminally covalently attached scFv comprising a scFv variable light domain, an scFv linker and a scFv variable heavy domain in either orientation (VH1-CH1-hinge-CH2-CH3-[optional linker]-VH2-scFv linker-VL2 or VH1-CH1-hinge-CH2-CH3-[optional linker]-VL2-scFv linker-VH2). This embodiment further utilizes a common light chain comprising a variable light domain and a constant light domain, that associates with the heavy chains to form two identical Fabs that bind CLDN6. As for many of the embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

The antibodies described herein provide mAb-scFv formats, where the CD3 binding domain sequences are as shown in FIG. 10A-10F or a variant thereof and the CLDN6 binding domain sequences are as shown in FIGS. 13-15 and 18 or a variant thereof.

In addition, the Fc domains of the mAb-scFv format comprise skew variants (e.g. a set of amino acid substitutions as shown in FIG. 1, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C: T366W/S354C), optionally ablation variants (including those shown in FIG. 3), optionally charged scFv linkers (including those shown in FIG. 5) and the heavy chain comprises pI variants (including those shown in FIG. 2).

In some embodiments, the mAb-scFv format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments include mAb-scFv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to CLDN6 as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to CLDN6 as outlined herein; and c) a common light chain comprising a variable light domain and a constant light domain.

In some embodiments, the mAb-scFv format includes skew variants, pI variants, ablation variants and FcRn variants. Accordingly, some embodiments include mAb-scFv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that

binds to CLDN6 as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that comprises the skew variants L368D/K370S, the pl variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to CLDN6 as outlined herein; and c) a common light chain comprising a variable light domain and a constant light domain.

4. 2+1 Fab₂-scFv-Fc Format

One heterodimeric scaffold that finds particular use in the antibodies described herein is the “2+1 Fab₂-scFv-Fc” format (also referred to in previous related filings as “Central-scFv format”) shown in FIG. 17B with an exemplary combination of a CD3 binding domain and two tumor target antigen (CLDN6) binding domains. In this embodiment, the format relies on the use of an inserted scFv domain thus forming a third antigen binding domain, wherein the Fab portions of the two monomers bind CLDN6 and the “extra” scFv domain binds CD3. The scFv domain is inserted between the Fc domain and the CH1-Fv region of one of the monomers, thus providing a third antigen binding domain. As described, CLDN6×CD3 bispecific antibodies having the 2+1 Fab₂-scFv-Fc format are potent in inducing redirected T cell cytotoxicity in cellular environments that express low levels of CLDN6. Moreover, as shown in the examples, CLDN6×CD3 bispecific antibodies having the 2+1 Fab₂-scFv-Fc format allow for the “fine tuning” of immune responses as such antibodies exhibit a wide variety of different properties, depending on the CLDN6 and/or CD3 binding domains used. For example, such antibodies exhibit differences in selectivity for cells with different CLDN6 expression, potencies for CLDN6 expressing cells, ability to elicit cytokine release, and sensitivity to soluble CLDN6. These CLDN6 antibodies find use, for example, in the treatment of CLDN6-associated cancers.

In this embodiment, one monomer comprises a first heavy chain comprising a first variable heavy domain, a CH1 domain (and optional hinge) and Fc domain, with a scFv comprising a scFv variable light domain, an scFv linker and a scFv variable heavy domain. The scFv is covalently attached between the C-terminus of the CH1 domain of the heavy constant domain and the N-terminus of the first Fc domain using optional domain linkers (VH1-CH1-[optional linker]-VH2-scFv linker-VL2-[optional linker including the hinge]-CH2-CH3, or the opposite orientation for the scFv, VH1-CH1-[optional linker]-VL2-scFv linker-VH2-[optional linker including the hinge]-CH2-CH3). The optional linkers can be any suitable peptide linkers, including, for example, the domain linkers included in FIG. 6. In some embodiments, the optional linker is a hinge or a fragment thereof. The other monomer is a standard Fab side (i.e., VH1-CH1-hinge-CH2-CH3). This embodiment further utilizes a common light chain comprising a variable light domain and a constant light domain, that associates with the heavy chains to form two identical Fabs that bind CLDN6. As for many of the embodiments herein, these constructs include skew variants, pl variants, ablation variants, additional Fc variants, etc. as desired and described herein.

In one embodiment, the 2+1 Fab₂-scFv-Fc format antibody includes an scFv with the VH and VL of a CD3 binding domain sequence depicted in FIG. 10 or a variant thereof. In one embodiment, the 2+1 Fab₂-scFv-Fc format antibody includes two Fabs having the VH and VL of a CLDN6 binding domain as shown in FIGS. 13-15 and 18 or a variant thereof.

In exemplary embodiments, the CLDN6 binding domain of the 2+1 Fab₂-scFv-Fc CLDN6×CD3 bispecific antibody includes the VH and VL CD3 binding domain sequences finding particular use in these embodiments include, but are not limited to, H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 or a variant thereof, as well as those depicted in FIG. 10 and those depicted in FIGS. 14 and 15 of WO2014/145806, hereby incorporated by reference including the Legends.

Any suitable CLDN6 ABD can be included in the 2+1 Fab₂-scFv-Fc format antibody, included those provided herein. CLDN6 ABDs that are of particular use in these embodiments include, but are not limited to, VH and VL domains selected from the group consisting of: H1_L1, H1.1_L1, H1.2_L1, H1.3_L1, H1.4_L1, H1.5_L1, H1.6_L1, H1.7_L1, H1.8_L1, H1.9_L1, H1.19_L1, H1.22_L1, H1.24_L1, H2_L1, H2.1_L1, H2.2_L1, H2.3_L1, H2.4_L1, H2.5_L1, H2.6_L1, H2.7_L1, H2.8_L1, H2.9_L1, H2.11_L1, H2.12_L1, H2.71_L1, H2.75_L1, H2.90_L1, H2.91_L1, H2.118_L1, H2.119_L1, H1_L1.1, H1.1_L1.1, H1.2_L1.1, H1.3_L1.1, H1.4_L1.1, H1.5_L1.1, H1.6_L1.1, H1.7_L1.1, H1.8_L1.1, H1.9_L1.1, H1.19_L1.1, H1.22_L1.1, H1.24_L1.1, H2_L1.1, H2.1_L1.1, H2.2_L1.1, H2.3_L1.1, H2.4_L1.1, H2.5_L1.1, H2.6_L1.1, H2.7_L1.1, H2.8_L1.1, H2.9_L1.1, H2.11_L1.1, H2.12_L1.1, H2.71_L1.1, H2.75_L1.1, H2.90_L1.1, H2.91_L1.1, H2.118_L1.1, H2.119_L1.1, H1_L1.4, H1.1_L1.4, H1.2_L1.4, H1.3_L1.4, H1.4_L1.4, H1.5_L1.4, H1.6_L1.4, H1.7_L1.4, H1.8_L1.4, H1.9_L1.4, H1.19_L1.4, H1.22_L1.4, H1.24_L1.4, H2_L1.4, H2.1_L1.4, H2.2_L1.4, H2.3_L1.4, H2.4_L1.4, H2.5_L1.4, H2.6_L1.4, H2.7_L1.4, H2.8_L1.4, H2.9_L1.4, H2.11_L1.4, H2.12_L1.4, H2.71_L1.4, H2.75_L1.4, H2.90_L1.4, H2.91_L1.4, H2.118_L1.4, H2.119_L1.4, H1_L1.7, H1.1_L1.7, H1.2_L1.7, H1.3_L1.7, H1.4_L1.7, H1.5_L1.7, H1.6_L1.7, H1.7_L1.7, H1.8_L1.7, H1.9_L1.7, H1.19_L1.7, H1.22_L1.7, H1.24_L1.7, H2_L1.7, H2.1_L1.7, H2.2_L1.7, H2.3_L1.7, H2.4_L1.7, H2.5_L1.7, H2.6_L1.7, H2.7_L1.7, H2.8_L1.7, H2.9_L1.7, H2.11_L1.7, H2.12_L1.7, H2.71_L1.7, H2.75_L1.7, H2.90_L1.7, H2.91_L1.7, H2.118_L1.7, H2.119_L1.7, H1_L1.16, H1.1_L1.16, H1.2_L1.16, H1.3_L1.16, H1.4_L1.16, H1.5_L1.16, H1.6_L1.16, H1.7_L1.16, H1.8_L1.16, H1.9_L1.16, H1.19_L1.16, H1.22_L1.16, H1.24_L1.16, H2_L1.16, H2.1_L1.16, H2.2_L1.16, H2.3_L1.16, H2.4_L1.16, H2.5_L1.16, H2.6_L1.16, H2.7_L1.16, H2.8_L1.16, H2.9_L1.16, H2.11_L1.16, H2.12_L1.16, H2.71_L1.16, H2.75_L1.16, H2.90_L1.16, H2.91_L1.16, H2.118_L1.16, H2.119_L1.16, H1_L1.18, H1.1_L1.18, H1.2_L1.18, H1.3_L1.18, H1.4_L1.18, H1.5_L1.18, H1.6_L1.18, H1.7_L1.18, H1.8_L1.18, H1.9_L1.18, H1.19_L1.18, H1.22_L1.18, H1.24_L1.18, H2_L1.18, H2.1_L1.18, H2.2_L1.18, H2.3_L1.18, H2.4_L1.18, H2.5_L1.18, H2.6_L1.18, H2.7_L1.18, H2.8_L1.18, H2.9_L1.18, H2.11_L1.18, H2.12_L1.18, H2.71_L1.18, H2.75_L1.18, H2.90_L1.18, H2.91_L1.18, H2.118_L1.18, H2.119_L1.18, H1_L1.19, H1.1_L1.19, H1.2_L1.19, H1.3_L1.19, H1.4_L1.19, H1.5_L1.19, H1.6_L1.19, H1.7_L1.19, H1.8_L1.19, H1.9_L1.19, H1.19_L1.19, H1.22_L1.19, H1.24_L1.19, H2_L1.19, H2.1_L1.19, H2.2_L1.19, H2.3_L1.19, H2.4_L1.19, H2.5_L1.19, H2.6_L1.19, H2.7_L1.19, H2.8_L1.19, H2.9_L1.19, H2.11_L1.19, H2.12_L1.19, H2.71_L1.19, H2.75_L1.19, H2.90_L1.19, H2.91_L1.19, H2.118_L1.19, H2.119_L1.19, H1_L1.21, H1.1_L1.21, H1.2_L1.21, H1.3_L1.21, H1.4_L1.21,

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H1.5_L1.21, H1.6_L1.21, H1.7_L1.21, H1.8_L1.21,
H1.9_L1.21, H1.19_L1.21, H1.22_L1.21, H1.24_L1.21,
H2_L1.21, H2.1_L1.21, H2.2_L1.21, H2.3_L1.21,
H2.4_L1.21, H2.5_L1.21, H2.6_L1.21, H2.7_L1.21,
H2.8_L1.21, H2.9_L1.21, H2.11_L1.21, H2.12_L1.21,
H2.71_L1.21, H2.75_L1.21, H2.90_L1.21, H2.91_L1.21,
H2.118_L1.2, H2.119_L1.21, H1_L1.22, H1.1_L1.22,
H1.2_L1.22, H1.3_L1.22, H1.4_L1.22, H1.5_L1.22,
H1.6_L1.22, H1.7_L1.22, H1.8_L1.22, H1.9_L1.22,
H1.19_L1.22, H1.22_L1.22, H1.24_L1.22, H2_L1.22,
H2.1_L1.22, H2.2_L1.22, H2.3_L1.22, H2.4_L1.22,
H2.5_L1.22, H2.6_L1.22, H2.7_L1.22, H2.8_L1.22,
H2.9_L1.22, H2.11_L1.22, H2.12_L1.22, H2.71_L1.22,
H2.75_L1.22, H2.90_L1.22, H2.91_L1.22, H2.118_L1.22,
H2.119_L1.22, H1_L1.23, H1.1_L1.23, H1.2_L1.23,
H1.3_L1.23, H1.4_L1.23, H1.5_L1.23, H1.6_L1.23,
H1.7_L1.23, H1.8_L1.23, H1.9_L1.23, H1.19_L1.23,
H1.22_L1.23, H1.24_L1.23, H2_L1.23, H2.1_L1.23,
H2.2_L1.23, H2.3_L1.23, H2.4_L1.23, H2.5_L1.23,
H2.6_L1.23, H2.7_L1.23, H2.8_L1.23, H2.9_L1.23,
H2.11_L1.23, H2.12_L1.23, H2.71_L1.23, H2.75_L1.23,
H2.90_L1.23, H2.91_L1.23, H2.118_L1.23, H2.119_L1.23,
H1_L1.27, H1.1_L1.27, H1.2_L1.27, H1.3_L1.27,
H1.4_L1.27, H1.5_L1.27, H1.6_L1.27, H1.7_L1.27,
H1.8_L1.27, H1.9_L1.27, H1.19_L1.27, H1.22_L1.27,
H1.24_L1.27, H2_L1.27, H2.1_L1.27, H2.2_L1.27,
H2.3_L1.27, H2.4_L1.27, H2.5_L1.27, H2.6_L1.27,
H2.7_L1.27, H2.8_L1.27, H2.9_L1.27, H2.11_L1.27,
H2.12_L1.27, H2.71_L1.27, H2.75_L1.27, H2.90_L1.27,
H2.91_L1.27, H2.118_L1.27, H2.119_L1.27, H1_L1.60,
H1.1_L1.60, H1.2_L1.60, H1.3_L1.60, H1.4_L1.60,
H1.5_L1.60, H1.6_L1.60, H1.7_L1.60, H1.8_L1.60,
H1.9_L1.60, H1.19_L1.60, H1.22_L1.60, H1.24_L1.60,
H2_L1.60, H2.1_L1.60, H2.2_L1.60, H2.3_L1.60,
H2.4_L1.60, H2.5_L1.60, H2.6_L1.60, H2.7_L1.60,
H2.8_L1.60, H2.9_L1.60, H2.11_L1.60, H2.12_L1.60,
H2.71_L1.60, H2.75_L1.60, H2.90_L1.60, H2.91_L1.60,
H2.118_L1.60, H2.119_L1.60, H1_L1.107, H1.1_L1.107,
H1.2_L1.107, H1.3_L1.107, H1.4_L1.107, H1.5_L1.107,
H1.6_L1.107, H1.7_L1.107, H1.8_L1.107, H1.9_L1.107,
H1.19_L1.107, H1.22_L1.107, H1.24_L1.107, H2_L1.107,
H2.1_L1.107, H2.2_L1.107, H2.3_L1.107, H2.4_L1.107,
H2.5_L1.107, H2.6_L1.107, H2.7_L1.107, H2.8_L1.107,
H2.9_L1.107, H2.11_L1.107, H2.12_L1.107,
H2.71_L1.107, H2.75_L1.107, H2.90_L1.107,
H2.91_L1.107, H2.118_L1.107, H2.119_L1.107,
H1_L1.114, H1.1_L1.114, H1.2_L1.114, H1.3_L1.114,
H1.4_L1.114, H1.5_L1.114, H1.6_L1.114, H1.7_L1.114,
H1.8_L1.114, H1.9_L1.114, H1.19_L1.114, H1.22_L1.114,
H1.24_L1.114, H2_L1.114, H2.1_L1.114, H2.2_L1.114,
H2.3_L1.114, H2.4_L1.114, H2.5_L1.114, H2.6_L1.114,
H2.7_L1.114, H2.8_L1.114, H2.9_L1.114, H2.11_L1.114,
H2.12_L1.114, H2.71_L1.114, H2.75_L1.114,
H2.90_L1.114, H2.91_L1.114, H2.118_L1.114,
H2.119_L1.114, H1_L1.187, H1.1_L1.187, H1.2_L1.187,
H1.3_L1.187, H1.4_L1.187, H1.5_L1.187, H1.6_L1.187,
H1.7_L1.187, H1.8_L1.187, H1.9_L1.187, H1.19_L1.187,
H1.22_L1.187, H1.24_L1.187, H2_L1.187, H2.1_L1.187,
H2.2_L1.187, H2.3_L1.187, H2.4_L1.187, H2.5_L1.187,
H2.6_L1.187, H2.7_L1.187, H2.8_L1.187, H2.9_L1.187,
H2.11_L1.187, H2.12_L1.187, H2.71_L1.187,
H2.75_L1.187, H2.90_L1.187, H2.91_L1.187,
H2.118_L1.187, H2.119_L1.187, H1_L1.189,
H1.1_L1.189, H1.2_L1.189, H1.3_L1.189, H1.4_L1.189,
H1.5_L1.189, H1.6_L1.189, H1.7_L1.189, H1.8_L1.189,
H1.9_L1.189, H1.19_L1.189, H1.22_L1.189,
H1.24_L1.189, H2_L1.189, H2.1_L1.189, H2.2_L1.189,

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H2.3_L1.189, H2.4_L1.189, H2.5_L1.189, H2.6_L1.189,
H2.7_L1.189, H2.8_L1.189, H2.9_L1.189, H2.11_L1.189,
H2.12_L1.189, H2.71_L1.189, H2.75_L1.189,
H2.90_L1.189, H2.91_L1.189, H2.118_L1.189,
H2.119_L1.189, H1_L2, H1.1_L2, H1.2_L2, H1.3_L2,
H1.4_L2, H1.5_L2, H1.6_L2, H1.7_L2, H1.8_L2, H1.9_L2,
H1.19_L2, H1.22_L2, H1.24_L2, H2_L2, H2.1_L2,
H2.2_L2, H2.3_L2, H2.4_L2, H2.5_L2, H2.6_L2, H2.7_L2,
H2.8_L2, H2.9_L2, H2.11_L2, H2.12_L2, H2.71_L2,
H2.75_L2, H2.90_L2, H2.91_L2, H2.118_L2 and
H2.119_L2.

In particular embodiments, the α CLDN6 ABD VH/VL pairs are selected from the group consisting of H1.9_L1.187, H1.24_L1.187, H2.91_L1.187 and H1.9_L1.187 or a variant thereof.

In addition, the Fe domains of the 2+1 Fab₂-scFv-Fc format comprise skew variants (e.g. a set of amino acid substitutions as shown in FIG. 1, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in FIG. 3), optionally charged scFv linkers (including those shown in FIG. 5) and the heavy chain comprises pl variants (including those shown in FIG. 2).

In some embodiments, the 2+1 Fab₂-scFv-Fc format antibody includes skew variants, pl variants, and ablation variants. Accordingly, some embodiments include 2+1 Fab₂-scFv-Fc formats that comprise: a) a first monomer (the Fab-scFv-Fc side) that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to CLDN6 as outlined herein, and an scFv domain that binds to CD3; b) a second monomer (the Fab-Fc side) that comprises the skew variants L368D/K370S, the pl variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with variable light domain of the common light chain, makes up an Fv that binds to CLDN6 as outlined herein; and c) a common light chain comprising the variable light domain and a constant light domain, where numbering is according to EU numbering. In some embodiments, the α CLDN6 VH_VL pairs are selected from the group consisting of H1.9_L1.187, H1.24_L1.187, H2.91_L1.187 and H1.9_L1.187 or a variant thereof.

In some embodiments, the 2+1 Fab₂-scFv-Fc format antibody includes skew variants, pl variants, ablation variants and FcRn variants. Accordingly, some embodiments include 2+1 Fab₂-scFv-Fc formats that comprise: a) a first monomer (the Fab-scFv-Fc side) that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to CLDN6 as outlined herein, and an scFv domain that binds to CD3; b) a second monomer (the Fab-Fc side) that comprises the skew variants L368D/K370S, the pl variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with variable light domain of the common light chain, makes up an Fv that binds to CLDN6 as outlined herein; and c) a common light chain comprising a variable light domain and

a constant light domain, where numbering is according to EU numbering. In some embodiments, the α CLDN6 VH_VL pairs are selected from the group consisting of H1.9_L1.187, H1.24_L1.187, H2.91_L1.187 and H1.9_L1.187 or a variant thereof. CD3 binding domain sequences finding particular use in these embodiments include, but are not limited to, H1.30_L1.47, H1.32_L1.47, H1.89_L1.47, H1.90_L1.47, H1.33_L1.47, H1.31_L1.47, L1.47_H1.30, L1.47_H1.30, L1.47_H1.32, L1.47_H1.89, L1.47_H1.90, L1.47_H1.33, and L1.47_H1.31 or a variant thereof.

FIGS. 8A-8C shows some exemplary Fc domain sequences that are useful with the 2+1 Fab₂-scFv-Fc format. The “monomer 1” sequences depicted in FIGS. 8A-8C typically refer to the Fc domain of the “Fab-Fc heavy chain” and the “monomer 2” sequences refer to the Fc domain of the “Fab-scFv-Fc heavy chain.” Further, FIG. 9 provides useful CL sequences that can be used with this format.

Exemplary anti-CLDN6 $\times$ anti-CD3 2+1 Fab₂-scFv-Fc format antibodies are depicted in FIGS. 21, 22, 59 and 60.

5. Central-Fv

One heterodimeric scaffold that finds particular use in the antibodies described herein is the Central-Fv format (FIG. 36I). In this embodiment, the format relies on the use of an inserted Fv domain (i.e., the central Fv domain) thus forming an “extra” third antigen binding domain, wherein the Fab portions of the two monomers bind a CLDN6 and the “extra” central Fv domain binds CD3. The “extra” central Fv domain is inserted between the Fc domain and the CH1-Fv region of the monomers, thus providing a third antigen binding domain (i.e., the “extra” central Fv domain), wherein each monomer contains a component of the “extra” central Fv domain (i.e., one monomer comprises the variable heavy domain and the other a variable light domain of the “extra” central Fv domain).

In this embodiment, one monomer comprises a first heavy chain comprising a first variable heavy domain, a CH1 domain, and Fc domain and an additional variable light domain. The light domain is covalently attached between the C-terminus of the CH1 domain of the heavy constant domain and the N-terminus of the first Fc domain using domain linkers (VH1-CH1-[optional linker]-VL2-hinge-CH2-CH3). The other monomer comprises a first heavy chain comprising a first variable heavy domain, a CH1 domain and Fc domain and an additional variable heavy domain (VH1-CH1-[optional linker]-VH2-hinge-CH2-CH3). The light domain is covalently attached between the C-terminus of the CH1 domain of the heavy constant domain and the N-terminus of the first Fc domain using domain linkers.

This embodiment further utilizes a common light chain comprising a variable light domain and a constant light domain, that associates with the heavy chains to form two identical Fabs that each bind an CLDN6. As for many of the embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

The antibodies described herein provide central-Fv formats, where the CD3 binding domain sequences are as shown in FIG. 10 or a variant thereof, and the CLDN6 binding domain sequences are as shown in FIGS. 13, 14 and 15 or a variant thereof.

6. One-Armed Central-scFv

One heterodimeric scaffold that finds particular use in the antibodies described herein is the one-armed central-scFv format (FIG. 36C). In this embodiment, one monomer comprises just an Fc domain, while the other monomer

includes a Fab domain (a first antigen binding domain), a scFv domain (a second antigen binding domain) and an Fc domain, where the scFv domain is inserted between the Fc domain and the Fc domain. In this format, the Fab portion binds one receptor target and the scFv binds another. In this format, either the Fab portion binds a CLDN6 and the scFv binds CD3 or vice versa.

In this embodiment, one monomer comprises a first heavy chain comprising a first variable heavy domain, a CH1 domain and Fc domain, with a scFv comprising a scFv variable light domain, an scFv linker and a scFv variable heavy domain. The scFv is covalently attached between the C-terminus of the CH1 domain of the heavy constant domain and the N-terminus of the first Fc domain using domain linkers, in either orientation, VH1-CH1-[optional domain linker]-VH2-scFv linker-VL2-[optional domain linker]-CH2-CH3 or VH1-CH1-[optional domain linker]-VL2-scFv linker-VH2-[optional domain linker]-CH2-CH3. The second monomer comprises an Fc domain (CH2-CH3). This embodiment further utilizes a light chain comprising a variable light domain and a constant light domain that associates with the heavy chain to form a Fab.

As for many of the embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

The antibodies described herein provide central-Fv formats where the CD3 binding domain sequences are as shown in FIG. 10 or a variant thereof, and the CLDN6 binding domain sequences are as shown in FIGS. 13, 14 and 15 or a variant thereof.

In addition, the Fc domains of the one-armed central-scFv format generally include skew variants (e.g. a set of amino acid substitutions as shown in FIG. 1, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in FIG. 3), optionally charged scFv linkers (including those shown in FIG. 5) and the heavy chain comprises pI variants (including those shown in FIG. 2).

In some embodiments, the one-armed central-scFv format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments of the one-armed central-scFv formats comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the light chain, makes up an Fv that binds to CLDN6 as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that includes an Fc domain having the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K; and c) a light chain comprising a variable light domain and a constant light domain.

In some embodiments, the one-armed central-scFv format includes skew variants, pI variants, ablation variants and FcRn variants. Accordingly, some embodiments of the one-armed central-scFv formats comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the light chain, makes up an Fv that binds to CLDN6 as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that

includes an Fc domain having the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and the FcRn variants M428L/N434S; and c) a light chain comprising a variable light domain and a constant light domain.

7. One-Armed scFv-mAb

One heterodimeric scaffold that finds particular use in the antibodies described herein is the one-armed scFv-mAb format (FIG. 36D). In this embodiment, one monomer comprises just an Fc domain, while the other monomer uses a scFv domain attached at the N-terminus of the heavy chain, generally through the use of a linker: VH-scFv linker-VL-[optional domain linker]-CH1-hinge-CH2-CH3 or (in the opposite orientation) VL-scFv linker-VH-[optional domain linker]-CH1-hinge-CH2-CH3. In this format, the Fab portions each bind CLDN6 and the scFv binds CD3. This embodiment further utilizes a light chain comprising a variable light domain and a constant light domain, that associates with the heavy chain to form a Fab. As for many of the embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

The antibodies described herein provide one-armed scFv-mAb formats, where the CD3 binding domain sequences are as shown in FIG. 10 or a variant thereof, and wherein the CLDN6 binding domain sequences are as shown in FIGS. 13, 14 and 15 or a variant thereof.

In addition, the Fc domains of the one-armed scFv-mAb format generally include skew variants (e.g. a set of amino acid substitutions as shown in FIG. 1, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in FIG. 3), optionally charged scFv linkers (including those shown in FIG. 5) and the heavy chain comprises pI variants (including those shown in FIG. 2).

In some embodiments, the one-armed scFv-mAb format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments of the one-armed scFv-mAb formats comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the light chain, makes up an Fv that binds to CLDN6 as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that includes an Fc domain having the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K; and c) a light chain comprising a variable light domain and a constant light domain.

In some embodiments, the one-armed scFv-mAb format includes skew variants, pI variants, ablation variants and FcRn variants. Accordingly, some embodiments one-armed scFv-mAb formats comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the light chain, makes up an Fv that binds to CLDN6 as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that includes an Fc domain having the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the abla-

tion variants E233P/L234V/L235A/G236del/S267K, and the FcRn variants M428L/N434S; and c) a light chain comprising a variable light domain and a constant light domain.

8. scFv-mAb

One heterodimeric scaffold that finds particular use in the antibodies described herein is the mAb-scFv format (FIG. 36E). In this embodiment, the format relies on the use of a N-terminal attachment of a scFv to one of the monomers, thus forming a third antigen binding domain, wherein the Fab portions of the two monomers bind CLDN6 and the "extra" scFv domain binds CD3.

In this embodiment, the first monomer comprises a first heavy chain (comprising a variable heavy domain and a constant domain), with a N-terminally covalently attached scFv comprising a scFv variable light domain, an scFv linker and a scFv variable heavy domain in either orientation ((VH1-scFv linker-VL1-[optional domain linker]-VH2-CH1-hinge-CH2-CH3) or (with the scFv in the opposite orientation) ((VL1-scFv linker-VH1-[optional domain linker]-VH2-CH1-hinge-CH2-CH3)). This embodiment further utilizes a common light chain comprising a variable light domain and a constant light domain that associates with the heavy chains to form two identical Fabs that bind CLDN6. As for many of the embodiments herein, these constructs include skew variants, pI variants, ablation variants, additional Fc variants, etc. as desired and described herein.

The antibodies described herein provide scFv-mAb formats, where the CD3 binding domain sequences are as shown in FIG. 10 or a variant thereof, and wherein the CLDN6 binding domain sequences are as shown in FIGS. 13, 14 and 15 or a variant thereof.

In addition, the Fc domains of the scFv-mAb format generally include skew variants (e.g. a set of amino acid substitutions as shown in FIG. 1, with particularly useful skew variants being selected from the group consisting of S364K/E357Q:L368D/K370S; L368D/K370S:S364K; L368E/K370S:S364K; T411T/E360E/Q362E:D401K; L368D/K370S:S364K/E357L, K370S:S364K/E357Q, T366S/L368A/Y407V:T366W and T366S/L368A/Y407V/Y349C:T366W/S354C), optionally ablation variants (including those shown in FIG. 3), optionally charged scFv linkers (including those shown in FIG. 5) and the heavy chain comprises pI variants (including those shown in FIG. 2).

In some embodiments, the scFv-mAb format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments include scFv-mAb formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to CLDN6 as outlined herein, and a scFv domain that binds to CD3; b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to CLDN6 as outlined herein; and c) a common light chain comprising a variable light domain and a constant light domain.

In some embodiments, the scFv-mAb format includes skew variants, pI variants, ablation variants and FcRn variants. Accordingly, some embodiments include scFv-mAb formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/

L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to CLDN6 as outlined herein; and a scFv domain that binds to CD3; b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a variable heavy domain that, with the variable light domain of the common light chain, makes up an Fv that binds to CLDN6 as outlined herein; and c) a common light chain comprising a variable light domain and a constant light domain.

9. Dual scFv Formats

The antibodies described herein also provide dual scFv formats as are known in the art (FIG. 36B). In this embodiment, the CLDN6×CD3 heterodimeric bispecific antibody is made up of two scFv-Fc monomers (both in either (VH-scFv linker-VL-[optional domain linker]-CH2-CH3) format or (VL-scFv linker-VH-[optional domain linker]-CH2-CH3) format, or with one monomer in one orientation and the other in the other orientation.

The antibodies described herein provide dual scFv formats where the CD3 binding domain sequences are as shown in FIG. 10A-10F or a variant thereof, and wherein the CLDN6 binding domain sequences are as shown in FIGS. 13-15 and 18 or a variant thereof. In some embodiments, the dual scFv format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments include dual scFv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, and a first scFv that binds either CD3 or CLDN6; and b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants

E233P/L234V/L235A/G236del/S267K, and a second scFv that binds either CD3 or CLDN6. In some embodiments, the dual scFv format includes skew variants, pI variants, ablation variants and FcRn variants. In some embodiments, the dual scFv format includes skew variants, pI variants, and ablation variants. Accordingly, some embodiments include dual scFv formats that comprise: a) a first monomer that comprises the skew variants S364K/E357Q, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a first scFv that binds either CD3 or CLDN6; and b) a second monomer that comprises the skew variants L368D/K370S, the pI variants N208D/Q295E/N384D/Q418E/N421D, the ablation variants E233P/L234V/L235A/G236del/S267K, the FcRn variants M428L/N434S and a second scFv that binds either CD3 or CLDN6.

10. Non-Heterodimeric Bispecific Antibodies

As will be appreciated by those in the art, the anti-CLDN6×anti-CD3 antibodies provided herein can also be included in non-heterodimeric bispecific formats (see FIG. 36J). In this format, the anti-CLDN6×anti-CD3 includes: 1) a first monomer comprising a VH1-CH1-hinge-CH2-CH3; 2) a second monomer comprising a VH2-CH1-hinge-CH2-CH3; 3) a first light chain comprising a VL1-CL; and 4) a second light chain comprising a VL2-CL. In such embodiments, the VH1 and VL1 form a first antigen binding domain and VH2 and VL2 form a second antigen binding domain. One of the first or second antigen binding domains binds CLDN6 and the other antigen binding domain binds CD3.

Any suitable CLDN6 binding domain and CD3 binding domain can be included in the anti-CLDN6×anti-CD3 anti-

body in the non-heterodimeric bispecific antibody format, including any of the CLDN6 binding domains and CD3 binding domains and related VHs and VLs provided herein or a variant thereof (see, e.g., FIGS. 10, 13-15 and 18).

11. Trident Format

In some embodiments, the bispecific antibodies described herein are in the “Trident” format as generally described in WO2015/184203, hereby expressly incorporated by reference in its entirety and in particular for the Figures, Legends, definitions and sequences of “Heterodimer-Promoting Domains” or “HPDs”, including “K-coil” and “E-coil” sequences. Tridents rely on using two different HPDs that associate to form a heterodimeric structure as a component of the structure. In this embodiment, the Trident format include a “traditional” heavy and light chain (e.g., VH1-CH1-hinge-CH2-CH3 and VL1-CL), a third chain comprising a first “diabody-type binding domain” or “DART®”, VH2-(linker)-VL3-HPD1 and a fourth chain comprising a second DART®, VH3-(linker)-(linker)-VL2-HPD2. The VH1 and VL1 form a first ABD, the VH2 and VL2 form a second ABD, and the VH3 and VL3 form a third ABD. In some cases, the second and third ABDs bind the same antigen, in this instance generally CLDN6, e.g., bivalently, with the first ABD binding a CD3 monovalently.

Any suitable CLDN6 binding domain and CD3 binding domain can be included in the anti-CLDN6×anti-CD3 antibody in the Trident bispecific antibody format, including any of the CLDN6 binding domains and CD3 binding domains and related VHs and VLs provided herein or a variant thereof (see, e.g., FIGS. 10, 13-15 and 18).

12. Monospecific, Monoclonal Antibodies

As will be appreciated by those in the art, the novel CLDN6 ABD sequences outlined herein can also be used in both monospecific antibodies (e.g., “traditional monoclonal antibodies”) or non-heterodimeric bispecific formats. Accordingly, in some embodiments, the antibodies described herein provide monoclonal (monospecific) antibodies comprising the 6 CDRs and/or the vh and vl sequences from the figures, generally with IgG1, IgG2, IgG3 or IgG4 constant regions, with IgG1, IgG2 and IgG4 (including IgG4 constant regions comprising a S228P amino acid substitution) finding particular use in some embodiments. That is, any sequence herein with a “H_L” designation can be linked to the constant region of a human IgG1 antibody.

Any suitable CLDN6 ABD can be included in the monospecific antibody, including any of the CLDN6 ABDs described herein. In some embodiments, the monospecific antibody is an CLDN6 monospecific antibody that has a VH₁VL₁ pairs selected from the group consisting of: H1₁L₁, H1.1₁L₁, H1.2₁L₁, H1.3₁L₁, H1.4₁L₁, H1.5₁L₁, H1.6₁L₁, H1.7₁L₁, H1.8₁L₁, H1.9₁L₁, H1.19₁L₁, H1.22₁L₁, H1.24₁L₁, H2₁L₁, H2.1₁L₁, H2.2₁L₁, H2.3₁L₁, H2.4₁L₁, H2.5₁L₁, H2.6₁L₁, H2.7₁L₁, H2.8₁L₁, H2.9₁L₁, H2.11₁L₁, H2.12₁L₁, H2.71₁L₁, H2.75₁L₁, H2.90₁L₁, H2.91₁L₁, H2.118₁L₁, H2.119₁L₁, H1₁L_{1.1}, H1.1_{1.1}L_{1.1}, H1.2_{1.1}L_{1.1}, H1.3_{1.1}L_{1.1}, H1.4_{1.1}L_{1.1}, H1.5_{1.1}L_{1.1}, H1.6_{1.1}L_{1.1}, H1.7_{1.1}L_{1.1}, H1.8_{1.1}L_{1.1}, H1.9_{1.1}L_{1.1}, H1.19_{1.1}L_{1.1}, H1.22_{1.1}L_{1.1}, H1.24_{1.1}L_{1.1}, H2_{1.1}L_{1.1}, H2.1_{1.1}L_{1.1}, H2.2_{1.1}L_{1.1}, H2.3_{1.1}L_{1.1}, H2.4_{1.1}L_{1.1}, H2.5_{1.1}L_{1.1}, H2.6_{1.1}L_{1.1}, H2.7_{1.1}L_{1.1}, H2.8_{1.1}L_{1.1}, H2.9_{1.1}L_{1.1}, H2.11_{1.1}L_{1.1}, H2.12_{1.1}L_{1.1}, H2.71_{1.1}L_{1.1}, H2.75_{1.1}L_{1.1}, H2.90_{1.1}L_{1.1}, H2.91_{1.1}L_{1.1}, H2.118_{1.1}L_{1.1}, H2.119_{1.1}L_{1.1}, H1_{1.4}L_{1.4}, H1.1_{1.4}L_{1.4}, H1.2_{1.4}L_{1.4}, H1.3_{1.4}L_{1.4}, H1.4_{1.4}L_{1.4}, H1.5_{1.4}L_{1.4}, H1.6_{1.4}L_{1.4}, H1.7_{1.4}L_{1.4}, H1.8_{1.4}L_{1.4}, H1.9_{1.4}L_{1.4}, H1.19_{1.4}L_{1.4}, H1.22_{1.4}L_{1.4}, H1.24_{1.4}L_{1.4}, H2_{1.4}L_{1.4}, H2.1_{1.4}L_{1.4}, H2.2_{1.4}L_{1.4}, H2.3_{1.4}L_{1.4}, H2.4_{1.4}L_{1.4}, H2.5_{1.4}L_{1.4}, H2.6_{1.4}L_{1.4}, H2.7_{1.4}L_{1.4},

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H2.8_L1.4, H2.9_L1.4, H2.11_L1.4, H2.12_L1.4, H2.71_L1.4, H2.75_L1.4, H2.90_L1.4, H2.91_L1.4, H2.118_L1.4, H2.119_L1.4, H1_L1.7, H1.1_L1.7, H1.2_L1.7, H1.3_L1.7, H1.4_L1.7, H1.5_L1.7, H1.6_L1.7, H1.7_L1.7, H1.8_L1.7, H1.9_L1.7, H1.19_L1.7, H1.22_L1.7, H1.24_L1.7, H2_L1.7, H2.1_L1.7, H2.2_L1.7, H2.3_L1.7, H2.4_L1.7, H2.5_L1.7, H2.6_L1.7, H2.7_L1.7, H2.8_L1.7, H2.9_L1.7, H2.11_L1.7, H2.12_L1.7, H2.71_L1.7, H2.75_L1.7, H2.90_L1.7, H2.91_L1.7, H2.118_L1.7, H2.119_L1.7, H1_L1.16, H1.1_L1.16, H1.2_L1.16, H1.3_L1.16, H1.4_L1.16, H1.5_L1.16, H1.6_L1.16, H1.7_L1.16, H1.8_L1.16, H1.9_L1.16, H1.19_L1.16, H1.22_L1.16, H1.24_L1.16, H2_L1.16, H2.1_L1.16, H2.2_L1.16, H2.3_L1.16, H2.4_L1.16, H2.5_L1.16, H2.6_L1.16, H2.7_L1.16, H2.8_L1.16, H2.9_L1.16, H2.11_L1.16, H2.12_L1.16, H2.71_L1.16, H2.75_L1.16, H2.90_L1.16, H2.91_L1.16, H2.118_L1.16, H2.119_L1.16, H1_L1.18, H1.1_L1.18, H1.2_L1.18, H1.3_L1.18, H1.4_L1.18, H1.5_L1.18, H1.6_L1.18, H1.7_L1.18, H1.8_L1.18, H1.9_L1.18, H1.19_L1.18, H1.22_L1.18, H1.24_L1.18, H2_L1.18, H2.1_L1.18, H2.2_L1.18, H2.3_L1.18, H2.4_L1.18, H2.5_L1.18, H2.6_L1.18, H2.7_L1.18, H2.8_L1.18, H2.9_L1.18, H2.11_L1.18, H2.12_L1.18, H2.71_L1.18, H2.75_L1.18, H2.90_L1.18, H2.91_L1.18, H2.118_L1.18, H2.119_L1.18, H1_L1.19, H1.1_L1.19, H1.2_L1.19, H1.3_L1.19, H1.4_L1.19, H1.5_L1.19, H1.6_L1.19, H1.7_L1.19, H1.8_L1.19, H1.9_L1.19, H1.19_L1.19, H1.22_L1.19, H1.24_L1.19, H2_L1.19, H2.1_L1.19, H2.2_L1.19, H2.3_L1.19, H2.4_L1.19, H2.5_L1.19, H2.6_L1.19, H2.7_L1.19, H2.8_L1.19, H2.9_L1.19, H2.11_L1.19, H2.12_L1.19, H2.71_L1.19, H2.75_L1.19, H2.90_L1.19, H2.91_L1.19, H2.118_L1.19, H2.119_L1.19, H1_L1.21, H1.1_L1.21, H1.2_L1.21, H1.3_L1.21, H1.4_L1.21, H1.5_L1.21, H1.6_L1.21, H1.7_L1.21, H1.8_L1.21, H1.9_L1.21, H1.19_L1.21, H1.22_L1.21, H1.24_L1.21, H2_L1.21, H2.1_L1.21, H2.2_L1.21, H2.3_L1.21, H2.4_L1.21, H2.5_L1.21, H2.6_L1.21, H2.7_L1.21, H2.8_L1.21, H2.9_L1.21, H2.11_L1.21, H2.12_L1.21, H2.71_L1.21, H2.75_L1.21, H2.90_L1.21, H2.91_L1.21, H2.118_L1.2, H2.119_L1.21, H1_L1.22, H1.1_L1.22, H1.2_L1.22, H1.3_L1.22, H1.4_L1.22, H1.5_L1.22, H1.6_L1.22, H1.7_L1.22, H1.8_L1.22, H1.9_L1.22, H1.19_L1.22, H1.22_L1.22, H1.24_L1.22, H2_L1.22, H2.1_L1.22, H2.2_L1.22, H2.3_L1.22, H2.4_L1.22, H2.5_L1.22, H2.6_L1.22, H2.7_L1.22, H2.8_L1.22, H2.9_L1.22, H2.11_L1.22, H2.12_L1.22, H2.71_L1.22, H2.75_L1.22, H2.90_L1.22, H2.91_L1.22, H2.118_L1.22, H2.119_L1.22, H1_L1.23, H1.1_L1.23, H1.2_L1.23, H1.3_L1.23, H1.4_L1.23, H1.5_L1.23, H1.6_L1.23, H1.7_L1.23, H1.8_L1.23, H1.9_L1.23, H1.19_L1.23, H1.22_L1.23, H1.24_L1.23, H2_L1.23, H2.1_L1.23, H2.2_L1.23, H2.3_L1.23, H2.4_L1.23, H2.5_L1.23, H2.6_L1.23, H2.7_L1.23, H2.8_L1.23, H2.9_L1.23, H2.11_L1.23, H2.12_L1.23, H2.71_L1.23, H2.75_L1.23, H2.90_L1.23, H2.91_L1.23, H2.118_L1.23, H2.119_L1.23, H1_L1.27, H1.1_L1.27, H1.2_L1.27, H1.3_L1.27, H1.4_L1.27, H1.5_L1.27, H1.6_L1.27, H1.7_L1.27, H1.8_L1.27, H1.9_L1.27, H1.19_L1.27, H1.22_L1.27, H1.24_L1.27, H2_L1.27, H2.1_L1.27, H2.2_L1.27, H2.3_L1.27, H2.4_L1.27, H2.5_L1.27, H2.6_L1.27, H2.7_L1.27, H2.8_L1.27, H2.9_L1.27, H2.11_L1.27, H2.12_L1.27, H2.71_L1.27, H2.75_L1.27, H2.90_L1.27, H2.91_L1.27, H2.118_L1.27, H2.119_L1.27, H1_L1.60, H1.1_L1.60, H1.2_L1.60, H1.3_L1.60, H1.4_L1.60, H1.5_L1.60, H1.6_L1.60, H1.7_L1.60, H1.8_L1.60, H1.9_L1.60, H1.19_L1.60, H1.22_L1.60, H1.24_L1.60,

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H2_L1.60, H2.1_L1.60, H2.2_L1.60, H2.3_L1.60, H2.4_L1.60, H2.5_L1.60, H2.6_L1.60, H2.7_L1.60, H2.8_L1.60, H2.9_L1.60, H2.11_L1.60, H2.12_L1.60, H2.71_L1.60, H2.75_L1.60, H2.90_L1.60, H2.91_L1.60, H2.118_L1.60, H2.119_L1.60, H1_L1.107, H1.1_L1.107, H1.2_L1.107, H1.3_L1.107, H1.4_L1.107, H1.5_L1.107, H1.6_L1.107, H1.7_L1.107, H1.8_L1.107, H1.9_L1.107, H1.19_L1.107, H1.22_L1.107, H1.24_L1.107, H2_L1.107, H2.1_L1.107, H2.2_L1.107, H2.3_L1.107, H2.4_L1.107, H2.5_L1.107, H2.6_L1.107, H2.7_L1.107, H2.8_L1.107, H2.9_L1.107, H2.11_L1.107, H2.12_L1.107, H2.71_L1.107, H2.75_L1.107, H2.90_L1.107, H2.91_L1.107, H2.118_L1.107, H2.119_L1.107, H1_L1.114, H1.1_L1.114, H1.2_L1.114, H1.3_L1.114, H1.4_L1.114, H1.5_L1.114, H1.6_L1.114, H1.7_L1.114, H1.8_L1.114, H1.9_L1.114, H1.19_L1.114, H1.22_L1.114, H1.24_L1.114, H2_L1.114, H2.1_L1.114, H2.2_L1.114, H2.3_L1.114, H2.4_L1.114, H2.5_L1.114, H2.6_L1.114, H2.7_L1.114, H2.8_L1.114, H2.9_L1.114, H2.11_L1.114, H2.12_L1.114, H2.71_L1.114, H2.75_L1.114, H2.90_L1.114, H2.91_L1.114, H2.118_L1.114, H2.119_L1.114, H1_L1.187, H1.1_L1.187, H1.2_L1.187, H1.3_L1.187, H1.4_L1.187, H1.5_L1.187, H1.6_L1.187, H1.7_L1.187, H1.8_L1.187, H1.9_L1.187, H1.19_L1.187, H1.22_L1.187, H1.24_L1.187, H2_L1.187, H2.1_L1.187, H2.2_L1.187, H2.3_L1.187, H2.4_L1.187, H2.5_L1.187, H2.6_L1.187, H2.7_L1.187, H2.8_L1.187, H2.9_L1.187, H2.11_L1.187, H2.12_L1.187, H2.71_L1.187, H2.75_L1.187, H2.90_L1.187, H2.91_L1.187, H2.118_L1.187, H2.119_L1.187, H1_L1.189, H1.1_L1.189, H1.2_L1.189, H1.3_L1.189, H1.4_L1.189, H1.5_L1.189, H1.6_L1.189, H1.7_L1.189, H1.8_L1.189, H1.9_L1.189, H1.19_L1.189, H1.22_L1.189, H1.24_L1.189, H2_L1.189, H2.1_L1.189, H2.2_L1.189, H2.3_L1.189, H2.4_L1.189, H2.5_L1.189, H2.6_L1.189, H2.7_L1.189, H2.8_L1.189, H2.9_L1.189, H2.11_L1.189, H2.12_L1.189, H2.71_L1.189, H2.75_L1.189, H2.90_L1.189, H2.91_L1.189, H2.118_L1.189, H2.119_L1.189, H1_L2, H1.1_L2, H1.2_L2, H1.3_L2, H1.4_L2, H1.5_L2, H1.6_L2, H1.7_L2, H1.8_L2, H1.9_L2, H1.19_L2, H1.22_L2, H1.24_L2, H2_L2, H2.1_L2, H2.2_L2, H2.3_L2, H2.4_L2, H2.5_L2, H2.6_L2, H2.7_L2, H2.8_L2, H2.9_L2, H2.11_L2, H2.12_L2, H2.71_L2, H2.75_L2, H2.90_L2, H2.91_L2, H2.118_L2 and H2.119_L2 or a variant thereof.

In particular monoclonal embodiments, the VH_VL pair is selected from the group consisting of H1.9_L1.187, H1.24_L1.187, H2.91_L1.187 and H1.9_L1.187 or a variant thereof.

E. Particular Embodiments of the Invention

The invention specifically provides 1+1 and 2+1 formats that bind CD3 and CLDN6.

1. 1+1 Format

In particular 1+1 format embodiments, the α CLDN6 ABD is the Fab and has the VH_VL pair H1.9_L1.187 and the α CD3 ABD is a scFv selected from the group consisting of anti-CD3 H1.30_L1.47, anti-CD3 H1.32_L1.47, anti-CD3 H1.89_L1.47, anti-CD3 H1.90_L1.47, anti-CD3 H1.33_L1.47, anti-CD3 H1.31_L1.47, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.32, anti-CD3 L1.47_H1.89, anti-CD3 L1.47_H1.90, anti-CD3 L1.47_H1.33, and anti-CD3 L1.47_H1.31.

In particular 1+1 format embodiments, the α CLDN6 ABD is the Fab and has the VH_VL pair H1.24_L1.187, and

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the α CD3 ABD is a scFv selected from the group consisting of anti-CD3 H1.30_L1.47, anti-CD3 H1.32_L1.47, anti-CD3 H1.89_L1.47, anti-CD3 H1.90_L1.47, anti-CD3 H1.33_L1.47, anti-CD3 H1.31_L1.47, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.32, anti-CD3 L1.47_H1.89, anti-CD3 L1.47_H1.90, anti-CD3 L1.47_H1.33, and anti-CD3 L1.47_H1.31.

In particular 1+1 format embodiments, the α CLDN6 ABD is the Fab and has the VH_VL pair H2.91_L1.187 and the α CD3 ABD is a scFv selected from the group consisting of anti-CD3 H1.30_L1.47, anti-CD3 H1.32_L1.47, anti-CD3 H1.89_L1.47, anti-CD3 H1.90_L1.47, anti-CD3 H1.33_L1.47, anti-CD3 H1.31_L1.47, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.32, anti-CD3 L1.47_H1.89, anti-CD3 L1.47_H1.90, anti-CD3 L1.47_H1.33, and anti-CD3 L1.47_H1.31.

In particular 1+1 format embodiments, the α CLDN6 ABD is the Fab and has the VH_VL pair H1.9_L1.187. and the α CD3 ABD is a scFv selected from the group consisting of anti-CD3 H1.30_L1.47, anti-CD3 H1.32_L1.47, anti-CD3 H1.89_L1.47, anti-CD3 H1.90_L1.47, anti-CD3 H1.33_L1.47, anti-CD3 H1.31_L1.47, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.32, anti-CD3 L1.47_H1.89, anti-CD3 L1.47_H1.90, anti-CD3 L1.47_H1.33, and anti-CD3 L1.47_H1.31.

2. 2+1 Format

In particular 2+1 format embodiments, the α CLDN6 ABD is the Fab and has the VH_VL pair H1.9_L1.187 and the α CD3 ABD is a scFv selected from the group consisting of anti-CD3 H1.30_L1.47, anti-CD3 H1.32_L1.47, anti-CD3 H1.89_L1.47, anti-CD3 H1.90_L1.47, anti-CD3 H1.33_L1.47, anti-CD3 H1.31_L1.47, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.32, anti-CD3 L1.47_H1.89, anti-CD3 L1.47_H1.90, anti-CD3 L1.47_H1.33, and anti-CD3 L1.47_H1.31.

In particular 2+1 format embodiments, the α CLDN6 ABD is the Fab and has the VH_VL pair H1.24_L1.187, and the α CD3 ABD is a scFv selected from the group consisting of anti-CD3 H1.30_L1.47, anti-CD3 H1.32_L1.47, anti-CD3 H1.89_L1.47, anti-CD3 H1.90_L1.47, anti-CD3 H1.33_L1.47, anti-CD3 H1.31_L1.47, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.32, anti-CD3 L1.47_H1.89, anti-CD3 L1.47_H1.90, anti-CD3 L1.47_H1.33, and anti-CD3 L1.47_H1.31.

In particular 2+1 format embodiments, the α CLDN6 ABD is the Fab and has the VH_VL pair H2.91_L1.187 and the α CD3 ABD is a scFv selected from the group consisting of anti-CD3 H1.30_L1.47, anti-CD3 H1.32_L1.47, anti-CD3 H1.89_L1.47, anti-CD3 H1.90_L1.47, anti-CD3 H1.33_L1.47, anti-CD3 H1.31_L1.47, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.32, anti-CD3 L1.47_H1.89, anti-CD3 L1.47_H1.90, anti-CD3 L1.47_H1.33, and anti-CD3 L1.47_H1.31.

In particular 2+1 format embodiments, the α CLDN6 ABD is the Fab and has the VH_VL pair H1.9_L1.187. and the α CD3 ABD is a scFv selected from the group consisting of anti-CD3 H1.30_L1.47, anti-CD3 H1.32_L1.47, anti-CD3 H1.89_L1.47, anti-CD3 H1.90_L1.47, anti-CD3 H1.33_L1.47, anti-CD3 H1.31_L1.47, anti-CD3 L1.47_H1.30, anti-CD3 L1.47_H1.30, anti-CD3

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L1.47_H1.32, anti-CD3 L1.47_H1.89, anti-CD3 L1.47_H1.90, anti-CD3 L1.47_H1.33, and anti-CD3 L1.47_H1.31.

V. Nucleic Acids of the Invention

The disclosure further provides nucleic acid compositions encoding the anti-CLDN6 antibodies provided herein, including, but not limited to, anti-CLDN6 $\times$ anti-CD3 bispecific antibodies and CLDN6 monospecific antibodies.

As will be appreciated by those in the art, the nucleic acid compositions will depend on the format and scaffold of the heterodimeric protein. Thus, for example, when the format requires three amino acid sequences, such as for the 1+1 Fab-scFv-Fc format (e.g. a first amino acid monomer comprising an Fc domain and a scFv, a second amino acid monomer comprising a heavy chain and a light chain), three nucleic acid sequences can be incorporated into one or more expression vectors for expression. Similarly, some formats (e.g. dual scFv formats such as disclosed in FIG. 1) only two nucleic acids are needed; again, they can be put into one or two expression vectors.

As is known in the art, the nucleic acids encoding the components of the antibodies described herein can be incorporated into expression vectors as is known in the art, and depending on the host cells used to produce the heterodimeric antibodies described herein. Generally the nucleic acids are operably linked to any number of regulatory elements (promoters, origin of replication, selectable markers, ribosomal binding sites, inducers, etc.). The expression vectors can be extra-chromosomal or integrating vectors.

The nucleic acids and/or expression vectors of the antibodies described herein are then transformed into any number of different types of host cells as is well known in the art, including mammalian, bacterial, yeast, insect and/or fungal cells, with mammalian cells (e.g. CHO cells), finding use in many embodiments.

In some embodiments, nucleic acids encoding each monomer and the optional nucleic acid encoding a light chain, as applicable depending on the format, are each contained within a single expression vector, generally under different or the same promoter controls. In embodiments of particular use in the antibodies described herein, each of these two or three nucleic acids are contained on a different expression vector. As shown herein and in 62/025,931, hereby incorporated by reference, different vector ratios can be used to drive heterodimer formation. That is, surprisingly, while the proteins comprise first monomer:second monomer:light chains (in the case of many of the embodiments herein that have three polypeptides comprising the heterodimeric antibody) in a 1:1:2 ratio, these are not the ratios that give the best results.

The heterodimeric antibodies described herein are made by culturing host cells comprising the expression vector(s) as is well known in the art. Once produced, traditional antibody purification steps are done, including an ion exchange chromatography step. As discussed herein, having the pIs of the two monomers differ by at least 0.5 can allow separation by ion exchange chromatography or isoelectric focusing, or other methods sensitive to isoelectric point. That is, the inclusion of pI substitutions that alter the isoelectric point (pI) of each monomer so that such that each monomer has a different pI and the heterodimer also has a distinct pI, thus facilitating isoelectric purification of the "1+1 Fab-scFv-Fc" and "2+1" heterodimers (e.g., anionic exchange columns, cationic exchange columns). These substitutions also aid in the determination and monitoring of

any contaminating dual scFv-Fc and mAb homodimers post-purification (e.g., IEF gels, cIEF, and analytical IEX columns).

VI. Biological and Biochemical Functionality of the Heterodimeric Bispecific Antibodies

Generally the bispecific CLDN6×CD3 antibodies described herein are administered to patients with cancer, and efficacy is assessed, in a number of ways as described herein. Thus, while standard assays of efficacy can be run, such as cancer load, size of tumor, evaluation of presence or extent of metastasis, etc., immuno-oncology treatments can be assessed on the basis of immune status evaluations as well. This can be done in a number of ways, including both in vitro and in vivo assays.

VII. Treatments

Once made, the compositions of the antibodies described herein find use in a number of applications. CLDN6 is highly expressed in renal cell carcinoma, accordingly, the heterodimeric compositions of the antibodies described herein find use in the treatment of such CLDN6 positive cancers.

VIII. Antibody Compositions for In Vivo Administration

Formulations of the antibodies used in accordance with the antibodies described herein are prepared for storage by mixing an antibody having the desired degree of purity with optional pharmaceutically acceptable carriers, excipients or stabilizers (Remington's Pharmaceutical Sciences 16th edition, Osol, A. Ed. [1980]), in the form of lyophilized formulations or aqueous solutions.

IX. Administrative Modalities

The antibodies and chemotherapeutic agents described herein are administered to a subject, in accord with known methods, such as intravenous administration as a bolus or by continuous infusion over a period of time.

X. Treatment Modalities

In the methods described herein, therapy is used to provide a positive therapeutic response with respect to a disease or condition. By "positive therapeutic response" is intended an improvement in the disease or condition, and/or an improvement in the symptoms associated with the disease or condition. For example, a positive therapeutic response would refer to one or more of the following improvements in the disease: (1) a reduction in the number of neoplastic cells; (2) an increase in neoplastic cell death; (3) inhibition of neoplastic cell survival; (5) inhibition (i.e., slowing to some extent, preferably halting) of tumor growth; (6) an increased patient survival rate; and (7) some relief from one or more symptoms associated with the disease or condition.

Positive therapeutic responses in any given disease or condition can be determined by standardized response criteria specific to that disease or condition. Tumor response can be assessed for changes in tumor morphology (i.e., overall tumor burden, tumor size, and the like) using screening techniques such as magnetic resonance imaging (MRI) scan, x-radiographic imaging, computed tomographic (CT) scan, bone scan imaging, endoscopy, and tumor biopsy

sampling including bone marrow aspiration (BMA) and counting of tumor cells in the circulation.

In addition to these positive therapeutic responses, the subject undergoing therapy may experience the beneficial effect of an improvement in the symptoms associated with the disease.

Treatment according to the disclosure includes a "therapeutically effective amount" of the medicaments used. A "therapeutically effective amount" refers to an amount effective, at dosages and for periods of time necessary, to achieve a desired therapeutic result.

A therapeutically effective amount may vary according to factors such as the disease state, age, sex, and weight of the individual, and the ability of the medicaments to elicit a desired response in the individual. A therapeutically effective amount is also one in which any toxic or detrimental effects of the antibody or antibody portion are outweighed by the therapeutically beneficial effects.

A "therapeutically effective amount" for tumor therapy may also be measured by its ability to stabilize the progression of disease. The ability of a compound to inhibit cancer may be evaluated in an animal model system predictive of efficacy in human tumors.

Alternatively, this property of a composition may be evaluated by examining the ability of the compound to inhibit cell growth or to induce apoptosis by in vitro assays known to the skilled practitioner. A therapeutically effective amount of a therapeutic compound may decrease tumor size, or otherwise ameliorate symptoms in a subject. One of ordinary skill in the art would be able to determine such amounts based on such factors as the subject's size, the severity of the subject's symptoms, and the particular composition or route of administration selected.

Dosage regimens are adjusted to provide the optimum desired response (e.g., a therapeutic response). For example, a single bolus may be administered, several divided doses may be administered over time or the dose may be proportionally reduced or increased as indicated by the exigencies of the therapeutic situation. Parenteral compositions may be formulated in dosage unit form for ease of administration and uniformity of dosage. Dosage unit form as used herein refers to physically discrete units suited as unitary dosages for the subjects to be treated; each unit contains a predetermined quantity of active compound calculated to produce the desired therapeutic effect in association with the required pharmaceutical carrier.

The specification for the dosage unit forms of the disclosure are dictated by and directly dependent on (a) the unique characteristics of the active compound and the particular therapeutic effect to be achieved, and (b) the limitations inherent in the art of compounding such an active compound for the treatment of sensitivity in individuals.

The efficient dosages and the dosage regimens for the bispecific antibodies described herein depend on the disease or condition to be treated and may be determined by the persons skilled in the art.

An exemplary, non-limiting range for a therapeutically effective amount of an bispecific antibody used in the antibodies described herein is about 0.1-100 mg/kg.

All cited references are herein expressly incorporated by reference in their entirety.

Whereas particular embodiments of the disclosure have been described above for purposes of illustration, it will be appreciated by those skilled in the art that numerous varia-

tions of the details may be made without departing from the invention as described in the appended claims.

EXAMPLES

A. Example 1: Engineering α CLDN6 $\times\alpha$ CD3 Bispecific Antibodies

Sequences for CD3 binding domains having different CD3 binding affinities are depicted in FIG. 10. Sequences for CLDN6 binding domains are depicted in FIGS. 14-15. A number of formats for α CLDN6 $\times\alpha$ CD3 bispecific antibodies (bsAbs) were conceived, illustrative formats for which are outlined below and in FIG. 17.

One such format is the 1+1 Fab-scFv-Fc format which comprises a single-chain Fv ("scFv") covalently attached to a first heterodimeric Fc domain, a heavy chain variable region (VH) covalently attached to a complementary second heterodimeric Fc domain, and a light chain (LC) transfected separately so that a Fab domain is formed with the variable heavy domain.

Another format is the 2+1 Fab₂-scFv-Fc format which comprises a VH domain covalently attached to a CH1 domain covalently attached to an scFv covalently attached to a first heterodimeric Fc domain (VH-CH1-scFv-Fc), a VH domain covalently attached to a complementary second heterodimeric Fc domain, and a LC transfected separately so that Fab domains are formed with the VH domains.

DNA encoding chains of the α CLDN6 $\times\alpha$ CD3 bsAbs were generated by standard gene synthesis followed by isothermal cloning (Gibson assembly) or subcloning into a pTT5 expression vector containing fusion partners (e.g. domain linkers as depicted in FIG. 6 and/or backbones as depicted in FIGS. 7-9). DNA was transfected into HEK293E cells for expression. Sequences for illustrative α CLDN6 $\times\alpha$ CD3 bsAbs (based on binding domains as described above) in the 1+1 Fab-scFv-Fc format and in the 2+1 Fab₂-scFv-Fc format are depicted respectively in FIGS. 19-22.

B. Example 2: Engineering CLDN6 Binding Domains for Enhanced Selectivity

The claudin family of proteins include numerous other claudins. Jukes-Cantor distance of various claudin sequences to CLDN6 were determined as follows: CLDN9 (0.32), CLDN4 (0.51), CLDN3 (0.52), CLDN5 (0.64), CLDN8 (0.8), and CLDN17 (0.82). In comparison to CLDN6, CLDN9 is more highly expressed in healthy tissues (e.g. the cervix and the esophagus), so cross-reactivity of CLDN6 therapeutic with CLDN9 could lead to off-target toxicity. However in addition to having Jukes-Cantor Distance off just 0.32 and 96% identity, CLDN6 and CLDN9 differ by only 3 residues in their extracellular loops (as depicted in FIG. 12); therefore, it is a significant challenge to develop an antibody capable of binding CLDN6 selectively over CLDN9. Accordingly, this section describes identification and engineering of suitable CLDN6 binding domains (with minimal cross-reactivity) suitable for use in the α CLDN6 $\times\alpha$ CD3 bsAbs of the invention. Additionally, a prior art CLDN6 binding domain described in U.S. Pat. No. 10,233,253 was used as a comparator (in a bivalent mAb format as in XENP26863 and in a 1+1 Fab-scFv-Fc bsAb with CD3 High scFv as in XENP26849; sequences depicted in FIG. 18).

The following experiments used to screen CLDN6 binding domains were generally performed as follows.

HEK293T or HEK293E cells were respectively stably or transiently transfected to express CLDN6, CLDN9, or other targets of interest. 48 hours post transfection, cells were harvested and incubated with dilutions of the indicated test articles for 1 hour at 4° C. Next, cells were washed and stained with a secondary antibody (typically anti-human Fc AlexaFluor647) for 1 hour at 4° C., followed by two more washes. Binding was then assessed via flow cytometry.

1. Identification of CLDN6 Binding Domains Selective for CLDN6 Over CLDN9

Six murine anti-CLDN6 mAbs were investigated for their binding to CLDN6 and CLDN9, as well to CLDN3 and CLDN4 which are also members of the claudin family. FIG. 23 depicts binding of the mAbs to transfected cells. Specificity for targets was determined by comparing EC50 values and AUC (area under the binding curve). The data show that clones C6-10, C6-15, and C6-21 were not selective for CLDN6 over CLDN9. Additionally, clone C6-10 even showed some binding to CLDN4. Notably, clones C6-11, C6-24, and C6-30 demonstrated selectivity for CLDN6 over CLDN9 and little to no binding to CLDN3 and CLDN4.

2. Humanization of CLDN6 Binding Domains

The variable regions of C6-11, C6-24, and C6-30 were humanized using string content optimization (see, e.g., U.S. Pat. No. 7,657,380, issued Feb. 2, 2010). For each clone, two humanized variable heavy domains (i.e. H1 and H2) and two humanized variable light domains (i.e. L1 and L2) were engineered and paired (i.e. H1L1, H1L2, H2L1, H2L2). Sequences for humanized C6-30 variable domains and bivalent mAbs based on the domains are depicted respectively in FIGS. 14 and 16 (as XENP34218, XENP34219, XENP34220, and XENP34221). Binding of the humanized clones to CLDN3, CLDN4, CLDN6, and CLDN9 were investigated to determine whether humanization affected their binding profiles and are depicted in FIGS. 24-26. All humanized variants of clone C6-30 and C6-24 demonstrated binding profiles equivalent to their murine precursor. Unexpectedly, all humanized variants of clone C6-11 unfavorably demonstrated increased binding to CLDN4.

3. Engineering to Remove Degradation Liable Residues

The sequence for C6-30 was investigated for degradation liable residues. Heavy chain CDR2 included D52/P52a as an isomerization motif; N54/G55 as a deamidation motif; and Q64/G75 as a deamidation motif. Accordingly, a library was made with mutations at these residues to investigate whether the liability could be removed without impacting on CLDN6 binding and selectivity. FIG. 27 depicts the variants investigated as well as their binding to CLDN6 and CLDN9 as generally described above. The data show that many of the variants that maintained selectivity had decreased binding to CLDN6. Notably, G55A did maintain selectivity with minimal loss to CLDN6 binding.

4. Characterization of CLDN6 Binding Domains for Cross-Reactivity in the Context of a Bispecific Antibody

Prototype α CLDN6 $\times\alpha$ CD3 bispecific antibodies (bsAbs) in both the 1+1 Fab-scFv-Fc format and the 2+1 Fab₂-scFv-Fc format were engineered and produced (as described in Example 1) using the humanized CLDN6 binding domains described above and the CD3 binding domains as described in Example 1.

For ease of clinical development (e.g. by investigating the therapeutics in model animals), it is useful for the binding domains to be cross-reactive for mouse and/or cynomolgus antigen. Therefore, the prototype α CLDN6 $\times\alpha$ CD3 bsAbs were investigated for their binding to human CLDN6, cyno-

molgus CLDN6, and murine CLDN6. As shown in FIG. 28, each of the prototype bsAbs were able to human, cynomolgus, and mouse CLDN6.

Next, the prototype bsAbs were investigated for their binding to confirm that formatting the CLDN6 binding domains as bispecific antibodies did not affect their selectivity for CLDN6 over CLDN9. Selectivity was determined according to CLDN9/CLDN6 EC50 ratio. As shown in FIG. 29, each of the bispecific antibodies maintained selectivity for CLDN6 over CLDN9, although the ones based on C6-30 were more selective. Notably as well, while humanized C6-24 demonstrated similar selectivity in both the 1+1 Fab-scFv-Fc format and the 2+1 Fab₂-scFv-Fc format, the humanized C6-30 variants demonstrated further enhanced selectivity in the 2+1 Fab₂-scFv-Fc format.

Additionally, binding to CLDN6 I143V isotype was investigated, as this variant is found in ~30% of the population, and the data show that each of the variant was able to bind the I143V isotype.

5. Engineering for Enhanced CLDN6 Selectivity

Humanized C6-30 (both H1L1 and H2L1) which already demonstrated good selectivity, especially in the context of 2+1 Fab₂-scFv-Fc, was engineered to enhance selectivity. In a first round of engineering, single point mutations were introduced into either the variable heavy or the variable light domain resulting in a library of ~300 variants formatted as bivalent mAbs. Each mAb was screened for binding to CLDN6 and CLDN9 (on transfected cells) at a single concentration of 30 µg/ml. Most of point mutations had a similar impact on both CLDN6 and CLDN9 (e.g. improved binding to both or reduced binding to both). Nonetheless, there were several point mutations which were found to reduce binding to CLDN9 while maintaining or even improving binding to CLDN6 (as indicated by a skew toward higher CLDN6 MFI values and lower CLDN9 MFI values; circled in FIG. 30). The binding of these variants to CLDN6 and CLDN9 were re-investigated at various concentrations in two separate experiments, data depicting EC50 and CLDN9/CLDN6 EC50 ratio for which are depicted in FIG. 31. It should be noted that binding data from the experiments cannot be compared head-to-head as antigen density on the transfected cells vary between experiments. Nonetheless, several variants were enhanced in selectivity in comparison to the parental humanized clone (i.e. XENP34218_H1L1 and XENP34220_H2L1) and in comparison to comparator XENP26863. While some of the variants such as XENP35865 having H2_L1.60 were weaker CLDN6 binders in comparison to the parental humanized clones, they were also much weaker CLDN9 binders providing drastically enhanced selectivity for CLDN6 over CLDN9. Interestingly, several variants such as XENP35101 and XENP35102 respectively having H2.8_L1 and H2.9_L1 resulted in drastically enhanced selectivity for CLDN9 over CLDN6.

Next, favorite variant heavy domains (H1.9, H1.22, H1.24, H2.3, H2.9, H2.12, H2.90, H2.91, and H2.118) and favorite variant light domains (L1.187 and L1.189) from the first round of engineering that provided the best improvement in CLDN6 selectivity were combined to create a new library of 16 variants. The binding of these variants to CLDN6 and CLDN9 were investigated as described above, data for which are shown in FIG. 32. The data in FIG. 33 depicts the EC50 and AUC for each of the test articles as well as the CLDN9/CLDN6 EC50 ratio and CLDN6/CLDN9 AUC ratio as indicators of selectivity. Each of the 16 combination variants were enhanced in selectivity in comparison to the parental humanized clone (e.g.

XENP36956 having H1.22_L1.87 and a CLDN9/CLDN6 EC50 ratio of 44.94 in comparison to XENP34218 having H1L1 and a CLDN9/CLDN6 EC50 ratio of 23.03). Notably, combining favorite variant heavy domains and favorite variant light domain did not always enhance selectivity in comparison to the parental variant (e.g. XENP36956 having H1.22_L1.187 and a CLDN9/CLDN6 EC50 ratio of 44.94 in comparison to XENP36972 having H1_L1.187 and a CLDN9/CLDN6 EC50 ratio of 52.64). Finally, 10 of the 16 combination variants demonstrated superior selectivity in comparison to comparator XENP26863 based on CLDN9/CLDN6 EC50 ratio; and 16/16 combination variants demonstrated superiority based on CLDN6/CLDN9 AUC ratio. It should be noted that variants combining variable heavy domain mutations were also investigated, but these variants did not outperform the clones with only one mutation per variable heavy domain. It should also be noted that the ratios should not be the only determinant of enhanced selectivity. For example, although XENP36956 having H1.9_L1.187 and CLDN9/CLDN6 EC50 of 44.94 appears to be slightly less selective than control XENP26863, XENP36956 had a much higher EC50 for CLDN9 binding in comparison to XENP26863 (29584 ng/ml in comparison to 2146 ng/ml) while maintaining a reasonable EC50 for CLDN6 binding.

6. Investigating Engineered CLDN6 Binding Domains in the Context of Bispecific Antibodies

Finally, several of the preferred combination variants were incorporated into 2+1 Fab₂-scFv-Fc CD3 bsAbs to investigate whether the bispecific format affected their selectivity. Surprisingly as depicted in FIG. 34A, many of the 2+1 Fab₂-scFv-Fc bispecific antibodies including XENP37231 having C6-30_H1.24_L1.187, XENP37227 having C6-30_H1.9_L1.187, and XENP37233 having C6-30_H2.91_L1.187 demonstrated enhanced CLDN6 binding in comparison to corresponding bivalent mAb having the same variant CLDN6 binding domain; however, the C6-30_H1.22_L1.89 variant actually demonstrated decreased binding in the bispecific format in comparison to the monospecific bivalent format while several other variants demonstrated similar binding in both the bispecific format and the monospecific bivalent format. Additionally surprising as depicted in FIG. 34B, several of the 2+1 Fab₂-scFv-Fc bispecific antibodies including XENP37227 comprising C6-30_H1.9_L1.187 variant and XENP37231 having C6-30_H1.24_L1.187 variant demonstrated reduced CLDN9 binding in comparison to corresponding bivalent mAb having the same variant CLDN6 binding domain.

7. Investigating Additional Off-Target Binding

To ascertain cross-reactivity to other proteins in the claudin family, binding of C6-24_H1L1, C6-30_H1L1, and C6-30_H2L1 to CLDN5, CLDN8, and CLDN17 was investigated as generally described above. Data as depicted in FIG. 35 show that none of the clones were cross-reactive for the additional claudins investigated. Additionally, C6-30 variants were screened alongside XENP37217 (a CLDN6×CD3 bispecific using a comparator CLDN6 binding domain, the sequence for which is depicted in FIG. 60) for binding to CLDN3, CLDN4, CLDN8, and CLDN17. 293 cells were transiently transfected with CLDN3, CLDN4, CLDN8, or CLDN17. Cells were plated and each test article was added to each cell line at a 100 µg/ml dose. After 1 hour incubation at 4° C., cells were washed and a secondary AF647 antibody was added. Cells were incubated at 4° C. for another hour, followed by additional washing and then analysis by flow cytometry. For each of these different claudin family members, the comparator molecule XENP37217 showed higher binding than the test articles utilizing the C6-30 binding

domains. XENP37217 showed particularly strong binding for CLDN8 and CLDN17, producing an MFI more than an order of magnitude higher than the C6-30 variants, as depicted in FIG. 61. This highlights the potential of CLDN6 binding domains having specificity for binding not only CLDN6 but also other claudin family members.

C. Example 3: Engineering α CLDN6 $\times$ α CD3 bsAbs to Optimize Redirected T Cell Cytotoxicity on CLDN6-Expressing Cells

α CLDN6 $\times$ α CD3 bsAbs were engineered with various tuned CLDN6 and CD3 binding domains and in different bispecific formats and produced as generally described in Example 1 to optimize redirected T cell cytotoxicity (RTCC), CLDN6 selectivity, and potential therapeutic index.

1. α CLDN6 $\times$ α CD3 bsAbs with Bivalent CLDN6 Binding are Enhanced in Redirected T Cell Cytotoxicity (RTCC) on CLDN6+ Cells

α CLDN6 $\times$ α CD3 bsAbs were engineered using C6-24_H1L1, C6-30_H1L1, and C6-30_H2L1 in 1+1 Fab-scFv-Fc and 2+1 Fab₂-scFv-Fc formats to investigate the impact of avidity (i.e. monovalent vs. bivalent binding of CLDN6 antigen). PA-1 cells (CLDN6hi; 1.1×10^6 CLDN6 density) were incubated with T cells isolated from human PBMCs at a 10:1 effector:target ratio for 48 hours. Data showing RTCC activity (as indicated by percentage dead cells and T cell activation) are depicted in FIGS. 37-38. The 2+1 Fab₂-scFv-Fc constructs showed 20-100 fold lower EC50 in comparison to 1+1 constructs. Additionally, potency shifts between the 2+1 vs. 1+1 construct were ~half-log more for C6-30_H2L1 in comparison to C6-30_H1L1. Collectively, the data indicate that bsAbs with bivalent CLDN6 binding are enhanced in RTCC on CLDN6+ target cells.

2. α CLDN6 $\times$ α CD3 bsAbs with Bivalent CLDN6 Binding and Lower Affinity CD3 Binding are Enhanced in Selectivity for CLDN6 Over CLDN9

Next, α CLDN6 $\times$ α CD3 bsAbs having C6-30_H2L1 and either high affinity CD3_High or lower affinity CD3_High-Int #1 and in either 1+1 Fab-scFv-Fc or 2+1 Fab₂-scFv-Fc formats were investigated for their selectivity for CLDN6+ target cells. HUTU-80 cells (CLDN6+) or HEK293-TREX cells stably transfected to express CLDN9 were incubated with T cells purified from human PBMCs at a 10:1 effector:target ratio for 48 hours. Comparator bsAb XENP26849 was also used. Data showing RTCC activity (as indicated by percentage dead cells and T cell activation) are depicted in FIGS. 39-40. Surprisingly, induction of RTCC activity on HUTU-80 cells by bsAbs having high affinity CD3_High only differed slightly than by bsAbs having lower affinity CD3_High-Int #1. However, consistent with Example 3A, the 2+1 format enabled much more potent RTCC activity in comparison to the 1+1 format. Notably, 2+1 Fab₂-scFv-Fc bsAb having the lower affinity CD3_High-Int #1 induced RTCC on off-target CLDN9+ cells less potently than 2+1 bsAbs having high affinity CD3_High, indicating that using lower affinity CD3 binding domains in the context of a 2+1 Fab₂-scFv-Fc format favorably improves selectivity for CLDN6 over CLDN9 expressing cells. Further, each of the bsAbs based on C6-30 demonstrated much weaker induction of RTCC on CLDN9 expressing cells in comparison to comparator bsAb XENP26849.

3. α CLDN6 $\times$ α CD3 bsAb Activity Correlates with CLDN6 Antigen Density and CD3 Binding Affinity

Next, RTCC activity on cell lines having different CLDN6 densities was investigated. 30K PA-1 (1.1×10^6 CLDN6

density), OV-90 (195K CLDN6 density), NEC-8 (175K CLDN6 density), and COV-318 (11K CLDN6 density) cells were incubated with T cells purified from human PBMCs at a 10:1 E:T ratio for 48 hours. RTCC activity (as indicated by percentage dead cells and T cell activation) are depicted in FIGS. 41-44. The data show that cell killing activity and T cell activation correlates with CLDN6 antigen density and affinity of the CD3 binding domain. Selectivity for higher CLDN6 antigen density over lower CLDN6 antigen density suggests potential reduced off-target killing of healthy tissue expressing low levels of CLDN6 in a clinical setting and improved therapeutic index.

4. Tuning α CLDN6 $\times$ α CD3 bsAb activity and selectivity

α CLDN6 $\times$ α CD3 bsAbs were further tuned by incorporating the lead selectivity-tuned CLDN6 binding domains (as described in Example 2) into the 2+1 Fab₂-scFv-Fc format with CD3_High-Int #1.

30K HUTU-80 (CLDN6+) or HEK293TREX cells stably transfected to express CLDN9 were incubated with T cells purified from human PBMC and test articles for 48 hours. In a first experiment, 10:1 effector:target ratio was used, data for which are depicted in FIG. 45. In a second experiment, 1:1 effector:target ratio was used, data for which are depicted in FIG. 46. Consistent with the data in Example 2.6, the bsAbs having selectivity-engineered CLDN6 binding domains demonstrated modulated activity on CLDN6+ cells and/or modulated activity on CLDN9+ cells in comparison to bsAbs having the parental C6-30_H1L1 and C6-30_H2L1. In one instance, XENP37233 demonstrated similar activity on CLDN9+ cells but significantly enhanced activity on CLDN6+ cells in comparison to the parental clones. In another instance, XENP37227 demonstrated reduced activity on CLDN9+ cells and enhanced activity on CLDN6+ cells (in comparison to the parental clones. In yet another instance, XENP37231 demonstrated little to no activity on CLDN9+ cells but slightly reduced activity on CLDN6+ cells in comparison to the parental clones. Each of these bsAbs may be suitable depending on whether clinical development favors enhanced activity on CLDN6+ cells, reduced activity on CLDN9+ cells, or preferably a balance of the two. Additionally in the second experiment, activation of T cells as indicated by CD69 expression, CD107a expression, and cytokine (IFN γ , IL-2, and TNF α) secretion was also determined as depicted in FIGS. 47-51, and the observations are consistent with the cell killing data.

FIG. 52 additionally depicts overlay of RTCC activity on HUTU-80 vs. on HEK293TREX expressing CLDN9 cells. The data show that each bsAb can be dosed at high concentrations to achieve efficacious killing of CLDN6+ cells while avoiding killing of CLDN9+ cells.

In another experiment, the RTCC activity of the bsAbs described above on PA-1 cells were compared to XENP37630 (sequences for which are depicted in FIG. 59) having the same CLDN6 binding domain as XENP37227 (i.e. C6-30_H1.9_L1.187) and lower affinity CD3 scFv (i.e. CD3_High-Int #2). Cell kill and T cell activation (as indicated by CD25 and CD107a expression) are depicted in FIG. 53 and show that each of the 4 bsAbs achieved efficacious target cell killing at high concentrations, with potencies ranked as XENP37223 (highest potency)>XENP37227>XENP37630>XENP37231 (lowest potency).

D. Example 4: Further Characterization of α CLDN6 $\times$ α CD3 bsAbs of the Invention

Based on the above in vitro experiments, several bsAbs were selected for further analysis in vivo. These antibodies

were further engineered with Xtend Fc (M428L/N434S) to enhanced serum half-life, illustrative sequences for which are depicted in FIGS. 22 and 59 as XENP37547 (Xtend analog to XENP37233), XENP37545 (Xtend analog to XENP37231), XENP37634 (Xtend analog to XENP37630), and XENP37541 (Xtend analog to XENP37227).

1. α CLDN6 $\times\alpha$ CD3 bsAbs Kill Cells Representative of CLDN6 in Tumors

In order to confirm the ability of α CLDN6 $\times\alpha$ CD3 bsAbs to effectively target cell lines expressing levels of CLDN6 antigen that are biologically relevant to tumor cells, IHC was conducted on biopsy cores of ovarian cancers and were qualitatively scored on a scale of 0-3 with 0 representing little to no target antigen expression and 3 representing high target antigen expression (herein referred to as IHC score). While there was significant variation among different types of ovarian cancers, many had IHC scores in the 1-2 range, representing approximately 200 k or fewer CLDN6 antigens per cell. For this reason, PA-1 cells were genetically engineered to express CLDN6 at a lower range of densities. Engineered PA-1 cell lines were established to express either 47 k, 55 k, 77 k, 105 k, 212 k, or 228 k CLDN6 antigens per cell approximately. A cell binding experiment was performed in which XENP37541 was incubated with each of the lower density PA-1 cell lines. Cell lines were then washed, stained with secondary antibody, and washed again before measuring binding by flow cytometry. As seen in FIG. 54, the level of binding is dependent on the density of CLDN6 expression, but XENP37541 is still capable of binding to cells with lower levels of CLDN6 expression, even down to 47 k CLDN6 antigens per cell.

Next, α CLDN6 $\times\alpha$ CD3 bsAbs XENP37541, XENP37634, and XENP37545, along with XENP32140 (a negative control RSV $\times$ CD3 bsAb), were tested for their ability to induce RTCC on these lower CLDN6 density PA-1 cell lines. A 1:1 effector:target cell ratio was used, along with

the test articles at the range of concentrations indicated in FIG. 55. After a 72 hour incubation, results were measured as depicted in FIG. 55, and showed that XENP37541, XENP37634, and XENP37545 are all able to effectively induce RTCC in a CLDN6 density dependent manner even at a lower CLDN6 expression levels. The results showed that XENP37541 had the strongest potency and lowest EC50 values across all CLDN6 densities.

2. α CLDN6 $\times\alpha$ CD3 bsAbs Enhance Allogeneic Anti-Tumor Effect of T Cells In Vivo

NOD SCID gamma (NSG) mice (n=10) were engrafted intradermally with 10×10^6 PA-1 cells (in Matrigel) on Day -7. On Day 0, mice were engrafted intraperitoneally with 5×10^6 human PBMCs. Mice were then treated on Days 0, 7, 14, and 21 with XENP37233, XENP37227, XENP37630, or XENP37231 at 0.3, 1.0, or 3.0 mg/kg or PBS control. Tumor volume was measured by caliper three times per week (data for which are shown in FIG. 56) and blood was drawn to investigate lymphocyte expansion (data not shown). The data show that each of the 4 bsAbs were able to induce anti-tumor effect at all concentrations. By Day 14, each of the bispecific antibodies enhanced anti-tumor activity in comparison to PBS control. Notably, lower potency molecules XENP37630 and XENP37231 demonstrated some dose dependency in anti-tumor activity.

3. α CLDN6 $\times\alpha$ CD3 bsAbs are Well Tolerated and Demonstrate Good Pharmacokinetic Profile in Cynomolgus Monkeys

Animals (n=1) were dosed with XENP37547, XENP37545, XENP37634, and XENP37541 at 1 $\times$, 10 $\times$, 30 $\times$, and 60 $\times$ dose on Day 0. Blood was drawn over time to determine concentration of drug in serum over time, data for which are depicted in FIGS. 57-58. The data show that the bsAbs have an in vivo half-life of ~2 weeks and was dose dependent. Additionally, the bsAbs were generally well-tolerated as indicated by clinical observations (e.g. of body weight; data not shown).

SEQUENCE LISTING

```
Sequence total quantity: 580
SEQ ID NO: 1          moltype = AA length = 20
FEATURE              Location/Qualifiers
REGION               1..20
                     note = Description of Artificial Sequence: Synthetic peptide
REGION               1..20
                     note = Positive Charged scFv Linkers +H
source               1..20
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 1
GKPGSGKPGS GKPGSGKPGS                               20

SEQ ID NO: 2          moltype = AA length = 5
FEATURE              Location/Qualifiers
REGION               1..5
                     note = Description of Artificial Sequence: Synthetic peptide
REGION               1..5
                     note = See specification as filed for detailed description
                           of substitutions and preferred embodiments
source               1..5
                     mol_type = protein
                     organism = synthetic construct

SEQUENCE: 2
GGGGS                                                    5

SEQ ID NO: 3          moltype = AA length = 5
FEATURE              Location/Qualifiers
REGION               1..5
                     note = Description of Artificial Sequence: Synthetic peptide
REGION               1..5
```

-continued

	note = See specification as filed for detailed description of substitutions and preferred embodiments	
source	1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 3 GSGGS		5
SEQ ID NO: 4 FEATURE REGION	moltype = AA length = 4 Location/Qualifiers 1..4 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..4 note = See specification as filed for detailed description of substitutions and preferred embodiments	
source	1..4 mol_type = protein organism = synthetic construct	
SEQUENCE: 4 GGGS		4
SEQ ID NO: 5 FEATURE REGION	moltype = AA length = 5 Location/Qualifiers 1..5 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5 note = Domain linker	
source	1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 5 GGGGS		5
SEQ ID NO: 6 FEATURE REGION	moltype = AA length = 15 Location/Qualifiers 1..15 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..15 note = Positive Charged scFv Linkers Gly-Ser 15	
source	1..15 mol_type = protein organism = synthetic construct	
SEQUENCE: 6 GGGSGGGGS GGGGS		15
SEQ ID NO: 7 FEATURE REGION	moltype = AA length = 18 Location/Qualifiers 1..18 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..18 note = Positive Charged scFv Linkers Whitlow linker	
source	1..18 mol_type = protein organism = synthetic construct	
SEQUENCE: 7 GSTSGSGKPG SGEGSTKG		18
SEQ ID NO: 8 FEATURE REGION	moltype = AA length = 14 Location/Qualifiers 1..14 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14 note = Positive Charged scFv Linkers 6paxA_1 (+A)	
source	1..14 mol_type = protein organism = synthetic construct	
SEQUENCE: 8 IRPRAIGGSK PRVA		14
SEQ ID NO: 9 FEATURE REGION	moltype = AA length = 15 Location/Qualifiers 1..15 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..15 note = Positive Charged scFv Linkers +B	
source	1..15 mol_type = protein organism = synthetic construct	
SEQUENCE: 9		

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GKGGSGKGGG	GKGGG	15
SEQ ID NO: 10	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..15	
	note = Positive Charged scFv Linkers +C	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 10		
GKGGSGKGGG	GKGGG	15
SEQ ID NO: 11	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..15	
	note = Positive Charged scFv Linkers +D	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 11		
GKGGSGKGGG	GKGGG	15
SEQ ID NO: 12	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..15	
	note = Positive Charged scFv Linkers +E	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 12		
GKGGSGKGGG	GKGGG	15
SEQ ID NO: 13	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..15	
	note = Positive Charged scFv Linkers +F	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 13		
GKGGSGKGGG	GKGGG	15
SEQ ID NO: 14	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..15	
	note = Positive Charged scFv Linkers +G	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 14		
GKPGSGKPGS	GKPGS	15
SEQ ID NO: 15	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..20	
	note = scFv Linker	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 15		
GKPGSGKPGS	GKPGSGKPGS	20
SEQ ID NO: 16	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..20	

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source	note = Positive Charged scFv Linkers +I 1..20 mol_type = protein organism = synthetic construct	
SEQUENCE: 16 GKGKSGKGKS GKGKSGKGKS		20
SEQ ID NO: 17 FEATURE REGION	moltype = AA length = 20 Location/Qualifiers 1..20 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..20 note = Negative Charged scFv Linkers Gly-Ser 20	
source	1..20 mol_type = protein organism = synthetic construct	
SEQUENCE: 17 GGGGSGGGGS GGGSGGGGS		20
SEQ ID NO: 18 FEATURE REGION	moltype = AA length = 14 Location/Qualifiers 1..14 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14 note = Negative Charged scFv Linkers 3hsc_2 (-A)	
source	1..14 mol_type = protein organism = synthetic construct	
SEQUENCE: 18 STAGDTHLGG EDFD		14
SEQ ID NO: 19 FEATURE REGION	moltype = AA length = 15 Location/Qualifiers 1..15 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..15 note = Negative Charged scFv Linkers -B	
source	1..15 mol_type = protein organism = synthetic construct	
SEQUENCE: 19 GEGSGEGGS GEGGS		15
SEQ ID NO: 20 FEATURE REGION	moltype = AA length = 15 Location/Qualifiers 1..15 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..15 note = Negative Charged scFv Linkers -C	
source	1..15 mol_type = protein organism = synthetic construct	
SEQUENCE: 20 GEGSGGEGGS GEGGS		15
SEQ ID NO: 21 FEATURE REGION	moltype = AA length = 15 Location/Qualifiers 1..15 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..15 note = Negative Charged scFv Linkers -D	
source	1..15 mol_type = protein organism = synthetic construct	
SEQUENCE: 21 GGGESGGGES GGGES		15
SEQ ID NO: 22 FEATURE REGION	moltype = AA length = 15 Location/Qualifiers 1..15 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..15 note = Negative Charged scFv Linkers -E	
source	1..15 mol_type = protein organism = synthetic construct	
SEQUENCE: 22 GEGESGEGES GEGES		15

-continued

SEQ ID NO: 23	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..15	
	note = Negative Charged scFv Linkers -F	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 23		
GGGESGGEGS GEGGS		15
SEQ ID NO: 24	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..20	
	note = Negative Charged scFv Linkers -G	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 24		
GEGESGEGES GEGESGEGES		20
SEQ ID NO: 25	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..15	
	note = scFv Linker	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 25		
GGGGSGGGGS GGGGS		15
SEQ ID NO: 26	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..20	
	note = scFv Linker	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 26		
GGGGSGGGGS GGGSGGGGS		20
SEQ ID NO: 27	moltype = AA length = 18	
FEATURE	Location/Qualifiers	
REGION	1..18	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..18	
	note = scFv Linker	
source	1..18	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 27		
GSTSGSGKPG SGEGSTKG		18
SEQ ID NO: 28	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..19	
	note = scFv Linker	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 28		
PRGASKGSA SQTGSAPGS		19
SEQ ID NO: 29	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..19	
	note = scFv Linker	
source	1..19	

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SEQUENCE: 29	mol_type = protein	
GTAAAGAGAA GGAAAGAAG	organism = synthetic construct	19
SEQ ID NO: 30	moltype = AA length = 19	
FEATURE	Location/Qualifiers	
REGION	1..19	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..19	
	note = scFv Linker	
source	1..19	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 30		
GTSGSSGSGS GSGSGGGG		19
SEQ ID NO: 31	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
REGION	1..10	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..10	
	note = Domain linker	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 31		
GGGSGGGGS		10
SEQ ID NO: 32	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..15	
	note = Domain linker	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 32		
GGGSGGGGS GGGGS		15
SEQ ID NO: 33	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..20	
	note = Domain linker	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 33		
GGGSGGGGS GGGSGGGGS		20
SEQ ID NO: 34	moltype = AA length = 25	
FEATURE	Location/Qualifiers	
REGION	1..25	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..25	
	note = Domain linker	
source	1..25	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 34		
GGGSGGGGS GGGSGGGGS GGGGS		25
SEQ ID NO: 35	moltype = AA length = 30	
FEATURE	Location/Qualifiers	
REGION	1..30	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..30	
	note = Domain linker	
source	1..30	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 35		
GGGSGGGGS GGGSGGGGS GGGSGGGGS		30
SEQ ID NO: 36	moltype = AA length = 35	

-continued

FEATURE	Location/Qualifiers	
REGION	1..35	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..35	
	note = Domain linker	
source	1..35	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 36		
GGGSGGGGS GGGSGGGGS GGGSGGGGS GGGGS		35
SEQ ID NO: 37	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = Domain linker	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 37		
GGGGA		5
SEQ ID NO: 38	moltype = AA length = 10	
FEATURE	Location/Qualifiers	
REGION	1..10	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..10	
	note = Domain linker	
source	1..10	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 38		
GGGAGGGGA		10
SEQ ID NO: 39	moltype = AA length = 15	
FEATURE	Location/Qualifiers	
REGION	1..15	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..15	
	note = Domain linker	
source	1..15	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 39		
GGGAGGGGA GGGGA		15
SEQ ID NO: 40	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..20	
	note = Domain linker	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 40		
GGGAGGGGA GGGAGGGGA		20
SEQ ID NO: 41	moltype = AA length = 25	
FEATURE	Location/Qualifiers	
REGION	1..25	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..25	
	note = Domain linker	
source	1..25	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 41		
GGGAGGGGA GGGAGGGGA GGGGA		25
SEQ ID NO: 42	moltype = AA length = 30	
FEATURE	Location/Qualifiers	
REGION	1..30	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..30	
	note = Domain linker	

-continued

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source          1..30
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 42
GGGGAGGGGA GGGGAGGGGA GGGGAGGGGA          30

SEQ ID NO: 43    moltype = AA length = 35
FEATURE         Location/Qualifiers
REGION          1..35
                 note = Description of Artificial Sequence: Synthetic
                 polypeptide
REGION          1..35
                 note = Domain linker
source          1..35
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 43
GGGGAGGGGA GGGGAGGGGA GGGGAGGGGA GGGGA    35

SEQ ID NO: 44    moltype = AA length = 30
FEATURE         Location/Qualifiers
REGION          1..30
                 note = Description of Artificial Sequence: Synthetic
                 polypeptide
REGION          1..30
                 note = Domain linker 30AA-linker
source          1..30
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 44
DPALVHQRPA PPGGGGSGGG GSGGGGSGGG          30

SEQ ID NO: 45    moltype = AA length = 5
FEATURE         Location/Qualifiers
REGION          1..5
                 note = Description of Artificial Sequence: Synthetic peptide
REGION          1..5
                 note = Domain linker
source          1..5
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 45
GKPGS          5

SEQ ID NO: 46    moltype = AA length = 25
FEATURE         Location/Qualifiers
REGION          1..25
                 note = Description of Artificial Sequence: Synthetic peptide
REGION          1..25
                 note = Domain linker
source          1..25
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 46
GKPGSGKPGS GKPGSGKPGS GKPGS          25

SEQ ID NO: 47    moltype = AA length = 30
FEATURE         Location/Qualifiers
REGION          1..30
                 note = Description of Artificial Sequence: Synthetic
                 polypeptide
REGION          1..30
                 note = Domain linker
source          1..30
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 47
GKPGSGKPGS GKPGSGKPGS GKPGSGKPGS          30

SEQ ID NO: 48    moltype = AA length = 5
FEATURE         Location/Qualifiers
REGION          1..5
                 note = Description of Artificial Sequence: Synthetic peptide
REGION          1..5
                 note = Domain linker
source          1..5
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 48

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GGGES	5
SEQ ID NO: 49	
FEATURE	moltype = AA length = 9
REGION	Location/Qualifiers
	1..9
	note = Description of Artificial Sequence: Synthetic peptide
REGION	1..9
	note = Domain linker "half hinge"
source	1..9
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 49	
KTHTCPPCP	9
SEQ ID NO: 50	
FEATURE	moltype = AA length = 15
REGION	Location/Qualifiers
	1..15
	note = Description of Artificial Sequence: Synthetic peptide
REGION	1..15
	note = Domain linker "full hinge C220S variant"
source	1..15
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 50	
EPKSSDKTHT CPPCP	15
SEQ ID NO: 51	
FEATURE	moltype = AA length = 19
REGION	Location/Qualifiers
	1..19
	note = Description of Artificial Sequence: Synthetic peptide
REGION	1..19
	note = Domain linker "flex half hinge"
source	1..19
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 51	
GGGSGGGGS KTHTCPPCP	19
SEQ ID NO: 52	
FEATURE	moltype = AA length = 19
REGION	Location/Qualifiers
	1..19
	note = Description of Artificial Sequence: Synthetic peptide
REGION	1..19
	note = Domain linker "charged half hinge1"
source	1..19
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 52	
GKPGSGKPGS KTHTCPPCP	19
SEQ ID NO: 53	
FEATURE	moltype = AA length = 14
REGION	Location/Qualifiers
	1..14
	note = Description of Artificial Sequence: Synthetic peptide
REGION	1..14
	note = Domain linker "charged half hinge2"
source	1..14
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 53	
GKPGSKTHTC PPCP	14
SEQ ID NO: 54	
FEATURE	moltype = AA length = 329
REGION	Location/Qualifiers
	1..329
	note = Description of Artificial Sequence: Synthetic polypeptide
REGION	1..329
	note = 1 + 1 Fab-scFv-Fc Backbone 1 Fab-Fc Side
source	1..329
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 54	
ASTKGPSVFAP LAPSSKSTSG GTAALGCLVK DYFPEPTVS WNSGALTSGV HTFPAVLQSS	60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP	120
SVFLFPPKPK DTLMISRTP VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS	180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIETISK AKGQPREPQV YTLPPSREEM	240
TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWE	300
QGDVFSQSVH HEALHNHYTQ KSLSLSPGK	329

SEQ ID NO: 55	moltype = AA length = 231	
FEATURE	Location/Qualifiers	
REGION	1..231	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..231	
	note = 1 + 1 Fab-scFv-Fc Backbone 1 scFv-Fc Side	
source	1..231	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 55		
EPKSSDKTHT CPDPPAPPVA	GPSVFLFPPK PKDTLMISRT	PEVTCVVVDV KHEDPEVKFN 60
WYVDGVEVHN AKTKPREEQY	NSTYRVSVSL TVLHQDWLNG	KEYKCKVSNK ALPAPIEKTI 120
SKAKGQPREP QVYTLPPSRE	QMTKNQVKLT CLVKGFPYPSD	IAVEWESNGQ PENNYKTPPP 180
VLDSDGSFFL YSKLTVDKSR	WQQGNVFSKS VMHEALHNNHY	TQKSLSLSPG K 231
SEQ ID NO: 56	moltype = AA length = 329	
FEATURE	Location/Qualifiers	
REGION	1..329	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..329	
	note = 1 + 1 Fab-scFv-Fc Backbone 2 Fab-Fc Side	
source	1..329	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 56		
ASTKGPSVFP LAPSSKSTSG	GTAALGCLVK DYFPEPVTVS	WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT	YICNVNHNKPS DTKVDKKVEP	KSCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK DTLMISRTPE	VTCVVVDVVKH EDPEVKFNWY	VDGVEVHNK TKPREEEYNS 180
TYRVVSVLTV LHQDWLNGKE	YKCKVSNKAL PAPIEKTISK	AKGQPREPQV YTLPPSREEM 240
TKNQVSLTCD VSGFYPSDIA	VEWESDGGQE NNYKTPPPVL	DSGGSFFLYS KLTVDKSRWE 300
QGDVFSCSVM HEALHNNHYTQ	KSLSLSPGK	329
SEQ ID NO: 57	moltype = AA length = 231	
FEATURE	Location/Qualifiers	
REGION	1..231	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..231	
	note = 1 + 1 Fab-scFv-Fc Backbone 2 scFv-Fc Side	
source	1..231	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 57		
EPKSSDKTHT CPDPPAPPVA	GPSVFLFPPK PKDTLMISRT	PEVTCVVVDV KHEDPEVKFN 60
WYVDGVEVHN AKTKPREEQY	NSTYRVSVSL TVLHQDWLNG	KEYKCKVSNK ALPAPIEKTI 120
SKAKGQPREP QVYTLPPSRE	EMTKNQVKLT CLVKGFPYPSD	IAVEWESNGQ PENNYKTPPP 180
VLDSDGSFFL YSKLTVDKSR	WQQGNVFSKS VMHEALHNNHY	TQKSLSLSPG K 231
SEQ ID NO: 58	moltype = AA length = 329	
FEATURE	Location/Qualifiers	
REGION	1..329	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..329	
	note = 1 + 1 Fab-scFv-Fc Backbone 3 Fab-Fc Side	
source	1..329	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 58		
ASTKGPSVFP LAPSSKSTSG	GTAALGCLVK DYFPEPVTVS	WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT	YICNVNHNKPS DTKVDKKVEP	KSCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK DTLMISRTPE	VTCVVVDVVKH EDPEVKFNWY	VDGVEVHNK TKPREEEYNS 180
TYRVVSVLTV LHQDWLNGKE	YKCKVSNKAL PAPIEKTISK	AKGQPREPQV YTLPPSREEM 240
TKNQVSLTCE VSGFYPSDIA	VEWESDGGQE NNYKTPPPVL	DSGGSFFLYS KLTVDKSRWE 300
QGDVFSCSVM HEALHNNHYTQ	KSLSLSPGK	329
SEQ ID NO: 59	moltype = AA length = 231	
FEATURE	Location/Qualifiers	
REGION	1..231	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..231	
	note = 1 + 1 Fab-scFv-Fc Backbone 3 scFv-Fc Side	
source	1..231	
	mol_type = protein	
	organism = synthetic construct	

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SEQUENCE: 59
 EPKSSDKTHT CPPCPAPPVA GPSVFLFPPK PKDTLMISRT PEVTCVVVDV KHEDPEVKFN 60
 WYVDGVEVHN AKTKPREEQY NSTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI 120
 SKAKGQPREP QVYTLPPSRE EMTKNQVKLT CLVKGFPYPSD IAVEWESNGQ PENNYKTTTP 180
 VLDSGGSFFL YSKLTVDKSR WQQGNVFSCS VMHEALHNHY TQKSLSLSPG K 231

SEQ ID NO: 60 moltype = AA length = 329
 FEATURE Location/Qualifiers
 REGION 1..329
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..329
 note = 1 + 1 Fab-scFv-Fc Backbone 4 Fab-Fc Side
 source 1..329
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 60
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 180
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
 TENEVSLTCL VKGFYPSDIA VEWESDGQPE NNYKTTTPVL DSDGSFFLYS KLEVDKSRWE 300
 QGDVFSCSVM HEALHNHYTQ KSLSLSPGK 329

SEQ ID NO: 61 moltype = AA length = 231
 FEATURE Location/Qualifiers
 REGION 1..231
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..231
 note = 1 + 1 Fab-scFv-Fc Backbone 4 scFv-Fc Side
 source 1..231
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 61
 EPKSSDKTHT CPPCPAPPVA GPSVFLFPPK PKDTLMISRT PEVTCVVVDV KHEDPEVKFN 60
 WYVDGVEVHN AKTKPREEQY NSTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI 120
 SKAKGQPREP QVYTLPPSRE EMTKNQVSLT CLVKGFPYPSD IAVEWESNGQ PENNYKTTTP 180
 VLDSGGSFFL YSKLTVDKSR WQQGNVFSCS VMHEALHNHY TQKSLSLSPG K 231

SEQ ID NO: 62 moltype = AA length = 329
 FEATURE Location/Qualifiers
 REGION 1..329
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..329
 note = 1 + 1 Fab-scFv-Fc Backbone 5 Fab-Fc Side
 source 1..329
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 62
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 180
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSRDEL 240
 TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWE 300
 QGDVFSCSVM HEALHNHYTQ KSLSLSPGK 329

SEQ ID NO: 63 moltype = AA length = 231
 FEATURE Location/Qualifiers
 REGION 1..231
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..231
 note = 1 + 1 Fab-scFv-Fc Backbone 5 scFv-Fc Side
 source 1..231
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 63
 EPKSSDKTHT CPPCPAPPVA GPSVFLFPPK PKDTLMISRT PEVTCVVVDV KHEDPEVKFN 60
 WYVDGVEVHN AKTKPREEQY NSTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI 120
 SKAKGQPREP QVYTLPPSRD QLTKNQVKLT CLVKGFPYPSD IAVEWESNGQ PENNYKTTTP 180
 VLDSGGSFFL YSKLTVDKSR WQQGNVFSCS VMHEALHNHY TQKSLSLSPG K 231

SEQ ID NO: 64 moltype = AA length = 329
 FEATURE Location/Qualifiers
 REGION 1..329
 note = Description of Artificial Sequence: Synthetic
 polypeptide

-continued

REGION 1..329
 note = 1 + 1 Fab-scFv-Fc Backbone 6 Fab-Fc Side
 source 1..329
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 64
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
 SVFLFPPKPK DTLMSRTP ETCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYAS 180
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
 TKNQVSLTCD VSGFYPSDIA VEWESDQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 300
 QGDVFSCSV HEALHNHYTQ KSLSLSPGK 329

SEQ ID NO: 65 moltype = AA length = 231
 FEATURE Location/Qualifiers
 REGION 1..231
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..231
 note = 1 + 1 Fab-scFv-Fc Backbone 6 scFv-Fc Side
 source 1..231
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 65
 EPKSSDKTHT CPPCPAPPVA GPSVFLFPPK PKDTLMISRT PEVTCVVVDV KHEDPEVKFN 60
 WYVDGVEVHN AKTKPREEQY ASTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI 120
 SKAKGQPREP QVYTLPPSRE QMTKNQVQLT CLVKGFPYPS IAVEWESNGQ PENNYKTTPP 180
 VLDSGDSFPL YSKLTVDKSR WQQGNVFSVS VMHEALHNHY TQKSLSLSPG K 231

SEQ ID NO: 66 moltype = AA length = 329
 FEATURE Location/Qualifiers
 REGION 1..329
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..329
 note = 1 + 1 Fab-scFv-Fc Backbone 7 Fab-Fc Side
 source 1..329
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 66
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
 SVFLFPPKPK DTLMSRTP ETCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYSS 180
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
 TKNQVSLTCD VSGFYPSDIA VEWESDQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 300
 QGDVFSCSV HEALHNHYTQ KSLSLSPGK 329

SEQ ID NO: 67 moltype = AA length = 231
 FEATURE Location/Qualifiers
 REGION 1..231
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..231
 note = 1 + 1 Fab-scFv-Fc Backbone 7 scFv-Fc Side
 source 1..231
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 67
 EPKSSDKTHT CPPCPAPPVA GPSVFLFPPK PKDTLMISRT PEVTCVVVDV KHEDPEVKFN 60
 WYVDGVEVHN AKTKPREEQY SSTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI 120
 SKAKGQPREP QVYTLPPSRE QMTKNQVQLT CLVKGFPYPS IAVEWESNGQ PENNYKTTPP 180
 VLDSGDSFPL YSKLTVDKSR WQQGNVFSVS VMHEALHNHY TQKSLSLSPG K 231

SEQ ID NO: 68 moltype = AA length = 327
 FEATURE Location/Qualifiers
 REGION 1..327
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..327
 note = 1 + 1 Fab-scFv-Fc Backbone 8 Fab-Fc Side
 source 1..327
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 68
 ASTKGPSVFP LAPCSRSTSE STAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
 GLYSLSSVVT VPSSSLGTKT YTCNVNHHKPS DTKVDKRVES KYGPPCPPCP APEFLGGPSV 120
 FLFPPKPKDT LMISRTPEVT CVVVDVSQED PEVQFNWYVD GVEVHNAKTK PREEEFNSTY 180
 RVVSVLTVLH QDWLNGKEYK CKVSNKGLPS SIEKTISKAK GQPREPQVYT LPPSQEEMTK 240
 NQVSLTCDVS GFYPSDIAVE WESDQGPENN YKTTPPVLD DGSFFLYSRL TVDKSRWEEG 300

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DVFSCSVMHE ALHNHYTQKS LSLSLGK 327

SEQ ID NO: 69 moltype = AA length = 229
 FEATURE Location/Qualifiers
 REGION 1..229
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..229
 note = 1 + 1 Fab-scFv-Fc Backbone 8 scFv-Fc Side
 source 1..229
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 69
 ESKYGPPCPP CPAPEFLGGP SVFLFPPKPK DTLMISRTPE VTCVVVDVSQ EDPEVQFNWY 60
 VDGVEVHNAK TKPREEQFNS TYRVVSVLTV LHQDWLNGKE YKCKVSNKGL PSSIEKTISK 120
 AKGQPREPQV YTLPPSQEQM TKNQVKLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL 180
 DSDGSFFLYS RLTVDKSRWQ EGNVFSCSVM HEALHNHYTQ KSLSLSLGK 229

SEQ ID NO: 70 moltype = AA length = 326
 FEATURE Location/Qualifiers
 REGION 1..326
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..326
 note = 1 + 1 Fab-scFv-Fc Backbone 9 Fab-Fc Side
 source 1..326
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 70
 ASTKGPSVFP LAPCSRSTSE STAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
 GLYSLSSVVT VPSSNFGTQT YTCNVDHKPS DTKVDKTVR KCCVECPPCP APPVAGPSVF 120
 LFPKPKDITL MISRTPEVTC VVVDVSHEDP EVQFNWYVDG VEVHNAKTKP REEFNSTFR 180
 VVSVLTVVHQ DWLNGKEYKC KVSNGKGLPAP IEKTISKTKG QPREPQVYTL PPSREEMTKN 240
 QVSLTCDVSG FYPSDIAVEW ESDGQPENNY KTTTPMLDSD GSFFLYSKLT VDKSRWEQGD 300
 VFSCSVMHEA LHNHYTQKSL SLSPGK 326

SEQ ID NO: 71 moltype = AA length = 228
 FEATURE Location/Qualifiers
 REGION 1..228
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..228
 note = 1 + 1 Fab-scFv-Fc Backbone 9 scFv-Fc Side
 source 1..228
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 71
 ERKCCVECP CPAPPVAGPS VFLEPPKPKD TLMISRTPEV TCVVVDVSHE DPEVQFNWYV 60
 DGVVEVHNAK KPREEQFNST FRVSVLTVV HQDWLNGKEY KCKVSNKGLP APIEKTISK 120
 KGQPREPQVY TLPPSREQMT KNQVKLTCLV KGFYPSDIAV EWESNGQPEN NYKTTTPMLD 180
 SDGSFFLYSK LTVDKSRWQQ GNVFSCSVMH EALHNHYTQK SLSLSPGK 228

SEQ ID NO: 72 moltype = AA length = 326
 FEATURE Location/Qualifiers
 REGION 1..326
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..326
 note = 1 + 1 Fab-scFv-Fc Backbone 10 Fab-Fc Side
 source 1..326
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 72
 ASTKGPSVFP LAPCSRSTSE STAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
 GLYSLSSVVT VPSSNFGTQT YTCNVDHKPS DTKVDKTVR KCCVECPPCP APPVAGPSVF 120
 LFPKPKDITL MISRTPEVTC VVVDVKHEDP EVQFNWYVDG VEVHNAKTKP REEFNSTFR 180
 VVSVLTVVHQ DWLNGKEYKC KVSNGKGLPAP IEKTISKTKG QPREPQVYTL PPSREEMTKN 240
 QVSLTCDVSG FYPSDIAVEW ESDGQPENNY KTTTPMLDSD GSFFLYSKLT VDKSRWEQGD 300
 VFSCSVMHEA LHNHYTQKSL SLSPGK 326

SEQ ID NO: 73 moltype = AA length = 228
 FEATURE Location/Qualifiers
 REGION 1..228
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..228
 note = 1 + 1 Fab-scFv-Fc Backbone 10 scFv-Fc Side
 source 1..228
 mol_type = protein

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organism = synthetic construct

SEQUENCE: 73
ERKCCVECP CPAPPVAGPS VFLEPPKPKD TLMISRTPEV TCVVVDVKHE DPEVQFNWYV 60
DGVEVHNAKT KPREEQFNST FRVSVLTVV HQDWLNGKEY KCKVSNKGLP APIEKTISK 120
KGQPREPQVY TLPPSRQMT KNQVKLTCLV KGFYPSDIAV EWESNGQPEN NYKTTPPMLD 180
SDGSFFLYSK LTVDKSRWQQ GNVFSCSVMH EALHNHYTQK SLSLSPGK 228

SEQ ID NO: 74      moltype = AA  length = 329
FEATURE           Location/Qualifiers
REGION           1..329
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION           1..329
                  note = 1 + 1 Fab-scFv-Fc Backbone 11 Fab-Fc Side
source           1..329
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 74
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
TKNQVSLTCD VSGFYPSDIA VEWESDQQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 300
QGDVFSCSVL HEALSHYTQ KSLSLSPGK 329

SEQ ID NO: 75      moltype = AA  length = 231
FEATURE           Location/Qualifiers
REGION           1..231
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION           1..231
                  note = 1 + 1 Fab-scFv-Fc Backbone 11 scFv-Fc Side
source           1..231
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 75
EPKSSDKTHT CPPCPAPPVA GPSVFLFPPK PKDTLMISRT PEVTCVVVDV KHEDPEVKFN 60
WYVDGVEVHN AKTKPREEQY NSTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI 120
SKAKGQPREP QVYTLPPSRE QMTKNQVQKL CLVKGFPYPS IAVEWESNGQ PENNYKTTPP 180
VLDSGSPFL YSKLTVDKSR WQQGNVFSCS VLHEALHSHY TQKSLSLSPG K 231

SEQ ID NO: 76      moltype = AA  length = 329
FEATURE           Location/Qualifiers
REGION           1..329
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION           1..329
                  note = 1 + 1 Fab-scFv-Fc Backbone 12 Fab-Fc Side
source           1..329
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 76
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
TKNQVSLTCD VAGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ 300
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 329

SEQ ID NO: 77      moltype = AA  length = 231
FEATURE           Location/Qualifiers
REGION           1..231
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION           1..231
                  note = 1 + 1 Fab-scFv-Fc Backbone 12 scFv-Fc Side
source           1..231
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 77
ERKSSDKTHT CPPRPAPPVA GPSVFLFPPK PKDTLMISRT PEVTCVVVDV KHEDPEVKFK 60
WYVDGVEVHN AKTKPREEQY NSTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI 120
SKAKGQPREP QVYTLPPSRE QMTKNQVQKL CLVKGFPYPS IAVEWESNGQ PENNYKTTPP 180
VLDSGSPFL YSKLTVDKSR WQQGNVFSCS VMHEALHNHY TQKSLSLSPG K 231

SEQ ID NO: 78      moltype = AA  length = 329
FEATURE           Location/Qualifiers
REGION           1..329
                  note = Description of Artificial Sequence: Synthetic

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REGION                polypeptide
                      1..329
note = 2 + 1 Fab2-scFv-Fc Backbone 1 Fab-Fc Side
source                1..329
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 78
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK DTLMSRTP EVCVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 300
QGDVFSCSVM HEALHNHYTQ KSLSLSPGK                                     329

SEQ ID NO: 79         moltype = AA length = 216
FEATURE              Location/Qualifiers
REGION               1..216
note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION               1..216
note = 2 + 1 Fab2-scFv-Fc Backbone 1 Fab-scFv-Fc Side
source               1..216
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 79
APPVAGPSVF LFPPKPKDTL MISRTPEVTC VVVDVKHEDP EVKFNWYVDG VEVHNAKTKP 60
REEQYNSTYR VVSVLTVLHQ DWLNGKEYKC KVSNAKALPAP IEKTISKAKG QPREPQVYTL 120
PPSRQMTKN QVKLTCLVKG FYPSDIAVEW ESNQGPENNY KTTTPVLDSD GSFFLYSKLT 180
VDKSRWQQGN VFSCSVMHEA LHNHYTQKSL SLSPGK                               216

SEQ ID NO: 80         moltype = AA length = 329
FEATURE              Location/Qualifiers
REGION               1..329
note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION               1..329
note = 2 + 1 Fab2-scFv-Fc Backbone 2 Fab-Fc Side
source               1..329
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 80
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK DTLMSRTP EVCVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 300
QGDVFSCSVM HEALHNHYTQ KSLSLSPGK                                     329

SEQ ID NO: 81         moltype = AA length = 216
FEATURE              Location/Qualifiers
REGION               1..216
note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION               1..216
note = 2 + 1 Fab2-scFv-Fc Backbone 2 Fab-scFv-Fc Side
source               1..216
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 81
APPVAGPSVF LFPPKPKDTL MISRTPEVTC VVVDVKHEDP EVKFNWYVDG VEVHNAKTKP 60
REEQYNSTYR VVSVLTVLHQ DWLNGKEYKC KVSNAKALPAP IEKTISKAKG QPREPQVYTL 120
PPSREEMTKN QVKLTCLVKG FYPSDIAVEW ESNQGPENNY KTTTPVLDSD GSFFLYSKLT 180
VDKSRWQQGN VFSCSVMHEA LHNHYTQKSL SLSPGK                               216

SEQ ID NO: 82         moltype = AA length = 329
FEATURE              Location/Qualifiers
REGION               1..329
note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION               1..329
note = 2 + 1 Fab2-scFv-Fc Backbone 3 Fab-Fc Side
source               1..329
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 82
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK DTLMSRTP EVCVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240

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TKNQVSLTCE VSGFYPSDIA VEWESDGQPE NNYKTTPPV L DSDGSFFLYS KLTVDKSRWE 300
 QGDVFSCSV M HEALHNHYTQ KSLSLSPGK 329

SEQ ID NO: 83 moltype = AA length = 216
 FEATURE Location/Qualifiers
 REGION 1..216
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..216
 note = 2 + 1 Fab2-scFv-Fc Backbone 3 Fab-scFv-Fc Side
 source 1..216
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 83
 APPVAGPSVF LFPPKPKDTL MISRTPEVTC VVVDVKHEDP EVKFNWYVDG VEVHNAKTKP 60
 REEQYNSTYR VVSVLTVLHQ DWLNGKEYKC KVS NKALPAP IEKTISKAKG QPREPQVYTL 120
 PPSREEMTKN QVKLTCLVKG FYPSDIAVEW ESNGQPENNY KTTTPVLDSD GSFFLYSKLT 180
 VDKSRWQQGN VFSCSVMHEA LHNHYTQKSL SLSPGK 216

SEQ ID NO: 84 moltype = AA length = 329
 FEATURE Location/Qualifiers
 REGION 1..329
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..329
 note = 2 + 1 Fab2-scFv-Fc Backbone 4 Fab-Fc Side
 source 1..329
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 84
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 180
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
 TENEVSLTCL VKGFYPSDIA VEWESDGQPE NNYKTTPPV L DSDGSFFLYS KLEVDKSRWE 300
 QGDVFSCSV M HEALHNHYTQ KSLSLSPGK 329

SEQ ID NO: 85 moltype = AA length = 216
 FEATURE Location/Qualifiers
 REGION 1..216
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..216
 note = 2 + 1 Fab2-scFv-Fc Backbone 4 Fab-scFv-Fc Side
 source 1..216
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 85
 APPVAGPSVF LFPPKPKDTL MISRTPEVTC VVVDVKHEDP EVKFNWYVDG VEVHNAKTKP 60
 REEQYNSTYR VVSVLTVLHQ DWLNGKEYKC KVS NKALPAP IEKTISKAKG QPREPQVYTL 120
 PPSREEMTKN QVSLTCLVKG FYPSDIAVEW ESNGQPENNY KTTTPVLDSD GSFFLYSKLT 180
 VDKSRWQQGN VFSCSVMHEA LHNHYTQKSL SLSPGK 216

SEQ ID NO: 86 moltype = AA length = 329
 FEATURE Location/Qualifiers
 REGION 1..329
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..329
 note = 2 + 1 Fab2-scFv-Fc Backbone 5 Fab-Fc Side
 source 1..329
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 86
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 180
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSRDEL 240
 TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTPPV L DSDGSFFLYS KLTVDKSRWE 300
 QGDVFSCSV M HEALHNHYTQ KSLSLSPGK 329

SEQ ID NO: 87 moltype = AA length = 216
 FEATURE Location/Qualifiers
 REGION 1..216
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..216
 note = 2 + 1 Fab2-scFv-Fc Backbone 5 Fab-scFv-Fc Side
 source 1..216

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mol_type = protein
organism = synthetic construct

SEQUENCE: 87
APPVAGPSVF LFPPKPKDTL MISRTPEVTC VVVDVKHEDP EVKFNWYVDG VEVHNAKTKP 60
REEQYNSTYR VVSVLTVLHQ DWLNGKEYKC KVSNAKALPAP IEKTISKAKG QPREPQVYTL 120
PPSRDQLTKN QVKLTCLVKG FYPSDIAVEW ESNQOPENNY KTTTPVLDSG GSFFLYSKLT 180
VDKSRWQQGN VFSCSVMEHA LHNHYTQKSL SLSPGK 216

SEQ ID NO: 88 moltype = AA length = 329
FEATURE Location/Qualifiers
REGION 1..329
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..329
note = 2 + 1 Fab2-scFv-Fc Backbone 6 Fab-Fc Side
source 1..329
mol_type = protein
organism = synthetic construct

SEQUENCE: 88
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYAS 180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 300
QGDFVSCSVMEHA LHNHYTQKSL SLSPGK 329

SEQ ID NO: 89 moltype = AA length = 216
FEATURE Location/Qualifiers
REGION 1..216
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..216
note = 2 + 1 Fab2-scFv-Fc Backbone 6 Fab-scFv-Fc Side
source 1..216
mol_type = protein
organism = synthetic construct

SEQUENCE: 89
APPVAGPSVF LFPPKPKDTL MISRTPEVTC VVVDVKHEDP EVKFNWYVDG VEVHNAKTKP 60
REEQYASTYR VVSVLTVLHQ DWLNGKEYKC KVSNAKALPAP IEKTISKAKG QPREPQVYTL 120
PPSRQMTKN QVKLTCLVKG FYPSDIAVEW ESNQOPENNY KTTTPVLDSG GSFFLYSKLT 180
VDKSRWQQGN VFSCSVMEHA LHNHYTQKSL SLSPGK 216

SEQ ID NO: 90 moltype = AA length = 329
FEATURE Location/Qualifiers
REGION 1..329
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..329
note = 2 + 1 Fab2-scFv-Fc Backbone 7 Fab-Fc Side
source 1..329
mol_type = protein
organism = synthetic construct

SEQUENCE: 90
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYSS 180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 300
QGDFVSCSVMEHA LHNHYTQKSL SLSPGK 329

SEQ ID NO: 91 moltype = AA length = 216
FEATURE Location/Qualifiers
REGION 1..216
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..216
note = 2 + 1 Fab2-scFv-Fc Backbone 7 Fab-scFv-Fc Side
source 1..216
mol_type = protein
organism = synthetic construct

SEQUENCE: 91
APPVAGPSVF LFPPKPKDTL MISRTPEVTC VVVDVKHEDP EVKFNWYVDG VEVHNAKTKP 60
REEQYSSTYR VVSVLTVLHQ DWLNGKEYKC KVSNAKALPAP IEKTISKAKG QPREPQVYTL 120
PPSRQMTKN QVKLTCLVKG FYPSDIAVEW ESNQOPENNY KTTTPVLDSG GSFFLYSKLT 180
VDKSRWQQGN VFSCSVMEHA LHNHYTQKSL SLSPGK 216

SEQ ID NO: 92 moltype = AA length = 329
FEATURE Location/Qualifiers
REGION 1..329

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note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..329
note = 2 + 1 Fab2-scFv-Fc Backbone 8 Fab-Fc Side
source 1..329
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 92
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWE 300
QGDVFSQSVL HEALHSHYTQ KSLSLSPGK 329

SEQ ID NO: 93      moltype = AA length = 216
FEATURE           Location/Qualifiers
REGION 1..216
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..216
note = 2 + 1 Fab2-scFv-Fc Backbone 8 Fab-scFv-Fc Side
source 1..216
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 93
APPVAGPSVF LFPPKPKDTL MISRTPEVTC VVVDVKHEDP EVKFNWYVDG VEVHNAKTKP 60
REEQYNSTYR VVSVLTVLHQ DWLNGKEYKC KVSNAKALPAP IEKTISKAKG QPREPQVYTL 120
PPSREQMTKN QVKLTCLVKG FYPSDIAVEW ESNGQPENNY KTTTPVLDSG GSFFLYSKLT 180
VDKSRWQQGN VFSCSVLHEA LHSHTYQKSL SLSPGK 216

SEQ ID NO: 94      moltype = AA length = 329
FEATURE           Location/Qualifiers
REGION 1..329
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..329
note = 2 + 1 Fab2-scFv-Fc Backbone 9 Fab-Fc Side
source 1..329
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 94
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 60
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 120
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 180
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 240
TKNQVSLTCD VAGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 300
QGNVFSQSVL HEALHSHYTQ KSLSLSPGK 329

SEQ ID NO: 95      moltype = AA length = 216
FEATURE           Location/Qualifiers
REGION 1..216
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..216
note = 2 + 1 Fab2-scFv-Fc Backbone 9 Fab-scFv-Fc Side
source 1..216
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 95
APPVAGPSVF LFPPKPKDTL MISRTPEVTC VVVDVKHEDP EVKFKWYVDG VEVHNAKTKP 60
REEQYNSTYR VVSVLTVLHQ DWLNGKEYKC KVSNAKALPAP IEKTISKAKG QPREPQVYTL 120
PPSREQMTKN QVKLTCLVKG FYPSDIAVEW ESNGQPENNY KTTTPVLDSG GSFFLYSKLT 180
VDKSRWQQGN VFSCSVMHEA LHNHHTYQKSL SLSPGK 216

SEQ ID NO: 96      moltype = AA length = 107
FEATURE           Location/Qualifiers
REGION 1..107
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..107
note = Constant Light Domain - Kappa
source 1..107
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 96
RTVAAPSVFI FPPSDEQLKS GTASVVCLLN NFYPREAKVQ WKVDNALQSG NSQESVTEQD 60
SKDSTYLSLS TLTLKADYE KHKVYACEVT HQGLSSPVTK SFNRGEC 107

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SEQ ID NO: 97      moltype = AA  length = 106
FEATURE           Location/Qualifiers
REGION            1..106
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..106
                  note = Constant Light Domain - Lambda
source            1..106
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 97
GQPKAAPSVT LFPSSSEELQ ANKATLVCLI SDFYPGAVTV AWKADSSPVK AGVETTPSK 60
QSNNKYAASS YLSLTPEQWK SHRSYSCQVT HEGSTVEKTV APTECS 106

SEQ ID NO: 98      moltype = AA  length = 254
FEATURE           Location/Qualifiers
REGION            1..254
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..254
                  note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv scFv (VHVL)
source            1..254
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 98
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWVRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGTL 120
VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSTVSPG GTVTLTCGSS TGAVTTSNYA 180
NWVQQKPGKS PRGLIGGTNK RAPGVPARFS GSLGGKAAL TISGAQPEDE ADYYCALWYS 240
NHWVFGGGTK LTVL 254

SEQ ID NO: 99      moltype = AA  length = 254
FEATURE           Location/Qualifiers
REGION            1..254
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..254
                  note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv scFv (VLVH)
source            1..254
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 99
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTKRAPGV 60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVLG KPGSGKPGSG 120
KPGSGKPGSE VQLVESGGGL VQPGGSLRLS CAASGFTFST YAMNWVRQAP GKGLEWVGRI 180
RSKYNNYATY YADSVKGRFT ISRDDSKNTL YLQMNSLRAE DTAVYYCVRH GNFGDSYVSW 240
FAYWGQGTLV TVSS 254

SEQ ID NO: 100     moltype = AA  length = 125
FEATURE           Location/Qualifiers
REGION            1..125
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..125
                  note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv Variable
                  Heavy (vh) Domain
source            1..125
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 100
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWVRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGTL 120
VTVSS 125

SEQ ID NO: 101     moltype = AA  length = 5
FEATURE           Location/Qualifiers
REGION            1..5
                  note = Description of Artificial Sequence: Synthetic peptide
REGION            1..5
                  note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv vhCDR1
source            1..5
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 101
TYAMN 5

SEQ ID NO: 102     moltype = AA  length = 19
FEATURE           Location/Qualifiers
REGION            1..19

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REGION	note = Description of Artificial Sequence: Synthetic peptide 1..19	
source	note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv vhCDR2 1..19 mol_type = protein organism = synthetic construct	
SEQUENCE: 102		
RIRSKYNNYA TYYADSVKG		19
SEQ ID NO: 103	moltype = AA length = 14 Location/Qualifiers	
FEATURE		
REGION	1..14 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14 note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv vhCDR3	
source	1..14 mol_type = protein organism = synthetic construct	
SEQUENCE: 103		
HGNFGDSYVS WFAV		14
SEQ ID NO: 104	moltype = AA length = 109 Location/Qualifiers	
FEATURE		
REGION	1..109 note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..109 note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv Variable Light (v1) Domain	
source	1..109 mol_type = protein organism = synthetic construct	
SEQUENCE: 104		
QAVVTQEPST TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV		60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVL		109
SEQ ID NO: 105	moltype = AA length = 14 Location/Qualifiers	
FEATURE		
REGION	1..14 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14 note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv vlCDR1	
source	1..14 mol_type = protein organism = synthetic construct	
SEQUENCE: 105		
GSSTGAVTTS NYAN		14
SEQ ID NO: 106	moltype = AA length = 7 Location/Qualifiers	
FEATURE		
REGION	1..7 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7 note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv vlCDR2	
source	1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 106		
GTNKRAP		7
SEQ ID NO: 107	moltype = AA length = 9 Location/Qualifiers	
FEATURE		
REGION	1..9 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9 note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv vlCDR3	
source	1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 107		
ALWYSNHWV		9
SEQ ID NO: 108	moltype = AA length = 20 Location/Qualifiers	
FEATURE		
REGION	1..20 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..20 note = CD3 High - [anti-CD3]_H1.30_L1.47_scFv Linker	
source	1..20	

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mol_type = protein
organism = synthetic construct

SEQUENCE: 108
GKPGSGKPGS GKPGSGKPGS 20

SEQ ID NO: 109
FEATURE
REGION
REGION
REGION
source
SEQUENCE: 109
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWVRQA PGKGLEWVGR IRSKANNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGT 120
VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSTLTVSPG GTVTLTCGSS TGAVTTSNYA 180
NWVQQKPGKS PRGLIGGTNK RAPGVPARFS GSLGGKAAL TISGAQPEDE ADYYCALWYS 240
NHWVFGGGTK LTVL 254

SEQ ID NO: 110
FEATURE
REGION
REGION
REGION
source
SEQUENCE: 110
QAVVTQEPSL TVSPGGTTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV 60
PARFSGSLLG GKAAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVLG KPGSGKPGSG 120
KPGSGKPGSE VQLVESGGGL VQPGGSLRLS CAASGFTFST YAMNWVRQAP GKGLEWVGRI 180
RSKANNYATY YADSVKGRFT ISRDDSKNTL LYLQMNSLRAE DTAVYYCVRH GNFSDSYVSW 240
FAYWGQGTILV TVSS 254

SEQ ID NO: 111
FEATURE
REGION
REGION
REGION
source
SEQUENCE: 111
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWVRQA PGKGLEWVGR IRSKANNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGT 120
VTVSS 125

SEQ ID NO: 112
FEATURE
REGION
REGION
REGION
source
SEQUENCE: 112
TYAMN 5

SEQ ID NO: 113
FEATURE
REGION
REGION
REGION
source
SEQUENCE: 113

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RIRSKANNYA TYYADSVKG 19

SEQ ID NO: 114 moltype = AA length = 14
 FEATURE Location/Qualifiers
 REGION 1..14
 note = Description of Artificial Sequence: Synthetic peptide
 REGION 1..14
 note = CD3 High-Int #1 - [anti-CD3]_H1.32_L1.47_scFv vHCDR3
 source 1..14
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 114
 HGNFGDSYVS WFAY 14

SEQ ID NO: 115 moltype = AA length = 109
 FEATURE Location/Qualifiers
 REGION 1..109
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..109
 note = CD3 High-Int #1 - [anti-CD3]_H1.32_L1.47_scFv Variable Light (vL) Domain
 source 1..109
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 115
 QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV 60
 PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVL 109

SEQ ID NO: 116 moltype = AA length = 14
 FEATURE Location/Qualifiers
 REGION 1..14
 note = Description of Artificial Sequence: Synthetic peptide
 REGION 1..14
 note = CD3 High-Int #1 - [anti-CD3]_H1.32_L1.47_scFv vLCDR1
 source 1..14
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 116
 GSSTGAVTTS NYAN 14

SEQ ID NO: 117 moltype = AA length = 7
 FEATURE Location/Qualifiers
 REGION 1..7
 note = Description of Artificial Sequence: Synthetic peptide
 REGION 1..7
 note = CD3 High-Int #1 - [anti-CD3]_H1.32_L1.47_scFv vLCDR2
 source 1..7
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 117
 GTNKRAP 7

SEQ ID NO: 118 moltype = AA length = 9
 FEATURE Location/Qualifiers
 REGION 1..9
 note = Description of Artificial Sequence: Synthetic peptide
 REGION 1..9
 note = CD3 High-Int #1 - [anti-CD3]_H1.32_L1.47_scFv vLCDR3
 source 1..9
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 118
 ALWYSNHWV 9

SEQ ID NO: 119 moltype = AA length = 20
 FEATURE Location/Qualifiers
 REGION 1..20
 note = Description of Artificial Sequence: Synthetic peptide
 REGION 1..20
 note = CD3 High-Int #1 - [anti-CD3]_H1.32_L1.47_scFv Linker
 source 1..20
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 119
 GKPGSGKPGS GKPGSGKPGS 20

SEQ ID NO: 120 moltype = AA length = 254
 FEATURE Location/Qualifiers

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REGION          1..254
                 note = Description of Artificial Sequence: Synthetic
                 polypeptide
REGION          1..254
                 note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv scFv
                 (VHVL)
source          1..254
                 mol_type = protein
                 organism = synthetic construct

SEQUENCE: 120
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDEYVS WFAYWGQGT 120
VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA 180
NWVQQKPGKS PRGLIGGTNK RAPGVPARFS GSLGGKAAL TISGAQPEDE ADYYCALWYS 240
NHWVFGGGTK LTVL 254

SEQ ID NO: 121      moltype = AA length = 254
FEATURE            Location/Qualifiers
REGION            1..254
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION            1..254
                   note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv scFv
                   (VLVH)
source            1..254
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 121
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV 60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVLG KPGSGKPGSG 120
KPGSGKPGSE VQLVESGGGL VQPGGSLRLS CAASGFTFST YAMNWRQAP GKGLEWVGRI 180
RSKYNNYATY YADSVKGRFT ISRDDSKNTL YLQMNSLRAE DTAVYYCVRH GNFGDEYVSW 240
FAYWGQGTIV TVSS 254

SEQ ID NO: 122      moltype = AA length = 125
FEATURE            Location/Qualifiers
REGION            1..125
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION            1..125
                   note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv
                   Variable Heavy (vh) Domain
source            1..125
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 122
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDEYVS WFAYWGQGT 120
VTVSS 125

SEQ ID NO: 123      moltype = AA length = 5
FEATURE            Location/Qualifiers
REGION            1..5
                   note = Description of Artificial Sequence: Synthetic peptide
REGION            1..5
                   note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv vhCDR1
source            1..5
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 123
TYAMN 5

SEQ ID NO: 124      moltype = AA length = 19
FEATURE            Location/Qualifiers
REGION            1..19
                   note = Description of Artificial Sequence: Synthetic peptide
REGION            1..19
                   note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv vhCDR2
source            1..19
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 124
RIRSKYNNYA TYYADSVKG 19

SEQ ID NO: 125      moltype = AA length = 14
FEATURE            Location/Qualifiers
REGION            1..14
                   note = Description of Artificial Sequence: Synthetic peptide
REGION            1..14

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source	note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv vhCDR3 1..14 mol_type = protein organism = synthetic construct	
SEQUENCE: 125 HGNFGDEYVS WFAV		14
SEQ ID NO: 126 FEATURE REGION	moltype = AA length = 109 Location/Qualifiers 1..109 note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..109 note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv Variable Light (v1) Domain	
source	1..109 mol_type = protein organism = synthetic construct	
SEQUENCE: 126 QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVL		60 109
SEQ ID NO: 127 FEATURE REGION	moltype = AA length = 14 Location/Qualifiers 1..14 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14 note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv v1CDR1	
source	1..14 mol_type = protein organism = synthetic construct	
SEQUENCE: 127 GSSTGAVTTS NYAN		14
SEQ ID NO: 128 FEATURE REGION	moltype = AA length = 7 Location/Qualifiers 1..7 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7 note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv v1CDR2	
source	1..7 mol_type = protein organism = synthetic construct	
SEQUENCE: 128 GTNKRAP		7
SEQ ID NO: 129 FEATURE REGION	moltype = AA length = 9 Location/Qualifiers 1..9 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9 note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv v1CDR3	
source	1..9 mol_type = protein organism = synthetic construct	
SEQUENCE: 129 ALWYSNHWV		9
SEQ ID NO: 130 FEATURE REGION	moltype = AA length = 20 Location/Qualifiers 1..20 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..20 note = CD3 High-Int #2 - [anti-CD3]_H1.89_L1.47_scFv Linker	
source	1..20 mol_type = protein organism = synthetic construct	
SEQUENCE: 130 GKPGSGKPGS GKPGSGKPGS		20
SEQ ID NO: 131 FEATURE REGION	moltype = AA length = 254 Location/Qualifiers 1..254 note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..254 note = CD3 High-Int #3 - [anti-CD3]_H1.90_L1.47_scFv scFv (VHVL)	
source	1..254	

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mol_type = protein
organism = synthetic construct

SEQUENCE: 131
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYVCVR HGNFGDPYVS WFAYWGQGTL 120
VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA 180
NWWQKPGKS PRGLIGGTNK RAPGVPARFS GSLLGKKAAL TISGAQPEDE ADYYCALWYS 240
NHWVFGGGTK LTVL 254

SEQ ID NO: 132      moltype = AA length = 254
FEATURE            Location/Qualifiers
REGION             1..254
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..254
                   note = CD3 High-Int #3 - [anti-CD3]_H1.90_L1.47_scFv scFv
                   (VLVH)
source             1..254
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 132
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV 60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVLG KPGSGKPGSG 120
KPGSGKPGSE VQLVESGGGL VQPGGSLRLS CAASGFTFST YAMNWRQAP GKGLEWVGRI 180
RSKYNNYATY YADSVKGRFT ISRDDSKNTL YLQMNSLRAE DTAVYVCVRH GNFGDPYVSW 240
FAYWGQGTLV TVSS 254

SEQ ID NO: 133      moltype = AA length = 125
FEATURE            Location/Qualifiers
REGION             1..125
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..125
                   note = CD3 High-Int #3 - [anti-CD3]_H1.90_L1.47_scFv
                   Variable Heavy (vh) Domain
source             1..125
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 133
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYVCVR HGNFGDPYVS WFAYWGQGTL 120
VTVSS 125

SEQ ID NO: 134      moltype = AA length = 5
FEATURE            Location/Qualifiers
REGION             1..5
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..5
                   note = CD3 High-Int #3 - [anti-CD3]_H1.90_L1.47_scFv vHCDR1
source             1..5
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 134
TYAMN 5

SEQ ID NO: 135      moltype = AA length = 19
FEATURE            Location/Qualifiers
REGION             1..19
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..19
                   note = CD3 High-Int #3 - [anti-CD3]_H1.90_L1.47_scFv vHCDR2
source             1..19
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 135
RIRSKYNNYA TYYADSVKG 19

SEQ ID NO: 136      moltype = AA length = 14
FEATURE            Location/Qualifiers
REGION             1..14
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..14
                   note = CD3 High-Int #3 - [anti-CD3]_H1.90_L1.47_scFv vHCDR3
source             1..14
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 136
HGNFGDPYVS WFAY 14

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NHWVFGGGTK LTVL 254

SEQ ID NO: 143 moltype = AA length = 254
 FEATURE Location/Qualifiers
 REGION 1..254
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..254
 note = CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv scFv (VLVH)
 source 1..254
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 143
 QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV 60
 PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVLG KPGSGKPGSG 120
 KPGSGKPGSE VQLVESGGGL VQPGGSLRLS CAASGFTFST YAMNWRQAP GKGLEWVGRI 180
 RSKYNNYATY YADSVKGRFT ISRDDSKNTL YLQMNSLRAE DTAIVYCVRH GNFGDSYVSW 240
 FDYWGQGLV TVSS 254

SEQ ID NO: 144 moltype = AA length = 125
 FEATURE Location/Qualifiers
 REGION 1..125
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..125
 note = CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv Variable Heavy (vh) Domain
 source 1..125
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 144
 EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT 60
 YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAIVYCVR HGNFGDSYVS WFDYWGQGL 120
 VTSS 125

SEQ ID NO: 145 moltype = AA length = 5
 FEATURE Location/Qualifiers
 REGION 1..5
 note = Description of Artificial Sequence: Synthetic peptide
 REGION 1..5
 note = CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv vhCDR1
 source 1..5
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 145
 TYAMN 5

SEQ ID NO: 146 moltype = AA length = 19
 FEATURE Location/Qualifiers
 REGION 1..19
 note = Description of Artificial Sequence: Synthetic peptide
 REGION 1..19
 note = CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv vhCDR2
 source 1..19
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 146
 RIRSKYNNYA TYYADSVKG 19

SEQ ID NO: 147 moltype = AA length = 14
 FEATURE Location/Qualifiers
 REGION 1..14
 note = Description of Artificial Sequence: Synthetic peptide
 REGION 1..14
 note = CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv vhCDR3
 source 1..14
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 147
 HGNFGDSYVS WFDY 14

SEQ ID NO: 148 moltype = AA length = 109
 FEATURE Location/Qualifiers
 REGION 1..109
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..109
 note = CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv

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source          Variable Light (v1) Domain
                1..109
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 148
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTKRAPGV 60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVL 109

SEQ ID NO: 149      moltype = AA length = 14
FEATURE            Location/Qualifiers
REGION             1..14
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..14
                   note = CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv v1CDR1
source             1..14
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 149
GSSTGAVTTS NYAN 14

SEQ ID NO: 150      moltype = AA length = 7
FEATURE            Location/Qualifiers
REGION             1..7
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..7
                   note = CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv v1CDR2
source             1..7
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 150
GTNKRAP 7

SEQ ID NO: 151      moltype = AA length = 9
FEATURE            Location/Qualifiers
REGION             1..9
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..9
                   note = CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv v1CDR3
source             1..9
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 151
ALWYSNHWV 9

SEQ ID NO: 152      moltype = AA length = 20
FEATURE            Location/Qualifiers
REGION             1..20
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..20
                   note = CD3-Intermediate - [anti-CD3]_H1.33_L1.47_scFv Linker
source             1..20
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 152
GKPGSGKPGS GKPGSGKPGS 20

SEQ ID NO: 153      moltype = AA length = 254
FEATURE            Location/Qualifiers
REGION             1..254
                   note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION             1..254
                   note = CD3 Low - [anti-CD3]_H1.31_L1.47_scFv scFv (VHVL)
source             1..254
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 153
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMSWVRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYYCVR HGNFGDSYVS WFAYWGQGTL 120
VTSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA 180
NWVQQKPGKS PRGLIGGTNK RAPGVPARFS GSLGGKAAL TISGAQPEDE ADYYCALWYS 240
NHWVFGGGTK LTVL 254

SEQ ID NO: 154      moltype = AA length = 254
FEATURE            Location/Qualifiers
REGION             1..254
                   note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION             1..254

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source          note = CD3 Low - [anti-CD3]_H1.31_L1.47_scFv scFv (VLVH)
                1..254
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 154
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTKRAPGV 60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVLG KPGSGKPGSG 120
KPGSGKPGSE VQLVESGGGL VQPGGSLRLS CAASGFTFST YAMSWVRQAP GKGLEWVGRI 180
RSKYNNYATY YADSVKGRFT ISRDDSKNTL YLQMNSLRAE DTAVYVCVRH GNFGDSYVSW 240
FAYWGQGTLLV TVSS 254

SEQ ID NO: 155      moltype = AA length = 125
FEATURE            Location/Qualifiers
REGION             1..125
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..125
                   note = CD3 Low - [anti-CD3]_H1.31_L1.47_scFv Variable Heavy
                   (vh) Domain
source             1..125
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 155
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMSWVRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYVCVR HGNGDSYVS WFAYWGQGTLL 120
VTSS 125

SEQ ID NO: 156      moltype = AA length = 5
FEATURE            Location/Qualifiers
REGION             1..5
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..5
                   note = CD3 Low - [anti-CD3]_H1.31_L1.47_scFv vhCDR1
source             1..5
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 156
TYAMS 5

SEQ ID NO: 157      moltype = AA length = 19
FEATURE            Location/Qualifiers
REGION             1..19
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..19
                   note = CD3 Low - [anti-CD3]_H1.31_L1.47_scFv vhCDR2
source             1..19
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 157
RIRSKYNNYA TYYADSVKG 19

SEQ ID NO: 158      moltype = AA length = 14
FEATURE            Location/Qualifiers
REGION             1..14
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..14
                   note = CD3 Low - [anti-CD3]_H1.31_L1.47_scFv vhCDR3
source             1..14
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 158
HGNGDSYVS WFAY 14

SEQ ID NO: 159      moltype = AA length = 109
FEATURE            Location/Qualifiers
REGION             1..109
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..109
                   note = CD3 Low - [anti-CD3]_H1.31_L1.47_scFv Variable Light
                   (vl) Domain
source             1..109
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 159
QAVVTQEPSL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTKRAPGV 60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVL 109

SEQ ID NO: 160      moltype = AA length = 14

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FEATURE	Location/Qualifiers	
REGION	1..14	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..14	
	note = CD3 Low - [anti-CD3]_H1.31_L1.47_scFv vLCDR1	
source	1..14	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 160		
GSSTGAVTTS NYAN		14
SEQ ID NO: 161	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7	
	note = CD3 Low - [anti-CD3]_H1.31_L1.47_scFv vLCDR2	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 161		
GTNKRAP		7
SEQ ID NO: 162	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9	
	note = CD3 Low - [anti-CD3]_H1.31_L1.47_scFv vLCDR3	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 162		
ALWYSNHWV		9
SEQ ID NO: 163	moltype = AA length = 20	
FEATURE	Location/Qualifiers	
REGION	1..20	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..20	
	note = CD3 Low - [anti-CD3]_H1.31_L1.47_scFv Linker	
source	1..20	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 163		
GKPGSGKPGS GKPGSGKPGS		20
SEQ ID NO: 164	moltype = AA length = 220	
FEATURE	Location/Qualifiers	
REGION	1..220	
	note = Human CLDN6 sequence sp~6747	
source	1..220	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 164		
MASAGMQILG VVLTLGWVN GLVSCALPMW KVTAFIGNSI VVAQVWEGW WMSCVVQSTG		60
QMCKVYDSL LALPQDLQAA RALCVIALLV ALFGLLVYLA GAKCTTCVEE KDSKARLVLT		120
SGIVFVISGV LTLIPVCWTA HAIIRDFYNP LVAEAQKREL GASLYLGWAA SGLLLGGGL		180
LCCTCPSGGS QGPSHYMARY STSAPAISRG PSEYPTKNYV		220
SEQ ID NO: 165	moltype = AA length = 53	
FEATURE	Location/Qualifiers	
REGION	1..53	
	note = Human CLDN6 sequence, N-terminal extracellular domain sp~6747[<i>m</i>]29-81	
source	1..53	
	mol_type = protein	
	organism = Homo sapiens	
SEQUENCE: 165		
MWKVTAFIGN SIVVAQVVWE GLWMSCVVQS TGQMCKVYD SLLALPQDLQ AAR		53
SEQ ID NO: 166	moltype = AA length = 23	
FEATURE	Location/Qualifiers	
REGION	1..23	
	note = Human CLDN6 sequence, C-terminal extracellular domain sp~6747[<i>m</i>]138-160	
source	1..23	
	mol_type = protein	
	organism = Homo sapiens	

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SEQUENCE: 166
 WTAHAIIRDF YNPLVAEAQK REL 23

SEQ ID NO: 167 moltype = AA length = 219
 FEATURE Location/Qualifiers
 REGION 1..219
 note = Mouse CLDN6 sequence spZ262
 source 1..219
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 167
 MASTGLQILG IVLTLLGWVN ALVSCALPMW KVTAFIGNSI VVAQMVWEGE WMSCVVQSTG 60
 QMQCKVYDSL LALPQDLQAA RALCVVTLI VLLGLLVYLA GAKCTTCVED RNSKSRLVLI 120
 SGIIFVISGV LTLIPVCWTA HSIIQDFYNP LVADAQKREL GASLYLGWAA SGLLLGGGL 180
 LCCACSSGGT QGPRHYMACY STSVPHSRGP SEYPTKNYV 219

SEQ ID NO: 168 moltype = AA length = 53
 FEATURE Location/Qualifiers
 REGION 1..53
 note = Mouse CLDN6 sequence, N-terminal extracellular
 domain spZ262[\m]29-81
 source 1..53
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 168
 MWKVTAFIGN SIVVAQMVWE GLWMSCVVQS TGQMCKKYD SLLALPQDLQ AAR 53

SEQ ID NO: 169 moltype = AA length = 26
 FEATURE Location/Qualifiers
 REGION 1..26
 note = Mouse CLDN6 sequence, C-terminal extracellular
 domain spZ262[\m]138-163
 source 1..26
 mol_type = protein
 organism = Mus musculus

SEQUENCE: 169
 WTAHSIIQDF YNPLVADAQK RELGAS 26

SEQ ID NO: 170 moltype = AA length = 220
 FEATURE Location/Qualifiers
 REGION 1..220
 note = Macaca fascicularis CLDN6 sequence (predicted)
 trQ0B0
 source 1..220
 mol_type = protein
 organism = Macaca fascicularis

SEQUENCE: 170
 MASAGMQLG VVLTLLGWVN GLVSCALPMW KVTAFIGNSI VVAQVVWEGE WMSCVVQSTG 60
 QMQCKVYDSL LALPQDLQAA RALCVIALLV ALFGLLVYLA GAKCTTCVEE KDSKARLVLT 120
 SGIVFVISGV LTLIPVCWTA HAIIRDFYNP LVAEQKREL GASLYLGWAA SGLLLGGGL 180
 LCCTCPSGGS RGP SHYMARY STSAPAIRG PSEYPTKNYV 220

SEQ ID NO: 171 moltype = AA length = 53
 FEATURE Location/Qualifiers
 REGION 1..53
 note = Macaca fascicularis CLDN6 sequence, N-terminal
 extracellular domain (predicted) trQ0B0[29]-81
 source 1..53
 mol_type = protein
 organism = Macaca fascicularis

SEQUENCE: 171
 MWKVTAFIGN SIVVAQVVWE GLWMSCVVQS TGQMCKKYD SLLALPQDLQ AAR 53

SEQ ID NO: 172 moltype = AA length = 23
 FEATURE Location/Qualifiers
 REGION 1..23
 note = Macaca fascicularis CLDN6 sequence, C-terminal
 extracellular domain (predicted) trQ0B0[13]8-160
 source 1..23
 mol_type = protein
 organism = Macaca fascicularis

SEQUENCE: 172
 WTAHAIIRDF YNPLVAEAQK REL 23

SEQ ID NO: 173 moltype = AA length = 220
 FEATURE Location/Qualifiers
 REGION 1..220
 note = cldn6
 source 1..220

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mol_type = protein
organism = Homo sapiens

SEQUENCE: 173
MASAGMQILG VVLTLLGWVN GLVSCALPMW KVTAFIGNSI VVAQVVWEGW WMSCVVQSTG 60
QMCKKVYDSL LALPQDLQAA RALCVIALLV ALFGLLVYLA GAKCTTCVEE KDSKARLVLT 120
SGIVFVISGV LTLIPVCWTA HAIIRDFYNP LVAEAKKREL GASLYLGWAA SGLLLLGGGL 180
LCCTCPSGGG QGPHSHMARY STSAPASIRG PSEYPTKNYV 220

SEQ ID NO: 174      moltype = AA length = 217
FEATURE            Location/Qualifiers
REGION             1..217
                   note = cldn9
source             1..217
                   mol_type = protein
                   organism = Homo sapiens

SEQUENCE: 174
MASTGLELLG MTLAVLGWLG TLVSCALPLW KVTAFIGNSI VVAQVVWEGW WMSCVVQSTG 60
QMCKKVYDSL LALPQDLQAA RALCVIALLV ALLGLLVAIT GAQCTTCVED EGAKARIVLT 120
AGVILLLAGI LVLIPVCWTA HAIQDFYNP LVAEALKREL GASLYLGWAA AALLMLGGGL 180
LCCTCPPPQV ERPRGPRLGY SIPSRSASG LDKRDYV 217

SEQ ID NO: 175      moltype = AA length = 120
FEATURE            Location/Qualifiers
REGION             1..120
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..120
                   note = mC6-30[CLDN6] Variable heavy (vh) domain
source             1..120
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 175
EVQLQQSGPE LVKPGASVKI SCKTSGYTFT EYTMHWVKQS PGKSLEWIGG IDPNNGNTHY 60
NQKFKGKATL TVDKSSSTAY MELRSLTSED SAVYYCARIY YFGRLYFDWF GAGTTVTVSS 120

SEQ ID NO: 176      moltype = AA length = 5
FEATURE            Location/Qualifiers
REGION             1..5
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..5
                   note = mC6-30[CLDN6] vhCDR1
source             1..5
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 176
EYTMH 5

SEQ ID NO: 177      moltype = AA length = 17
FEATURE            Location/Qualifiers
REGION             1..17
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..17
                   note = mC6-30[CLDN6] vhCDR2
source             1..17
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 177
GIDPNNGNTH YNQKFKG 17

SEQ ID NO: 178      moltype = AA length = 11
FEATURE            Location/Qualifiers
REGION             1..11
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..11
                   note = mC6-30[CLDN6] vhCDR3
source             1..11
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 178
IYYFGRLYFD F 11

SEQ ID NO: 179      moltype = AA length = 107
FEATURE            Location/Qualifiers
REGION             1..107
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..107
                   note = mC6-30[CLDN6] Variable light (vl) domain
source             1..107

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mol_type = protein
organism = synthetic construct

SEQUENCE: 179
DIQMTQSSSS FSVSLGDRVT ITKASEDIY NRLAWYQQKP GNVPRLLISG ATSLETGVPS 60
RFGSGSGSKD YTLSITSLQT EDVTYYCQQ YWSSPLTFGG GTKLEIK 107

SEQ ID NO: 180      moltype = AA length = 11
FEATURE            Location/Qualifiers
REGION             1..11
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..11
                   note = mC6-30 [CLDN6] v1CDR1
source             1..11
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 180
KASEDIYNRL A 11

SEQ ID NO: 181      moltype = AA length = 7
FEATURE            Location/Qualifiers
REGION             1..7
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..7
                   note = mC6-30 [CLDN6] v1CDR2
source             1..7
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 181
GATSLET 7

SEQ ID NO: 182      moltype = AA length = 9
FEATURE            Location/Qualifiers
REGION             1..9
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..9
                   note = mC6-30 [CLDN6] v1CDR3
source             1..9
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 182
QQYWSSPLT 9

SEQ ID NO: 183      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..449
                   note = XENP34243 mC6-30 Heavy Chain
source             1..449
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 183
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDSNNGNTHY 60
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYFDWF GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLISRTPV VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 184      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP34243 mC6-30 Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 184
DIQMTQSPSS LSASVGDRVT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSKD YTFTISSLQP EDIATYYCQQ YWSGPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 185      moltype = AA length = 120

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FEATURE	Location/Qualifiers	
REGION	1..120	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..120	
	note = C6-30[CLDN6]_H1 Variable heavy (vh) domain	
source	1..120	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 185		
QVQLVQSGAE VKKPGASVKV	SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY	60
NQKFQGRVTI TVDKSASTAY	MELSSLRSED TAVYYCARIY YFGRLYFDFW GAGTLVTVSS	120
SEQ ID NO: 186	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = C6-30[CLDN6]_H1 vhCDR1	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 186		
EYTMH		5
SEQ ID NO: 187	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..17	
	note = C6-30[CLDN6]_H1 vhCDR2	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 187		
GIDPNNGNTH YNQKFQG		17
SEQ ID NO: 188	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_H1 vhCDR3	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 188		
IYYFGRLYFD F		11
SEQ ID NO: 189	moltype = AA length = 120	
FEATURE	Location/Qualifiers	
REGION	1..120	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..120	
	note = C6-30[CLDN6]_H2 Variable heavy (vh) domain	
source	1..120	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 189		
EVQLVQSGAE VKKPGESLRI	SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY	60
NQKFQGHVTI SVDKSISTAY	LQWSSLKASD TAMYCARIY YFGRLYFDFW GAGTLVTVSS	120
SEQ ID NO: 190	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = C6-30[CLDN6]_H2 vhCDR1	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 190		
EYTMH		5
SEQ ID NO: 191	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	

-continued

REGION	1..17	
	note = C6-30[CLDN6]_H2 vhCDR2	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 191		
GIDPNNGNTH YNQKFQG		17
SEQ ID NO: 192	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_H2 vhCDR3	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 192		
IYYFGRLYFD F		11
SEQ ID NO: 193	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
REGION	1..107	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..107	
	note = C6-30[CLDN6]_L1 Variable light (v1) domain	
source	1..107	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 193		
DIQMTQSPSS LSASVGDRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS	60	
RFSGSGSGKD YFTISSLQP EDIATYYCQQ YWSSPLTFGG GTKVEIK	107	
SEQ ID NO: 194	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_L1 vlCDR1	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 194		
QASEDIYNRL A		11
SEQ ID NO: 195	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7	
	note = C6-30[CLDN6]_L1 vlCDR2	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 195		
GATSLET		7
SEQ ID NO: 196	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9	
	note = C6-30[CLDN6]_L1 vlCDR3	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 196		
QQYWSSPLT		9
SEQ ID NO: 197	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
REGION	1..107	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..107	
	note = C6-30[CLDN6]_L2 Variable light (v1) domain	
source	1..107	
	mol_type = protein	

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                                organism = synthetic construct
SEQUENCE: 197
DIQMTQSPDS LAVSLGERAT INCKASEDIY NRLAWYQQKP GQVPKLLISG ATSLETGVPS 60
RFGSGSGGKD YLTITSSLQA EDVAVYYCQQ YWSSPLTFGG GTKVEIK 107

SEQ ID NO: 198      moltype = AA length = 11
FEATURE            Location/Qualifiers
REGION             1..11
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..11
                   note = C6-30[CLDN6]_L2 vLCDR1
source             1..11
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 198
KASEDIYNRL A 11

SEQ ID NO: 199      moltype = AA length = 7
FEATURE            Location/Qualifiers
REGION             1..7
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..7
                   note = C6-30[CLDN6]_L2 vLCDR2
source             1..7
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 199
GATSLET 7

SEQ ID NO: 200      moltype = AA length = 9
FEATURE            Location/Qualifiers
REGION             1..9
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..9
                   note = C6-30[CLDN6]_L2 vLCDR3
source             1..9
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 200
QQYWSSPLT 9

SEQ ID NO: 201      moltype = AA length = 120
FEATURE            Location/Qualifiers
REGION             1..120
                   note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION             1..120
                   note = C6-30[CLDN6]_H1.1 Variable heavy (vh) domain
source             1..120
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 201
QVQLVQSGAE VKKPGASVKV SCKTSGYTFE EYTMHWVRQA PGQSLEWMGG ISPNNGNTHY 60
NKKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLVTVSS 120

SEQ ID NO: 202      moltype = AA length = 5
FEATURE            Location/Qualifiers
REGION             1..5
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..5
                   note = C6-30[CLDN6]_H1.1 vhCDR1
source             1..5
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 202
EYTMH 5

SEQ ID NO: 203      moltype = AA length = 17
FEATURE            Location/Qualifiers
REGION             1..17
                   note = Description of Artificial Sequence: Synthetic peptide
REGION             1..17
                   note = C6-30[CLDN6]_H1.1 vhCDR2
source             1..17
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 203
GISPNNGNTH YNKKFQG 17

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SEQ ID NO: 204	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_H1.1 vHCDR3	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 204		
IYYFGRLYFD F		11
SEQ ID NO: 205	moltype = AA length = 120	
FEATURE	Location/Qualifiers	
REGION	1..120	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..120	
	note = C6-30[CLDN6]_H1.2 Variable heavy (vh) domain	
source	1..120	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 205		
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG INPNNGNTHY 60		
NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLTVVSS 120		
SEQ ID NO: 206	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = C6-30[CLDN6]_H1.2 vHCDR1	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 206		
EYTMH		5
SEQ ID NO: 207	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..17	
	note = C6-30[CLDN6]_H1.2 vHCDR2	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 207		
GINPNNGNTH YNQKFQG		17
SEQ ID NO: 208	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_H1.2 vHCDR3	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 208		
IYYFGRLYFD F		11
SEQ ID NO: 209	moltype = AA length = 120	
FEATURE	Location/Qualifiers	
REGION	1..120	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..120	
	note = C6-30[CLDN6]_H1.3 Variable heavy (vh) domain	
source	1..120	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 209		
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDSNNGNTHY 60		
NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLTVVSS 120		
SEQ ID NO: 210	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	

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REGION	note = Description of Artificial Sequence: Synthetic peptide 1..5	
source	note = C6-30[CLDN6]_H1.3 vhCDR1 1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 210		
EYTMH		5
SEQ ID NO: 211	moltype = AA length = 17 Location/Qualifiers	
FEATURE		
REGION	1..17 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..17 note = C6-30[CLDN6]_H1.3 vhCDR2	
source	1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 211		
GIDSNNGNTH YNQKFQG		17
SEQ ID NO: 212	moltype = AA length = 11 Location/Qualifiers	
FEATURE		
REGION	1..11 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11 note = C6-30[CLDN6]_H1.3 vhCDR3	
source	1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 212		
IYYFGRLYPD F		11
SEQ ID NO: 213	moltype = AA length = 120 Location/Qualifiers	
FEATURE		
REGION	1..120 note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..120 note = C6-30[CLDN6]_H1.4 Variable heavy (vh) domain	
source	1..120 mol_type = protein organism = synthetic construct	
SEQUENCE: 213		
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDANNGNTHY 60		
NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLVTVSS 120		
SEQ ID NO: 214	moltype = AA length = 5 Location/Qualifiers	
FEATURE		
REGION	1..5 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5 note = C6-30[CLDN6]_H1.4 vhCDR1	
source	1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 214		
EYTMH		5
SEQ ID NO: 215	moltype = AA length = 17 Location/Qualifiers	
FEATURE		
REGION	1..17 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..17 note = C6-30[CLDN6]_H1.4 vhCDR2	
source	1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 215		
GIDANNGNTH YNQKFQG		17
SEQ ID NO: 216	moltype = AA length = 11 Location/Qualifiers	
FEATURE		
REGION	1..11 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11 note = C6-30[CLDN6]_H1.4 vhCDR3	
source	1..11 mol_type = protein	

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organism = synthetic construct

SEQUENCE: 216
IYYFGRLYFD F 11

SEQ ID NO: 217
FEATURE
REGION
    moltype = AA length = 120
    Location/Qualifiers
    1..120
    note = Description of Artificial Sequence: Synthetic
    polypeptide
    REGION
    1..120
    note = C6-30[CLDN6]_H1.5 Variable heavy (vh) domain
    source
    1..120
    mol_type = protein
    organism = synthetic construct

SEQUENCE: 217
QQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPQNGNTHY 60
NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYFDFW GAGTLTVVSS 120

SEQ ID NO: 218
FEATURE
REGION
    moltype = AA length = 5
    Location/Qualifiers
    1..5
    note = Description of Artificial Sequence: Synthetic peptide
    REGION
    1..5
    note = C6-30[CLDN6]_H1.5 vhCDR1
    source
    1..5
    mol_type = protein
    organism = synthetic construct

SEQUENCE: 218
EYTMH 5

SEQ ID NO: 219
FEATURE
REGION
    moltype = AA length = 17
    Location/Qualifiers
    1..17
    note = Description of Artificial Sequence: Synthetic peptide
    REGION
    1..17
    note = C6-30[CLDN6]_H1.5 vhCDR2
    source
    1..17
    mol_type = protein
    organism = synthetic construct

SEQUENCE: 219
GIDPQNGNTH YNQKFQG 17

SEQ ID NO: 220
FEATURE
REGION
    moltype = AA length = 11
    Location/Qualifiers
    1..11
    note = Description of Artificial Sequence: Synthetic peptide
    REGION
    1..11
    note = C6-30[CLDN6]_H1.5 vhCDR3
    source
    1..11
    mol_type = protein
    organism = synthetic construct

SEQUENCE: 220
IYYFGRLYFD F 11

SEQ ID NO: 221
FEATURE
REGION
    moltype = AA length = 120
    Location/Qualifiers
    1..120
    note = Description of Artificial Sequence: Synthetic
    polypeptide
    REGION
    1..120
    note = C6-30[CLDN6]_H1.6 Variable heavy (vh) domain
    source
    1..120
    mol_type = protein
    organism = synthetic construct

SEQUENCE: 221
QQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNSGNTHY 60
NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYFDFW GAGTLTVVSS 120

SEQ ID NO: 222
FEATURE
REGION
    moltype = AA length = 5
    Location/Qualifiers
    1..5
    note = Description of Artificial Sequence: Synthetic peptide
    REGION
    1..5
    note = C6-30[CLDN6]_H1.6 vhCDR1
    source
    1..5
    mol_type = protein
    organism = synthetic construct

SEQUENCE: 222
EYTMH 5

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SEQ ID NO: 223	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..17	
	note = C6-30[CLDN6]_H1.6 vhCDR2	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 223		
GIDPNSGNTH YNQKFAQ		17
SEQ ID NO: 224	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_H1.6 vhCDR3	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 224		
IYYFGRLYFD F		11
SEQ ID NO: 225	moltype = AA length = 120	
FEATURE	Location/Qualifiers	
REGION	1..120	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..120	
	note = C6-30[CLDN6]_H1.7 Variable heavy (vh) domain	
source	1..120	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 225		
QVQLVQSGAE VKKPGASVKV	SCKTSGYTFT EYTMHWVRQA	PGQSLEWMGG IDPNQGNTHY 60
NQKPGQGRVTI TVDKSASTAY	MELSSLRSED TAVYYCARIY	YFGRLYFDWF GAGTLVTSS 120
SEQ ID NO: 226	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = C6-30[CLDN6]_H1.7 vhCDR1	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 226		
EYTMH		5
SEQ ID NO: 227	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..17	
	note = C6-30[CLDN6]_H1.7 vhCDR2	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 227		
GIDPNQGNTH YNQKFAQ		17
SEQ ID NO: 228	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_H1.7 vhCDR3	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 228		
IYYFGRLYFD F		11
SEQ ID NO: 229	moltype = AA length = 120	
FEATURE	Location/Qualifiers	
REGION	1..120	
	note = Description of Artificial Sequence: Synthetic	

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REGION          polypeptide
                  1..120
note = C6-30[CLDN6]_H1.8 Variable heavy (vh) domain
source          1..120
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 229
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNDNTHY 60
NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYFDFW GAGTLTVSS 120

SEQ ID NO: 230    moltype = AA length = 5
FEATURE          Location/Qualifiers
REGION          1..5
note = Description of Artificial Sequence: Synthetic peptide
REGION          1..5
note = C6-30[CLDN6]_H1.8 vhCDR1
source          1..5
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 230
EYTMH                                                    5

SEQ ID NO: 231    moltype = AA length = 17
FEATURE          Location/Qualifiers
REGION          1..17
note = Description of Artificial Sequence: Synthetic peptide
REGION          1..17
note = C6-30[CLDN6]_H1.8 vhCDR2
source          1..17
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 231
GIDPNNDNTH YNQKFQG                                                    17

SEQ ID NO: 232    moltype = AA length = 11
FEATURE          Location/Qualifiers
REGION          1..11
note = Description of Artificial Sequence: Synthetic peptide
REGION          1..11
note = C6-30[CLDN6]_H1.8 vhCDR3
source          1..11
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 232
IYYFGRLYFD F                                                    11

SEQ ID NO: 233    moltype = AA length = 120
FEATURE          Location/Qualifiers
REGION          1..120
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION          1..120
note = C6-30[CLDN6]_H1.9 Variable heavy (vh) domain
source          1..120
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 233
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNANTHY 60
NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYFDFW GAGTLTVSS 120

SEQ ID NO: 234    moltype = AA length = 5
FEATURE          Location/Qualifiers
REGION          1..5
note = Description of Artificial Sequence: Synthetic peptide
REGION          1..5
note = C6-30[CLDN6]_H1.9 vhCDR1
source          1..5
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 234
EYTMH                                                    5

SEQ ID NO: 235    moltype = AA length = 17
FEATURE          Location/Qualifiers
REGION          1..17
note = Description of Artificial Sequence: Synthetic peptide
REGION          1..17
note = C6-30[CLDN6]_H1.9 vhCDR2
source          1..17

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SEQUENCE: 235	mol_type = protein	
GIDPNNANTH YNQKFQG	organism = synthetic construct	17
SEQ ID NO: 236	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_H1.9 vhCDR3	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 236		
IYYFGRLYFD F		11
SEQ ID NO: 237	moltype = AA length = 120	
FEATURE	Location/Qualifiers	
REGION	1..120	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..120	
	note = C6-30[CLDN6]_H1.19 Variable heavy (vh) domain	
source	1..120	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 237		
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60		
NQKFQGRVTI TVDKSASTAY MELSSLRSED TATYYCARIY YFGRLYPFDW GAGTLVTVSS 120		
SEQ ID NO: 238	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = C6-30[CLDN6]_H1.19 vhCDR1	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 238		
EYTMH		5
SEQ ID NO: 239	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..17	
	note = C6-30[CLDN6]_H1.19 vhCDR2	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 239		
GIDPNNGNTH YNQKFQG		17
SEQ ID NO: 240	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_H1.19 vhCDR3	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 240		
IYYFGRLYFD F		11
SEQ ID NO: 241	moltype = AA length = 120	
FEATURE	Location/Qualifiers	
REGION	1..120	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..120	
	note = C6-30[CLDN6]_H1.22 Variable heavy (vh) domain	
source	1..120	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 241		
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60		

NOKFOGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIL YFGRLYFDEFW GAGTLVTVSS 120

```

SEQ ID NO: 242      multype = AA   length = 5
FEATURE             Location/Qualifiers
REGION              1..5
                    note = Description of Artificial Sequence: Synthetic peptide
REGION              1..5
                    note = C6-30[CLDN6]_H1.22 vhCDR1
source              1..5
                    mol_type = protein
                    organism = synthetic construct

```

SEQUENCE: 242
EYTMH

```

SEQ ID NO: 243      multype = AA  length = 17
FEATURE             Location/Qualifiers
REGION              1..17
                    note = Description of Artificial Sequence: Synthetic peptide
REGION              1..17
                    note = C6-30[CLDN6]_H1.22 vhCDR2
source              1..17
                    mol_type = protein
                    organism = synthetic construct

```

SEQUENCE: 243
GIDPNNGNTH YNOKFOG

```

SEQ ID NO: 244      multype = AA  length = 11
FEATURE             Location/Qualifiers
REGION              1..11
                    note = Description of Artificial Sequence: Synthetic peptide
REGION              1..11
                    note = C6-30[CLDN6]_H1.22  vhCDR3
source              1..11
                    mol_type = protein
                    organism = synthetic construct

```

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SEQUENCE: 244
ILYFGRLYFD F

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SEQ ID NO: 245      multype = AA  length = 120
FEATURE             Location/Qualifiers
REGION              1..120
                    note = Description of Artificial Sequence: Synthetic
                        polypeptide
REGION              1..120
                    note = C6-30[CLDN6]_H1.24 Variable heavy (vh) domain
source              1..120
                    mol_type = protein
                    organism = synthetic construct

```

SEQUENCE: 245

QVQLVQSGAE VKKPGASVKV	SCKTSGYTFT EYTMHWVRQA	PGQSLEWMGG IDPNNGNTHY	60
NKQFOGRVTI TVDKSASTAY	MELSSLRSED TAVYYCARIY	YLGRLYFDFW GAGTLVTVSS	120

```

SEQ ID NO: 246      multype = AA  length = 5
FEATURE             Location/Qualifiers
REGION              1..5
                    note = Description of Artificial Sequence: Synthetic peptide
REGION              1..5
                    note = C6-30[CLDN6]_H1.24 vhCDR1
source              1..5
                    mol_type = protein
                    organism = synthetic construct

```

SEQUENCE: 246
EYTMH 5

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SEQ ID NO: 247      moltype = AA  length = 17
FEATURE             Location/Qualifiers
REGION              1..17
                    note = Description of Artificial Sequence: Synthetic peptide
REGION              1..17
                    note = C6-30[CLDN6]_H1.24 vhCDR2
source              1..17
                    mol_type = protein
                    organism = synthetic construct

```

SEQUENCE: 247
GIDPNNGNTH YNOKFOG 17

```
SEQ ID NO: 248      moltype = AA  length = 11
FEATURE             Location/Qualifiers
REGION              1..11
```

-continued

REGION	note = Description of Artificial Sequence: Synthetic peptide 1..11	
source	note = C6-30[CLDN6]_H1.24 vhCDR3 1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 248		
IYYLGRLYFD F		11
SEQ ID NO: 249	moltype = AA length = 120	
FEATURE	Location/Qualifiers	
REGION	1..120	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..120	
	note = C6-30[CLDN6]_H2.1 Variable heavy (vh) domain	
source	1..120 mol_type = protein organism = synthetic construct	
SEQUENCE: 249		
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG ISPNNGNTHY 60		
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120		
SEQ ID NO: 250	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = C6-30[CLDN6]_H2.1 vhCDR1	
source	1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 250		
EYTMH		5
SEQ ID NO: 251	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..17	
	note = C6-30[CLDN6]_H2.1 vhCDR2	
source	1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 251		
GISPNNGNTH YNQKFQG		17
SEQ ID NO: 252	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_H2.1 vhCDR3	
source	1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 252		
IYYFGRLYFD F		11
SEQ ID NO: 253	moltype = AA length = 120	
FEATURE	Location/Qualifiers	
REGION	1..120	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..120	
	note = C6-30[CLDN6]_H2.2 Variable heavy (vh) domain	
source	1..120 mol_type = protein organism = synthetic construct	
SEQUENCE: 253		
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG ITPNNGNTHY 60		
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120		
SEQ ID NO: 254	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = C6-30[CLDN6]_H2.2 vhCDR1	

-continued

source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 254		
EYTMH		5
SEQ ID NO: 255	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..17	
	note = C6-30[CLDN6]_H2.2 vhCDR2	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 255		
GITPNNGNTH YNQKFQG		17
SEQ ID NO: 256	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_H2.2 vhCDR3	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 256		
IYYFGRLYFD F		11
SEQ ID NO: 257	moltype = AA length = 120	
FEATURE	Location/Qualifiers	
REGION	1..120	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..120	
	note = C6-30[CLDN6]_H2.3 Variable heavy (vh) domain	
source	1..120	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 257		
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDSNNGNTHY 60		
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYFDWF GAGTLVTVSS 120		
SEQ ID NO: 258	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = C6-30[CLDN6]_H2.3 vhCDR1	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 258		
EYTMH		5
SEQ ID NO: 259	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..17	
	note = C6-30[CLDN6]_H2.3 vhCDR2	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 259		
GIDSNNGNTH YNQKFQG		17
SEQ ID NO: 260	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_H2.3 vhCDR3	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 260		
IYYFGRLYFD F		11

SEQ ID NO: 261	moltype = AA length = 120	
FEATURE	Location/Qualifiers	
REGION	1..120	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..120	
	note = C6-30[CLDN6]_H2.4 Variable heavy (vh) domain	
source	1..120	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 261		
EVQLVQSGAE VKKPGESLRI	SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDGNNGNTHY	60
NQKFQGHVTI SVDKSISTAY	LQWSSLKASD TAMYICARIY YFGRLYDFDW GAGTLVTVSS	120
SEQ ID NO: 262	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = C6-30[CLDN6]_H2.4 vhCDR1	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 262		
EYTMH		5
SEQ ID NO: 263	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..17	
	note = C6-30[CLDN6]_H2.4 vhCDR2	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 263		
GIDGNNGNTH YNQKFQG		17
SEQ ID NO: 264	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_H2.4 vhCDR3	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 264		
IYYFGRLYFD F		11
SEQ ID NO: 265	moltype = AA length = 120	
FEATURE	Location/Qualifiers	
REGION	1..120	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..120	
	note = C6-30[CLDN6]_H2.5 Variable heavy (vh) domain	
source	1..120	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 265		
EVQLVQSGAE VKKPGESLRI	SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPQNGNTHY	60
NQKFQGHVTI SVDKSISTAY	LQWSSLKASD TAMYICARIY YFGRLYDFDW GAGTLVTVSS	120
SEQ ID NO: 266	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = C6-30[CLDN6]_H2.5 vhCDR1	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 266		
EYTMH		5
SEQ ID NO: 267	moltype = AA length = 17	
FEATURE	Location/Qualifiers	

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REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..17	
	note = C6-30[CLDN6]_H2.5 vhCDR2	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 267		
GIDPQNGNTH YNQKFQG		17
SEQ ID NO: 268	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_H2.5 vhCDR3	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 268		
IYYFGRLYFD F		11
SEQ ID NO: 269	moltype = AA length = 120	
FEATURE	Location/Qualifiers	
REGION	1..120	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..120	
	note = C6-30[CLDN6]_H2.6 Variable heavy (vh) domain	
source	1..120	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 269		
EVQLVQSGAE VKKPGESLRI SCKTSGYTFE EYTMHWVRQM PGKSLEWMGG IDPNDGNTHY 60		
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYDFDW GAGTLTVSS 120		
SEQ ID NO: 270	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = C6-30[CLDN6]_H2.6 vhCDR1	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 270		
EYTMH		5
SEQ ID NO: 271	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..17	
	note = C6-30[CLDN6]_H2.6 vhCDR2	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 271		
GIDPNDGNTH YNQKFQG		17
SEQ ID NO: 272	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_H2.6 vhCDR3	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 272		
IYYFGRLYFD F		11
SEQ ID NO: 273	moltype = AA length = 120	
FEATURE	Location/Qualifiers	
REGION	1..120	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..120	
	note = C6-30[CLDN6]_H2.7 Variable heavy (vh) domain	

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source                1..120
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 273
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNQGNTHY 60
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYFDFW GAGTLVTVSS 120

SEQ ID NO: 274        moltype = AA  length = 5
FEATURE              Location/Qualifiers
REGION              1..5
                      note = Description of Artificial Sequence: Synthetic peptide
REGION              1..5
                      note = C6-30[CLDN6]_H2.7  vhCDR1
source              1..5
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 274
EYTMH                                                         5

SEQ ID NO: 275        moltype = AA  length = 17
FEATURE              Location/Qualifiers
REGION              1..17
                      note = Description of Artificial Sequence: Synthetic peptide
REGION              1..17
                      note = C6-30[CLDN6]_H2.7  vhCDR2
source              1..17
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 275
GIDPNQGNTH YNQKFQG                                         17

SEQ ID NO: 276        moltype = AA  length = 11
FEATURE              Location/Qualifiers
REGION              1..11
                      note = Description of Artificial Sequence: Synthetic peptide
REGION              1..11
                      note = C6-30[CLDN6]_H2.7  vhCDR3
source              1..11
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 276
IYYFGRLYFD F                                               11

SEQ ID NO: 277        moltype = AA  length = 120
FEATURE              Location/Qualifiers
REGION              1..120
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..120
                      note = C6-30[CLDN6]_H2.8  Variable heavy (vh) domain
source              1..120
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 277
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNDNTHY 60
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYFDFW GAGTLVTVSS 120

SEQ ID NO: 278        moltype = AA  length = 5
FEATURE              Location/Qualifiers
REGION              1..5
                      note = Description of Artificial Sequence: Synthetic peptide
REGION              1..5
                      note = C6-30[CLDN6]_H2.8  vhCDR1
source              1..5
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 278
EYTMH                                                         5

SEQ ID NO: 279        moltype = AA  length = 17
FEATURE              Location/Qualifiers
REGION              1..17
                      note = Description of Artificial Sequence: Synthetic peptide
REGION              1..17
                      note = C6-30[CLDN6]_H2.8  vhCDR2
source              1..17
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 279

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-continued

GIDPNNNDNTH YNQKFQG	17
SEQ ID NO: 280	
FEATURE	moltype = AA length = 11
REGION	Location/Qualifiers
	1..11
	note = Description of Artificial Sequence: Synthetic peptide
REGION	1..11
	note = C6-30[CLDN6]_H2.8 vhCDR3
source	1..11
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 280	
IYYFGRLYFD F	11
SEQ ID NO: 281	
FEATURE	moltype = AA length = 120
REGION	Location/Qualifiers
	1..120
	note = Description of Artificial Sequence: Synthetic polypeptide
REGION	1..120
	note = C6-30[CLDN6]_H2.9 Variable heavy (vh) domain
source	1..120
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 281	
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNANTHY 60	
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120	
SEQ ID NO: 282	
FEATURE	moltype = AA length = 5
REGION	Location/Qualifiers
	1..5
	note = Description of Artificial Sequence: Synthetic peptide
REGION	1..5
	note = C6-30[CLDN6]_H2.9 vhCDR1
source	1..5
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 282	
EYTMH	5
SEQ ID NO: 283	
FEATURE	moltype = AA length = 17
REGION	Location/Qualifiers
	1..17
	note = Description of Artificial Sequence: Synthetic peptide
REGION	1..17
	note = C6-30[CLDN6]_H2.9 vhCDR2
source	1..17
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 283	
GIDPNNANTH YNQKFQG	17
SEQ ID NO: 284	
FEATURE	moltype = AA length = 11
REGION	Location/Qualifiers
	1..11
	note = Description of Artificial Sequence: Synthetic peptide
REGION	1..11
	note = C6-30[CLDN6]_H2.9 vhCDR3
source	1..11
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 284	
IYYFGRLYFD F	11
SEQ ID NO: 285	
FEATURE	moltype = AA length = 120
REGION	Location/Qualifiers
	1..120
	note = Description of Artificial Sequence: Synthetic polypeptide
REGION	1..120
	note = C6-30[CLDN6]_H2.11 Variable heavy (vh) domain
source	1..120
	mol_type = protein
	organism = synthetic construct
SEQUENCE: 285	
EVQLVQSGAE VKKPGESLRI SCKTSGYSFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60	
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120	
SEQ ID NO: 286	
	moltype = AA length = 5

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FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = C6-30[CLDN6]_H2.11 vhCDR1	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 286		
EYTMH		5
SEQ ID NO: 287	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..17	
	note = C6-30[CLDN6]_H2.11 vhCDR2	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 287		
GIDPNNNGNTH YNQKFQG		17
SEQ ID NO: 288	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_H2.11 vhCDR3	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 288		
IYFGRLYFD F		11
SEQ ID NO: 289	moltype = AA length = 120	
FEATURE	Location/Qualifiers	
REGION	1..120	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..120	
	note = C6-30[CLDN6]_H2.12 Variable heavy (vh) domain	
source	1..120	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 289		
EVQLVQSGAE VKKPGESLRI SCKTSGYDFT EYTMHWVRQM PGKSLEWMGG IDPNNNGNTHY 60		
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120		
SEQ ID NO: 290	moltype = AA length = 5	
FEATURE	Location/Qualifiers	
REGION	1..5	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5	
	note = C6-30[CLDN6]_H2.12 vhCDR1	
source	1..5	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 290		
EYTMH		5
SEQ ID NO: 291	moltype = AA length = 17	
FEATURE	Location/Qualifiers	
REGION	1..17	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..17	
	note = C6-30[CLDN6]_H2.12 vhCDR2	
source	1..17	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 291		
GIDPNNNGNTH YNQKFQG		17
SEQ ID NO: 292	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_H2.12 vhCDR3	

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source                1..11
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 292
IYFGRLYFD F                                11

SEQ ID NO: 293        moltype = AA length = 120
FEATURE              Location/Qualifiers
REGION               1..120
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION               1..120
                      note = C6-30[CLDN6]_H2.71 Variable heavy (vh) domain
source                1..120
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 293
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDLNNGNTHY 60
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYFDWF GAGTLVTVSS 120

SEQ ID NO: 294        moltype = AA length = 5
FEATURE              Location/Qualifiers
REGION               1..5
                      note = Description of Artificial Sequence: Synthetic peptide
REGION               1..5
                      note = C6-30[CLDN6]_H2.71 vHCDR1
source                1..5
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 294
EYTMH                                5

SEQ ID NO: 295        moltype = AA length = 17
FEATURE              Location/Qualifiers
REGION               1..17
                      note = Description of Artificial Sequence: Synthetic peptide
REGION               1..17
                      note = C6-30[CLDN6]_H2.71 vHCDR2
source                1..17
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 295
GIDLNNGNTH YNQKFQG                                17

SEQ ID NO: 296        moltype = AA length = 11
FEATURE              Location/Qualifiers
REGION               1..11
                      note = Description of Artificial Sequence: Synthetic peptide
REGION               1..11
                      note = C6-30[CLDN6]_H2.71 vHCDR3
source                1..11
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 296
IYFGRLYFD F                                11

SEQ ID NO: 297        moltype = AA length = 120
FEATURE              Location/Qualifiers
REGION               1..120
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION               1..120
                      note = C6-30[CLDN6]_H2.75 Variable heavy (vh) domain
source                1..120
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 297
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPHNGNTHY 60
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYFDWF GAGTLVTVSS 120

SEQ ID NO: 298        moltype = AA length = 5
FEATURE              Location/Qualifiers
REGION               1..5
                      note = Description of Artificial Sequence: Synthetic peptide
REGION               1..5
                      note = C6-30[CLDN6]_H2.75 vHCDR1
source                1..5
                      mol_type = protein
                      organism = synthetic construct

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-continued

SEQUENCE: 298
EYTMH 5

SEQ ID NO: 299 moltype = AA length = 17
FEATURE Location/Qualifiers
REGION 1..17
 note = Description of Artificial Sequence: Synthetic peptide
REGION 1..17
 note = C6-30[CLDN6]_H2.75 vhCDR2
source 1..17
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 299
GIDPHNGNTH YNQKFQG 17

SEQ ID NO: 300 moltype = AA length = 11
FEATURE Location/Qualifiers
REGION 1..11
 note = Description of Artificial Sequence: Synthetic peptide
REGION 1..11
 note = C6-30[CLDN6]_H2.75 vhCDR3
source 1..11
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 300
IYYFGRLYFD F 11

SEQ ID NO: 301 moltype = AA length = 120
FEATURE Location/Qualifiers
REGION 1..120
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..120
 note = C6-30[CLDN6]_H2.90 Variable heavy (vh) domain
source 1..120
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 301
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNLNTHY 60
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYFDWF GAGTLVTVSS 120

SEQ ID NO: 302 moltype = AA length = 5
FEATURE Location/Qualifiers
REGION 1..5
 note = Description of Artificial Sequence: Synthetic peptide
REGION 1..5
 note = C6-30[CLDN6]_H2.90 vhCDR1
source 1..5
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 302
EYTMH 5

SEQ ID NO: 303 moltype = AA length = 17
FEATURE Location/Qualifiers
REGION 1..17
 note = Description of Artificial Sequence: Synthetic peptide
REGION 1..17
 note = C6-30[CLDN6]_H2.90 vhCDR2
source 1..17
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 303
GIDPNNLNTH YNQKFQG 17

SEQ ID NO: 304 moltype = AA length = 11
FEATURE Location/Qualifiers
REGION 1..11
 note = Description of Artificial Sequence: Synthetic peptide
REGION 1..11
 note = C6-30[CLDN6]_H2.90 vhCDR3
source 1..11
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 304
IYYFGRLYFD F 11

SEQ ID NO: 305 moltype = AA length = 120
FEATURE Location/Qualifiers

-continued

REGION 1..120
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..120
 note = C6-30[CLDN6]_H2.91 Variable heavy (vh) domain

source 1..120
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 305
 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNFNTHY 60
 NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYICARIY YFGRLYFDFW GAGTLTVSS 120

SEQ ID NO: 306 moltype = AA length = 5
 FEATURE Location/Qualifiers
 REGION 1..5
 note = Description of Artificial Sequence: Synthetic peptide

REGION 1..5
 note = C6-30[CLDN6]_H2.91 vHCDR1

source 1..5
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 306
 EYTMH 5

SEQ ID NO: 307 moltype = AA length = 17
 FEATURE Location/Qualifiers
 REGION 1..17
 note = Description of Artificial Sequence: Synthetic peptide

REGION 1..17
 note = C6-30[CLDN6]_H2.91 vHCDR2

source 1..17
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 307
 GIDPNNFNTH YNQKFQG 17

SEQ ID NO: 308 moltype = AA length = 11
 FEATURE Location/Qualifiers
 REGION 1..11
 note = Description of Artificial Sequence: Synthetic peptide

REGION 1..11
 note = C6-30[CLDN6]_H2.91 vHCDR3

source 1..11
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 308
 IYYFGRLYFD F 11

SEQ ID NO: 309 moltype = AA length = 120
 FEATURE Location/Qualifiers
 REGION 1..120
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..120
 note = C6-30[CLDN6]_H2.118 Variable heavy (vh) domain

source 1..120
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 309
 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
 NQKKQGHVTI SVDKSISTAY LQWSSLKASD TAMYICARIY YFGRLYFDFW GAGTLTVSS 120

SEQ ID NO: 310 moltype = AA length = 5
 FEATURE Location/Qualifiers
 REGION 1..5
 note = Description of Artificial Sequence: Synthetic peptide

REGION 1..5
 note = C6-30[CLDN6]_H2.118 vHCDR1

source 1..5
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 310
 EYTMH 5

SEQ ID NO: 311 moltype = AA length = 17
 FEATURE Location/Qualifiers
 REGION 1..17
 note = Description of Artificial Sequence: Synthetic peptide

REGION 1..17

-continued

source	note = C6-30[CLDN6]_H2.118 vhCDR2 1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 311 GIDPNNNGNTH YNQKKQG		17
SEQ ID NO: 312 FEATURE REGION	moltype = AA length = 11 Location/Qualifiers 1..11 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11 note = C6-30[CLDN6]_H2.118 vhCDR3	
source	1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 312 IYYFGRLYFD F		11
SEQ ID NO: 313 FEATURE REGION	moltype = AA length = 120 Location/Qualifiers 1..120 note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..120 note = C6-30[CLDN6]_H2.119 Variable heavy (vh) domain	
source	1..120 mol_type = protein organism = synthetic construct	
SEQUENCE: 313 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60 NQKFEHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLTVSS 120		
SEQ ID NO: 314 FEATURE REGION	moltype = AA length = 5 Location/Qualifiers 1..5 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..5 note = C6-30[CLDN6]_H2.119 vhCDR1	
source	1..5 mol_type = protein organism = synthetic construct	
SEQUENCE: 314 EYTMH		5
SEQ ID NO: 315 FEATURE REGION	moltype = AA length = 17 Location/Qualifiers 1..17 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..17 note = C6-30[CLDN6]_H2.119 vhCDR2	
source	1..17 mol_type = protein organism = synthetic construct	
SEQUENCE: 315 GIDPNNNGNTH YNQKFEG		17
SEQ ID NO: 316 FEATURE REGION	moltype = AA length = 11 Location/Qualifiers 1..11 note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11 note = C6-30[CLDN6]_H2.119 vhCDR3	
source	1..11 mol_type = protein organism = synthetic construct	
SEQUENCE: 316 IYYFGRLYFD F		11
SEQ ID NO: 317 FEATURE REGION	moltype = AA length = 107 Location/Qualifiers 1..107 note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..107 note = C6-30[CLDN6]_L1.1 Variable light (vl) domain	
source	1..107 mol_type = protein organism = synthetic construct	

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FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9	
	note = C6-30[CLDN6]_L1.4 v1CDR3	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 324		
QQYWSSPLT		9
SEQ ID NO: 325	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
REGION	1..107	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..107	
	note = C6-30[CLDN6]_L1.7 Variable light (v1) domain	
source	1..107	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 325		
DIQMTQSPSS LSASVGDRVIT ITCQASEDIY SRLAWYQQKP GKVPKLLISG ATSLETGVPS	60	
RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIK	107	
SEQ ID NO: 326	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_L1.7 v1CDR1	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 326		
QASEDIYSRL A		11
SEQ ID NO: 327	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7	
	note = C6-30[CLDN6]_L1.7 v1CDR2	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 327		
GATSLET		7
SEQ ID NO: 328	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9	
	note = C6-30[CLDN6]_L1.7 v1CDR3	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 328		
QQYWSSPLT		9
SEQ ID NO: 329	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
REGION	1..107	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..107	
	note = C6-30[CLDN6]_L1.16 Variable light (v1) domain	
source	1..107	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 329		
DIQMTQSPSS LSASVGDRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISA ATSLETGVPS	60	
RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIK	107	
SEQ ID NO: 330	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	

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REGION	1..11	
	note = C6-30[CLDN6]_L1.16 vLCDR1	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 330		
QASEDIYNRL A		11
SEQ ID NO: 331	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7	
	note = C6-30[CLDN6]_L1.16 vLCDR2	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 331		
AATSLET		7
SEQ ID NO: 332	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9	
	note = C6-30[CLDN6]_L1.16 vLCDR3	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 332		
QQYWSSPLT		9
SEQ ID NO: 333	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
REGION	1..107	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..107	
	note = C6-30[CLDN6]_L1.18 Variable light (vl) domain	
source	1..107	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 333		
DIQMTQSPSS LSASVGDRVT ITCQASEDIY NRLAWYQQKP GKVPKLLISG TTSLETGVPS	60	
RPSGSGSGKD YFTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIK	107	
SEQ ID NO: 334	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_L1.18 vLCDR1	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 334		
QASEDIYNRL A		11
SEQ ID NO: 335	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7	
	note = C6-30[CLDN6]_L1.18 vLCDR2	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 335		
GTTSLET		7
SEQ ID NO: 336	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9	
	note = C6-30[CLDN6]_L1.18 vLCDR3	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	

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SEQUENCE: 336
QQYWSSPLT 9

SEQ ID NO: 337 moltype = AA length = 107
FEATURE Location/Qualifiers
REGION 1..107
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..107
note = C6-30[CLDN6]_L1.19 Variable light (vl) domain
source 1..107
mol_type = protein
organism = synthetic construct

SEQUENCE: 337
DIQMTQSPSS LSASVGRVT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ASSLETGVPS 60
RFGSGSGSKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIK 107

SEQ ID NO: 338 moltype = AA length = 11
FEATURE Location/Qualifiers
REGION 1..11
note = Description of Artificial Sequence: Synthetic peptide
REGION 1..11
note = C6-30[CLDN6]_L1.19 v1CDR1
source 1..11
mol_type = protein
organism = synthetic construct

SEQUENCE: 338
QASEDIYNRL A 11

SEQ ID NO: 339 moltype = AA length = 7
FEATURE Location/Qualifiers
REGION 1..7
note = Description of Artificial Sequence: Synthetic peptide
REGION 1..7
note = C6-30[CLDN6]_L1.19 v1CDR2
source 1..7
mol_type = protein
organism = synthetic construct

SEQUENCE: 339
GASSLET 7

SEQ ID NO: 340 moltype = AA length = 9
FEATURE Location/Qualifiers
REGION 1..9
note = Description of Artificial Sequence: Synthetic peptide
REGION 1..9
note = C6-30[CLDN6]_L1.19 v1CDR3
source 1..9
mol_type = protein
organism = synthetic construct

SEQUENCE: 340
QQYWSSPLT 9

SEQ ID NO: 341 moltype = AA length = 107
FEATURE Location/Qualifiers
REGION 1..107
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..107
note = C6-30[CLDN6]_L1.21 Variable light (vl) domain
source 1..107
mol_type = protein
organism = synthetic construct

SEQUENCE: 341
DIQMTQSPSS LSASVGRVT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATNLETGVPS 60
RFGSGSGSKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIK 107

SEQ ID NO: 342 moltype = AA length = 11
FEATURE Location/Qualifiers
REGION 1..11
note = Description of Artificial Sequence: Synthetic peptide
REGION 1..11
note = C6-30[CLDN6]_L1.21 v1CDR1
source 1..11
mol_type = protein
organism = synthetic construct

SEQUENCE: 342
QASEDIYNRL A 11

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SEQ ID NO: 343	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7	
	note = C6-30[CLDN6]_L1.21 v1CDR2	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 343		
GATNLET		7
SEQ ID NO: 344	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9	
	note = C6-30[CLDN6]_L1.21 v1CDR3	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 344		
QQYWSSPLT		9
SEQ ID NO: 345	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
REGION	1..107	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..107	
	note = C6-30[CLDN6]_L1.22 Variable light (v1) domain	
source	1..107	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 345		
DIQMTQSPSS LSASVGRVT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATQLETGVPS		60
RFGSGSGGKD YFTISSLPQ EDIATYYCQQ YWSSPLTFGG GTKVEIK		107
SEQ ID NO: 346	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_L1.22 v1CDR1	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 346		
QASEDIYNRL A		11
SEQ ID NO: 347	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7	
	note = C6-30[CLDN6]_L1.22 v1CDR2	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 347		
GATQLET		7
SEQ ID NO: 348	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9	
	note = C6-30[CLDN6]_L1.22 v1CDR3	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 348		
QQYWSSPLT		9
SEQ ID NO: 349	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
REGION	1..107	
	note = Description of Artificial Sequence: Synthetic polypeptide	

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REGION 1..107
 note = C6-30[CLDN6]_L1.23 Variable light (vl) domain
 source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 349
 DIQMTQSPSS LSASVGDRVT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSRETGVPS 60
 RFGSGSGGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIK 107

SEQ ID NO: 350 moltype = AA length = 11
 FEATURE Location/Qualifiers
 REGION 1..11
 note = Description of Artificial Sequence: Synthetic peptide
 REGION 1..11
 note = C6-30[CLDN6]_L1.23 vlCDR1
 source 1..11
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 350
 QASEDIYNRL A 11

SEQ ID NO: 351 moltype = AA length = 7
 FEATURE Location/Qualifiers
 REGION 1..7
 note = Description of Artificial Sequence: Synthetic peptide
 REGION 1..7
 note = C6-30[CLDN6]_L1.23 vlCDR2
 source 1..7
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 351
 GATSRET 7

SEQ ID NO: 352 moltype = AA length = 9
 FEATURE Location/Qualifiers
 REGION 1..9
 note = Description of Artificial Sequence: Synthetic peptide
 REGION 1..9
 note = C6-30[CLDN6]_L1.23 vlCDR3
 source 1..9
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 352
 QQYWSSPLT 9

SEQ ID NO: 353 moltype = AA length = 107
 FEATURE Location/Qualifiers
 REGION 1..107
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..107
 note = C6-30[CLDN6]_L1.27 Variable light (vl) domain
 source 1..107
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 353
 DIQMTQSPSS LSASVGDRVT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLESGVPS 60
 RFGSGSGGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIK 107

SEQ ID NO: 354 moltype = AA length = 11
 FEATURE Location/Qualifiers
 REGION 1..11
 note = Description of Artificial Sequence: Synthetic peptide
 REGION 1..11
 note = C6-30[CLDN6]_L1.27 vlCDR1
 source 1..11
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 354
 QASEDIYNRL A 11

SEQ ID NO: 355 moltype = AA length = 7
 FEATURE Location/Qualifiers
 REGION 1..7
 note = Description of Artificial Sequence: Synthetic peptide
 REGION 1..7
 note = C6-30[CLDN6]_L1.27 vlCDR2
 source 1..7
 mol_type = protein

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organism = synthetic construct

SEQUENCE: 355
GATSLES 7

SEQ ID NO: 356
FEATURE
REGION
note = Description of Artificial Sequence: Synthetic peptide
REGION
note = C6-30[CLDN6]_L1.27 v1CDR3
source
mol_type = protein
organism = synthetic construct

SEQUENCE: 356
QQYWSSPLT 9

SEQ ID NO: 357
FEATURE
REGION
note = Description of Artificial Sequence: Synthetic
polypeptide
REGION
note = C6-30[CLDN6]_L1.60 Variable light (v1) domain
source
mol_type = protein
organism = synthetic construct

SEQUENCE: 357
DIQMTQSPSS LSASVGDRVT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RPSGSGSGKD YFTTISLQP EDIATYYCQQ LWSSPLTFGG GTKVEIK 107

SEQ ID NO: 358
FEATURE
REGION
note = Description of Artificial Sequence: Synthetic peptide
REGION
note = C6-30[CLDN6]_L1.60 v1CDR1
source
mol_type = protein
organism = synthetic construct

SEQUENCE: 358
QASEDIYNRL A 11

SEQ ID NO: 359
FEATURE
REGION
note = Description of Artificial Sequence: Synthetic peptide
REGION
note = C6-30[CLDN6]_L1.60 v1CDR2
source
mol_type = protein
organism = synthetic construct

SEQUENCE: 359
GATSLET 7

SEQ ID NO: 360
FEATURE
REGION
note = Description of Artificial Sequence: Synthetic peptide
REGION
note = C6-30[CLDN6]_L1.60 v1CDR3
source
mol_type = protein
organism = synthetic construct

SEQUENCE: 360
QQLWSSPLT 9

SEQ ID NO: 361
FEATURE
REGION
note = Description of Artificial Sequence: Synthetic
polypeptide
REGION
note = C6-30[CLDN6]_L1.107 Variable light (v1) domain
source
mol_type = protein
organism = synthetic construct

SEQUENCE: 361
DIQMTQSPSS LSASVGDRVT ITCQASEDIV NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RPSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIK 107

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SEQ ID NO: 362	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_L1.107 v1CDR1	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 362		
QASEDIVNRL A		11
SEQ ID NO: 363	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7	
	note = C6-30[CLDN6]_L1.107 v1CDR2	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 363		
GATSLET		7
SEQ ID NO: 364	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9	
	note = C6-30[CLDN6]_L1.107 v1CDR3	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 364		
QQYWSSPLT		9
SEQ ID NO: 365	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
REGION	1..107	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..107	
	note = C6-30[CLDN6]_L1.114 Variable light (vl) domain	
source	1..107	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 365		
DIQMTQSPSS LSASVGDRVT	ITCQASEDIY LRLAWYQQKP GKVPKLLISG ATSLETGVPS	60
RFSGSGSGKD YFTTISLQP	EDIATYYCQQ YWSSPLTFGG GTKVEIK	107
SEQ ID NO: 366	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_L1.114 v1CDR1	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 366		
QASEDIYLRL A		11
SEQ ID NO: 367	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7	
	note = C6-30[CLDN6]_L1.114 v1CDR2	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 367		
GATSLET		7
SEQ ID NO: 368	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	

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REGION	1..9	
	note = C6-30[CLDN6]_L1.114 v1CDR3	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 368		
QQYWSSPLT		9
SEQ ID NO: 369	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
REGION	1..107	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..107	
	note = C6-30[CLDN6]_L1.187 Variable light (v1) domain	
source	1..107	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 369		
DIQMTQSPSS LSASVGRVT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS	60	
RFGSGSGGKD YFTTISLQP EDIATYYCQQ YWSAPLTFGG GTKVEIK	107	
SEQ ID NO: 370	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_L1.187 v1CDR1	
source	1..11	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 370		
QASEDIYNRL A		11
SEQ ID NO: 371	moltype = AA length = 7	
FEATURE	Location/Qualifiers	
REGION	1..7	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..7	
	note = C6-30[CLDN6]_L1.187 v1CDR2	
source	1..7	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 371		
GATSLET		7
SEQ ID NO: 372	moltype = AA length = 9	
FEATURE	Location/Qualifiers	
REGION	1..9	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..9	
	note = C6-30[CLDN6]_L1.187 v1CDR3	
source	1..9	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 372		
QQYWSAPLT		9
SEQ ID NO: 373	moltype = AA length = 107	
FEATURE	Location/Qualifiers	
REGION	1..107	
	note = Description of Artificial Sequence: Synthetic polypeptide	
REGION	1..107	
	note = C6-30[CLDN6]_L1.189 Variable light (v1) domain	
source	1..107	
	mol_type = protein	
	organism = synthetic construct	
SEQUENCE: 373		
DIQMTQSPSS LSASVGRVT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS	60	
RFGSGSGGKD YFTTISLQP EDIATYYCQQ YWSGPLTFGG GTKVEIK	107	
SEQ ID NO: 374	moltype = AA length = 11	
FEATURE	Location/Qualifiers	
REGION	1..11	
	note = Description of Artificial Sequence: Synthetic peptide	
REGION	1..11	
	note = C6-30[CLDN6]_L1.189 v1CDR1	
source	1..11	

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mol_type = protein
organism = synthetic construct

SEQUENCE: 374
QASEDIYNRL A 11

SEQ ID NO: 375
FEATURE
REGION
REGION
source
mol_type = AA length = 7
Location/Qualifiers
1..7
note = Description of Artificial Sequence: Synthetic peptide
1..7
note = C6-30[CLDN6]_L1.189 v1CDR2
1..7
mol_type = protein
organism = synthetic construct

SEQUENCE: 375
GATSLET 7

SEQ ID NO: 376
FEATURE
REGION
REGION
source
mol_type = AA length = 9
Location/Qualifiers
1..9
note = Description of Artificial Sequence: Synthetic peptide
1..9
note = C6-30[CLDN6]_L1.189 v1CDR3
1..9
mol_type = protein
organism = synthetic construct

SEQUENCE: 376
QQYWSGPLT 9

SEQ ID NO: 377
FEATURE
REGION
REGION
source
mol_type = AA length = 449
Location/Qualifiers
1..449
note = Description of Artificial Sequence: Synthetic
polypeptide
1..449
note = XENP34218 Heavy Chain - C6-30[CLDN6]_H1
1..449
mol_type = protein
organism = synthetic construct

SEQUENCE: 377
QVQLVQSGAE VKKPGASVKV SCKTSGYFTT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
NQKFPQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYFDFW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVS SVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 378
FEATURE
REGION
REGION
source
mol_type = AA length = 214
Location/Qualifiers
1..214
note = Description of Artificial Sequence: Synthetic
polypeptide
1..214
note = XENP34218 Light Chain - C6-30[CLDN6]_L1
1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 378
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFSGSGSGKD YFTTISLQPD EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 379
FEATURE
REGION
REGION
source
mol_type = AA length = 449
Location/Qualifiers
1..449
note = Description of Artificial Sequence: Synthetic
polypeptide
1..449
note = XENP34219 Heavy Chain - C6-30[CLDN6]_H1
1..449
mol_type = protein
organism = synthetic construct

SEQUENCE: 379
QVQLVQSGAE VKKPGASVKV SCKTSGYFTT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
NQKFPQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYFDFW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 180

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GLYSLSVVTV VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMSRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 380 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP34219 Light Chain - C6-30[CLDN6]_L2
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 380
 DIQMTQSPDS LAVSLGERAT INKASEDIY NRLAWYQQKP GQVPKLLISG ATSLETGVPS 60
 RFGSGSGGKD YTLTISSLQA EDVAVYVCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 381 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..449
 note = XENP34220 Heavy Chain - C6-30[CLDN6]_H2
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 381
 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
 NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS 180
 GLYSLSVVTV VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMSRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 382 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP34220 Light Chain - C6-30[CLDN6]_L1
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 382
 DIQMTQSPSS LSASVGRDVT ITCQASEDIY NRLAWYQQKP GQVPKLLISG ATSLETGVPS 60
 RFGSGSGGKD YFTTISSLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 383 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..449
 note = XENP34221 Heavy Chain - C6-30[CLDN6]_H2
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 383
 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
 NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS 180
 GLYSLSVVTV VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMSRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 384 moltype = AA length = 214

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FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214
 note = XENP34221 Light Chain - C6-30[CLDN6]_L2

source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 384

DIQMTQSPDS	LAVSLGERAT	INCKASEDIY	NRLAWYQQKP	GQVPKLLISG	ATSLETGVPS	60
RFSGSGSGKD	YTLTISSLQA	EDVAVYYCQQ	YWSSPLTFGG	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLNNFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYSLSSSTLT	180
LSKADYEKHK	VYACEVTHQG	LSSPVTKSPN	RGEC			214

SEQ ID NO: 385 moltype = AA length = 449

FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..449
 note = XENP35044 Heavy Chain - C6-30[CLDN6]_H2

source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 385

EVQLVQSGAE	VKKPGESLRI	SCKTSGYTFT	EYTMHWVRQM	PGKSLEWMGG	IDPNNGNTHY	60
NQKFQGHVTI	SVDKSISTAY	LQWSSLKASD	TAMYYCARIY	YFGRLYPFDW	GAGTLVTVSS	120
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	NTKVDKKVEP	KSCDKTHTCP	PCPAPPVAGP	240
SVFLFPPKPK	DTLMISRTPE	VTCTVVDVKH	EDPEVKFNWY	VDGVEVHNAK	TKPREEQYNS	300
TYRVVSVLTV	LHQDWLNGKE	YKCKVSNKAL	PAPIEKTISK	AKGQPREPQV	YTLPPSREEM	360
TKNQVSLTCL	VKGFPYPSDIA	VEWESNGQPE	NNYKTTPPVL	DSGGSFFLYS	KLTVDKSRWQ	420
QGNVFSCSVM	HEALHNHYTQ	KSLSLSPGK				449

SEQ ID NO: 386 moltype = AA length = 209

FEATURE Location/Qualifiers
 REGION 1..209
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..209
 note = XENP35044 Light Chain - C6-30[CLDN6]_L1.1

source 1..209
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 386

DIQMTQSPSS	LSASVGDVRT	ITCQASQDIY	NRLAWYQQKP	GKVPKLLISG	ATSLETGVPS	60
RFSGSGSGKD	YTFTISSLQP	EDIATYYCQQ	YWSSPLTFGG	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLNNFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYSLSSSTLT	180
LSKADYEKHK	VYACEVTHQG	LSSPVTKSF				209

SEQ ID NO: 387 moltype = AA length = 449

FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..449
 note = XENP35047 Heavy Chain - C6-30[CLDN6]_H2

source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 387

EVQLVQSGAE	VKKPGESLRI	SCKTSGYTFT	EYTMHWVRQM	PGKSLEWMGG	IDPNNGNTHY	60
NQKFQGHVTI	SVDKSISTAY	LQWSSLKASD	TAMYYCARIY	YFGRLYPFDW	GAGTLVTVSS	120
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	NTKVDKKVEP	KSCDKTHTCP	PCPAPPVAGP	240
SVFLFPPKPK	DTLMISRTPE	VTCTVVDVKH	EDPEVKFNWY	VDGVEVHNAK	TKPREEQYNS	300
TYRVVSVLTV	LHQDWLNGKE	YKCKVSNKAL	PAPIEKTISK	AKGQPREPQV	YTLPPSREEM	360
TKNQVSLTCL	VKGFPYPSDIA	VEWESNGQPE	NNYKTTPPVL	DSGGSFFLYS	KLTVDKSRWQ	420
QGNVFSCSVM	HEALHNHYTQ	KSLSLSPGK				449

SEQ ID NO: 388 moltype = AA length = 214

FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214
 note = XENP35047 Light Chain - C6-30[CLDN6]_L1.4

source 1..214

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mol_type = protein
organism = synthetic construct

SEQUENCE: 388
DIQMTQSPSS LSASVGRVIT ITCQASEDVY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 389 moltype = AA length = 449
FEATURE Location/Qualifiers
REGION 1..449
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..449
note = XENP35050 Heavy Chain - C6-30[CLDN6]_H2
source 1..449
mol_type = protein
organism = synthetic construct

SEQUENCE: 389
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLPPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDNLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 390 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..214
note = XENP35050 Light Chain - C6-30[CLDN6]_L1.7
source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 390
DIQMTQSPSS LSASVGRVIT ITCQASEDIY SRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 391 moltype = AA length = 449
FEATURE Location/Qualifiers
REGION 1..449
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..449
note = XENP35059 Heavy Chain - C6-30[CLDN6]_H2
source 1..449
mol_type = protein
organism = synthetic construct

SEQUENCE: 391
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLPPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDNLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 392 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..214
note = XENP35059 Light Chain - C6-30[CLDN6]_L1.16
source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 392
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISA ATSLETGVPS 60
RFGSGSGSKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

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SEQ ID NO: 393      moltype = AA  length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..449
                   note = XENP35061 Heavy Chain - C6-30[CLDN6]_H2
source             1..449
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 393
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
NQKFGQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYFDFW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNKKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 394      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP35061 Light Chain - C6-30[CLDN6]_L1.18
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 394
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQKP GKVPKLLISG TTSLETGVPS 60
RFGSGSGGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 395      moltype = AA  length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..449
                   note = XENP35062 Heavy Chain - C6-30[CLDN6]_H2
source             1..449
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 395
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
NQKFGQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYFDFW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNKKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 396      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP35062 Light Chain - C6-30[CLDN6]_L1.19
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 396
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQKP GKVPKLLISG ASSLETGVPS 60
RFGSGSGGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 397      moltype = AA  length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..449

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source          note = XENP35064 Heavy Chain - C6-30[CLDN6]_H2
                1..449
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 397
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLISRTP EDEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDNLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 398      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP35064 Light Chain - C6-30[CLDN6]_L1.21
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 398
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATNLETGVPS 60
RFGSGSGSKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 399      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..449
                   note = XENP35065 Heavy Chain - C6-30[CLDN6]_H2
source             1..449
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 399
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLISRTP EDEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDNLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 400      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP35065 Light Chain - C6-30[CLDN6]_L1.22
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 400
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATQLETGVPS 60
RFGSGSGSKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 401      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..449
                   note = XENP35066 Heavy Chain - C6-30[CLDN6]_H2
source             1..449
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 401
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120

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ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 402      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP35066 Light Chain - C6-30[CLDN6]_L1.23
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 402
DIQMTQSPSS LSASVGDRVT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSRETGVPS 60
RFGSGSGSKD YTFTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 403      moltype = AA  length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..449
                   note = XENP35070 Heavy Chain - C6-30[CLDN6]_H2
source             1..449
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 403
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
NQKFQGHVTI SVDKSIATAY LQWSSLKASD TAMYYCARIY YFGRLYFDWF GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 404      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP35070 Light Chain - C6-30[CLDN6]_L1.27
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 404
DIQMTQSPSS LSASVGDRVT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLESGVPS 60
RFGSGSGSKD YTFTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 405      moltype = AA  length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..449
                   note = XENP35085 Heavy Chain - C6-30[CLDN6]_H1.1
source             1..449
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 405
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG ISPNNGNTHY 60
NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYFDWF GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

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SEQ ID NO: 406 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP35085 Light Chain - C6-30[CLDN6]_L1
 source 1..214
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 406
 DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 407 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..449
 note = XENP35086 Heavy Chain - C6-30[CLDN6]_H1.2
 source 1..449
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 407
 QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLLEWGG INPNNGNTHY 60
 NQKPFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS 180
 GLYSLSVVTV VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKHTTCP PCPAPPVAGP 240
 SVFLPPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 408 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP35086 Light Chain - C6-30[CLDN6]_L1
 source 1..214
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 408
 DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 409 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..449
 note = XENP35087 Heavy Chain - C6-30[CLDN6]_H1.3
 source 1..449
 mol_type = protein
 organism = synthetic construct

 SEQUENCE: 409
 QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLLEWGG IDSNNGNTHY 60
 NQKPFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS 180
 GLYSLSVVTV VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKHTTCP PCPAPPVAGP 240
 SVFLPPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 410 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP35087 Light Chain - C6-30[CLDN6]_L1

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source                1..214
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 410
DIQMTQSPSS LSASVGDVRT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSPN RGEC 214

SEQ ID NO: 411        moltype = AA length = 449
FEATURE              Location/Qualifiers
REGION               1..449
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION               1..449
                      note = XENP35088 Heavy Chain - C6-30[CLDN6]_H1.4
source                1..449
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 411
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDANNGNTHY 60
NQKFPQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFPYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 412        moltype = AA length = 214
FEATURE              Location/Qualifiers
REGION               1..214
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION               1..214
                      note = XENP35088 Light Chain - C6-30[CLDN6]_L1
source                1..214
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 412
DIQMTQSPSS LSASVGDVRT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSPN RGEC 214

SEQ ID NO: 413        moltype = AA length = 449
FEATURE              Location/Qualifiers
REGION               1..449
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION               1..449
                      note = XENP35089 Heavy Chain - C6-30[CLDN6]_H1.5
source                1..449
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 413
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPQNGNTHY 60
NQKFPQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFPYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 414        moltype = AA length = 214
FEATURE              Location/Qualifiers
REGION               1..214
                      note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION               1..214
                      note = XENP35089 Light Chain - C6-30[CLDN6]_L1
source                1..214
                      mol_type = protein
                      organism = synthetic construct

SEQUENCE: 414
DIQMTQSPSS LSASVGDVRT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180

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LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 415      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..449
                   note = XENP35090 Heavy Chain - C6-30[CLDN6]_H1.6
source             1..449
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 415
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNSGNTHY 60
NQKFGQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVSVELTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 416      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP35090 Light Chain - C6-30[CLDN6]_L1
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 416
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 417      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..449
                   note = XENP35091 Heavy Chain - C6-30[CLDN6]_H1.7
source             1..449
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 417
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNQGNTHY 60
NQKFGQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVSVELTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 418      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP35091 Light Chain - C6-30[CLDN6]_L1
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 418
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 419      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide

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REGION 1..449
 note = XENP35092 Heavy Chain - C6-30[CLDN6]_H1.8
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 419
 QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLLEWMGG IDPNNDNTHY 60
 NQKQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYFDFW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLPPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 420 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP35092 Light Chain - C6-30[CLDN6]_L1
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 420
 DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQK GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 421 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..449
 note = XENP35093 Heavy Chain - C6-30[CLDN6]_H1.9
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 421
 QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLLEWMGG IDPNNANTHY 60
 NQKQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYFDFW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLPPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 422 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP35093 Light Chain - C6-30[CLDN6]_L1
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 422
 DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQK GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 423 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..449
 note = XENP35094 Heavy Chain - C6-30[CLDN6]_H2.1
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 423
 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLLEWMGG ISPNNGNTHY 60

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NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPDFW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 424 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP35094 Light Chain - C6-30[CLDN6]_L1
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 424
 DIQMTQSPSS LSASVGDRTV ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 425 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..449
 note = XENP35095 Heavy Chain - C6-30[CLDN6]_H2.2
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 425
 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG ITPNNGNTHY 60
 NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPDFW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 426 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP35095 Light Chain - C6-30[CLDN6]_L1
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 426
 DIQMTQSPSS LSASVGDRTV ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 427 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..449
 note = XENP35096 Heavy Chain - C6-30[CLDN6]_H2.3
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 427
 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDSNNGNTHY 60
 NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPDFW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

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SEQ ID NO: 428 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..214
 note = XENP35096 Light Chain - C6-30[CLDN6]_L1
source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 428
DIQMTQSPSS LSASVGDRVT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSKD YTFTISSLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 429 moltype = AA length = 449
FEATURE Location/Qualifiers
REGION 1..449
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..449
 note = XENP35097 Heavy Chain - C6-30[CLDN6]_H2.4
source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 429
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDGNNGNTHY 60
NQKFGQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPPKP DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVH HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 430 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..214
 note = XENP35097 Light Chain - C6-30[CLDN6]_L1
source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 430
DIQMTQSPSS LSASVGDRVT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSKD YTFTISSLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 431 moltype = AA length = 449
FEATURE Location/Qualifiers
REGION 1..449
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..449
 note = XENP35098 Heavy Chain - C6-30[CLDN6]_H2.5
source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 431
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPQNGNTHY 60
NQKFGQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPPKP DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVH HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 432 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
REGION 1..214

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source          note = XENP35098 Light Chain - C6-30[CLDN6]_L1
                1..214
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 432
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGGKD YFTTISSLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEHKK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 433      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION            1..449
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..449
                  note = XENP35099 Heavy Chain - C6-30[CLDN6]_H2.6
source            1..449
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 433
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNDGNTHY 60
NQKFGQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLTVVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS 180
GLYSLSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKHTTCP PCPAPPVAGP 240
SVFLPPPKPK DTLMSRTPTE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVSVLTIV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 434      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP35099 Light Chain - C6-30[CLDN6]_L1
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 434
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGGKD YFTTISSLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEHKK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 435      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION            1..449
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..449
                  note = XENP35100 Heavy Chain - C6-30[CLDN6]_H2.7
source            1..449
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 435
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNQGNTHY 60
NQKFGQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLTVVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS 180
GLYSLSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKHTTCP PCPAPPVAGP 240
SVFLPPPKPK DTLMSRTPTE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVSVLTIV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 436      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP35100 Light Chain - C6-30[CLDN6]_L1
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 436
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGGKD YFTTISSLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120

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SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 437      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..449
                   note = XENP35101 Heavy Chain - C6-30[CLDN6]_H2.8
source             1..449
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 437
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNDNTHY 60
NQKFGQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLPPPKPK DTLISRTP EDEPVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 438      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP35101 Light Chain - C6-30[CLDN6]_L1
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 438
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 439      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..449
                   note = XENP35102 Heavy Chain - C6-30[CLDN6]_H2.9
source             1..449
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 439
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNANTHY 60
NQKFGQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLPPPKPK DTLISRTP EDEPVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 440      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP35102 Light Chain - C6-30[CLDN6]_L1
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 440
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 441      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic

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REGION          polypeptide
                1..449
                note = XENP35865 Heavy Chain - C6-30[CLDN6]_H2
source          1..449
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 441
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
NQKFGQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVSVELTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 442      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP35865 Light Chain - C6-30[CLDN6]_L1.60
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 442
DIQMTQSPSS LSASVGDRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGGKD YTFTISSLQP EDIATYYCQQ LWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 443      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION            1..449
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..449
                  note = XENP35883 Heavy Chain - C6-30[CLDN6]_H1.19
source            1..449
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 443
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLWMMGG IDPNNGNTHY 60
NQKFGQGRVTI TVDKSASTAY MELSSLSRSED TATYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVSVELTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 444      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP35883 Light Chain - C6-30[CLDN6]_L1
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 444
DIQMTQSPSS LSASVGDRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGGKD YTFTISSLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK STYLSSTLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 445      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION            1..449
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..449
                  note = XENP35886 Heavy Chain - C6-30[CLDN6]_H1.22
source            1..449
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 445

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QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLSEWMGG IDPNNGNTHY 60
 NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIL YFGRLYPFDW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSVVTV VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMSRTP ETCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSV M HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 446 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP35886 Light Chain - C6-30[CLDN6]_L1
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 446
 DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQK GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YFTFISSLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 447 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..449
 note = XENP35888 Heavy Chain - C6-30[CLDN6]_H1.24
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 447
 QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLSEWMGG IDPNNGNTHY 60
 NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YLGRLYPFDW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSVVTV VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMSRTP ETCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSV M HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 448 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP35888 Light Chain - C6-30[CLDN6]_L1
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 448
 DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQK GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YFTFISSLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 449 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..449
 note = XENP35890 Heavy Chain - C6-30[CLDN6]_H2.11
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 449
 EVQLVQSGAE VKKPGESLRI SCKTSGYSFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
 NQKFQGHVTI SVDKSIATY LQWSSLKASD TAMYVCARIY YFGRLYPFDW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSVVTV VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMSRTP ETCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420

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QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 450 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP35890 Light Chain - C6-30[CLDN6]_L1
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 450
 DIQMTQSPSS LSASVGDVRT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 451 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..449
 note = XENP35891 Heavy Chain - C6-30[CLDN6]_H2.12
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 451
 EVQLVQSGAE VKKPGESLRI SCKTSGYDFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
 NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYFDFW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSVVTV VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTP ETCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 452 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP35891 Light Chain - C6-30[CLDN6]_L1
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 452
 DIQMTQSPSS LSASVGDVRT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 453 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..449
 note = XENP35979 Heavy Chain - C6-30[CLDN6]_H2
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 453
 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
 NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYFDFW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSVVTV VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTP ETCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 454 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide

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REGION 1..214
 note = XENP35979 Light Chain - C6-30[CLDN6]_L1.187
 source 1..214
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 454
 DIQMTQSPSS LSASVGDVRT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFGSGSGGKD YTFTISSLQP EDIATYYCQQ YWSAPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 455 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..449
 note = XENP35981 Heavy Chain - C6-30[CLDN6]_H2
 source 1..449
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 455
 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
 NQKFGQHVIT SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPPKP DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 456 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP35981 Light Chain - C6-30[CLDN6]_L1.189
 source 1..214
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 456
 DIQMTQSPSS LSASVGDVRT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFGSGSGGKD YTFTISSLQP EDIATYYCQQ YWSGPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 457 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..449
 note = XENP35929 Heavy Chain - C6-30[CLDN6]_H2
 source 1..449
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 457
 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
 NQKFGQHVIT SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPPKP DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 458 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP35929 Light Chain - C6-30[CLDN6]_L1.107
 source 1..214
 mol_type = protein
 organism = synthetic construct
 SEQUENCE: 458
 DIQMTQSPSS LSASVGDVRT ITCQASEDIV NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60

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RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 459 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..449
 note = XENP35936 Heavy Chain - C6-30[CLDN6]_H2
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 459
 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
 NQKFGQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYFDFW GAGTLTVTSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSVVVT VPSSSLGTQT YICNVNHHKPS NTKVDKKVEP KSCDKHTTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 460 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP35936 Light Chain - C6-30[CLDN6]_L1.114
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 460
 DIQMTQSPSS LSASVGRVIT ITCQASEDIY LRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 461 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..449
 note = XENP36021 Heavy Chain - C6-30[CLDN6]_H2.71
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 461
 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDLNNGNTHY 60
 NQKFGQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYFDFW GAGTLTVTSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSVVVT VPSSSLGTQT YICNVNHHKPS NTKVDKKVEP KSCDKHTTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 462 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP36021 Light Chain - C6-30[CLDN6]_L1
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 462
 DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 463 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449

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note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..449
note = XENP36022 Heavy Chain - C6-30[CLDN6]_H2.72
source 1..449
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 463
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDANNGNTHY 60
NQKFQGHVTI SVDKSIISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 464      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..214
note = XENP36022 Light Chain - C6-30[CLDN6]_L1
source 1..214
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 464
DIQMTQSPSS LSASVGDVRT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 465      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION 1..449
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..449
note = XENP36025 Heavy Chain - C6-30[CLDN6]_H2.75
source 1..449
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 465
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPHNGNTHY 60
NQKFQGHVTI SVDKSIISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 466      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..214
note = XENP36025 Light Chain - C6-30[CLDN6]_L1
source 1..214
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 466
DIQMTQSPSS LSASVGDVRT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 467      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION 1..449
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..449
note = XENP36040 Heavy Chain - C6-30[CLDN6]_H2.90
source 1..449
      mol_type = protein
      organism = synthetic construct

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SEQUENCE: 467

EVQLVQSGAE	VKKPGESLRI	SCKTSGYTFT	EYTMHWVRQM	PGKSLEWMGG	IDPNNLNTHY	60
NQKFQGHVTI	SVDKSISTAY	LQWSSLKASD	TAMYYCARIY	YFGRLYPFDW	GAGTLVTVSS	120
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	NTKVDKKVEP	KSCDKTHTCP	PCPAPPVAGP	240
SVFLFPPKPK	DTLMISRTPE	VTCTVVDVKH	EDPEVKFNWY	VDGVEVHNAK	TKPREEQYNS	300
TYRIVSVLTV	LHQDWLNGKE	YKCKVSNKAL	PAPIEKTISK	AKGQPREPQV	YTLPPSREEM	360
TKNQVSLTCL	VKGFPYPSDIA	VEWESNGQPE	NNYKTTTPVL	DSDGSFFLYS	KLTVDKSRWQ	420
QGNVFSCSV	HEALHNHYTQ	KSLSLSPGK				449

SEQ ID NO: 468 moltype = AA length = 214

FEATURE Location/Qualifiers

REGION 1..214

 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214

 note = XENP36040 Light Chain - C6-30[CLDN6]_L1

source 1..214

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 468

DIQMTQSPSS	LSASVGRVDT	ITCQASEDIY	NRLAWYQQKP	GKVPKLLISG	ATSLETGVPS	60
RFGSGSGGKD	YTFTISSLQP	EDIATYYCQQ	YWSSPLTFGG	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYKHKH	VYACEVTHQG	LSSPVTKSFN	RGEC			214

SEQ ID NO: 469 moltype = AA length = 449

FEATURE Location/Qualifiers

REGION 1..449

 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..449

 note = XENP36041 Heavy Chain - C6-30[CLDN6]_H2.91

source 1..449

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 469

EVQLVQSGAE	VKKPGESLRI	SCKTSGYTFT	EYTMHWVRQM	PGKSLEWMGG	IDPNNFNTHY	60
NQKFQGHVTI	SVDKSISTAY	LQWSSLKASD	TAMYYCARIY	YFGRLYPFDW	GAGTLVTVSS	120
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	NTKVDKKVEP	KSCDKTHTCP	PCPAPPVAGP	240
SVFLFPPKPK	DTLMISRTPE	VTCTVVDVKH	EDPEVKFNWY	VDGVEVHNAK	TKPREEQYNS	300
TYRIVSVLTV	LHQDWLNGKE	YKCKVSNKAL	PAPIEKTISK	AKGQPREPQV	YTLPPSREEM	360
TKNQVSLTCL	VKGFPYPSDIA	VEWESNGQPE	NNYKTTTPVL	DSDGSFFLYS	KLTVDKSRWQ	420
QGNVFSCSV	HEALHNHYTQ	KSLSLSPGK				449

SEQ ID NO: 470 moltype = AA length = 214

FEATURE Location/Qualifiers

REGION 1..214

 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214

 note = XENP36041 Light Chain - C6-30[CLDN6]_L1

source 1..214

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 470

DIQMTQSPSS	LSASVGRVDT	ITCQASEDIY	NRLAWYQQKP	GKVPKLLISG	ATSLETGVPS	60
RFGSGSGGKD	YTFTISSLQP	EDIATYYCQQ	YWSSPLTFGG	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYKHKH	VYACEVTHQG	LSSPVTKSFN	RGEC			214

SEQ ID NO: 471 moltype = AA length = 449

FEATURE Location/Qualifiers

REGION 1..449

 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..449

 note = XENP36065 Heavy Chain - C6-30[CLDN6]_H2.118

source 1..449

 mol_type = protein

 organism = synthetic construct

SEQUENCE: 471

EVQLVQSGAE	VKKPGESLRI	SCKTSGYTFT	EYTMHWVRQM	PGKSLEWMGG	IDPNNGNTHY	60
NQKKQGHVTI	SVDKSISTAY	LQWSSLKASD	TAMYYCARIY	YFGRLYPFDW	GAGTLVTVSS	120
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	NTKVDKKVEP	KSCDKTHTCP	PCPAPPVAGP	240
SVFLFPPKPK	DTLMISRTPE	VTCTVVDVKH	EDPEVKFNWY	VDGVEVHNAK	TKPREEQYNS	300
TYRIVSVLTV	LHQDWLNGKE	YKCKVSNKAL	PAPIEKTISK	AKGQPREPQV	YTLPPSREEM	360

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TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPV L DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSV M HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 472 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP36065 Light Chain - C6-30[CLDN6]_L1
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 472
 DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQK GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 473 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..449
 note = XENP36066 Heavy Chain - C6-30[CLDN6]_H2.119
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 473
 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
 NQKFEGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLPPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLT V LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPV L DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSV M HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 474 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP36066 Light Chain - C6-30[CLDN6]_L1
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 474
 DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQK GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 475 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..449
 note = XENP36956 Heavy Chain - C6-30[CLDN6]_H1.22
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 475
 QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
 NQKFQQRVTI TVDKSASTAY MELSSLRSED TAVYYCARIL YFGRLYPFDW GAGTLTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLPPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLT V LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPV L DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSV M HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 476 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic

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REGION          polypeptide
                  1..214
note = XENP36956 Light Chain - C6-30[CLDN6]_L1.187
source          1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 476
DIQMTQSPSS LSASVGDVRT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSKD YFTTISLQP EDIATYYCQQ YWSAPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 477    moltype = AA length = 449
FEATURE          Location/Qualifiers
REGION          1..449
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION          1..449
note = XENP36960 Heavy Chain - C6-30[CLDN6]_H1.22
source          1..449
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 477
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
NQKFKQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIL YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLISRTP ETCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VENESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 478    moltype = AA length = 214
FEATURE          Location/Qualifiers
REGION          1..214
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION          1..214
note = XENP36960 Light Chain - C6-30[CLDN6]_L1.189
source          1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 478
DIQMTQSPSS LSASVGDVRT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSKD YFTTISLQP EDIATYYCQQ YWSGPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 479    moltype = AA length = 449
FEATURE          Location/Qualifiers
REGION          1..449
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION          1..449
note = XENP36968 Heavy Chain - C6-30[CLDN6]_H2.3
source          1..449
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 479
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDSNNGNTHY 60
NQKFKQGHVTI SVDKSIATY LQWSSLKASD TAMYVCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLISRTP ETCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VENESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 480    moltype = AA length = 214
FEATURE          Location/Qualifiers
REGION          1..214
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION          1..214
note = XENP36968 Light Chain - C6-30[CLDN6]_L1.189
source          1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 480

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DIQMTQSPSS LSASVGDVRT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YTFTISSLQP EDIATYYCQQ YWSGPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 481 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..449
 note = XENP36963 Heavy Chain - C6-30[CLDN6]_H1.9
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 481
 QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNANTHY 60
 NQKFGQGRVTI TVDKSASTAY MELSSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 482 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP36963 Light Chain - C6-30[CLDN6]_L1.189
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 482
 DIQMTQSPSS LSASVGDVRT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YTFTISSLQP EDIATYYCQQ YWSGPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 483 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..449
 note = XENP36964 Heavy Chain - C6-30[CLDN6]_H2.3
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 483
 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDSNNGNTHY 60
 NQKFGQGHVTI SVDKSISTAY LQWSSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 484 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP36964 Light Chain - C6-30[CLDN6]_L1.187
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 484
 DIQMTQSPSS LSASVGDVRT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YTFTISSLQP EDIATYYCQQ YWSAPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 485 moltype = AA length = 449
 FEATURE Location/Qualifiers

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REGION 1..449
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..449
note = XENP36965 Heavy Chain - C6-30[CLDN6]_H2.12

source 1..449
mol_type = protein
organism = synthetic construct

SEQUENCE: 485

EVQLVQSGAE VKKPGESLRI	CKTSGYDFT EYTMHWVRQM	PGKSLEWMGG IDPNNGNTHY	60
NQKFQGHVTI SVDKSISTAY	LQWSSLKASD TAMYYCARIY	YFGRLYFDFW GAGTLVTVSS	120
ASTKGPSVFP LAPSSKSTSG	GTAALGCLVK DYFPEPVTVS	WNSGALTSGV HTFPAVLQSS	180
GLYSLSSVVT VPSSSLGTQT	YICNVNHHKPS NTKVDKKVEP	KSCDKTHTCP PCPAPPVAGP	240
SVFLFPPKPK DTLNISRTPE	VTCVVVDVKH EDPEVKFNWY	VDGVEVHNAK TKPREEQYNS	300
TYRVSVLTV LHQDWLNGKE	YKCKVSNKAL PAPIEKTISK	AKGQPREPQV YTLPPSREEM	360
TKNQVSLTCL VKGFYPSDIA	VEWESNGQPE NNYKTTPPVL	DSGGSFFLYS KLTVDKSRWQ	420
QGNVFSCSVM HEALHNHYTQ	KSLSLSPGK		449

SEQ ID NO: 486 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214
note = XENP36965 Light Chain - C6-30[CLDN6]_L1.187

source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 486

DIQMTQSPSS LSASVGRVDT	ITCQASEDIY NRLAWYQQKP	GKVPKLLISG ATSLETGVPS	60
RFSGSGSGKD YFTTISLQP	EDIATYYCQQ YWSAPLTFGG	GTKVEIKRTV AAPSVFIFPP	120
SDEQLKSGTA SVVCLLNIFY	PREAKVQWKV DNALQSGNSQ	ESVTEQDSKD STYLSSTLT	180
LSKADYEKKH VYACEVTHQG	LSSPVTKSFN RGEC		214

SEQ ID NO: 487 moltype = AA length = 449
FEATURE Location/Qualifiers
REGION 1..449
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..449
note = XENP36957 Heavy Chain - C6-30[CLDN6]_H1.24

source 1..449
mol_type = protein
organism = synthetic construct

SEQUENCE: 487

QVQLVQSGAE VKKPGASVKV	CKTSGYTFT EYTMHWVRQA	PGQSLEWMGG IDPNNGNTHY	60
NQKFQGRVTI TVDKSASTAY	MELSSLRSED TAVYYCARIY	YLGRLYFDFW GAGTLVTVSS	120
ASTKGPSVFP LAPSSKSTSG	GTAALGCLVK DYFPEPVTVS	WNSGALTSGV HTFPAVLQSS	180
GLYSLSSVVT VPSSSLGTQT	YICNVNHHKPS NTKVDKKVEP	KSCDKTHTCP PCPAPPVAGP	240
SVFLFPPKPK DTLNISRTPE	VTCVVVDVKH EDPEVKFNWY	VDGVEVHNAK TKPREEQYNS	300
TYRVSVLTV LHQDWLNGKE	YKCKVSNKAL PAPIEKTISK	AKGQPREPQV YTLPPSREEM	360
TKNQVSLTCL VKGFYPSDIA	VEWESNGQPE NNYKTTPPVL	DSGGSFFLYS KLTVDKSRWQ	420
QGNVFSCSVM HEALHNHYTQ	KSLSLSPGK		449

SEQ ID NO: 488 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214
note = XENP36957 Light Chain - C6-30[CLDN6]_L1.187

source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 488

DIQMTQSPSS LSASVGRVDT	ITCQASEDIY NRLAWYQQKP	GKVPKLLISG ATSLETGVPS	60
RFSGSGSGKD YFTTISLQP	EDIATYYCQQ YWSAPLTFGG	GTKVEIKRTV AAPSVFIFPP	120
SDEQLKSGTA SVVCLLNIFY	PREAKVQWKV DNALQSGNSQ	ESVTEQDSKD STYLSSTLT	180
LSKADYEKKH VYACEVTHQG	LSSPVTKSFN RGEC		214

SEQ ID NO: 489 moltype = AA length = 449
FEATURE Location/Qualifiers
REGION 1..449
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..449
note = XENP36958 Heavy Chain - C6-30[CLDN6]_H1.19

source 1..449
mol_type = protein

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organism = synthetic construct

SEQUENCE: 489
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
NQKFQGRVTI TVDKSASTAY MELSSLRSED TATYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLISRTP ETCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 490      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION              1..214
                    note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..214
                    note = XENP36958 Light Chain - C6-30[CLDN6]_L1.187
source              1..214
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 490
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSGKD YFTTISLQP EDIATYYCQQ YWSAPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 491      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION              1..449
                    note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..449
                    note = XENP36959 Heavy Chain - C6-30[CLDN6]_H1.9
source              1..449
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 491
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNANTHY 60
NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLISRTP ETCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 492      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION              1..214
                    note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..214
                    note = XENP36959 Light Chain - C6-30[CLDN6]_L1.187
source              1..214
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 492
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSGKD YFTTISLQP EDIATYYCQQ YWSAPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 493      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION              1..449
                    note = Description of Artificial Sequence: Synthetic
                      polypeptide
REGION              1..449
                    note = XENP36961 Heavy Chain - C6-30[CLDN6]_H1.24
source              1..449
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 493
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YLGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLISRTP ETCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300

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TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 494 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP36961 Light Chain - C6-30[CLDN6]_L1.189
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 494
 DIQMTQSPSS LSASVGDRTV ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFGSGSGSKD YFTFTISLQP EDIATYYCQQ YWSGPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 495 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..449
 note = XENP36962 Heavy Chain - C6-30[CLDN6]_H1.19
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 495
 QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
 NQKFQGRVTI TVDKSASTAY MELSSLRSED TATYYCARIY YFGRLYPFDW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 496 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP36962 Light Chain - C6-30[CLDN6]_L1.189
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 496
 DIQMTQSPSS LSASVGDRTV ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFGSGSGSKD YFTFTISLQP EDIATYYCQQ YWSGPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK STYLSSTLT 180
 LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 497 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..449
 note = XENP36966 Heavy Chain - C6-30[CLDN6]_H2.91
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 497
 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNFNTHY 60
 NQKFQGHVTI SVDKSIATY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 498 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214

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note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..214
note = XENP36966 Light Chain - C6-30[CLDN6]_L1.187
source 1..214
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 498
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGGKD YFTTISLQP EDIATYYCQQ YWSAPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEHKK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 499      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION 1..449
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..449
note = XENP36967 Heavy Chain - C6-30[CLDN6]_H2.118
source 1..449
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 499
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
NQQKQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYFDWF GAGTLTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSVVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 500      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..214
note = XENP36967 Light Chain - C6-30[CLDN6]_L1.187
source 1..214
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 500
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGGKD YFTTISLQP EDIATYYCQQ YWSAPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEHKK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 501      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION 1..449
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..449
note = XENP36969 Heavy Chain - C6-30[CLDN6]_H2.12
source 1..449
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 501
EVQLVQSGAE VKKPGESLRI SCKTSGYDFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
NQQKQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYFDWF GAGTLTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSVVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
QGNVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 502      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..214
note = XENP36969 Light Chain - C6-30[CLDN6]_L1.189
source 1..214
      mol_type = protein
      organism = synthetic construct

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SEQUENCE: 502
DIQMTQSPSS LSASVGDRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSKD YFTTISLQP EDIATYYCQQ YWSGPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 503      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION            1..449
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..449
                  note = XENP36970 Heavy Chain - C6-30[CLDN6]_H2.91
source            1..449
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 503
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNFNTHY 60
NQKFKQGHVTI SVDKSIATAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLISRTPTE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDVLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ 420
QQNVFSCSVH HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 504      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP36970 Light Chain - C6-30[CLDN6]_L1.189
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 504
DIQMTQSPSS LSASVGDRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSKD YFTTISLQP EDIATYYCQQ YWSGPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 505      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION            1..449
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..449
                  note = XENP36972 Heavy Chain - C6-30[CLDN6]_H1
source            1..449
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 505
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
NQKFKQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLISRTPTE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
TYRVVSVLTV LHQDVLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTPPVVL DSDGSFFLYS KLTVDKSRWQ 420
QQNVFSCSVH HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 506      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP36972 Light Chain - C6-30[CLDN6]_L1.187
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 506
DIQMTQSPSS LSASVGDRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSKD YFTTISLQP EDIATYYCQQ YWSAPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSLT 180
LSKADYKHKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 507      moltype = AA length = 449

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FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..449
 note = XENP36971 Heavy Chain - C6-30[CLDN6]_H2.118
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 507
 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
 NQKKQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSVVTV VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVSVSLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 508 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP36971 Light Chain - C6-30[CLDN6]_L1.189
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 508
 DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFGSGSGGKD YFTTISLQPD EDIATYYCQQ YWSGPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 509 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..449
 note = XENP36973 Heavy Chain - C6-30[CLDN6]_H1
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 509
 QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
 NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSVVTV VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEQYNS 300
 TYRVSVSLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCL VKGFYPSDIA VEWESNGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWQ 420
 QGNVFSCSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 510 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP36973 Light Chain - C6-30[CLDN6]_L1.189
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 510
 DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFGSGSGGKD YFTTISLQPD EDIATYYCQQ YWSGPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 511 moltype = AA length = 446
 FEATURE Location/Qualifiers
 REGION 1..446
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..446
 note = XENP26863 Heavy Chain - mAb206-LCC[CLDN6]_H0
 source 1..446

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mol_type = protein
organism = synthetic construct

SEQUENCE: 511
EVQLQQSGPE LVKPGASMKI SCKASGYSFT GYTMNWVKQS HGKNLEWIGL INPYNGGTIY 60
NQKFKGKATL TVDKSSSTAY MELLSLTSED SAVYYCARDY GFVLDYWGQG TTLTVSSAST 120
KGPSVFPLAP SSKSTSGGTA ALGCLVKDYF PEPVTVSWNS GALTSGVHTF PAVLQSSGLY 180
SLSSVVTGPS SSGLTQTYIC NVNHKPSNTK VDKKVEPKSC DKHTCTPPCP APPVAGPSVF 240
LFPKPKDTL MISRTPEVTC VVVDVKHEDP EVKFNWYVDG VEVHNAKTKP REEQYNSTYR 300
VSVLTVLHQ DWLNGKEYKC KVSINKALPAP IEKTISKAKG QPREPQVYTL PPSREEMTKN 360
QVSLTCLVKG FYPDSIAVEW ESNQGPENNY KTTTPVLDSD GSFFLYSKLT VDKSRWQQGN 420
VFSCSVMHEA LHNHYTQKSL SLSPGK 446

SEQ ID NO: 512      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP26863 Light Chain - mAb206-LCC[CLDN6]_L0
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 512
QIVLTQSPAI MSASPGEKVT ITCASSSSVS YLHWFQQKPG TSPKLWVYST SNLPSGVPAR 60
FGSGSGSTSY SLTISRMEAE DAATYYCQQR SIYPPWTFGG GTKLEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 513      moltype = AA length = 446
FEATURE            Location/Qualifiers
REGION             1..446
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..446
                   note = XENP26849 Chain 1 - mAb206-LCC[CLDN6]_H0_IgG1_pI(-)_
                   Isosteric_A_/PVA_/S267K/L368D/K370S
source             1..446
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 513
EVQLQQSGPE LVKPGASMKI SCKASGYSFT GYTMNWVKQS HGKNLEWIGL INPYNGGTIY 60
NQKFKGKATL TVDKSSSTAY MELLSLTSED SAVYYCARDY GFVLDYWGQG TTLTVSSAST 120
KGPSVFPLAP SSKSTSGGTA ALGCLVKDYF PEPVTVSWNS GALTSGVHTF PAVLQSSGLY 180
SLSSVVTGPS SSGLTQTYIC NVNHKPSDTK VDKKVEPKSC DKHTCTPPCP APPVAGPSVF 240
LFPKPKDTL MISRTPEVTC VVVDVKHEDP EVKFNWYVDG VEVHNAKTKP REEQYNSTYR 300
VSVLTVLHQ DWLNGKEYKC KVSINKALPAP IEKTISKAKG QPREPQVYTL PPSREEMTKN 360
QVSLTCDVSG FYPDSIAVEW ESDGQPENNY KTTTPVLDSD GSFFLYSKLT VDKSRWEQGD 420
VFSCSVMHEA LHNHYTQKSL SLSPGK 446

SEQ ID NO: 514      moltype = AA length = 485
FEATURE            Location/Qualifiers
REGION             1..485
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..485
                   note = XENP26849 Chain 2 -
                   [ANTI-CD3]_H1.30_L1.47_scFv (GKPGS)4_Fc (216)
                   _IgG1_C220S/PVA_/S267K/S364K/E357Q
source             1..485
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 514
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWVRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLR AEDTAVYYCVR HGNFGDSYVS WFAYWGQGT 120
VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSTLTVSPG GTVTLTCGSS TGAVTTSNYA 180
NHWVQKPGKS PRGLIGGTNK RAPGVPARFS GSLLGKKAAL TISGAQPEDE ADYYCALWYS 240
NHWVFGGKTK LTVLEPKSSD KTHCTPPCPA PPVAGPSVFL FPKPKDTLM ISRTPEVTCV 300
VVDVKHEDPE VKFNWYVDGV EVHNAKTKP REEQYNSTYRV VSVLTVLHQD WLNKEYKCK 360
VSNKALPAPI EKTISKAKGQ PREPQVYTL PPSREEMTKNQ VKLTCLVKG FYPDSIAVEWE 420
SNGQPENNYK TTPPVLDSDG SFPLYSKLTV DKSRWQQGNV FSCSVMHEAL HNHYTQKSLS 480
LSPGK 485

SEQ ID NO: 515      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP26849 Chain 3 - mAb206-LCC[CLDN6]_L0

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source          1..214
                mol_type = protein
                organism = synthetic construct

SEQUENCE: 515
QIVLTQSPAI MSASPGEKVT ITCASSSSVS YLHWFQQKPG TSPKLWVYST SNLPSGVPAR 60
FGSGSGSTSY SLTISRMEAE DAATYYCQQR SIYPWPWTFGG GTKLEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSPN RGEC 214

SEQ ID NO: 516      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION            1..449
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..449
                  note = XENP34229 Chain 1 - C6-30[CLDN6]_H1_IgG1_PVA_/S267K
                  IgG1_pI(-) _Isosteric_A_PVA_/S267K/L368D/K370S
source            1..449
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 516
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLIEWMGG IDPNNGNTHY 60
NQKFPQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLTVTSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSTVTP VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMSRTPTE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCD VSGFYPSDIA VEWESDQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 420
QGDVFSQSVSM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 517      moltype = AA length = 485
FEATURE            Location/Qualifiers
REGION            1..485
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..485
                  note = XENP34229 Chain 2 -
                  [ANTI-CD3]_L1.47_H1.30_scFv(GKPGS)4_Fc(216)
                  _IgG1_C220S/PVA_/S267K/S364K/E357Q
source            1..485
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 517
QAVVTQEPSEL TVSPGGTVTL TCGSSTGAVT TSNYANWVQQ KPGKSPRGLI GGTNKRAPGV 60
PARFSGSLLG GKAALTISGA QPEDEADYYC ALWYSNHWVF GGGTKLTVLG KPGSGKPGSG 120
KPGSGKPGSE VQLVESGGGL VQPGGSLRLS CAASGFTFST YAMNWRQAP GKGLEWVGRI 180
RSKYNNYATY YADSVKGRFT ISRDDSKNTL YLQMNSLRAE DTAVYYCVRH GNFGDSYVSW 240
FAYWQGQTLV TVSSEPKSSD KTHTCPPCPA PPVAGPSVFL FPPKPKDTLM ISRTPEVTCV 300
VVDVKHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV VSVLTVLHQD WLNKEYKCK 360
VSNKALPAPI EKTISKAKGQ PREPQVYTL PPSREQMTKNQ VKLTCVLKGF YPSDIAVEWE 420
SNGQPENNYK TTPPVLDSDG SFFLYSKLTV DKSRWQQGNV FSCSVMEAL HNHYTQKSL 480
LSPGK 485

SEQ ID NO: 518      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP34229 Chain 3 - C6-30[CLDN6]_L1
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 518
DIQMTQSPSS LSASVGRDVT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSKD YTFITSLQEP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSSLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSPN RGEC 214

SEQ ID NO: 519      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION            1..449
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..449
                  note = XENP34637 Chain 1 -
                  C6-30[CLDN6]_H2_L1_IgG1_pI(-) _Isosteric_A
                  _PVA_/S267K/L368D/K370S
source            1..449

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mol_type = protein
organism = synthetic construct

SEQUENCE: 519
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYVCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWE 420
QGDVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 520      moltype = AA length = 485
FEATURE            Location/Qualifiers
REGION             1..485
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION             1..485
                    note = XENP34637 Chain 2 -
                    [ANTI-CD3]_H1.30_L1.47_scFv (GKPGS) 4_Fc (216)
                    _IgG1_C220S/PVA_/S267K/S364K/E357Q
source             1..485
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 520
EVQLVESGGG LVQPGGSLRL SCAASGFTFS TYAMNWRQA PGKGLEWVGR IRSKYNNYAT 60
YYADSVKGRF TISRDDSKNT LYLQMNSLRA EDTAVYVCVR HGNFGDSYVS WFAYWGQGTL 120
VTVSSGKPGS GKPGSGKPGS GKPGSQAVVT QEPSLTVSPG GTVTLTCGSS TGAVTTSNYA 180
NWVQQKPGKS PRGLIGGTNK RAPGVPARFS GSLLGGKAAL TISGAQPEDE ADYYCALWYS 240
NHWVFGGGTK LTVLEPKSSD KTHTCPPCPA PPVAGPSVFL FPPKPKDTLM ISRTPEVTCV 300
VVDVKHEDPE VKFNWYVDGV EVHNAKTKPR EEQYNSTYRV VSVLTVLHQD WLNGKEYKCK 360
VSNKALPAPI EKTISKAKGQ PREPQVYTL PPSREQMTKNQ VKLTCLVKGK YPSDIAVEWE 420
SNGQPENNYK TTPPVLDSG SFFLYSKLTV DKSRWQQGNV FSCSVMHEAL HNHYTKSLS 480
LSPGK 485

SEQ ID NO: 521      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION             1..214
                    note = XENP34637 Chain 3 - C6-30[CLDN6]_L1
source             1..214
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 521
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFSGSGSGKD YFTTISLQEP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 522      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION             1..449
                    note = XENP35385 Chain 1 - C6-30[CLDN6]_H1_IgG1_PVA_/S267K
                    IgG1_pI(-) _Isosteric_A_PVA_/S267K/L368D/K370S
source             1..449
                    mol_type = protein
                    organism = synthetic construct

SEQUENCE: 522
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYVCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWE 420
QGDVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 523      moltype = AA length = 485
FEATURE            Location/Qualifiers
REGION             1..485
                    note = Description of Artificial Sequence: Synthetic
                    polypeptide
REGION             1..485
                    note = XENP35385 Chain 2 -

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[ANTI-CD3]_L1.47_H1.32_scFv(GKPGS)4_Fc(216)
_IgG1_C220S/PVA_/S267K/S364K/E357Q

source 1..485
mol_type = protein
organism = synthetic construct

SEQUENCE: 523

QAVVTQEP	SL	TVSPGGTVTL	TCGSSTGAVT	TSNYANWVQQ	KPGKSPRGLI	GGTNKRAPGV	60
PARFSGSL	LLG	GKAALTISGA	QPEDEADYYC	ALWYSNHVWF	GGGKLTVLG	KPGSGKPGSG	120
KPGSGKPG	SGE	VQLVESGGGL	VQPGGSLRLS	CAASGFTFST	YAMNWVRQAP	GKGLEWVGRI	180
RSKANNYAT	Y	YADSVKGRFT	ISRDDSKNTL	YLQMNSLRAE	DTAVYYCVRH	GNFGDSYVSW	240
FAYWGQGT	LV	TVSSEPKSSD	KTHTCPPCPA	PPVAGPSVFL	FPPKPKDTLM	ISRTPEVTCV	300
VVDVKHED	PE	VKFNWYVDGV	EVHNAKTKPR	EEQYNSTYRV	VSVLTVLHQD	WLNKEYKCK	360
VSNKALPAP	I	EKTISKAKGQ	PREPQVYTL	PSREQMTKNQ	VKLTCLVKGF	YPSDIAVEWE	420
SNGQPENNY	K	TTPPVLDSDG	SFFLYSKLTV	DKSRWQQGNV	FSCSVMHEAL	HNHYTQKSLS	480
LSPGK							485

SEQ ID NO: 524 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..214
note = XENP35385 Chain 3 - C6-30[CLDN6]_L1
source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 524

DIQMTQSP	SS	LSASVGDRT	ITQASEDIY	NRLAWYQQKP	GKVPKLLISG	ATSLETGVPS	60
RFSGSGSG	KD	YTFTISSLPQ	EDIATYYCQQ	YWSSPLTFGG	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSG	TA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYEKK	H	VYACEVTHQG	LSSPVTKSFN	RGEC			214

SEQ ID NO: 525 moltype = AA length = 449
FEATURE Location/Qualifiers
REGION 1..449
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..449
note = XENP35387 Chain 1 - C6-30[CLDN6]_H2_IgG1_PVA_/S267K_IgG1_pI(-)_Isosteric_A_PVA_/S267K/L368D/K370S
source 1..449
mol_type = protein
organism = synthetic construct

SEQUENCE: 525

EVQLVQSG	AE	VKPKGESLRI	CKTSGYTFT	EYTMHWVRQM	PGKSLEWMGG	IDPNNGNTHY	60
NQKFQGHV	TI	SVDKSISTAY	LQWSSLKASD	TAMYVCARIY	YFGRLYPFDW	GAGTLVTVSS	120
ASTKGPSV	FP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTPPAVLQSS	180
GLYSLSSV	VT	VPSSSLGTQT	YICNVNHKPS	DTKVDKKEVP	KSCDKTHTCP	PCPAPPVAGP	240
SVFLFPPK	PK	DTLMISRTP	YTCVVVDVKH	EDPEVKFNWY	VDGVEVHNAK	TKPREEEYNS	300
TYRVVSVL	TV	LHQDWLNKE	YKCKVSNKAL	PAPIEKTISK	AKGQPREPQV	YTLPPSREEM	360
TKNQVSLT	CD	VSGFYPSDIA	VEWESDGP	QPE	NNYKTTPPVL	DSGGSFFLYS	420
QGDVFS	CSVM	HEALHNHYTQ	KSLSLSPGK				449

SEQ ID NO: 526 moltype = AA length = 485
FEATURE Location/Qualifiers
REGION 1..485
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..485
note = XENP35387 Chain 2 - [ANTI-CD3]_L1.47_H1.32_scFv(GKPGS)4_Fc(216)_IgG1_C220S/PVA_/S267K/S364K/E357Q
source 1..485
mol_type = protein
organism = synthetic construct

SEQUENCE: 526

QAVVTQEP	SL	TVSPGGTVTL	TCGSSTGAVT	TSNYANWVQQ	KPGKSPRGLI	GGTNKRAPGV	60
PARFSGSL	LLG	GKAALTISGA	QPEDEADYYC	ALWYSNHVWF	GGGKLTVLG	KPGSGKPGSG	120
KPGSGKPG	SGE	VQLVESGGGL	VQPGGSLRLS	CAASGFTFST	YAMNWVRQAP	GKGLEWVGRI	180
RSKANNYAT	Y	YADSVKGRFT	ISRDDSKNTL	YLQMNSLRAE	DTAVYYCVRH	GNFGDSYVSW	240
FAYWGQGT	LV	TVSSEPKSSD	KTHTCPPCPA	PPVAGPSVFL	FPPKPKDTLM	ISRTPEVTCV	300
VVDVKHED	PE	VKFNWYVDGV	EVHNAKTKPR	EEQYNSTYRV	VSVLTVLHQD	WLNKEYKCK	360
VSNKALPAP	I	EKTISKAKGQ	PREPQVYTL	PSREQMTKNQ	VKLTCLVKGF	YPSDIAVEWE	420
SNGQPENNY	K	TTPPVLDSDG	SFFLYSKLTV	DKSRWQQGNV	FSCSVMHEAL	HNHYTQKSLS	480
LSPGK							485

SEQ ID NO: 527 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214

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note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..214
note = XENP35387 Chain 3 - C6-30[CLDN6]_L1
source 1..214
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 527
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 528      moltype = AA length = 449
FEATURE            Location/Qualifiers
REGION 1..449
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..449
note = XENP34233 Chain 1 -
      C6-30[CLDN6]_H1_IgG1_pI(-)_Isosteric_A_PVA
      _/S267K/L368D/K370S
source 1..449
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 528
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
NOKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 300
TYRVVSVLTV LHQDNLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCD VSGFYPSDIA VENESDGPPE NNYKTPPVL DSDGSFFLYS KLTVDKSRWE 420
QGDVLFSCVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 529      moltype = AA length = 722
FEATURE            Location/Qualifiers
REGION 1..722
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..722
note = XENP34233 Chain 2 -
      C6-30[CLDN6]_H1_Fab_(G4S)2_[ANTI-CD3]_L1.47
      _H1.30_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E3
      57Q
source 1..722
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 529
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
NOKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCGGGGSGG GGSQAVVTQE 240
PSLTVSPGGT VTLTCGSGTG AVTTSNYANW VQQKPGKSPR GLIGGTNKRK PGVPARFSGS 300
LLGGAALTI SGAQPEDEAD YYCALWYSNH WVFGGGKLT VLGKPGSGKP GSGKPGSGKP 360
GSEVQLVESG GGLVQPGGSL RLSCAASGFT FSTYAMNWVR QAPGKGLEWV GRIRSKYNNY 420
ATYYADSVKG RFTISRDSK NTLYLQMSNL RAEDTAVYYC VRHGNFGDSY VSWFAYWQGQ 480
TLVTVSSGGG GSGGGGSKTH TCPPCPAPPV AGPSVFLFPP KPKDTLMISR TPEVTCVVVD 540
VKHEDPEVKF NWYVDGVEVH NAKTKPREEQ YNSTYRVVSV LTVLHQDNLN GKEYKCKVSN 600
KALPAPIEKT ISKAKGQPRE PQVYTLPPSR EQMTKNQVKL TCLVKGFYPS DIAVEWESNG 660
QPENNYKTPP PVLDSGGSFF LYSKLTVDKS RWQQGNVFSC SVMHEALHNNH YTKSLSLSP 720
GK 722

SEQ ID NO: 530      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic
      polypeptide
REGION 1..214
note = XENP34233 Chain 3 - C6-30[CLDN6]_H1
source 1..214
      mol_type = protein
      organism = synthetic construct

SEQUENCE: 530
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

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SEQ ID NO: 531      moltype = AA  length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..449
                   note = XENP34638 Chain 1 -
                   C6-30[CLDN6]_H2_IgG1_pI(-)_Isosteric_A_PVA
                   _/S267K/L368D/K370S
source             1..449
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 531
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
NQKFGQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 180
GLYSLSVVTV VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 300
TYRVSIVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCD VSGFYPSDIA VEWESDGPQE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWE 420
QGDVFSQSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 532      moltype = AA  length = 722
FEATURE            Location/Qualifiers
REGION             1..722
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..722
                   note = XENP34638 Chain 2 -
                   C6-30[CLDN6]_H2_(G4S)2_[ANTI-CD3]_L1.47_H1.30
                   _scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q
source             1..722
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 532
EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
NQKFGQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 180
GLYSLSVVTV VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCGGGGSGG GGSQAVVTQE 240
PSLTVSPGGT VTLTCGSSTG AVTTSNYANW VQOKPGKSPR GLIGGTNKRA PGVPARFSGS 300
LLGGKAALTI SGAQPEDEAD YYCALWYSNH WVFGGGTKLT VLGKPGSGKP GSGKPGSGKP 360
GSEVQLVESG GGLVQPGGSL RLSCAASGFT FSTYAMNWVR QAPGKGLEWV GRIRSKYNNY 420
ATYYADSVKG RFTISRDDSK NTLYLQMNSL RAEDTAVYYC VRHGNFGDSY VSWPAYWGQG 480
TLVTVSSGGG GSGGGGSKTH TCPPCPAPPV AGPSVFLPPP KPKDTLMISR TPEVTCVVVD 540
VKHEDPEVKF NQYVDGVEVH NAKTKPREEQ YNSTYRVVSV LTVLHQDWLN GKEYKCKVSN 600
KALPAPIEKT ISKAKGQPRE PQVYTLPPSR EQMTKNQVKL TCLVKGFYPS DIAVEWESNG 660
QPENNYKTPP PVLDSGGSFF LYSKLTVDKS RWQQGNVFSC SVMHEALHNN YTQKSLSLSP 720
GK 722

SEQ ID NO: 533      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP34638 Chain 3 - C6-30[CLDN6]_H2
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 533
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGGKD YTFTISSLPQ EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 534      moltype = AA  length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..449
                   note = XENP35386 Chain 1 -
                   C6-30[CLDN6]_H1_IgG1_pI(-)_Isosteric_A
                   _PVA_/S267K/L368D/K370S
source             1..449
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 534
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLWMGG IDPNNGNTHY 60

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NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPDFW GAGTLTVVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTP ETCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 420
 QGDVFCSCVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 535 moltype = AA length = 722
 FEATURE Location/Qualifiers
 REGION 1..722
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..722
 note = XENP35386 Chain 2 -
 C6-30[CLDN6]_H1_L1_Fab_(G4S)2_Fc(222)
 _IgG1_PVA_/S267K/S364K/E357Q
 source 1..722
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 535
 QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
 NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPDFW GAGTLTVVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCGGGSGG GGSQAVVTQE 240
 PSLTVSPGGT VTLTCGSSTG AVTTSNYANW VQKKPGKSPR GLIGGTNKRA PGVPAFSGS 300
 LLGGKAALTI SGAQPEDEAD YYCALWYSNH WVFGGGT KLT VLGKPGSGKP GSGKPGSGKP 360
 GSEVQLVESG GGLVQPGGSL RLSCAASGFT FSTYAMNWRV QAPGKGLEWV GRIRSKANNY 420
 ATYYADSVKG RFTISRDDSK NTLYLQMNLS RAEDTAVYYC VRHGNFGDSY VSWFAYWGQG 480
 TLVTVSSGGG GSGGGGSKTH TCPPCPAPPV AGPSVFLFPP KPKDTLMISR TPEVTCVVVD 540
 VKHEDPEVKF NWYVDGVVEV NAKTKPREEQ YNSTYRVVSV LTVLHQDWLN GKEYCKVSN 600
 KALPAPIEKT ISKAKGQPRE PQVYTLPPSR EQMTKNQVKL TCLVKGFYPS DIAVEWESNG 660
 QPENNYKTPP PVLDSGGSFF LYSKLTVDKS RWQQGNVFSC SVMHEALHNN YTQKSLSLSP 720
 GK 722

SEQ ID NO: 536 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP35386 Chain 3 - C6-30[CLDN6]_H1
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 536
 DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RPSGSGSGKD YFTFTISLQP EDIATYYCQQ YWSSPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 537 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..449
 note = XENP35388 Chain 1 -
 C6-30[CLDN6]_H2_IgG1_pI(-)_Isosteric
 _A_PVA_/S267K/L368D/K370S
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 537
 EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY 60
 NQKFQGHVTI SVDKSIATAY LQWSSLKASD TAMYYCARIY YFGRLYPDFW GAGTLTVVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTP ETCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 420
 QGDVFCSCVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 538 moltype = AA length = 722
 FEATURE Location/Qualifiers
 REGION 1..722
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..722

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note = XENP35388 Chain 2 -
C6-30[CLDN6]_H2_L1_Fab_(G4S)2_Fc(222)_IgG1
PVA/S267K/S364K/E357Q

source 1..722
mol_type = protein
organism = synthetic construct

SEQUENCE: 538

EVQLVQSGAE VKKPGESLRI SCKTSGYTFT EYTMHWVRQM PGKSLEWMGG IDPNNGNTHY	60
NQKFQGHVTI SVDKSISTAY LQWSSLKASD TAMYYCARIY YFGRLYPFDW GAGTLVTVSS	120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS	180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCGGGGSGG GGSQAVVTQE	240
PSLTVSPGGT VTLTCGSSTG AVTTSNYANW VQOKPGKSPR GLIGGTNKRA PGVPARFSGS	300
LLGGKAALTI SGAQPEDEAD YYCALWYSNH WVFGGGTKLT VLGKPGSGKP GSGKPGSGKP	360
GSEVQLVESG GGLVQPGGSL RLSCAASGFT FSTYAMNWVR QAPGKGLEWV GRIRSKANNY	420
ATYYADSVKG RFTISRDSK NTLYLQMNLS RAEDTAVYYC VRHGNFGDSY VSWFAYWGQG	480
TLVTVSSGGG GSGGGGSKTH TCPPCPAPPV AGPSVFLFPP KPKDTLMISR TPEVTCVVVD	540
VKHEDPEVKF NWWVDGVEVH NAKTKPREEQ YNSTYRVVSV LTVLHQDWLN GKEYCKKVSN	600
KALPAPIEKT ISKAKGQPRE PQVYTLPPSR EQMTKNQVKL TCLVKGFYPS DIAVEWESNG	660
QPENNYKTPP PVLDSGDSFF LYSKLTVDKS RWQQGNVFSC SVMHEALHNN YTQKSLSLSP	720
GK	722

SEQ ID NO: 539 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..214
note = XENP35388 Chain 3 - C6-30[CLDN6]_H2
source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 539

DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS	60
RFGSGSGGKD YFTTISLQP EDIATYYCQQ YWSSPLTEGG GTKVEIKRTV AAPSVFIFPP	120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT	180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEC	214

SEQ ID NO: 540 moltype = AA length = 449
FEATURE Location/Qualifiers
REGION 1..449
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..449
note = XENP37227 Chain 1 -
C6-30[CLDN6]_H1.9_IgG1(-)_Isosteric_
_A_PVA_/S267K/L368D/K370S
source 1..449
mol_type = protein
organism = synthetic construct

SEQUENCE: 540

QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNANTHY	60
NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLVTVSS	120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS	180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP	240
SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS	300
TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM	360
TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE	420
QGDVFCSCVM HEALHNNHYTQ KSLSLSPGK	449

SEQ ID NO: 541 moltype = AA length = 722
FEATURE Location/Qualifiers
REGION 1..722
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..722
note = XENP37227 Chain 2 -
C6-30[CLDN6]_H1.9_(G4S)2_[ANTI-CD3]_L1.47
_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E3
57Q
source 1..722
mol_type = protein
organism = synthetic construct

SEQUENCE: 541

QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNANTHY	60
NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLVTVSS	120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPFAVLQSS	180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCGGGGSGG GGSQAVVTQE	240
PSLTVSPGGT VTLTCGSSTG AVTTSNYANW VQOKPGKSPR GLIGGTNKRA PGVPARFSGS	300
LLGGKAALTI SGAQPEDEAD YYCALWYSNH WVFGGGTKLT VLGKPGSGKP GSGKPGSGKP	360

-continued

GSEVQLVESG GGLVQPGGSL RLSCAASGFT FSTYAMNWVR QAPGKGLEWV GRIRSKANNY 420
 ATYYADSVKG RFTISRDDSK NTLYLQMNSL RAEDTAVYYC VRHGNFGDSY VSWFAYWGQG 480
 TLVTVSSGGG GSGGGGSKTH TCPPCAPPV AGPSVFLFPP KPKDTLMISR TPEVTCVVVD 540
 VKHEDPEVKF NWYVDGVEVH NAKTKPREEQ YNSTYRVVSV LTVLHQDWLN GKEYCKKVS 600
 KALPAPIEKT ISKAKGQPRE PQVYTLPPSR EQMTKNQVKL TCLVKGFYPS DIAVEWESNG 660
 QPENNYKTTT PVLDSDGSFF LYSKLTVDKS RWQQGNVFSC SVMHEALHNNH YTKSLSLSP 720
 GK 722

SEQ ID NO: 542 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP37227 Chain 3 - C6-30[CLDN6]_H1.9
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 542
 DIQMTQSPSS LSASVGDRVT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFGSGSGSKD YFTTISLQP EDIATYYCQQ YWSAPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNMFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSPN RGEC 214

SEQ ID NO: 543 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..449
 note = XENP37228 Chain 1 -
 C6-30[CLDN6]_H1.19_IgG1_pI(-)_Isosteric
 _A_PVA_/S267K/L368D/K370S
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 543
 QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
 NQKFPQGRVTI TVDKSASTAY MELSSLRSED TATYYCARIY YFGRLYFDFW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPPKF DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWE 420
 QGDVFSCSVM HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 544 moltype = AA length = 722
 FEATURE Location/Qualifiers
 REGION 1..722
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..722
 note = XENP37228 Chain 2 -
 C6-30[CLDN6]_H1.19_(G4S)2_[ANTI-CD3]_L1.47
 _H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E3
 57Q
 source 1..722
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 544
 QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
 NQKFPQGRVTI TVDKSASTAY MELSSLRSED TATYYCARIY YFGRLYFDFW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCGGGGSGG GGSQAVVTQE 240
 PSLTVSPGGT VTLTCSSTG AVTTSNYANW VQQKPGKSPR GLIGGTNKRA PGVPARFSGS 300
 LLGGKAALTI SGAQPEDEAD YYCALWYSNH WVFGGGT KLT VLGKPGSGKP GSGKPGSGKP 360
 GSEVQLVESG GGLVQPGGSL RLSCAASGFT FSTYAMNWVR QAPGKGLEWV GRIRSKANNY 420
 ATYYADSVKG RFTISRDDSK NTLYLQMNSL RAEDTAVYYC VRHGNFGDSY VSWFAYWGQG 480
 TLVTVSSGGG GSGGGGSKTH TCPPCAPPV AGPSVFLFPP KPKDTLMISR TPEVTCVVVD 540
 VKHEDPEVKF NWYVDGVEVH NAKTKPREEQ YNSTYRVVSV LTVLHQDWLN GKEYCKKVS 600
 KALPAPIEKT ISKAKGQPRE PQVYTLPPSR EQMTKNQVKL TCLVKGFYPS DIAVEWESNG 660
 QPENNYKTTT PVLDSDGSFF LYSKLTVDKS RWQQGNVFSC SVMHEALHNNH YTKSLSLSP 720
 GK 722

SEQ ID NO: 545 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide

-continued

REGION 1..214
 note = XENP37228 Chain 3 - C6-30[CLDN6]_H1.19
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 545
 DIQMTQSPSS LSASVGDVRT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSAPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 546 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..449
 note = XENP37229 Chain 1 -
 C6-30[CLDN6]_H1.22_IgG1_pI(-)_Isosteric
 _A_PVA_/S267K/L368D/K370S

source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 546
 QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
 NQKPFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIL YFGRLYFDFW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTPE VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 300
 TYRVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 420
 QGDVFSQSVN HEALHNHYTQ KSLSLSPGK 449

SEQ ID NO: 547 moltype = AA length = 722
 FEATURE Location/Qualifiers
 REGION 1..722
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..722
 note = XENP37229 Chain 2 -
 C6-30[CLDN6]_H1.22_(G4S)2_[ANTI-CD3]_L1.47
 _H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E3
 57Q

source 1..722
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 547
 QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
 NQKPFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIL YFGRLYFDFW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTPPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCGGGGSGG GGSQAVVTQE 240
 PSLTVSPGGT VTLTCGSGTG AVTTSNYANW VQKPKGKSPR GLIGGTNKRA PGVPARFSGS 300
 LLGGKAALTI SGAQPEDEAD YYCALWYSNH WVFGGGTKLT VLGKPGSGKP GSGKPGSGKP 360
 GSEVQLVESG GGLVQPGGSL RLSCAASGFT FSTYAMNWVR QAPGKGLEWV GRIRSKANNY 420
 ATYYADSVKG RFTISRDDSK NTLYLQMNSL RAEDTAVYYC VRHGNFGDSY VSWFAYWGQG 480
 TLVTVSSGGG GSGGGGSKTH TCPPCPAPPV AGPSVFLFPP KPKDTLMISR TPEVTCVVVD 540
 VKHEDPEVKF NWYVDGVEVH NAKTKPREEQ YNSTYRVSV LTVLHQDWLN GKEYKCKVSN 600
 KALPAPIEKT ISKAKGQPRE PQVYTLPPSR EQMTKNQVKL TCLVKGFYPS DIAVEWESNG 660
 QPENNYKTPP PVLSDGSFF LYSKLTVDKS RWQQGNVFSC SVMHEALHNN YTQKSLSLSP 720
 GK 722

SEQ ID NO: 548 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214
 note = XENP37229 Chain 3 - C6-30[CLDN6]_H1.22

source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 548
 DIQMTQSPSS LSASVGDVRT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RFSGSGSGKD YFTTISLQP EDIATYYCQQ YWSAPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 549 moltype = AA length = 449
 FEATURE Location/Qualifiers

-continued

REGION 1..449
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..449
note = XENP37230 Chain 1 -
C6-30[CLDN6]_H1.22_IgG1_pI(-)_Isosteric
_A_PVA_/S267K/L368D/K370S

source 1..449
mol_type = protein
organism = synthetic construct

SEQUENCE: 549

QVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTMHWVRQA	PGQSLEWMGG	IDPNNGNTHY	60
NQKFQGRVTI	TVDKSASTAY	MELSSLRSED	TAVYYCARIL	YFGRLYPFDW	GAGTLVTVSS	120
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	DTKVDKKVEP	KSCDKHTTCP	PCPAPPVAGP	240
SVFLPPPKPK	DTLMISRTPE	VTCVVVDVKH	EDPEVKFNWY	VDGVEVHNAK	TKPREEEYNS	300
TYRVVSVLTV	LHQDNLNGKE	YKCKVSNKAL	PAPIEKTISK	AKGQPREPQV	YTLPPSREEM	360
TKNQVSLTCD	VSGFYPSDIA	VEWESDGPQE	NNYKTPPVVL	DSGGSFFLYS	KLTVDKSRWE	420
QGDVFCSSVM	HEALHNHYTQ	KSLSLSPGK				449

SEQ ID NO: 550 moltype = AA length = 722
FEATURE Location/Qualifiers
REGION 1..722
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..722
note = XENP37230 Chain 2 -
C6-30[CLDN6]_H1.22_(G4S)2_[ANTI-CD3]_L1.47
_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q

source 1..722
mol_type = protein
organism = synthetic construct

SEQUENCE: 550

QVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTMHWVRQA	PGQSLEWMGG	IDPNNGNTHY	60
NQKFQGRVTI	TVDKSASTAY	MELSSLRSED	TAVYYCARIL	YFGRLYPFDW	GAGTLVTVSS	120
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	NTKVDKKVEP	KSCGGGGSGG	GGSQAVVTQE	240
PSLTVSPGGT	VTLTCSSTG	AVTTSNYANW	VQQKPGKSPR	GLIGGTNKR	PGVPARFSGS	300
LLGGKAALTI	SGAQPEDEAD	YYCALWYSNH	WVFGGGTKLT	VLGKPGSGKP	SGKPGSGGKP	360
GSEVQLVESG	GGLVQPGGSL	RLSCAASGFT	FSTYAMNWVR	QAPGKGLEWV	GRIRSKANNY	420
ATYYADSVKG	RFTISRDDSK	NTLYLQMNSL	RAEDTAVYYC	VRHGNFGDSY	VSWFAYWGQG	480
TLVTVSSGGG	GSGGGGSKTH	TCPPCPAPPV	AGPSVFLFPP	KPKDTLMISR	TPEVTCVVVD	540
VKHEDPEVKF	NWYVDGVEVH	NAKTKPREEQ	YNSTYRVVSV	LTVLHQDWLN	GKEYCKCKVSN	600
KALPAPIEKT	ISKAKGQPRE	PQVYTLPPSR	EQMTKNQVKL	TCLVKGFYPS	DIAREWESNG	660
QPENNYKTTT	PVLDSGGSFF	LYSKLTVDKS	RWQQGNVFSC	SVMHEALHNH	YTQKSLSLSP	720
GK						722

SEQ ID NO: 551 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214
note = XENP37230 Chain 3 - C6-30[CLDN6]_H1.22

source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 551

DIQMTQSPSS	LSASVGRDVT	ITCQASEDIY	NRLAWYQQKP	GKVPKLLISG	ATSLETGVPS	60
RFSGSGSGKD	YTFTISSLQP	EDIATYYCQQ	YWSGPLTFGG	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYEKKH	VYACEVTHQG	LSSPVTKSFN	RGEC			214

SEQ ID NO: 552 moltype = AA length = 449
FEATURE Location/Qualifiers
REGION 1..449
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..449
note = XENP37231 Chain 1 -
C6-30[CLDN6]_H1.24_IgG1_pI(-)_Isosteric
_A_PVA_/S267K/L368D/K370S

source 1..449
mol_type = protein
organism = synthetic construct

SEQUENCE: 552

QVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTMHWVRQA	PGQSLEWMGG	IDPNNGNTHY	60
NQKFQGRVTI	TVDKSASTAY	MELSSLRSED	TAVYYCARIY	YLGRLYFDFW	GAGTLVTVSS	120

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ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSSVVT	VPSSSLGTQT	YICNVNHNKPS	DTKVDKKVEP	KSCDKTHTCP	PCPAPPVAGP	240
SVFLFPPKPK	DTLMISRTP	VTCTVVVDVKH	EDPEVKFNWY	VDGVEVHNAK	TKPREEEYNS	300
TYRVVSVLTV	LHQDWLNGKE	YKCKVSNKAL	PAPIEKTISK	AKGQPREPQV	YTLPPSREEM	360
TKNQVSLTCD	VSGFYPSDIA	VEWESDGQPE	NNYKTPPVVL	DSDGSFFLYS	KLTVDKSRWE	420
QGDVFSQSV	HEALHNHYTQ	KSLSLSPGK				449

SEQ ID NO: 553 moltype = AA length = 722
 FEATURE Location/Qualifiers
 REGION 1..722
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..722
 note = XENP37231 Chain 2 -
 C6-30[CLDN6]_H1.24_(G4S)2_[ANTI-CD3]_L1.47
 _H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E3
 57Q
 source 1..722
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 553

QVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTMHWVRQA	PGQSLEWMGG	IDPNNGNTHY	60
NQKFPQGRVTI	TVDKSASTAY	MELSSSLRSED	TAVYYCARIY	YLGRLYFDFW	GAGTLTVTSS	120
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSSVVT	VPSSSLGTQT	YICNVNHNKPS	NTKVDKKVEP	KSCGGGSGSG	GGQAQVVTQE	240
PSLTVSPGGT	VTLTCSSTG	AVTTSNYANW	VQQKPGKSPR	GLIGGTNKRA	PGVPAFSGS	300
LLGGKAALTI	SGAQPEDEAD	YYCALWYSNH	WVFGGGTKLT	VLGKPGSGKP	SGKPGSGKP	360
GSEVQLVESG	GGLVQPGGSL	RLSCAASGFT	FSTYAMNWVR	QAPGKGLEWV	GRIRSKANNY	420
ATYYADSVKG	RFTISRDDSK	NTLYLQMNSL	RAEDTAVYYC	VRHGNFGDSY	VSWFAYWGQG	480
TLVTYSSGGG	GSGGGGSKTH	TCPPCPAPPV	AGPSVFLFPP	KPKDTLMISR	TPEVTCVVVD	540
VKHEDPEVKF	NWYVDGVEVH	NAKTKPREEQ	YNSTYRVVSV	LTVLHQDWLN	GKEYCKKVS	600
KALPAPIEKT	ISKAKGQPRE	PQVYTLPPSR	EQMTKNQVKL	TCLVKGFYPS	DIAVEWESNG	660
QPENNYKTP	PVLDSGGSFF	LYSKLTVDKS	RWQQGNVFSC	SVMHEALHNN	YTQKSLSLSP	720
GK						722

SEQ ID NO: 554 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..214
 note = XENP37231 Chain 3 - C6-30[CLDN6]_H1.24
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 554

DIQMTQSPSS	LSASVGRDVT	ITCQASEDIY	NRLAWYQQKP	GKVPKLLISG	ATSLETGVPS	60
RFSGSGSGKD	YTFTISSLQP	EDIATYYCQQ	YWSAPLTFGG	GTKVEIKRTV	AAPSVPFIPP	120
SDEQLKSGTA	SVVCLLNNFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYSLSSLT	180
LSKADYEKHK	VYACEVTHQG	LSSPVTKSFN	RGEC			214

SEQ ID NO: 555 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..449
 note = XENP37232 Chain 1 -
 C6-30[CLDN6]_H1.24_IgG1_pI(-)_Isosteric
 _A_PVA_/S267K/L368D/K370S
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 555

QVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTMHWVRQA	PGQSLEWMGG	IDPNNGNTHY	60
NQKFPQGRVTI	TVDKSASTAY	MELSSSLRSED	TAVYYCARIY	YLGRLYFDFW	GAGTLTVTSS	120
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSSVVT	VPSSSLGTQT	YICNVNHNKPS	DTKVDKKVEP	KSCDKTHTCP	PCPAPPVAGP	240
SVFLFPPKPK	DTLMISRTP	VTCTVVVDVKH	EDPEVKFNWY	VDGVEVHNAK	TKPREEEYNS	300
TYRVVSVLTV	LHQDWLNGKE	YKCKVSNKAL	PAPIEKTISK	AKGQPREPQV	YTLPPSREEM	360
TKNQVSLTCD	VSGFYPSDIA	VEWESDGQPE	NNYKTPPVVL	DSDGSFFLYS	KLTVDKSRWE	420
QGDVFSQSV	HEALHNHYTQ	KSLSLSPGK				449

SEQ ID NO: 556 moltype = AA length = 722
 FEATURE Location/Qualifiers
 REGION 1..722
 note = Description of Artificial Sequence: Synthetic
 polypeptide
 REGION 1..722

-continued

note = XENP37232 Chain 2 -
C6-30[CLDN6]_H1.24_(G4S)2_[ANTI-CD3]_L1.47
_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E3
57Q

source 1..722
mol_type = protein
organism = synthetic construct

SEQUENCE: 556

QVQLVQSGAE	VKKPGASVKV	SCKTSGYTFT	EYTMHWVRQA	PGQSLEWMGG	IDPNNGNTHY	60
NQKFQGRVTI	TVDKSASTAY	MELSSLRSED	TAVYYCARIY	YLGRLYFDWF	GAGTLVTVSS	120
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	NTKVDKKVEP	KSCGGGGSGG	GGSQAVVTQE	240
PSLTVSPGGT	VTLTCGSSTG	AVTTSNYANW	VQQKPGKSPR	GLIGGTNKRA	PGVPARFSGS	300
LLGGKAALTI	SGAQPEDEAD	YYCALWYSNH	WVFGGGTKLT	VLGKPGSGKP	GSGKPGSGKP	360
GSEVQLVESG	GGLVQPGGSL	RLSCAASGFT	FSTYAMNWVR	QAPKGLEWV	GRIRSKANNY	420
ATYYADSVKG	RFTISRDDSK	NTLYLQMNSL	RAEDTAVYYC	VRHGNFGDSY	VSWFAYWGQG	480
TLVTVSSGGG	GSGGGGSKTH	TCPPCPAPPV	AGPSVFLPPP	KPKDTLMISR	TPEVTCVVVD	540
VKHEDPEVKF	NWYVDGVEVH	NAKTKPREEQ	YNSTYRVVSV	LTVLHQDWLN	GKEYKCKVSN	600
KALPAPIEKT	ISKAKGQPRE	PQYVTLPPSR	EQMTKNQVKL	TCLVKGFYPS	DIAVEWESNG	660
QPENNYKTTT	PVLDSGGSFF	LYSKLTVDKS	RWQQGNVFSC	SVMHEALHNN	YTQKSLSLSP	720
GK						722

SEQ ID NO: 557 moltype = AA length = 214
FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..214
note = XENP37232 Chain 3 - C6-30[CLDN6]_H1.24
source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 557

DIQMTQSPSS	LSASVGRVIT	ITCQASEDIY	NRLAWYQQKP	GKVPKLLISG	ATSLETGVPS	60
RFGSGSGGKD	YTFTISSLQP	EDIATYYCQQ	YWSGPLTFGG	GTKVEIKRTV	AAPSVFIFPP	120
SDEQLKSGTA	SVVCLLNIFY	PREAKVQWKV	DNALQSGNSQ	ESVTEQDSKD	STYLSSTLT	180
LSKADYEKHK	VYACEVTHQG	LSSPVTKSFN	RGEC			214

SEQ ID NO: 558 moltype = AA length = 449
FEATURE Location/Qualifiers
REGION 1..449
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..449
note = XENP37233 Chain 1 -
C6-30[CLDN6]_H2.91_IgG1_pI(-)_Isosteric
_A_PVA_/S267K/L368D/K370S
source 1..449
mol_type = protein
organism = synthetic construct

SEQUENCE: 558

EVQLVQSGAE	VKKPGESLRI	SCKTSGYTFT	EYTMHWVRQM	PGKSLEWMGG	IDPNNFNTHY	60
NQKFQGHVTI	SVDKSISTAY	LQWSSLKASD	TAMYYCARIY	YFGRLYFDWF	GAGTLVTVSS	120
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	DTKVDKKVEP	KSCDKTHTCP	PCPAPPVAGP	240
SVFLFPPKPK	DTLMISRTPE	VTCVVVDVKH	EDPEVKFNWY	VDGVEVHNAK	TKPREEEYNS	300
TYRVSVLTV	LHQDWLNGKE	YKCKVSNKAL	PAPIEKTISK	AKGQPREPQV	YTLPPSREEM	360
TKNQVSLTCD	VSGFYPSDIA	VEWESDGQPE	NNYKTTTPVL	DSGGSFFLYS	KLTVDKSRWE	420
QGDVFSQSV	HEALHNNHTQ	KSLSLSPGK				449

SEQ ID NO: 559 moltype = AA length = 722
FEATURE Location/Qualifiers
REGION 1..722
note = Description of Artificial Sequence: Synthetic polypeptide
REGION 1..722
note = XENP37233 Chain 2 -
C6-30[CLDN6]_H2.91_(G4S)2_[ANTI-CD3]_L1.47
_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E3
57Q
source 1..722
mol_type = protein
organism = synthetic construct

SEQUENCE: 559

EVQLVQSGAE	VKKPGESLRI	SCKTSGYTFT	EYTMHWVRQM	PGKSLEWMGG	IDPNNFNTHY	60
NQKFQGHVTI	SVDKSISTAY	LQWSSLKASD	TAMYYCARIY	YFGRLYFDWF	GAGTLVTVSS	120
ASTKGPSVFP	LAPSSKSTSG	GTAALGCLVK	DYFPEPVTVS	WNSGALTSGV	HTFPAVLQSS	180
GLYSLSSVVT	VPSSSLGTQT	YICNVNHKPS	NTKVDKKVEP	KSCGGGGSGG	GGSQAVVTQE	240
PSLTVSPGGT	VTLTCGSSTG	AVTTSNYANW	VQQKPGKSPR	GLIGGTNKRA	PGVPARFSGS	300

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LLGGKAALTI  SGAQPEDEAD  YYCALWYSNH  WVFGGGTKLT  VLGKPGSGKP  GSGKPGSGKP  360
GSEVQLVESG  GGLVQPGGSL  RLSCAASGFT  FSTYAMNWVR  QAPGKGLEWV  GRIRSKANNY  420
ATYYADSVKG  RFTISRDDSK  NTLYLQMNLS  RAEDTAVYYC  VRHGNFGDSY  VSWFAYWGQG  480
TLVTVSSGGG  GSGGGGSKTH  TCPPCPAPPV  AGPSVFLFPP  KPKDTLMISR  TPEVTCVVVD  540
VKHEDPEVKF  NWWVDGVEVH  NAKTKPREEQ  YNSTYRVVSV  LTVLHQDWLN  GKEYKCKVSN  600
KALPAPIEKT  ISKAKGQPRE  PQVYTLPPSR  EQMTKNQVKL  TCLVKGFYPS  DIAVEWESNG  660
QPENNYKTTT  PVLDSGDSFF  LYSKLTVDKS  RWQQGNVFSC  SVMHEALHSH  YTQKSLSLSP  720
GK                                                    722

SEQ ID NO: 560      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION             1..214
                  note = XENP37233 Chain 3 - C6-30[CLDN6]_H2.91
source             1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 560
DIQMTQSPSS  LSASVGDVRT  ITCQASEDIY  NRLAWYQQKP  GKVPKLLISG  ATSLETGVPS  60
RFSGSGSGKD  YFTTISLQEP  EDIATYYCQQ  YWSAPLTFGG  GTKVEIKRTV  AAPSVFIFPP  120
SDEQLKSGTA  SVVCLLNIFY  PREAKVQWKV  DNALQSGNSQ  ESVTEQDSKD  STYLSSTLT  180
LSKADYEKKH  VYACEVTHQG  LSSPVTKSFN  RGEC                    214

SEQ ID NO: 561      moltype = AA  length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION             1..449
                  note = XENP37547 Chain 1 -
                  C6-30[CLDN6]_H2.91_IgG1_pI(-)_Isosteric
                  _A_PVA_/S267K/L368D/K370S
source             1..449
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 561
EVQLVQSGAE  VKKPGESLRI  SCKTSGYTFT  EYTMHWVRQM  PGKSLEWMGG  IDPNNFNTHY  60
NQKFQGHVTI  SVDKSIATAY  LQWSSLKASD  TAMYYCARIY  YFGRLYPFDW  GAGTLVTVSS  120
ASTKGPSVFP  LAPSSKSTSG  GTAALGCLVK  DYFPEPVTVS  WNSGALTSGV  HTPPAVLQSS  180
GLYSLSSVVT  VPSSSLGTQT  YICNVNHKPS  DTKVDKKVEP  KSCDKTHTCP  PCPAPPVAGP  240
SVFLPPPKPK  DTLMISRTPE  VTCVVVDVKH  EDPEVKFNWY  VDGVEVHNAK  TKPREEEYNS  300
TYRVSVSLTV  LHQDWLNGKE  YKCKVSNKAL  PAPIEKTISK  AKGQPREPQV  YTLPPSREEM  360
TKNQVSLTCD  VSGFYPSPDIA  VEWESDQPE  NNYKTTPPVL  DSDGSFFLYS  KLTVDKSRWE  420
QGDVFSQSVL  HEALHSHYTQ  KSLSLSPGK                    449

SEQ ID NO: 562      moltype = AA  length = 722
FEATURE            Location/Qualifiers
REGION             1..722
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION             1..722
                  note = XENP37547 Chain 2 -
                  C6-30[CLDN6]_H2.91_(G4S)2_[anti-CD3]_L1.47_
                  H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_
                  _/S267K/S364K/E357Q/M428L/N434S
source             1..722
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 562
EVQLVQSGAE  VKKPGESLRI  SCKTSGYTFT  EYTMHWVRQM  PGKSLEWMGG  IDPNNFNTHY  60
NQKFQGHVTI  SVDKSIATAY  LQWSSLKASD  TAMYYCARIY  YFGRLYPFDW  GAGTLVTVSS  120
ASTKGPSVFP  LAPSSKSTSG  GTAALGCLVK  DYFPEPVTVS  WNSGALTSGV  HTPPAVLQSS  180
GLYSLSSVVT  VPSSSLGTQT  YICNVNHKPS  NTKVDKKVEP  KSCGGGSGSG  GGSQAVVTQE  240
PSLTVSPGGT  VTLTCGSSTG  AVTTSNYANW  VQQKPGKSPR  GLIGGTNKRA  PGVPARFSGS  300
LLGGKAALTI  SGAQPEDEAD  YYCALWYSNH  WVFGGGTKLT  VLGKPGSGKP  GSGKPGSGKP  360
GSEVQLVESG  GGLVQPGGSL  RLSCAASGFT  FSTYAMNWVR  QAPGKGLEWV  GRIRSKANNY  420
ATYYADSVKG  RFTISRDDSK  NTLYLQMNLS  RAEDTAVYYC  VRHGNFGDSY  VSWFAYWGQG  480
TLVTVSSGGG  GSGGGGSKTH  TCPPCPAPPV  AGPSVFLFPP  KPKDTLMISR  TPEVTCVVVD  540
VKHEDPEVKF  NWWVDGVEVH  NAKTKPREEQ  YNSTYRVVSV  LTVLHQDWLN  GKEYKCKVSN  600
KALPAPIEKT  ISKAKGQPRE  PQVYTLPPSR  EQMTKNQVKL  TCLVKGFYPS  DIAVEWESNG  660
QPENNYKTTT  PVLDSGDSFF  LYSKLTVDKS  RWQQGNVFSC  SVLHEALHSH  YTQKSLSLSP  720
GK                                                    722

SEQ ID NO: 563      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                  note = Description of Artificial Sequence: Synthetic

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REGION          polypeptide
                  1..214
note = XENP37547 Chain 3 - C6-30[CLDN6]_H2.91
source          1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 563
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSKD YFTFTISLQP EDIATYYCQQ YWSAPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGECC 214

SEQ ID NO: 564    moltype = AA length = 449
FEATURE          Location/Qualifiers
REGION          1..449
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION          1..449
note = XENP37545 Chain 1 -
                  C6-30[CLDN6]_H1.24_IgG1_pI(-)_Isosteric_A
                  _PVA_/S267K/L368D/K370S
source          1..449
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 564
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
NQKPFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YLGRLYFDFW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
SVFLFPPKPK DTLMISRTP VTCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 300
TYRVSVELTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
TKNQVSLTCD VSGFYPSDIA VEWESDGGQE NNYKTTTPVL DSDGSFFLYS KLTVDKSRWE 420
QGDVFSCSV LHEALSHYTQ KSLSLSPGK 449

SEQ ID NO: 565    moltype = AA length = 722
FEATURE          Location/Qualifiers
REGION          1..722
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION          1..722
note = XENP37545 Chain 2 -
                  C6-30[CLDN6]_H1.24_(G4S)2_[anti-CD3]_L1.47_H1
                  .32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/
                  S267K/S364K/E357Q/M428L/N434S
source          1..722
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 565
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNGNTHY 60
NQKPFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YLGRLYFDFW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCGGGSGG GGSQAVVTQE 240
PSLTVSPGGT VTLTCGSSGT AVTTSNYANW VQQKPGKSPR GLIGGTNKRA PGVPARFSGS 300
LLGGKAALTI SGAQPEDEAD YYCALWYSNH WVFVGGGTKLT VLGKPGSGKP GSGKPGSGKP 360
GSEVQLVESG GGLVQPGGSL RLSCAASGFT FSTYAMNWVR QAPGKGLEWV GRIRSKANNY 420
ATYYADSVKG RFTISRDDSK NTLYLQMNSL RAEDTAVYYC VRHGNFGDSY VSWFAYWGQG 480
TLVTVSSGGG GSGGGGSKTH TCPPCPAPPV AGPSVFLFPP KPKDTLMISR TPEVTCVVVD 540
VKHEDPEVKF NNYVDGVEVH NAKTKPREEQ YNSTYRVVSV LTVLHQDWLN GKEYKCKVSN 600
KALPAPIEKT ISKAKGQPRE PQVYTLPPSR EQMTKNQVKL TCLVKGFYPS DIAVEWESNG 660
QPENNYKTPP PVLDSGGSFF LYSKLTVDKS RWQQGNVFSC SVLHEALHSH YTKSLSLSP 720
GK 722

SEQ ID NO: 566    moltype = AA length = 214
FEATURE          Location/Qualifiers
REGION          1..214
note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION          1..214
note = XENP37545 Chain 3 - C6-30[CLDN6]_H1.24
source          1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 566
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGSKD YFTFTISLQP EDIATYYCQQ YWSAPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGECC 214

SEQ ID NO: 567    moltype = AA length = 449

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FEATURE Location/Qualifiers
REGION 1..449
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..449
note = XENP37541 Chain 1 - C6-30[CLDN6]_H1.9_IgG1_pI(-)_Isosteric_A_PVA_/S267K/L368D/K370S

source 1..449
mol_type = protein
organism = synthetic construct

SEQUENCE: 567

QVQLVQSGAE VKKPGASVKV	CKTSGYTFT EYTMHWVRQA	PGQSLWMMGG IDPNNANTHY	60
NQKFQGRVTI TVDKSASTAY	MELSSLRSED TAVYYCARIY	YFGRLYPFDW GAGTLVTVSS	120
ASTKGPSVFP LAPSSKSTSG	GTAALGCLVK DYFPEPVTVS	WNSGALTSGV HTPPAVLQSS	180
GLYSLSSVVT VPSSSLGTQT	YICNVNHKPS DTKVDKKVEP	KSCDKHTTCP PCPAPPVAGP	240
SVFLFPPKPK DTLMISRTPE	VTCVVVDVKH EDPEVKFNWY	VDGVEVHNAK TKPREEEYNS	300
TYRVVSVLTV LHQDNLNGKE	YKCKVSNKAL PAPIEKTISK	AKGQPREPQV YTLPPSREEM	360
TKNQVSLTCD VSGFYPSDIA	VEWESDGQPE NNYKTTTPVL	DSGGSFFLYS KLTVDKSRWE	420
QGDVFSQSVL HEALHSHYTQ	KSLSLSPGK		449

SEQ ID NO: 568 moltype = AA length = 722

FEATURE Location/Qualifiers
REGION 1..722
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..722
note = XENP37541 Chain 2 - C6-30[CLDN6]_H1.9_(G4S)2_[anti-CD3]_L1.47_H1.32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q/M428L/N434S

source 1..722
mol_type = protein
organism = synthetic construct

SEQUENCE: 568

QVQLVQSGAE VKKPGASVKV	CKTSGYTFT EYTMHWVRQA	PGQSLWMMGG IDPNNANTHY	60
NQKFQGRVTI TVDKSASTAY	MELSSLRSED TAVYYCARIY	YFGRLYPFDW GAGTLVTVSS	120
ASTKGPSVFP LAPSSKSTSG	GTAALGCLVK DYFPEPVTVS	WNSGALTSGV HTPPAVLQSS	180
GLYSLSSVVT VPSSSLGTQT	YICNVNHKPS NTKVDKKVEP	KSCGGGGSGG GGSQAVVTQE	240
PSLTVSPGGT VTLTCGSSTG	AVTTSNYANW VQKPKGKSPR	GLIGGTNKRA PGVPARFSGS	300
LLGGKAALTI SGAQPEDEAD	YYCALWYSNH WVFGGGTKLT	VLGKPGSGKP GSGKPGSGKP	360
GSEVQLVESG GGLVQPGGSL	RLSCAASGFT FSTYAMNWVR	QAPGKGLEWV GRIRSKANNY	420
ATYYADSVKG RFTISRDDSK	NLTLYLQMNSL RAEDTAVYYC	VRHGNFGDSY VSWFAYWGQG	480
TLVTVSSGGG GSGGGGSKTH	TCPPCPAPPV AGPSVFLPPP	KPKDTLMISR TPEVTCVVVD	540
VKHEDPEVKF NWYVDGVEVH	NAKTKPREEQ YNSTYRVVSV	LTVLHQDNLN GKEYCKKVSN	600
KALPAPIEKT ISKAKGQPRE	PQYVTLPPSR EQMTKNQVKL	TCLVKGFYPS DIAVEWESNG	660
QPENNYKTPP PVLDSGGSFF	LYSKLTVDKS RWQQGNVFSC	SVLHEALHSH YTKSLSLSP	720
GK			722

SEQ ID NO: 569 moltype = AA length = 214

FEATURE Location/Qualifiers
REGION 1..214
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..214
note = XENP37541 Chain 3 - C6-30[CLDN6]_H1.9

source 1..214
mol_type = protein
organism = synthetic construct

SEQUENCE: 569

DIQMTQSPSS LSASVGRVIT	ITCQASEDIY NRLAWYQQKP	GKVPKLLISG ATSLETGVPS	60
RFGSGSGGKD YTFTISSLPQ	EDIATYYCQQ YWSAPLTFGG	GTKVEIKRTV AAPSVFIFPP	120
SDEQLKSGTA SVVCLLNIFY	PREAKVQWKV DNALQSGNSQ	ESVTEQDSKD STYLSSTLT	180
LSKADYEKHK VYACEVTHQG	LSSPVTKSFN RGEC		214

SEQ ID NO: 570 moltype = AA length = 449

FEATURE Location/Qualifiers
REGION 1..449
note = Description of Artificial Sequence: Synthetic polypeptide

REGION 1..449
note = XENP37630 Chain 1 - C6-30[CLDN6]_H1.9_IgG1_pI(-)_Isosteric_A_PVA_/S267K/L368D/K370S

source 1..449
mol_type = protein
organism = synthetic construct

SEQUENCE: 570

QVQLVQSGAE VKKPGASVKV	CKTSGYTFT EYTMHWVRQA	PGQSLWMMGG IDPNNANTHY	60
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NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPDFW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTP ETCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 420
 QGDVFCSCVM HEALHNYHTQ KSLSLSPGK 449

SEQ ID NO: 571 moltype = AA length = 722
 FEATURE Location/Qualifiers
 REGION 1..722
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..722
 note = XENP37630 Chain 2 -
 C6-30[CLDN6]_H1.9_(G4S)2_[anti-CD3]_L1.47_H1.
 89_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q
 source 1..722
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 571
 QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNANTHY 60
 NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPDFW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCGGGSGG GGSQAVVTQE 240
 PSLTVSPGGT VTLTCSSTG AVTTSNYANW VQKPKGKSPR GLIGGTNKRA PGVPAFSGS 300
 LLGGKAALTI SGAQPEDEAD YYCALWYSNH WVFGGGT KLT VLKPGSGKPGSGK 360
 GSEVQLVESG GGLVQPGGSL RLSCAASGFT FSTYAMNWVR QAPGKGLEWV GRIRSKYNNY 420
 ATYYADSVKG RFTISRDSK NTLYLQMNLS RAEDTAVYYC VRHGNFGDEY VSWFAYWGQG 480
 TLVTYSSGGG GSGGGGSKTH TCPPCPAPPV AGPSVFLFPP KPKDTLMISR TPEVTCVVVD 540
 VKHEDPEVKF NWYVDGVEVH NAKTKPREEQ YNSTYRVVSV LTVLHQDWLN GKEYCKVSN 600
 KALPAPIEKT ISKAKGQPRE PQVYTLPPSR EQMTKNQVKL TCLVKGFYPS DIAVEWESNG 660
 QPENNYKTPP PVLDSGGSFF LYSKLTVDKS RWQQGNVFSC SVMHEALHNN YTQKSLSLSP 720
 GK 722

SEQ ID NO: 572 moltype = AA length = 214
 FEATURE Location/Qualifiers
 REGION 1..214
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..214
 note = XENP37630 Chain 3 - C6-30[CLDN6]_H1.9
 source 1..214
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 572
 DIQMTQSPSS LSASVGDRTV ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
 RPSGSGSGKD YTFITISLQP EDIATYYCQQ YWSAPLTFGG GTKVEIKRTV AAPSVFIFPP 120
 SDEQLKSGTA SVVCLLNNFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYSLSTLT 180
 LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEC 214

SEQ ID NO: 573 moltype = AA length = 449
 FEATURE Location/Qualifiers
 REGION 1..449
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..449
 note = XENP37634 Chain 1 -
 C6-30[CLDN6]_H1.9_IgG1_pI(-)_Isosteric_A_PVA
 _/S267K/L368D/K370S/M428L/N434S
 source 1..449
 mol_type = protein
 organism = synthetic construct

SEQUENCE: 573
 QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNANTHY 60
 NQKFQGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPDFW GAGTLVTVSS 120
 ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
 GLYSLSSVVT VPSSSLGTQT YICNVNHKPS DTKVDKKVEP KSCDKTHTCP PCPAPPVAGP 240
 SVFLFPPKPK DTLMISRTP ETCVVVDVKH EDPEVKFNWY VDGVEVHNAK TKPREEEYNS 300
 TYRVVSVLTV LHQDWLNGKE YKCKVSNKAL PAPIEKTISK AKGQPREPQV YTLPPSREEM 360
 TKNQVSLTCD VSGFYPSDIA VEWESDGQPE NNYKTTPPVL DSDGSFFLYS KLTVDKSRWE 420
 QGDVFCSCVL HEALHSHYTQ KSLSLSPGK 449

SEQ ID NO: 574 moltype = AA length = 722
 FEATURE Location/Qualifiers
 REGION 1..722
 note = Description of Artificial Sequence: Synthetic polypeptide
 REGION 1..722

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note = XENP37634 Chain 2 -
      C6-30[CLDN6]_H1.9_(G4S)2_[anti-CD3]_L1.47_H1.
      89_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q/
      M428L/N434S
source      1..722
            mol_type = protein
            organism = synthetic construct

SEQUENCE: 574
QVQLVQSGAE VKKPGASVKV SCKTSGYTFT EYTMHWVRQA PGQSLEWMGG IDPNNANTHY 60
NQKFKGRVTI TVDKSASTAY MELSSLRSED TAVYYCARIY YFGRLYPFDW GAGTLVTVSS 120
ASTKGPSVFP LAPSSKSTSG GTAALGCLVK DYFPEPVTVS WNSGALTSGV HTFPAVLQSS 180
GLYSLSSVVT VPSSSLGTQT YICNVNHKPS NTKVDKKVEP KSCGGGGSGG GGSQAVVTQE 240
PSLTVSPGGT VTLTCGSSTG AVTTSNYANW VQOKPGKSPR GLIGGTNKR PGVPAFSGS 300
LLGGKAALTI SGAQPEDEAD YYCALWYSNH WVFGGGKTLL VLGKPGSGKP GSGKPGSGKP 360
GSEVQLVESG GGLVQPGGSL RLSCAASGFT FSTYAMNWR QAPKGLEWV GRIRSKYNNY 420
ATYYADSVKG RFTISRDDSK NTLYLQMNSL RAEDTAVYYC VRHGNFGDEY VSWFAYWGQG 480
TLVTVSSGGG GSGGGGSKTH TCPPCPAPPV AGPSVFLPPP KPKDTLMISR TPEVTCVVVD 540
VKHEDPEVKF NWYVDGVEVH NAKTKPREEQ YNSTYRVVSV LTVLHQDWLN GKEYKCKVSN 600
KALPAPIEKT ISKAKGQPRE PQVYTLPPSR EQMTKNQVKL TCLVKGFYPS DIAVEWESNG 660
QPENNYKTTT PVLDSGGSFF LYSKLTVDKS RWQQGNVFSC SVLHEALHSH YTKSLSLSP 720
GK 722

SEQ ID NO: 575      moltype = AA length = 214
FEATURE            Location/Qualifiers
REGION            1..214
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..214
                  note = XENP37634 Chain 3 - C6-30[CLDN6]_H1.9
source            1..214
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 575
DIQMTQSPSS LSASVGRVIT ITCQASEDIY NRLAWYQQKP GKVPKLLISG ATSLETGVPS 60
RFGSGSGGKD YTFTISSLQP EDIATYYCQQ YWSAPLTFGG GTKVEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSKD STYLSSTLT 180
LSKADYEKHK VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 576      moltype = AA length = 723
FEATURE            Location/Qualifiers
REGION            1..723
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..723
                  note = XENP37217 Chain 1-
                  Comp_CH02[CLDN6]_H0_Fab_(G4S)2_SP34_L1.47_H1.
                  32_scFv(GKPGS)4_(G4S)2_Fc(222)_IgG1_PVA_/S267K/S364K/E357Q
source            1..723
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 576
EVQVEQSGLE LVKPGASVKI SCKTSGYTFT EYTMHWVKQS HGKSLEWIGG INPNNGNTRY 60
NQKFKDKATL TVDKSSRTAY MELHTLTSED SAVYYCARCG DYDLFFFAY WGQGLVTVTS 120
AASTKGPSVF PLAPSSKSTS GGTAALGCLV KDYFPEPVTV SWNSGALTSG VHTFPAVLQS 180
SGLYSLSSVV TVPSSSLGTQ TYICNVNHKP SNTKVDKKVE PKSCGGGGSG GGSQAVVTQ 240
EPSLTVSPGG TVTLTCGSST GAVTTSNYAN WVQOKPGKSP RGLIGGTNKR APGVPAFSG 300
SLLGGKAALT ISGAQPEDEA DYYCALWYSN HWVFGGKTLL TVLGKPGSGK PGSGKPGSGK 360
PGSEVQLVES GGLVQPGGSS LRLSCAASGF TFSTYAMNWR RQAPKGLEW VGRIRSKANN 420
YATYYADSVK GRFTISRDDS KNTLYLQMNS LRAEDTAVYY CVRHGNFGDS YVSWFAYWGQ 480
GTLVTVSSGG GSGGGGSKT HTCPPCPAPP VAGPSVFLFP KPKDTLMIS RTPEVTCVVV 540
DVKHEDPEVK FNWYVDGVEV HNAKTKPREE QNSTYRVVS VLTVLHQDWL NGKEYKCKVS 600
NKALPAPIEK TISKAKGQPR EPQVYTLPPS REQMTKNQVK LTCLVKGFYP SDIAVEWESN 660
GPENNYKTTT PPVLDSGGSF FLYSKLTVDK SRWQQGNVFS CSVMEALHN HYTKSLSLSP 720
PGK 723

SEQ ID NO: 577      moltype = AA length = 450
FEATURE            Location/Qualifiers
REGION            1..450
                  note = Description of Artificial Sequence: Synthetic
                  polypeptide
REGION            1..450
                  note = XENP37217_Comp_CH02[CLDN6]_H0_Fab
                  -IgG1_pI(-)_Isosteric_A_PVA_/S267KL368DK370S
source            1..450
                  mol_type = protein
                  organism = synthetic construct

SEQUENCE: 577
EVQVEQSGLE LVKPGASVKI SCKTSGYTFT EYTMHWVKQS HGKSLEWIGG INPNNGNTRY 60
NQKFKDKATL TVDKSSRTAY MELHTLTSED SAVYYCARCG DYDLFFFAY WGQGLVTVTS 120

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AASTKGPSVF PLAPSSKSTS GGTAALGCLV KDYFPEPVTV SWNSGALTSG VHTFPAVLQS 180
SGLYSLSSVV TVPSSSLGTQ TYICNVNHPK SDTKVDKKVE PKSCDKTHTC PPCPAPPVAG 240
PSVFLFPPPK KDTLMISRTPEVTCVVVDVK HEDPEVKFNW YVDGVEVHNA KTKPREEEYN 300
STYRVVSVLT VLHQDWLNGK EYKCKVSNKA LPAPIEKTIS KAKGQPREPQ VYTLPPSREE 360
MTKNQVSLTC DVSGFYPSDI AVEWESDQQP ENNYKTPPV LDSDGSFFLY SKLTVDKSRW 420
EQGDVFSCSV MHEALHNHYT QKSLSLSPGK 450

SEQ ID NO: 578      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP37217_Comp_CH02 [CLDN6]_L0_Light Chain
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 578
DIVMTQFQKF MSTSVGDRVS VTCKASQNVG TNVANYQQKP GQSPEALISS ASSRFSGVDP 60
RFTGSGSGTD FTLTITNVQS EDLADYFCQQ YNSFPFTFGS GTELEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEK 214

SEQ ID NO: 579      moltype = AA  length = 449
FEATURE            Location/Qualifiers
REGION             1..449
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..449
                   note = XENP38084 Chain 1-_Comp_CH02 [CLDN6]_H0_Heavy Chain
source             1..449
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 579
EVQVEQSGLE LVKPGASVKI SCKTSGYTFT EYTMHWVKQS HGKSLEWIGG INPNNGNTRY 60
NQKFKDKATL TVDKSSRTAY MELHTLTSED SAVYYCARCG DYDLFFFAY WGGTLVTVS 120
AASTKGPSVF PLAPSSKSTS GGTAALGCLV KDYFPEPVTV SWNSGALTSG VHTFPAVLQS 180
SGLYSLSSVV TVPSSSLGTQ TYICNVNHPK SNTKVDKKVE PKSCDKTHTC PPCPAPELRG 240
GPKVFLFPPK PKDTLMISRT PEVTCVVVDV SHEDPEVKFN WYVDGVEVHN AKTKPREEQY 300
ASTYRVVSVL TVLHQDWLNG KEYKCKVSNK ALPAPIEKTI SKAKGQPREP QVYTLPPSRE 360
EMTKNQVSLT CLVKGFIYPSD IAVEWESNGQ PENNYKTPPV YLSDGSFFLY YSKLTVDKSR 420
WQQGNVFSVS VMHEALHNHY TQESLSLSP 449

SEQ ID NO: 580      moltype = AA  length = 214
FEATURE            Location/Qualifiers
REGION             1..214
                   note = Description of Artificial Sequence: Synthetic
                   polypeptide
REGION             1..214
                   note = XENP38084 Chain 2-_Comp_CH02 [CLDN6]_H0_L0
source             1..214
                   mol_type = protein
                   organism = synthetic construct

SEQUENCE: 580
DIVMTQFQKF MSTSVGDRVS VTCKASQNVG TNVANYQQKP GQSPEALISS ASSRFSGVDP 60
RFTGSGSGTD FTLTITNVQS EDLADYFCQQ YNSFPFTFGS GTELEIKRTV AAPSVFIFPP 120
SDEQLKSGTA SVVCLLNIFY PREAKVQWKV DNALQSGNSQ ESVTEQDSK STYLSSTLT 180
LSKADYEKKH VYACEVTHQG LSSPVTKSFN RGEK 214

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What is claimed is:

1. A bispecific antibody comprising:

a) a first monomer comprising:

i) an anti-CD3 single chain variable fragment (scFv) comprising the amino acid sequence of SEQ ID NO: 98; and

ii) a first variant human IgG1 Fc domain,

b) a second monomer comprising:

i) a variable heavy domain (VH); and

ii) a second variant human IgG1 Fc domain; and

c) a light chain comprising a variable light domain (VL), wherein said VH and said VL form a means for binding an extracellular domain of human claudin 6 (CLDN6).

2. A bispecific antibody comprising:

a) a first monomer comprising, from N- to C-terminus, an anti-CD3 single chain variable fragment (scFv) com-

prising the amino acid sequence of SEQ ID NO: 98, and a first variant human IgG1 Fc domain;

b) a second monomer comprising, from N- to C-terminus, VH-CH1-hinge-second variant human IgG1 Fc domain; and

c) a light chain comprising a variable light domain (VL), wherein said VH and said VL form a means for binding an extracellular domain of human claudin 6 (CLDN6).

3. The bispecific antibody according to claim 1, wherein said first variant human IgG1 Fc domain comprises the amino acid substitutions T366S, L368A, and Y407V, and said second variant human IgG1 Fc domain comprises the amino acid substitution T366W, wherein numbering is according to the EU Index as in Kabat.

4. The bispecific antibody according to claim 2, wherein said first variant human IgG1 Fc domain comprises the

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amino acid substitutions T366S, L368A, and Y407V, and said second variant human IgG1 Fc domain comprises the amino acid substitution T366W, wherein numbering is according to the EU Index as in Kabat.

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